GALILEO FOC – SYSTEM SUPPORT

System Baseline Design Definition Document
Part 1: Logical Architecture

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CHANGE RECORDS

ISSUE	DATE	§ CHANGE RECORDS
1.0	29/10/2010	First version of the document for the FOC contract delivered as part of formal SQR1A-KP2 Milestone. This version is an evolution of former IOV DDF1 (ref. GAL-RPT-ALS-SYST-R-0063 issue 12) in line with FOC TRB v1.1.
1.1	14/03/2011	Version updated for FOC System REKP in line with FOC TRB v1.2
2.0	31/07/2015	Version updated for System DCDR addressing delta design changes from S-CDR onward as impacting logical architecture.
2.1draft	08/10/2015	Intermediate issue for SdCDR Readiness check-point.
2.1	20/11/2015	Update for System DCDR kick-off.
3.0	31/05/2016	Section 4.3.2: removal of blended mode CV/TWSTFT. Section 4.6.2.2: add of GAL-DCN-TASI-SYST-X-1009 in SFC5.2.1 description Section 4.4.2.5.1 updated as per SDCDR Action P1_NW-199 related to RID System-199 Section 4.6.3.2.1.2: update for removal of SIS OFF Line monitoring (not implemented in the MGF)
4.0	23/09/2016	Correction of some editorial issues, obsolete statements, misplacement of STRB2.0 tags, “Galileo Global Component” replaced by “Galileo Core Infrastructure” Removal of ERIS function (§5.6) Update of section 4.9.2, 4.8.3: Provision of Galileo orbital data for SAR user community (DCNs GAL-DCN-ASTG-SYST-X-14165, -14198, -1055)
5.0	28/09/2018	Diagrams from MODAF model added in §6, allocation of functional architecture to physical architecture. Update for DCN-1116 PHM configuration (STRB 2.0.4) Introduction of 7th TTC, LEOP in GCC Different further updates and corrections (Reference Documentation, etc).
6.0	11/03/2019	Update of Applicable Documentation Update of DCN implementation delayed to SB2.1 after GMS 3.0 CDR (SB2.0), with creation of a new hashtag #STRB2.1_change • Ranging Online Monitoring : GAL-DCN-TASI-SYST-X-0945 • AUTOMATED SHS SETTING BY SISE MONITORING: GAL-DCN-TASI-SYST-X-0989

		• SV INSERTION IN GMS: GAL-DCN-TASI-SYST-X-0952, GAL-DCN-TASI-SYST-X-0891 • Received Power Level Monitoring: GAL-DCN-TASI-SYST-X-0946 • QoS S-Band Filtering: GAL-DCN-TASI-SYST-X-0863 (GCS), GAL-DCN-TASI-SYST-X-0810 (GCS-SS ICD), GAL-DCN-TASI-SYST-X-0809 (GMS) • CS Separate Encryption Management: GAL-DCN-TASI-SYST-X-0738 (GCS GMS ICD), GAL-DCN-TASI-SYST-X-0875 (GMSREQ) , GAL-DCN-TASI-SYST-X-0970 (GCSREQ) • Provision of Uplinked Nav message to GSC: GAL-DCN-TASI-SYST-X-1021, GAL-DCN-TASI-SYST-X-1023 • Implementation of NeQuick Iono Model in GCS/GMS: GAL-DCN-TASI-SYST-X-0972 (GMSREQ), GAL-DCN-TASI-SYST-X-0973 (GCSREQ) and GAL-DCN-TASI-SYST-X-0779 (GCS GMS ICD)

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1. INTRODUCTION

1.1 Scope of the document

This document is the System Design Definition File Logical Architecture volume (SYSBSLDEF part 1 also referred as DDF1). It describes the functional architecture of the Galileo FOC System as specified in the Galileo System Technical Requirement Baseline (STRB) [AD.00] aiming at implementing Galileo Mission Requirements specified in the Mission Requiremets Document (MRD) [AD.50].

The mapping between System Design Definition File and STRB is provided below:

- **DDF1** (Design Definition File – Logical architecture – **This document**) includes design changes up to STRB2.0 to describe the system concepts from the logical perspective.
  STRB2.0 DCNs are identified with the hashtag #STRB2.0_change, and STRB2.1 DCNs are identified with the hashtag #STRB2.1_change inside this document. These changes are further described in STRB2.0/STRB2.1 Requirement Justification File SYS-RD.70 and are subject to S-DCDR review, as such are tackled in present document to provide an anticipation of logical design impact, assuming such changes will be endorsed at Program Level and flown down in FOC2 Builds. The hashtag #STRB2.1_change has been introduced to document the outcome of the flowdown of STRB2.0 DCNs to Segments (at Segment CDRs), where some DCNs have been postponed to SB 2.1.

- **DDF2** (Design Definition File – Physical architecture) includes design changes up to STRB1.5 to describe the associated physical implementation. It does not contain the physical description of STRB1.5.1 and 2.0 since it is not yet contracted at segment level (thus no segment physical implementation available).

The DDF1 content is directly traced to the system applicable requirement documents (See Applicable Documents list) plus STRB 2.0 DCNs on top of them as stated above.

1.2 Scope of Application

This document is generated as part of Galileo Project Phase CDE1 and applicable to all teams in charge of designing and validating the Galileo System. It is used during all the phases of design, development up to the acceptance and hand-over of the system to the customer.

1.3 Functional Analysis Approach

In the frame of the Galileo design phase, a functional analysis has been implemented at system level to achieve the following objectives:

- To ensure a proper flow-down of mission requirements (MRD) into system requirements (GSRD)
- To define a system functional architecture satisfying GSRD requirements
- To ensure that GSRD requirements are coherently translated into segment requirements and interface documents
- To set up a complete and integrated data model linking the design documents and the requirement documents to demonstrate the previous objectives.

Practically speaking, this was implemented as follow during the IOV phase:

- **Up to PDR: Top-down approach:** where the System Design team builds up a functional architecture (list of functions and signals exchanged among them) derived from the input requirements.

- **Between PDR and CDR: Bottom-up approach**: where System Design team consolidate the functional architecture based on lower level (segment) functional analysis and on inter-segment interface consolidation process.

  This logic was applied in IOV phase leading to agreed STRB1.2 which has successfully passed the system CDR.

For what concerns the consolidated phase between CDR and SdCDR: a new system design phase was led, interleaving the top-down and bottom-up approaches:

- Since CDR and up to the establishment of STRB1.5.1, changes were driven by the compliance against the MRD as well as segment DCP (Design Change Proposal). These intermediate STRB were minor in terms of system function implementation and in general consisted of regularisation between system design and implementation at segment level.
- For what concerns the FOC2 changes submitted at DCDR, changes are driven either:
  - In a top-down approach, by system needs:
    - **Targeting the compliance against MRD issue 7.1 and issue 7.2 and resolution of issues identified as Lesson-learnt from previous phases** (IOVR lesson-learnt): To do that a set of issues have been identified and tracked at System level, called STI (System Technical Issues), which are then analysed for resolution and may imply design changes to solve it.
  - In a bottom up approach by the need to accommodate at system level
    - Design change proposal raised by N-1 (Segment's DCP Design Change Proposal).
    - Various operability improvements.

The next scheme summarises this logic:

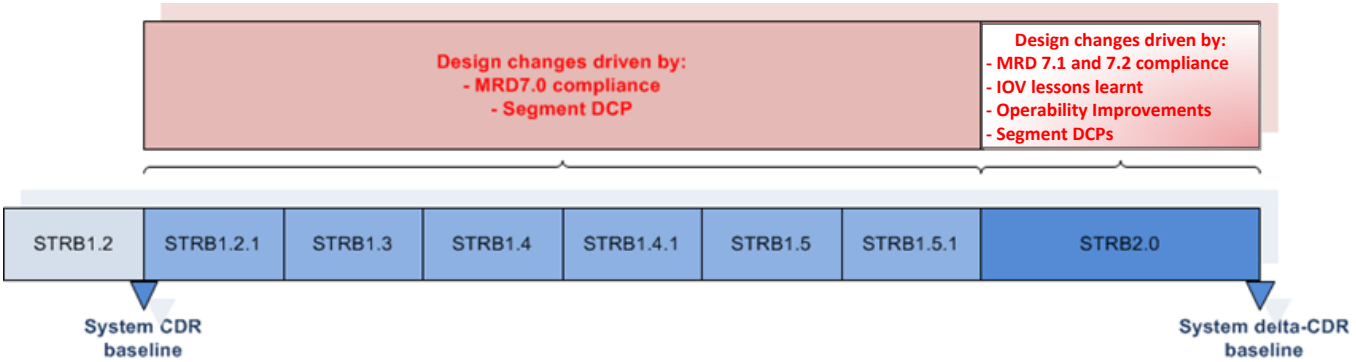


Figure 1: Design changes logic since CDR up to SdCDR

The top-down functional analysis started back in phases A and B was based on a hierarchical tree functional structure as follows:

- Top level service functions are identified by system engineers from the input requirement document (GSRD)
- These services functions are decomposed into lower level System Functional Chains
- These SFC's call lower level (segment) functions

Therefore the functional architecture is composed of three functional layers: the high level (mission), the SFC (system level) and the segment functions on which SFCs are build, as shown in Figure 2:

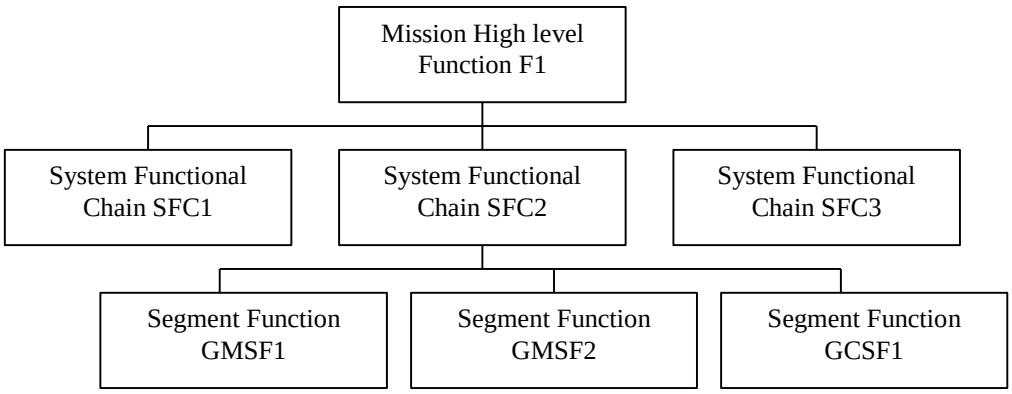


Figure 2: System Architecture Functional Tree

1.3.1 Functional Tree Level 1: Mission high Level Functions

Mission High Level functions are set corresponding to the services provided by the Galileo Core System to the end users ("Mission Services") or to external service providers ("Support to" services), as specified in top level applicable document [GSRD]

GSRD analyses concluded to identify the following services

- SERV1 - Provide Open Service (OS): positioning and timing services.
- SERV2 - Provide Safety of Life (SoL): Integrity service: ***due to SoL removal specified in MRD7.1, this high-level mission function is removed. The numbering is however kept for traceability purposes.***
- SERV3 - Provide Public Regulated Service (PRS): positioning, timing and security services.
- SERV4 - Support to Commercial Service (CS).
- SERV5 - Support to External Region Integrity Service (ERIS): ***due to ERIS removal, this high-level mission function is removed [AD.50]. The numbering is however kept for traceability purposes. The EDBS mission requirements are allocated to SERV4- Support to Commercial Service (CS).***
- SERV6 - Provide Search and Rescue Service (SAR): ***"Provide SAR" replaced the previous terminology "Support SAR", considering the SAR as a Galileo mission service [AD.50].***

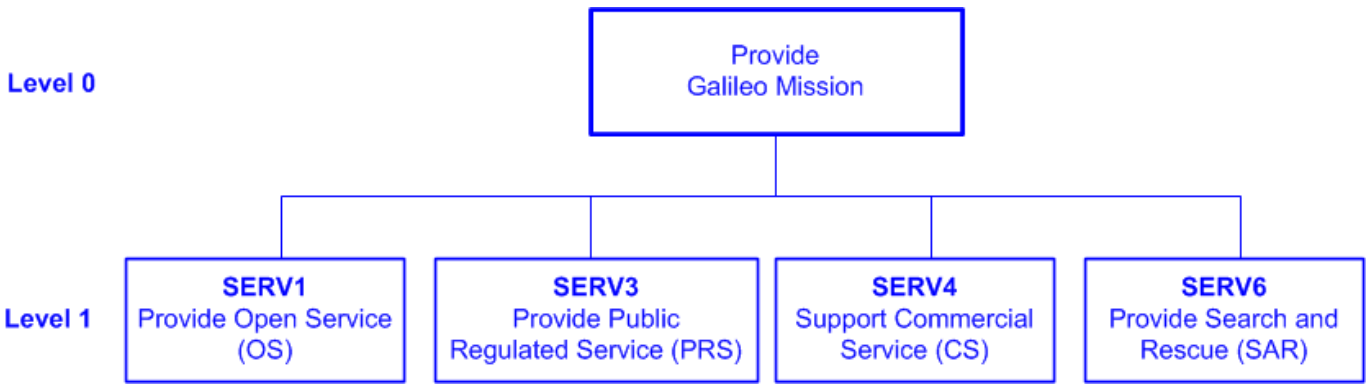


Figure 3: Level 1 Functional Tree

For further details on service description, refer to section 3.3.

1.3.2 Functional Tree level 2: System Functional Chains (SFC)

The System Functional Chains (SFC) are transversal functions generally involving more than one segment.

The list of SFCs has been consolidated in order to ensure the full coverage of the functional needs to provide the six services listed in the previous bullet. Generally, a service is derived into 3 types of SFCs:

- Service SFC (yellow boxes): which are in charge of providing the services to the end-users and/or to external service providers.
- Enabling SFC (green boxes): which are in charge of monitoring and controlling the system in order to bring it and maintain it in states and modes which allow the system to have the Service SFC running with the required level of performance and safety.
- Security SFC (blue boxes): which are in charge of protecting the system components and the services.

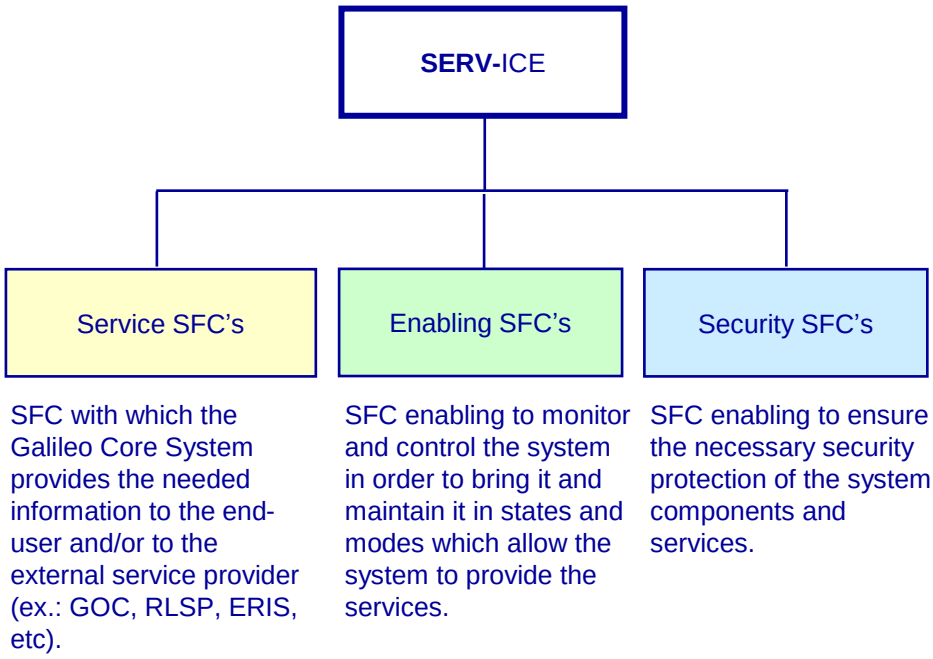


Figure 4: Types of SFC Contributing to the Provision of Services

The list of SFCs which have been identified to support each service and their corresponding interfaces is provided in section 3.3. For detailed description of each SFC content, refer to dedicated section 4 (System Functional Chain Description).

1.3.3 Functional Tree Level 3: Segment Functions and Signals

The Segment Functions are functions allocated to a specific segment (namely GMS, GCS, SSgt, TUS, External Entities). These functions are the building blocks for each SFC.

These functions have been translated in requirements for the segments. These requirements are included in GMSREQ, GCSREQ, SSREQ and related internal or external Interfaces, (provided in the list of applicable documents). Segment functions are described in the dedicated section 5 (Segment Function Description).

1.4 System Functional Model Description

The approach for functional architecture consolidation presented in the previous section allowed defining a design fulfilling the input requirement (GSRD) and in line with segment implementation.

The purpose of this section is to give more details regarding the way this design has been completely and coherently specified though a set of requirement documents and ICDs.

1.4.1 System and Segment Function Modeling

Functions need to be implemented in a model to check in an exhaustive way the completeness of the design (from a functional point of view) and the presence of any possible inconsistencies of the signals exchanged by the segment functions.

This modeling was implemented using the SDL-tau tool: The functions defined in the SDL Tau are fully listed and described in section 5, and traced to the segment requirements and the data-dictionary (see Annex A and B for details).

For Galileo the resulting SDL model (first level) is shown in the Figure 5 below:

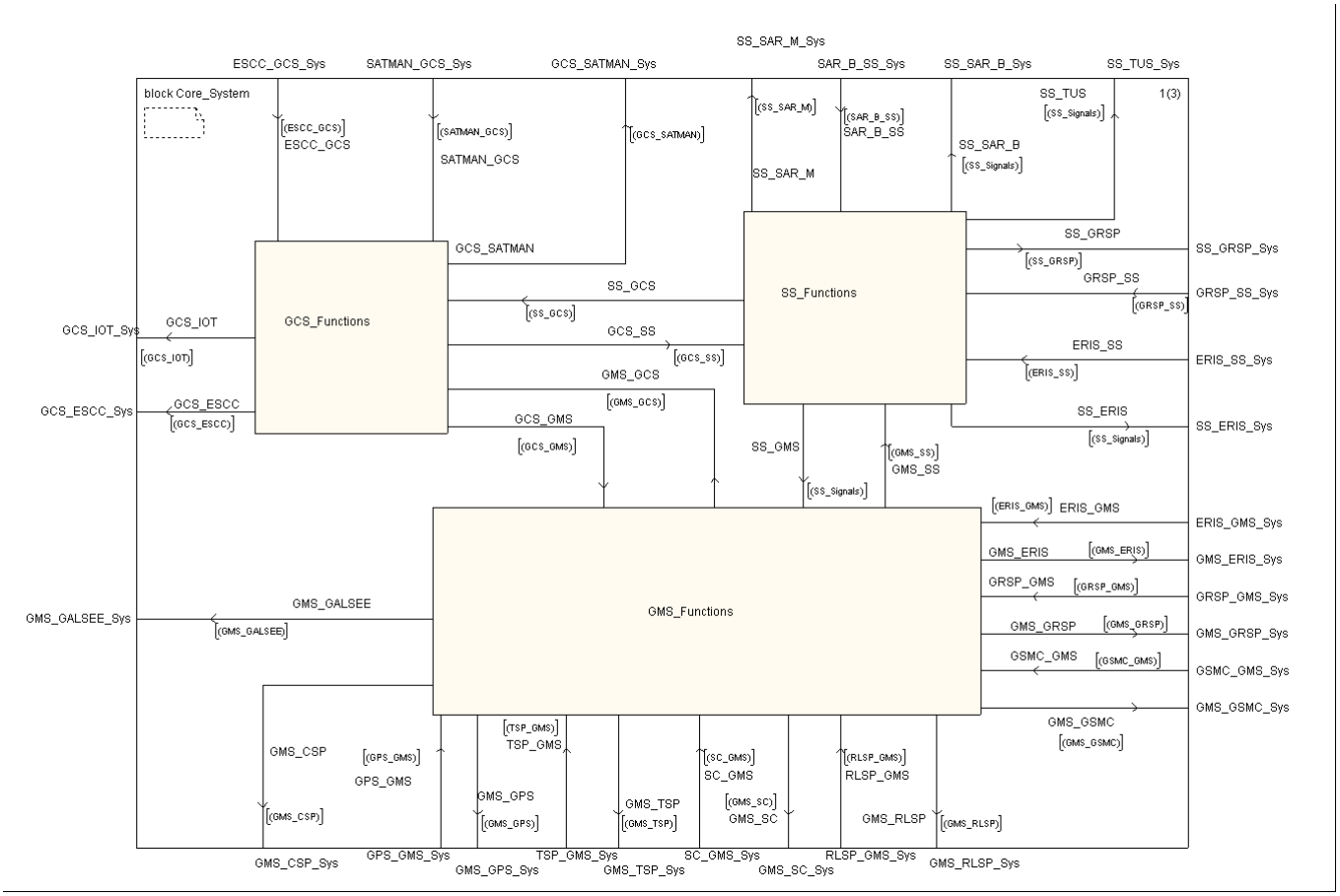


Figure 5: Galileo FOC first level SDL model

The following figures show the way (including colour coding) the functions modelled with SDL are described in this document.

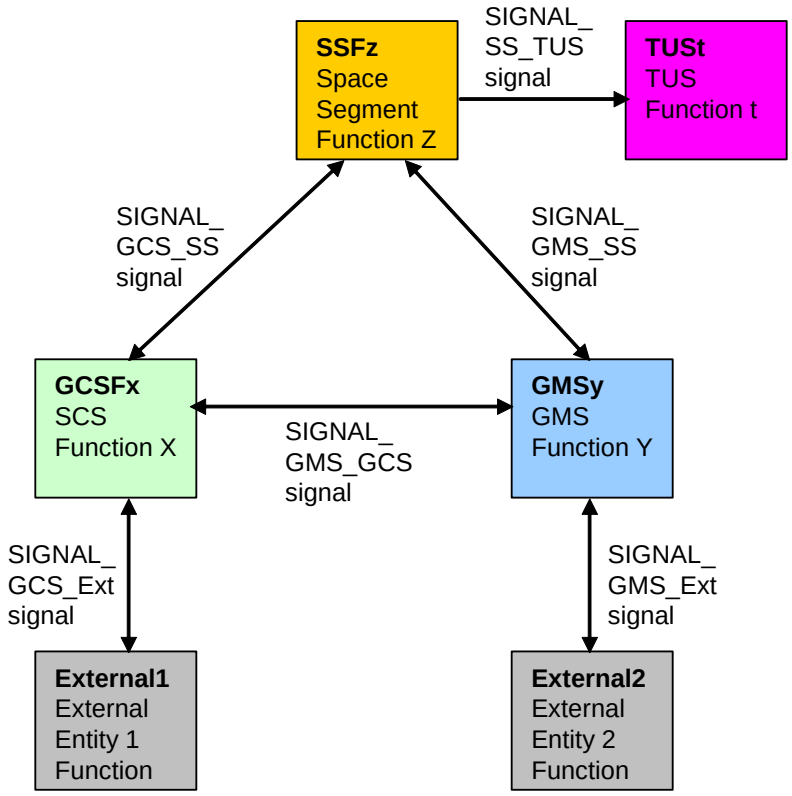


Figure 6: Links between Functional Block Diagram (Segment Functions) and Relative Color Coding

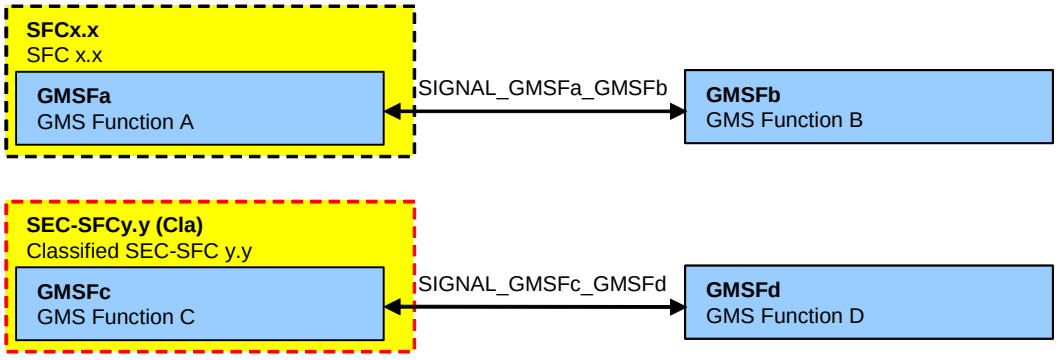


Figure 7: Links between Functional Block Diagram (SFC) and Relative Color Coding

Note: *In the functional block diagrams of each SFC, the following convention is adopted in order to ease the understanding of the functional allocation to segments (see Figure 6):*

- *GCS functions are represented in box highlighted in green*
- *GMS functions are represented in box highlighted in blue*
- *SS functions are represented in box highlighted in orange*
- *External Entity functions are represented in box highlighted in grey*
- *TUS functions are represented in box highlighted in rose*
- *Other SFCs are represented in bow highlighted in yellow and surrounded by dashed line (black dashed line for non-Classified SFC, red dashed line for Classified SFCs): see Figure 7.*

1.4.2 Traceability using DOORS

DOORS is a tool dedicated to system engineering enabling to organize documents into a data model. This data model allows managing the system requirements (including interfaces) and their links to system design definition and justification file.

The complete design documentation has been imported in DOORS. In DOORS, the system DDF1 is composed of three modules:

- **SYSBSLDEF1-SFC module:** Containing the description of the System Functional Chains including a textual description and a functional block diagram. It includes the link to GSRD and segment level requirements.

- **SYSBSLDEF1-Segment Functions module**: Containing the description of the Segment functions used in the System Functional Chains and defined in the SDL model. This description includes:
    - a textual description linked to the same modules as SYSBSLDEF part 1 module with the same types of links
    - a link to the SDL Tau block diagrams describing the functions context.

- **SignalDB module (also called Data-Dictionary):** Containing the description of the signals between all the segment functions as defined in the SDL model.

Note: The system DDF2 is composed of one module:

- **SYSBSLDEF2**: containing
    - The description of the physical architecture and of the Galileo System constraints,
    - The mapping of segment functions defined in the system functional architecture as described in SYSBSLDEF part 1 onto physical elements of the system,
    - The description of the physical implementation of the system functional chains (as identified in SYSBSLDEF part 1). This description includes static and dynamic views.

This description is summarized in the following figure:

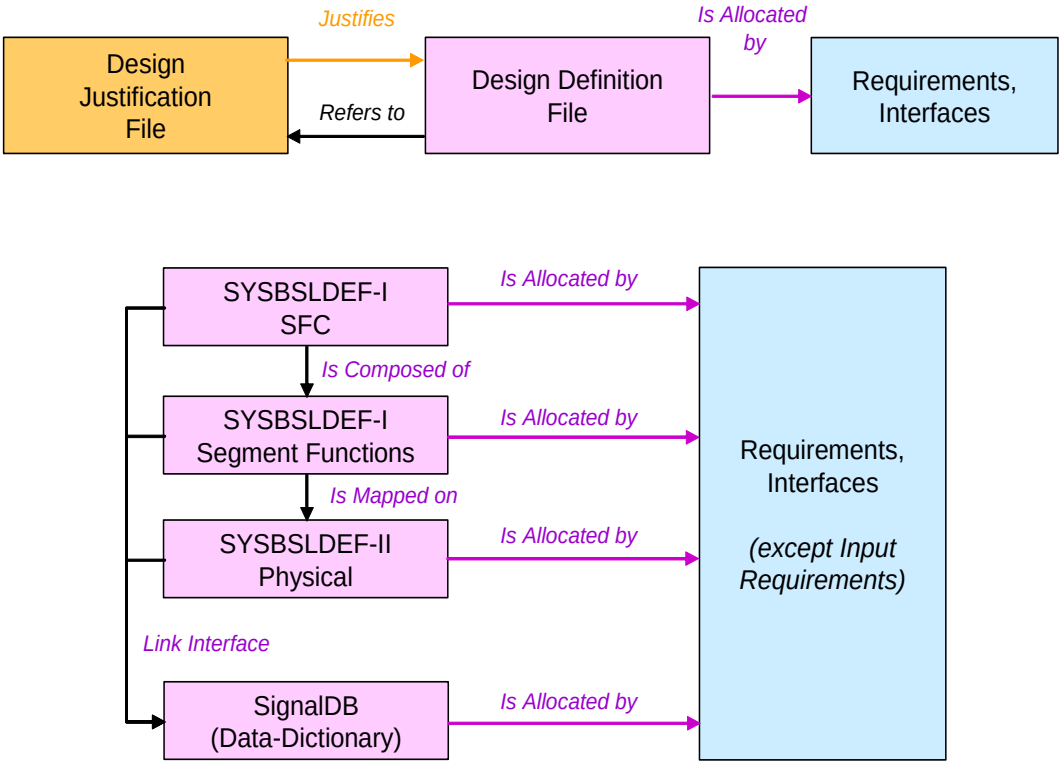


Figure 8: SYSBSLDEF Part 1 Modules DOORS Context

This complete traceability ensures that all functions and associated signals have been correctly flowed down.
On the other hand, this traceability allows to make sure that if segment functions are properly implemented the system fulfils GSRD requirements.

1.5 Document Structure

This document is composed of following main sections:

Section 3: Galileo System Overview: this section recalls the main characteristics of the Galileo System including:

- Galileo System Description and Boundaries: this section gives a high level presentation of the main components of the Galileo System and Interfaces with External Entities
- Galileo Mission Phases
- Galileo Services: this section intends to describe the services offered by the Galileo System to the end users.

Section 4: Galileo System Functional Chains Description: this section includes the detail functional description of all the System Functional Chains needed to provide the different Galileo Services.

Section 5: Segment function description: this section includes the description of the segment functions identified at system level in the system functional architecture, and used in the system functional chains. The description includes the outputs of the functions, the associated design constraints and the functional context of the function.

2. APPLICABLE AND REFERENCE DOCUMENTS

2.1 Applicable Documents

AD#	Title	Document No.	Issue
AD.00	FOC TRB 2.0	GAL-LI-ESA-SYST-X-3849	1.0
AD.01	GSRD	GAL-REQ-ESA-SYST-X-0011	7.0
AD.02	TUSREQ	GAL-REQ-ESA-REC-X-0025	2.7
AD.03	GMSREQ	GAL-REQ-GLI-SYST-A-0165	15.0
AD.04	FOC SSREQ	GAL-REQ-ESA-SYST-X-0116	1.8
AD.05	GCSREQ	GAL-REQ-GLI-SYST-A-0019	17.3
AD.06	FOC SISICD	GAL-ICD-ESA-SYST-X-0027	2.2
AD.07	GMS SS ICD	GAL-ICD-GLI-SYST-A/0014	16.4
AD.08	GCS GMS ICD	GAL-ICD-GLI-SYST-A-0166	16.9
AD.09	FOC GCS SS ICD	GAL-ICD-ESA-SYST-X-0030	2.6
AD.10	SS Ext IRD	GAL-IRD-ESA-SYST-X-00314	2.0
AD.11	GMS Ext IRD	GAL-IRD-ESA-SYST-X-0031	2.0
AD.12	GCS Ext IRD	GAL-IRD-GLI-SYST-A-0371	9.2
AD.13	GMS to TSP ICD	GAL-ICD-GLI-SYST-A/0677	8.0
AD.14	GMS to GRSP ICD	GAL-ICD-GLI-SYST-A/0679	8.0
AD.15	GMS to GGTO ICD	GAL-ICD-GLI-SYST-A/0678	8.0
AD.17	GMS to SC ICD	GAL-ICD-GLI-SYST-A/0686	9.0
AD.19	SAR-Beacon to SSeg to SAR-MEOLUT ICD (COSSARGALICD Vol 2)	GAL-ICD-ESA-SYST-X-0089	1.1
AD.20	GMS to SAR RLSP ICD (COSSARGAL ICD, Vol.3)	GAL-ICD-GLI-GMS-A/0375	10.8
AD.21	SS to SLR ICD	GAL-ICD-GLI-SYST-I-0726	3.0
AD.22	GCS ESCC ICD	GAL-ICD-GLI-SYST-A/0716	3.7
AD.23	GCS SATMAN ICD	GAL-ICD-GLI-SYST-A-0717	4.6
AD.24.	GMS to CSP ICD	GAL-ICD-GLI-SYST-A/1028	4.0
AD.25	GSS Site IRD	GSA-GAL-HSP-IRD-237012	2.0
AD.26	ULS Site IRD	GSA-GAL-HSP-IRD-237012	2.0
AD.27	TTC Site IRD	GSA-GAL-HSP-IRD-237014	2.0
AD.28	REDU Site IRD	ESA-DTEN-NG-REQ/02788	2.5
AD.29	FOC System Verification Requirements	GAL-REQ-ESA-SYST-X/0017	2.1
AD.30	LEOP Requirement Document	GAL-REQ-ESA-OPS-X/0003	2.1
AD.31	GCS to IOT ICD	ESA-DTEN-NG-ICD/03147	1.5
AD.32	IOT Services Requirements Document	GAL-REQ-ESA-OPS-X/0025	1.0
AD.33	GCC Site IRD	GSA-GAL-HSP-IRD-237015	2.0
AD.34	GMS to GSMC ICD	GAL-ICD-GLI-SYST-A/1029	3.5
AD.35	Operations Service Requirement Document (OSRD)	GAL-REQ-GLI-SYST-A-0023	3.2
AD.36	Galileo FOC Launch Services Technical Requirements	GAL-REQ-ESA-LNS-X/0005	1.2
AD.37	Galileo Ground Segment Integration Standards	GAL-SPE-ESA-SYST-X/0093	3.0

AD#	Title	Document No.	Issue
AD.39	IOT Security Service Requirements Document	GAL-REQ-ASTD-IOT-R-0019	2.2
AD.40	GDDN ERD	GAL-REQ-GLI-SYST-A/10021	2.5
AD.41	FOC-2 Requirements Engineering Plan and Specifications	GAL-MGT-ESA-SYST-X-3102	1.0
AD.42	BBKME ICD	GAL-ICD-GLI-SYST-A/0376-x	6.1
AD.43	Galileo Segments to GDDN ICD	ESA-DTEN-NG-ICD/03764	2.8
AD.45	Galileo Constellation Geometry Requirements	GAL-REQ-ESA-SYST-X-1241	1.0
AD.46	SAR GS REQ	GAL-REQ-ESA-SYST-X-0538	1.3
AD.48	ESA FOC2 roadmap	GAL-TN-ESA-SYST-X-2486	3.0
AD.50	Galileo Mission Requirements Document	MRD	7.2

Table 1: List of Applicable Documents

2.2 Reference Documents

SYS-RD#	Title	Document No.	Issue
SYS-RD.1	FOC System Design Definition File part I Functional architecture	GAL-RPT-TASI-SYST-X-0005	2.1
SYS-RD.2	FOC System Design Definition File part I Physical architecture	GAL-RPT-TASI-SYST-X-0006	2.1
SYS-RD.3	FOC System Design IOV Justification File	GAL-RPT-ALS-SYST-R-0005	13.1
SYS-RD.4	System Performance Budget File	GAL-TN-GLI-SYST-I-0006	20.0
SYS-RD.6	FOC MOCD	GAL-MGT-ESA-OPS-X/0026	2.3
SYS-RD.7	Classified Annex to Baseline System Design Definition Document - Part 1: Logical Architecture	GAL-DDF-CIMC-SEC-X/5185x	2.1
SYS-RD.8	Classified Annex to Baseline System Design Definition Document - Part 2: Physical Architecture	GAL-DDF-CIMC-SEC-X/5186x	2.1
SYS-RD.10	SIS Flags usage	GAL-TN-ESA-SYST-X-2069	1.3
SYS-RD.12	Annex D (Commercial Service) of Cryptographic Algorithms Definition and Requirements	GAL-REQ-GLI-SYST-A/0599	1.10
SYS-RD.14	Classified Annex of FOC MOCD	GAL-MGT-ESA-SEC-X-0629x	1.0
SYS-RD.16	Navigation Message Definition and Justification	GAL-TN-ALS-SYST-I-0659	2.0
SYS-RD.23	System States and Modes and System Start-Up	GAL-TN-ALS-SYST-A-0653	2.0
SYS-RD.29	Operations concept for PL IOT and SC commissioning using GCS	GAL-TN-ALS-SYST-R-0658	2.0
SYS-RD.30	LEOP Services DDVP	GAL-PL-CNES-LEOP-A/0007	1.3
SYS-RD.38	IOT DDJF	GAL-JF-INM-IOT-R/0001	3.1
SYS-RD.47	SISICD Justification File	GAL-TN-ASTG-SYST-A-0027	1.3
SYS-RD.54	TTCF, GSS and ULS Sites List	GAL-TN-GLI-SYST-I-1223	1.0
SYS-RD.69	System Start-Up	GAL-TN-ALS-SYST-A-0693	3.0
SYS-RD.70	Requirement Justification File	GAL-DJF-TASI-SYST-X-0002	1.2
SYS-RD.71	GST-UTC Offset Interface Options	GAL-TN-ESA-SYST-X/267	1.3
SYS-RD.72	GGTO Implementation and Operations	GAL-TN-ESA-SYST-X/2494	3.0
SYS-RD.73	Galileo Single Frequency Ionosheric Model for User Receivers	Issue 1.1, June 2015. From European Union 2015	1.1
SYS-RD.74	Galileo System Architecture Configuration	GAL-TN-ESA-SYST-X-1075	5.3
SYS-RD.76	GCC split concept configuration	GAL-TN-ESA-SYST-X-2662	1.0
SYS-RD.77	Galileo FOC SAR Ground Segment Input to Element Requirement Documents (SGSERD)	GAL-REQ-ESA-SYST-X-0572	1.7
SYS-RD.78	SGS DDJF	GAL-DJF-CAP-SGS-X-0036	1.4
SYS-RD 79	Early Services Search and Rescue Service Operational Interfaces	GAL-ICD-ESA-OPS-X/0734	1.0
SYS-RD.80	Galileo FOC – System Support IOV SAR Link Budget	GAL-TN-TASI-SYST-X-0116	1.0
SYS-RD.81	SGS Performance Budget File	GAL-PBF-CAP-SGS-X/0034	1.3
SYS-RD.82	SAR Ground Segment SGS Definition file (DDF)	GAL-DDF-CAP-SGS-X/0012	1.5
SYS-RD.83	SAR/Galileo MEOSAR Reference Beacon	GAL-REQ-ESA-SYST-X/1532	2.0

SYS-RD#	Title	Document No.	Issue
	Requirements		
SYS-RD.84	SAR/Galileo SGSKPI Collection Platform Requirements	GAL-REQ-ESA-SYST-X/1954	2.0
SYS-RD.85	Trade-off on FOC Return Link Service Provision	GAL-TN-TASI-SYST-X-0240	1.0
SYS-RD.86	SAR Return Link System Description Document	GAL-TN-ESA-SYST-X/2686	1.0
SYS-RD.87	Constellation Deployment and Maintenance plan	GAL-PL-ESA-SYST-X-0478	4.0
SYS-RD.88	GDDN Design Documentation	GAL-DDF-BT-OPS5-X-0001	1.9
SYS-RD.89	Use of External Data Broadcast Service for OS navigation message authentication Technical assessment	GAL-TN-TASI-SYST-X-0222	3.0
SYS-RD.90	TN#02: Availability of Galileo Commercial Service Data Dissemination as specified in EC-GAL-CR-18 on GMRD	GAL-TN-ASTG-SYST-X-13782	1.0
SYS-RD.91	CR 19 I/NAV Optimization Technical Assessment	GAL-TN-ASTG-SYST-X-13754	1.1
SYS-RD.92	Implementation of Separated E6 Pilot and E6 data encryption	GAL-TN-ASTG-SYST-X-13767	1.0
SYS-RD.93	RAMS contribution to System Availability	GAL-TN-TASI-SYST-X-0060	6.0
SYS-RD.94	OD analysis to support DCN0763 to GCSREQ-26588	ESA-DTEN-NG-MEMO-0014362 dated 9/03/2011	
SYS-RD.95	TTCF Network Sizing	GAL-TN-TASI-SYST-X-0185	1.2
SYS-RD.96	System Performance Budget File Classified	GAL-TN-ESA-SYST-X-0396-x	4.0
SYS-RD.97	SAR Annex to PBF/DJF	GAL-TN-ASTG-SYST-X-12126	3.0
SYS-RD.98	CCN6 Technical Description	GAL-TN-CAP-SGS-X/0823	
SYS-RD.99	GSC Requirements for SAR/Galileo Service	GAL-EC-SAR-REQ-005	
SYS-RD.100	SIS Flags Operations Concept	GAL-TN-TASI-SYST-X-0266	3.0

Table 2: List of System Reference documents

3. GALILEO SYSTEM OVERVIEW

3.1 Galileo high-level system description

The detailed physical description of the Galileo system functions is provided in SYSBSLDEF part II (SYS-RD.1).
The Galileo Core System is composed of the following segments:

- **Space Segment (SSEG):** This segment is responsible for the provision of the satellites (while system is responsible for the constellation deployment strategy and replenishment). Once operational the satellites are responsible for disseminating the messages necessary to the Galileo users. It is composed of 24 MEO operational satellites and 6 in flight spares.
- **Ground Control Segment (GCS):** This segment is responsible for ensuring the monitoring and control of the Galileo constellation. It is implemented in the 2 GCC, which can share this task and 7 remote TTC stations spread around the world. Since FOC2 it is possible to add temporarily other external TTC stations for satellites in LEOP.
  The 6th TTC Papeete has been deployed in Galileo system build 1.5.1. In FOC2, a 7th TTC facility has been added in Kourou, leading to two TTC stations being deployed in Kourou.
  The sizing of the TTCF network is justified in [SYS.RD.95]
- **Ground Mission Segment (GMS):** This segment is mainly responsible for generating Galileo main services mission data (OS and PRS), for managing their proper dissemination to the users via the space segment and for monitoring and controlling the overall ground segment (i.e. GCS and GMS). In addition, the GMS supports the multiplexing and dissemination of data generated by service facilities:
    - Real-time dissemination of CS data via Galileo satellites and service monitoring, plus generation of reports on Galileo services and status.
- Quasi real-time (15 minutes) dissemination of SAR RLM messages and service monitoring

In the Galileo system build 1.5.0, the GMS is composed of 2 instances each one implemented in the 2 GCC, 5 remote uplink sites each one comprising 2 uplink stations stations (ULS) and 15 ground sensor stations (GSS) network:

- 8 sites are dual GSS sites with 2 chains: each chain consisting of 1 antenna + 1 GRCP + 1 GRCN.
- 7 sites are single GSS sites with 1 chain: each chain consisting of 1 antenna + 1 GRCP + 1 GRCN.

The next Galileo system builds increase the number of ULS antenna as well as GSS receivers.

3.2 Galileo Mission Phases

Given the complexty of the Galileo System, a gradual provision of Galileo services is foreseen deploying the space and ground resources necessary to support the required system functions in incremental manner.
This has brought to the concept of System Builds, e.g. a series of discrete system configurations, deployed over several years, implementing the various functionalities defined by staggered System Technical Requirment Baselines (STRB), approaching gradually the identified final System configuration
System performance and operability will therefore improve in discrete steps along the different staggered Galilo system builds. This logic is described in the Galileo FOC2 roadmap [AD.48].

The 2 main phases of the Galileo program are described below:

- **In-Orbit validation (IOV) Phase:** Its scope is to early validate/verify Galileo (segments, operations and service provision) before the deployment of the overall system. The IOV concept is based on the definition of a reduced architecture. The IOV results will finalise the definition and development of the segments and operations.

- **Full Deployment Phase:** Its scope is to complete the deployment of Galileo to reach a Full Operational Capability (FOC).

3.3 Galileo Services

The Galileo Core System offers the following services to the end users :
- SERV1 - Provide Open Service (OS): positioning and timing services.
- SERV3 - Provide Public Regulated Service (PRS): positioning, timing and security services.
- SERV4 - Support to Commercial Service (CS).
- SERV6 - Provide Search and Rescue Service (SAR)

Remarks:
- SERV2 - Provide Safety of Life (SoL): Integrity service: ***due to SoL removal specified in MRD7.1, this high-level mission function is removed. The numbering is however kept for traceability purposes.***
- SERV5 - Support to External Region Integrity Service (ERIS): ***due to ERIS removal specified in MRD7.1, this high-level mission function is removed. The numbering is however kept for traceability purposes. The EDBS mission requirements are allocated to SERV4- Support to Commercial Service (CS).***

3.3.1 SERV1 - Open Service (OS)

3.3.1.1 Overview

Open Services (OS) are positioning and timing services provided free of charge to end-users. The end-user is provided by ranging signals and navigation messages broadcast by the Galileo constellation over the following frequencies: E5a, E5b and E1.

The following categories of users are envisaged:
- Single carrier OS (SC-OS) users: These users use only one frequency, namely E5a or E5b or E1B/C (E1B carrying the data while E1C being the pilot channel) to perform ranging measurements and receive navigation data. As a consequence they compensate broadcast group delay and ionospheric propagation errors using a model of which the parameters are broadcast in the navigation data.
- Dual carrier OS (DC-OS) users: These users receive simultaneously the frequency pair E5a/E1B or E5b/E1B. They do not need any additional broadcast data to remove the ionospheric propagation errors. A specific type of receiver called laboratory receiver is specified for timing application.

The OS performance is characterized in terms of:
- Accuracy of the position, velocity and/or time estimates,
- Availability of the service with the specified accuracy performance,

The specified performance is provided in the following table for the various kinds of users as specified in GSRD [AD.01]. The effective performances achievable by the designed system are indicated in bracket and justified in SPBF [SYS-RD.04].

	OS					
	Single carrier			Dual carrier		Timing & frequency
	E5a	E5b	E1B/C	E5a-E1B/C	E5b-E1B/C	
Positioning Horizontal accuracy (95%)	24 m *(29.5m)*	24 m *(28.2m)*	15 m *(15.9m)*	4 m	4m	-
Positioning Vertical accuracy (95%)	35 m *(54.8m)*	35 m *(50.4m)*	35 m *(37.1m)*	8 m	8m	-
Velocity Accuracy (95 %)	20 cm/s *(28.7)*	20 cm/s *(28.7)*	20 cm/s *(28.7)*	20 cm/s	20 cm/s	
Time Accuracy (95 %) Timing Laboratory (Fixed User)	-			-		30 ns 95 % of the time over any 24 hours of operation period
Frequency Accuracy (calibration Laboratory Receiver)						3x10-13
Mean Availability	99.5%			99.5%		99.5%

Table 3: Open Service Performance

3.3.1.2 Functional Implementation

Figure 9 shows all the System Functional Chains (SFC) needed to perform the OS (SERV1):

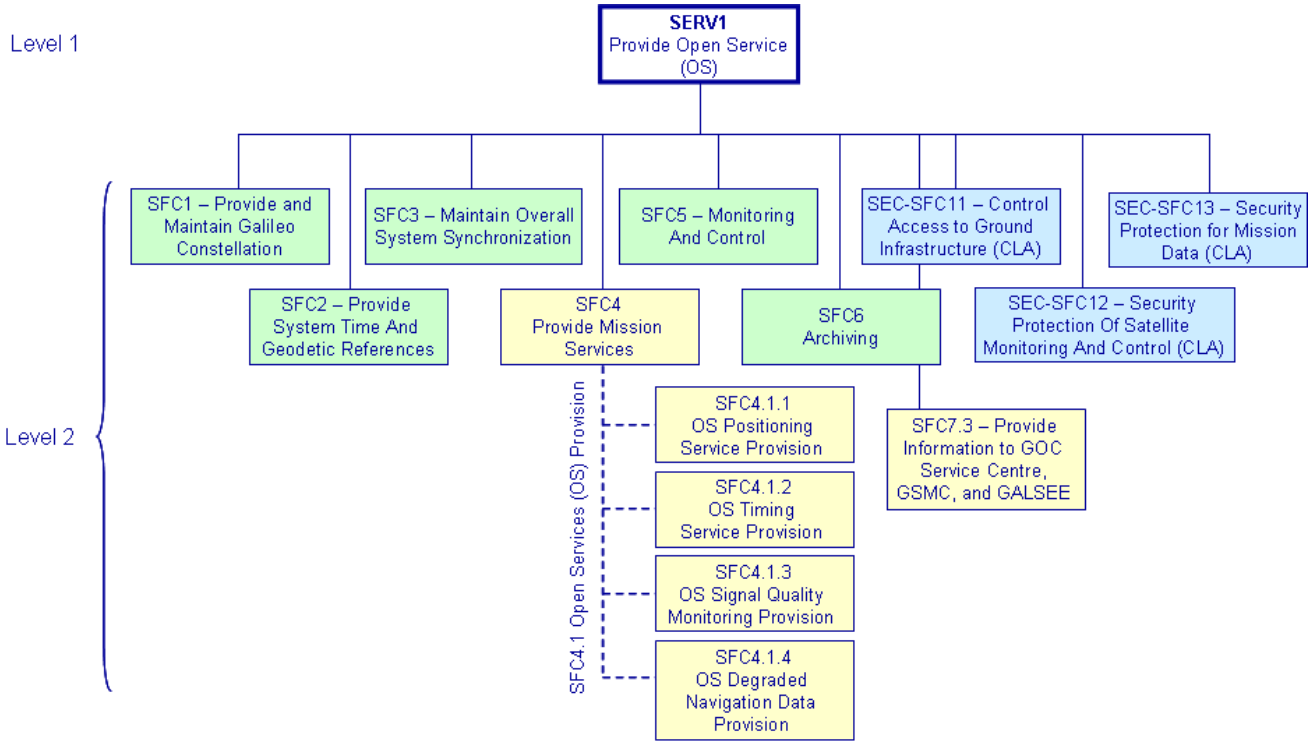


Figure 9: Level 2 Functional Tree of SERV1 Open Service

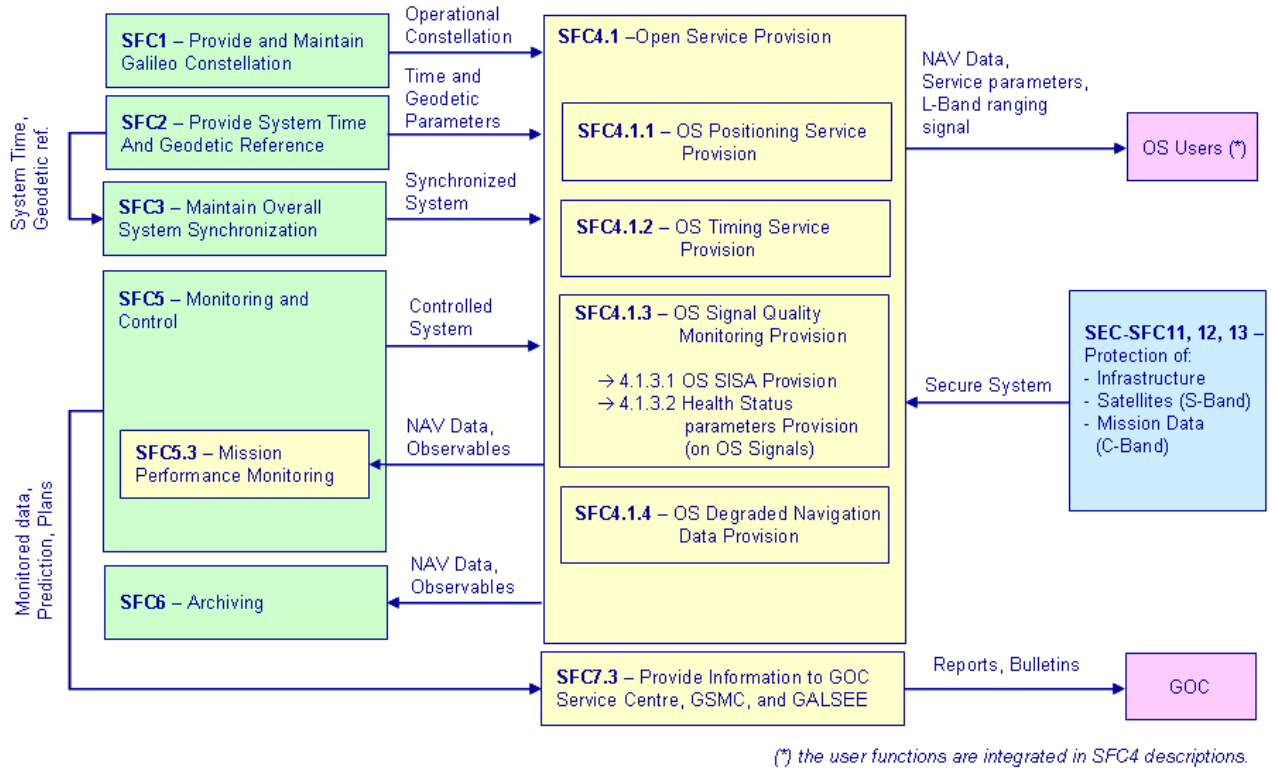


Figure 10: SFCs involved in Open Service Provision

3.3.2 SERV2 - Safety of Life (SoL)

This service was descoped in MRD7.1.

3.3.3 SERV3 - Public Regulated Service (PRS)

3.3.3.1 Overview

The Public Regulated Services (PRS) are positioning and timing services (same as OS) provided using E6A and E1A frequency. The access to this service is controlled by encryption.

Remark: the provision of integrity information in PRS service was descoped in MRD7.1.

The following categories of users are envisaged:

- Single carrier PRS (SC-PRS) users: These users use only one frequency (E6A or E1A) to perform ranging measurements and receive navigation data. As a consequence they compensate broadcast group delay and ionospheric propagation errors using a model of which the parameters are broadcast in the navigation data. The performance for these users is specified in a so-called “rural pedestrian” environment.
- Dual carrier PRS (DC-PRS) users: These users simultaneously receive E6A and E1A. They do not need any additional broadcast data to remove the ionospheric propagation errors. The performance for these users is specified in a so-called “aeronautical” environment.

The performance of the PRS services is provided in the following table as specified in GSRD [AD.01].

The effective performances achievable by the designed system are indicated in bracket and justified in SPBF [SYS-RD.04].

	PRS		
	Single carrier		Dual carrier
	E1A	E6A	E1A-E6A
Positioning Horizontal accuracy (95%)	15m *(15.2m)*	24m *(28.3m)*	6.5 m
Positioning Vertical accuracy (95%)	35m *(35.5m)*	35m *(41.5m)*	12 m
Velocity Accuracy (95 %)	20 cm/s *(23.1cm/s)*	20 cm/s *(23.1cm/s)*	20 cm/s
Mean Availability	99.5%	99.5%	99.5%

Table 4: Public Regulated Service Performance

3.3.3.2 Functional Implementation

Figure 11 shows all the System Functional Chains (SFC) needed to perform the PRS Service (SERV3):

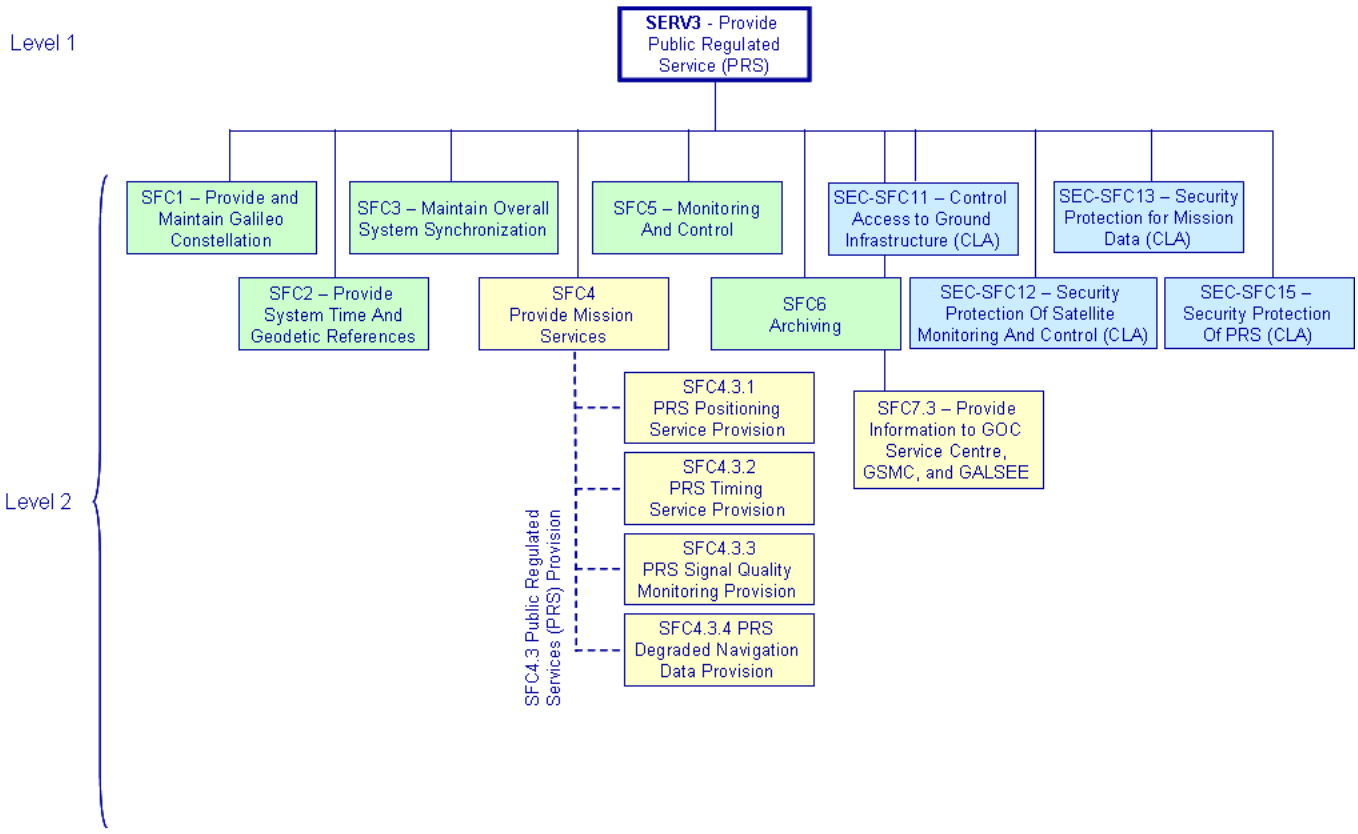


Figure 11: Level 2 Functional Tree of SERV3 Public Regulated Service

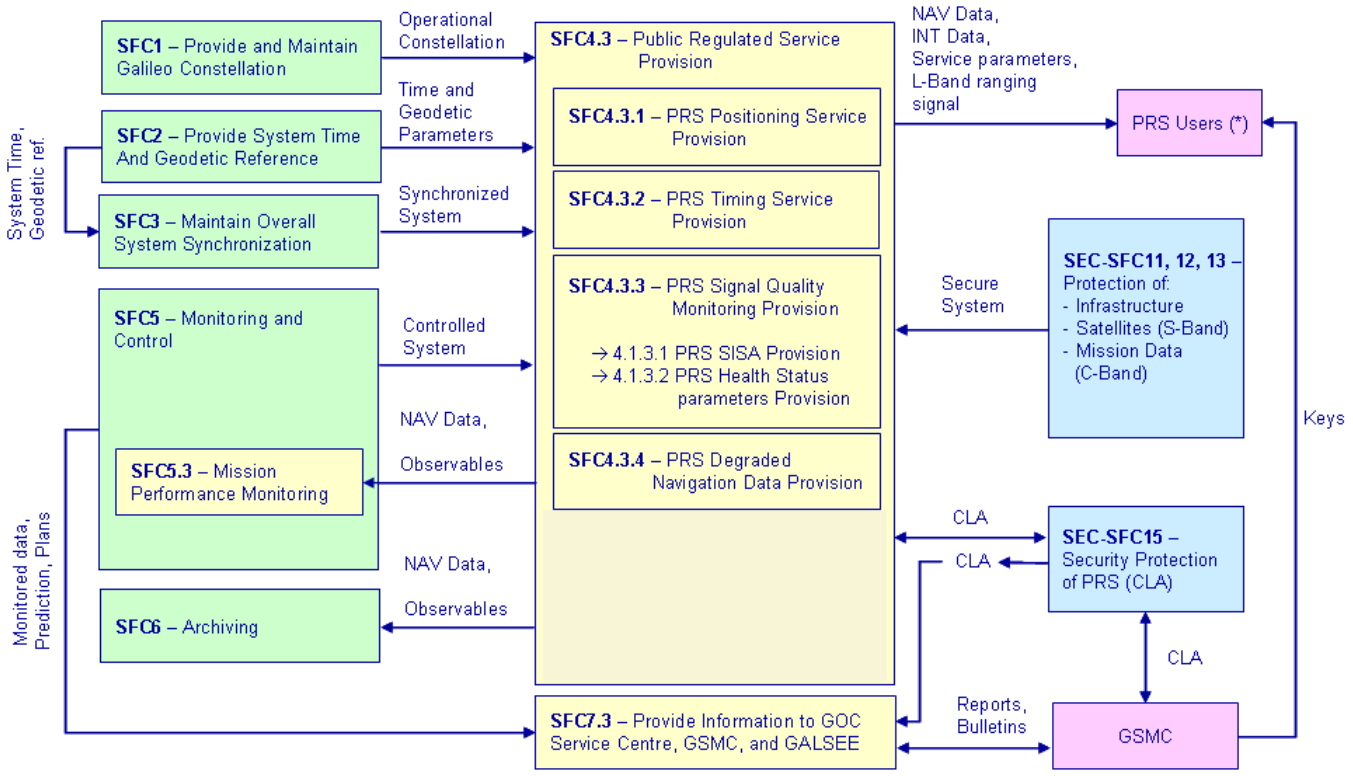


Figure 12: SFCs involved in Public Regulated Service Provision

3.3.4 SERV4 - Support to Commercial Service (CS)

3.3.4.1 Overview

The support to commercial service is not provided by the Galileo Core System itself to the end users. Indeed, the Galileo Core System provides a support to external commercial service providers (gathered in one physical entity called the Galileo Service Center - GSC). As such, the Galileo Core System provides a support service to the CS service providers.
The commercial service support" service can be split in the following sub-services:

- Disseminating commercial service data received from commercial service providers in a way that the access to the data is "controllable" in order to allow fees to be levied (means by the external commercial service providers).
- Provide ranging capability on a dedicated E6B signal. The Galileo Core System also ensures the key management of the encrypted ranging code.
- Provide information to the GSC.

CS dissemination latency and availability requirements have been introduced in the frame of EC CR-18, which is assessed in [SYS-RD.90].
The new requirements [AD.50] are:

- a maximum dissemination latency by Galileo Core Infrastructure of 6 seconds (from when the last bit of the message leave the GSC to when it is received by the users);
  - CS coverage by:

- 1 satellite at an elevation of 30º or higher broadcasting external data, with a 98% minimum availability in the latitude range [-60, +60],
- 1 satellite at an elevation of 20º or higher broadcasting external data, with a 98% minimum availability worldwide,
- 2 satellites at an elevation of 10º or higher broadcasting external data, with a 98% minimum availability worldwide, and
- 4 satellites at an elevation of 5º or higher broadcasting external data, with a 90%minimum availability worldwide and 98% average availability worldwide.

Note that the requirements above were not entirely endorsed in the GSRDissue 7.0 [AD.01], where a slight different availability and CS coverage targets was proposed, e.g.:

- 2 satellites at an elevation of 10º or higher, with a 99% availability worldwide
- 1 satellite at an elevation of 20º or higher, with a 99% availability worldwide

In both cases, the CS coverage performance is expected to be reached with the planned 5 ULS sites equipped with 4 antenna each as already envisaged in the STRB 2.0.

No specific performance is requested today on the CS ranging accuracy. Potential additional navigation data are considered part of Commercial Data that will be provided directly by Commercial Service

3.3.4.2 Functional Implementation

Figure 13 shows all the System Functional Chains (SFC) needed to perform the support to Commercial Service (SERV4):

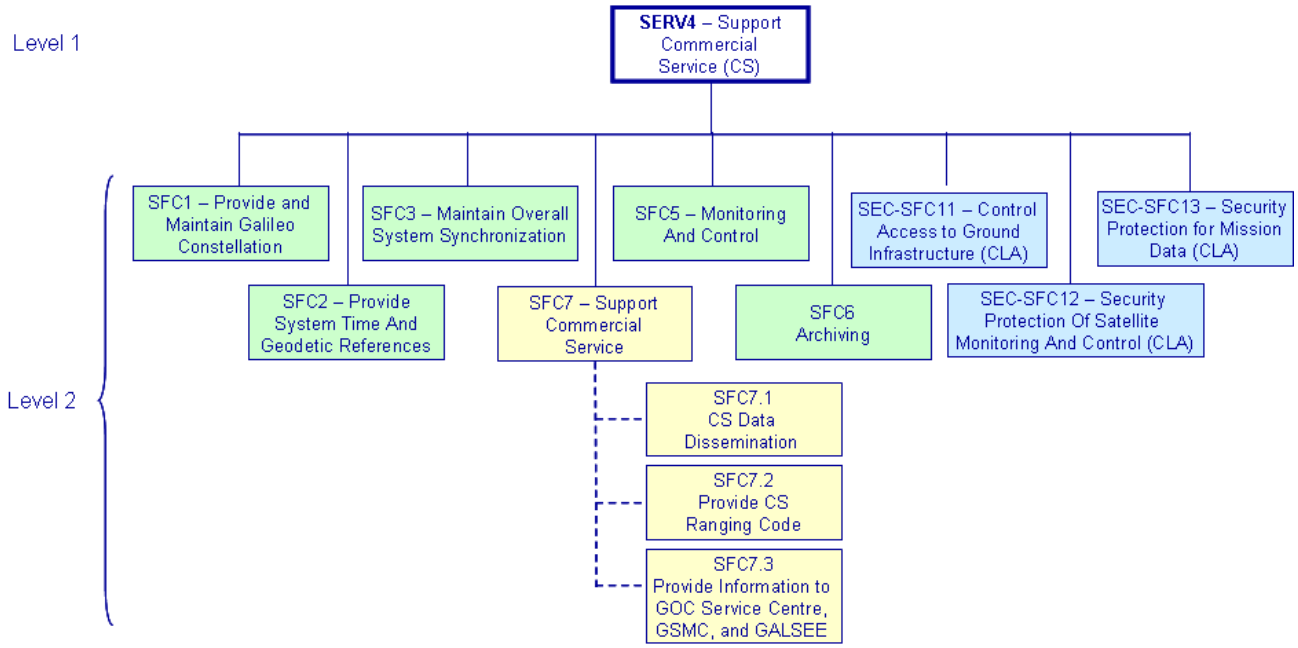


Figure 13: Level 2 Functional Tree of SERV4 Support to Commercial Service

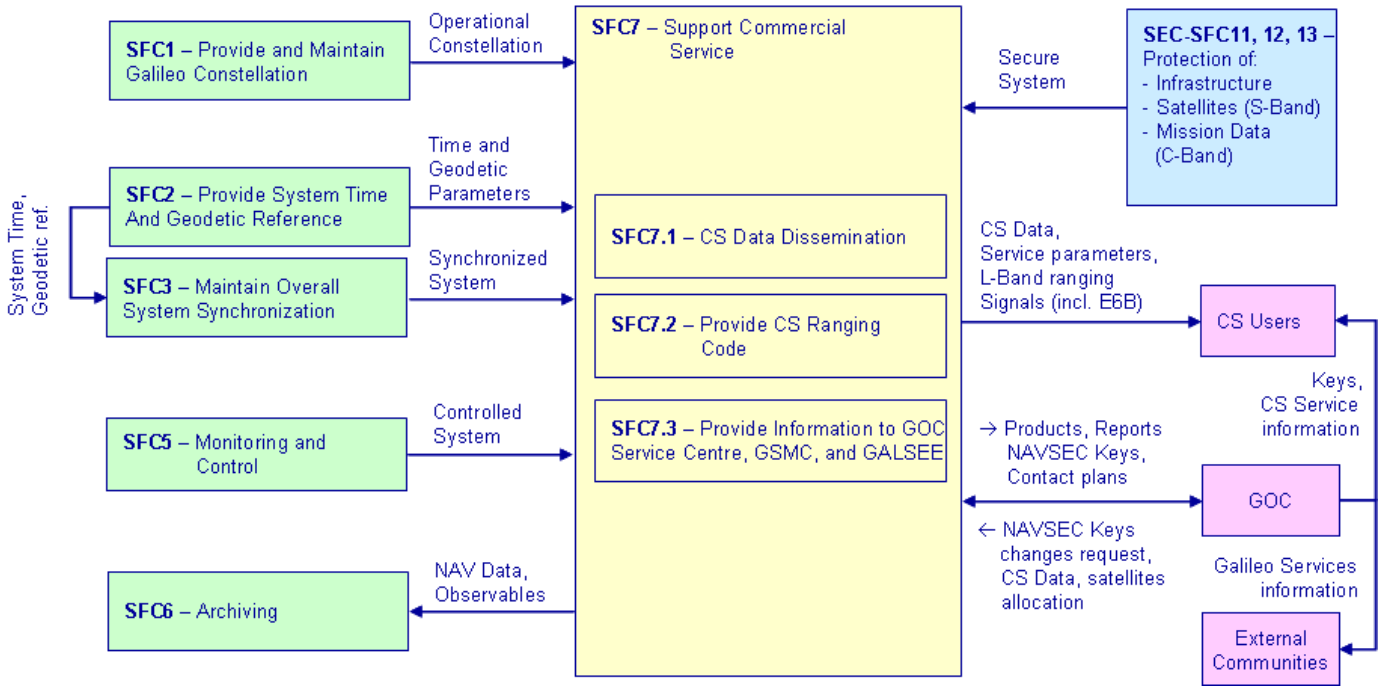


Figure 14: SFCs involved in Support to CS Provision

3.3.5 SERV5 - Support to External Region Integrity Service (ERIS)

This service was descoped in MRD7.1.

3.3.6 SERV6 - Provide Search and Rescue (SAR)

3.3.6.1 Overview

The SAR service is provided by the Galileo Core systems to a Search and Rescue service provider. This support consists in:

- Ensuring the forward link (relay) by the Galileo Satellites of SAR messages emitted by distress beacons to so-called MEOLUT.
- Ensuring the transmission of SAR return link messages provided by a return link service provider (RLSP) to the SAR users.

The “Support to SAR” service is therefore split in two sub-services:

- Forward link SAR service.
- Return link SAR service.

3.3.6.2 Functional Implementation

Figure 15 shows all the System Functional Chains (SFC) needed to perform the provide SAR Service (SERV6):

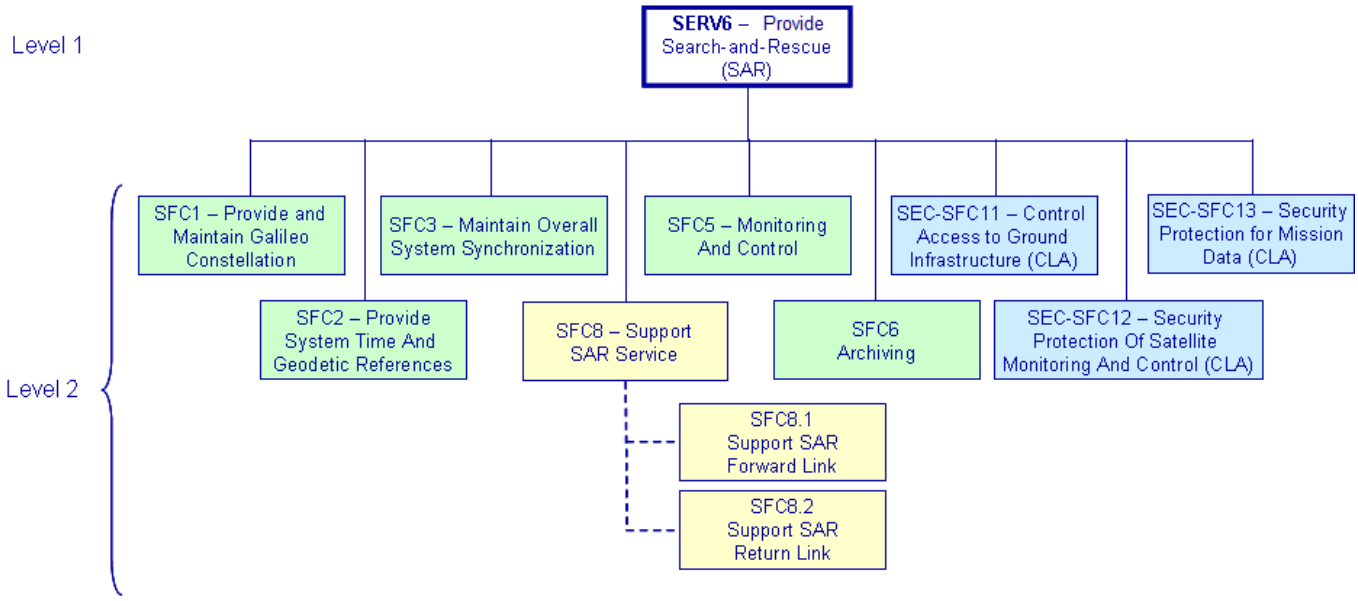


Figure 15: Level 2 Functional Tree of SERV6 Provide Search-and-Rescue

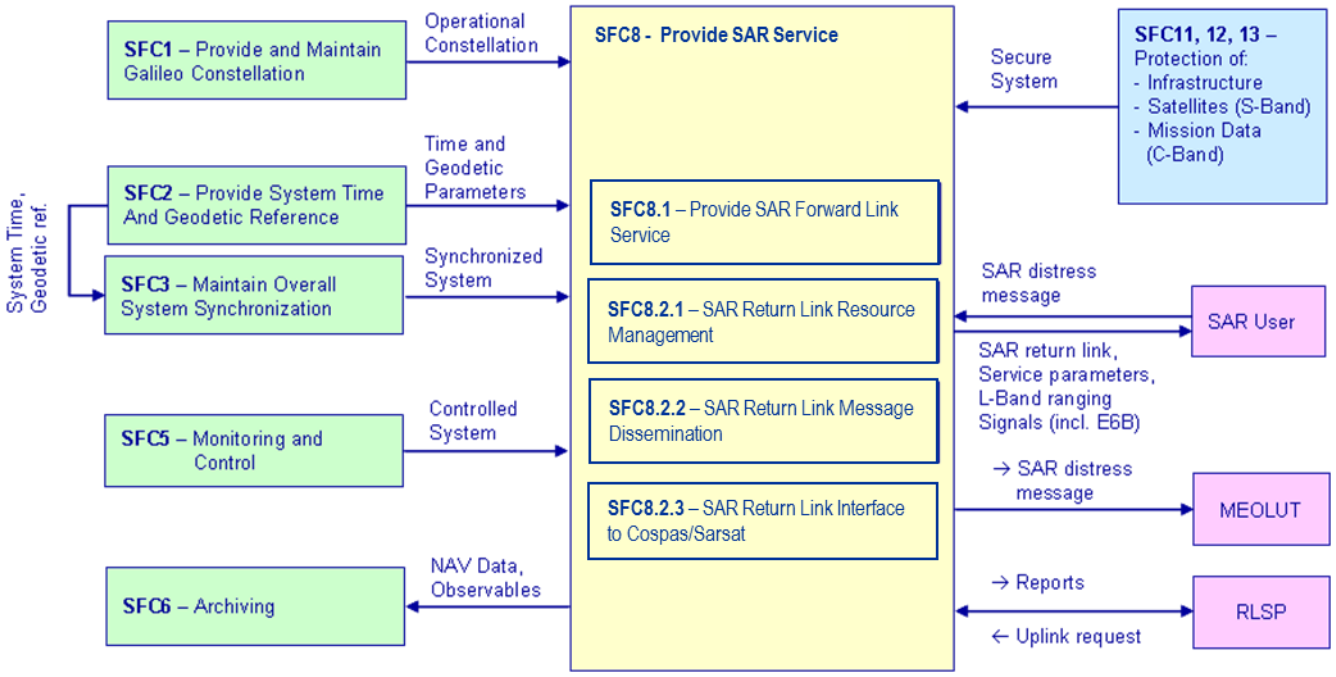


Figure 16: SFCs involved in SAR Provision

The SAR/Galileo Forward and Return Link services shall guarantee the following performance.

SAR/Galileo Forward link service Performance						
Capacity	**Coverage**	**detection probability**	**2D independent Localization probability**	**Independent Beacon Localisation Accuracy**	**Availability**	**Processing anomalies**
450 active beacons	European SAR/Galileo Coverage Area (GSRD-3268)	99% within 10 minutes	98% within 10 minutes	<5 Km with 95% confidence	99.5% over any one year	<1 every 10000 events

SAR/Galileo Return link service Performance			
Capacity	**Delivery Time**	**Coverage**	**Availability**
600 bits per minute on each satellite	within 15 minutes from reception by the Galileo Ground Segment	Global Earth coverage	>99% of the time over the nominal operational lifetime of the system

Table 5 : SAR/Galileo service performance

The performances achievable by the designed system are reported in a dedicated document [SYS.RD-97]

4. SYSTEM FUNCTIONAL CHAIN DESCRIPTION

4.1 Top Level Functional Tree

The table hereunder presents the list of the System Functional Chains, with sub functions:

SFC ID	System Functional Chain
SFC1	**Provide and Maintain the Galileo Constellation**
SFC1.1	Provide Nominal Constellation
SFC1.2	Maintain Nominal Constellation
SFC2	**Generate System Time and Geodetic Reference**
SFC2.1	Generate Galileo System Time
SFC2.2	Provide GPS-Galileo Time-Offset
SFC2.3	Contribute to Geodetic Reference Determination
SFC3	**Maintain Overall System Timing Synchronization**
SFC3.1	Satellite Times Synchronization
SFC3.1.1	Payload Frequency Reference Steering
SFC3.1.2	Satellite LGST Steering
SFC3.1.3	Satellite Clocks Monitoring and Switching
SFC3.1.4	Platform Timing Parameters Provision
SFC3.2	GMS Times Synchronization
SFC3.3	GCS Times Synchronization
SFC4	**Provide Mission Services (OS, PRS)**
SFC4.1	Open Services (OS) Provision
SFC4.1.1	OS Positioning Service Provision
SFC4.1.2	OS Timing Service Provision
SFC4.1.3	OS Signal Quality Monitoring Provision
SFC4.1.3.1	OS Signal-In-Space Accuracy (SISA) Provision
SFC4.1.3.2	Health Status Parameters Provision (on OS Signals)
SFC4.1.4	OS Degraded Navigation Data Provision
SFC4.3	Public Regulated Services (PRS) Provision
SFC4.3.1	PRS Positioning Service Provision
SFC4.3.2	PRS Timing Service Provision
SFC4.3.3	PRS Signal Quality Monitoring Provision
SFC4.3.3.1	PRS Signal-In-Space (SISA) Provision
SFC4.3.3.2	Health Status Parameters Provision (on PRS Signals)
SFC4.3.4	PRS Degraded Navigation Data Provision
SFC4.3.6	PRS Security Service Provision (Uncla)
SFC5	**Monitoring and Control**
SFC5.1	Ground Asset Monitoring and Control
SFC5.1.1	Ground Elements Monitoring and Control
SFC5.1.2	Ground Network Management
SFC5.2	Satellite Monitoring and Control
SFC5.2.1	On Line Satellite Monitoring and Control (incl. uncla security req.)
SFC5.2.2	Off Line Satellite Monitoring and Control
SFC5.3	Mission Performance Monitoring
SFC5.3.1	Mission Performance On Line Monitoring
SFC5.3.2	Mission Performance Off Line Monitoring
SFC5.4	Mission Planning Provision
SFC6	**Archiving**
SFC6.1	Archive Migration
SFC6.2	Archive Retrieval
SFC7	**Support Commercial Service**
SFC7.1	CS Data Dissemination

SFC ID	System Functional Chain
SFC7.2	Provide CS Ranging Code
SFC7.3	Provide Information to GOC Service Centre, GSMC, and GALSEE
SFC8	**Provide SAR Service**
SFC8.1	Provide SAR Forward Link Service
SFC8.2	Provide SAR Return Link Service
SFC8.2.1	SAR Return Link Resource Management
SFC8.2.2	SAR Return Link Message Dissemination
SFC8.2.3	SAR Return Link Interface to Cospas/Sarsat
SFC9	**Support EDBS Service**
SFC9.1	EDBS Management
SFC9.2	EDBS Dissemination
SFC9.2.1	EDBS Uplink
SFC9.2.2	ERIS Direct Uplink Inhibition
SFC10	**Deleted**
SEC-SFC11 (Cla)	**Control Access to the Ground Infrastructures**
SEC-SFC11.1	Security protection of the Galileo networks
SEC-SFC11.1.1	Protection of Intersite Internal Connection (Real-time and Non-Real-time)
SEC-SFC11.1.2	Protection of External Real Time Connection
SEC-SFC11.1.3	Protection of External Non Real Time Connection
SEC-SFC11.1.4	Network Security Management
SEC-SFC11.1.5	Network Security Device Key Management
SEC-SFC11.1.6	Protection of Inter GCC Connection
SEC-SFC11.1.7	Network Security Device M&C Dataflow Protection and Communication
SEC-SFC11.1.8	Intra-Site Network Protection
SEC-SFC11.2	Security protection of ground infrastructure
SEC-SFC11.2.1	Site Alarming and Physical Intrusion Detection System
SEC-SFC11.2.2	Physical Access Control
SEC-SFC11.3	Access control
SEC-SFC11.3.1	User Access control
SEC-SFC11.3.2	Asset Access control
SEC-SFC11.4	Information dataflow protection
SEC-SFC11.4.1	Anti-malware management
SEC-SFC12 (Cla)	**Security Protection Of Satellite Monitoring And Control**
SEC-SFC12.1	TT&C Data Flow Protection
SEC-SFC12.3	Satellite Key Management
SEC-SFC13 (Cla)	**Security Protection For Mission Data**
SEC-SFC13.1	C-Band Data Flow Protection
SEC-SFC13.3	Key management
SEC-SFC14 (Cla)	**Security Protection Of SoL**
SEC-SFC14.1	SoL Data Flow Protection
SEC-SFC14.2	Remote SM Management
SEC-SFC14.3	Key Management for SoL
SEC-SFC15 (Cla)	**Security Protection Of PRS**
SEC-SFC15.1	PRS Data Flow Protection
SFC15.1.1	NAVSEC PRS Protection
SFC15.1.2	COMSEC PRS MSG Protection
SFC15.1.3	COMSEC PRS OTAR Protection
SEC-SFC15.3	Key Management for PRS
SEC-SFC15.4	GSF Security Data Protection
SEC-SFC15.5	PRS Security Redundancy

SFC ID	System Functional Chain
SEC-SFC16 (Cla)	**PRS Access Management**
SEC-SFC16.1	PRS Users Management
SEC-SFC16.2	PRS Receivers Management (including secondary channel)
SEC-SFC16.3	PRS Orders Management (including secondary channel)
SEC SFC16.4	UNCLA PRS data Management through Secondary Channel
SEC-SFC17 (Cla)	**System and PRS Service Monitoring**
SEC SFC 17.1	PRS Service Performance Monitoring
SEC-SFC18 (Cla)	**Galileo SECURITY Monitoring and Management**
SEC-SFC18.1	Galileo Security Operational Intelligence Facility Monitoring and Management
SEC-SFC 18.1.1	Critical Assets Auditing and Monitoring
SEC-SFC 18.1.2	Cyber Security Processing
SEC-SFC 18.1.3	Cyber Security Monitoring, Control & Reporting
SEC-SFC 18.1.4	Forensics - Storage & analysis (offline)
SEC-SFC18.2	Security Events Monitoring and Management
SEC-SFC 18.2.1	UNCLA Security Events Centralization
SFC 5.1.1	Ground elements monitoring and control
SFC 5.1.2	Ground Network Management
SFC 5.2.1	On Line Satellite Monitoring and Control
SFC 6.3	Event Log collection and forwarding to GSMC
SEC-SFC 18.2.2	CLA Security Events Centralization
SEC-SFC 18.2.2.1	Satellite Monitoring and Control Remote SU Management
SEC-SFC 18.2.2.2	Mission Data Remote SU Management
SEC-SFC 18.2.2.3	PRS Data Remote SU Management
SEC-SFC 18.2.2.4	CLA Sec Event PKMF Centralization
SEC-SFC 18.2.3	UNCLA and CLA Sec Events Distribution to GSMC
SEC-SFC 18.2.4	Security Events Analysis and Management in GSMC (T&V Tool)
SEC-SFC18.3	Jamming Detection and Management

Table 6: System Functional Chain List

The following sections are organized in SFCs. For each one a presentation of the top level function is associated to second level functional analyses.
The Cla SEC-SFCs are addressed in the classified Annex of the DDF.

4.2 SFC1 Provide and Maintain the Galileo Constellation

4.2.1 Introduction

4.2.1.1 General Description

This SFC is responsible for the establishment of the Galileo satellite constellation and for the maintenance of this constellation over the programme lifetime. The functionality provided via this SFC can be broken down into the following sub-SFCs:

SFC1.1 Provide Nominal Constellation

This SFC provides all the functionality to launch and commission each of the Galileo satellites into the required position in the constellation. This includes:

1. Launch services;
2. LEOP operations
3. Handover of the satellites to the GCS
4. IOT (including Satellite platform and payload commissioning)
5. Satellite and GMS configuration to add new satellites to the constellation service provisioning purposes

Note that the fine-positioning manoeuvres will be executed from the GCS during the IOT, using the manoeuvre parameters calculated by the LEOP FDF, whereas the IOT Station will use the Orbit and Attitude data calculated by the GCS FDF.

At the end of the IOT phase, the reference orbits and biases will be available: these constitute an output of this SFC towards SFC5.2, which will use them to follow the dynamic evolution of the satellites and plan the station keeping manoeuvres.

SFC1.2 Maintain Nominal Constellation

This SFC provides all the functionality to maintain the nominal Galileo constellation and includes:

1. Station-keeping manoeuvres;
2. Re-phasing of in-orbit spares;
3. Constellation replenishment through the activation of on-ground spares;
4. Constellation renewal through the introduction of new satellites;
5. Disposal of satellites into graveyard orbits at their end of life.

The first level functional tree for the Provide and Maintain the Galileo Constellation SFC is shown below.

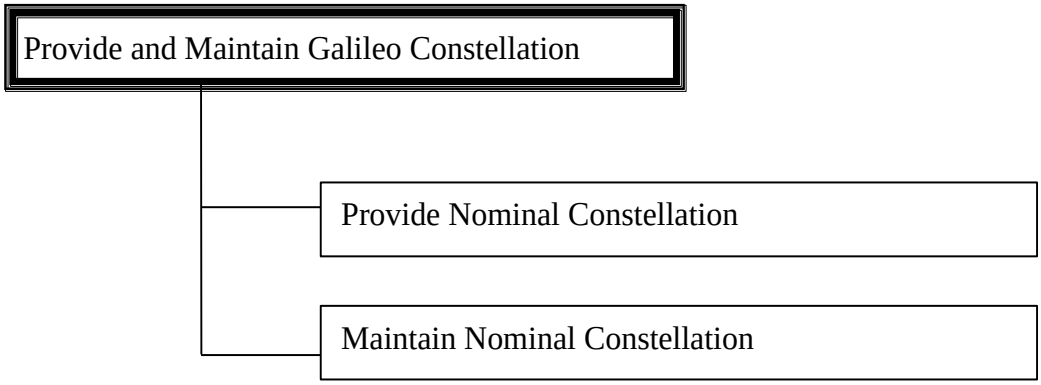


Figure 17: SFC1 Functional Tree

4.2.1.2 Definitions

Launcher Services Provider	Launcher Services Provider A Launcher Services Provider is an entity that will place a number of satellites into the target orbit by direct injection. There will be a different provider for every launcher (e.g. Starsem for Soyuz, Arianespace for Ariane 5).
ESCC	External Satellite Control Centre The ESCC is a physical facility external to the Galileo system comprising a satellite control centre and a network of ground TT&C stations. The ESCC may be used to perform specific functions in support of the Galileo mission, including LEOP, as well as the provision of additional ground TT&C resources in support of special operations (e.g. precise Orbit Determination in support of fine positioning manoeuvres, serious satellite outages, Galileo TT&C station failures, specific contingencies requiring extended TT&C coverage, etc)
LEOP	Launch and Early Orbit Phase: This phase covers all operations of the spacecraft (including orbit manoeuvres) as required up to the point where it can be formally handed over to the Galileo System (to perform IOT – see below). The LEOP phase is considered complete once the spacecraft has reached the drift orbit (with the solar arrays deployed and the spacecraft operating in normal mode) and all criteria for formal satellite acceptance have been met.
IOT	In-Orbit Testing Testing applied to the satellite in orbit following launch and the activation of an in-orbit spare. Typically a subset of the testing campaign previously performed on ground will be performed with the aim of confirming that no degradation to the satellite performances has occurred since on-ground acceptance. It includes Platform Commissioning, Clear IOT and Secure IOT. The IOT site will be equipped to perform unclassified Payload IOT measurements. During IOT the satellite M&C will be performed by the GCC-GCS.
GSC	GNSS Service Centre (GSC) The GSC deals with a range of issues related to the operation of the Galileo system. In particular, the GOC SC deals with all the administrative, legal and commercial issues related to the provision of Galileo services. In order to achieve this, it interacts with the GCC through the GMS which provides technical reports and technical data resulting from the daily functioning of the system. This includes reports related to the current status of all satellites within the constellation.

4.2.1.3 Assumptions

This SFC has been written under the following assumptions:
1. the IOT will be performed by the GCC-GCS + IOT;
2. the Platform Commissioning will be performed from the GCC-GCS, using the available TTC stations;

3. the fine-positioning manoeuvres will be performed by the GCS during the execution of the IOT (i.e. interleaved with IOT operations), using manoeuvre parameters calculated by the LEOP FDF
4. GCS has the capability to perform spacecraft M&C during IOT using either its own GCS TTCF in either Spread spectrum or ESA standard mode, or the external LEOP TTC stations, which only support TT&C operations in ESA standarad mode. The baseline is support IOT operations using the GCS TTCF in Spread Sprectrum mode, with the LEOP TTC stations available in case of need.
   IOT operations are usually performed when the spacecraft is simultaneous visibile from a S-Band station (for the transmission of TC and reception of TM data) and the IOT C band and L-Band station (for the transmission of the C band uplink and reception of the L band downliank signals in response to the transmitted TC).

4.2.2 SFC1.1 Provide Nominal Constellation

4.2.2.1 General Description

Galileo nominal constellation is designed to provide 30 satellites in a Walker 24/3/1 constellation. In the initial deployment of the full Galileo constellation each of the 3 orbital planes will include up to two spare satellites.
The system will ensure that all satellites have nominally circular orbits with a ground-track repeat cycle of 17 orbits in 10 days and a nominal inclination of 56 degrees.

The system provides the functionality to ensure that the initial biases on the orbit parameters of each satellite in the Walker constellation are optimized to satisfy tolerances for a 12 years timeframe during which one orbit keeping manoeuvre per satellite will be performed.

In order to support this strategy, the system will ensure that the initial orbit accuracy is high, with the semi-major axis of the satellite orbit correct to within ± 5m.

In addition, the initial orbit will be established such that for each satellite:

- The absolute out-of-plane position of any spacecraft with respect to the RAAN of the designated reference constellation orbit plane shall be better than +/-1.0 degrees

- The absolute out-of-plane position of any spacecraft with respect to the reference constellation inclination of 56 degrees shall be better than +/-2.0 degrees.

- The absolute along track orbit keeping between any spacecraft and its designated operational orbit slot in the same orbit plane shall be better than a user-definable value specified in degrees (nominally +/- 2.0 degrees).

In order to support this activity, the system will provide the following services:
- launchers services,
- LEOP services, after separation from the launch vehicle
- In-orbit testing (IOT) services

The Ground Segment will provide the following functions:

- LEOP services, satellite Commissioning and satellite in-orbit acceptance testing (IOT);
- Satellite post-acceptance testing;
- Contingency Operations;
- Planning of constellation replenishment.

Relocation of orbit spares will allow minimising unexpected satellite outages, although some careful strategy is expected to be put in place to carefully assess the overall constellation health status and ensure that sufficient spares are placed in orbit as far as possible to anticipate possible satellite losses.

For what concerns the choice of orbital parameters, it must be noted that the precise values of the target orbital parameters will be established by analysing the various Constellation Deployment strategies and Station-Keeping strategies.

Once the final Station-Keeping Strategy is agreed, the target orbits and biases will be communicated to the Launch Service Provider. LEOP servicesare provided by the Galileo system itself, supported by an external TT&C station network (**#STRB2.0_change: GAL-DCN-TASI-SYST-X-1131)**.

The choice of a specific Constellation Deployment Strategy will be driven by performance trade-offs to optimise early availability of services as a function of initial (IOV) and intermediate constellation geometry (transition to FOC) and for maintenance and disposal.

4.2.2.2 Launcher Services

To launch the satellites into orbit, the system will be able to make use of a choice of the available medium and heavy lift launcher systems including Soyuz and Ariane 5 (remark: Proton has been removed in **#STRB2.0_change: GAL-DCN-TASI-SYST-X-0853**):
The orbit dynamics after separation of the S/C from the launcher has been analyzed in detail in order to derive a separation strategy which guarantees collision avoidance between the satellites and between the satellites and the launcher.

The number of S/C injected into the orbit is for the big launchers ARIANE 5-ES:

- **4 S/C for AR5-ES:** The S/C are located symmetrically at one orbital plane so that the angular distance between adjacent satellites is nearly 90 degrees for launches L8 to L10 (up to GSAT2022)

The small launcher Soyuz is capable to inject two to two satellites into the target orbit. A single launch option is also possible where only collision avoidance between S/C and launcher must be considered, however, it could be foreseen for satellite replacement purposes:

- **SOYUZ:** 2 S/C in one orbital plane for launches L1 to L7 (GSAT0101 to GSAT 0104, GSAT0201 to GSAT 0206, GSAT0208 to GSAT 02011)

Batch 3 satellites (from GSAT0301 onward) are started to be launched at L11 at a rate of 2 satellites per launch. (**#STRB2.0_change: GAL-DCN-TASI-SYST-X-1131)**

The deployment analysis for multiple launches covers the trajectory dynamics beginning with the end of the separation process so that the deployment strategy guarantees short term and long term collision avoidance where only the separation process itself is not included. The dynamics of the relative translational and rotational motion during the separation process can only be performed with detailed knowledge of the separation mechanism.
As part of the separation procedure, the satellites in one level are separated simultaneously since no collision risk exists directly after separation due to the big angular distance (90 deg for 4 S/C, 120 deg for 3 S/C, 180 deg for 2 S/C). All analyses have been performed for a separation velocity of vs = 1 m/sec of the satellites w.r.t. the launcher. Results for other values of vs can be derived easily by scaling (distances ~ vs).

In particular, the Launcher Services will provide the functionality to:

- conduct the launch operations needed to inject the Galileo satellites into the required orbit
- obtain over flight permission, drop zone permission and announcing the launch to the international community

- ensure the successful separation between the satellite(s) and the launcher (Dispenser)

In order to support different constellation deployment strategies, the Launcher Services will ensure availability of different types of launchers, which can be customized to accommodate multi-spacecraft launches of Galileo IOV satellites.
The current baseline foresees:

- in IOV phase, nominally two Soyuz-Fregat ST launch vehicles achieving the following injection accuracies:
  - Semi-major axis: ± 100 km
  - Orbit eccentricity: ± 0.001
  - Inclination: ± 0.12 deg
  - RAAN: ± 0.12 deg

- in FOC phase, a combination of Soyuz-Fregat ST launch vehicles and presumably Ariane 5 ES.
- Transition time from S/C and launcher separation to the achievement of the final orbit, with the required accuracy, within 35 days, including time necessary for fine positioning, which can be as long as 8 days, and IOT operations performed in parallel on a single satellite at a time.
- Placing the Galileo satellites into the required orbits by direct injection (availability of an upper stage with re-start capability). The Galileo satellites shall start the initial switch on sequence from launch vehicle event and/or separation signals attaining a stable state with solar arrays deployed, without the need for ground commands.

Note: During the Galileo IOV phase, the baseline is that both spacecraft (in one level in the dispenser) are separated simultaneously since no collision risk exists directly after separation due to the large angular separation, i.e. 180o for 2 spacecraft. [N.B. for Galileo in general, depending on the number of satellites launched simultaneously, it will be necessary to perform the separation of the spacecraft from the launch vehicle dispenser sequentially with time delays between spacecraft from different levels]. For collision avoidance purposes, separation is performed with the Fregat upper stage in a 3-axis stabilised mode.

The required launcher attitude is constrained at separation and re-orientation (rotation about the roll axis) of the launcher is required so that the required direction of the relative velocity vectors at separation is achieved. The reference frame (U, V, W) is related to the launcher position in the orbit at the time of separation and is defined as follows:

- U: radius vector with origin at the Earth centre and passing through the launcher CoG
- V: perpendicular to U in the orbit plane (in flight direction)
- W: perpendicular to the orbit plane (right handed system)

The relative S/C velocity vectors after separation are:

- +V direction for one S/C.
- -V direction for the other

The requirements for the relative velocity vectors are:

- relative separation velocity: 1. ms-1 ≤ v ≤ 1.5 m/sec (TBC) (LSREQ-212)
- accuracy of relative velocity (3-sigma): delta-v / v = +/- 0.05.
- orientation error of velocity vector (half-cone, 3-sigma) = 5°, (including attitude errors of the launcher and errors in the separation process)

The tip-off rates of the spacecraft will be within the following ranges:

- longitudinal angular rate, about separation direction (S/C z-axis): +/- 2 °/sec.
- transverse angular rate, S/C x and y-axes+/- 2 °/sec (TBC) (LSREQ-211)

Note that this angle definition is valid for both single launch configurations (horizontal or vertical). Single launch horizontal configuration is when satellite +Z axis is along the launcher +X axis (longitudinal one), vertical configuration is when the satellite +X axis is along the launcher +X axis.

The main perturbation effect of attitude thruster activities on the trajectory motion is caused by the rate damping directly after separation. The maximum duration for rate damping is about 30 minutes which is a short time compared to the orbital period (14.1 hours). Therefore the perturbation effect can be described in a good approximation by a variation of the separation velocity vector of the spacecraft. The separation direction is along the satellite Z-axis, so that the maximum angular rates will occur about the transverse satellite axes (X- and Y-axes).

For all Galileo launch configurations, including IOV, deployment strategies have been derived which minimize the collision risk between the satellites and between the satellites and the launcher.

The Launcher Service Provider (Launch Authority) will provide to the LEOP operations team all the necessary information on the satellite separation and injection parameters.

The separation information file will contain data depending on the launcher authority. Typical data are:

- the vector state (position and velocity of the launcher, in an agreed reference frame (e.g. ITRF or ICRF))
- time (UTC)
- position and velocities of satellites at separation, and related covariance parameters (3sigma), TBD minutes after separation

4.2.2.3 LEOP Services

The LEOP starts with taking control of the launched GALILEO satellites after their separation from the launch vehicle until the positioning of the satellites in their final orbital slot.

The Galileo satellites are designed to accommodate all functionality necessary to support the constellation deployment phase.
In particular this includes:

- The satellite reaction control system and fuel budget necessary according to fulfil the constellation orbit keeping strategy (see above),
- The launcher interface capability to impress specific relative velocities satellites of a multiple launch at separation from the launch vehicle toavoid collisions .
- M&C and ranging TT&C capabilities, including:
  - the ability to configure the TT&C data protection functions; the ability to monitor and control multiple satellites in parallel during a single launch
- Spacecraft autonomy and monitoring capabilities to ensure spacecraft health and capability to detect and isolate on board failures.

The Launch & early operations phase (LEOP) starts at s/c separation from the launch vehicle and ends when the orbit drift start manoeuvres are completed and drift phase is started (to be completed in 5 days). It will be carried out under the direct control of the accredited GCC(GCS).

During this phase the satellite will go through the following operations sequence :

- INIT Sequence (at s/c separation event);
- Sun Acquisition Mode (entered autonomously from successful INIT Sequence)
- Earth Acquisition Mode (entered on Ground TC)
- Orbit Change Mode (entered on Ground TC, for correction of orbit injection dispersion & stn acq manoeuvres)
- Normal Mode (entered on Ground TC)

Notes:

1. Part of the P/F Commissioning tests and activities will/may be performed during LEOP.
2. The exact duration of the station acquisition/drift phase is dependent on final S/C orbital phasing/location and the planned drift rate, and it will therefore differ for each satellite.
3. Worst case drift phase for an individual Galileo satellite is nominally 10 days, but likely to be longer depending on drift/station acquisition strategy adopted to satisfy/optimize dual-satellite IOT operational constraints.
4. P/F Commissioning & PL IOT will/may start before the satellite reaches its final operational orbit (need for drift stop manoeuvres will need to be considered).

The LEOP will be scalable to comply with the FOC launch scenario required to complete the Galileo Constellation.

In order to minimize the number of station keeping manoeuvres to be performed during the satellite's lifetime, the LEOP Services will have to achieve the following accuracy, at the end of fine positioning:

- the phasing angle, u(k)i or its equivalent orbit vector element, shall be better than +/- 0.002° (3 sigma).
- the phasing angle drift rate, D(k)i or its equivalent orbit vector element, shall be better than 1.6 * 10-4 deg/day (3-sigma)
- the across track orbit angle shall be:
    - +/-0.01° for RAAN, RAAN(k)
    - +/-0.01° for inclination, i(k)

where:
k=orbital plan,
i = satellite number.

The execution of LEOP Services requires specific functionalities implemented in the GCS. Global and continuous satellite M&C coverage, is ensured by a suitable additional network of TTC stations in order to continue in parallel the operation of the Galileo constellation.
The GCC-GCS will include the means to interface with the Launch Base.
LEOP will be concluded with the start of the drift manoeuvre (Out-Plane & In-Plane) and transition to Nominal Mode.

4.2.2.4 Specific LEOP functionalities

In order to add LEOP satellite operation capabilities to the Galileo system, the GCC-GCS functions as well as the Galileo network capabilities are needed to be extended.
The GCC_GCS performs the M&C of two additional LEOP S/C in parallel to other ongoing S/C operations. All spacecrafts in LEOP require continuous parallel contacts which can only be ensured by using additional TT&C stations on top of those used for continuously ongoing Galileo constellation operations. This operational task requires extended planning functionalities and operational preparation facilities for creation and archiving of LEOP operational products within the GCC-GCS.

From a network point of view, the GCC-GCS is connected, via the GDDN, to 2 External Network Operations Centers (ENOC-1, ENOC-2) and the Launch Site. Every ENOC ensures the connectivity with up to 2 TT&C stations.

Specifically, the GCC-GCS can perform during LEOP:

- Orbit determination of all satellites separated from the same launch vehicle, either based on the expected position or on data from launch service provider
- A preliminary assessment of the injection orbit accuracy, within 20 minutes after spacecraft separation
- Computation of an optimised manoeuvre sequence to reach a arget position by minimising the propellant consumption
- Calculation of relative distances for collision avoidance (satellites and launcher upper stage)
- Simulation and validation of LEOP operations
- Estimations of delta-v (propellant budget)
- Calculation of specific Flight dynamics data during LEOP: orbit manoeuvre executions, satellite mode transitions, Sun/Moon eclipse windows
- External TT&C station data and their satellite visibility windows processing
- Calculate & control the spacecraft orbit and attitude in the following MEO orbit range:
    - Semi-major axis 29600+/-10000 km;
    - Eccentricity 0<e<0.4;
    - Inclination 44<i<68 degrees.
- Orbit determination with an accuracy as follows, for LEOP fine positioning, the last orbital manoeuvre:
    - Semi Major axis <= 2m,
    - Eccentricity <= 5*10^-6,
    - Inclination <=0.001 degrees,
    - RAAN <=0.001 deg.
- Network data exchange with 2 ENOCs and launch site

4.2.2.5 Satellite commissioning and IOT operations

After successful fine positioning at the end of the LEOP, the newly launched spacecrafts will undergo some tests to confirm that the satellite integrity and ensure that no degradation has occurred since on-ground acceptance.

The initial tests will be identified as Satellite commissioning tests, covering both Platform and Payload, and will be based on selecting specific units by TC and observing the corresponding TM parameters. Satellite checkout and Initial Platform and Payload commissioning tests will be based on S-Band TC&TM only, without using the IOT facilities in Redu.

The purpose of IOT tests is to demonstrate that the satellite units have survived launch conditions. This is achieved performing simple switch on tests, where possible avoiding any unnecessary operational risk. Demonstration of all backup or non-nominal modes is not within the scope of platform Commissioning tests.

The IOT operations campaign comprises two distinct parts:

1. Non-secure IOT operations, with the objective of verifying payload in orbit performance, based on unclassified data and signal measurements
2. Secure IOT operations (SIOT), with the objective of commissioning the on board security units, is a classified information, not addressed in this document.

The system will start IOT operations already during the drift phase (i.e. before the satellite has reached its nominal orbital position). This shall allow the IOT campaign to be completed within 5 days after the arrival of the satellite at its nominal orbital position The Overall duration of LEOP and IOT Phase (performed in parallel) will be completed within maximum 35 days.
Notes:

- The IOT test campaign will take into account the time constraints associated to satellite atomic clock stabilisation in the definition of the payload tests, in order to complete the IOT within the planned 35 days. These time constraints coming from the atomic clock user manual are recalled hereafter:
  - RAFS:
    - Allow a RAFS warm-up time of 4 hours.
    - Frequency stability after 48 hours is above 1000s.
    - Allow a RAFS Frequency stabilisation time of 4 weeks.
  - PHM:
    - Allow a PHM warm-up time of 20 hours (completion of the 6th step of start-up sequence).
    - Allow additional PHM turn-on time of 24 hours after warm-up time to reach a stability of approx. 10000seconds
    - Allow a PHM Frequency drift performance time of 1 week after the PHM_ON command has been received.
- The entire IOT campaign will require the GCS providing S-band TT&C monitoring and control while PL IOT tests are conducted from the IOT station facilities in Redu. Off line and real time coordination will be therefore required to keep both sides synchronised. Near real time TM availability at the IOT Station is foreseen to:
  - confirm PL configurations during the measurements / message uplink,
  - support trouble-shooting activities associated to unexpected behaviour,
  - use automated IOT procedure triggering by S/C TM parameters;
  - Relate measurements taken at Redu facility with on board Payload configuration.

The IOT Services Provider will provide the following functionality:

- Communication with the GALILEO Payload through C, L and UHF Bands
- Definition and execution of all the relevant IOT Mission procedures
- Uplink of mission data in C-Band
- Uplink of SAR beacon signals in UHF-band
- Reception of navigation data in L-band
- Reception of SAR signals in L-band
- Performing and evaluating PL IOT tests by using a dedicated test bench and measurement system
- Local archive of near real time TM parameters provided by the GCS and performed measurements

The GALILEO IOT Services will support the GALILEO P/L IOT operations on a single satellite at a time within a plan to optimise the overall schedule to support IOT operations of each satellite from the same launch.
It is foreseen that the IOT Services will complete IOT operations within 35 days for a maximum of two satellites in the same launch.
The IOT site in Redu has only 1 Antenna per type (C-Band, L-Band, UHF) and therefore can only perform measurements and message uplink operations for one satellite at a time. The baseline is to perform IOT operations sessions on a single satellite basis, however the duration of each session is limited by the satellite visibility, for both measurement purposes (i.e. the visibility of a given satellite from Redu is far from 24 hrs) and TC/TM purposes (i.e. the visibility of a given satellite from a given TT&C

station is far from 24 hrs), also considering that, at any given time, a given TT&C station can be used either for M&C of an operational satellite or for M&C of a satellite undergoing IOT.
The conclusion is that, for a 4 satellites launch, the IOT duration will be greater than 35 days, however less than 35 x 4 days, due to the fact that while Satellite 1 is not visible, Satellite 2 is visible and therefore the "dead time slots" present for 1 satellite will be utilized to perform IOT measurements of other satellites, in alternating fashion.
Note that from a State Probability point of view, the urgency to complete the IOT operations is essentially applicable to the first satellite of a replacement launch (in which case it is rather unlikely that 4 satellites are launched together: a single or dual launch is the more likely case for replacement launch).

The IOT Services functional breakdown is summarized hereafter:

- Facilities
  - IOT Test Control Centre:
    - control of the execution of the P/L IOT Services specific part of the Planned IOT Operational Procedures
    - observation of the execution of the P/L IOT Flight Procedures executed from the GCS, by means of near-real time TM monitoring)\
    - centralized data archiving
  - IOT Station facilities:
    - Uplink Navigation Data in C-Band using Spread-Spectrum Mode
    - Uplink of a SAR Test Signal in UHF-Band
    - Reception of the Navigation Data (including SAR Distress message) in L-Band using Spread-Spectrum Mode
    - Reception of the SAR Forward link downlink signal (Test Signal)
  - Interconnection Media:
    - Networks
    - Voice Loops
  - IOT to GCS interface:
    - Reception of Satellite Telemetry in near-real-time (of the satellite undergoing IOT)
    - Reception of Telecommand History for the Satellite undergoing IOT
    - Reception of Orbit data
  - Analysis Functions:
    - Test Preparation (preparation of IOT operational procedures)
    - Test Execution (automated execution of IOT operational procedures, display of satellite TM received from GCS, display results of automated IOT measurements)
    - Test Results Analysis (retrieval of archived data, analysis of retrieved data, Test Report preparation
  - IOT Measurements
    - Functional Tests
    - RF level Tests
    - Baseband level Tests
    - IOT test reports, IOT data and IOT results archiving
- Engineering Services
- Operations Services

The IOT Site will be equipped with a SCOS-in-a-box system, in order to receive, process and display, Telemetry received in near-real-time from the GCS.
The GCS will send to the IOT Site:

- Near-Real-Time spacecraft Telemetry
- TC History
- FDF predictions of orbit vectors for the relevant satellite
- Orbit data in TLE format

The IOT Site will also receive the relevant versions of the SRDB, by off-line media.

4.2.2.6 Satellite and GMS configuration for service provision

After a newly launched satellite has reached its final orbital slot and has successfully undergone all IOT tests, the satellite and the GMS need to be re-configured in order to add the newly launched satellite to the constellation for service provisioning purposes.
At satellite level, SVID for new SV (and therefore PRN codes) is defined before launch to allow execution of key loading tests and it is necessary to:

- Synchronize the on board NSGU time with the ground reference timescale (GST). The specific SCAT function of the MUCF is used for this purpose

At GMS level, it is necessary to update the constellation configuration information, in order to assign a previously unused family of PRS codes to the SVID chosen to identify the newly launched satellite. The updated configuration information is distributed to the relevant GMS elements, so that all the algorithms can start using the data broadcast by the new satellite.
See DDF2 [SYS.RD-02] dynamic scenario #SFC1.1-8 for further details.
#STRB2.1_change: GAL-DCN-TASI-SYST-X-0891 and GAL-DCN-TASI-SYST-X-0952

4.2.2.7 SFC Functional Block Diagram

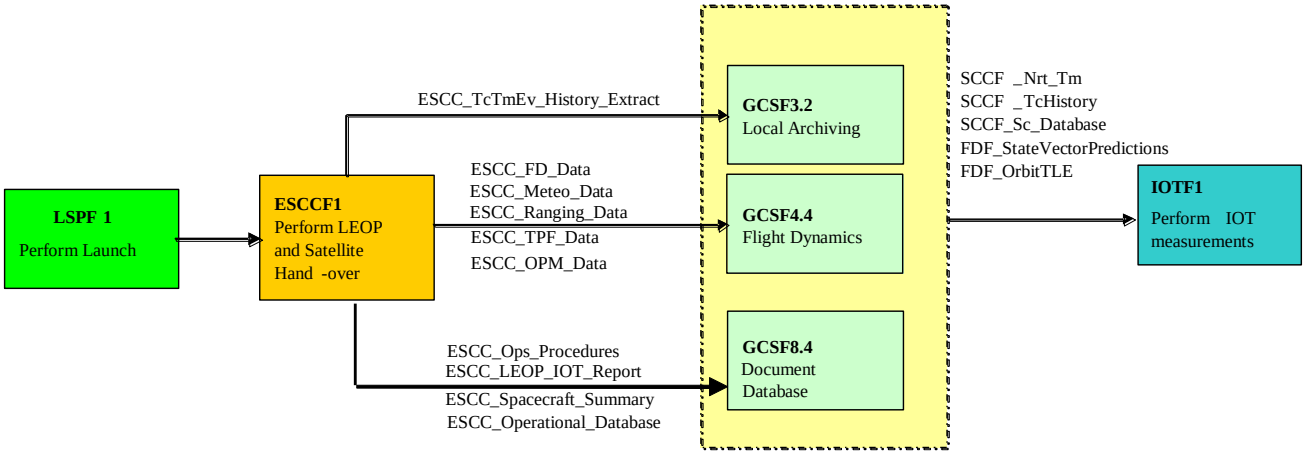


Figure 18: SFC1.1 Provide Nominal Constellation Functional Block Diagram

4.2.3 SFC1.2 Maintain Nominal Constellation

4.2.3.1 SFC Description

4.2.3.1.1 General Description

This SFC provides the functionality to maintain the nominal Galileo constellation of 24 operational satellites plus all in-orbit spares.
This functional chain illustrates how the system provides all of the functionality required to maintain and replenish the Galileo satellite constellation in line with a system lifetime of 20 years.
This is achieved in the following way:

1. Maintenance of the constellation through the:

- Re-phasing and activation of in-orbit spare satellites (nominally two per plane),
- Launch and activation of on-ground spare satellites. The functionality to launch, commission and perform in-orbit testing of these spare satellites is provided by SFC1.1.

2. Replenishment of the constellation through the construction of new satellites as and when required.

Note: the impact of spare satellite management (number of on-ground and in-orbit spare satellites, passed time from call-up until the spare satellite become operational, launcher strategy, etc.) on the system availability and RAMS performances during the Galileo system lifetime (20 years) are fully detailed in the related RAMS documentation: see dedicated RAMS contribution to System Availability document [SYS-RD.93] and associated performance impact analyses in SPBF [SYS-RD.4].

4.2.3.1.2 Re-phasing of in-orbit spares

Each in-orbit spare satellite can be re-positioned at any position within the orbital plane (+ /-180 deg).
The system will allow the re-phasing of an in-orbit spare such that the satellite will arrive in its operational orbital position (constellation slot) within an average of 11.41 days, if two in-spare satellites are available, or 13.6 days, if only one in-spare satellite is available, from call-up.

In the case of a manoeuvre interruption due to gas bubble ingestion within the thrusters, the re-phasing may be delayed by up to 2 days.

Then the in-orbit spares will be capable of achieving nominal operational performance within 1 day of arrival at their new operational orbital position.

Spare Galileo satellites will be fully on and transmitting as the nominal satellites in the constellation.
Navigation products will be generated by ground segment and up-linked to the spare satellites.. When a spare satellite is being re-phased (i.e. under manouever), it will not be able to transmit navigation signals..

The system will provide functionality to minimize the risk of collisions between a satellite and other constellation elements during a re-phasing manoeuvre.
If, for whatever reason, it is not possible to move a satellite that has failed or shows unacceptable performance degradation to a graveyard orbit, the system will re-phase a co-planar spare satellite up to a 5º relative anomaly with respect to the degraded (unmovable) satellite.

Note: in addition to the spare satellite re-phasing capability, the fuel budget is dimensioned so that any nominal Galileo satellite is capable of being re-positioned twice during its operational lifetime to a location 22.5º away in the same orbital plane within a period of 10 days.
This scenario is defined for cases in which no spare satellites are available at the time one satellite in a slot failed, and launching new spare satellites is delayed; in such a case it may be possible to re-position one of the neighbouring satellites halfway towards the failed satellite to reduce the gap in the Constellation until a new satellite is deployed in orbit.

4.2.3.1.3 Activation of on-ground spares

The Galileo satellites are designed to be able to be stored on the ground for up to 5 years.

The system will support the deployment of new satellites into their nominal orbital positions within 35 days of injection into orbit (including in-orbit testing).

For on-ground spare activation, the system will provide the following performance:

- For a single launch, it will be possible to deploy a spare satellite in either a spare location (half way between two nominal slots, and 180 away from an already deployed spare satellite) or in a nominal slot where a satellite has already failed, within 35 days of injection
- For a dual launch, it will be possible to deploy the first spare satellite in the nearest gap between orbital positions and the second one 180 degrees apart from the first one, within 35 days of injection.

4.2.3.1.4 Constellation Replenishment

In support of constellation replenishment, the system will send daily reports to the GSC providing a summary of the constellation status. This provides the GOC with the necessary information to initiate the contractual process to build replenishment satellites. (Note that this functionality is provided as part of SFC7.3 Provide Information to Service Centres).
For constellation replenishment (i.e. once in-orbit and on-ground spares are exhausted), the first 2 on ground spares will be available for deployment within 5 months of call-up.

Following this, any subsequent on ground spares will be available for launch in less than 5 months of call-up or 14 months after the last spare was launched (whichever duration is the larger).

In case of failure of a replenishment launch of 2 or more satellites, the system has the capability to conduct another recovery launch within maximum 2 years.

Marging against a replenishmenet satellite launch failure can be built maintaining up to 2 spares satellite per orbital plane.
It must be also noted that the design of the satellites might change during the constellation lifetime (20 years), which could imply, in specific cases, a faster replenishment of some satellites with respect to the original plan.

The system will provide all functionality necessary for the planning and execution of orbit manoeuvres in support of constellation maintenance and replenishment (see sections 3.3.2 and 3.3.4 above). This functionality is covered in detail in SFC5.4 (Planning Provision) and SFC5.2 (Satellite Monitoring and Control - both on-line in SFC5.2.1 and off-line in SFC5.2.2). In particular, this includes:

- Ground segment functions including:
  - Planning,
  - Flight dynamics,
  - Ranging.
- Satellite functions (including attitude and orbit control, ranging and satellite M&C).

The launch and in-orbit testing of all replenishment satellites will be performed according to the functionality described in SFC1.1.

The decision to launch a new spare would in general require defining:

1. initial deployment parameters (e.g., target phase value in the plane, arrival time)
2. Target operational orbit and the associated manoeuvres to reach it. This is performed running the dedicated FDF function, in which main input data are the reference orbit, the current operational orbit, the thrusters' characteristics and the operational constrains. The outputs will be the theoretical manoeuvres to be performed and the new propagated orbit that takes into account the computed manoeuvres.

At completion of the deployment operations the GCS Flight Dynamic Function will verify that all the station keeping requirements are fulfilled.

4.2.3.1.5 Satellite Disposal / Graveyard Orbit

The system will provide the functionality to dispose of Galileo satellites at the end of their operational lifetime by moving them to graveyard orbits in which they are clear of other operational satellites (both Galileo and other known external satellites).

The Ground Segment will produce the Flight Dynamics data necessary to compute the manoeuvre parameters which must be used to move the dead satellite to the graveyard orbit.
All Satellites will have the functionality and fuel allocated to be moved to a graveyard orbit, to turn off all equipment and to ensure that they do not generate any RF pollution.
The satellite disposal orbit will be located above the operational MEO orbit with a minimum re-boost altitude of 300 km.

Note: A collision between a satellite in the graveyard orbit and an operational satellite is extremely unlikely even in the case where the graveyard perigee reaches the operational orbit due to the following reasons:

- There is a differential precession rate in the RAAN between operational and graveyard orbit due to the difference in the semi-major axis. This differential precession rate is about 0.33 deg/year so that the RAAN values of both planes differ by about 6.6° after 20 years.
- Therefore a collision can only occur at the intersection line of both orbit planes if the satellites have the same altitude at this point.

Once transferred into the graveyard orbit and checks from ground are completed, the satellite will be put in safe passive state with all radiating elements switched off, to become effectively an in orbit debris, which will be tracked by the NORAD command centre in USA (this being out of the perimeter of the Galileo System).

4.2.3.2 SFC1.2 Functional Block Diagram

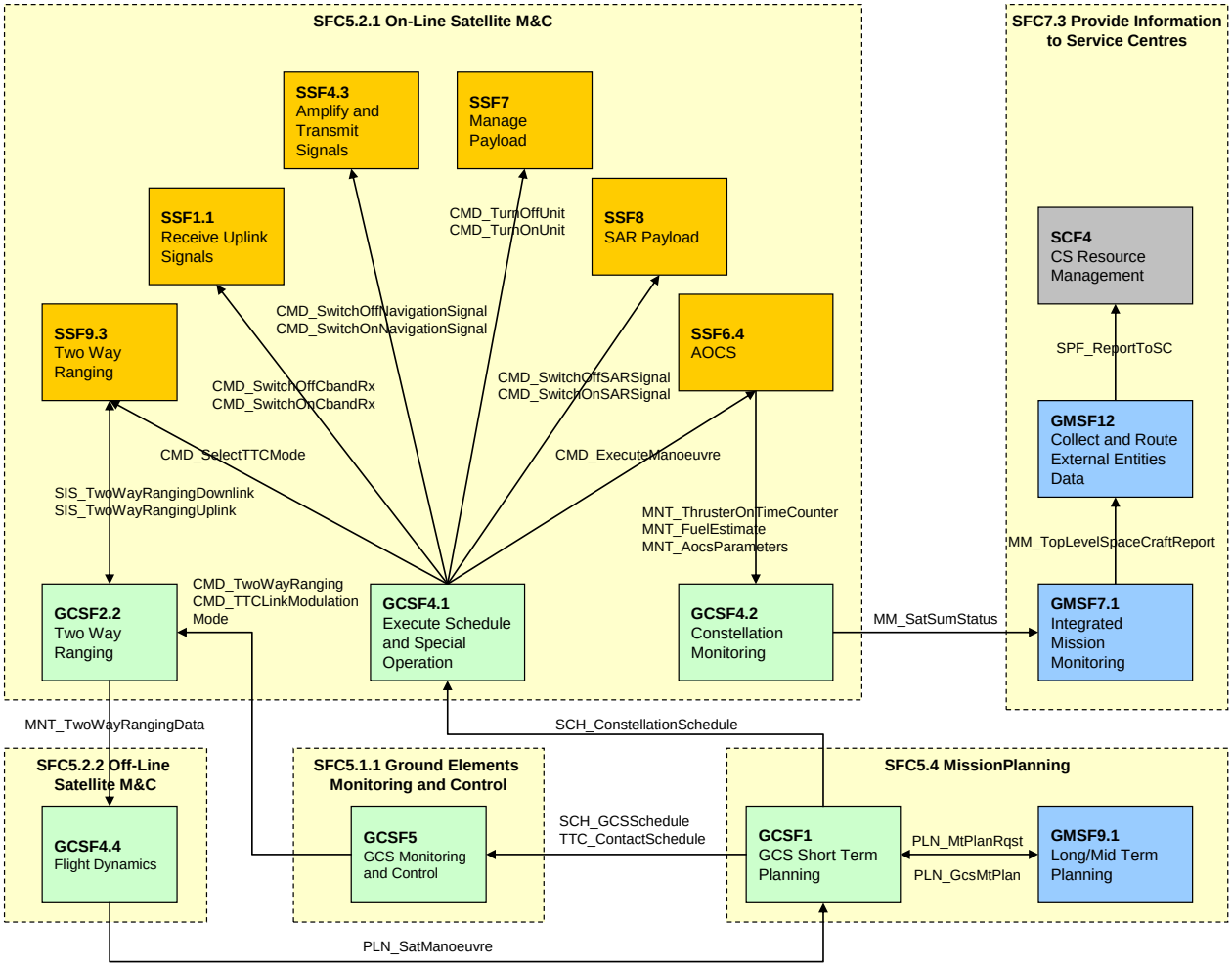


Figure 19: SFC1.2 Maintain Nominal Constellation Functional Block Diagram

4.3 SFC2 Generate System Time and Geodetic Reference

This System Functional Chain is in charge of providing all the means and associated processes needed by the system to generate and maintain the Galileo Timing Parameters and Geodetic Reference Frame, with proper interface with dedicated External Entities.
This includes three sub-SFCs corresponding to the generation of the following parameters:

- Galileo System Time (GST) in coordination with External Time Service Provider
- Galileo to GPS Time Offset (GGTO) in coordination with External GPS/USNO
- Geodetic Reference Frame (GTRF) in coordination with External GRSP

The first level functional tree for the “Mission Service Provision” SFC is then as shown below:

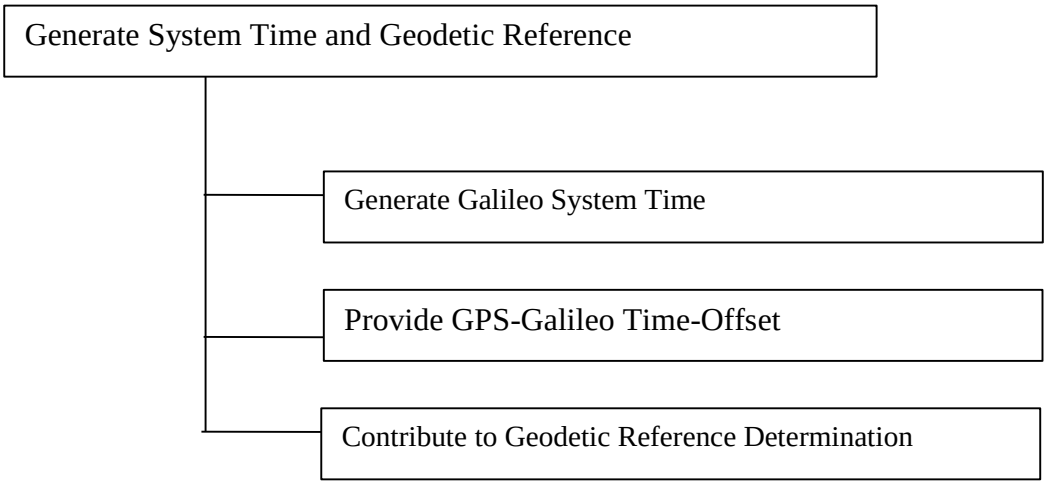


Figure 20 : SFC2 Functional tree (1st level)

4.3.1 SFC2.1 Generate Galileo System Time

The purpose of this SFC is to provide all timing functionalities needed by the Galileo System to support the provision of the mission services. This includes:

1. Navigation time keeping: GST provision for the OS and PRS navigation services
2. Metrological time keeping: GSTphysical realisation (GST(MC)), GST(MC) to UTC (modulo 1 second) steering and GST to UTC conversion parameters generation needed to support the timing service.
3. GST provision for system overall synchronization

It is important to note that the MRD specification (MRD-260/885) of the timing service performance and availability are conditioned to the availability and quality of the TSP products. Therefore, it is essential to ensure that the TSP product quality and availability is covered by appropriate TSP Design and Performance requirements in-line with the MRD timing availability and accuracy targets.

4.3.1.1 Definitions

This section provides definitions of the different timing parameters carried by this SFC.

- **GST:** The Galileo System Time(GST) serves as a stable time reference for Galileo System supporting the navigation (mainly the determination of Galileo Satellite Time Correction data), UTC dissemination and GPS interoperability function. The GST is a continuous coordinate time scale in a geocentric reference frame, which will be steered towards UTC modulo 1 second.

- **GST(MC):** this is the physical realization of GST at each GCC. GST(MC) is defined at a fixed and clearly identified physical point of the electrical path at PTF output of the timing signal used as reference by the GSS.

EAL: Stands for Free Atomic time, a free running timescale which is generated by the Bureau International des Poids et Measures (BIPM) in Sevres, France by averaging the data collected from more than 200 atomic time and frequency standards located at about 50 timing laboratories worldwide. Note: EAL is BIPM internal timescale.
TAI: International Atomic Time represents BIPM internal atomic timescale formed by calibrating EAL against a small number of primary caesium standards that generate the SI second. TAI is a continuous timescale, thus no leap seconds are inserted. TAI forms a basis for UTC timescale, please refer below.
UTC: Coordinated Universal Time (UTC) is the international atomic time scale that serves as the basis for all civil timekeeping. UTC is a 24-hour timekeeping system. The hours, minutes, and seconds expressed by UTC represent the time-of-day at the Earth's prime meridian (0° longitude) located near Greenwich, England. UTC is calculated by BIPM.
UTC runs at the same frequency as TAI. However, it differs from TAI by an integer number of seconds. This difference changes when leap seconds occur. When necessary, leap seconds are added to UTC on either June 30 or December 31 (note the leap second could be either positive or negative), please refer also to definition of UT1.
UT1: The UT1 is UT (Universal Time) corrected for polar motion. UT was originally a scientific name for GMT (Greenwich Mean Time). GMT is a former time scale for civil timekeeping based on mean solar time at Greenwich, now superseded by UTC. UT1 is influenced by the Earth rotation which is not a constant throughout the years. Consequently to keep the UTC within 0.9 seconds of UT1 a leap second is inserted. Then the decision as to when a leap second is needed is taken by the IERS (International Earth Rotation Service).

> Note: Where the reference to TAI is given in the context of GST steering or determination of its time and frequency offset, this shall be understood as UTC modulo 1 second.
> Coordinated Universal Time (UTC), maintained by the BIPM, is the time scale that forms the basis for the coordinated dissemination of standard frequencies and time signals. The UTC scale is adjusted by the insertion of leap seconds to ensure approximate agreement with the time derived from the rotation of the Earth. These leap seconds are inserted on the advice of the International Earth Rotation and Reference Systems Service (IERS), see also above.

The next figure explains the relation between the UTC, TAI and UT1 parameters:

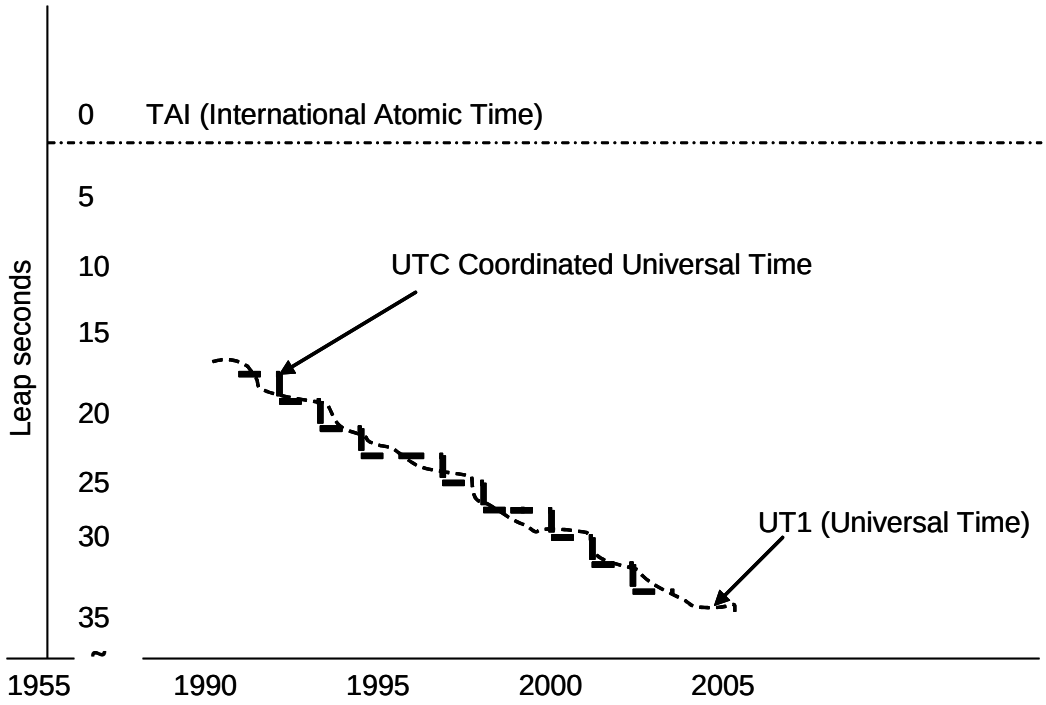


Figure 21 : Relation between the UTC, TAI and UT1

4.3.1.2 SFC Description

4.3.1.2.1 Navigation time keeping: GST provision for the navigation services

The navigation timekeeping functional chain is responsible for generating and maintaining a Galileo autonomous and stable timescale called GST, needed as reference to support the Galileo mission services (navigation positioning).

The navigation timekeeping functionality is provided by the Galileo Mission Segment including:

- The generation and maintenance of the GST stable timescale (GMSF4.3): The GST is generated from frequency reference generated by on-ground highly stable atomic clocks (Maser and Cesium). In addition the satellite clocks will be also used to extend the on-ground ensemble to obtain even more resilient time scale.

- The monitoring and control of on-ground atomic clocks.

During the nominal operations, the maintenance of the GST will be handled by the GMSF4.3. GST stability, crucial for the navigation functionality of the system, will be ensured by combining the two kinds of PTF clocks/frequency standards (Cesium and Active Hydrogen Masers) together.
The A-HM with the very good short term stability, however exhibiting rather large linear frequency drift in a long term (for observation times above one day, in general weeks/months) will be steered to the PTF ensemble of the Cesium clocks (exhibit better long term stability). By doing so the A-HM linear frequency drift will be effectively removed and its short term performance preserved.
As mentioned above, satellite clocks are considered in the GST generation process.
When the TSP interface is operational, the physical GST referred at PTF Maser output (GST-MC) is steered to UTC (modulo 1 s) using TSP inputs, as well as PTF clock ensemble. The TSP corrections are applied in such a way to maintain the GST stability performance required for navigation purposes. In

GST autonomous mode (i.e when TSP corrections are not available or reliable) the GST-MC is only corrected using the PTF clock ensemble measurements.

Within the scope of the Navigation timekeeping, GST generated by GMS needs to comply with the following implementation and performance constraints:

- The GST is generated by GMS autonomously from any other time scale: the key driver is the stability and knowledge of the autonomous time scale used as reference for the navigation positioning services. This stability has a direct impact on the positioning performance of the user.
- In case of lack of TSP steering corrections, the GMS shall not accumulate for a period of 10 days an incremental uncertainty of the GST- UTC (modulo 1 second) offset greater than 20 ns to 95% confidence level with respect to the last GST- UTC (modulo 1 second) offset provided by TSP. This excludes contributions due to the uncertainties on TSP corrections.
- The GST(MC) short term frequency stability, in term of Allan-Deviation, shall meet the stability requirements given in the following table:

τ (s)	ADEV
1	3 E-13
10	4 E-14
100	7 E-15
1000	4 E-15
10000	2 E-15

Table 7 : GST(MC) frequency stability

During IOV, a permanent non-compliance has been stated at GMS level against GST_MC stability (see GAL-RFD-GMS-000209 iss. 2),as quantified in the following table :

Integration time , tau (s)	**GMSREQ spec**	**Deviation**
1	3E-13	3.6E-13
10	4E-14	6E-14
100	7E-15	1.3E-14
1000	4E-15	5.0E-15
10000	2E-15	1E-15

These GST(MC) stability deviations from the OSPF assumptions on GST stability for satellite time correction computation are negligible. This is quantified in the following table, which shows the impact of GST-MC deviations on OSPF assumptions at the integration time of interest for computation of time correction for satellite RAFS and satellite AHM:

Integration time , tau (s)	**OSPF assumption**	**GST(MC) stability considering the proposed deviation**	**GST stability (combining GST(MC) and PTF GRC contributions) considering the proposed deviation**
100min	1.0E-14	3.2E-15	1.03E-14
8h	1.3E-14	3.2E-15	1.32E-14

The stability of GST(MC) physical realization shall be lower than 4.3E-15 in terms of Allan standard deviation in any 24 hours, including the contribution due to corrections for steering towards UTC (modulo 1 second).

The GST performance/stability requirements are discussed in the corresponding section of the SPBF.

4.3.1.2.2 Metrological Timekeeping: GST(MC)-UTC steering for the purpose of the timing service (provision of UTC conversion parameters)

The metrological timekeeping functional chain is responsible for:

- Steering the GST(MC) generated by the Galileo navigation timekeeping function to universal time scale (UTC), used as reference by timing service users.
- Provide GST to UTC conversion parameters to the timing service users to allow them to retrieve the UTC from the GST.

The Metrological timekeeping function is provided by TSP with support of GMS (GMSF4.3) with a substantial support of an external entity (the Time Service Provider). As opposed to the Navigation timekeeping where the preservation of the stable timescale is the most important task, the aim of the Metrological timekeeping is to steer the GST(MC) towards UTC (modulo 1 second) to maintain the time offset between the GST(MC) and UTC (modulo 1 second) as close as possible, without hindering GST(MC) frequency stability and to determine the GST-UTC offset with the required level of uncertainty

Note: To maintain the GST physical offset within 50 ns and the knowledge of GST to UTC (module 1 s) within 26 ns (95%)

4.3.1.2.2.1 GST(MC) to UTC Modulo 1 second Steering

The GST(MC) to UTC Modulo 1 Second steering is performed by GMS based on steering corrections generated by an external entity, the Time Service Provider (TSP).

a) Information provided by GMS to TSP: In order to generate the steering corrections, the GMS (GMSF4.3) provides to TSP the following information:

- Clock measurement data, consisting of the measurements of each of the clocks from the PTF ensemble (4xCs, 2xA-HM).
- Access to GST(MC): The GMS to TSP interface allows the TSP to accessing to the GST(MC) through CV (daily) and TWSTFT(every two hours)This allows the TSP to compare the GST(MC) and other UTC(k) timescales. The following time transfer techniques will be used:
- Galileo-CV and GPS-CV (Common in View): time transfer techniques based on Galileo-GPS combined receivers relying on Galileo and GPS Signal-In-Space.

These receivers are deployed on both GMS, to collect the data that are post processed by TSP following the CV standard code: Receivers deployed in GMS (GPS, Galileo and GPS/Galileo combined) will be connected to GST(MC). Receivers deployed in TSP will be connected to the corresponding UTC(k). In this way measurements collected at each side of interface will be referred to the respective system time. The measurements provide results in the form of GST(MC)-UTC(K).
TWSTFT (Two Way Satellite Time and Frequency Transfer): TWSTFT will serve as a primary method to provide the GST(MC) comparison results to the TSP. As opposed to CV, the reception of the SIS is not required. TWSTFT is based on the synchronous exchange of the radio frequency signals with modulated timing information over a geostationary communication satellite between the two stations of which the clocks are being compared. Each pair of stations working always in parallel need to strictly follow the

schedule denoting the slot during which the RF signals are exchanged. This schedule is provided to GMS by TSP via an operational interface.

The resulting data are forwarded by GMS to TSP via a dedicated interface. Galileo data are combined with the data from UTC(k) also gathered by TSP in order to generate results in the form of GST(MC)-UTC(K). TSP uses the results of the TWSTFT and CV measurements to assess in a "real time" GST(MC) performance with respect to other timescales (UTC(K), BIPM). The results of these comparisons are fed into the GST(MC) to UTC (modulo 1 second) steering algorithm to generate the corresponding steering corrections.

Feedback on the steering corrections actually applied by GMS.

Notes:

- Handling of the data addressed to TSP is done via a GMS catalogue based external web server.
- In addition the TWSTFT and CV will be used for GGTO determination and the PTF to PTF synchronization.

b) Information provided by TSP to GMS:

- GST(MC)-TAI frequency steering corrections: TSP generates the GST(MC) to UTC (modulo 1 second) steering corrections based on the products received from GMS, UTC(K) and BIPM. The steering corrections are defined in terms of in terms of normalized frequency steering correction and time of application. They are needed to meet the following GST accuracy implementation constraint:

  o The offset between the GST and UTC (modulo 1 second) shall be kept within the 50ns for 95% of any period of one year.

- The GST(MC)-TAI time offset phase and frequency offset prediction and their uncertainties. These data are needed to meet the following GST-UTC offset prediction performance:

  o The offset between UTC (modulo 1 second) and GST shall be known with a maximum uncertainty of 28 ns with 2-sigma confidence level, assuming the estimation of UTC six weeks in advance.

- Feedback on the GST(MC) performance including the information on the particular clocks from the PTF ensemble.
- The UTC-TAI as an entire number of leap seconds
- Results of monitoring the Galileo SIS regarding the validation of UTC dissemination performance

Notes:

- the 28ns figure includes both GMS and TSP contributions to the GST(MC)- UTC (modulo 1 second) offset uncertainty. Based on the allocation as per GMSREQ and GMSextIRD the TSP contribution is limited to 13ns (1σ): please refer to the corresponding section of the SPBF [SYS-RD.04].

- GMS shall ensure that application of the steering corrections does not affect the provision of the Galileo Services, consequently the steering products needs to be validated before their application. The steering corrections are applied 1:1 after validation: should the received steering correction be larger than a configurable threshold, is truncated to the threshold value and still implemented. More precisely, the steering is estimated to be not larger than the 24h stability of the master AHM, i.e. ca. 1 to 4E-15 (stochastic component); as a maximum limit, a threshold of 1E-14 is implemented. The PTF baseline foresees also two additional checks: if the TSP correction ADEV or the difference between the TSP correction and the one computed vs. PTF Cs ensemble exceed configurable thresholds, a warning to GACF is raised, but the corrections are still implemented.
- The GMS shall monitor the GST performance including:

o Long-term analysis of GST stability using as reference UTC computed by TSP .
o Accuracy of GST- UTC(modulo 1 second) Offset using as reference UTC computed by TSP .
o OD&TS accuracy through off-line comparison of OD&TS results with external data provided by TSP and GSRP.

An improvement of the interface between PTF and TSP is planned for FOC2 by sending additional information about PTF operational events from GMS to TSP.
#STRB2.0_change: GAL-DCN-TASI-SYST-X-0924 and GAL-DCN-ASTG-SYST-X-13617
The TSP will adapt the determination of GST-UTC offset estimate taking into account the specific GMS operational events. This new functionalities includes:

- the transmission to TSP of information such as PTF operational mode, GST ADEV degradation, GST phase jump, GST frequency jump, Clock ADEV degradation, Clock phase jump and Clock frequency jump The TSP will adapt the determination of GST-UTC offset estimate taking into account the specific GMS operational events.

- Generation of the new signal TIME_PtfAnomalyData from SPF to automatically notify the TSP when a clock anomaly is detected within the PTF.
  - Provision to TSP in automatic mode of absolute frequency correction value applied by the PTF master AHM (new signal TIME_AbsFreqCorrApllied) and PTF switch.
  - Complement of the operational interface to support the exchange of relevant operational events (CNF_PtfStatus): information with impact on the availability of system data provision to TSP
    - TWSTFT station calibration changes
    - GPS receiver calibration changes,
    - Replacement or change of calibration values in the measurement system
    - Picostepper. Replacement or change of calibration values in the picostepper
    - Interface problems: Interface failures between GMS and TSP (VPN Issues, DNS Errors, data batches transfer not completed...)

#STRB2.0_change: GAL-DCN-ASTG-SYST-X-13922: improvement of timing monitoring

The TSP provides to GMS the results of monitoring of Galileo SIS regarding the validation of UTC dissemination service. This includes regular forwarding to GMS, on hourly basis, of a performance report (TIME_UtcMntReport) on GST-UTC conversion parameters disseminated by Galileo.

The validation report is generated by TSP using Galileo SIS measurements carried out through a Galileo timing receiver deployed at a UTC laboratory and referenced to local UTC physical realization by means of 1PPS and 10MHz signals.

The differences between the UTC retrieved through Galileo timing receiver (UTC(user)) and UTC physical realization at the hosting laboratory (UTC(k)) are collected and passed to the TSP on an hourly basis along with the GST-UTC conversion parameters decoded by the Galileo timing receiver from the Galileo SIS reception. The TSP will estimate the offset between GST(User) and the UTC prediction computed by the TSP and use these [UTC(User)-UTC] values to compute performance measures based on a statistical analysis over the past 24 hours. The TSP will then forward the last hourly values and the statistical analysis to the GMS on an hourly basis by means of the signal (TIME_UtcMntReport).

4.3.1.2.2.2 GST to UTC conversion parameters provision for the Galileo timing service

In order to support the timing service, the Galileo system provides GST to UTC conversion parameters to the users (including UTC-TAI offset referred to as the leap seconds). These conversion parameters such as A0, A1, etc are generated by the GMS based on the products provided by the TSP.
Regarding performance, the GST to UTC conversion parameters generated by GMS allow Galileo user to achieve the following:

- Timing accuracy: a user equipped with a Standard Timing/Calibration Laboratory Receiver will be able to predict UTC to 28 ns for 95% of any 24 hours of operation operation, without any other aids, when the TSP interface defined in Req. GSRD-2437 is operational.

  The GMS contribution to the uncertainty of the GST- UTC (modulo 1 second) shall not exceed 9 ns (95% confidence level), excluding PTF-TSP synchronization link uncertainty contributions external to GMS.
  The offset between GST(MC) and TAI shall be determined by the TSP with a maximum uncertainty of 26 ns to 95% confidence level assuming the estimation of TAI six weeks in advance, excluding the GST(MC) -TAI offset uncertainty contributions due to GMS.

- Frequency accuracy: Based on the Galileo SIS and the GST-UTC conversion parameters, a user equipped with a Standard Timing/Calibration Laboratory Receiver will be able to determine frequency (a posteriori) with an accuracy (expressed as a normalized frequency offset relative to UTC, averaging over one day) better than 3 x 10-13 (2-sigma).

  The GMS contribution to the GST frequency accuracy shall not exceed 3.6E-14 (95% confidence level) averaged over 24 hours, excluding the contributions due to PTF-TSP synchronization link uncertainty external to GMS.
  TSP contribution to GST(MC) frequency accuracy (expressed in terms of normalized GST (MC) frequency offset relative to UTC frequency) shall be better than 5.4E-14 (95% confidence level) in any period of 24 hours, excluding the normalized GST(MC) -UTC frequency offset contribution due to GMS.

For FOC2 it is planned that the GST-UTC Conversion parameters Ao and A1 disseminated to the users will be set to dummy values to indicate that they are unavailable, see SYS-RD.71 (**#STRB2.0_change: GAL-DCN-ASTG-SYST-X-13884, GAL-DCN-ASTG-SYST-X-13885, GAL-DCN-TASI-SYST-X-0960 and GAL-DCN-ASTG-SYST-X-13894 : definition of dummy GST-UTC value)**

TSP will set the GST-UTC offset parameters to dummy (i.e.: GST-UTC offsets=.-10^{-30} s and GST-UTC offset drifts= -10^{-50}) in the following cases:

- occurrence of out-of-range (wrt their maxInclusive values) in the TIME_GstUtcOffset signal sent by TSP to GMS or
- the GST-UTC offset predictions are deemed not reliable by the TSP for any reason (see SYS-RD.71).

The dummy values for GST-UTC Conversion parameters Ao and A1 shall consist of a binary sequence of one digit (SYS-RD.71).

Notes:

- The dummy corresponds to small negative values;
- The all one's sequence adopted to signal dummy Ao and A1 values is made to provide backward compatibility. However, this has a (low) impact at user level, as the corresponding decimal value (Ao-=.-10^{-30} s and A1=-.-10^{-50}) can also occur in nominal operations.

Leap second information are always disseminated, even when the Ao and A1 coefficients are set to dummy. Indeed, the leap second information is needed for navigation purposes, to allow the user receivers to time-tag the position solution.

The GMS will deploy dummy value for Ao and A1 coefficients if at least one of the following events occurs:

- No GST-UTC offset prediction from TSP is available for more than 10 day (GMS long autonomy) **(#STRB2.0_change : GAL-DCN-TASI-SYST-X-0960**)
- Occurrence of a GMS Timing HMI
- GST-UTC offset sets to dummy by TSP due to occurrence of out-of-range value or unreliable GST-UTC offset estimate
- Occurrence of critical alerts in the GMS infrastructure involved in the application of GST-UTC offset data provided by TSP.

In case the GST-UTC offset prediction from TSP is unavailable for not more than 10 days (GMS short autonomy), the GMS disseminates the latest available GST-UTC prediction provided by TSP (**#STRB2.0_change : GAL-DCN-TASI-SYST-X-0960**)

#STRB2.0_change : GAL-DCN-TASI-SYST-X-0986 : automation of provision of timing products

Normally, the GST-UTC dissemination takes place in automatic way, to avoid the risk of potential operator error. However, the operator is provided with the means to disable operation, to provide manual GST-UTC dissemination.

Notes:

- As stated above, when enabled, an automatic GST-UTC dissemination takes place when the GST-UTC offset provided by TSP is set to dummy. In this case a warning to the operator is also raised.
- A warning to the GMS operator will be also raised when the estimated GST-UTC(SIS) accuracy exceeds a configurable threshold, to support GMS mission operation(according to **#STRB2.0_change GAL-DCN-TASI-SYST-X-0949**). However, in this case the automatic dissemination of dummy GST-UTC offset is not foreseen.

In addition, regular on line trend monitoring aof GST-UTC offset is foreseen at GMS level, to allow early identification of degraded GST-UTC offset accuracy performance and an operational interface with TSP for recovery .

Note: The validity time (48 hours) for full performance of GST-UTC conversion parameters will be defined in the Service Definition Document. The GST-UTC performance are not guaranteed if the age-of-date of GST-UTC offset message exceeds 48 hours.

In order to ensure the correct computation of GST-UTC offset at user level, the following constraints apply to the weekly parameters disseminated in the GST-UTC conversion parameters message:

- the absolute value of the difference between untruncated WN and WNot values shall not exceed 127
- when Dt_{LS} and Dt_{LSF} differ, the absolute value of the difference between the untruncated WN and WN_{LSF} values shall not exceed 127.

GST-UTC Conversion weekly parameters WN, WNot, and WN_{LSF} are defined in SISICD-1423.

4.3.1.2.3 GST provision for System overall synchronization

The GST needs to be distributed with in the Galileo in order to ensure the system overall synchronization (see dedicated SFC3.2&SFC3.3) The GST distribution relies on the following interfaces:

- 1 pps (1 pulse per second), derived directly from the master clock output, referred to as GST(MC), together with the 5/10 MHz frequency signals output from the master clock, used for the synchronization of local elements;
- IRIG-B: Used for the timing synchronization of distant elements, however within the GCC boundaries. The time codes originally developed by the Inter-Range Instrumentation Group (IRIG), is now used in

government, military and commercial fields. There are many formats and several modulation schemes, but they are typically amplitude modulated on an audio sine wave carrier. The most common version is IRIG-B, which sends day of year, hour, minute, and second data on a 1 kHz carrier frequency, with an update rate of once per second. The time code generators will be fed with the reference signals derived directly from GST(MC)

- NTP: Network Time Protocol used for the timing synchronization of the distant elements over TCP/IP networks.

Note: As mentioned above, GST is used to synchronise elements internal to Galileo system, whereas External entities synchronise themselves to UTC

4.3.1.3 SFC Functional Block Diagram

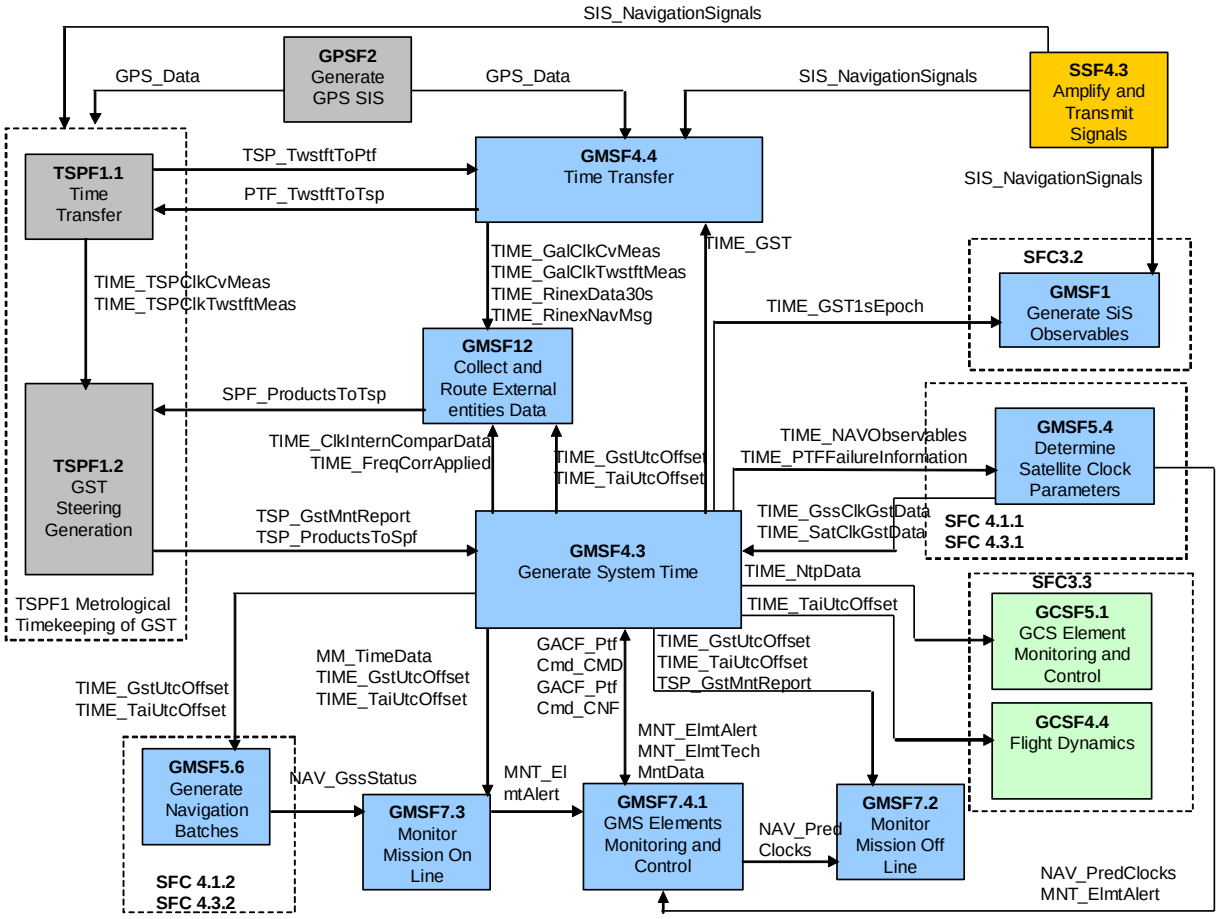


Figure 22 : SFC2.1 Functional Block Diagram

4.3.2 SFC2.2 Provide GPS-Galileo Time-Offset

Galileo system has been designed to be compatible with the current GNSS, and in particular with the GPS.
In order to further enhance the interoperability between Galileo and GPS, both systems agreed to determine jointly the intersystem time bias (referred to as GGTO Galileo-GPS Time Offset) between the respective system times and to propagate this information to their users.

Note: DDF1 description captures the baseline described in the latest applicable issue of the GMS-GGTO ICD [AD.15]

4.3.2.1 SFC description

This SFC is responsible for generating in coordination with GPS the GGTO (GPS-Galileo Time Offset) parameters needed by the OS and PRS positioning service users, who would like to use jointly GPS and Galileo System for positioning determination.

The GGTO is generated on Galileo-GPS combined receiver (SIS based determination) technique; the use of combined receiver for GGTO determination has been specified for FOC ([see GMSREQ-11751 and relevant ESA-DTEN-NG-DCN/03575 v1.3.

Notes:

- To compute a position operating with the combined Galileo-GPS receiver without GGTO, it is necessary to resolve 5 unknowns (one more than with a single constellation): user position (x, y, z), GST-receiver offset and GPS-receiver offset. A total of 5 satellites in view from both constellations are then needed to compute a position. In case GGTO is available, the position can be obtained with four satellites in view.

The following performance constraints apply to the function responsible for GGTO determination:

- The GMS contribution to the accuracy of the offset between GST and GPS Time (modulo 1 s) shall be less than 3.5 ns with 2-sigma confidence level over any 24 hours when the GGTO is determined through combined Galileo/GPS receiver.
- The stability of GGTO, expressed as an Alan deviation, is better than 2x10e-14 over any one day.
- The validity of computed Galileo/GPS time Offset shall be 1day.

Note: The GGTO uncertainty requirement specified in GSRD-3123 of 5 ns (2-sigmas) cannot be met with the time transfer driven GGTO determination, please refer to the corresponding section of the SPBF.

In particular, a GGTO performances requirement of 12.6 ns/24 hours (2-sigmas) with common view link has been specified at GMS level (see GMSREQ-11753) .

The GGTO accuracy budgets by using Combined Galileo-GPS receiver is given in SYSPBF.

A GGTO coordination process with GPS is carried out before GGTO dissemination by comparing against a configurable threshold the difference between the GGTO value determined by Galileo and the GGTO value determined by GPS System (GMSREQ-11053). The threshold used for GGTO coordination check on the delta between the GGTO computed by Galileo and USNO (TIME_GgtoDelta) will be set to a default value of 10 ns (see SYS-RD.72), that is two times the GGTO accuracy specified in GSRD (5 ns at 95% percentile).

The GGTO validation process also implies a check on the availability/validity of GGTO determined by Galileo. As a result of this process, either validated GGTO or dummy GGTO can be disseminated.

In particular, the disseminated GGTO will be set to dummy in case one of the following events occurs (nominal GGTO configuration, see SYS-RD.72):

- There is no GGTO value available at PTF output
- There is an alarm in the infrastructure involved in the computation of Galileo GGTO

- The difference between GGTO computed by GALILEO and USNO (TIME_GgtoDelta) is not monitored.
- The delta between GGTO computed by GALILEO and USNO (TIME_GgtoDelta) exceeds a configurable threshold and the coordination process with GGTO is not able to solve the discrepancy between PTF GGTO and USNO GGTO.

The above-mentioned list of events triggering the dissemination of dummy GGTO is applicable to nominal system configuration, when the GGTO computation is active at USNO.

#STRB2.0_change: GAL-DCN-TASI-SYST-X-986: automation of GGTO provision

Normally, the GGTO coordination validation process and the GGTO dissemination take place in automatic way, to avoid the risk of potential operator error. However, the operator is provided with the means to disable both automatic GGTO coordination validation and automatic GGTO dissemination, to provide manual operation.

In this case the threshold for the check on the delta between the GGTO computed by Galileo and USNO (TIME_GgtoDelta) will be set to a default value in the order of 10 ns, see SYS-RD.72.

Until the computation of GGTO at USNO is not active, an interim dummy GGTO policy is put in place. In this case, both automatic GGTO coordination validation process and automatic GGTO dissemination are disabled.

In order to ensure the correct computation of GGTO at user level, the difference between the truncated week number WNOG in the disseminated GGTO and untruncated Galileo week number WN is operationally kept below 31.

4.3.2.1.1 Galileo-GPS combined receiver driven determination: FOC baseline

The GGTO determination based on the use of a Galileo-GPS combined receive is the primary method for FOC, as it guarantees better GGTO accuracy performance with time-transfer methods.

a) GGTO determination

The GGTO combined receiver driven determination is a SIS based method which is relying on the simultaneous reception of the Galileo and GPS signal in space.

A combined receiver is deployed on each GPS and GMS side. On each side the combined receivers collect the observables from both Galileo and GPS SIS transmitted by Galileo and GPS satellites. The GGTO is automatically generated by receiver based on those measurements. No further processing or the exchange of the measurements results is required.

b) GGTO coordination process

The GGTO result obtained with combined receiver needs to be coordinated with the GPS first, before it can be adopted by GMS for the inclusion into the Galileo navigation messages.

Note:
- Handling of the data addressed to GPS/USNO will be done via a GMS catalogue based external web server.
- The interface between GPS and GMS allows also the exchange of the data related to the interface status and contingency information (see SFC7.3). The data are routed to GPS with GMSF12.

4.3.2.2 SFC Functional Block Diagram

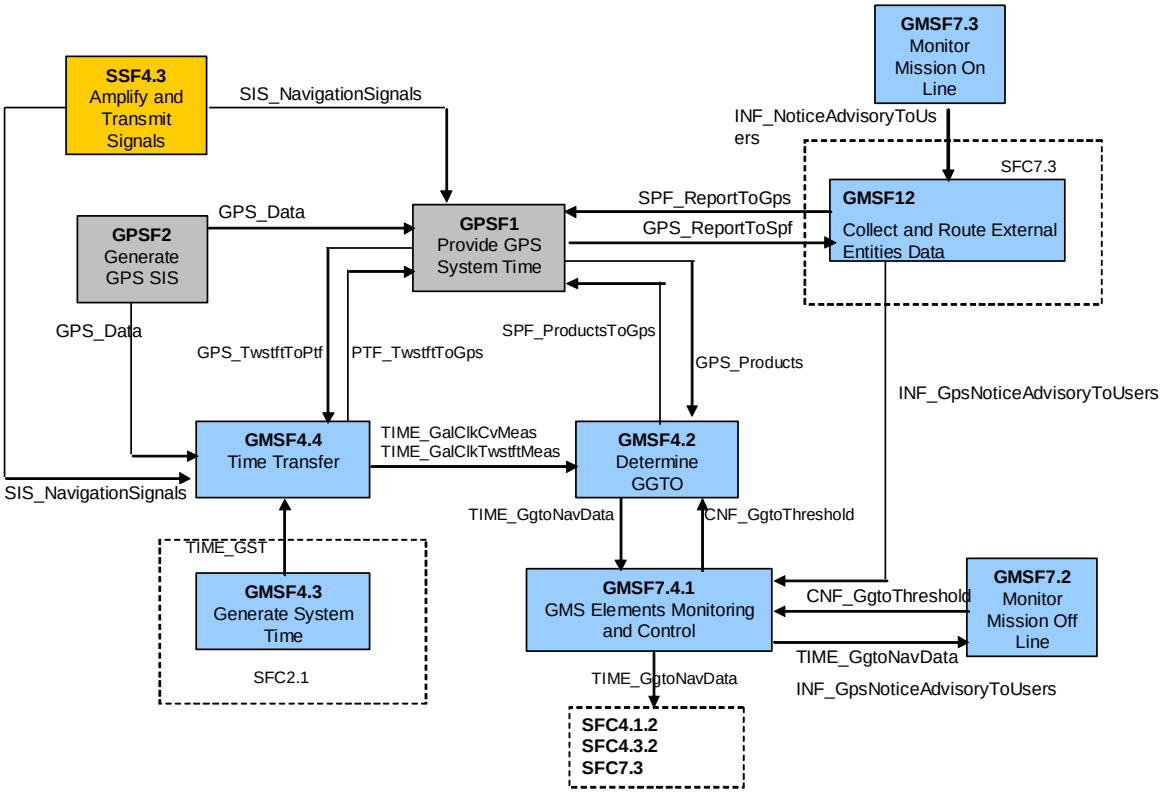


Figure 23 : SFC2.2 Functional Block Diagram

4.3.3 SFC2.3 Contribute to Geodetic Reference Determination

The SFC is responsible for:

- Supporting the generation and maintenance of the Galileo Terrestrial Reference Frame
- Supporting the SLR campaigns and provision of the SLR data to GMS
- Providing non GTRF related products consumed by both GMS and GCS
- Providing independent validation of a number of GMS products

It is important to note that the MRD specification of the geodetic service performance and availability are conditioned to the availability and quality of the GRSP products. Therefore, it is essential to ensure that the GRSP product quality and availability is covered by appropriate GRSP Design and Performance requirements in-line with the MRD geodetic availability and accuracy targets.

4.3.3.1 Definitions

Geodetic reference system: A geodetic reference system is a coordinate system to which positioning and timing application refers to. Such a system is defined by:

- An origin (as close as possible to the Centre of mass of the Earth),
- 3 orthogonal Cartesian axes, oriented opportunely,

- An associated ellipsoid, chosen to best fit a global geoid,
- Models, algorithms and constants that contribute to its realization.

The term reference frame is addressed to a specific realization of a reference system, referred to a given epoch.

- **Galileo Terrestrial Reference Frame:** The GTRF is the geodetic reference system forming the basis of all related calculations inside the Galileo system. The GTRF is determined by GRSP, with support of GMS, autonomously from other terrestrial reference frames.

  GRSP provides 3D Cartesian coordinates and velocities of the GSS tracking antennas in GTRF (GSS coordinates). This set of coordinates and velocities define the GTRF.

  Note: Autonomous should be understood as self-sufficient /sovereign, nevertheless tightly linked to ITRF in order to meet the GTRF definition and corresponding performance constraints.

4.3.3.2 SFC Description

4.3.3.2.1 Support the generation and maintenance of the Galileo Terrestrial Reference Frame

The Galileo Terrestrial Reference Frame (GTRF) is determined by the Geodetic Reference Service Provider (GRSP) with the support of GMS.

GTRF realization is based on the globally distributed network of Galileo Sensor Stations (GSS.) and it is generated according to the following process:

a) GMS Raw data collection, provision of data to the GRSP
GMS is responsible for collecting and providing to the GRSP the following:

- the code and phase pseudo-range measurements (raw data) from all Galileo and GPS satellites in view collected by the GSS network (including the files containing the Galileo navigation messages).

Note raw data are supplied to GSRP in e RINEX Format.

- -other GTRF related information needed by the GRSP to realize the GTRF, including:
  - Initial sensor station antennas co-ordinates obtained during the installation / re-installation to an accuracy of 5cm (95 % confidence level) with respect to the ITRF.
  - Measurements taken by all antennas at all available GSS stations from all frequencies with a sampling rate as specified in the corresponding ICD.
  - Calibration of all GSS receiver antennas including absolute phase center variation tables (PCV) for all applicable frequencies
  - Calibration of satellite antenna including absolute PCV tables for all applicable frequencies. The update frequency of these data is to be specified in more detail in the GMS to GRSP ICD [AD.14].

Notes:

- Exhaustive list of the data to be provided by GMS to GRSP is given in the corresponding ICD.
- Handling of the data addressed to GRSP will be done via a GMS catalogue based external web server.

An improvement of the interface between GMS and GSRP is planned for future Galileo evolution by sending additional information from GMS to GSRP.
#STRB2.0_change: GAL-DCN-ASTG-SYST-X-13618

This includes:

- Provision to GRSP of the GMS static satellite and start-up GSS configuration (CNF_SatCfgDataStatic).which is required for the analysis/troubleshooting of the system behaviour without the need to rely on operator action

- Complement of the Operational Interface to support the exchange of relevant Operational Events related information with impact on the availability of system data provision to GRSP. In particular for information regarding the GSS Timing Mode (report a change of the time realisation mode configured at the GSS), GMS-GRSP I/F failures (VPN Issue, DNS error, Data batch not received complete in due time, …) or the GRSP-GMS Web interface failure

- Modification of the GRP_Alarm schema file in order to include the Severity concept for the management of Earth Rotation Parameters alarms, especially their unavailability, and updating of the enumerated according to the insertion of new interfaces as per first bullet.

b) GRSP products generation

Based on the data provided by GMS, the GRSP is responsible for computing the precise GSS coordinates. GRSP is required to perform the periodical recalculation of GSS coordinates to ensure that the GTRF is calibrated to ITRF to within 3 cm (2 sigma).

The positions of all GSS's are computed/maintained by embedding the GSS's into a larger network of IGS stations and solving the station coordinates simultaneously for all stations of the whole network. This allows computing the calibration necessary to align the GTRF to the latest ITRF realization.

GRSP provides the following GTRF and transformation data

- The coordinates and velocity of all GSS antennas, and related information
- The transformation matrices and procedures between GTRF and ITRF
- The physical model of the Earth to be used in the transformation of the Keplerian parameters into geocentric Cartesian coordinates

GRSP also provides the following additional geodetic parameters supporting the GTRF determination and maintenance:

- Earth rotation parameters
- Solar radiation intensity constant
- GSS coordinates and velocities of GSS stations

Within the scope of GTRF generation task, GRSP provides also to the GMS transformation matrixes to allow for the conversion between GTRF and ITRF in its latest realization.

In addition, GRSP is required to generate:

- -Periodically (weekly) validation reports for the GTRF estimated station position and estimated velocity
- -Alarm if the station coordinates solution deviates from the official GTRF solution by more than 10centimeters in any of the three directions according to AD.14 section 7.3.11. The alarm is raised as soon the threshold is exceeded.

Note:

FOCGMSEIFREQ-2023 specifies that the alarm is generated when the deviation vecotor exceeds a 20 cm in 3D per week. This is not in line with GMS-GRSP ICD section 7.3.1.11, where it is stated that an alarm shall be generated if the GSS station coordinates solution deviates from the official GTRF solution by more than 10 centimetres in any of the three directions.
#STRB2.0_change: GAL-DCN-ASTG-SYST-X-13901
The discrepancy has been resolved by GAL-DCN-ASTG-SYST-X-13901 aligning FOCGMSEIFREQ-2023 to the single direction threshold (10 cm) specified in the GMSGRSP ICD.

c) GRSP products usage
GMS uses the data provided by GRSP to calibrate OD&TS processes and to optimize the OD&TS configuration.

4.3.3.2.2 Support the SLR campaigns and provision of the SLR data to GMS

Satellite Laser Ranging is a high-precision two-way ranging technique allowing the determination of satellite orbit parameters with high precision (down to millimeter accuracy).

The Galileo SLR campaigns are performed by dedicated ILRS entity upon GMS request.

The results of such campaigns are distributed:
- to Scientific Community involved in geodetic studies
- to Geodetic Reference Service Provider
- to GMS for system calibration

a) SLR campaign request and inputs

GMS can request a SLR campaign to the ILRS via the GRSP.
In order to perform such campaign, the ILRS needs to receive from the Galileo system (via the GRSP interface) information on the Galileo constellation. This includes:

- Information provided by the GCS and forwarded by GMS: spacecraft configuration parameters
- manoeuvres announcements, and orbit predictions (in CPF format)

Note: in order to be used by the ILRS, the orbit predictions accuracy provided by the GCS shall comply with the following performance:

- 9 m radial
- 165 m along-track
- 15 m cross-track.

Note : the stated orbit predictions accuracy expressed in terms of error on radial, along track and cross-track components is consistent with orbit prediction accuracy specified for CGS (***GCSREQ-26588***) in terms of error on semi-major axis, eccentricity inclination, RAAN an phasing angle (see SYS-RD.94 justifying DCN-TASI-SYST-X-0763 (STRB1.5)).

- Constellation status from the GMS: constellation planning and satellite status. GMS has the capability to enable/disable the dissemination of the orbit predictions, mainly to inform the GRSP of any satellite anomalies that could impede SLR tracking.

b) SLR campaign and products

The ILRS campaign relies on a dedicated network of SLR ground stations, measuring their distance to the Galileo satellites.

The SLR measurements accuracy is better than 2 cm RMS.
The resulting measurements are provided to GRSP and automatically forwarded to GMS on a daily basis.
GMS will use a quality indicator to select and discard SLR data.

c) SLR measurements usage

SLR data are used for:
1) off-line calibration of MSF algorithms (including reference ODTS) and accuracy monitoring
2) Independent validation of GRTF alignment to ITRF
3) GMS orbit independent validation

For the purpose of off-line calibration of MSF algorithms and accuracy monitoring (Use Case 1), the relevant MSF algorithms are configured to accept as input SLR measurements, as well as L-band GSS measurements. The relevant combined processing is carried out by applying a relative weighting between GSS measurements and SLR measurements. GMS products can be achieved by using only GSS measurement by unweighting of SLR measurements.

The independent validation of GTRF (Use Case 2) is achieved by computing the SLR measurement residuals having configured the GSS coordinates with the new values to be validated. The SLR measurement residuals are computed per satellite as difference between the computed satellite slant range toward SLR station site derived using only GSS measurements and compensated SLR measurements.

The independent validation of MSF reference (Use Case 3) is achieved by computing the SLR measurement residuals in the same way as per Use Case 2, but deriving the computed satellite.-SLR station site slant range using the current GSS coordinates.

The triggering frequency of the three use cases are summarized in the following table:

Use Case 1	Use Case 2	Use Case 3
• A few months after a new satellite is inserted in GMS solution. • Quarterly, for alignment of what KPI' s defines for operations. • After every event that put a satellite out of the GMS operational configuration (maneuver and other satellite outages). • Detection of GMS solution degradation.	• Every time an updated GTRF is provided by GSRP.	• Daily , as this use case is foreseen to reduce the latency (one week)of MSF reference orbit validation provided by GSRP.

4) **Offline OD&TS calibration and accuracy monitoring**

The SLR measurement are optionally used as additional input to the Galileo off-line orbit determination function to calibrate and monitor the accuracy of the solution obtained with other data (normally L-band GSS measurements).

The GSS and SLR data are combined in the off-line orbit determination process by applying a relative weighting between GSS and SLR data. The off-line ODTS solution using only GSS data is obtained by unweighting (i.e. setting to dummy) the SLR data

5) **Independent verification of GTRF alignment to ITRF**

The GTRF determined by the GRSP has to be aligned with ITRF to ensure inter-operability with other geodesy systems. The GRSP has a requirement to compute GTRF so that the GSS stations coordinates provided to GMS are no more than 3 cm (2 sigmas)

- GRSP validates internally its products before delivering them to GMS. However, GMS is also required to independently verify GTRF alignment upon reception of a new GTRF version.
- This verification at GMS level has to be carried out using the following data:
  - L-band observation obtained by the GSS network.
  - GSS coordinates in the GRTF (new GTRF version) provided by the GRSP.
  - SLR measurements provided by the GRSP.
- SLR stations in ITRF coordinates provided by GSRP
- The verification can be obtained through the following steps performed at GMS level:
  - The SLR station coordinates received from GRSP are converted from ITRF to GTRF
  - The ODTS process is run at GMS using the new GSS coordinates and GSS observables only.
  - The SLR residuals are computed as difference between SLR measurements converted in GTRF and slant range from SLR stations (in GTRF) to the orbit determined using GSS observables only
  - If the computed SLR residuals exceed a configurable threshold (default value : 10 cm rms), the GRTF is flagged as not valid. The orbit and clock solution determined by the GMS are sent GSRP for further investigation.

6) **Independent verification of GMS reference orbit**

The GMS is able to send its reference orbits, clocks and ERP solutions to GRSP for validation. However, GRSP has to wait for its final products to be finalised, which a two-weeks (TBC) latency.
The use of SLR measurements, with a latency of one day, would allow the GMS to get an autonomous check of its reference orbit independently from weekly validation at GSRP.

This check at GMS level is carried out using the following data:

- L-band observation obtained by the GSS network.
- GSS coordinates in the GRTF available at GMS
- SLR measurements provided by the GRSP on daily basis
- SLR stations in ITRF coordinates provided by GSRP on daily basis

The verification can be obtained through the following steps performed at GMS level:

- The SLR station coordinates received from GRSP are converted from ITRF to GTRF
- The ODTS process is run at GMS using the available GSS coordinates and GSS observables only.

- The SLR residuals are computed as difference between SLR measurements converted in GTRF and slant range from SLR stations (in GTRF) to the orbit determined using GSS observables only

If the computed SLR residuals exceed a configurable threshold (default value : 10 cm rms), the orbit and clock solution determined by the GMS is flagged as not valid, and an investigation started.

4.3.3.2.3 Provide the non GTRF related products consumed by both GMS and GCS

The GRSP is responsible to retrieve scientific data required by the GMS off-line mission calibration functions and GCS Flight Dynamic functions

This includes the following data:

- Earth rotation parameters (daily) meeting following accuracy:
  - Earth Pole offsets: 3.6 mas (95%), worst over two days
  - UT1-UTC: 0.54 ms (67%), worst over two days
  - Celestial Intermediate Pole offsets: 1.4 mas (95%), worst over two days
- Ionospheric electron densities as obtained from the IRI2012model
- NORAD TLE data

Notes:

- in FOC2, Solar Flux information is removed from the data supplied by GSRP, as GMS uses a simplified SRP Cannonball model (**#STRB2.0_change: GAL-DCN-ASTG-SYST-X-13720)**
- The Ionospheric electron densities derived from IRI 2012 model are used by GCS flight dynamics function (GCSF4.4) only as alternative to the NeQuick model provided by GMS navigation data determination function, should the latter not be available.

The NeQuick model is used by default to reduce Galileo dependency from external data.
(#STRB2.1_change: GAL-DCN-TASI-SYST-X-0972, GAL-DCN-TASI-SYST-X-0973 and GAL-DCN-TASI-SYST-X-0779 Implementation of Nequick in FDF).

4.3.3.2.4 Provide Independent validation of a number of GMS products

GRSP as part of its support to Galileo System, provides an independent validation of products generated by the offline mission monitoring function (GMSF7.2) (orbits, clocks and ERPs). Following validation reports are provided weekly:

- Validation reports on the satellite and station clock estimates
- Validation reports on the satellite orbit estimates
- Validation reports on the Earth Rotation Parameters estimates

4.3.3.3 SFC Functional Block Diagram

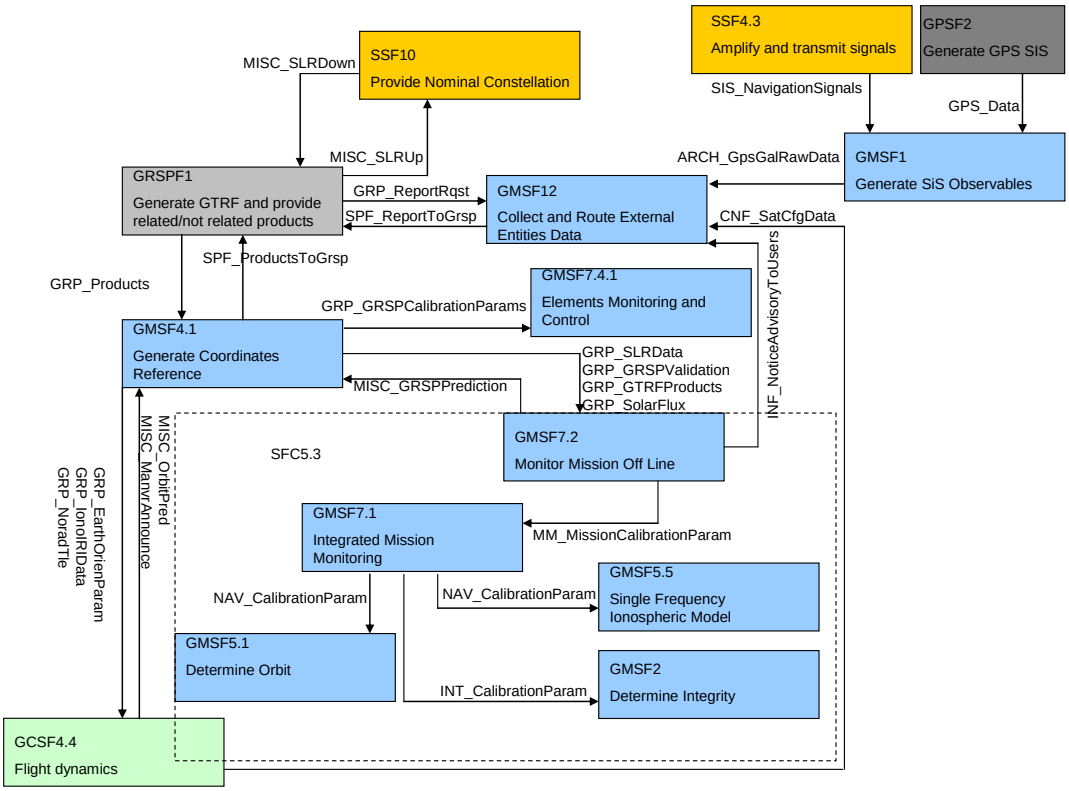


Figure 24 : SFC2.3 Functional Block Diagram

4.3.3.4 Transfer of the GTRF realization from the IOV phase to the FOC phase

In the transition phase from IOV to FOC the GPS data would still be needed and have a large weight as long as:

- the number of Galileo satellites in orbit
- and the number of globally distributed GSS and non-GSS stations equipped with Galileo receivers
- and the number of Galileo receiving sites contained in the ITRF (including reliable velocities)

will not allow for an accurate Galileo only solution with a reliable alignment of the frame to the ITRF.

Before IOV only GPS satellites will be available.

During the IOV phase only four Galileo satellites are operated, whereas after IOV an increasing number of Galileo satellites will be present. During this time the GTRF will be still a combination of GPS and Galileo data..
In order to provide a reliable alignment to the ITRF, a sufficient number of Galileo stations needs to be part of the ITRF network. Note that a Galileo only solutions could be produced only assuming a large number of GSS stations (>100).

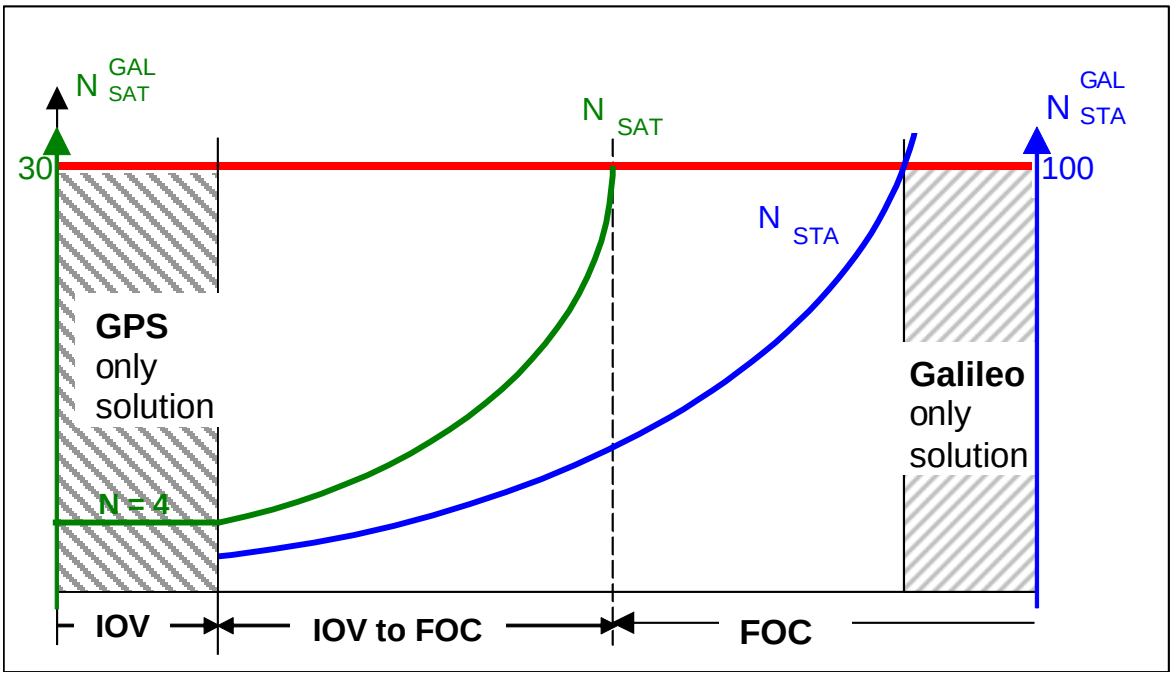


Figure 25 : Schematic overview for the realisation of GTRF during the transition phase IOV to FOC

The previous figure shows that the start of the Galileo only phase requires a minimum number of Galileo stations as well of Galileo satellites in order to achieve a proper alignment to the ITRF.

4.4 SFC3 Maintain Overall System Timing Synchronisation

In a satellite navigation system, the timing constraint is one of the key issues to achieve the mission performance. The accuracy of the timing parameters generated and the overall synchronisation of the components of the systems have a direct impact on the user positioning, timing and integrity solution performances.

The Maintain Overall System Synchronisation SFC is then responsible for providing all required functionality to:

- generate and maintain the different local timing parameters with associated performances, based on the provision of the GST as common time base.
  Important note: the generation and steering of the GST is described in a dedicated SFC: SFC2.1
- Synchronise the elements involved in different mission chains

4.4.1 Introduction

4.4.1.1 Synchronisation constraint per service

The synchonisation constraints per service are fully justified in dedicated Justification Topic JT-20 in IOV DJF SYS-RD.3.

4.4.1.2 Main Synchronisation functions

Considering the constraints described in JT-20 of IOV DJF SYS-RD.3, the following system synchronisation functionalities are required:

- Synchronisation functions to support Galileo OS and PRS services including:
- Generation of a common Galileo System Time (GST) base steered to international Time scale (UTC, TAI…): see SFC2.
- Synchronisation of Navigation Determination and Dissemination chain functions to this common GST to support OS and PRS positioning and timing services. In that frame, the purpose of the SFC3 is to support the following needs:
  - Synchronisation of GMS Navigation Determination and Dissemination chain functions needed to achieved the ranging accuracy specified to the GMS OD&TS functions (orbit and clock parameters)
  - Synchronisation of Satellite Local realization of GST to the GST to allow the user/GMS to compute positioning and timing solution. This includes
  - Generation of proper satellite timing parameters (TOW and WN).
  - Synchronisation of the satellites wrt GST to allow the user to retrieve its own clock offset wrt GST and all satellite clocks wrt to GST

  Note: In navigation applications, the ranging accuracy achievable highly depends on the accuracy of the frequency reference and the LGST. Any drift (and rate of drift) of the frequency reference w.r.t a fixed value (10.23MHz), and any residual offset between the Satellite LGST and the ground GST are monitored by on-ground OD&TS functions. Then they are indicated to the user in a dedicated clock correction model disseminated in the navigation message (see SFC4.1).
- Synchronisation of Dissemination chain functions to this common GST to support the dissemination of external time-critical data
  - Synchronisation of mission monitoring and control functions to support the above mission functions including:
    - Synchronisation of Ground element monitoring and control functions (see SFC3.2, 3.3 and 5.1).

- Synchronisation of Satellite monitoring and control functions for TC time tagging and TM stamping (see SFC3.1 and 5.2). This synchronisation is needed
  - At GCS level for TC tagging
  - At Satellite level for TM stamping and TC application: see SFC3.1.4
- Synchronisation of Mission monitoring functions: see SFC5.3

The following figure summarises the timing constraints for each high level functional chain, and the timing exchanges between them:

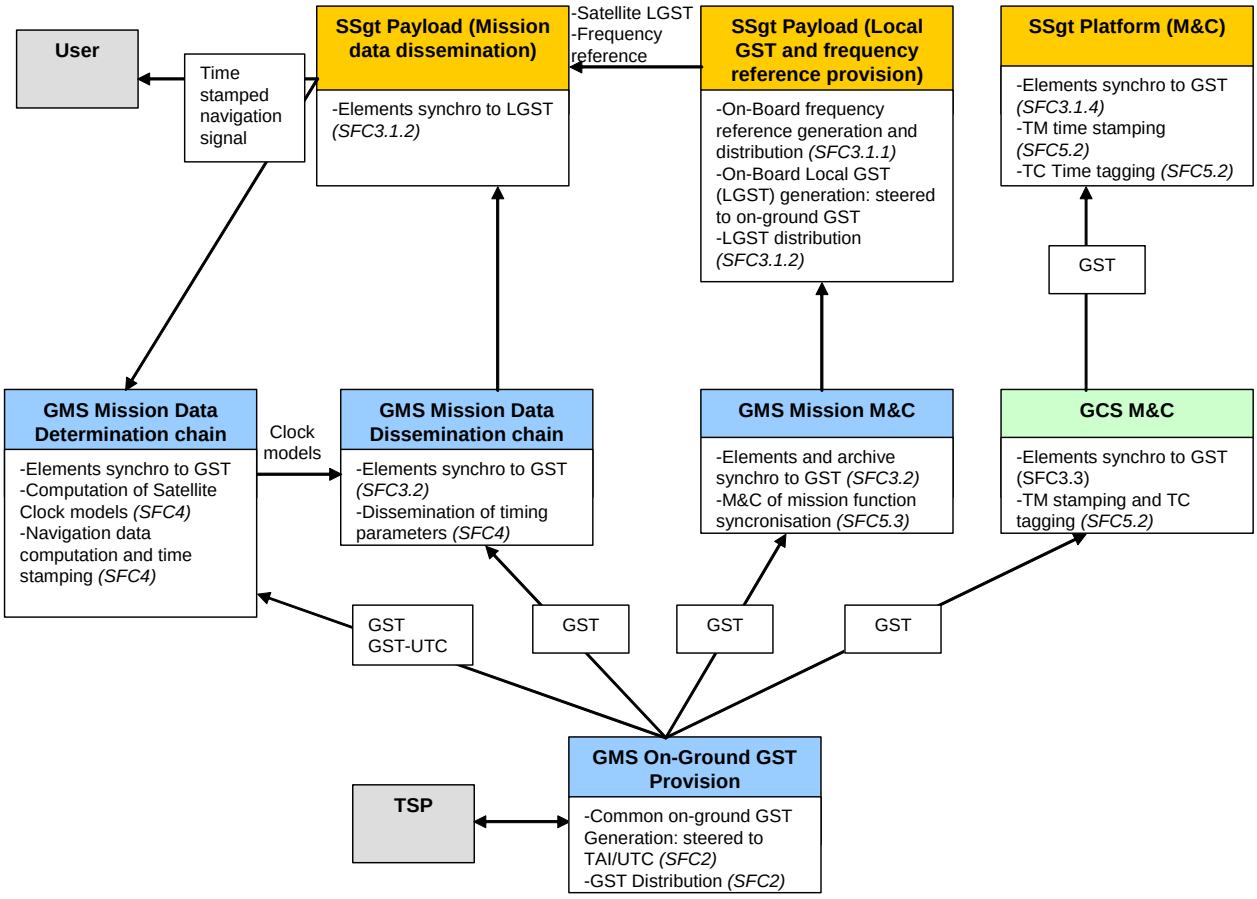


Figure 26 : SFC3 overall timing architecture

The provision of the synchronisation and timing parameters needed at each segment level, namely within space, GMS and GCS segments is shown in the first level functional tree below.

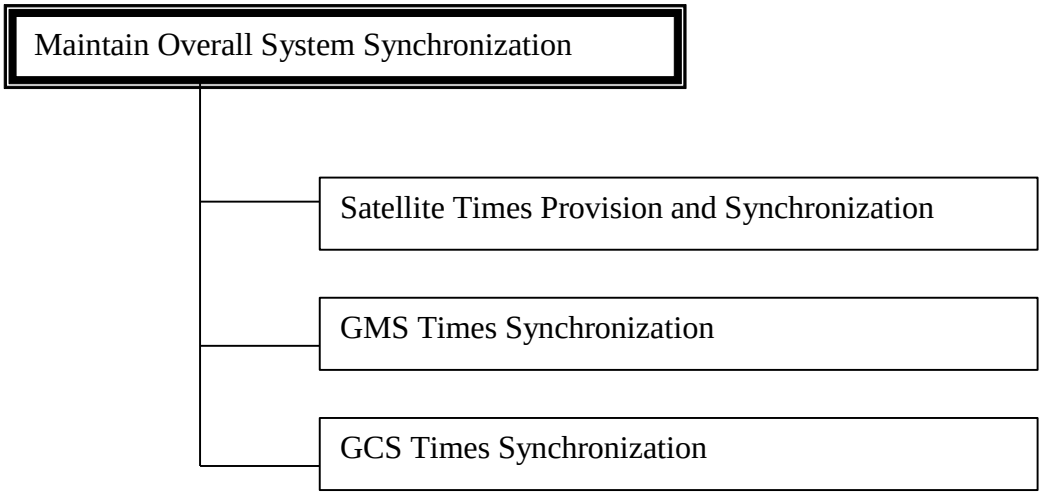


Figure 27: SFC3 Functional Tree (1st level)

4.4.2 SFC3.1 Satellite Times Provision and Synchronisation

4.4.2.1 SFC3.1 High Level Description

The Satellite Time Provision and Synchronisation functional chain is in charge of providing all required functionality to generate and maintain the different on-board timing parameters.

Several types of timing parameters are defined on-board each satellite to support different applications:
- Timing parameters for navigation applications:
- On-board frequency reference defining a frequency reference source.
- Payload on-board time or “satellite Local GST” defining a satellite common time base
- Timing parameters for Satellite Monitoring and Control:
- ICDU frequency
- Platform on-board time
- Additional time for security application:
  - PLSU Frequency reference and time including:
    - 1 time for PRS PRN code generation: this time is directly synchronized to the LGST received from NSGU by pulse per second (PPS).
    - 1 time for payload STC/STM management autonomously generated and synchronized by ground STC.
- PFSU Frequency reference and time: generated autonomously for platform STC/STM management and syncrhonised by ground STC.

Each type of parameter has specific accuracy performance constraints.

The sub SFC’s 3.1.1, 3.1.2 and 3.1.3 are in charge of generating and maintaining the timing parameters used for navigation application:
- SFC3.1.1: Maintenance of the offset between LGST and GST within a specified range with monitoring and control of the payload frequency reference (including initial frequency synchronization)
- SFC3.1.2: Maintenance of the offset between LGST and GST within a specified range with monitoring and control of the satellite LGST
- SFC3.1.3: Monitoring and control of the satellite atomic clocks, and in particular switching control.
- SFC3.1.4: in charge of generating and steering the platform time used for command time-tagging

Note: PLSU and PFSU time steering are functionally part of SFC3.1.1, 3.1.2 and respectively 3.1.4. See also details in DD2 physical description part.

The 2nd level functional tree for the “Satellite Times Synchronisation and provision” SFC is shown below:

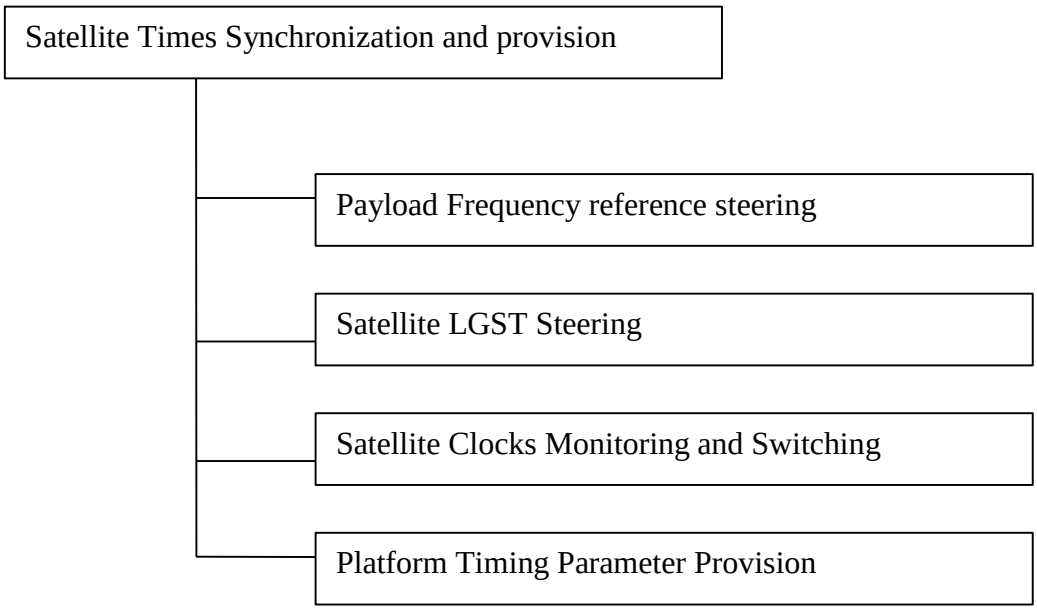


Figure 28: SFC3.1 Functional Tree (2nd level)

Note: the provision and maintenance of timing parameters for security applications are detailed in dedicated classified SFCs.

4.4.2.2 Definitions

4.4.2.2.1 On-Board Timing parameters for Navigation Applications

To support navigation application, each Galileo Satellite generates a frequency reference derived from on-board atomic clocks. Based on this frequency reference, each satellite derives a local realisation of the Galileo System Time (LGST).

Payload frequency reference definition: Very stable frequency reference produced on-board at Payload level. This frequency is:

- derived by the CMCU from the satellite primary atomic clock frequency (10MHz)
- initially steered to 10.23MHz as seen from the ground (including relativistic effect)

characterized by its long term frequency drift and its frequency stability (stability is characterised by Quality factor and Allan Deviation as detailed in Atomic Clock physical description section in DDF2).
modelled to the end user with its phase, frequency and frequency drift
The frequency reference is used:

- By the NSGU to generate a common satellite time base (satellite LGST)
- By the NSGU to generate the navigation signals (ranging codes and carriers) for L-Band downlink
- By the Payload Security Units to generate the PRS ranging codes
- By the C-Band receiver (MISREC) generate the ranging code to be synchronised with the uplink C-Band signals for acquisition
- By the SAR payload to generate SAR downlink signals.

Satellite Local GST definition: Local realization of the GST (LGST) derived from the payload frequency reference (time is measured by the number of oscillations, i.e. by a frequency count). This time is generated by the NSGU for the Galileo payload. It is used as reference for:

- Message downlink synchronisation: the start epoch of transmission of downlink message pages is synchronised with a Time-of-Transmission (exactly synchronised with the LGST 1 second epoch).
- L-Band Message Time Stamping: a counter composed of Week Number (WN) and Time-of-Week (ToW) indicating the LGST 1 second value. This value is downlinked periodically in the message (the 1-second epoch value indicating the start epoch when the navigation page starts to be downlinked).

4.4.2.2.2 On-Board Timing parameters For Monitoring and Control

To support the monitoring and control of the Galileo satellite, a frequency reference and a dedicated time are generated by the platform.

Platform frequency reference definition: frequency reference produced on-board at Platform level (ICDU), derived from ICDU internal clocks. The frequency reference is used to generate a platform time to manage TMs and TCs.

Platform Time definition (ICDU On-Board Time or "IOBT"): Counter derived from the platform frequency reference. This time is generated by the ICDU to provide an autonomous time for the platform in case the payload is not available (LGST not available). It can be slaved to the NSGU PPS time (counter) when available to avoid time drift. It is used as reference for:

- Timestamp of telemetry packet
- Time tagged Telecommands application time

The functional chain 3.1.4 is in charge of generating and maintaining these timing parameters used for Monitoring and Control application.

4.4.2.3 SFC3.1.1 Payload Frequency Reference Steering

4.4.2.3.1 SFC description

The purpose of this system functional chain is to generate and steer the satellite frequency reference to be used within the payload for navigation applications (see section 3.3.2.1). This steering is ensured by proper monitoring of this frequency reference and potential adjustment with dedicated frequency steps.

Preliminary note: Absolute vs. Relative frequency steps definition: in the system requirement baseline, the frequency steps can be specified in absolute or relative value. The relationship between those values is the following:

$$F_{RELATIVE} = \frac{F_{ABSOLUTE}}{F_{REFERENCE}}$$

where $F_{Reference}$ = 10.23MHz

Monitoring of the primary clock and satellite LGST:
The GMS OD&TS function (GMSF5.4) is responsible for restitution of the frequency reference parameters of the primary clock from the L-Band SIS of each satellite. These parameters represent a real time picture of the primary clock behaviour at each epoch (Note that they are different from the clock model parameters transmitted to the user that are supposed to be valid for certain duration). These parameters are composed of:

- A Phase (a0 parameter) indicating the instantaneous LGST-GST phase offset A Frequency (a1 parameter) indicating the instantaneous offset LGST frequency and GST frequency
- A frequency drift (a2 parameter) indicating the drift of the LGST frequency wrt GST frequency.

The GMS on-line mission monitoring function (GMSF7.3) is then responsible for analysing these parameters. In particular, this function is responsible for:

- Synchronise the on-board frequency reference at initialization to 10.23MHz reference (with possibility to optimize this value)
- Setting a specific threshold in order to ensure that the LGST-GST offset remains within a specified range [-18ms, 62ms].

Note: The allowed LGST-GST offset of [-18ms, 62ms]. (instead of *[-18ms, 80ms]*) *has been aligned to GMSREQ-11514. .The setting of the threshold by mission monitoring operators will ensure that this offset can be corrected by the computed clock models as specified in SISICD [AD.6]. In particular, considering that a0 parameter sizing allows to correct an initial bias of +/-62ms maximum, this threshold shall be finally set to ensure that at the start of validity of the clock model, the offset shall be within [-18ms, 62ms].*

Frequency Reference Adjustment Command
At satellite initialization or in case the LGST-GST offset reaches the specific threshold, the GMS on-line mission monitoring function (GMSF7.3) shall request an adjustment of the on-board frequency reference in the Mission planning (GMSF9 and GCSF1).
This adjustment is characterized by frequency steps applied to the frequency reference directly. The absolute frequency step size is an integer multiple of 0.002046 mHz: see the Mid-Term plan parameters in GCSGMSICD [AD.8]: *Satellite Primary Clock synchronisation request & parameters for on-board clock frequency adjustment*)
The satellite M&C functions of the GCS are then responsible for sending a dedicated S-Band TC to the Satellite to adjust the frequency reference: CMD_AdjustFrequencyReference.

Note: a dedicated procedure is defined to ensure that such command does not create any satellite outage discontinuity at user level: see SFC5.2

Satellite Frequency Reference Adjustment Application
When instructed with a dedicated TC, the satellite function responsible for generating the payload frequency reference (SSF5.1) shall apply the frequency step requested.
Two different methods can be implemented for the frequency reference adjustment:

- **Coarse steering method:** consists in adjusting the Payload frequency reference applying a single frequency step. This method can be used to:
  - Initially synchronize the on-board frequency reference to 10.23MHz (with the possibility to optimize this initial value in order to maximize the initial time before the first adjustment: see scenario #SFC3.1-1 in DDF2 [SYS-RD.2])).
  - Adjust the LGST-GST offset in one shot when a certain limit is reached.

Smooth steering method: consists in adjusting periodically the Payload frequency reference value applying in a coherent way small but frequent frequency steps to maintain the LGST-GST offset within the specified range.
See method trade-off in IOV DJF SYS-RD.3.

Synchronisation execution feedback
When the command has been executed, a dedicated telemetry is downlinked to the GCS. The GCS constellation monitoring function (GCSF4.2) is then responsible for informing the mission Monitoring function of the GMS (GMSF7.1) of the execution status of the command.

4.4.2.3.2 SFC Functional Block Diagram

The following block diagram shows the functional architecture of the Frequency reference adjustment functional chain.
Note that in this block diagram, the resynchronisation request and frequency adjustments apply to the primary clock only.

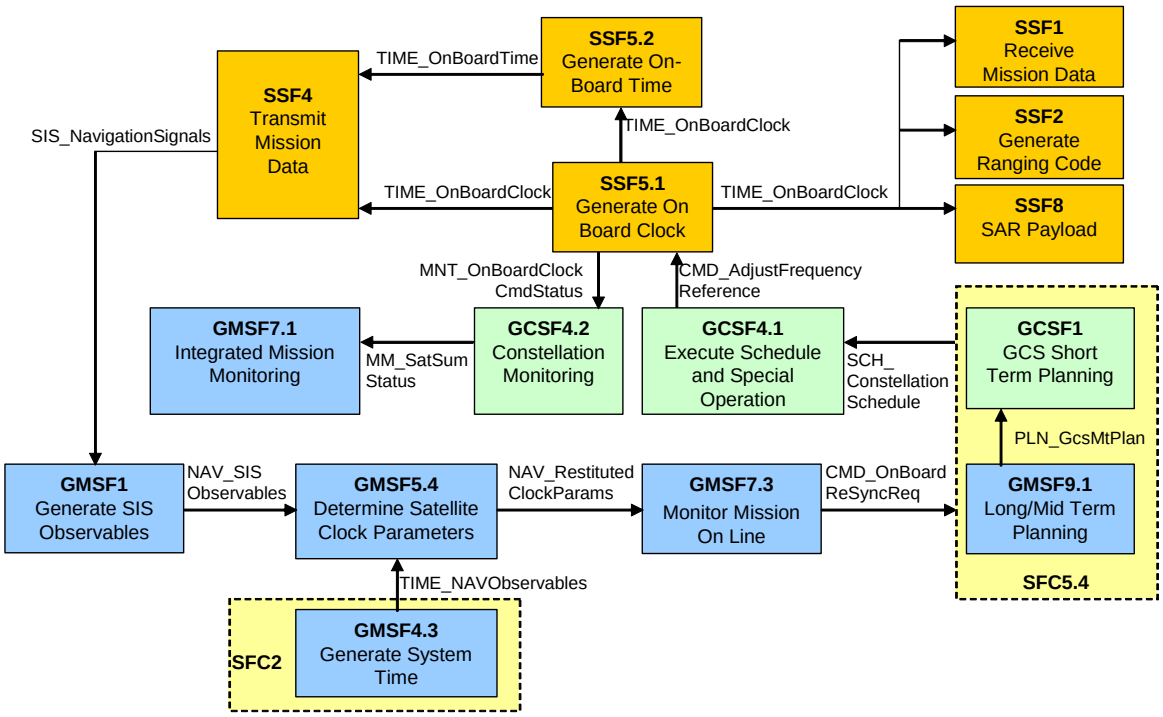


Figure 29: SFC3.1.1 Functional Block Diagram

4.4.2.4 SFC3.1.2 Satellite LGST Steering

4.4.2.4.1 SFC description

The purpose of this system functional chain is to generate and steer the satellite LGST to be used within the payload for navigation applications (see section ý3.3.2.1).
This steering is ensured by proper monitoring of this LGST and potential adjustment with dedicated time steps.

Monitoring of the satellite LGST:

- The GMS OD&TS function (GMSF5.4) is responsible for restituting the frequency reference parameters of the primary clock from the L-Band SIS of each satellite. In that case, these parameters are composed of:
  - The instantaneous LGST-GST phase offset retrieved from the ranging signal.
  - The TOW and WN parameters generated by the satellite and transmitted in the navigation message (L-Band Message Time Stamping).
- The GMS on-line mission monitoring function (GMSF7.3) is then responsible for analysing these parameters. In particular, this function is responsible for setting a specific threshold in order to ensure that the LGST-GST offset remains within a specified range [-18ms, 62 ms].

Note: The allowed LGST-GST offset of [-18ms, 62ms],(instead of [-18ms, 80ms) has been aligned with GMSREQ-11514. An LGST-LST offset of 62 ms, *as this is the maximum range that can be corrected by the clock mode specified in SISICD.*
Following the removal of integrity form Galileo, the possibility to relax the satellite clock physical synchronization from [-18 ms, 62 ms] to [±62 ms] is under study. This would allow an optimization of the time interval between successive satellite clock physical re-synchronization without affecting navigation data message design, at expense of modest degradation of latency performance for Galileo EDBS.

On-Board LGST Adjustment Command

At satellite initialization (see dedicated initialization dynamic scenario in DDF2 [SYS-RD.2]) or in case the LGST-GST offset reaches a Predefined Threshold, the GMS on-line mission monitoring function (GMSF7.3) shall request an adjustment of the on-board LGST in the Mission planning (GMSF9 and GCSF1).This adjustment is characterized by:

- time steps applied to the NSGU time register directly. The time adjustment enables to correct +/-0.5 s with a resolution step of 100ns: see the Mid-Term plan parameters in GCSGMSICD [AD.8], *Satellite on-board time of transmission synchronisation request and parameter for time adjustment*.
- TOW and WN adjustment values (only during initialization) to set up the Band Message Time Stamping after payload switch-on.

The satellite M&C functions of the GCS are responsible for sending a dedicated S-Band TC to the Satellite to adjust the on-board LGST: CMD_AdjustSatelliteTime.

Note: a dedicated procedure is defined to ensure that such command does not create any continuity event at user level: see SFC5.2

Satellite LGST Adjustment Application

When instructed with a dedicated TC, the satellite function responsible for generating the satellite LGST (SSF5.2) shall apply the time offset directly to the NSGU time register, and correct (at initialization only) the on-board TOW and WN values.

Synchronisation execution feedback

When the command has been executed, a dedicated telemetry is downlinked to the GCS. The GCS constellation monitoring function (GCSF4.2) is then responsible for informing the mission Monitoring function of the GMS (GMSF7.1) of the execution status of the command.

4.4.2.4.2 SFC Functional Block Diagram

The following block diagram shows the functional architecture of the Satellite LGST steering functional chain.

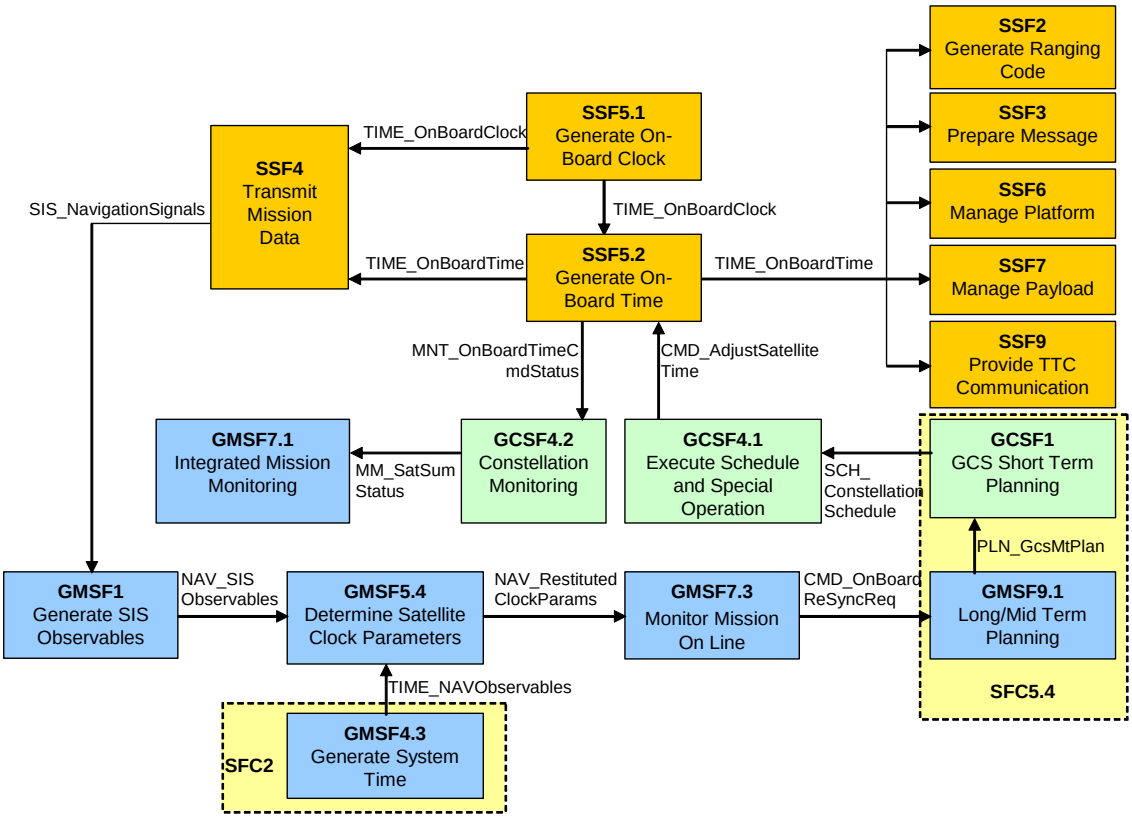


Figure 30: SFC3.1.2 Functional Block Diagram

Note: the functional block diagram is quite similar to the SFC3.1.1, except that the commands applied from the Ground segment to the Satellite apply on the time directly.

4.4.2.5 SFC3.1.3 Satellite Clocks Monitoring and Switching

4.4.2.5.1 SFC description

The satellite Clocks monitoring and control SFC 3.1.1 is in charge of managing the on-board atomic clock redundancies (clock monitoring, activations or switching), to comply with the availability and reliability requirements.

Preliminary notes:
"Clock behavior" means the behavior of the 10.23MHz frequency reference signal derived from the atomic clocks (there is no functionality on-board or on-ground to monitor the output of the atomic clock itself).

1) Clock configuration initial conditions: in this SFC, it is assumed that the following initial condition for clock configuration is ensured on board the satellite:
- a primary clock is ON and delivers a proper primary frequency reference;
- a hot backup clock is ON and deliver a proper hot backup frequency reference (except in configurations III and IV below);
- 2 clocks are OFF and in cold backup mode.

2) Clock configuration constraints: due to power, thermal and clock management constraints on board the satellite, 4 possible operational clock configurations are defined for the navigation payload (**#STRB2.0_change: GAL-DCN-TASI-SYST-X-1116)**:

Configuration	Number of PHM			Number of RAFS	
	ON step 6	ON step 3	Stay Alive	ON	OFF
I	1	0	1	1	1
II	1	1	0	1	1
III	1	1	0	0	2
IV	1	0	1	0	2

Table 8 – On board Clock configurations

PHMs are kept switched-on to prevent the risk that a PHM cannot be re-started after being switched-off for a long time.

3) Clock performance constraints: In normal conditions the PHM is set as default primary clock due to its better stability performance. The use of RAFS as primary clock is allowed during primary clock switching due to either primary PHM clock failure or maintenance of cold redundant PHM clock (GAL-DCN-ESA-SS-X/0003, CHID12). Functionalities at satellite level are specified to allow primary clock seamless switching between Prime and Hot redundant clocks (possible only for initial configuration I, II). This allows a transition PHM – RAFS – PHM inside the Fault Free SIS definition (see GAL-DCN-ESA-SS-X/0300) without causing an interruption of the relevant SIS data processing by the OSPF

.
The following functionalities are foreseen depending on the type of clock monitored:

- Primary clock monitoring and control (direct monitoring):

  In nominal configuration, the system is responsible for monitoring the behavior of the primary clock on board.

- Monitoring of the primary clock behavior per satellite

  The GMS OD&TS function (GMSF5.4) is responsible for restituting the frequency reference parameters of the primary clock directly from observables of the L-Band SIS of each satellite. The GMS on-line mission monitoring function (GMSF7.3) is then responsible for comparing these parameters to ground model.

- On-Board primary clock switch request. A primary clock switch is needed when:
- a significant degradation of the primary clock frequency stability is observed, which cannot be controlled (it is not possible to steer the primary clock frequency with frequency or time adjustments to remain compliant with the mission performances).

  Under these conditions, the GMS on-line mission monitoring function (GMSF7.3) shall request a switch of the primary clock on-board in the Mission planning (GMSF9 and GCSF1): see the Mid-Term plan parameters in GCSGMSICD [AD.8]: *Satellite Primary to Backup Clock switch request*.

- The cold backup PHM clock maintenance is needed. In this case the primary clock switch request can be manually planned by an operator.
- Satellite primary clock switch command.

  Based on mission planning, the satellite M&C function of the GCS is responsible for commanding the clock switch on-board the satellite (CMD_ClockSwitch, CMD_TurnOffUnits).

  In order to comply with satellite clock configuration constraints (see preliminary notes), the following sequencing shall be followed to switch the clock in case of PHM1 failure:

- Switchover Prime and Hot redundant clocks. The failed S-PHM1 clock becomes the hot backup and the RAFS clock (RAFS1) becomes the new prime clock (switch Event1, in configuration I or II) or set the backup PHM to step 6 (in configuration III or IV).
- Switch off the “failed” clock S-PHM1.
- Switch on the cold backup clock S-PHM2. This is the new hot backup clock, which needs some time to warm up and stabilise to reach nominal performance (only in configuration I or II of Table 8)
  Upon completion of PHM2 stabilisation, switch the current primary clock RAFS1 to S-PMH2 that becomes the new primary clock (Switch Event 2, in configuration I or II of Table 8).

The sequencing for primary clock switching in case of maintenance of cold back up S-PHM2 shall as follows (valid for starting configuration I or II of Table 8):

- Switchover Prime and Hot redundant clocks. The S-PHM1 clock becomes the hot backup clock and the RAFS1 clock becomes the new master clock (Switch Event 1)
- Switch off the “old” primary clock (S-PHM1)
- Switch on the cold backup clock S-PHM2 (that becomes hence the new hot backup), to undergo maintenance.
- Upon completion of S-PHM2 maintenance, switch off S-PHM2 and switch on S-PHM1 for its stabilisation.
- Upon completion of PHM1 stabilisation, switchover Prime and Hot Backup clocks so that RAFS1 is set as Hot backup and S-PMH1 becomes the new primary clock (Switch Event 2).

- Switch execution feedback

  When the command has been executed, a dedicated telemetry is downlinked to the GCS. The GCS constellation monitoring function (GCSF4.2) is then responsible for informing the mission Monitoring function of the GMS (GMSF7.1) of the execution status of the command.

- Hot backup clock vs. primary clock monitoring and control (indirect monitoring):

  In nominal configuration, the system is also responsible for monitoring the behavior of the hot backup clock on board.
  The hot backup clock monitoring is performed in two operational modes, namely:

  - “Operational mode”: this mode is foreseen to ensure a suitable level of redundancy in case of primary clock failure. In this mode the hot backup clock monitoring is carried out using on-line telemetry data. A satellite hot backup clock switching request will be triggered in case an anomalous behaviour is detected.
  - “Analysis Mode”: this mode is foreseen to provide hot back up clock long-term characterisation to support satellite clock technology development. In this mode the hot backup clock monitoring is carried out using historical data retrieved by the ground central archieve

  The steps to provide hot backup clock monitoring in operational mode are as follows:

  a) Monitoring of the hot backup clock behavior per satellite

     - The Satellite function in charge of maintaining the on-board timing parameters (SSF5.1) is responsible for monitoring on-board the hot backup clock (as the frequency reference derived from the backup clock is not used to generate any signal in the L-Band link, it cannot be observed directly).
     - The monitoring of the back up clock is performed through the provision of Phase Comparison Measurement Samples: they consist in comparing at regular time intervals the phase of the 10.23MHz frequency reference derived from the hot backup

clock to the one derived from the primary clock. This results in a collection of phase samples that is downlinked regularly to the ground segment via Telemetry link.
- These Phase Comparison Measurement Samples TM are then forwarded periodically by the GCS to the GMS mission monitoring functions.
- The GMS off-line mission monitoring function (GMSF7.2) is then responsible for analysing the Phase Comparison measurement samples to retrieve the hot backup clock behavior. This behavior is then compared to ground model.

b) On-Board hot backup clock switch request

A backup clock substitution is needed if a significant degradation of the hot backup clock frequency stability is observed (note that it is not possible to adjust the frequency reference derived from the backup clock).
In that case, the GMS off-line mission monitoring function (GMSF7.2) shall request the substitution of the hot backup clock on-board in the Mission planning (GMSF9 and GCSF1): see the Mid-Term plan parameters in GCSGMSICD [AD.8]*: Satellite Hot Backup clock to Cold Backup clock switch request*.

c) Satellite hot backup clock substitution command

- Based on mission planning, the satellite M&C functions of the GCS are responsible for managing the switch of the hot backup clock on-board the satellite with proper TCs (CMD_ClockSwitch, CMD_TurnOffUnits)
- In order to comply with satellite clock configuration constraints (see preliminary notes), the following sequencing shall be followed for hot backup clock substitution:
  - Switch off the hot back up clock
  - Switch on a cold backup clock
  - Switch from the “old” backup clock to the new one

d) Switch execution feed back

When the command has been executed, a dedicated telemetry is downlinked to the GCS. The GCS constellation monitoring function (GCSF4.2) is then responsible for informing the mission Monitoring function of the GMS (GMSF7.1) of the execution status of the command.

4.4.2.5.2 SFC Functional Block Diagram

The following functional block diagram highlights the two links (L-Band SIS and S-Band TM) used to monitor the behavior of the clocks (resp. primary clock model and backup clock phase comparison).

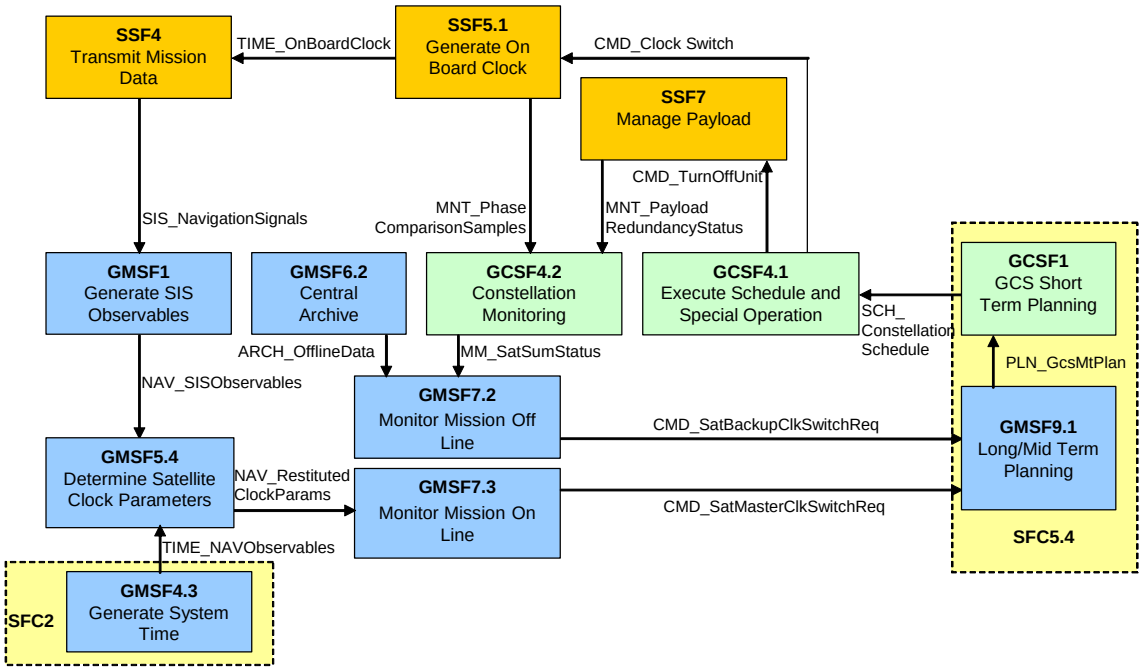


Figure 31: SFC3.1.3 Functional Block Diagram

Satellite Clocks Monitoring and Switching: Future Evolution :

#STRB2.0_change: GAL-DCN-TASI-SYST-X-0909 and GAL-DCN-TASI-SYST-X-0919

The current satellite clock switching applied in IOV, now renamed as coarse or “legacy” clock switching, implies a discontinuity in the provision of Galileo navigation services, Commercial service and SAR service..

As per current design, the GMS performs two actions as answer for the satellite clock switching, namely:

- removes the satellite from the contact plan. No uplinks take places to the satellite in any epoch within start and end of the announced switch event (from 4 to 8 hours).
- flags the satellite as invalid (SISA set to NAPA) for any message produced 100 minutes before the start of the switch event,
- Stops the generation of navigation messages for all services (OS, PRS, SAR, CS) when the current epoch is within the end and start of announced switch event. Once the signal is restarted, the first message is available not later than 4 hours after the end time with a valid DVS but with SISA set to NAPA for up to 50 hours with respect to the announced end of the event.

Additionally, the GCS flags the satellite as inoperative (DVS =Working Without Guarantee - WWG) before the maintenance

The satellite coarse switching applied in IOV is not consistent with Galileo service availability requirements, as the cold redundant S-PHM requires periodic maintenance.

each maintenance operation implies two switch events, each one leading to stopping the navigation message for period up to 8 hours and the transmission of navigation message with SISA set to NAPA for a period of about 46 hours. This represents around 0.2% of system availability or 1% if SISA is considered.

A significant improvement of service availability is planned for FOC satellites for which the capability to perform seamless clock switching between primary clock and hot redundant clock has been already specified and successfully tested at satellite level (see GAL-DCN-ESA-SYST-X-0300).

A future evolution is planned for GMS and GCS to provide seamless satellite clock switching avoiding service discontinuity, as well as a refined satellite clock coarse switching procedures limiting the impact on service availability wrt IOV.

Thus, two satellite clocks monitoring and control events are planned in Galileo future evolution, namely :

- Satellite seamless clock switching,
- Satellite coarse clock switching ("legacy clock switching)

Seamless clock switching is only possible when one RAFS (and one PHM) is switched on (**#STRB2.0_change: GAL-DCN-TASI-SYST-X-1116)**, i.e. when the initial configuration corresponds to configuration I or II (cf. SFC 3.1.3 description).

1) Satellite Seamless Clock Switching

The satellite seamless clock switching is obtained by ensuring that the switching between the satellite primary and hot redundant clocks occurs in a phase-coherent and frequency-coherent way, within tolerated limits. Under these conditions, the GMS continuously determines and disseminates valid navigation message with SISA set in the monitored status (valid SISA not set to NAPA). This does not consider false alarms or satellite clock discontinuity higher than the ones expected for seamless clock switch provided that the occurrence of such problems is in line with navigation message determination availability.

A degraded SISA will be disseminated during the seamless clock switch exposure time (see definition at point b above), as the primary/hot back clock frequency offset drift coefficient (a2) cannot be computed during that time, and the measurement arc for computation of primary/hot back up clock phase offset coefficient (a0 coefficient) and frequency offset coefficient (a1 t) will be temporarily affected by the clock offset discontinuity tolerated for seamless switching.
The Seamless Clock Switching Degraded SISA will be based on an a-priori characterization that should ensure the expected Confidence Level during the GMS exposure time to the clock seamless switching.
Upon expiration of exposure time to satellite clock seamless switching, the GMS will disseminate the Initial SISA until the MSF is capable to provide the new HSISA for that satellite.

The ranging accuracy will cope with the tolerated (small) discontinuity of clock phase and frequency, and a full discontinuity of frequency drift.

1. Seamless Clock Switching Degraded SISA

An inflated SISA, referred to as Seamless Clock Degraded SISA is defined and disseminated during seamless clock switching event.
The SISA degradation is due to the fact that the primary/hot back up clock frequency offset drift coefficient (a2) cannot be evaluated during clock switching event and the primary/back up clock phase offset coefficient (ao) and frequency offset coefficient (a1) are affected by the (small) clock phase/frequency offset discontinuities allowed for seamless clock switching.

2. GMS exposure time to the clock seamless switching

-

In case of seamless clock switching, the Seamless Clock Switching Degraded SISA is disseminated during the GMS exposure time to the clock seamless switching that shall be understood as follows:

- any epoch where the 100 minutes prediction of the navigation message goes beyond the start time of the clock seamless switching
- any epoch where the clock seamless switching is affecting the last 24 hours data of the OSPF long arc used to compute satellite clock frequency drift (a2 coefficient).

2) Satellite Coarse Clock Switching in FOC

Satellite coarse clock switching is maintained for IOV satellites. The evolution of this kind of switching is the same as for IOV, except for SISA management which is aligned with the approach adopted seamless switching.
In particular, the SISA unavailability is mitigated at expense of a small degradation of SISA performance by setting SISA to a valid value (even though degraded) four hours after the end of switch event instead of 50 hours.
So, the satellite coarse clock switching in FOC implies:

- Removal of the satellite from the contact plan. No uplinks take places to the satellite in any epoch within end start and end of announced switch event.
- Dissemination of the Coarse Clock Switching Degraded SISA 4 hours after the end of the switch event, until the GMS is not exposed to the coarse clock switching exposure time (see definition at point d above).

1. Coarse Clock Switching Degraded SISA

   During FOC an inflated SISA, referred to as Coarse Clock Switching Degraded SISA is defined and disseminated during coarse clock switching event.
   The SISA degradation is due to the fact that the hot back up clock frequency drift coefficient (a2) cannot be evaluated during clock switching event and the hot back up clock phase error (ao) and frequency error add up.

2. GMS exposure time to the clock coarse switching

   In case of coarse satellite clock switching , the Coarse Clock Switching Degraded SISA is disseminated 4 hours after the end of the event until the expiration of GMS exposure time to the clock coarse switching shall be understood as follows:

   - any epoch where the clock coarse switching is affecting the last 24 hours data of the OSPF long arc used to compute satellite clock frequency drift(a2 coefficient) as per FOC1 design.

4.4.2.6 SFC3.1.4 Platform Timing Parameters Provision

4.4.2.6.1 SFC Description

The SFC3.1.4 is responsible for providing the timing parameters needed by the platform to time-tag the TCs and time-stamp the TMs.

a) Platform Frequency Reference Generation
The platform generates its own frequency reference (independently from the payload) with dedicated atomic clocks.

b) Platform Time Generation

A dedicated platform time is generated from the platform frequency reference in order to:
- time-tag telecommands
- run application software
- time stamp telemetries

This platform time is synchronised to the Local GST with an accuracy better than 1 ms. When the Payload time is available and synchronized (see SFC3.1.1 and 3.1.2), the platform time is slaved to it to avoid significant degradation of the synchronisation.

c) Platform Time monitoring and Control by GCS
The GCS Constellation Monitoring Function (GCSF4.2) is responsible for correlating the platform time with the GST for the TM stamping and TC tagging. The monitoring and the command for GST and platform time synchronisation is done via TM/TC processing. This synchronisation shall be better than +/-5ms

4.4.2.6.2 SFC Functional Block Diagram

The following functional block diagram describes the platform timing parameters functionalities.

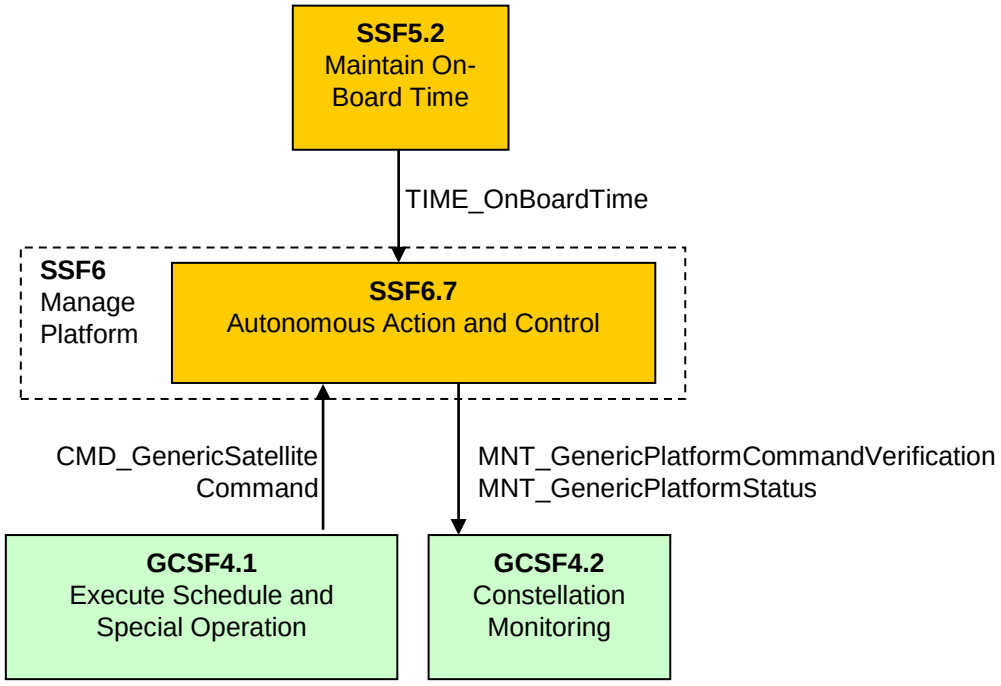


Figure 32: SFC3.1.4 Functional Block Diagram

4.4.3 SFC3.2 GMS Times Synchronisation

4.4.3.1 SFC description

This SFC is responsible for providing the needed functionalities to allow a synchronisation of the GMS segment timing parameters to the Galileo System Time (GST).
The generation and maintenance of the GST is described in dedicated SFC2.1.
This SFC describes the distribution of this GST to GMS functional chains directly impacted by the accuracy of the synchronization to the GST.

- Synchronisation of GMS Navigation Determination and Dissemination chain functions to support OS and PRS positioning and timing functions. In that case, the ranging accuracy obtained by the GMS OD&TS functions (GMSF5) is directly driven by the synchronization accuracy to GST of these functions.
    - o Navigation and Local time synchronization: The GMS function responsible to Generate navigation SIS observables (GMSF1) provides the time stamping of the measurements with respect to a Local Time derived from the Atomic Clock in the GSS.

      Notes:

      - This synchronisation can be achieved via timing signal provided by NTP server or directly using the L-Band time stamped navigation signal.
      - The ranging accuracy is achievable providing a synchronization of satellites as described in SFC3.1
      - The GMS has the online capability on an operator command to reconfigure the GST reference to any configurable time reference point connected via GSS.

See also SFC3.2 GMS Times Synchronisation section in DDF2 (SFC3.2 Physical Static View).

4.4.3.2 SFC Functional Block Diagram

The following functional block diagram describes the GMS time synchronisation functionalities

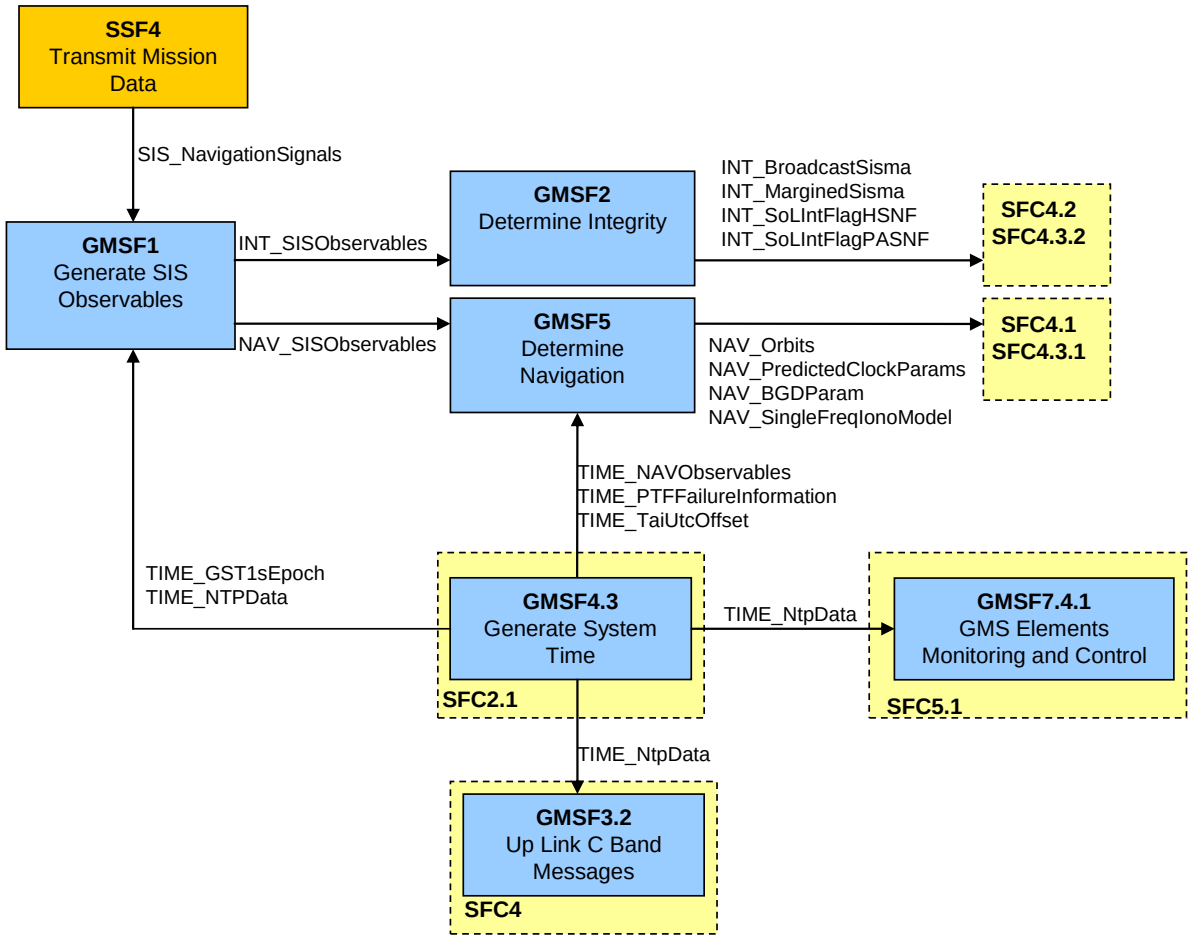


Figure 33: SFC3.2 Functional Block Diagram

4.4.4 SFC3.3 GCS Times Synchronization

4.4.4.1 SFC description

This SFC is responsible for providing the needed functionalities to:
- allow a synchronisation of the GCS segment timing parameters to the Galileo System Time
- retrieve UTC/TAI time parameters for specific flight dynamics applications

a) Initial synchronisation of the GCS time when GST is not available
When the GST generated by the GMS is not yet available (mainly during initialisation phase as some functionalities in GCS will be operational before the GMS Function responsible for generating the GST), the GCS will establish a temporary NTP Server synchronized through an appropriate device (Internet or GPS receiver).
Note that during IOV, it is foreseen that this synchronization is provided to UTC by the Hosting Entity.

b) GCS elements synchronisation to the GST:
- Provision of GST via NTP timing signal provided by GMS when available, for synchronisation to GSTof non-secure GCS elements.
The GMS·will continuously provide to the GCS KMF IRIG-B time code signals to each and every of the three GCS chains (OPE, VAL and INT). The GMS Function responsible for generating the GST (GMSF4.3) provides the needed interface to deliver the NTP timing reference to the GCS elements (see SFC2.1 for the GST generation functionalities)

- Provision of GST via IRIG timing signal provided by GMS when available, for GCS S-KMF synchronisation to GST.
The GMS Function responsible for generating the GST (GMSF4.3) provides the needed interface to deliver the IRIG timing reference to the GCS KMF (see SFC2.1 for the GST generation functionalities).

The GMS·GCS interface will allows the GMS to continuously provide IRIG-B time code signals to each and every of the three GCS chains (OPE, VAL and INT).

- Reception and Dissemination of Timing Reference at GCS level
Then the GCS provides the means to receive the timing reference provided by GMS and disseminate this to the elements of the GCS

c) Provision of UTC/TAI timing information for flight dynamics applications.
This includes the provision of TAI-UTC offset in term of leap second to allow the flight dynamic functions (GCSF4.4) to use the NORAD and SLR data.

4.4.4.2 SFC Functional Block Diagram

The following functional block diagram describes the GCS time synchronisation functionalities

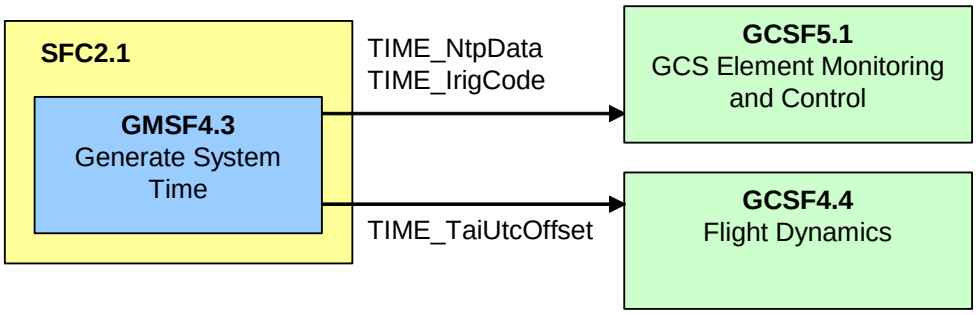


Figure 34: SFC3.3 Functional Block Diagram

In a satellite navigation system, the timing constraint is one of the key issues to achieve the mission performance. The accuracy of the timing parameters generated and the overall synchronisation of the components of the systems have a direct impact on the user positioning and timing solution performances.

The Maintain Overall System Synchronisation SFC is then responsible for providing all required functionality to:

- generate and maintain the different local timing parameters with associated performances, based on the provision of the GST as common time base.

  Important note: the generation and steering of the GST is described in a dedicated SFC: SFC2.1.

- Synchronise the elements involved in different mission chains

4.5 SFC4 Provide Mission Services (OS, PRS)

4.5.1 SFC4 Presentation

4.5.1.1 SFC4 high level description

This Service Functional Chain is in charge of providing all the means and associated processes needed by the end users to achieve the mission service objectives.
This includes two sub-SFCs corresponding to the core Mission services provided by the Galileo system: Open Service (OS), and Public Regulated Service (PRS).
Note: All SFC4 sub-SFCs can support different types of Galileo satellites at the same time, by proper system configuration.

The first level functional tree for the “Mission Service Provision” SFC is then as shown below:

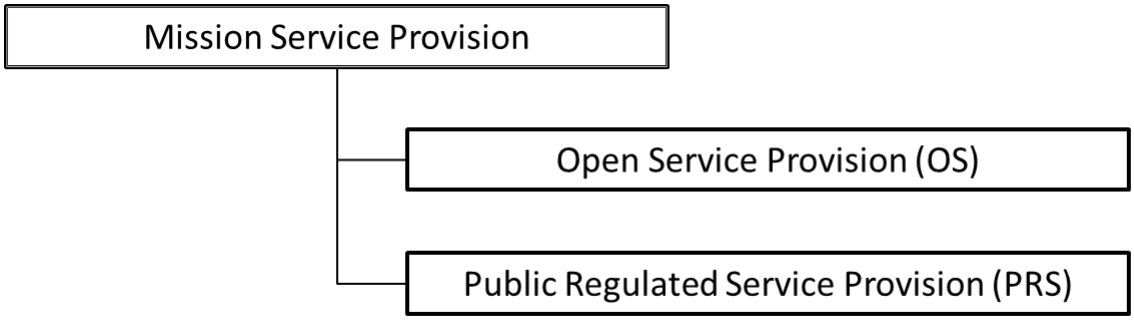


Figure 35 : SFC4 Functional tree (1st level)

The detail of each sub-SFC is given hereafter:

- SFC4.1: Open Services (OS), including:
  - SFC4.1.1: OS Positioning Service: responsible for the provision of the ranging signals and navigation data to allow the OS users to perform ranging measurements and compute positioning solution.
  - SFC4.1.2: OS Timing Service: responsible for providing timing data to retrieve the TAI, UTC precise times from GST.
  - SFC4.1.3: OS Signal Quality Monitoring Service: including the following sub SFCs:
    - SFC4.1.3.1: OS Signal-In-Space Accuracy (SISA) provision
    - SFC4.1.3.2: OS Health Status Flags provision.
  - SFC4.1.4: OS Degraded Positioning Service S-band provision.
- SFC4.2: Safety Of Life Integrity Service (SoL): This sub-SFC has been descoped.
- SFC4.3: Public Related Service (PRS), composed of the following sub SFCs, each of them including an encryption layer for access control.
  - SFC4.3.1: PRS Positioning Service: responsible for the provision of the ranging signals and navigation data to allow the PRS users to perform ranging measurements and compute positioning solution, with specific control access.
  - SFC4.3.2: PRS Timing Service: responsible for providing timing data to retrieve the TAI, UTC precise times from GST, with specific access control.
  - SFC4.3.3: PRS Signal Quality Monitoring Service: including the following sub SFCs:
    - SFC4.3.3.1: PRS Signal-In-Space Accuracy (SISA) provision
    - SFC4.3.3.2: PRS Health Status Flags and Satellite Health Status Flags provision.

- SFC4.3.4: PRS Degraded Positioning Service S-band provision.
- SFC4.3.5: PRS Integrity Service: This functionality has been de-scoped
- SFC4.3.6: PRS Security service: responsible for managing the access control to the PRS service, providing dedicated security data to the PRS users.

Each SFC section includes three sub-sections (when applicable) describing:

1. SFC description composed of :
   - Description of generation and dissemination of the respective mission data and signals.
   - Description of the processes needed by the end users to handle the disseminated signals and mission data, in order to achieve Galileo mission Service objectives.
2. Associated performance constraints, detailed in the System PBF.
3. SFC Functional Block Diagrams. This section describes functionally in two separate figures:
   - The service data generation and provision by the system.
   - The service data usage by the users.

4.5.1.2 Definitions

This section compiles the definitions of the services and main terms used in this SFC:

- **Positioning Service Definition:** A positioning service enables an end-user to estimate his position and velocity with a given accuracy.
- **Timing Service Definition:** A timing service enables an end-user to estimate the Coordinated Universal Time (UTC) with a given accuracy.
- **Open Service Definition**: Open Services (OS) are positioning and timing services provided free of charge to end-users. The end-user is provided by ranging signals and navigation messages broadcasted by the Galileo constellation over the following frequencies: E5a, E5b and E1B.

  The following categories of users are envisaged:
  - Single carrier OS users: These users use only one frequency (E5a or E5b or E1B/C) to perform ranging measurements and receive navigation data. As a consequence they compensate the group delay and ionospheric propagation errors using a model of which the parameters are broadcasted in the navigation data.
  - Dual carrier OS users: These users receive simultaneously E5a and E1B/C or E5b and E1B/C. They are able to compensate the group delay and ionospheric propagation errors without using any additional data.

  The OS performance is characterized in terms of:
  - Position, velocity and time estimation accuracy,
  - Availability

- **Public Related Service Definition**: The Public Regulated Services (PRS) are positioning and timing services provided using E6A and E1A frequencies. The access to this service is controlled by both signal and data encryption.
  The following categories of users are envisaged:
  - Single carrier PRS users for positioning: These users use only one frequency (E6A or E1A) to perform ranging measurements and receive navigation data. As a consequence they compensate the group delay and ionospheric propagation errors using a model of which the parameters are broadcasted in the navigation data.

Dual carrier PRS users for positioning and timing These users simultaneously receive E6A and E1A. They do not need any additional broadcast data to remove the ionospheric propagation errors.

4.5.2 SFC4.1 Open Services (OS) Provision

This functional chain is responsible for the Open Services provision, as indicated in the second level functional tree for the “Open Services Provision” SFC below below:

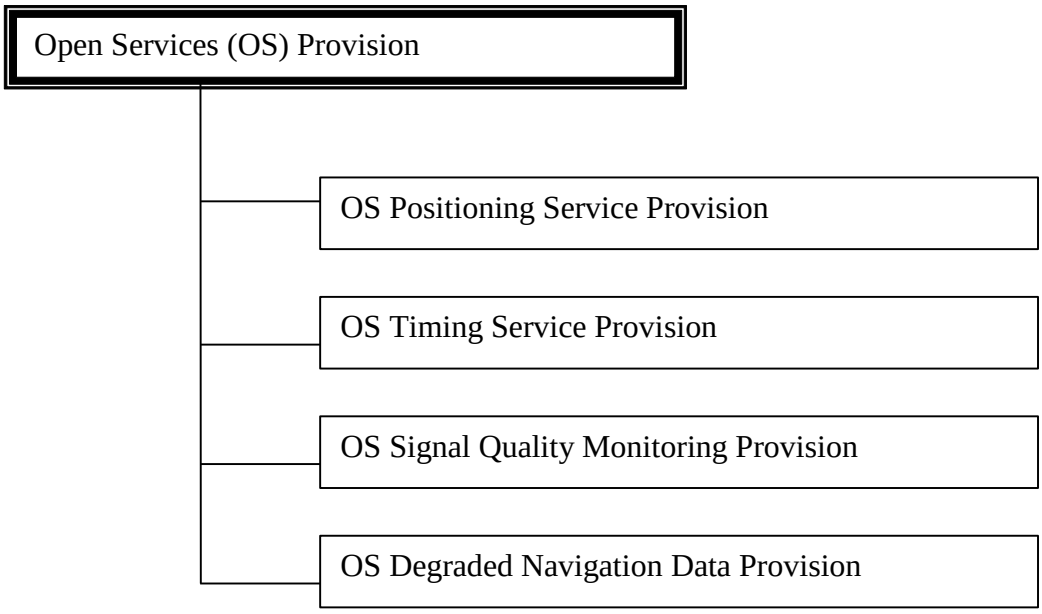


Figure 36: SFC4.1 Functional Tree (2nd level)

4.5.2.1 SFC4.1.1 OS Positioning Service Provision

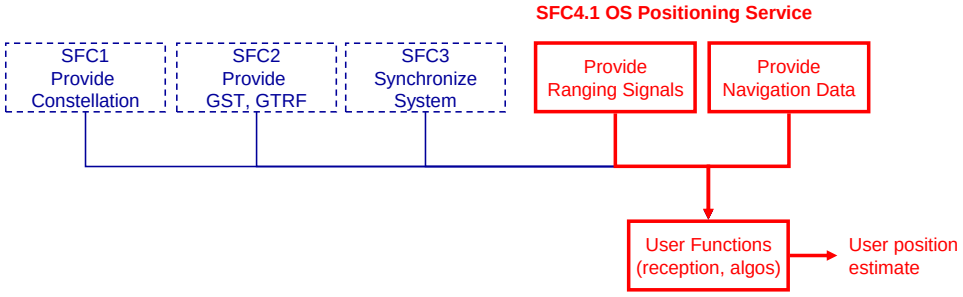
The OS positioning service consists in enabling the user to estimate his position and velocity.

4.5.2.1.1 SFC description

This functional chain is in charge of providing all the means and associated processes needed by the end user to compute a positioning solution, namely:

- OS ranging signals (E5a-I, E5a-Q, E5b-I, E5b-Q, E1B and E1C) to perform pseudorange measurements. The pseudo-range is the measured distance between the receiver and a satellite used in the navigation algorithm.
- Navigation data (ephemeris, almanacs, time and clock corrections). The navigation data to correct pseudorange measurements (clock corrections) and compute the user position (ephemeris and almanacs). The OS navigation data is transmitted on the E5a, E5b and E1B data channels.

This is indicated in the following functional chart:



The quality and the precision of both ranging signals and navigation data has a direct impact on the precision of the final position and velocity computed by the user.

4.5.2.1.1.1 OS Ranging Signal Provision by the system

The SFC4.1.1 is responsible for providing and maintaining the OS ranging signals to the user.

This includes the following main functionalities:

- OS ranging signal generation: generated by the satellite by modulating:
  - a navigation ranging code
  - and carriers generated from the following frequencies: E5 and E1.

  Note: The navigation ranging code can be modulated with a navigation message being a data channel or not modulated with a navigation message being a pilot channel.

- Ranging accuracy performance maintenance: this is ensured by:
  - GMS navigation determination chain to provide accurate navigation data (see next section)
  - GCS/GMS satellite monitoring and control to ensure the required satellite stability (see SFC 5.2).

4.5.2.1.1.2 Navigation Data Provision for OS Positioning

The SFC4.1.1 is responsible for providing the navigation data needed by the user to perform OS positioning.

This includes the following main functionalities:

a. OS positioning data computation: The navigation data is computed by a fully automated Mission Determination Chain composed of:
   - On ground navigation data determination functions (GMSF5): every 10 minutes, based on information provided by the GSS (GSS Raw Data) derived from the SIS (GMSF1). The following data type are generated:

     - Ephemeris (generated by GMSF5.1): The ephemeris are predicted satellite positions in GTRF composed by orbital parameters to be transmitted by the satellites.

     - Time and Clock corrections are predicted parameters to retrieve the offset between the on board time (LGST, time disseminated in the message) and the Galileo System Time (GST, reference for the entire system) (GMSF5.4). They are computed on a satellite per satellite basis for the frequency combinations (E5a-E1) and (E5b-E1).
       Additional time corrections are also generated specificaly for single frequency OS users like:
       - Broadcast Group Delay (BGD) (GMSF5.2)
       - Ionospheric model parameters (GMSF5.5).

       To maintain the Galileo interoperability with GPS within the time corrections are also computed and transmitted to the users the GPS-Galileo Time Offset (GGTO) see SFC2.3.

- Almanacs (generated by GMSF5.1): contains the ephemeris and satellite clock corrections in a reduced accuracy to inform the users about the approximate position of each satellite in the constellation.

The OS Navigation data is computed to be uplinked to the satellite through both C- and S-bands (see SFC4.1.4).

- Onboard navigation data determination functions (SSF3.1) including Local realization of the Galileo System Time (LGST) at satellite level.
  - LGST parameters generation: The LGST parameters (Time of Week (ToW) and Week Number (WN)) are generated by the satellite and maintained by the system (SFC3.1) with respect to its onboard clock.
  - LGST parameters dissemination: They are inserted in the OS navigation messages by the satellite (SSF3.2) before the message modulation (SSF4.2).

b. OS navigation message generation: All the above data types are structured into the navigation messages (GMSF5.6) and authenticated on ground (GMSF8.3) before their uplink to the satellite. The authentication is validated on board the satellite (SSF1.2)
Two kinds of OS navigation messages are generated:
   - F/NAV to be transmitted on the E5a data channel
   - I/NAV to be transmitted on the E5b and E1B data channels

c. OS navigation message uplink: The uplink scheduling algorithm (see SFC5.3.2) ensures that the navigation data broadcasted by the satellites are refreshed to cope with the navigation data validity time. The driven constraint is the satellites master clock (RAFS or PHM) stability.

d. OS navigation message dissemination: Once the data is uplinked, the navigation messages are stored (SSF1.3) and prepared (SSF3.2) to be modulated with the ranging codes (SSF3.3) and the OS signal carriers (SSF4.2).

Notes:
- The detail of each type of data and message structures is given in the System Data Dictionary and associated justification (see Annex A of DDF2).
- The ephemeris is unique for each satellite. It is computed wrt to the Apparent Center of Phase of the satellite L-Band Antenna (common to every frequency).
- Each satellite broadcasts its own ephemeris and clock and time corrections through the SIS.
- Each satellite will broadcast the almanacs for all the satellites in the constellation.

4.5.2.1.1.3 Positioning Solution Computation by the User

In the frame of the OS positioning service, the Standard Galileo Open-Service Receivers are in charge of the following functionalities:
- OS Ranging Signal Acquisition (TUSF1) and OS Navigation Data Retrieval (TUSF2)
- Pseudoranges measurements determination (TUSF5) including:
  - Compensation for satellite clock offset with respect to GST using the clock correction parameters
  - Tropospheric pre-processing compensation.
  - Compensation for satellite interfrequency bias for single carrier only using the relevant Broadcast Group Delay (BGD) parameters.
  - Ionosphere compensation
  - Relativistic effect corrections, due to:
    - orbit eccentricity based on algorithm specified in TUSREQ-22937 and associated Annex A

- geometrical range (Sagnac) effect based on algorithm specified in TUSREQ-22939 and associated Annex D.

  Note: Correction for Sagnac effect is adopted as the standard practice for navigation users The difference between using Sagnac correction wrt iteratively solving for the travel light-time correction is expected to be negligible considering the ODTS accuracy/noise. For further details, refer to dedicated TN "Relativistic Effects in Galileo System", ref. GAL-TN-ASTG-SYST-I-0016

- Position, Velocity and Time (PVT) solution computation (TUSF4) including:
  Positioning Determination: at a fixed rate of at least once per second using all valid pseudorange measurements (see SFC4.1.3.2). To be able to determine its position the receiver needs at least four valid pseudorange measurements.

For each of the valid pseudorange measurements, the user computes the satellites position in GTRF thanks to the ephemeris parameters.
The receiver performs then at each epoch an update of its initial PVT solution using all valid pseudorange measurements and computed satellites positions (Least Square Estimate). This user position is in the same reference as the satellites position, i.e. in GTRF.
A transformation is made to convert position vectors from GTRF coordinates to user Nord, Est, Down (NED) coordinates.

4.5.2.1.2 SFC associated Performance Constraints

The types of performances associated to the OS Positioning SFC are:

- Positioning accuracy obtained at user level: derived into a ranging accuracy and constellation geometry to be guaranteed by the System. The ranging accuracy depends on:
  - The quality of the signal (stability…) and associated modulation
  - The stability of the satellite position and clocks wrt the associated models (ephemeris and clock corrections) provided to the user in the navigation message.
  - Availability

See SPBF [SYS-RD.04] for detail derivation of each performance constraints.

4.5.2.1.3 SFC Functional Block Diagram

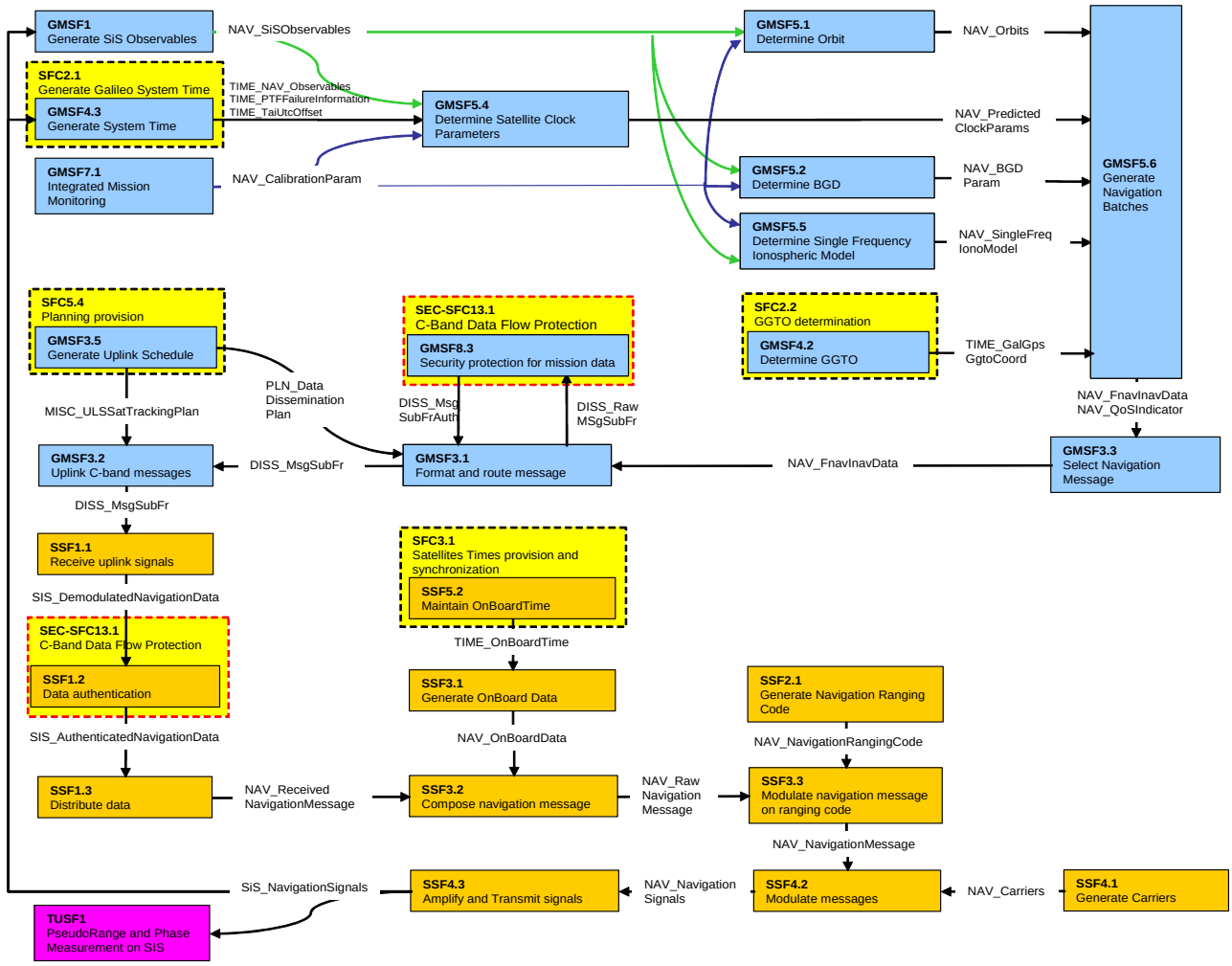


Figure 37: SFC4.1.1 Ranging Signal and Navigation Data Provision Functional Block Diagram

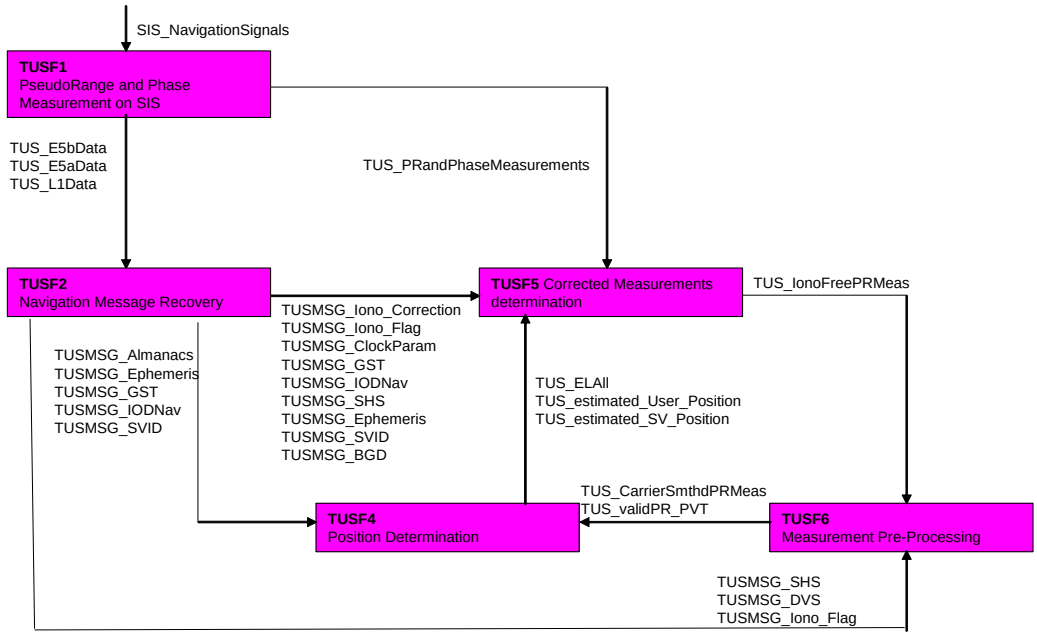


Figure 38 : SFC4.1.1 Ranging Signal and Navigation Data Usage Functional Block Diagram

4.5.2.2 SFC4.1.2 OS Timing Service Provision

4.5.2.2.1 SFC description

This functional chain is in charge of providing to the user the parameters and associated processes to predict the Coordinated Universal Time (UTC) based on the GST through conversion parameters.

4.5.2.2.1.1 Timing Parameters Provision by the System

As part of the OS Timing service SFC, the following data need to be provided:

- All data needed to compute a positioning solution (LGST Parameters, Time and clock corrections and ephemeris (see SFC4.1.1)), in order to determine the offset between the receiver time and GST.
- GST-UTC offset parameters:
    - GST-UTC offset generation: It is generated on ground (GMSF4.3) as described in the SFC2.1 related sections.
    - GST-UTC offset dissemination: It is disseminated within the OS navigation messages (GMSF5.6) as part of the OS data as described in SFC4.1.1.

4.5.2.2.1.2 Timing Solution Computation by the User

In the frame of the OS timing service, the Galileo Timing/Calibration Laboratory Satellite-only Test User Receiver is in charge of the following functions (TUSF7):

- Estimate GST with great accuracy and stability.
- Compute UTC.

4.5.2.2.2 SFC associated Performance Constraints

The type of performances associated to the OS Time solution computation SFC are:

- UTC realization accuracy
- Availability

See SPBF [SYS-RD.04] for detail derivation of each performance constraints.

4.5.2.2.3 SFC Functional Block Diagram

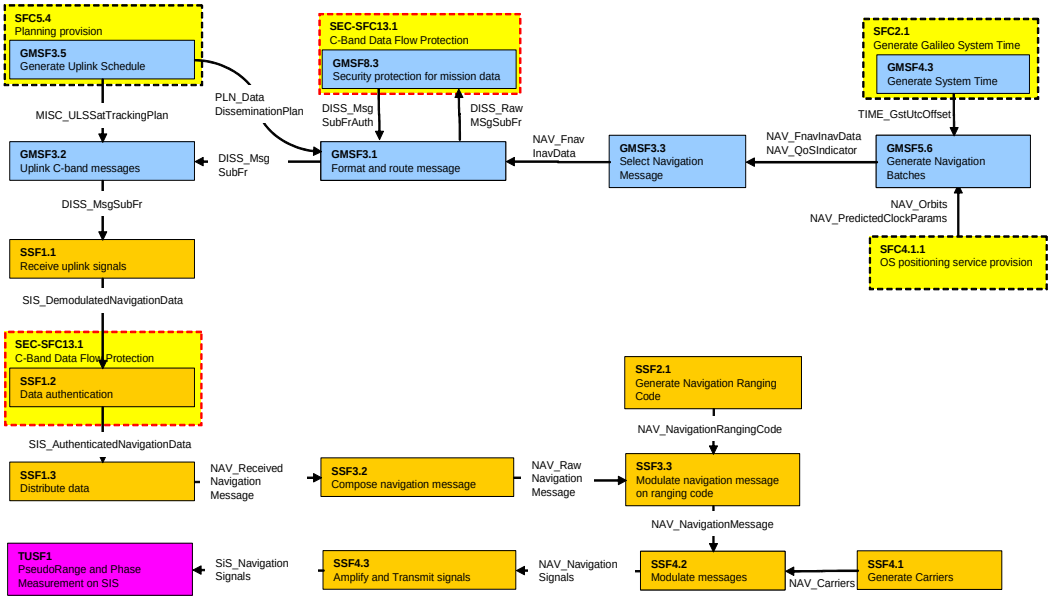


Figure 39: SFC4.1.2 Timing Data Provision Functional Block Diagram

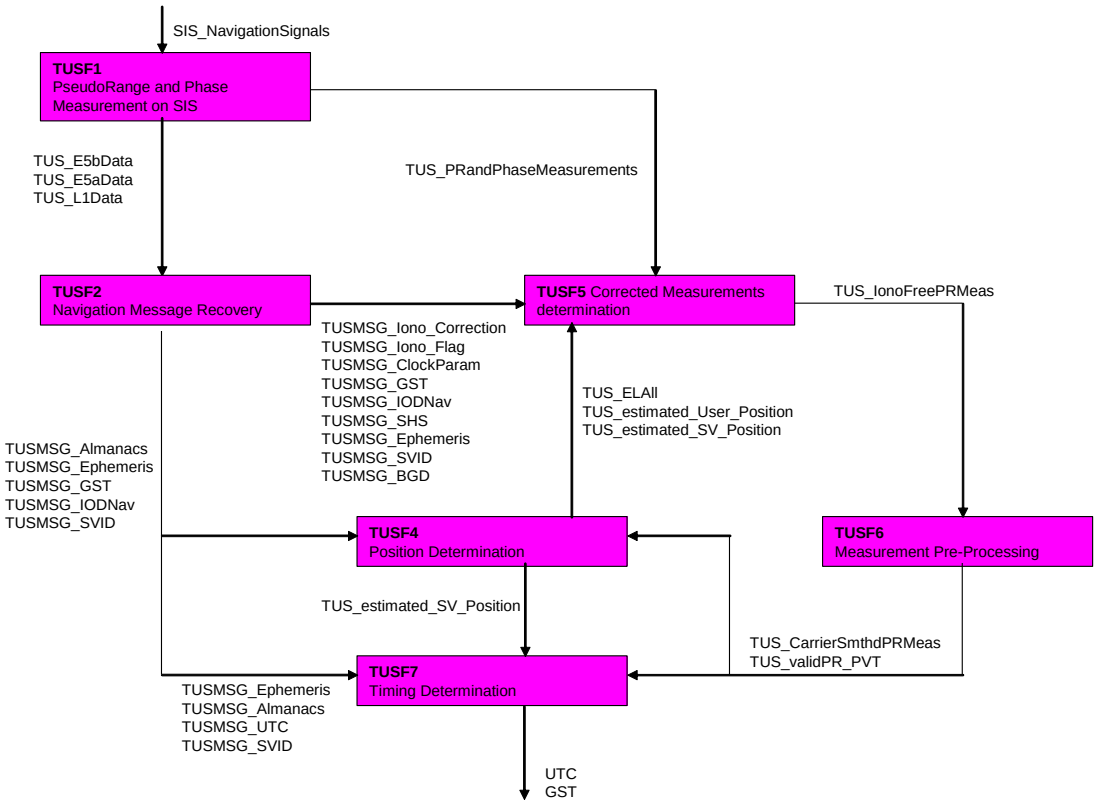


Figure 40 : SFC4.1.2 Data Usage Functional Block Diagram

4.5.2.3 SFC4.1.3 OS Signal Quality Monitoring Provision

The Signal Quality is indicated thanks to different parameters transmitted to the user:

- Signal In Space Accuracy (SISA): This parameter provides an indication to the user of the quality of the SIS broadcasted by satellites and in particular the quality of the ODTS outputs.
- Health Status including Signal Health Status and Data Validity Status

4.5.2.3.1 SFC4.1.3.1 OS Signal In Space Accuracy (SISA) Provision

4.5.2.3.1.1 SFC description

4.5.2.3.1.1.1 SISA Provision by the System

This SFC is in charge of generating and providing the Signal-In-Space Accuracy Parameter (SISA) as part of OS.

a. SISA Generation: SISA is generated (GMSF5.3) by a fully automated mission determination chain, based on information provided by:
   - GSS (GSS Raw Data) derived from the SIS (GMSF1)
   - SISA computation check: calibration parameters computed off-line (GMSF7.1) (see SFC5.3.2)
   - Planning information (SFC5.2)

   SISA is linked to a navigation data batch and generated every 10 minutes. It is computed on a satellite per satellite basis for the frequency combinations (E1B-E5a) and (E1B-E5b).

b. SISA dissemination: SISA is disseminated within the OS navigation messages. Therefore the SISA dissemination functional chain follows the same scheme as the SFC4.1.1.

Note: The detailed description of the SISA parameter is given in the System Data Dictionary and associated justification (see Annex A of DDF2).

SISA and Planning Events
For planned events implying satellite unavailability period, the SISA will be automatically set to NAPA for navigation messages whose validity duration is in the windows the event is announced to occur.

4.5.2.3.1.1.2 SISA Usage by the User

The SISA can be used by the OS positioning and timing users as an indication of the quality of the signal-in-space and in particular the quality of the ODTS output. ()

4.5.2.3.1.2 SFC associated Performance Constraints

#STRB2.0_change: GAL-DCN-TASI-SYST-X-0933
The (E1/E5a and E1/E5b) SISA is constrained by an upper bound value defined as in the following:

- 110 cm after T
- 85 cm with a probability higher than 0.5 after T + 9 days
- 85 cm after T + 18 days

where T is the first epoch of continuous fault free navigation signal.

with GSS Degraded atmospheric environment and at least 21 satellites of the nominal Galileo constellation providing fault free navigation signal; T being the first epoch of continuous fault free navigation signal.

See SPBF [SYS-RD.04] for details.

4.5.2.3.1.3 SFC Functional Block Diagram

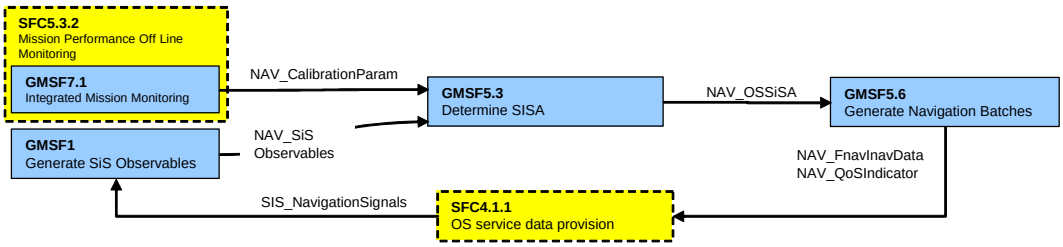


Figure 41 : SFC4.1.3.1 Functional Block Diagram

4.5.2.3.2 SFC4.1.3.2 Health Status Parameters Provision (on OS Signals)

4.5.2.3.2.1 SFC description

4.5.2.3.2.1.1 Health Status Flag Provision by the System

Two types of Health Status parameters are provided to the user:

- Signal Health Status flag
- Data validity Status flag

Signal health status (SHS):

The SHS will be used (STRB1.5.1) to inform the user of the status of the signal in space from a given satellite for each message type (i.e. F/NAV, G/NAV or I/NAV) and signal type (E5a-I, E5b-I, E6-A, E1A, E1-B).

The SHS can be set in two ways in the GMS:
- Manual setting by operator
- Automatic setting as results of planned events.

In case of differences between the automatic SHS and the operator configured SHS, the GMS shall uplink the most conservative value with the following order of precedence “In Test”, “Out of Service”, “Will be Out of Service“ and finally “OK” according to the next table

		SHS Manual setting			
		OK	**Will be out of service**	**Out of service**	**Test**
SHS automatic setting	**OK**	OK	Will be out of service	Out of service	Test
	Out of Service	Out of Service	Out of Service	Out of Service	Test
	Test	Test	Test	Test	Test

Specifically for planned events the tasks which lead to an automatic setting of SHS are sent by the GCS and the SHS setting will be according to the nature of the task. The following table provide a list of tasks which can be notified by GCS:

Task Originator	Task	SHS value during planned event
GCS MTP Request	Non-routine payload configuration	Out of service
	Satellite Master clock synchronization	Out of service
	Satellite on-board time of transmission synchronization	Out of service
	Satellite Master Clock Switch	Out of service
	Satellite hot backup to cold backup clock switch	OK
	Orbit manoeuvre	Out of service
	OBSM (requested by GCS)	Out of service
	TTCF Station unavailability	OK
	SAR unavailability	OK
	Satellite unavailability	Out of service
	SIS unavailable	Out of service

Note that for planned events, it is not requested to transmit “Signal Will be out of service” in order to warn the user that the signal will be unavailable in the future.

For un-planned events the SHS will be automatically triggered by the system per satellite and per service; (**#STRB2.1_change: GAL-DCN-TASI-SYST-X-0989**) that functionality will be based on the online monitoring of ranging accuracy (i.e. SISE) (**#STRB2.1_change: GAL-DCN-TASI-SYST-X-0945**). The SISE will be online monitored and the SHS will be automatically set to “Signal out of service” in case a violation of Not to Exceed threshold persist for more than a configurable time or the violation occur twice in a predetermined time period, thus mitigating the users exposure time to misleading information caused by major service failures.

Navigation Data Validity Status (DVS): the DVS is provided to the user according to the following processes:
DVS generation: The DVS can be set by the satellite to “Working without guarantee” in the following cases:

- Autonomously when:
  - The navigation data batch available at satellite level for dissemination is no more valid. This decision is based on onboard time out counters (SSF3.2) compared to a predefined threshold.
  - Or events are detected on board (SSF6.1) which affects navigation service performance (e.g. events generated by the Failure Detection, Isolation and Recovery (FDIR) functionality).
  - Or the S-Band telecommand up-link is lost for a predefined period.
- Or on reception of dedicated TC via S-band (SFC5.2).

DVS dissemination: This flag is disseminated within the OS navigation messages. The DVS is transmitted on F/NAV (E5a) and I/NAV (E5b and E1B) messages once per sub-frame.

4.5.2.3.2.1.2 Health Status Flag Usage by the User

The Health Status Flags are used by the users to determine valid satellite range measurements (TUSF6) that can be used for positioning or timing computation.

4.5.2.3.2.2 SFC Functional Block Diagram

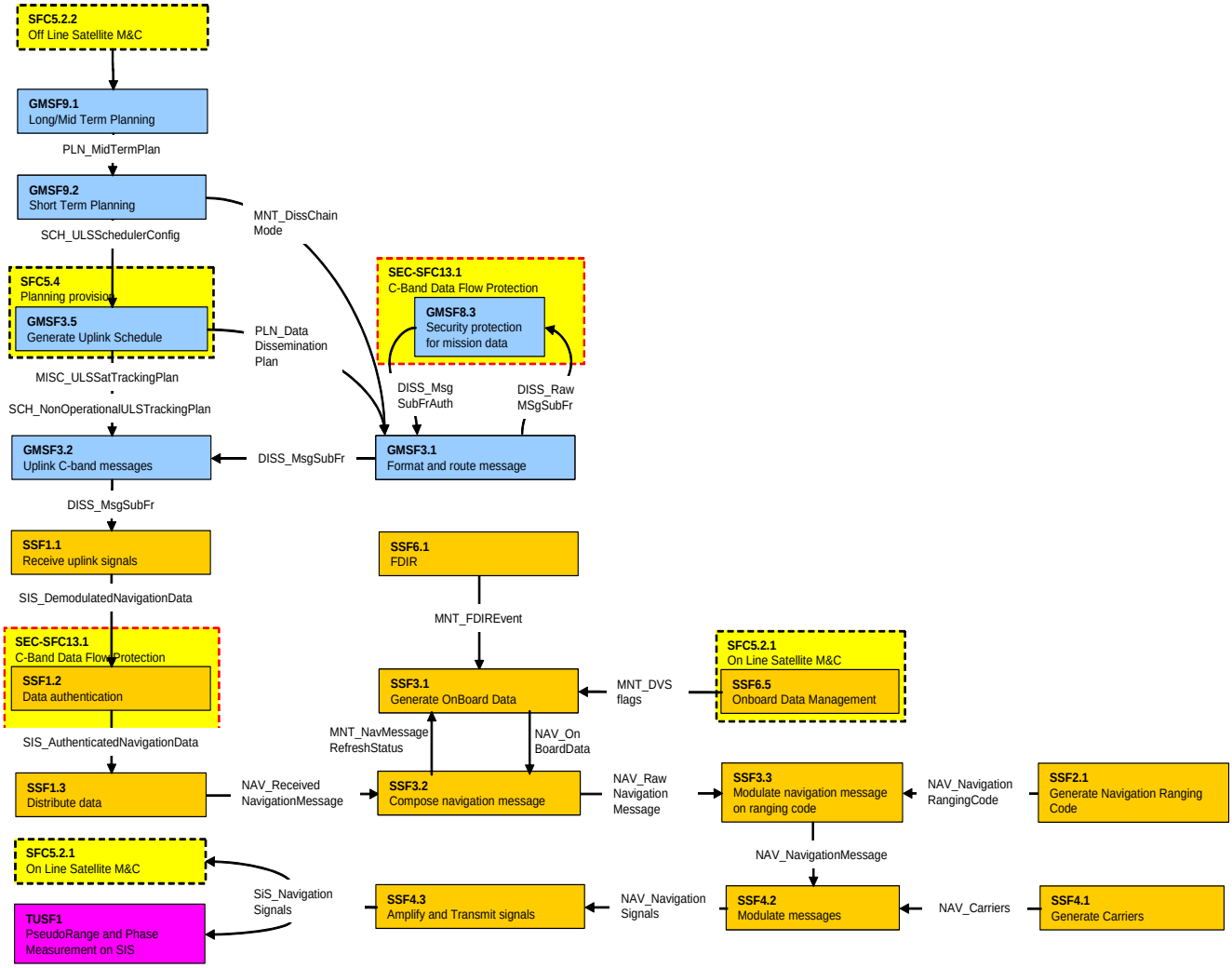


Figure 42 : SFC4.1.3.2 Functional Block Diagram

4.5.2.4 SFC4.1.4 OS Degraded Navigation Data Provision

4.5.2.4.1 SFC description

The purpose of this system functional chain is to support the graceful degradation of open navigation service. Navigation data is uplinked to the satellite via the S-band TT&C link and stored on-board the satellite. This data is then available to support a degraded mode of operation for the OS when no more up-to-date navigation data is available on the satellite (ie via the nominal C-band uplink).

4.5.2.4.1.1 Degraded Data Provision by the System

a. S-band Degraded data generation: The S-band degraded data is generated by the GMS (GMSF5.6) every 10 minutes and structured into both F/NAV and I/NAV pages. It has the same

format and contents as the OS navigation data described on the SFC4.1.1 with less precision but longer validity.

b. S-band degraded data dissemination: This data is encapsulated in dedicated TCs uplinked to the satellite via the GCS at each TT&C contact (ie nominally once per orbit).

The GCS will support the navigation mission services by up-linking the so called “Degraded” Navigation messages as part of nominal contact activity. This is used to back up the nominal message dissemination performed by GMS in C band, as messages uplinked in S-Band will remain valid longer than messages uplinked by GMS in C band. “Degraded” navigation messages can result in less accurate PVT solutions.

S-Band navigation messages are computed by GMS and transferred on line to the GCS within discrete files, together with the indication of their quality, for all Navigation services (i.e. I/NAV and F/NAV messages for Open service, G/NAV message for PRS navigation and G/NAV data 9 for PRS messages). To guarantee the independence of services the GCS will select the best messages per service basis within the files generated by GMS, pre-venting the uplink of invalid messages.
Note: the use of QoS as above explained will be implemented by STRB2.1 (**#STRB2.1_change: GAL-DCN-TASI-SYST-X-0863, GAL-DCN-TASI-SYST-X-0810, GAL-DCN-TASI-SYST-X-0809***)*

The satellite stores the navigation data on-board so that it is available for use in degraded conditions. When the C-band OS data is no longer valid the satellite will transmit the S-band OS navigation messages. These are processed in the same way as the OS navigation messages uplinked on the C-band and described in the SFC4.1.1.
The GCS reports to GMS the uplink status of the S-band degraded data.

4.5.2.4.1.2 Degraded Data Usage by the User

The usage by the user of the S-band degraded data is the same as described for the SFC4.1.1.

4.5.2.4.2 SFC Functional Block Diagram

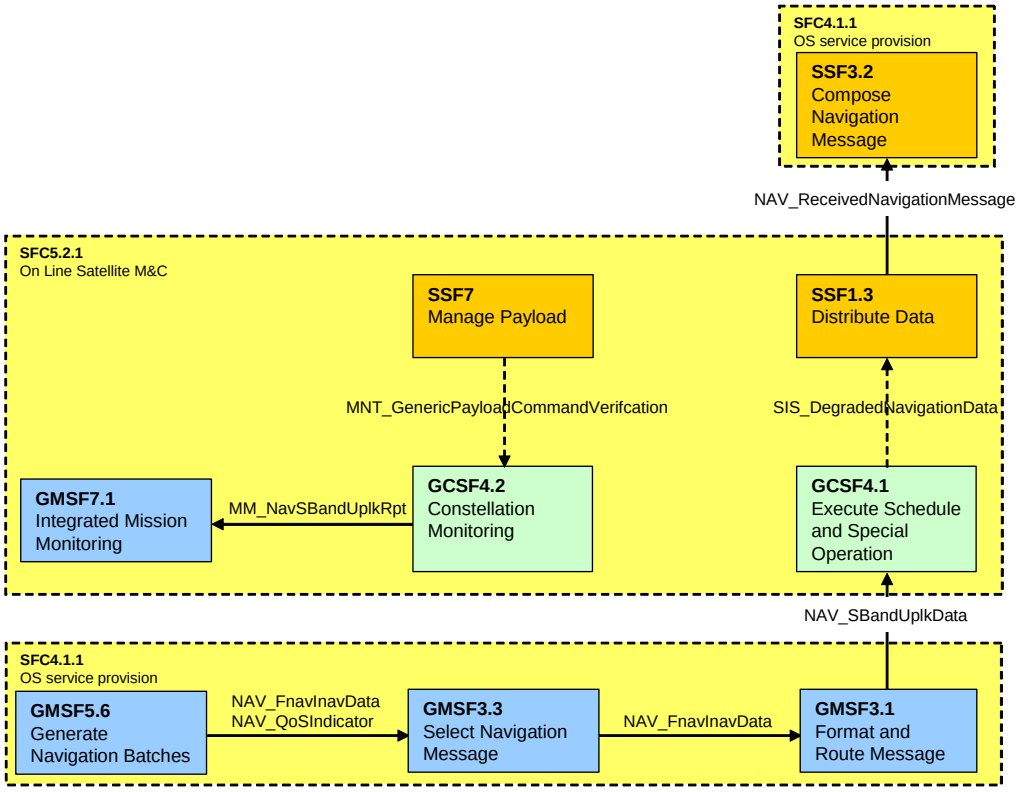


Figure 43 : SFC4.1.4 Functional Block Diagram

4.5.3 SFC4.3 Public Regulated Services (PRS) Provision

This functional chain is responsible for the PRS Services provision. This includes the generation, dissemination and usage of the PRS Service messages needed to perform the following PRS main applications:

- PRS Positioning (SFC4.3.1)
- PRS Timing (SFC4.3.2)
- PRS Signal Quality Monitoring Provision (SFC4.3.3)
- PRS Degraded Navigation Data Provision (SFC4.3.4)
- PRS Security (SFC4.3.6)

The performance of PRS service are reported in CLA DDF [SYS-RD.7] and CLA PBF [SYS-RD.96].

The second level functional tree for the "PRS Services Provision" SFC is shown below:

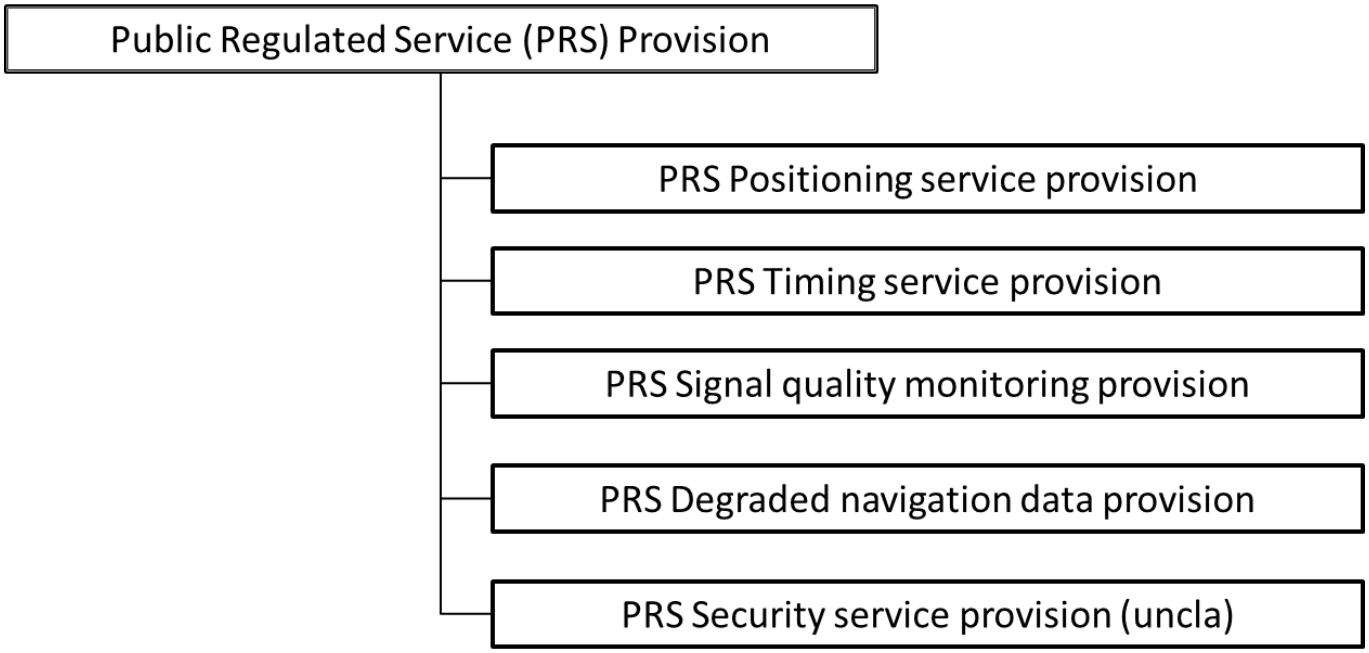


Figure 44: Figure: SFC4.3 Functional Tree (2nd level)

The functional chains involved in the PRS service provision are independent from the OS service and any other Radio Navigation Satellite System (RNSS).

All PRS services are protected with encryption mechanisms to limit the access to restricted users.

4.5.3.1 SFC4.3.1 PRS Positioning Service Provision

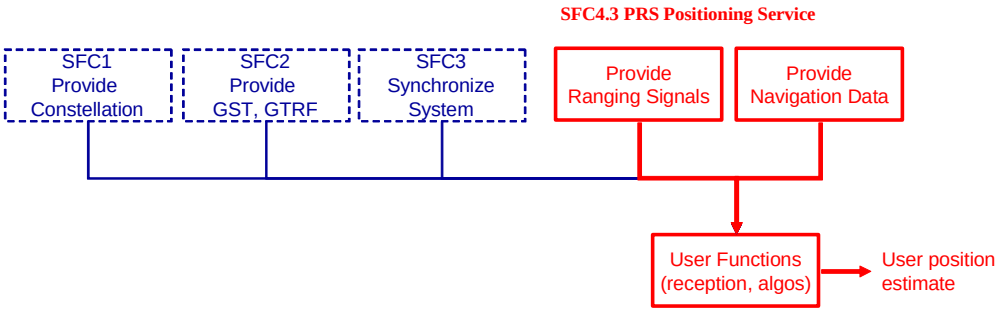
4.5.3.1.1 SFC description

Functional Architecture identical to the one described for OS SFC4.1.1 with additional security functions for Message protection layer.

This functional chain is in charge of providing all the means and associated processes needed by the end user to compute a positioning solution, namely:

- PRS ranging signals (E6A, E1A) to perform pseudorange measurements. The pseudo-range is the measured distance between the receiver and a satellite used in the navigation algorithm.
- Navigation data (ephemeris, almanacs, time and clock corrections. The navigation data to correct pseudorange measurements (clock corrections) and compute the user position (ephemeris and almanacs). The PRS navigation data is transmitted on the E6A and E1A data channels.

as indicated in the following functional chart:



Note: The quality and the precision of both ranging signals and navigation data has a direct impact on the precision of the final position and velocity computed by the user.

4.5.3.1.1.1 PRS Ranging Signal Provision by the system

The SFC4.3.1 is responsible for providing and maintaining the PRS ranging signals to the user.
This includes the following main functionalities:

- PRS ranging signal generation: generated by the satellite by modulating:

- a navigation ranging code modulated with a navigation message for the data channel
- and carriers generated from the following frequencies: E6A, E1A

4.5.3.1.1.2 Navigation Data Provision for PRS Positioning

The SFC4.3.1 is responsible for providing the navigation data needed by the user to perform PRS positioning.
This includes the following main functionalities:

a. PRS positioning data computation: The navigation Data Generation: the data are computed by a fully automated Mission Determination Chain every 10 minutes, based on information provided by the GSS (GSS Raw Data) derived from the SIS (GMSF1).

   As part of the PRS Positioning SFC, this includes the following data types:

   - Ephemeris: The ephemeris are predicted satellite positions composed by orbital parameters to be transmitted by the satellites (GMSF5.1).
   - Almanacs: contains the orbital parameters and satellite clock corrections in a reduced accuracy to inform to the users about the approximate position of each satellite in the constellation (GMSF5.1).
   - Time and Clock corrections are predicted parameters to retrieve the offset between the on board time (LGST, time disseminated in the message) and the Galileo System Time (GST, reference for the entire system) (GMSF5.4). They are computed on a satellite per satellite basis for the frequency combination (E1A-E6a).

     Some other time corrections are also generated for single frequency PRS users like:

     - Broadcast Group Delay (BGD) (GMSF5.2)
     - Ionospheric model parameters (GMSF5.5).

   To maintain the Galileo interoperability with GPS within the time corrections are also computed and transmitted to the users the GPS-Galileo Time Offset (GGTO): see SFC2.3.
   The PRS Navigation data is computed to be uplinked to the satellite through both C- and S-bands (see SFC4.3.4).

b. PRS navigation message generation: ll the above data types are structured into the PRS navigation messages (GMSF5.6), encrypted (see SEC-SFC15.1.2) and authenticated on ground

(GMSF8.3) before their uplink to the satellite. The authentication will be validated on board the satellite (SSF1.2).

c. PRS navigation message uplink: The uplink scheduling algorithm (see SFC5.3.2) ensures that the navigation data broadcast by the satellites are refreshed to cope with the navigation data validity time. The driven constraint is the satellites master clock (RAFS or PHM) stability
d. PRS navigation message dissemination: Once the data is uplinked, the navigation messages are stored (SSF1.3) and prepared (SSF3.2) to be modulated with the ranging codes (SSF3.3) and the PRS signal carriers (SSF4.2).

4.5.3.1.1.3 Positioning Solution Computation by the User

In the frame of the PRS positioning service, the Standard Galileo PRS Receivers are in charge of the following functionalities:
- PRS Ranging Signal Acquisition (TUSF1) and PRS Navigation Data Retrieval (TUSF2) after signal and navigation data decryption.
- Pseudoranges measurements determination (TUSF5) including:
    - o Compensation for satellite clock offset with respect to GST using the clock correction parameters
    - o Tropospheric pre-processing compensation.
    - o Compensation for satellite inter-frequency bias using the relevant Broadcast Group Delay (BGD) parameters (single carrier only).
    - o Ionosphere compensation.

- Position, Velocity and Time (PVT) solution computation (TUSF4) including:
    - o Positioning Determination: at a fixing rate of at least once per second using all valid measurements (see SFC4.1.3.2).

4.5.3.1.2 SFC associated Performance Constraints

The types of performances associated to the OS Positioning SFC are:
- Positioning accuracy obtained at user level: derived into a ranging accuracy and constellation geometry to be guaranteed by the System. This accuracy depends on:
    - the quality of the signal (stability…) and associated modulation (note: the constant-envelope modulation allows resistance to interference-jamming for PRS receivers).
    - the stability of the satellite position and clocks wrt the associated models (ephemeris and clock corrections) provided to the user in the navigation message.
    - Availability

See SPBF [SYS-RD.04] for detail derivation of each performance constraints.

4.5.3.1.3 SFC Functional Block Diagram

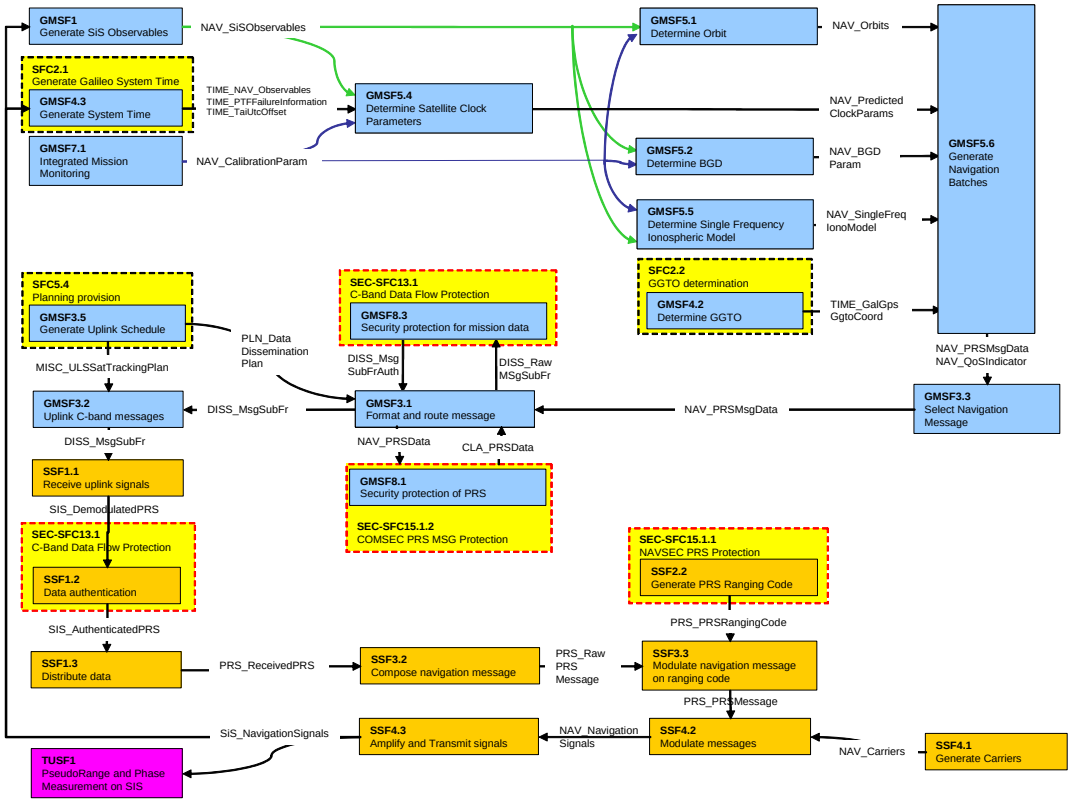


Figure 45 : SFC4.3.1 Data Provision Functional Block Diagram

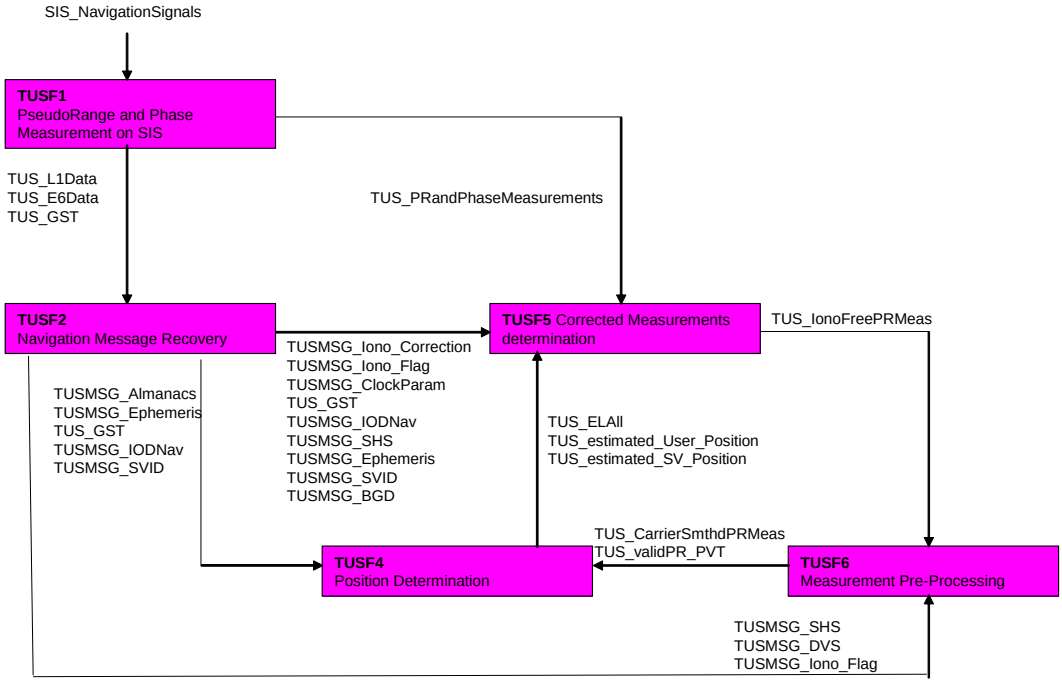


Figure 46 : SFC4.3.1 Data Usage Functional Block Diagram

4.5.3.2 SFC4.3.2 PRS Timing Service Provision

4.5.3.2.1 SFC description

Functional Architecture identical to the one described for OS SFC4.1.2 with additional security functions for Message protection layer.

This functional chain is in charge of providing to the PRS user the parameters and associated processes to predict the Coordinated Universal Time (UTC) based on the GST through conversion parameters generated by GMSF4.3.

4.5.3.2.1.1 Timing Parameters Provision by the System

As part of the OS Timing service SFC, the following data need to be provided:

- All data needed to compute a positioning solution (LGST Parameters, Time and clock corrections and ephemeris), in order to determine the offset between the receiver time and GST.

- GST-UTC offset parameters:
  a. GST-UTC offset generation: It is generated on ground (GMSF4.3) as described in the SFC2.1 related sections.
  b. GST-UTC offset dissemination: It is disseminated within the PRS navigation messages (GMSF5.6) as part of the PRS data as described in SFC4.3.1.

4.5.3.2.1.2 Timing Solution Computation by the User

In the frame of the PRS timing service, the Galileo Timing/Calibration Laboratory Satellite-only Test User Receiver is in charge of computing the UTC (TUSF7).
The processes to estimate and compute UTC are described in SFC4.1.2.

4.5.3.2.2 SFC associated Performance Constraints

The type of performances associated to the PRS Time solution computation SFC are:
- UTC realization accuracy
- Availability

See SPBF [SYS-RD.04] for detail derivation of each performance constraints.

4.5.3.2.3 SFC Functional Block Diagram

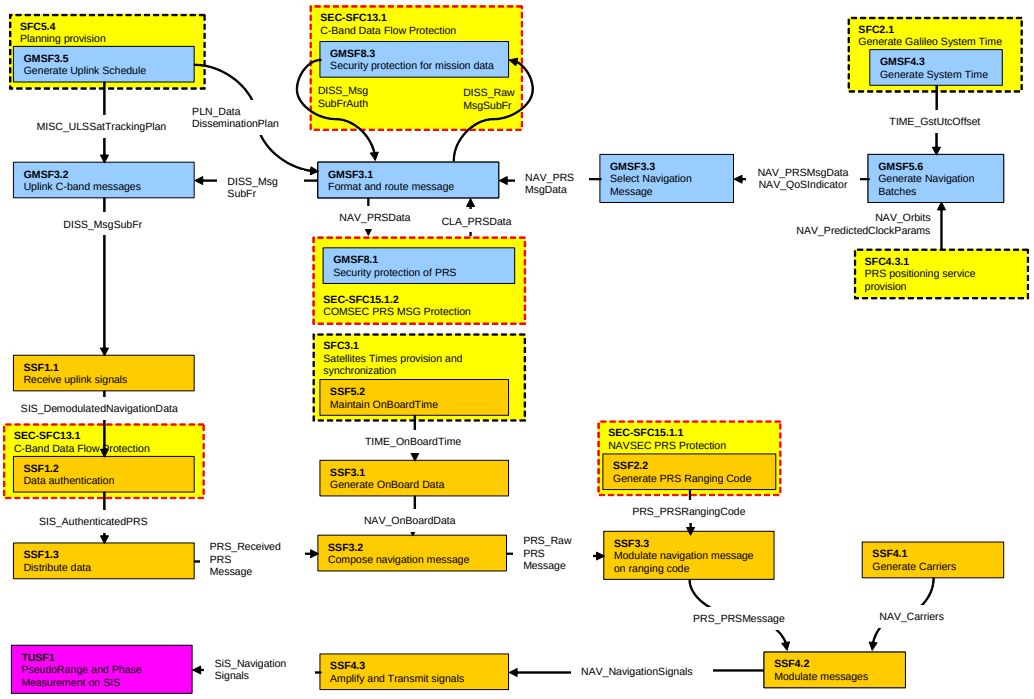


Figure 47 : SFC4.3.2 Data Provision Functional Block Diagram

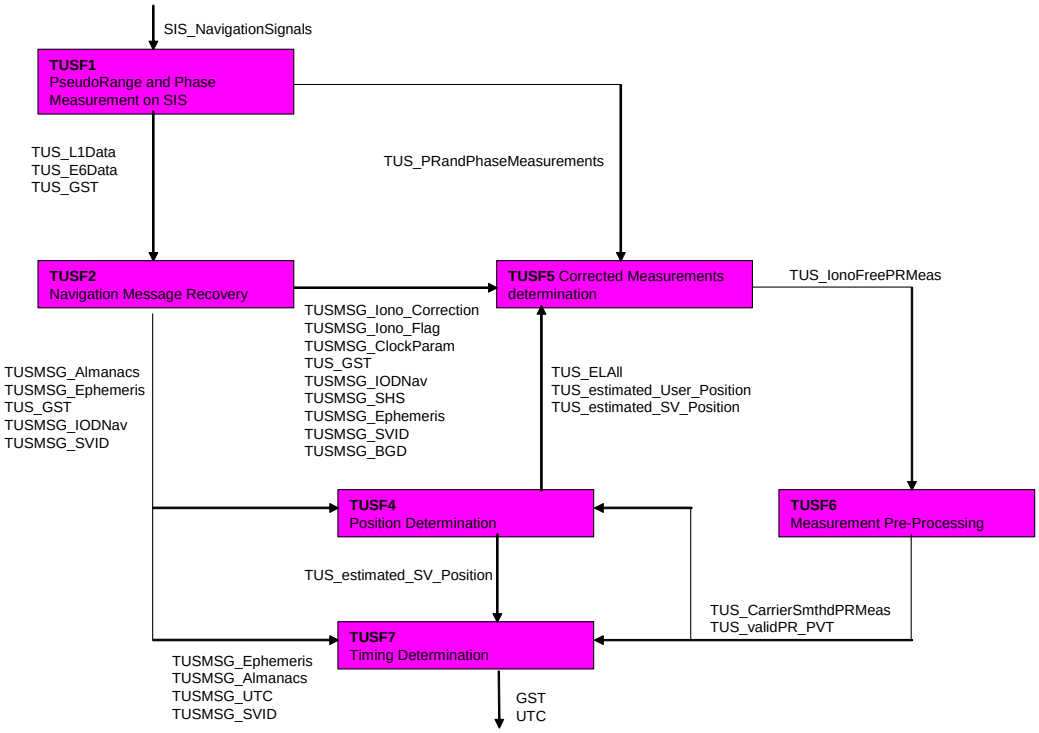


Figure 48 : Figure: SFC4.3.2 Data Usage Functional Block Diagram

4.5.3.3 SFC4.3.3 PRS Signal Quality Monitoring Provision

Functional Architecture identical to the one described for OS SFC4.1.3 with additional security functions.

The Signal Quality is indicated thanks to different parameters transmitted to the user:
- Signal In Space Accuracy (SISA)
- Health Status including:
  - Signal Health Status flag
  - Data Validity Status flag
  - Satellite Health Status flag and C-band link status flags

4.5.3.3.1 SFC4.3.3.1 PRS Signal In Space Accuracy (SISA) Provision

4.5.3.3.1.1 SFC description

4.5.3.3.1.1.1 SISA Provision by the System

This SFC is in charge for generating and providing the Signal-In-Space Accuracy parameter (SISA) to the PRS users.

a. SISA generation: SISA is generated (GMSF5.3) by a fully automated Mission determination chain based on information provided by:
   - § GSS (GSS Raw Data) derived from the SIS (GMSF1)
   - § SISA computation check: calibration parameters computed off-line (GMSF7.1) (see SFC5.3.2)
   - § Planning information (SFC5.2)

   SISA is linked to a navigation data batch and generated every 10 minutes. It is computed on a satellite per satellite basis for the frequency combination (E1A-E6A). For planned events, the SISA will be automatically set to NAPA for navigation messages whose validity duration is in the windows the event is announced to occur

b. SISA dissemination: SISA is disseminated within the PRS navigation messages. Therefore the SISA dissemination functional chain follows the same scheme as the SFC4.3.1.

Note: The detailed description of the SISA parameter is given in the System Data Dictionary and associated justification (see Annex A of DDF2).

SISA and Planning Events
For planned events implying satellite unavailability period, the SISA will be automatically set to NAPA for navigation messages whose validity duration is in the windows the event is announced to occur

4.5.3.3.1.1.2 SISA Usage by the User

The SISA can be used by the PRS positioning and timing users as an indication of the quality of the signal-in-space and in particular the quality of the ODTS output..

4.5.3.3.1.2 SFC associated Performance Constraints

The (E1/E6) SISA is constrained by an upper-bound value defined as in the following:
- 110 cm after T
- 85 cm with a probability higher than 0.5 after T + 9 days
- 85 cm after T + 18 days

where T is the first epoch of continuous fault free navigation signal.

with GSS Degraded atmospheric environment and at least 21 satellites of the nominal Galileo constellation providing fault free navigation signal; T being the first epoch of continuous fault free navigation signal.

See SPBF [SYS-RD.04] for detail derivation of each performance constraints.

4.5.3.3.1.3 SFC Functional Block Diagram

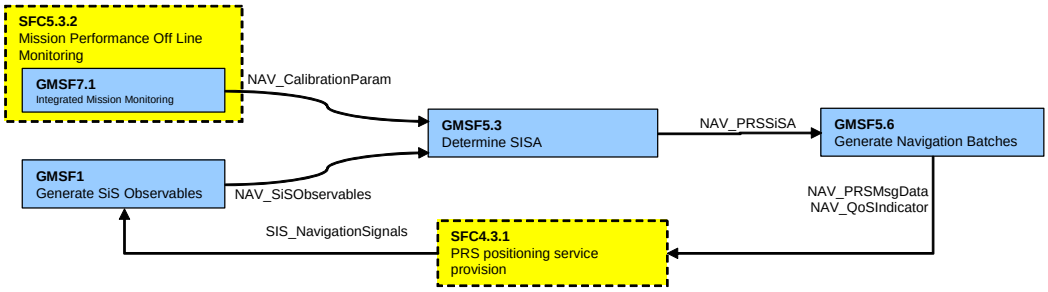


Figure 49 : SFC4.3.3.1 Functional Block Diagram

4.5.3.3.2 SFC4.3.3.2 Health Status Parameters Provision (on PRS Signals)

4.5.3.3.2.1 SFC Description

4.5.3.3.2.1.1 Health Status Flag Provision by the System

Two types of Health Status parameters are provided to the user:
§ Signal Health Status flag
§ Data Validity Status flag

Signal health status (SHS):

The SHS will be used to info the user of the status of the signal in space from a given satellite for each message type (i.e. F/NAV, G/NAV or I/NAV) and signal type (E5a-I, E5b-I, E6-A, E1A, E1-B).

The SHS can be set in two ways in the GMS:
- Manual setting by operator
- Automatic setting as results of planned events.

In case of differences between the automatic SHS and the operator configured SHS, the GMS shall uplink the most conservative value with the following order of precedence “In Test”, “Out of Service”, “Will be Out of Service“ and finally “OK” according to the next table

		SHS Manual setting			
		OK	**Will be out of service**	**Out of service**	**Test**
SHS automatic setting	**OK**	OK	Will be out of service	Out of service	Test
	Out of Service	Out of Service	Out of Service	Out of Service	Test
	Test	Test	Test	Test	Test

Specifically for planned events the tasks which lead to an automatic setting of SHS are sent by the GCS and the SHS setting will be according to the nature of the task. The following table provide a list of tasks which can be notified by GCS:

Task Originator	Task	SHS value during planned event
GCS MTP Request	Non-routine payload configuration	Out of service
	Satellite Master clock synchronization	Out of service
	Satellite on-board time of transmission synchronization	Out of service
	Satellite Master Clock Switch	Out of service
	Satellite hot backup to cold backup clock switch	OK
	Orbit manoeuvre	Out of service
	OBSM (requested by GCS)	Out of service
	TTCF Station unavailability	OK
	SAR unavailability	OK
	Satellite unavailability	Out of service
	SIS unavailable	Out of service

Note that for planned events, it is not requested to transmit “Signal Will be out of service” in order to warn the user that the signal will be unavailable in the future.

Navigation data validity status (DVS): the DVS is provided to the user according to the following processes:

a. DVS generation: The DVS can be set by the satellite to “Working without guarantee” in the following cases:

Autonomously when:

· The navigation data batch available at satellite level for dissemination is no more valid. This decision is based on onboard time out counters (SSF3.2) compared to a predefined threshold.

· Or events are detected on board (SSF6.1) which affects navigation service performance (e.g. events generated by the Failure Detection, Isolation and Recovery (FDIR) functionality).

· Or the S-Band telecommand up-link is lost for a predefined period.

Or on reception of dedicated TC via S-band (SFC5.2).

b. DVS dissemination: This flag is disseminated within the PRS navigation messages on the PRS signals E6A and E1A.

The above flags have the same functionalities as the ones described for the OS (see SFC4.1.3).
Within the PRS service two other status indicators are generated by the satellite. These two parameters are provided for system operation and monitoring purposes:

Satellite health status: This includes a status summary and status alert (e.g. Payload Summary FDIR Status, PHM Clock Health Status).

- Satellite Health Status generation: This status flag is generated on board the satellite (SSF6.2) based on internal satellite data.
- Satellite Health Status dissemination: It is disseminated as part of the PRS navigation message (SSF3.2).

C-band link status indicates the connection status of the Galileo channel
C-band link status generation: It is generated onboard the satellite (SSF1.1)
C-band link status dissemination: It is disseminated as part of the PRS navigation message (SSF3.2).

Notes:

- The Satellite Health Status is received by the GMS and send to the GCS for processing and to react in case of error detection. No specific use is foreseen at PRS user level.
- The C-band link status is received by the GMS and sent to the GCS. However no further processing is foreseen.

4.5.3.3.2.1.2 Health Status Flag Usage by the User

The Health Status Flags are used by the PRS user to determine valid satellite range measurements (TUSF6) that can be used for positioning or timing computation.

4.5.3.3.2.2 SFC associated Performance Constraints

Section Deleted

4.5.3.3.2.3 SFC Functional Block Diagram

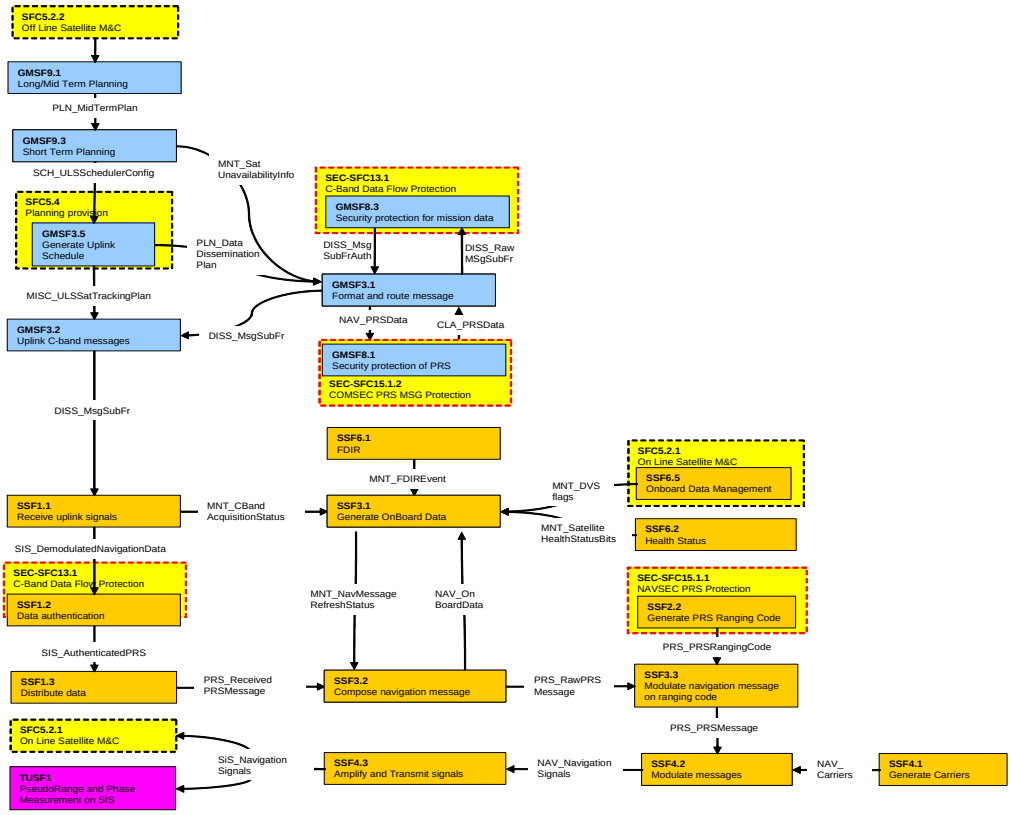


Figure 50 : SFC4.3.3.2 Data Provision Functional Block Diagram

4.5.3.4 SFC4.3.4 PRS Degraded Navigation Data Provision

The functional architecture for this SFC is basically identical to the one described for OS SFC4.1.4 but with:

- additional security functions for encryption of the G/NAV navigation data and
- for the uplink of the security data.

4.5.3.4.1 SFC description

The purpose of this system functional chain is to support the graceful degradation of PRS navigation service. Navigation data and security data (see SFC4.3.6 and related sections) are uplinked to the satellite via the S-band TT&C link and stored on-board the satellite. This data is then available to support a degraded mode of operation for the PRS when no more up-to-date navigation data is available on the satellite (ie via the nominal C-band uplink).

4.5.3.4.1.1 Degraded Data Provision by the System

<u>S-band Degraded data generation</u>

The S-band degraded data is generated by the GMS (GMSF5.6) and sent to the GCS (GCSF4.1) for uplinking to the satellites via the S-band TT&C link. It has exactly the same format and contents as the PRS navigation data described on the SFC4.3.1 but specifically structured (GMSF3.1) to be uplinked on the S-band link.

S-band degraded data dissemination.
In normal operating conditions, the navigation data is uplinked at every TT&C contact with the satellite (ie nominally once per orbit).
The GCS will support the navigation mission services by up-linking the so called “Degraded” Navigation messages as part of nominal contact activity. This is used to back up the nominal message dissemination performed by GMS in C band, as messages uplinked in S-Band will remain valid longer than messages uplinked by GMS in C band. “Degraded” navigation messages can result in less accurate PVT solutions.

S-Band navigation messages are computed by GMS and transferred on line to the GCS within discrete files, to-gether with the indication of their quality, for all Navigation services (i.e. I/NAV and F/NAV messages for Open service, G/NAV message for PRS navigation and G/NAV data 9 for PRS messages). To guarantee the independ-ence of services the GCS will select the best messages per service basis within the files generated by GMS, pre-venting the uplink of invalid messages.

Note: the use of QoS as above explained will be implemented by STRB2.1:
#STRB2.1_change: GAL-DCN-TASI-SYST-X-0863, GAL-DCN-TASI-SYST-X-0810, GAL-DCN-TASI-SYST-X-0809

The satellite stores the navigation data on-board so that it is available for use in case of system degraded conditions. These conditions occur when the time period on which the navigation data batches uplinked through the C-band were optimized is not longer valid. The switch between navigation batches for both C-band and S-band, are managed onboard thanks to the Time-of-Applicability (ToA) parameters associated to each batch (see SFC4.1.1 and SFC4.3.1 for more details).
When used the S-band PRS navigation messages are processed in the same way as the PRS navigation messages data uplinked on the C-band and described in the SFC4.3.1 and SFC4.3.6.

4.5.3.4.1.2 Degraded Data Usage by the User

The usage by the user of the S-band degraded data is the same as described for the SFC4.3.1.

4.5.3.4.2 SFC Functional Block Diagram

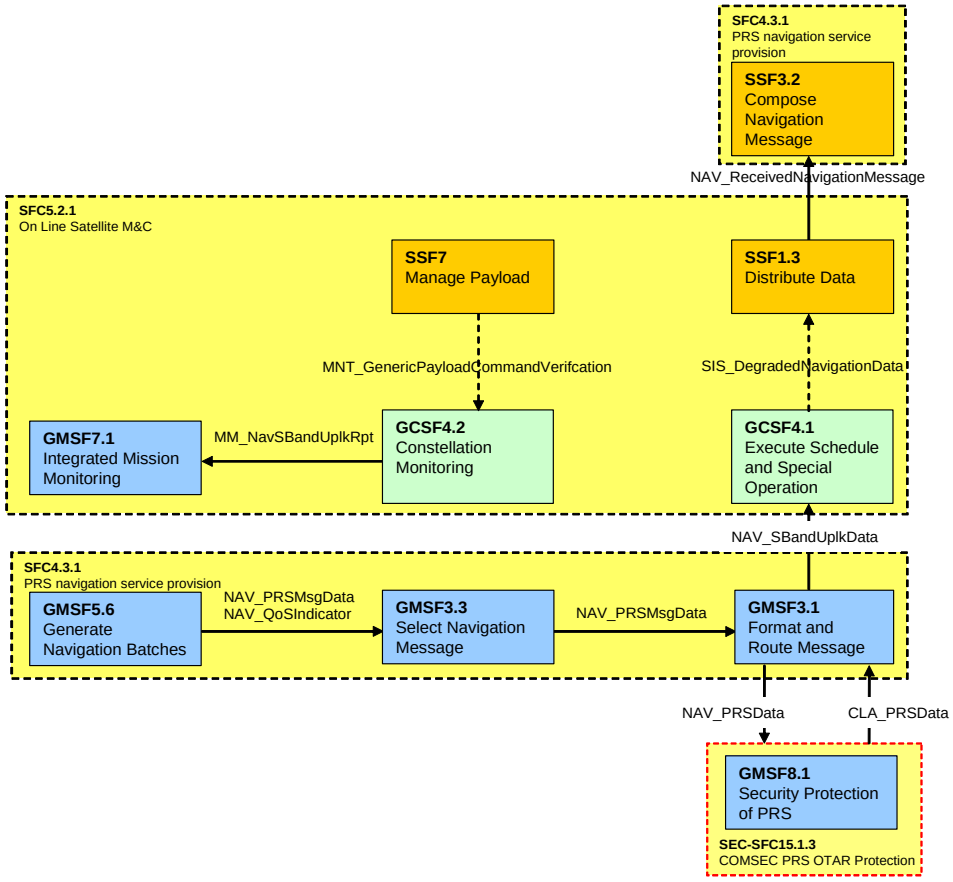


Figure 51 : SFC4.3.4 Data Provision Functional Block Diagram

4.5.3.5 SFC4.3.5 PRS Integrity Service Provision

Section deleted

4.5.3.6 SFC4.3.6 PRS Security Service Provision (Uncla)

4.5.3.6.1 SFC description

The PRS service includes the generation and provision of security data also called G/NAV-9. The generation and the contents of the security data is managed by GMSF8.1 and described in the SEC-SFC-13.

4.5.3.6.1.1 Security Service Provision by the System

The main functionalities of this SFC are as follows:

PRS security messages generation: The PRS security messages are generated by GMS (GMSF8.1) as a batch of data. The generation process is described in the SEC-SFC-13.

PRS security data uplink: The PRS security data is uplinked through the C-band (see SFC5.3.2) together with the PRS navigation messages.

PRS security data dissemination: The PRS security data is stored onboard the satellite (SSF1.3) together with an associated Time-of-Applicability (ToA). It is downlinked on the SIS (SSF3.2) as soon its applicability starts.

Notes:

- The PRS security data is also uplinked through the S-band TT&C link. This process is described in the SFC4.3.4.
- The PRS security data has a total variable batch size in number of messages depending on operational security needs.

4.5.3.6.1.2 Security Service Computation by the User

The PRS security service computation by the user is described in SEC-SFC15.1.2 (COMSEC PRS MSG Protection).

4.5.3.6.2 SFC Functional Block Diagram

The block diagram described below shows the dissemination process of the PRS security messages data through the C-band link.

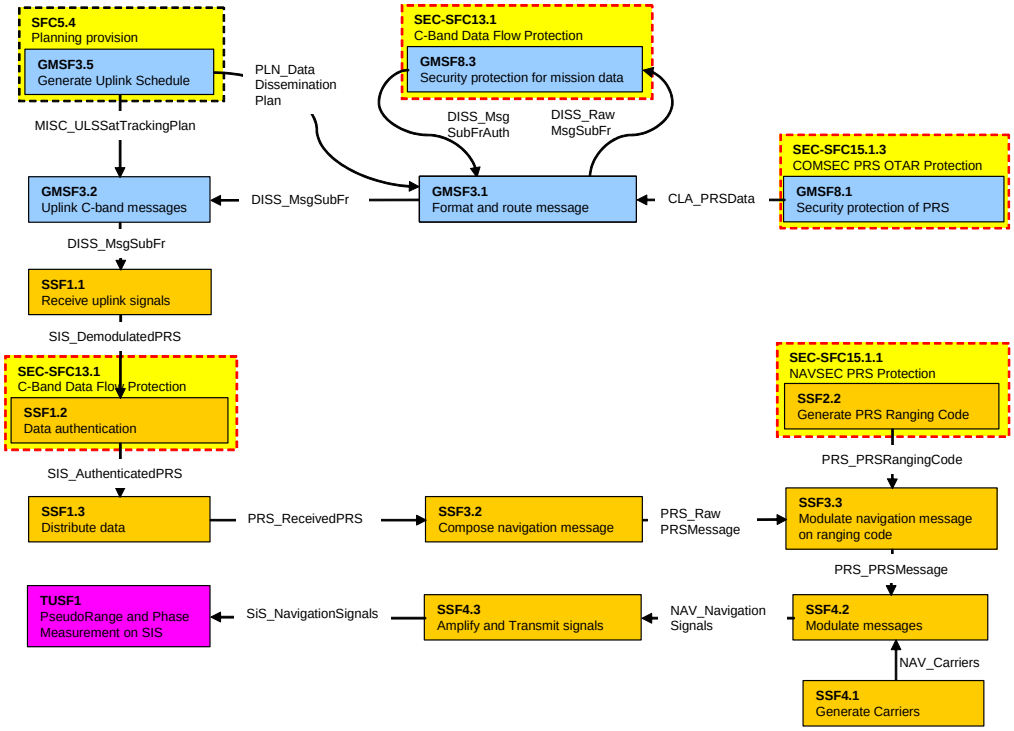


Figure 52 : SFC4.3.6 Functional Block Diagram

4.6 SFC5 Monitoring And Control (Ground, Satellite, Mission and Planning)

The Service Functional Chain 5 is in charge of providing all the means needed to perform the monitoring and control, and planning of the Galileo system. This includes ground assets M&C (including network), satellite M&C, Mission monitoring, and planning provision

The first level functional tree for the “Monitoring and Control” SFC is then as shown below:

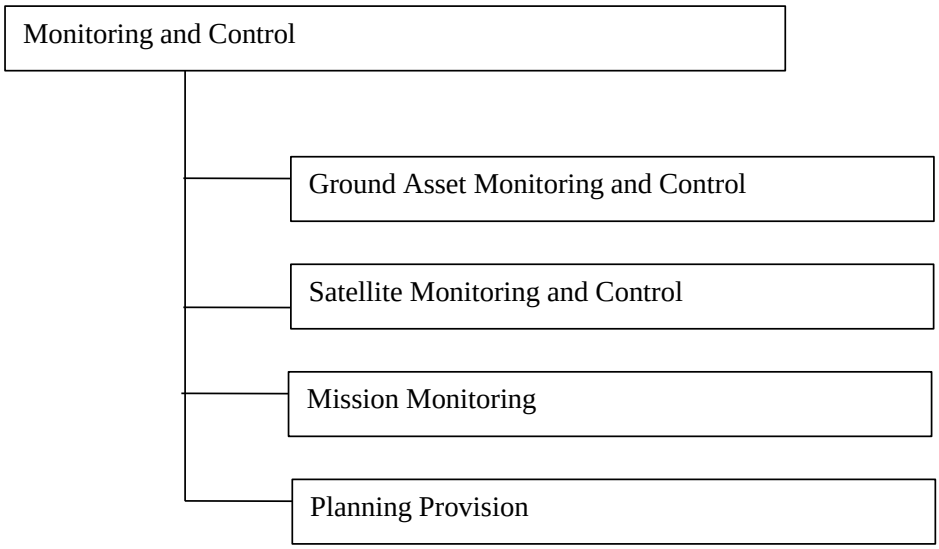


Figure 53: SFC5 Functional tree (1st level)

4.6.1 SFC5.1 Ground Assets Monitoring and Control

4.6.1.1 Introduction

The “Ground Assets Monitoring and Control” functional chain is part of the global mission monitoring functional chain. It gathers all the functions required to perform the monitoring and control of all the Galileo ground assets and uses resources of both ground segments, GMS and GCS.

This functional chain is mainly in charge of the following functions:

- allow the operator to routinely monitor the real time status of all the Galileo ground assets down to equipment (LRU) and software (SWRU) level.
- allow the operator to routinely perform technical control of all Galileo ground assets, including asset mode changes and configuration changes
- execute the schedules derived from the short term plans
- allow the operator to support ground segment contingencies by detecting and if possible diagnosing any possible failure and either recovering the assets directly using the control capability or configuring the Galileo ground assets for corrective maintenance

The Galileo ground assets monitoring and control function is organized in different operational levels:

- Level 0: For maintenance, via local dedicated MMI in each element,
- Level 1: For maintenance, via remote centralised MMI located at GCC,
- Level 2: For allowing the capability to monitor and control the GMS during normal operations, via remote centralised MMI located at GCC,

Note that a level-3 for monitoring and control operation is also specified. This level-3 allows the production of corrective and preventive maintenance planning to be executed by operations performed at level-2. This level-3 corresponds to integrated mission monitoring, as described in a dedicated SFC (SFC5.3).

In order to implement these operational levels, the following sub-functions are defined:
Ground Element monitoring and control, located inside each Galileo ground elements. This sub-function allows to perform:

- dedicated element level-0 operations in a stand alone mode, directly by local operators, through a local interface
- interface for dedicated element remote controlled level-1 and level-2 operations, from central monitoring and control

Central monitoring and control of the ground segments. This sub-function allows to perform at each segment level:

- level-1 operations for the corresponding elements within each segment, using remote interface and centralized resources
- interface for level-2 operations for the corresponding elements within each segment, from the overall monitoring and control function

Overall monitoring and control of the ground segments. This sub-function allows to perform level 2 operations, to interact with operators and to:

- execute predefined automatic schedule
- ensure high level control of elements
- ensure high level monitoring of the facilities
- follow operations progress
- grant the unique control of elements

This SFC interfaces with the following SFCs:

- planning provision (SFC5.4),
- mission monitoring (SFC5.3),
- Mission data provision (SFC4)

As shown below, the SFC5.1 is split into two sub-SFCs, each one dedicated to one aspect of the ground assets monitoring and control system functional chain:

- the first sub-SFC deals with ground elements M&C
- the second sub-SFC deals with ground network M&C, here called ground network management

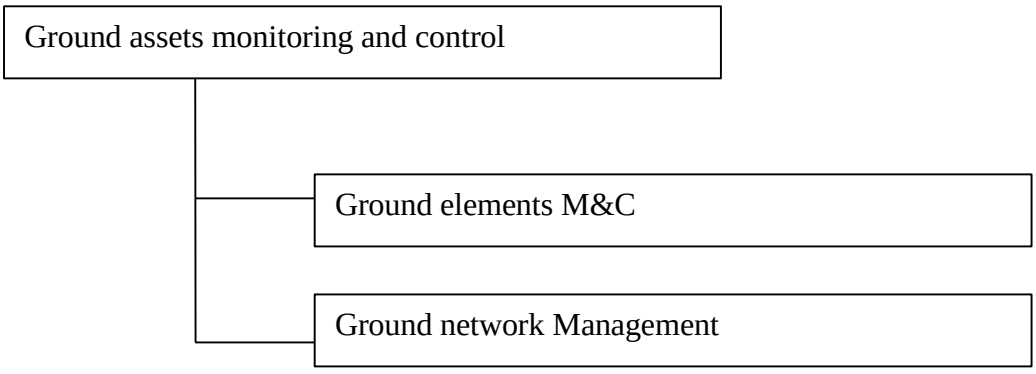


Figure 54 : SFC5.1 functional tree (2nd level)

4.6.1.2 SFC5.1.1 Ground Elements Monitoring and Control

4.6.1.2.1 SFC description

This SFC gathers all the segment functions necessary to perform the monitoring and control of all the Galileo ground elements, located both in the remote sites (TTCF stations, ULS and GSS) and in the GCC.

Ground elements monitoring and control consists in:

- monitoring the statuses of all elements to allow operators to:

  be aware of the health and configuration of these elements
  follow the correct execution of the ground segment operations scheduled from the short term planning
  perform operational analyses and reports

- controlling those elements in order to change their configuration as scheduled from the short term planning instructions or in case of element/equipment failure.

raising alarms in case of fault / failure of elements or part of elements
automatic and/or manual switching to internal redundant units in case of failure of an element / component
gathering hosting site information, including infrastructure alarms and meteorological data

- monitoring of remote GMS internal and external installations.

  This shall include at least:

- Monitoring of temperature, smoke/fire and access control alarms for the internal installations
- Monitoring of pressure, temperature, humidity, wind and rain for the external installations by a meteo sensor.
- Security monitoring e.g.,
- Local Rack and equipment Physical Security
- Local Rack Door Alarm
- Local Access Door Alarm
- Local Room/Area Intrusion Sensors

This SFC interfaces with the operator according to the following scheme:

- Elements' statuses and alarms are gathered at element level by the element M&C function (GMSF7.4.1 and GCSF5.1) and transferred to the central M&C function at segment level (GMSF7.4.2 and GCSF5.2). This latter function in turn transfers them to the overall M&C function (GMSF7.4.3), which is in charge to present them to the operators through relevant MMI.

Operator's or schedule execution macro-commands follow the inverse way, from overall M&C function to element M&C function

This SFC also interfaces with the following SFCs:

- GMS short term planning function (part of the SFC 5.4 - planning provision) for execution of all the GMS tasks included in the GMS schedules

GCS short term planning function (part of the SFC 5.4 - planning provision) for execution of all the GCS tasks included in the GCS schedules
Integrated mission monitoring to include all ground elements' monitoring information into the overall mission monitoring for level 3 operations (part of the SFC 5.3 - Mission performance monitoring)
Satellite monitoring and control to give this SFC the information that the back-up contact plan has been activated instead of the nominal one (SFC5.2)
Mission data provision (SFC 4)

4.6.1.2.2 SFC Functional Block Diagram

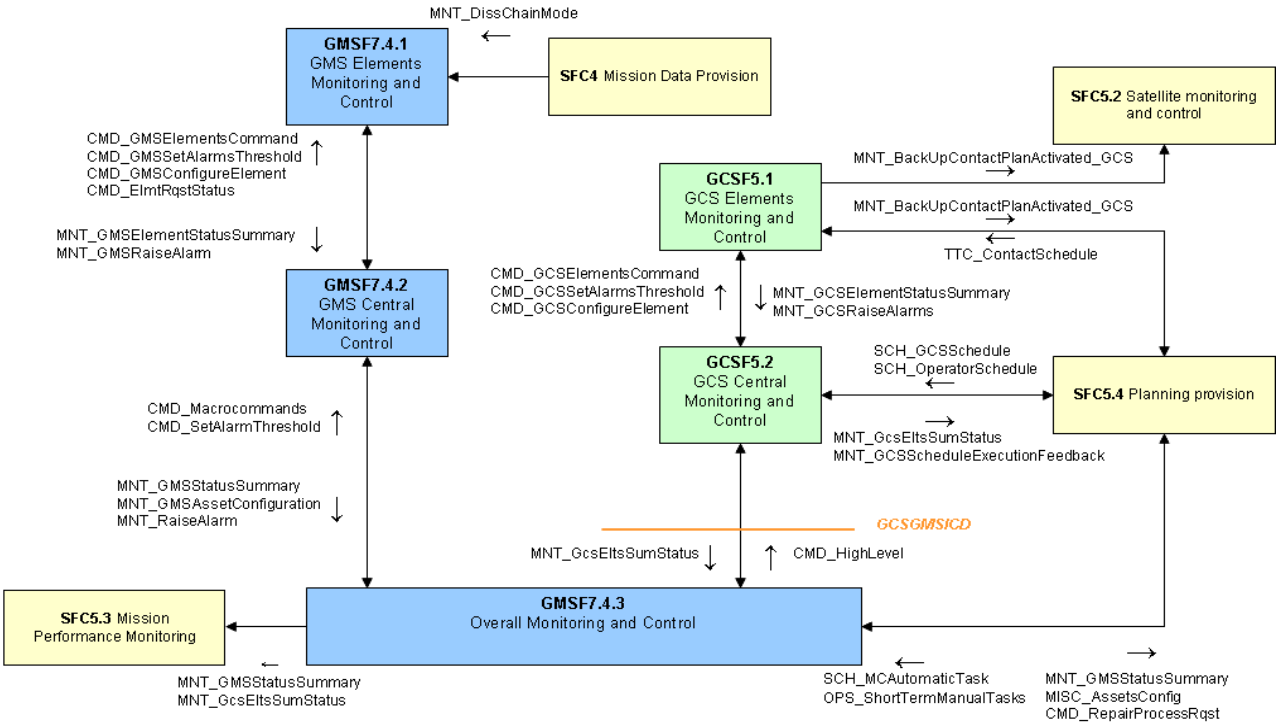


Figure 55 : SFC5.1.1 functional block diagram

4.6.1.3 SFC5.1.2 Ground Network Management

4.6.1.3.1 SFC description

This SFC includes all the segment functions necessary to perform the management, including monitoring and control, of the Galileo network elements (excluded CSP equipment of GDDN), located both in the remote sites and in the GCC.

The network management function consists in:

- network control, i.e. controlling the network to activate redundancy, manage priorities between data flows, SW updates.
- network equipment monitoring, i.e. gathering the statuses of all the network components to allow operators to be aware of their health and configuration
- network traffic monitoring for performance assessment. The following network performances are continuously computed: latency, integrity, continuity availability
- provide GDDN network performance computation to Communication Service Provider for communication service assessment

Network management is performed according to the following scheme:

Network equipment statuses and alarms are generated and gathered inside each segment by the GMS and GCS network M&C functions (GMSF7.4.4 and GCSF5.3).

These functions are also in charge of network performance measurement through traffic probes.
This information from both GMS and GCS (network equipment monitoring as well as network performance measurements) are sent to the network monitoring and control function, which is included

in the GMS (GMSF7.4.4). This function is in charge to assess the network performance, namely latency, integrity, continuity and availability. It interfaces to the operator through generation of alarms and warnings, reports, displays.

Function GMSF7.4.4 is in charge to send the relevant information to the CSP function CSPF1 for communication service fulfilment and quality assessment (this is performed through GMS function GMSF12, in charge of the management of the interface with external entities).

Network equipment monitoring information is then sent to the overall monitoring and control function (GMSF7.4.3), whereas performance assessment as well as network summary status information are passed to integrated mission monitoring function (GMSF7.1).

This SFC also interfaces with the following SFCs:

- GMS short term planning function (part of the SFC 5.4 - planning provision) for execution of all the GMS tasks included in the GMS schedules
- Integrated mission monitoring to include all ground elements' monitoring information into the overall mission monitoring for level 3 operations (part of the SFC 5.3 - Mission performance monitoring)

4.6.1.3.2 SFC Functional Block Diagram

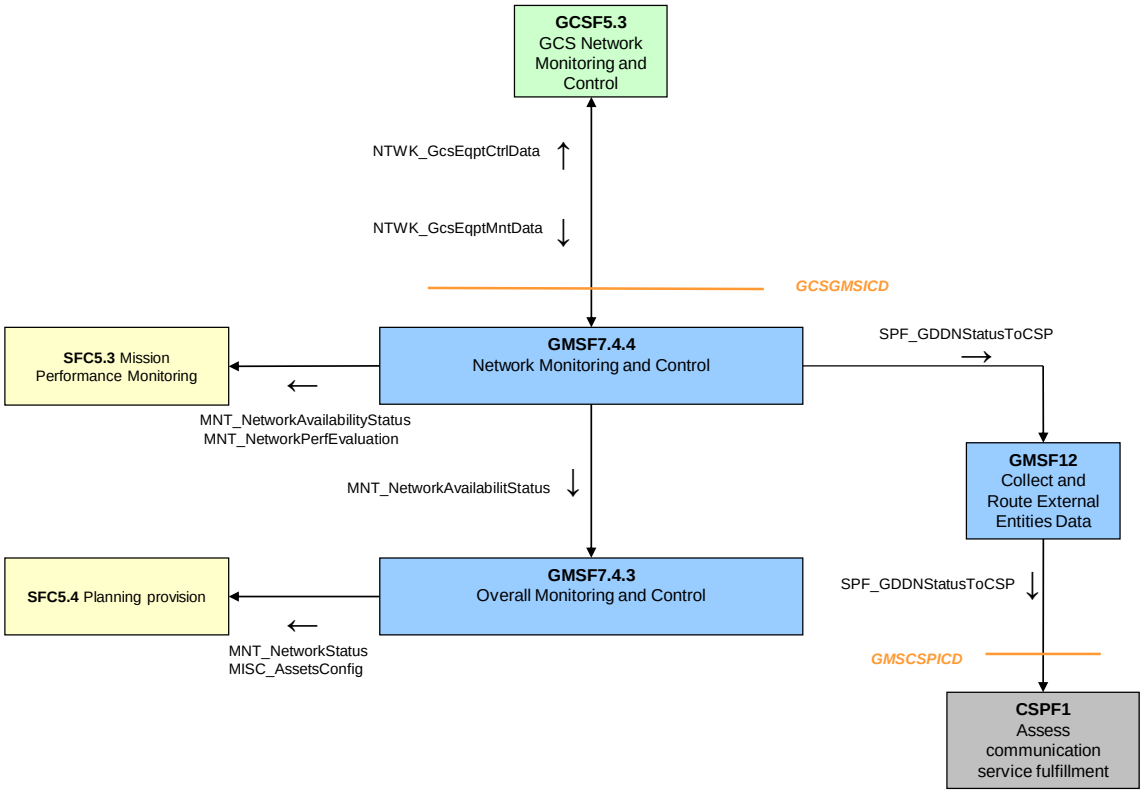


Figure 56 : SFC5.1.2 functional block diagram

4.6.2 SFC5.2 Satellite Monitoring and Control

4.6.2.1 Introduction

4.6.2.1.1 General Description

The satellite monitoring and control functions can be logically broken down into on-line and off-line functions as follows:

On-line satellite M&C:
These activities relate to the monitoring and control functions performed through direct interaction between the ground and the satellite to ensure the correct operation of the satellites. This includes monitoring and control via S-band (nominally once per orbit) and continuous satellite health status monitoring via L-band.
On-line satellite M&C encompasses both routine and special operations, with the latter including active responses to anomalies or events which could have a detrimental impact the operation of the system (ie contingencies). Note that in general the operation of the satellites, and the associated monitoring and control functions, will be fully automated. Exceptions to this are where manual intervention by the operator is required for operational reasons (eg for contingency cases).
The M&C functions addressed through the on-line satellite M&C functionality include the following (not exhaustive):
Transmission of individual spacecraft commands and/or command schedules (nominally once per orbit via S-band);

- Reception of spacecraft telemetry (nominally once per orbit via S-band and continuously via L-band);
- Management of on-board FDIR functionality (eg enabling / disabling FDIR functions such as autonomous entry into safe mode, redundancy switching, setting of associated parameters and thresholds etc);
- Management of the on-board telemetry packet stores (eg dumping to ground or deletion of a store)
- Controlling the operational mode of the S-band link (eg Spread spectrum or ESA standard mode)
- Execution of on-board software exchanges (eg requesting memory downloads, executing software uploads etc)

Off line satellite M&C:
These activities relate to system performance assessments, trend analyses and parameter tuning that are performed based on analysis of data retrieved previously from spacecraft constellation (and other sources) and stored in the monitoring and control archive.
Off-line satellite M&C activities encompass the following functions (not exhaustive):

- Analyses of telemetry data received from all satellite sub-systems (eg anomaly investigation, consistency, trends etc).
- Interaction with the Satellite Manufacturer (SATMAN - for on-board software updates, anomaly investigation etc)
- Orbit determination and analysis (eg for any deviations from predicted orbital position)
- Predictions of satellite events for input to long and short term planning
- Definition of manoeuvres required to maintain the satellite orbit
- Simulations performed in support of the above activities

The first level functional tree for the “Satellite Monitoring and Control” SFC is therefore shown below.

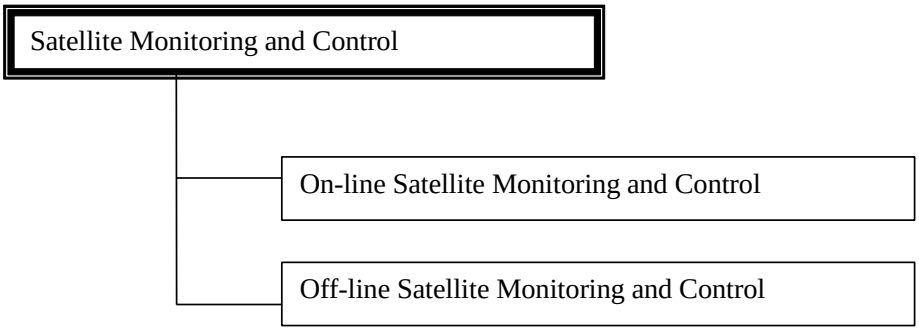


Figure 57 : SFC5.2 Functional Tree (2nd level)

Note that all functions related to monitoring of mission performances at satellite level (both on-line and off-line) are covered in detail within SFC5.3 (Mission Monitoring).

4.6.2.1.2 Definitions

Routine Operations — Refers to all nominal housekeeping operations carried out to maintain the system in its nominal state.

Special Operations — Refers to those operations that cannot be strictly included as part of standard Routine Operations. Specific examples of such operations are:

- LEOP and IOT operations,
- On board Software Management (OBSM) operations,
- S/C orbit manoeuvre operations,
- Contingency operations (related to failures in spacecraft constellation or ground segment),

Special operations are expected to occur rarely and to extend only over short periods.

CLEAR Mode — In the context of the S-band TT&C link, CLEAR mode refers to the link security mode where no encryption is applied at the transfer frame level.

PROTECTED Mode — In the context of the S-band TT&C link, PROTECTED mode refers to the link security mode where encryption is applied at the transfer frame level. This encryption is applied to all telecommands and telemetry sent between the ground and the spacecraft (including secure telecommand and secure telemetry).

ESA Standard Mode — ESA Standard Mode describes a Radio Frequency (RF) TT&C link mode of operation based upon the use of phased modulation (PM). A more detailed definition is provided in [AD/9]. This mode of operation is not intended for use during routine operations and will be used primarily during special operations (particularly during LEOP / IOT and contingency operations).

Spread Spectrum Mode — Spread Spectrum Mode describes a RF TT&C link mode of operation based upon the use of Direct Sequence Spread Spectrum technique (see [AD/9] for details). This mode of

operation is intended to be used during routine operation of the S-band link.

4.6.2.2 SFC5.2.1 On-line Satellite Monitoring and Control

4.6.2.2.1 SFC description

The purpose of this system functional chain is to provide monitoring and control functionality to ensure the correct operation of the Galileo satellites. Satellite M&C functionality is supported in the following ways:

Intermittently during S-band TT&C contact windows (nominally once per satellite orbit) where full monitoring and control functionality will be available.

Continuously via L-band navigation service signals. Satellite M&C data are included in the encrypted G/NAV message and concerns:

- C-band uplink channel status.
- Satellite health status. This comprises a limited set of telemetry data and FDIR alerts.

The above data are transmitted as soon as the payload is switched on, when dummy G/NAV messages are transmitted before the PRS service is operational.
Whereas the C-band uplink channel status is processed and displayed at GMS level, Satellite health status data are collected by the GMS and the forwarded them to the GCS for Monitoring purposes.

In general satellite M&C performed via the TT&C link will be entirely automated thereby minimising the need for real-time operator intervention. In this way the operators will be able to focus on non-routine events (e.g. on-board failures etc - see below).

This use of automation encompasses both:

- Commanding of the satellite - where all routine control of the satellite will be achieved through the use of pre-defined and carefully planned command schedules. The spacecraft design ensures that the on-board reception, processing and distribution rate can cope with the maximum TC uplink rate, without affecting the performances of other on-board software processes. The spacecraft has the capability of producing telemetry that allows the ground to unambiguously determine the spacecraft configuration; this information is used by the ground to determine the correct telecommand to be uplinked, in case of configuration-dependent telecommands.
- Monitoring of the satellite - where dissemination and processing of all telemetry (sent via both S-band and L-band) will be automated (e.g. limit checking, command verification, archiving etc).

The use of automation is specified to ensure that the M&C system is able to process autonomously TC and TM data without human intervention during routine operations. This is achieved through:

- Automated execution of planned satellite M&C activities through the use of validated operational procedures and the uplinking of pre-defined and verified command schedules. The command schedules will include time-tagged commands, uplinked during a contact period and executed during the non-visibility period. The on-board time-tag buffer is sized to allow routine operations to be conducted with one ground contact every two orbits (worst case)

The planned satellite activities are provided to this SFC as part of the GCS Short Term Plan (STP) output from the planning functions (see SFC5.4).The autonomy designed into the Galileo satellites which provides the capability to ensure no degradation below the nominal spacecraft performance even if no valid telecommand is received for two orbital periods (assuming no failure on-board).

The autonomy designed into the Galileo satellites which ensures that the satellite will perform autonomously on-board all AOCS controls and nominal manoeuvres battery charge, thermal control and system Failure Detection Isolation & Recovery Guarantees spacecraft survival, without irrevocable consequence, even if no valid telecommand is received for two orbits (assuming any single failure on-board).
Note that the spacecraft is designed to remain in Safe Mode until appropriate recovery commands are sent from the ground.

This functional chain is used to provide the following observability functions for routine satellite M&C:

- Reception of telemetry downloads (received via both S-band and L-band) regarding the health status of the satellite and all on-board units.
- Setting and reporting of all parameters used within the scope of the autonomous on-board operational capability including AOCS, FDIR and satellite mode management (e.g. alarm thresholds, attitude biases etc.). Note that the spacecraft is designed so that for every Register-Load telecommand, a corresponding Execute telecommand is defined, in order to allow read-back of uploaded critical data before allowing the spacecraft to use them.
- Archiving (in on-board packet stores) of all satellite telemetry generated outside ground TT&C contact and download on TC request from ground.
- Note that management of the on-board TM packet store is discussed in more detail in SFC6 (Archiving).
- Presentation of anomalies and alarms indicated via telemetry to the operators of the system.
- Verification of the acceptance and execution of all telecommands sent to each satellite via dedicated telemetry. The spacecraft design ensures that verification telemetry is available for all telecommands that cause a spacecraft configuration change, at most 3 seconds after telecommand execution completion. On the other hand, appropriate telemetry will be donwlinked indicating failure of correct telecommand execution.

Spacecraft maintenance tasks or telecommanding included in the GCS Short Term Plan (STP) or generated in response to an observed event are also supported by this functional chain. This includes the following maintenance tasks in particular:

- On-board clock management, including regular (e.g. every year) cold backup PHM maintenance as specified in SFC3.1.3.
- Other platform and payload configuration tasks (note that for every multi-data telecommand controlling equipment configuration is individually verifiable via the relevant equipment telemetry parameters)
- On-board software uploads and dumps
- Execution and evaluation of satellite attitude and orbit manoeuvres

This SFC also provides functionality in support of handling contingencies that occur during routine satellite M&C (e.g. due to un-planned or un-foreseen events). This includes the following functions:

- Re-planning to make use of spare time available in the existing contact window or in the contingency contact window provided in the short term plan.
- Implementation of contingency operational procedures (e.g. including direct manual commanding of the spacecraft). For example this enables investigative or recovery actions in response to anomalies identified during routine satellite M&C activities (see above).
- Note that the operational concept for the system limits this type of activity to the implementation of pre-defined (and well rehearsed) operational procedures in response to specific alarms (e.g. requesting diagnostic TM reports).
- Ground system capability to support special operations with up to 4 satellites in parallel, whilst ensuring that the execution of a special operation scenario does not affect routine satellite M&C.

- Automated actions executed on-board the satellite (ie FDIR) in response to anomalies, failures and errors and for which ground intervention is not required.
- Provision of telemetry by the satellite to help define the context of any autonomous recovery actions taken on-board and analysis on ground (e.g. event triggers, related event history etc).
- The FDIR algorithms provided on the satellite are designed to provide all support telemetry packets considered necessary for the ground analysis of the failure.

The system also supports the download of a pre-defined minimum set of telemetry packets to provide an unambiguous identification of the entered mode.

Note that all on-board fault management functions can be enabled and disabled from the ground.
In support of all of the functionality described above, the system will provide S-band TT&C up-links and downlinks in accordance with the following constraints:

- Support for 2 operational TTC modes (independent of whether Clear or Protected security is enabled):
  - Spread Spectrum
  - ESA Standard Mode
- The capability to set TTC mode by ground command or autonomously on-board by FDIR functionality
- Support for the following fixed frequencies when operating in spread spectrum mode:
  - A telemetry frequency of 2,225.025000 MHz in spread spectrum mode
  - A telecommand frequency of 2,048.877188 MHz in spread spectrum mode
- Support for the following frequency flexibility when operating in ESA standard mode:

Channel Number	TM Frequency (MHz)	TC Frequency (MHz)
1	2212.749000	2037.573038
2	2215.818000	2040.399075
3	2218.887000	2043.225113
4	2221.956000	2046.051150
5	2225.025000	2048.877188
6	2228.094000	2051.703225
7	2231.163000	2054.529263
8	2234.232000	2057.355300
9	2237.301000	2060.181338

- Support for two way ranging using the telecommand and telemetry frequencies quoted above
  - The satellites provide the capability to perform ranging using both Code Ranging and Tone Ranging techniques when operating in ESA Standard mode. This will enable the use of commercially available (ie external) TT&C services during initial operations (ie LEOP and IOT) and also for special operations. Depending on the Ground Station capability, two different tone frequencies can be used: 100 KHz or 400 KHz.
  - Note that the Galileo Ground Segment (GCS) is only able to support ESA Standard mode with code ranging (not tone ranging).
- Support for ranging in spread spectrum mode.
  - When performing ranging the spacecraft TT&C transponder will be configured to operate in coherent mode
  - The TT&C uplinks and downlinks will be able to be transmitted using either ‘right-hand’ or ‘left-hand’ circular polarization (ie RHCP or LHCP).

The spacecraft design ensures that the following functions/features are in accordance with the Ground Control Segment to Space Segment ICD ([AD/9]):

- Communication between GCS and spacecraft
- Telecommands packetization
- Reception and processing of a continuous uplink of Telecommands
- TC and TM services
- APID definition

Moreover, telecommands destined to different payloads and subsystems, will be distinguished by their APID, which will be the same as the APID of the telemetries from those subsystems. The APIDs are defined in the design phase and can only be changed in case of contingency.

In support of the secure operation of the Galileo system, the TT&C system will also provide security protection of both the telecommand and telemetry links. This protection will be applied at all times when the satellite is transmitting navigation signals.

This includes the following features:

CLEAR mode provides authentication but not encryption. This mode can be used during LEOP/IOT and also special operations. Note that this mode of operation is only compatible with ESA Standard RF mode.

PROTECTED mode provides authentication and encryption. This mode is used for nominal operations.

Note that this security functionality is described in detail in SEC-SFC-12.1 (TT&C Data Flow Protection), which can be considered as providing security functionality in support of this SFC.

(#STRB2.0_change: GAL-DCN-TASI-SYST-X-1009) The Payload and Platform Security function transmits its unencrypted health status irrespective of the security unit operational mode (clear mode/protected mode).

As described above, the TTC link provides 2 security modes and 2 RF modes. The table below summarises the relationship between these modes:

		Security Mode	
		PROTECTED	**CLEAR**
RF Mode	**Spread Spectrum**	Yes	Not Allowed
	ESA Standard	Yes	Yes

Note.The following system functionality is not fully covered here as they are the subject of separate dedicated SFCs:

- Off-line satellite M&C analysis functions (covered by SFC5.2.2).
- Planning functions (covered by SFC5.4)
- Uplinking of navigation data for degraded mode operations (see SFC4.1.4 and SFC4.3.4)
- Security functions (see SEC-SFCs 12, 13, and 15)

4.6.2.2.2 Evolution of OBSW to support future NSGU ICD related changes

The functionality foreseen in STRB 2.0 will allow the ground to support the NSGU ICD evolutions maintaining the constellation operational, without interruption of the service provided to the user.
#STRB2.0_change: GAL-DCN-TASI-SYST-X-00872

A change to a NSGU ICD will imply that new software versions are developed both for the on ground elements and for the on board NSGU. To preserve the service provision, the NSGU OBSW patch operation has to be performed one satellite at a time, as this is a mission impact operations during which the affected satellite will be not able to support the service.
A certain time (several days or weeks depending on GCS and Operations constraints) is needed to complete the OBSW patch operations for the entire constellation.
In the transient phase in which NSGU patch operations are ongoing, two sets of satellite are identified:

1. Satellites that have been already patched
2. Satellites that are still to be patched.

To keep the service operational, a new functionality is introduced on ground to support the generation of two different Navigation messages for both the above sets of satellites, allowing the operations team to keep changing at each iteration the set of satellites supporting the different Navigation message.

4.6.2.2.3 SFC Functional Block Diagram

The block diagram below summarises the functional elements involved in the on-line monitoring and control of the Galileo satellites:
Note that for the sake of clarity:

- The signals exchanged between on-board functions are further described in 1st Figure;
- the TCs specifically required in the system specifications to be exchanged between ground and space (represented in red) are fully detailed in 2nd Figure;
- the TMs specifically required in the system specifications to be exchanged between ground and space (represented in blue) are fully detailed in 3rd Figure;
- the interactions between this SFC and other system SFCs are summarised in last figure.

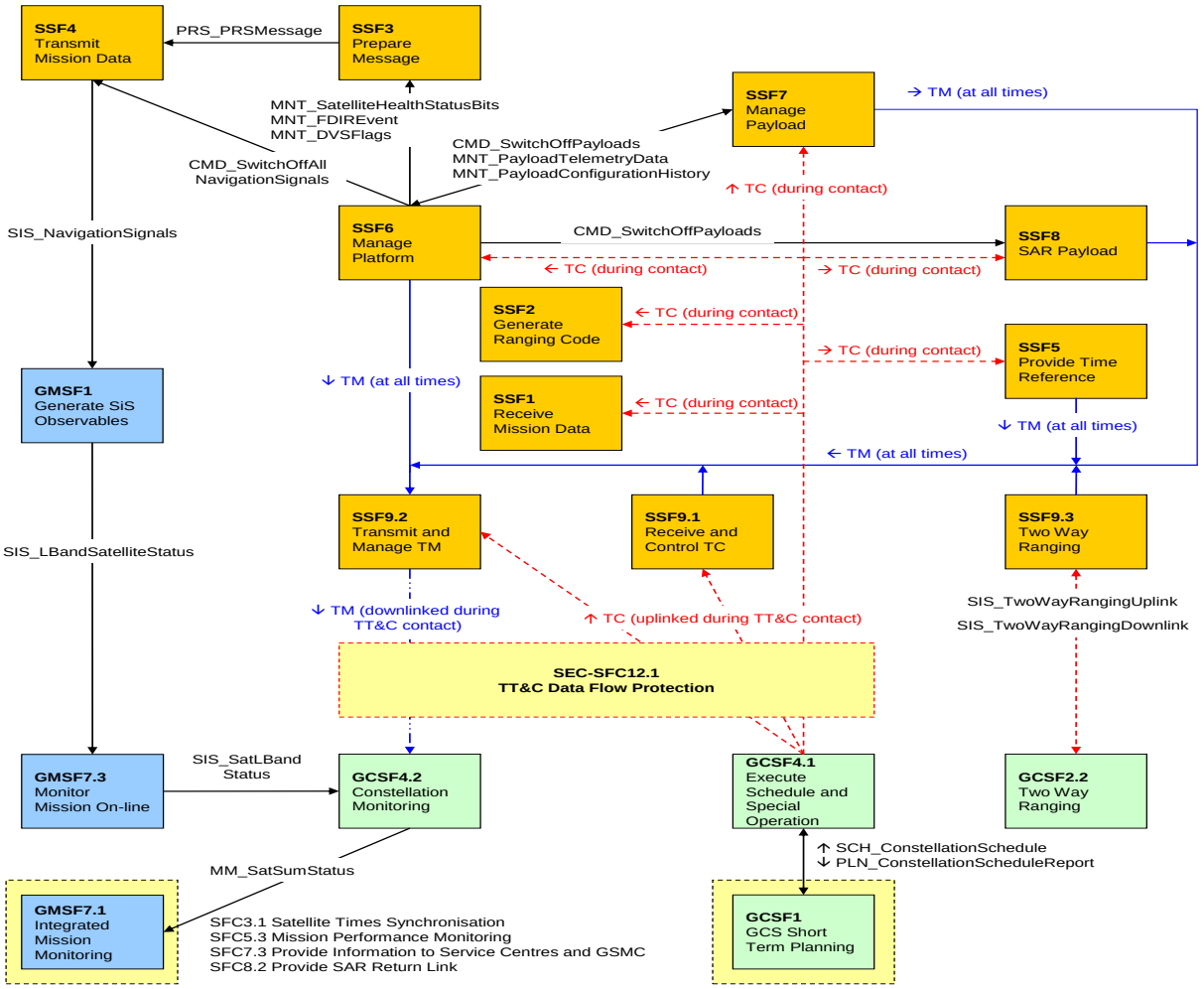


Figure 58 : SFC5.2.1 Functional Block Diagram

The figure below provides detail of the signals exchanged between the on-board functions in support of on-line satellite M&C. Note that this diagram does not include the telemetry signals that are continuously generated on-board (this is covered in figure below titled "Detail of telemetry generated on-board and downlinked to ground during TTC contact"). All of the signals shown can be exchanged at anytime (ie in or out of TTC contact) during nominal operation of the spacecraft.

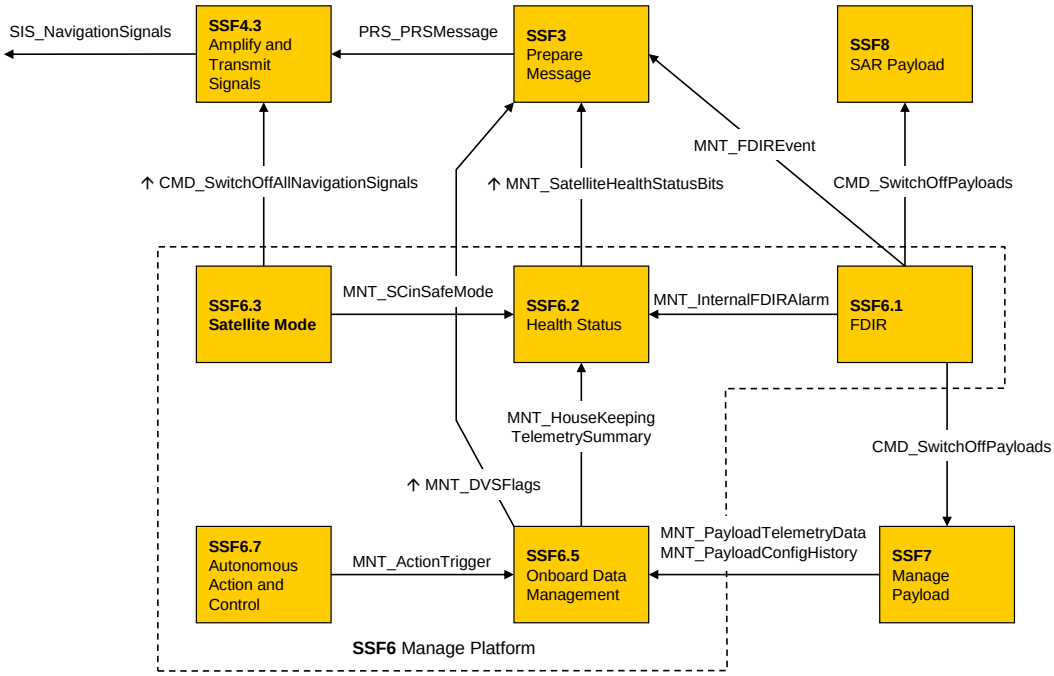


Figure 59 : Detail of on-board exchanges in support of on-line satellite M&C

The figure below provides detail of the telecommands that can be up-linked as part of the command schedule (or manually) during a TTC contact. Note that this figure does not include those commands related to the security functions on-board the spacecraft (covered in SEC-SFCs 12, 13 and 15). Note also that the commands shown are not exhaustive but are those for which there is direct traceability to system, segment or system interface requirements (see signal database).

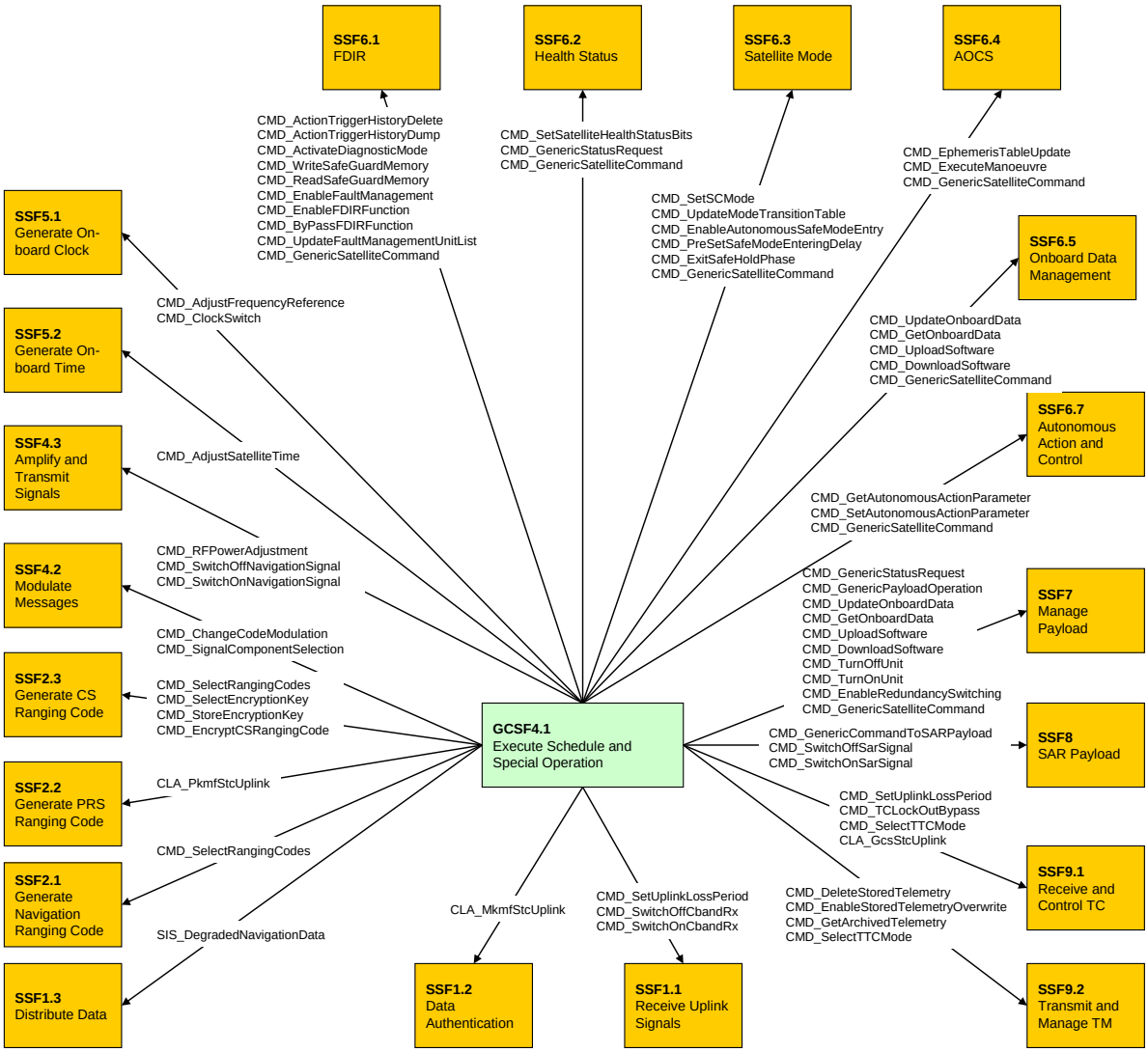


Figure 60 : Detail of commands uplinked to satellite during TTC contact

The figure below provides detail of the telemetry generated by each of the on-board functions in support of satellite M&C. As is illustrated in the figure, this telemetry is generated continuously and is archived on-board by the SSF9.2 Transmit and Manage TM function. However, during a TTC contact this telemetry is additionally downlinked to the ground. As the figure illustrates, dumps of archived telemetry may also be downlinked during a TTC contact. Note also that the telemetry signals shown are not exhaustive but are those for which there is direct traceability to system, segment or system interface requirements (see signal database).

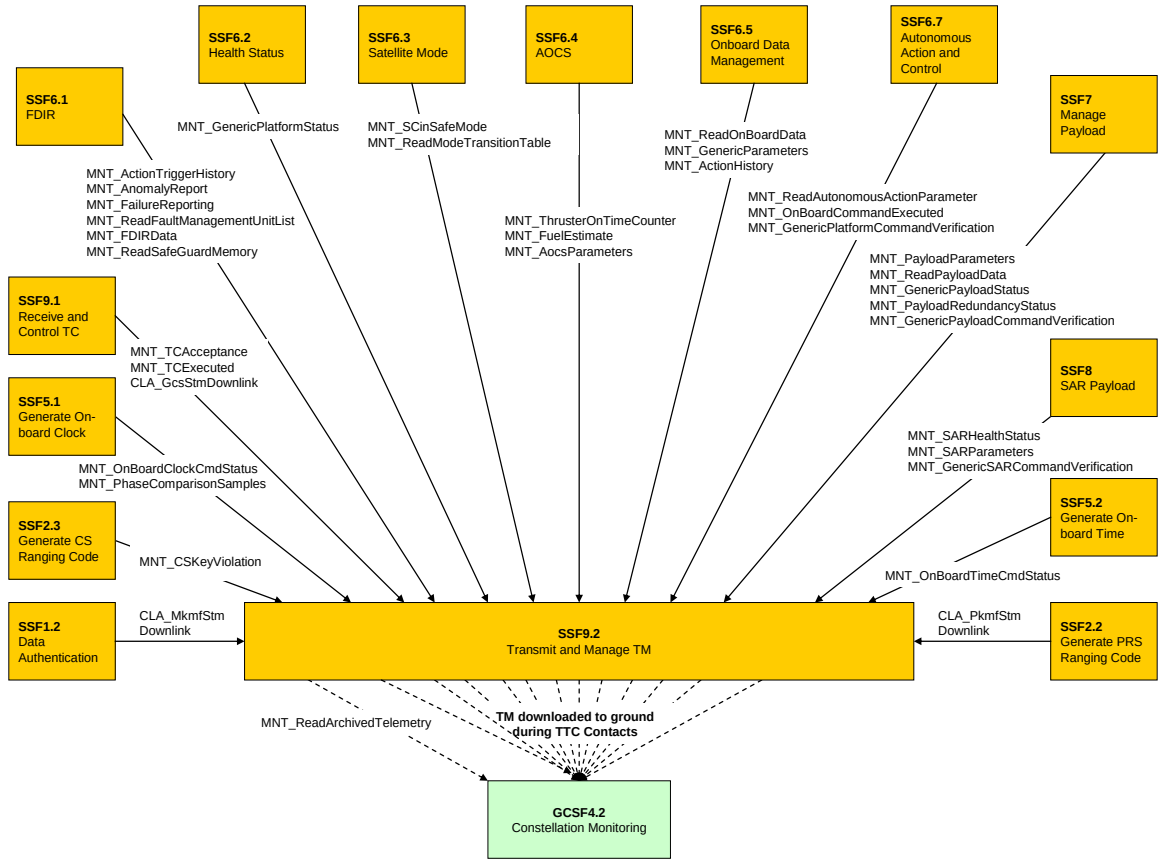


Figure 61 : Detail of telemetry generated on-board and downlinked to ground during TTC contact

As illustrated in Figure 6, this SFC also provides support to the following system functional chains through the uplinking and downlinking of telecommands and telemetry associated with:
Implementation of the constellation schedule as defined in the GCS Short Term Plan (SFC5.4)

- Navigation data uplink for degraded mode operations:
  - Uplink of navigation data for OS (SFC4.1.4)
  - Uplink of navigation data for PRS (SFC4.3.4)
- Support to commercial service (SFC7.2)
- Platform security unit monitoring and control:
  - Uplink / downlink of STC / STM to PFSU (for IOV satellites) or CSU (for FOC satellites) (SEC-SFC12.2)
  - Payload security unit monitoring and control:
    - Uplink / downlink of STC / STM to PLSU (for IOV satellites) or CSU (for FOC satellites) - mission part (SEC-SFC13.2)
    - Uplink / downlink of STC / STM to PLSU (for IOV satellites) or CSU (for FOC satellites)- PRS part (SEC-SFC15.4)

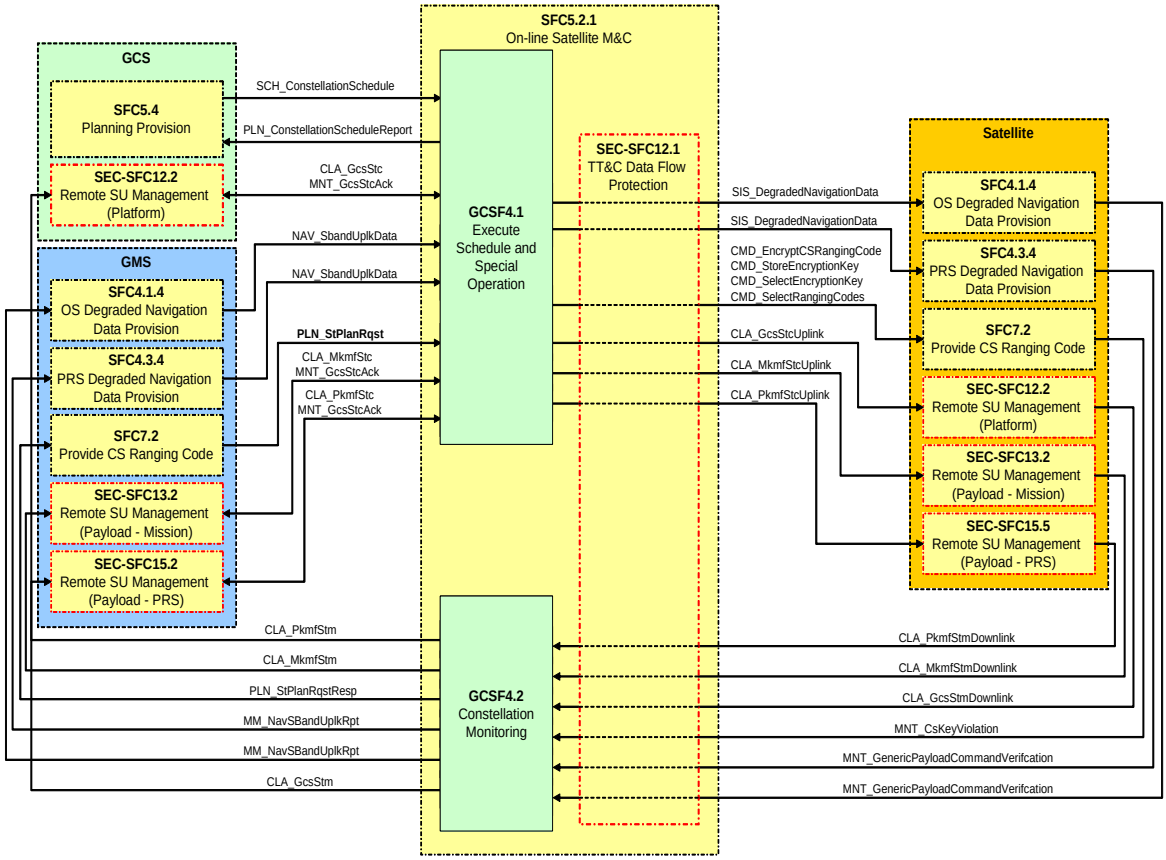


Figure 62 : Interactions with other system functional chains

4.6.2.2.4 Evolution of SFC5.2 for CS encryption

As the Galileo commercial service is expected to levy fees, uncla encryption keys are foreseen to protect the commercial service in order to allow the reception to subscribed users only (GMSF8.4 and related SFC7.2).
The GMS is responsible for the Commercial Keys management and when necessary, in coordination with the Service Centre, it supplies to the GCS planning tool with a planning request to uplink new CS Keys on board the spacecraft. CS keys are actually provided to the GCS as an input argument of the CS key management planning request.
However, GMS only gets back a feedback on the success or failure of the TC sent by GCS to uplink the CS keys. At any time, MKMF operators can only presume that the expected CS keys are correctly loaded on board, without having actual evidence of that. To date (see GAL-ICP-DCP-TAS-GMS-000428, issue 1.0) actual info on CS keys loaded on board is only exchanged from GCS to GMS operators offline, via operational means (i.e. mail, fax).
This is not deemed adequate for FOC considering

- the increasing number of satellites that is being deployed
- the key role that is being defining for the commercial service.

An improvement of the SFC5.2, Satellite Constellation Monitor and Control is therefore foreseen in STRB#2.1, in which an on line electronic interface between GCS and GMS (**#STRB2.1_change: GAL-DCN-TASI-SYST-X-0738 v.3.0 on GCSGMS ICD**) is introduced to allow the GCS to provide info on the current uncla CS keys (traffic and protection keys) selected on board each spacecraft extracted from TM data.

The CS key status info is retrieved by GCS from TM data collected issuing dedicated specific TCs (#STRB2.1_change: GAL-DCN-TASI-SYST-X-970 v1.0 on GCSREQ).
The CS Key Status report sent by GCS to GMS will contain the following information:

- NSGU status (NSGU A on, NSGU B on, both NSGU off)
- CS Key Type (Base, Traffic E6b, Traffic E6c, Protection)
  - List of Keys:
  - Key index
  - Key CRC
  - CS Active Key Index
  - Validity Flag (per data item)

The GMS uncla MKMF (#STRB2.1_change: GAL-DCN-TASI-SYST-X-875 on GMSREQ) will be in charge of processing such data and comparing it with the expected CS key status, warning the operator in case mismatches are found.
The NSGU info provided across the GCSGMS ICD may be only partial: it will be up to operators to follow the appropriate operational procedure to ensure that the necessary information is flown to the GMS and that a complete set of data is provided.
The sampling time of the telemetry is also included in the report, as well as a data validity field which shall be set to false in case the telemetry is not available in the SCCF archive.

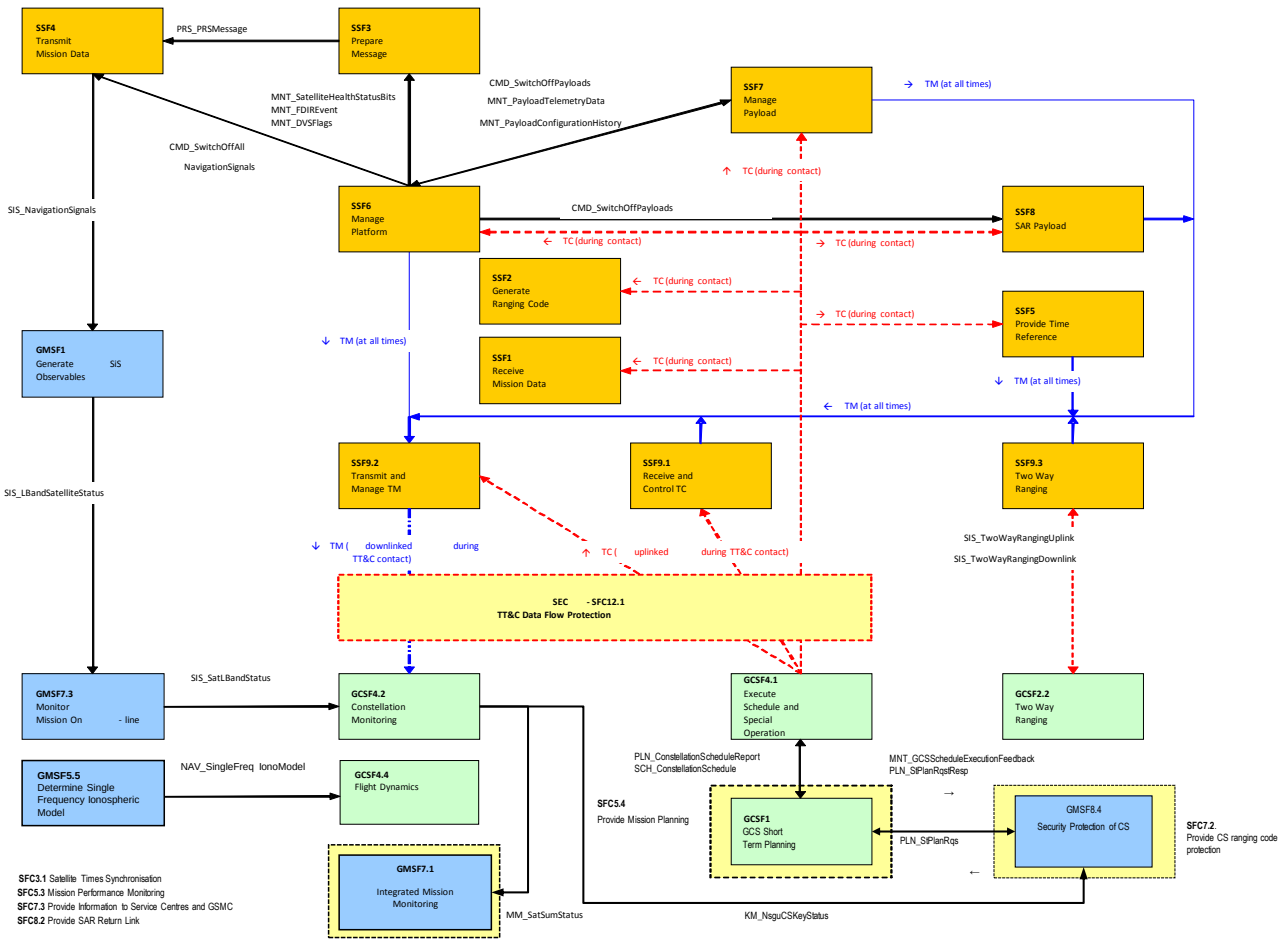


Figure 63 : SFC5.2.1 functional block diagram in STRB2.0

In addition to the above, the evolution of SFC5.2 for STRB#2.0 also foresees for the GMS the capability to send:

- from GMS SFC8.4, signal PLN_StPlanRqs, carrying GMS CS key management planning requests (**#STRB2.1_change: GAL-DCN-TASI-SYST-X-0738 v.3.0** on GCSGMS ICD) to the GCS (GCSF1), providing the possibility for separate encryption keys for each of the two frequencies (E6b, E6c) supporting the Commercial service (see GMSF8.4 and SFC7.2 for more details).
- **#STRB2.1_change: GAL-DCN-TASI-SYST-X-0972, GAL-DCN-TASI-SYST-X-0973 and GAL-DCN-TASI-SYST-X-0779:** from GMS SFC5.5, signal NAV_SingleFreq IonoModel, with the Ionospheric single frequency parameters computed by the NeQuick model to GCS flight dynamics function (GCSF4.4), to correct S-band ranging measurements for ionospheric delay. The NeQuick model is used by default, to reduce Galileo dependency from external data, i.e. IRI 2012 model (which can be used in alternative).

4.6.2.3 SFC5.2.2 Off-line Satellite Monitoring and Control

4.6.2.3.1 SFC description

The purpose of this system functional chain is to provide all off-line functions required to help ensure and maintain the health of each satellite in the Galileo constellation.
This includes the following specific aspects:

- Command schedule generation and validation (automatic translation of the GCS STP into a command schedule to be uplinked to the satellite at the next contact).
- Orbit determination functionality in the GCS based on either of the following:
  - Orbit vectors provided to the GCS by the GMS (based upon the L-band navigation message)
- Radiometric measurements taken in S Band from the Galileo TT&C ground stations (i.e. 2-way ranging).
- Flight dynamics support to the execution of attitude and orbit control manoeuvres. This includes the prediction of orbit events (eg eclipses, sensor blinding etc) and all associated inputs to planning. Inputs to these functions are provided by the orbit determination functions (see above) and also from the GMS (eg NORAD data, ionospheric data etc.
- Note that manoeuvres will be executed as part of on-line satellite monitoring and control (see SFC5.2.1).
- Off-line performance and trend analysis based on TM received from all spacecraft units during on-line satellite M&C (see SFC5.2.1). This includes as a minimum analysis of the following:
- Satellite AOCS subsystem:
    - Satellite attitude prediction and reconstruction
    - AOCS failure analysis
    - Routine analysis of current AOCS performance
    - Long term AOCS performance trend analysis
- Satellite antenna pointing performance
    - Thruster performance
  - Satellite end-of-life estimation (including fuel budget estimation).
  - Satellite anomaly and event investigation.
  - Analysis of environmental conditions as derived from on-board TM data to validate the orbit and clock estimations (see SFC 5.3)
- Support for on-board software maintenance upload and verification, including:
- Generation of inputs to the planning functions (OBSM tasks etc)
  - Validation on the ground of the OBSW updates to be uplinked
  - Conversion of validated SW into commands to be uplinked to the satellite
  - Verification of the on-board implementation of the SW update via telemetry

- Configuration management of satellite platform and payload on-board software for all satellites in the constellation.
- Support from the Satellite Manufacturer (SATMAN) for anomaly investigation and OBSW maintenance.
- Satellite database management for all satellites in the constellation including interaction with SATMAN for delivery of satellite TC/TM and flight dynamics databases as well as satellite operational procedures.
- Verification and validation through simulation in support of the following activities in particular:
  - OBSW updates
  - Orbit manoeuvres
  - Operational procedures
  - Anomaly simulation

Note: Constellation maintenance, renewal and satellite disposal is covered by SFC1.

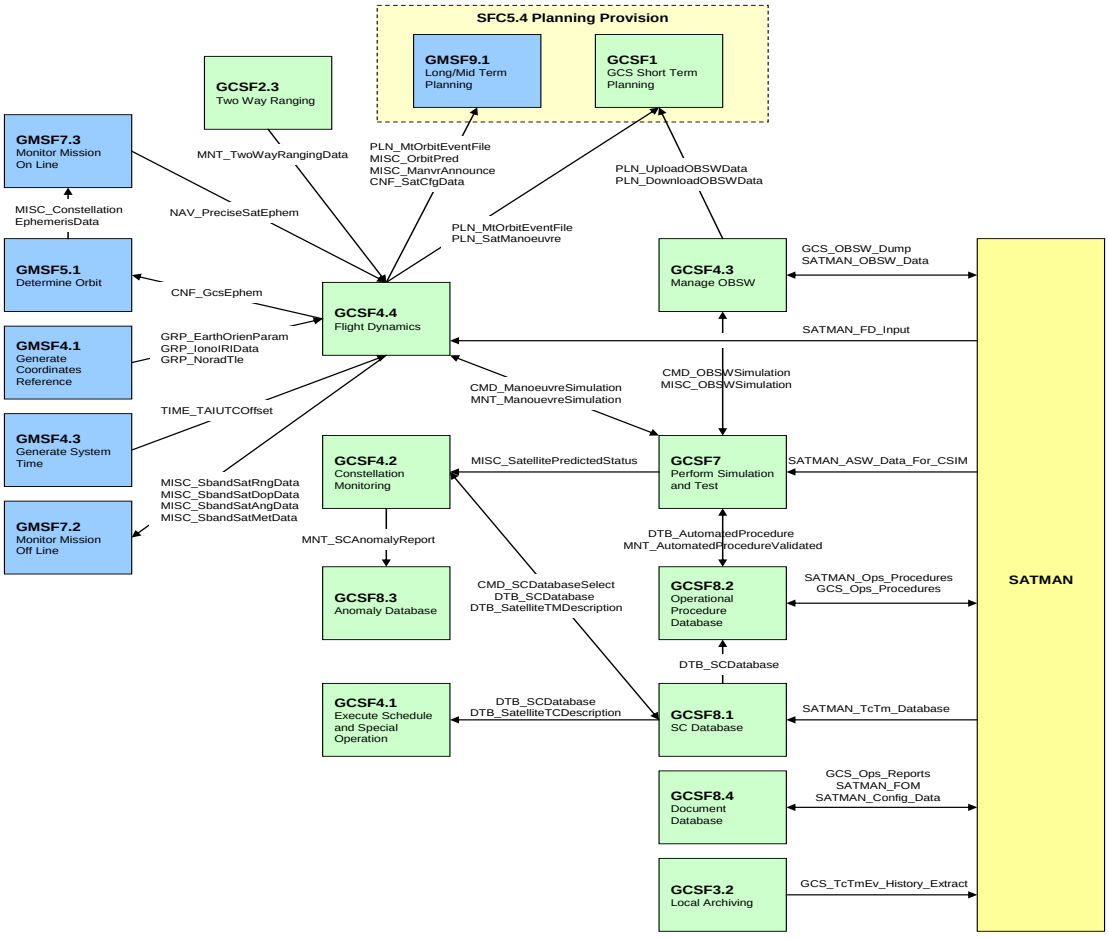


Figure 64: SFC5.2.2 Functional Block Diagram

4.6.3 SFC5.3 Mission Performance Monitoring

The Mission Monitoring Functional Chain is responsible for providing all required functionality to monitor and analyse all the parameters involved in the mission of the Galileo System.

Two kinds of mission monitoring are defined:
On-line mission monitoring, responsible for:

- Verifying in real time the service parameters critical from a mission performance point of view.
- Monitoring in real time the status and quality of elements involved in the mission determination and dissemination chain (see SFC4 for functional definition of these chains).

The purpose of the on-line monitoring function is to be able to raise potential alerts to be continuously displayed for the operators in real time. Based on this information, dedicated failure investigation and corrective actions can be triggered (defined by operational procedures) to ensure that system performances are still met.

Off-line mission monitoring, responsible for monitoring and analysing the parameters needed to guarantee a sufficient level of performance of the system. This includes the following tasks:

- Off line mission data provision for algorithms calibration
- Off line statistical mission performance analyses
- Off-line satellite and constellation performance analyses

The purpose of the off-line monitoring function is to be able to maintain the system in a configuration compliant with the mission performances, generating proper configuration parameters and reports.

The first level functional tree for the “Mission performance monitoring” SFC is shown below:

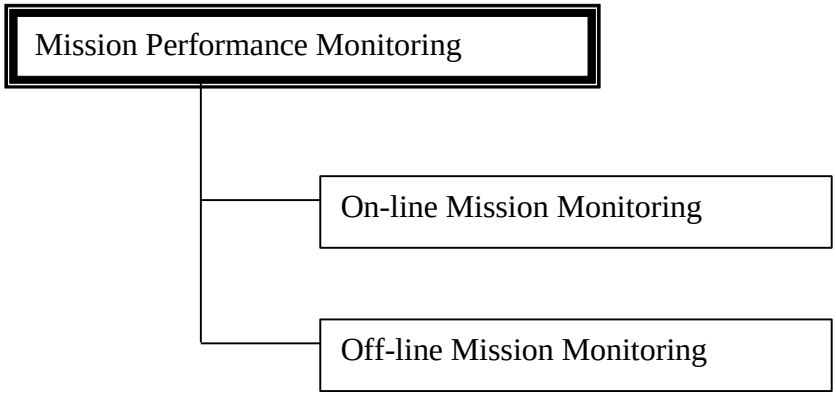


Figure 65 : SFC5.3 Functional Tree (1st level)

Notes:
a) The corrective actions and associated functionalities resulting of this monitoring and analyses are not described in this SFC. Refer to:

- SFC3.1 for Satellite clock correction
- SFC5.1 for ground element configuration
- SFC5.2 for satellite control
- SFC5.4 for planning validation and performance prediction

b) The mission monitoring of CS and SAR are not addressed here but in dedicated SFCs (SFC7, SFC8 and SFC9)
c) The process to generate reports and send them to external entities is addressed in dedicated SFC7.3

4.6.3.1 SFC5.3.1 Mission Performance On Line Monitoring

4.6.3.1.1 SFC description

The On-Line Mission Monitoring Functional Chain is responsible for providing all required functionality to monitor and analyse in real-time the parameters that are critical for the mission performance.

In particular, it shall provide the needed data for:

- On-line verification of the service performances (OS, and PRS) at user level;
- On-line verification of the mission data provision chain quality (mission data determination products, Ground Segment Resources monitoring)
- On-line satellite status monitoring (on-board master clock, satellite general status, satellite transmitted power).

The Integrated Mission Monitoring function is responsible for routing and displaying to the operators any data or alarms raised by the on-line monitoring functions.
In addition the on-line monitoring function is responsible for providing the GCS with:

- Satellite status
- Precise ephemeris computed by the GMS OD&TS function

4.6.3.1.1.1 On-line verification of service performances

In order to verify the performances of the Galileo services, the following service parameters (for OS and PRS services) are checked on-line (new functionalities added in FOC2):

- OS and PRS Positioning and Timing accuracy and availability at user level
  - Ranging accuracy per satellite and per service
  - BGD Accuracy per satellite and per service
  - UTC disseminated via Galileo SiS

#STRB2.1_change: GAL-DCN-TASI-SYST-X-0945, #STRB2.0_change: GAL-DCN-TASI-SYST-X-0949 and GAL-DCN-ASTG-SYST-X-13922

a) Service performance on-line monitoring
In order to support the on-line verification of these service performances, the on-line mission monitoring functions shall perform the following tasks:

- GSS positioning error measurement: based on SIS observables (SIS and navigation message), the positioning accuracy is measured at each GSS location. This accuracy can be characterized by the error between the GSS position computed by the positioning algorithm and the true known position.
  - User positioning and timing accuracy performance estimation for OS and User Positioning performance estimation for PRS services: based on SIS observables (SIS and navigation message), the position and timing accuracy achievable with standard user positioning algorithm (see SFC4) are estimated on worldwide coverage periodically.
  - SISA set to NAPA: in case the OSPF is not able to generate a valid Navigation Message (due to a Satellite Feared Event), for maximal 50 hours messages containing a SISA value set to “NAPA”, an updated IOD and the last Navigation Message are generated and uplinked.**(#STRB2.0_change: GAL-DCN-TASI-SYST-X-0910)**
- UTC(SIS) – UTC(from TSP) is monitored periodically and per service, enabling a configuration of a GST-UTC dummy value when a threshold is exceeded **(#STRB2.0_change: GAL-DCN-TASI-SYST-X-0949)**

Notes:
“Periodically” means configurable down to 1 minute
“Worldwide” coverage means that the estimation is performed on every points of a configurable grid (down to 1 degrees) representing the service coverage.

- User Positioning and Timing availability performance prediction for OS and User Positioning performance prediction for PRS services: based on SIS Observables (SIS and Navigation Message),

the system shall estimate periodically over a coming period of 10 days the percentage of time during which the user positioning and timing accuracy are achieved. This estimation is performed:

- with standard user positioning algorithm (see SFC4)
  - on every points of a configurable grid (down to 1deg) representing the service worldwide coverage;
- over a predicted period of 10days with a configurable refresh rate (configurable down to 5 minutes);
  - taking into account any planned outage identified in the Mid Term Plan over this prediction period.

- ***Ranging accuracy:*** based on GSS data, the System shall monitor the ranging accuracy by periodically measuring the SIS Error at worst user location of the current navigation message, per satellite and per service vs online reference orbit and clock computed by GMS
- ***BGD per satellite and per service:*** based on GSS data, the System shall monitor the BGD accuracy by periodically measuring the BGD Error of the current navigation message, per satellite and per service vs the online reference computed by GMS
- ***UTC Disseminated via Galileo SiS:*** The system shall periodically monitor the timing accuracy per service, based on UTC_SIS (i.e. the UTC disseminated via the GALILEO SIS) vs UTC measurement data from TSP. Implementation of new hourly signal TIME_UtcMntReport from TSP to GMS. **It is important to note that the MRD specification (260/885) of the timing service performance and availability are conditioned to the availability and quality of the TSP products. Therefore, it is essential to ensure that the TSP product quality and availability is covered by appropriate TSP Design and Performance requirements in-line with the MRD timing availability and accuracy targets.**

b) Service performance alerts and display
The service performances parameters above indicated are compared with performance targets characterised by configurable thresholds.

Each time a mission threshold is exceeded, dedicated on-line mission alarms are raised.

The Integrated Mission monitoring function is then responsible for routing and displaying to the operators any data or alarms raised by the on-line monitoring functions. This display shall be refreshed with a configurable rate.

c) Recovery procedures
Based on these data or alerts, failure investigation and repair processes can be initiated by operators.

4.6.3.1.1.2 On-line Check of the on-ground mission data provision chain Status

In order to verify the quality of the mission data determination and dissemination chains, two main aspects are checked in real-time:
a) Mission data determination chain products quality on-line monitoring
The Quality Of Service (QoS) provided by the navigation determination chain are displayed to the operators via the Integrated Mission Monitoring function. QoS indicator could be used to detect failures requiring dedicated recovery actions.

b) Mission data provision chain configuration on-line monitoring
The system configuration shall be monitored in real time to ensure that the current configuration enables to fulfil the system performances. For this, the Overall Monitoring and Control function of the GMS

(GMSF7.4.3) is responsible for sending to the Integrated Monitoring function a summary status of the elements of the ground segment.
This information will be used by the operator to locate and repair the anomaly.

4.6.3.1.1.3 On-line Monitoring of the satellite Status

a) **Satellite on-board master clock monitoring:** the GMS on-line mission monitoring function is responsible for analysing the data received from the OD&TS algorithms (restituted clock parameters), to monitor and control the satellite master clock. These data shall allow to detect any deviation or any abnormal satellite clock behaviour, and to send clock synchronization request to the GCS. This is further described in a dedicated SFC (SFC3.1).

Note: Satellite back-up clock is monitored off-line (see dedicated section)

b) Satellite transmitted power

#STRB2.1_change: new functionality GAL-DCN-TASI-SYST-X-0946. Based on measurments collected by the ground sensor network, the ground segment shall perform the online monitoring of the differential satellite transmitted power (i.e. difference between the current C/N0 measurements and the same measurements taken 10 days before, on the basis of integer multiples of the ground track repeat cycle), per satellite and per signal component, based on a configurable average of the differences over all GSS.
Note: Integer multiples of the ground track repeat cycle are to be considered in order to compute the differences under the same geometric conditions.

- **General satellite status from GCS to GMS:** The GCS constellation monitoring function (GCSF4.2) is responsible for transmitting periodically a summary of the satellite status to the GMS. This satellite status is displayed by the GMS integrated mission monitoring function to the operators. Based on this information, the operator can monitor the impact on the mission performances.

c) General satellite status from GMS to GCS

The GMS On-Line Mission Monitoring function is responsible for transmitting periodically to the GCS a satellite status composed of:

- A satellite status information based on GMS mission data processing (ODTS): this information allows to indicate to the GCS the status of the satellite from a mission performance point of view, namely, NAV_SatMntData: information regarding the satellite status derived from the ODTS determination processes

  Note that these parameters are provided for information only (as this status cannot be monitored directly by the GCS) and cannot be considered as safety messages by the GCS.

- Satellite L-Band Telemetry Status received in the L-Band message: this information is computed on-board the satellite and downlinked in L-Band and refreshed every minutes. This enables the GCS to have a high level but continuous summary the satellite health status. The detail of the satellite telemetry is then recovered at each S-Band contact: see SFC5.2.

d) Satellite Precise Ephemeris on-line provision to GCS

The GMS On-Line Mission Monitoring function is responsible for transmitting periodically the ephemeris computed by the OD&TS algorithm to the Flight dynamics functions of the GCS. These ephemeris will be used by the GCS nominally for orbit propagation (including manoeuvre preparation)
In case these ephemeris are not available, the above tasks rely on the ephemeris computed by the GCS flight dynamics functions based on S-Band ranging

4.6.3.1.2 SFC Functional Block Diagram

The following block diagram shows the functional architecture of the on-line mission monitoring functional chain:

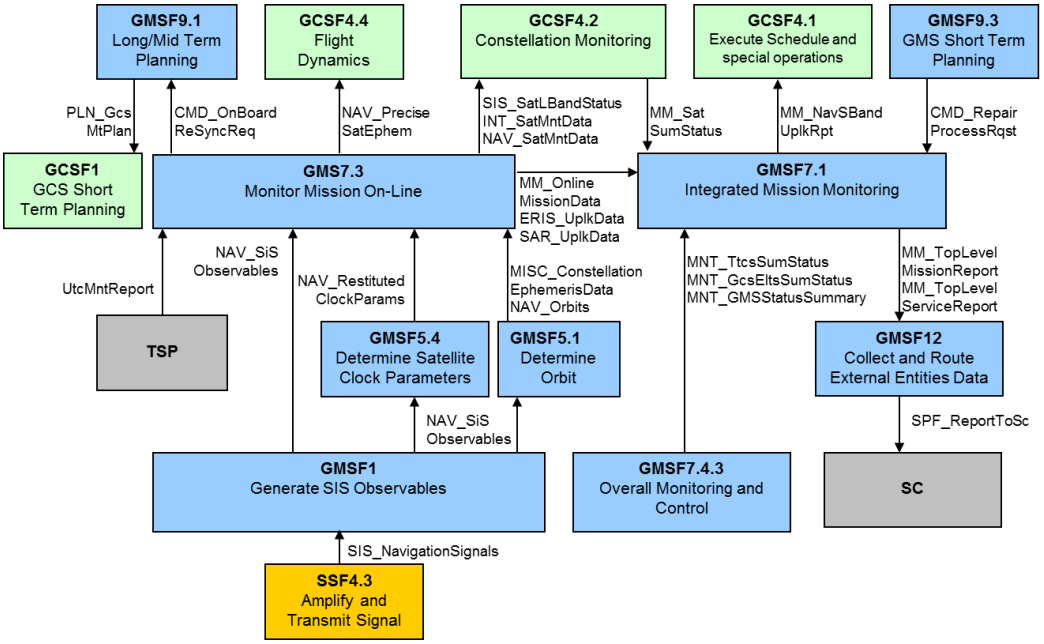


Figure 66 : SFC5.3.1 Functional Block Diagram

4.6.3.2 SFC5.3.2 Mission Performance Off Line Monitoring

4.6.3.2.1 SFC description

The Off-Line Mission Monitoring Functional Chain is responsible for providing all required functionality to monitor and analyse the parameters needed to guarantee a sufficient level of performance of the system with off-line processes.
In particular, it shall provide the needed data for:

- System calibration: support to algorithm and element parameters calibration
- Statistical mission performance analyses: support to report generation: see SFC7.3
- Off-line monitoring of satellite and constellation status including:
- Satellite on-board hot backup clock monitoring and control (see SFC3.1.2)
- Received power level on-ground
- Constellation long-term analyses to support manoeuvre planning

4.6.3.2.1.1 System Calibration

The off-line monitoring functional chain is responsible for:

- OD&TS parameters initialization and calibration
- Navigation determination chain parameters calibration
- Support to External entity products: GST and GRSP

a) Support to the OD&TS algorithms initialization and calibration

In order to support the calibration of the GMS OD&TS algorithm, the off-line monitoring functional chain is in charge of managing the following tasks:

- GMS OD&TS algorithm initialization: the initialization of the GMS OD&TS algorithm relies on the provision of satellite ephemeris data computed by the GCS flight Dynamics function (GCSF4.4)
- OD&TS algorithms accuracy analyses: the orbits and clock predictions produced by OD&TS shall be compared to off-line precise models based on SIS measurements and external sources support (e.g. TGVF). The GCS TM and TC packet data related to specific spacecraft domains are also used to correlate/validate orbit and clock estimations against environmental conditions on-board the spacecraft.
- The precise models for orbit determination shall be off-line computed using data from different sources:
  - SIS measurements and OD&TS a posteriori determination

- Satellite Laser Ranging (SLR) data
  - Long-term analysis of orbit stability

b) Support to the calibration of the parameters used in the navigation determination chain

The off-line monitoring function is in charge of monitoring the needed parameter to calibrate navigation determination chain parameters. This includes:

- the SISA parameter computed by the navigation determination chain based on:
  - SISA calibration parameters
  - Planning information to provide margins on SISA in case of satellite maintenance activity.

c) Support to GRSP products calibration

The offline estimation of SVs and clocks and ERP together with the pseudoranges received by the GSS network are provided to GRSP to contribute the generation and maintenance of the Galileo Terrestrial Reference Frame to reference products calculation. (see SFC2.3).

d) Support to TSP products calibration: GST parameters accuracy and stability check

The following GST parameters shall be off-line checked based on TSP inputs (see SFC2.1 and SFC2.2):

- GST stability long-term analysis
- GST-UTC offset accuracy analysis
- GGTO products

4.6.3.2.1.2 Statistical mission performance analyses

The off-line monitoring functional chain is responsible for the following off-line analyses:

a) Mission service performances statistical analyses:

The following mission performance parameters shall be off-line and periodically estimated on a worldwide coverage:

- Service Accuracy
- Service Availability

These analyses are based on archived data.

b) Performance Trend analyses

The following performance trend analyses are used to generate dedicated bulletin to the external world (see SFC7.3) including at least:

- ULS tracking performance
- GSS environments characterisation (multipath, interference monitoring) and availability
- Network performance

c) Verification of the SISA computation

The SISA parameter computation is off-line checked by analysing the real SISE distribution a posteriori.

d) Report Generation for External World

Reports containing the results of the above analyses are distributed to the mission and system management and external service centre (see SFC7.3).

4.6.3.2.1.3 Off-line Monitoring of the satellite and constellation status

a. Satellite on-board hot backup clock monitoring

The GMS off-line mission monitoring function is responsible for receiving the phase comparison samples received in the telemetry data and transmitted by the GCS, to monitor and control the satellite back up clock. These data shall allow:

- detecting any abnormal behaviour of this back up clock in order to send clock switching request to the GCS or
- sending hot backup clock re-synchronization request to keep the frequency derived from hot backup clock synchronized to the master frequency reference within a configurable range

This is further described in a dedicated SFC (SFC3.1.2).

b. Constellation long-term mission analyses

The GMS off-line mission monitoring function is responsible to lead constellation mission analyses in order to support the validation of manoeuvres potentially planned in the mission planning. These manoeuvres are initially requested by GCS and are validated in the mission planning to maintain the constellation in a nominal geometry.

c. EIRP

#STRB2.1_change: new functionality STRB2.0 GAL-DCN-TASI-SYST-X-0946

the system will offline- monitor the received power level on ground derived based on C/No parameters contained in the measurements as collected at each GSS. Specifically the GMS shall provide support to the satellite transmitted power analysis with a time series, per satellite and per signal component, of the difference between the current and previous C/N0 measurements, on a constellation repeat cycle basis (same geometry), for each GSS.

4.6.3.2.2 SFC Functional Block Diagram

The following block diagram shows the functional architecture of the off-line mission monitoring functional chain.

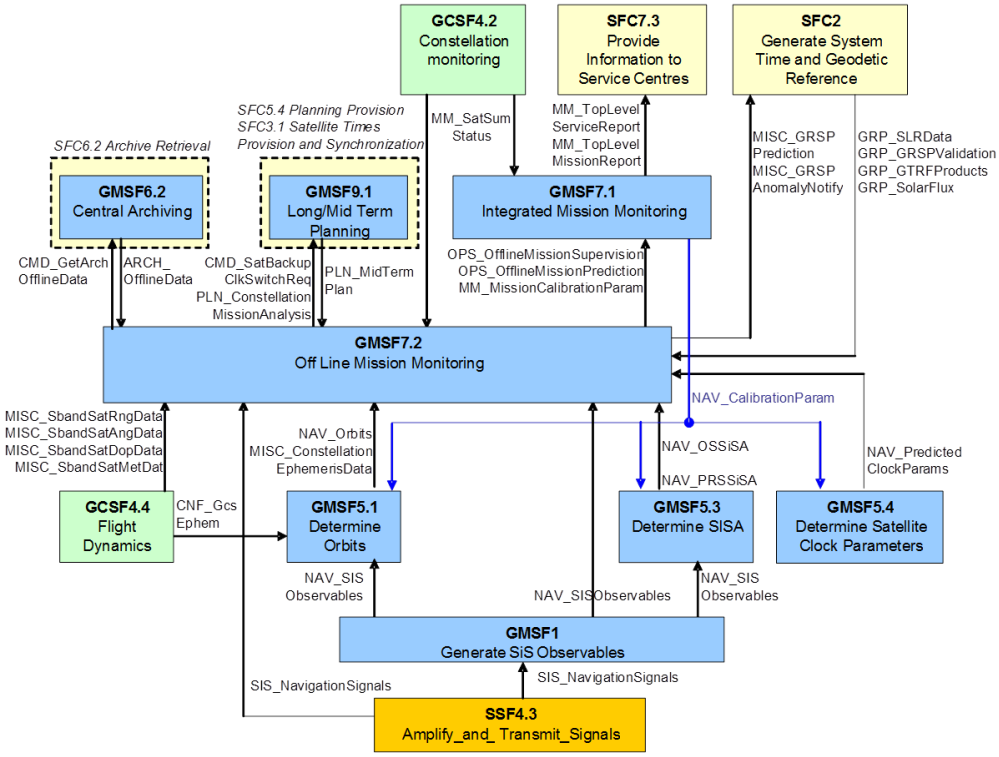


Figure 67 : SFC5.3.2 Functional Block Diagram

4.6.4 SFC5.4 Mission Planning Provision

The objective of the Mission Planning provision SFC is to enable planning both routine and special system operations for the following activities:

- Constellation management and satellite control
- Ground assets monitoring and control
- Mission service management

The mission planning includes the production of executable schedules for automation.

4.6.4.1 Definitions

This section recalls the main definition of planning and schedules as per the operational concept

Long-Term Plan: The long-term plan is the high level planning which provides input to lower, more detailed levels of the planning process, used to derive both the mid-term plan (plan refinement) and the short term plans. It typically covers a period of 1-6 months, with periodic updates, but can be extended to cover up to 12 months. The long term plan considers activities such as new launches, replenishment, G/S and S/C high level maintenance. Due to the long period to be planned, some inconsistencies or conflicts between activities may coexist (for example, exact start/end dates may not be known and longer temporal. slots could be reserved/specified for an activity).

Mid-Term Plan: The mid-term plan is essentially a refinement of an existing long-term plan which specifies a greater level of detail. It covers a period of up to 4 weeks, where activities planned are conflict-free (all the inconsistencies and overlaps are resolved).

Short-Term Plan: The Short Term plan is a refinement of the mid-term plan. It consists in a time ordered and conflict free plan with a fine granularity of the tasks to be executed based on the activities planned in the Mid Term plan, and integrating all tasks and events required to complete the planning of these activities. The short-term plan period is expected to be in the range of 1 to 10 days.

Schedules: Schedules are generated from the short-term plans and represent the last element in the planning chain prior to execution. The schedules can be divided in 3 main categories:

· Manual Operations: Timelines & Operators Tasks,
· Automatic Tasks: Executable Schedule for the Overall, Central and Element M&C functions,
· Ground to space contact schedules (ULS to satellite and TTC).

4.6.4.2 SFC Description

The mission planning functional chain is composed of two levels:

- The Long/Mid-Term Planning function (GMSF9.1) covers the overall system activities (i.e. of all segments) over a period of at least six months. The mid-term plan is a refinement of the long-term plan over a period of one to four weeks.
- The Short-Term Planning functions are done at GCS and GMS level (respectively GCSF1 and GMSF9.2) to plan and schedule the activities of the segments within the boundaries of the constraints imposed by the mid-term plan. The Short-Term planning functions produce the executable schedules.

The internal GMS exchanges which contribute to this functional chain are described in the sections relative to the segment functions: GMSF9.1 Long/Mid-Term Planning and GMSF9.2 Short-Term Planning. This section focuses on the exchange between GMS and GCS planning functions.

In nominal conditions (i.e. the system is operating nominally without failure), the Mission Planning SFC is articulated in the following tasks:

- ·Task 1 - Long/Mid-Term Planning inputs collection:
  - The Long/Mid-Term Planning function collects information on the system configuration and health status (including orbit event predictions) from the monitoring and control functions of GCS and GMS.
  - The Long/Mid-Term Planning function collects planning requests from GMS and from GCS.

    Note: It is expected that out of the GCS requests, only those having an impact on the mission are passed to GMS. All others are managed internally by the GCS short-term planning function (see JT-86 in IOV DJF SYS-RD.03).
- Task 2 - Verification of Long/Mid-Term Plan compatibility with the system mission performance and provision of service bulletin:

  - The Long/Mid-Term Planning function verifies that planning requests and the system configuration are compatible with providing the system mission performance. The needed information on the constellation are provided by the off-line mission analyses function (see SFC5.3).The resulting mid-term plan shall be checked for assuring that the service performances (as per on-line mission performances prediction) of Galileo are met during the foreseen activities.

    Note: The GMS is not in charge of issuing satellite manoeuvre planning requests as suggested by GMSREQ-9839. This is indeed the responsibility of GCS as shown in GCSGMSICD-262.

  - The Long/Mid-Term Planning function prepares bulletins to announce the planned unavailability of resources and/or periods when the services performance is not met. Such bulletins are then delivered to the external service centres more than 48hrs in advance of the planned unavailability periods.

- Task 3 - Provision of Mid-Term Plan to the GCS and GMS Short-Term Planning functions:
    - The Long/Mid-Term Planning function provides Mid-Term Plan to the GMS and GCS Short-Term Planning functions for the up-coming period of 1 to 4 weeks.

- Task 4 -Elaboration of Short-Term Plan and executable schedules by GCS and GMS - Mid-Term Plan Refinement:
    - The GCS and GMS Short-Term Planning functions provide feedback on the mid-term plan to the Long/Mid-Term plan function in order to inform about status of the requested task. In case some mid-term plan constraints cannot be satisfied, another iteration might be required.
    - The GCS and GMS Short-Term Planning functions elaborate short-term plans and executable schedules in line with the constraints contained in the mid-term plan and considering the system configuration for the short-term plan period.
      Note: The short-term plan period is expected to be in the range of 1 to 10 days.
    - The GCS and GMS planning operators can perform some adjustment at short-term plan level provided that they do not modify planning requests allocated by the mid-term plan.
    - The GCS Short-Term Plan includes the nominal TTC contact schedule as well as associated back-up contact schedules making use of alternate TTC stations. The activation of a back-up contact schedule is triggered by operators and does not take more than 10mn. The GCS automatically informs the GMS of the change.
      Note: The GCSGMSICD specifies that the TTC contact schedules are passed to GMS at each GCS STP update. However the activation of a back-up TTC schedule does not require the update of the STP and therefore is not communicated to GMS, which is not in line with GCSREQ-26096.
    - The GCS Short-Term Plan includes a configurable slot to allow sending STC at each routine S-Band contact:
      1. The total STC data to be released from the GCS-KMF TC stack is, under nominal conditions, of 15 Kbytes per S/C contact at TC frame level, including security overheads
      2. At GMS level secure tasks or activities are planned at each routine S Band contact, limiting the total STC data to be released from each GMS xKMF TC stack of 15 Kbytes (nominal), including security overheads.

      So the allocated bandwidth under nominal condition is dimensioned to 45 Kbytes per contact at TC frame level for the 3 KMFs (GCS-KMF, GMS-MKMF and GMS-PKMF).
      The GMS xKMF provide the means to display in real time the total data size of STC data loaded in the xKMF TC stacks and generate a warning in case of 15 Kbytes limit violation.
      This (soft) limit is defined for a short reference S band contact of 20 minutes, assuming a total of 180 Kbytes for the uplink at Transfer Frame level per Spacecraft in the contact.
      The shift leader will have the authority to coordinate the overall TC activity planned by classified and unclassified elements and grant permission to GMS xKMF operators to exceed the STC nominal limit taking into account the foreseen STC volume from all X-KMFs and the actual expected S band contact duration.
    - The GMS is able to transmit to the GCS, short-term planning requests to command the CS carrier ranging code keys. These requests are translated by GCS into specific telecommands (TC) with the parameters provided for uplink to each satellite. In return, the GCS provides the GMS with feedback on the status of the short-term planning request.
      Note: There is a problem of inconsistency between GMSREQ-10313 and GCSGMSICD (this issue will be resolved in the GCSGMS evolution, see section 4.6.4.6).
      Note: This exchange is further described in SFC7.2 Provision of CS Ranging Code.
    - The GMS short-term planning function is in charge of:

- Generating the C-Band ULS to Satellite Uplink Schedule composed of a ULS to satellite allocation plan and a ULS to satellite tracking plan. This schedule covers both operational and test operations (see evolution in section 4.6.4.7).

- Task 5 - Collection of Feedback on Short-Term Plan Execution:
  - the GCS Short-Term plan function provides feedbacks on the execution status (e.g., successful / failure) to the Long/Mid-Term Plan function.
  - The GMS generates reports on the planning execution.

  In case of contingencies (i.e. failure of one or more system components), re-planning might be necessary.

4.6.4.3 Evolution of Mission Service resumption after satellite maintenance activities

#STRB2.0_change: GAL-DCN-TASI-SYST-X-0889

After a spacecraft has been put out of service for more than 2 days (e.g. for an orbit manoeuvre, planned satellite unavailability, coarse on board clock switch, other maintenance activities) the GCS is required to provide to GMS the ephemeris necessary to resume the pseudo-range data processing for such satellite. To minimise the service unavailability a specific strategy has to be put in place for the provision of satellite ephemeris which allows the GSS to resume tracking as quick as possible with the minimum level of accuracy necessary:

- in case of a planned station keeping manoeuvre, not later than 4.5 hours after the end of the last manoeuvre burn, based on the predicted orbit vector
- in case of planned unavailability longer than 2 days, not later than 1 day after the end of the announced unavailability;
- in case of planned clock coarse switch longer than 2 days, not later than 1 day after the end of the announced clock event
- in case of a satellite-(re-)introduction, at the time the operation is planned.

Such quick GCS ephemeris is only used to resume GMS processing, which will take at least 50 hours to provide a new valid navigation message.
After an orbit manoeuvre, the GCS is expected to determine the orbit based on S Band measurements collected from the TTCF network over 4 orbits to ensure that the precise orbit positioning has been achieved and no other station keeping manoeuvre is required for several years.
In favourable cases, by the time the GCS FDF determines the precise orbit, no further correction manoeuvre is required and the GMS processing has been already resumed by nearly 2 days.
In case a further correction is required, GMS operations have to be interrupted again.

4.6.4.4 Orbit Manoeuvre planning request evolution

#STRB2.0_change: GAL-DCN-TASI-SYST-X-0855

The evolution for a planning request for orbit manoeuvre will identify a time slot, to be allocated within the MTP, covering the whole sequence of operations foreseen at GCS level including:

- preparatory activities to be performed before the actual orbit manoeuvre (e.g., Satellite / AOCS / FDIR SW reconfigurations, thrusters cat bed heating, attitude slew manoeuvres, etc.);
- post manoeuvre operations (e.g., again Satellite / AOCS / FDIR SW reconfigurations, attitude slew manoeuvres, ranging etc.);
- Series of thruster burns, when required;
- Time margins to allow for last minute adjustments of manoeuvre timing.
- time needed to compute the first satellite ephemerides needed to resume GMS operations.

This allows:

- GMS to plan all necessary activities before and after the block of manoeuvre operations (e.g. preparation for the out of service transition, restoration of mission services).

- Introducing some level of independence between MTP and GCS STP, thus avoiding useless MTP - GCS STP iterations caused by manoeuvre parameters changes.

4.6.4.5 MTP – GCS STP Planning interaction evolution

#STRB2.0_change: GAL-DCN-TASI-SYST-X-0855

The GMS MTP defines the plan of operations foreseen over a defined time span. For planned requests originated by the GCS the MTP provides information on how the request has been handled, e.g.:

- Planned when the task has been included in the MTP,
- Pending" if the task has not yet been included in the current MTP horizon
- Cancelled" in case the MTP operator has decided to delete the GCS request.

Upon GCS STP production, GMS is notified of the status of tasks included in the GCS STP which were included in the MTP (Accepted / Rejected).

Once the MTP is elaborated by GMS, status information is provided for each Accepted Task (e.g.: "Planned" if the operation has been included in the MTP, "Pending" if the GCS requested operations is not yet included in the present MTP, "Cancelled" in case the MTP operator has decided to delete the GCS request).

Tasks falling within the GCS STP time horizon are expected to be flagged by GCS as "Planned" during the GCS STP processing.

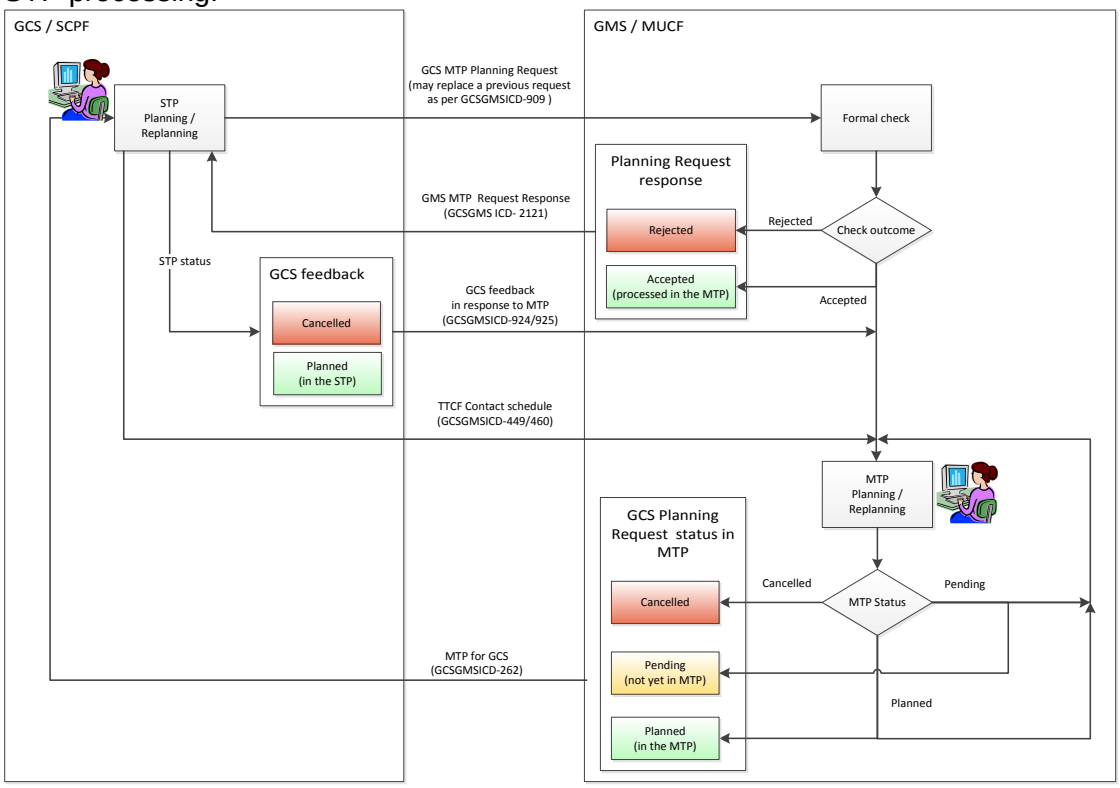


Figure 68 : GCS GMS Planning interactions

Since both MTP and STP are manually modifiable by operators, GCS STP and GMS MTP may be not aligned at a given time. To reach some stability and eventually the alignment at GCS and GMS level, restrictions apply on the GMS MTP Planning Request status transitions after receiving the GCS feedback in accordance to the following Table 9 : and Table 10 :, showing respectively the allowed and forbidden status transitions in the GMS MTP (columns 1 and 3), upon receiving the GCS STP feedback (column 2).

Planning Request status in the MTP before GCS feedback	GCS Feedback following GCS STP elaboration	Planning Request status in the MTP after GCS STP feedback
Cancelled	Cancelled	Cancelled
Pending	N/A	Pending
Pending	N/A	Planned
Pending	N/A	Cancelled
Planned	Planned	Planned
Planned	Planned	Cancelled

Table 9 : GMS MTP allowed Planning Request status transitions

Planning Request status in the MTP before GCS feedback	GCS Feedback following GCS STP elaboration	Planning Request status in the MTP after GCS STP feedback
Cancelled	Cancelled	Pending
Cancelled	Cancelled	Planned
Planned	Planned	Pending

Table 10 : GMS MTP forbidden Planning Request status transitions

4.6.4.6 Evolution for Planning Request for CS Encryption enabling / disabling

#STRB2.1_change: GAL-DCN-TASI-SYST-X-0738, GAL-DCN-TASI-SYST-X-0875 and GAL-DCN-TASI-SYST-X-0970

Enabling and Disabling the on board encryption of the Commercial Service signal is an operation requiring synchronization of the whole space constellation and of the entire GCSS network and requires coordination from the uncla MKMF.

The evolution of the GCSGMS ICD will resolve the existing inconsistency with GMSREQ-10313, providing the capability to plan all necessary activities at ground and space segment level to enable or disable the CS encryption.

Specific planning request will be sent from the GMS (uncla MKMF) to the GCS (SCPF) to Enable or Disable the CS encryption on board a satellite, including parameters specifying the frequency band on which the operation shall apply (e.g., E6b, E6c, Both).

Once accepted, such requests will become part of the GCS Short term plan and translated in TCs set to the spacecraft.

Feedbacks are sent from the GCS (SCPF) to the GMS (uncla MKMF):

- upon accepting the planning request in the GCS STP (PLN_MtPlanFdback)
- upon execution of the relevant task commands (PLN_MtPlanExec).

4.6.4.7 Evolution of planned GMS Tests activities

#STRB2.1_change: GAL-DCN-TASI-SYST-X-0891

The evolution of the GMS planning function will support “Manual” contacts in parallel to “Operational” contacts. A “Manual” contact will make possible planning, over a certain time window, maintenance and testing activities with a given spacecraft, allocating for that pass a specific ULS and/or GSS, for which the GMS planning function is expected to automatically identify the relevant visibility periods.

For the duration of the Manual contact, the involved Spacecraft, ULS and/or GSS will be detracted from the resources available to plan GMS operational contacts.

The navigation messages for a satellite involved in a Manual contact will have the Signal Health Status (SHS) Flag for all supported services set automatically to “Signal component currently In Test”.

4.6.4.8 MUCF tracking schedule distribution evolution for STRB1.5.1

The evolution of the MUCF tracking plan distribution foresees that upon accepting and validating the GMS STP, both the new and the current active tracking plan (which remains active until the start time of the new plan), are always simultaneously available to the Navigation dissemination function and to ULS.

This ensures that in case of a MGF or ULS reboot, it is possible to immediately resume operations without waiting for a new tracking plan distribution

4.6.4.9 New SV introduction in the GMS MTP planning context

The evolution of the MUCF GMOM/GMPS planning tool foresees the on line introduction of a Satellite in the planning context, without the need for complex off line reconfigurations and application restart.
Not all detailed operational information on the new satellite may be necessary for planning purposes, details shall be required only when needed (e.g., once the plan gets validated).

4.6.4.10 SFC Functional Block Diagram

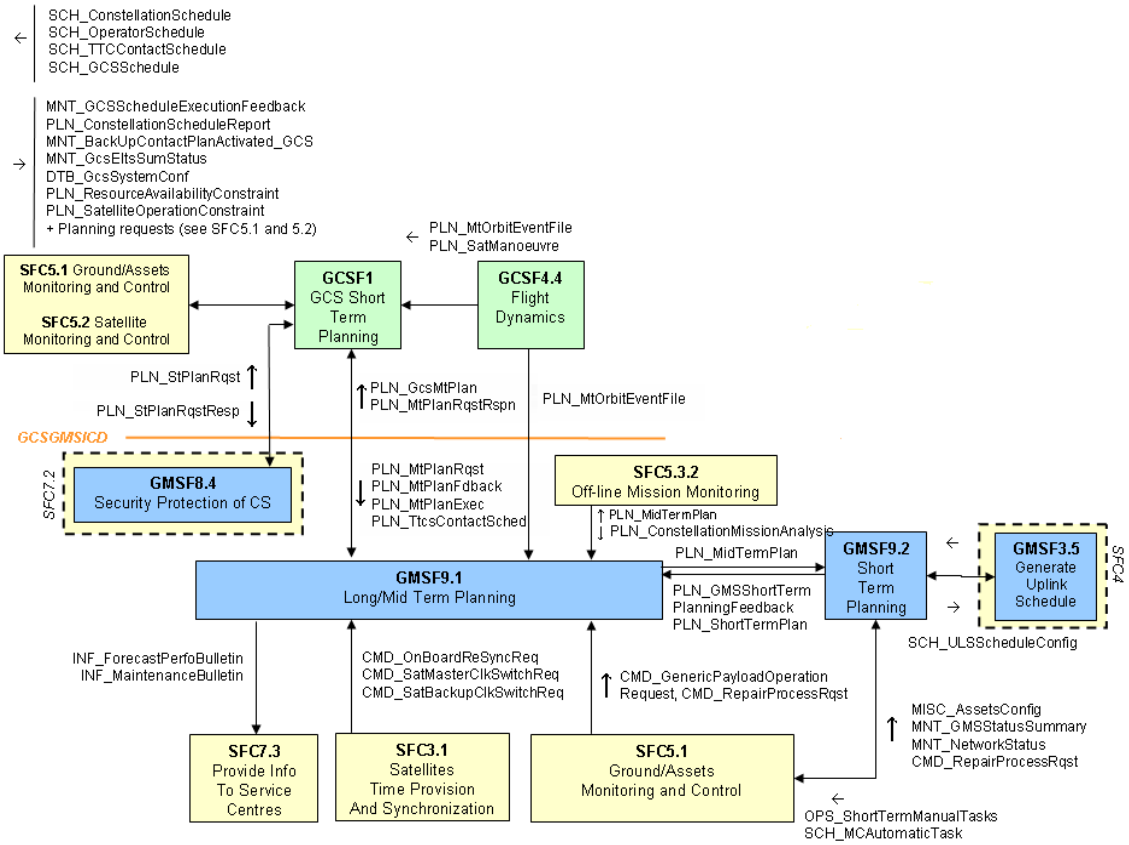


Figure 69 : SFC5.4 Mission Planning Functional Block Diagram

4.7 SFC6 Archiving

This functional chain is in charge of providing all the needed functionalities to archive and retrieve the following data types

- Constellation Management and Satellite Control data (GSRD 738)
- Navigation Processing and Control data (GSRD 744, 799)
- External Entities data (GSRD 759)

The archive strategy applied in the Galileo Ground Segment is basically independent at each GCC.

In each GCC data archival is performed at different levels:

- Short and medium term storages implemented in local/centralised temporary archives at segment and/ or element level
- Permanent long-term storage at system level: this is a central repository supported by the GMS GACF element.

Data to be archived includes among other

- Constellation M&C data, including ground and satellite data, event messages, logs, database, configuration data
- Navigation Processing and Control data, events, logs, configuration files, database
- External Entities data

Data are continuously collected into segment centralised archives or element archives. The transfer to the Long Term Archive (LTA) is performed through migration operations run on scheduled basis that transfer regularly (e.g., every day) the most recent data (e.g. extra data collected locally since the last migration) to the LTA.

In the final system configuration, the daily data volume expected to be archived in the GACF LTA can be up to 110 GB per day, leading to a total archive capacity of over 730 TB in the 20 years mission lifetime.

Short/medium term archives allow retrieving recent data directly from the local archive, without necessarily requiring a retrieve operation from the GMS LTA. In case historical data are not present in the local archive (typically because they are too old), a request for retrieving such data from the GMS LTA is initiated from the interested element.

The GACF LTA s deployed in both GCC and each archives data independently all data exchanged in each GCC. The GACF LTA is designed to be able to archive data for the entire Galileo mission lifetime on hard disks. Although no migration to off line media is necessary, it may be considered to un-mount disks containing very old data from the GACF archive in order to improve the access performance to most recent data. The decision to perform such an operation will require assessing access performance for mounted and un-mounted disks and the possibility to procure additional disk space.

The GACF LTA in each GCC comprises a full redundant local replica, with expandable capacity and can also support back up on Linear Tape-Open (LTO) 3 tapes capable of archiving up to 800 GB of compressed data.

Not that considering the rapid obsolescence of hard disks technology, the archive capacity has been presently limited to 256 TB (vs the target 730 TB covering the entire mission lifetime), with the idea of procuring more performant hard disks in a few years from now. The present capacity, depending on different assumptions on the constellation deployment speed, may be sufficient up store data for the calendar year comprised between 2019 and 2025 (see GAL-RFD-TAS-GMS-400003 v2.0).

An automated synchronisation between the GACF LTA in the two GCCs is scheduled every 24 hours to allow data generated only locally in a GCC to be stored also in the second GACF LTA. The archive process is automated in each GCC and concerns:

1. Real-time data flowing on the RT LAN (e.g. data coming from the GSSs). This is expected to be the dominant data flow for the archive sizing (about 80 GB per day in the full constellation configuration). Note that:
   - GSS RT data are transferred independently to both GCCs, where they are captured by the local GACF LTA.
   - Mission monitoring data transferred on RT LAN to GMS elements in one GCC are also internally cross-exchanged to the remote GCC internally by GSM and hence archived into the local GACF LTA of both GCCs by the local migration process.
   - Being the same RT data available at both GCCs, they are excluded from the inter GCC GACF LTA synchronisation process.
2. M&C data flows transferred via SNMP between GMS elements and GACF and data migrated from GCS and GMS elements to GACF LTA. These data are managed by a GACF which is configured to periodically collect and migrate them to the GACF LTA

   This class of data include Spacecraft constellation TM and TC data, expected not to exceed 10 GB per day. In nominal conditions the migration is local within each GCC. In case of a GACF LTA failure, it is possible to configure the migration to the remote GACF LTA

   The inter GCC GACF LTA synchronisation ensures that data that are only stored at one GCC are also replicated in the remote GCA LTA. Specific measures are taken to ensuring that data are not duplicated in the archive. For example when the same data flow is exchanged between several elements it is archived in the GCAF LTA only once.

   Since constellation TM and TC data are internally synchronized by GCS (nominally Satellite data are transferred to the remote SCCF at end of each Satellite pass), the same constellation data are migrated from the GCS to the local GACF LTA of each GCC. Therefore in nominal conditions this data does not need to be replicated across GCCs by the GACF LTA synchronisation process.
3. Non Real Time (NRT) data exchanged with no GACF involvement. These also include data generated as result of some manual (typically off line) activities carried out by operators. GACF does not automatically capture data flows on the NRT LAN, as these are expected to be migrated in the GACF LTA under control of the source element. Also for this data type, replication is foreseen when necessary via the inter GCC GACF LTA synchronization process.

   Note that presently some files flowing from GCS to MUCF (e.g. MM_NavSBandUplkRpt, PLN_MtPlanExec,PLN_MtPlanFdback, PLN_MtPlanRqst, PLN_MtOrbitEventFile and PLN_TtcsContactSched,), are not included as part of the GCS controlled migration to GACF LTA and therefore not archived.

It is worth observing that in case it is necessary to re-populate one of the two GACF LTA with significant amount of data (e.g. after a failure), specific actions have to be taken (e.g. transferring the data via tapes) to avoid risks that the GACF LTA synchronisation could hang up placing an excessive demand over the Inter GCC link.

Further details on the archive process are described in SYS-RD.2.

This SFC is composed of two sub-SFCs:
- Archive Migration: data storage process
- Archive Retrieval: data retrieval process from the archive

The first level functional tree for the “Archive” SFC is shown below:

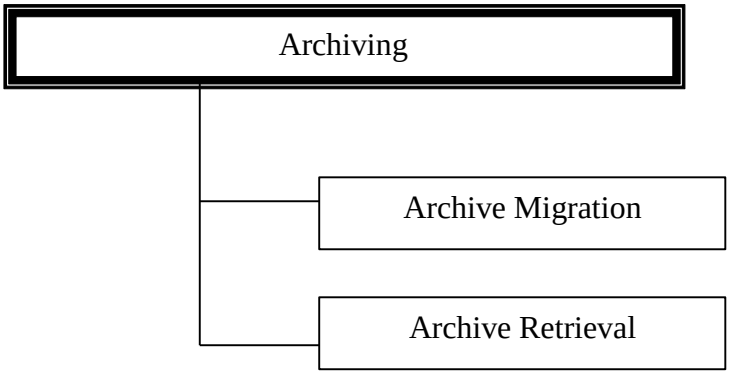


Figure 70 : SFC6 Functional Tree (1st level)

4.7.1 SFC6.1 Archive Migration

4.7.1.1 SFC description

This functional chain is in charge of archiving all the data produced or received by the Galileo system.
Archive Data Types
The Galileo Ground Segment archives the following data types:

- Constellation Management and Satellite Control data: including Databases, TM/TCs, Operational Data, Flight Dynamics
- Navigation Processing and Control data
- Services Performance Monitoring (and Prediction) data and results,
- M&C data including:

  Database
  Events and message logs
  Operator Actions log-files
  Ground assets and network M&C data
  All main processes related data

- All the data exchanged between GMS elements across the GMS networks, unless already recorded otherwise in the central repository,
- All the data exchanged with other external systems across the GMS networks, unless already recorded otherwise in the central repository
- Special data needed to implement the search and rescue function
- Configuration data and software parameters
- Reports and Audit Files

GCS and GMS element events log files are collected by GMS and forwarded to the Galileo Security Monitoring Centre (GSMC) for the analysis of possible security threats.
Events are all categorised in accordance to a common Security Event Dictionary and common xml format.
Unclassified events generated by GCS elements are collected in a GCS centralised server, from where they are migrated to GMS. To reduce the latency, two different priorities are defined for the migration to the GMS, based on the applicable event category, with polling intervals configurable between 0 minutes (immediate) and 24 hours.
GCS and GMS Classified events are collected directly by the GMS PKMF, which is eventually in charge of forwarding both CLA and Uncla events to the GSMC.
Archive types: As per the monitoring and control functional chain, the archive is functionally (and physically) structured in a hierarchical way. Different types of archives are thus identified depending on the type of data to be stored and the duration of their storage. In the Galileo ground infrastructure, the following archives are identified:

- Local archive at element level: the GMS and GCS elements archive locally configuration data, database needed for the element processing functions, working storage areas and where applicable historical Monitoring and Control data storage covering a rather limited time span.
- Satellite local Archive: each satellite shall provide a local TM buffer to store the on-board processed telemetry before downlink to the GCS. It is needed due to the non-continuous downlink between spacecraft and control centre.

  Notes:

  - The satellite local archive size is consistent with a frequency of one ground contact every two orbits at a nominal TM sampling rate.
  - Provided it does not violate any security requirements, it is possible to patch /dump a contiguous memory area by TC specifying start address and number of words or start and end memory address.
- Central Archive at Segment level (GCS, GMS). These are intermediate storage areas acting as intermediate segment data repository supporting:
  - Archival need for multiple elements of the same segment
  - Archival of configuration and processing data (e.g. configuration files, databases, working storage)
  - Mid-term archival of Monitoring and Control data and quick retrieval of historical Monitoring and Control data. The available storage capacity for historical data is usually configured as cyclical buffer covering a time span larger than for the local archive.
- Global Long Term Archive (LTA) for the whole system.
- The GMS is responsible for the central permanent archiving of all data for the entire system lifetime, with the following functionalities:
  - Modular and expandable storage capabilities
  - Permanent data repository (supporting archival for the entire mission life)
  - Reliable storage and resilience to failures
  - Management of archived data supporting retrieval requests.
- Security data of security elements (in particular keys and security audit events) are archived in dedicated CLA archives of both GCSs and GMSs of GCCs for a minimum time of 20 years.

Archiving process: Data Migration process from central archives to global archive:
The Galileo Ground infrastructure provides the means to allow the transfer from the GMS and the GCS of all the data to the permanent data archive, avoiding data duplication.
When the GCS data are not directly acquired by a centralised GMS M&C function, GCS data are migrated from GCS local / central archives to the GMS LTA on regular GCS requests. The GMS answer by a confirmation file containing the transfer status of the migration.
Satellite TM. The following TM types are foreseen:

- Real time TM acquired during S band contacts
- Deferred TM stored in the on board Satellite local archive in periods that are outside ground contact and downloaded to ground on TC request during an S band contact. The on-board buffer has a 48 hours capability managed with 2 processes:

  automatic overwriting of the on-board buffer is enabled and disabled on TC request,
  deletion of the complete on-board buffer is performed on TC request.
  The GCS is responsible for consolidating real time and deferred TM packets storing them within the GCS local archive sorted in chronological order.

A data and software backup technology is also implemented in the Ground Control Centre with data storage equipment, e.g. LTO-tape and replicated synchronised hard disks, tape, CD, or DVD, for permanent off-line backup.

4.7.1.2 SFC Functional Block Diagram

The following figure shows the functional block diagram of the functions and the signal exchanged in the frame of the archiving SFC.
Note: The detail of the exchanges at satellite level is presented in a dedicated figure after.

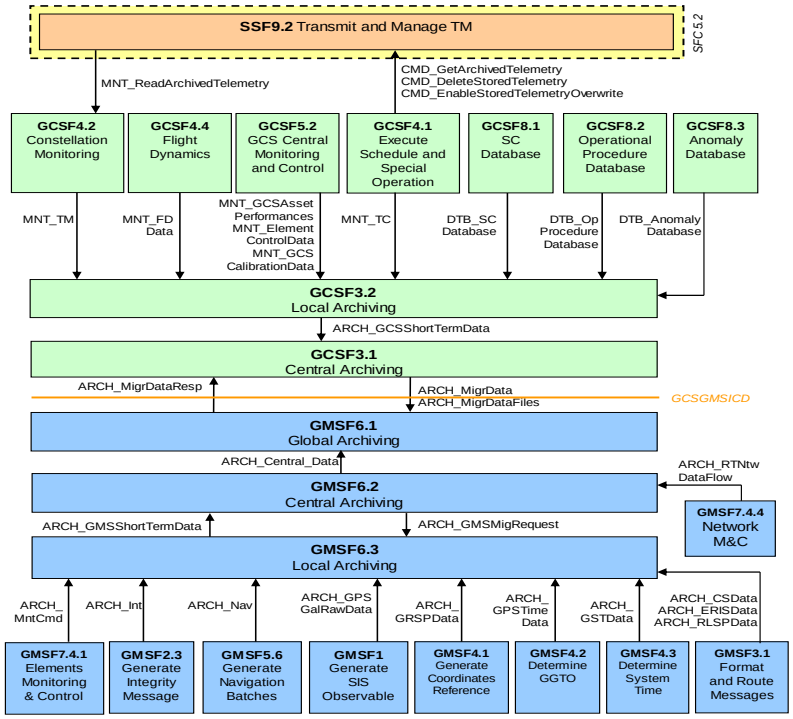


Figure 71 : SFC6.1 Functional Block Diagram

4.7.2 SFC6.2 Archive Retrieval

4.7.2.1 SFC description

The Ground Control Segments have the capability to retrieve archived data for replay and off line analysis:
- replay for display of historical satellite TM&TC
- replay to perform tests and training in a representative off-line environment
- off line analysis to verify mission service performances and mission operational parameters a posteriori
- off line analysis of satellite constellation
- off line analysis and replay of ground element control and mission events

The retrieval and replay process supports configuration management and catalogue functions allowing selective retrieval capabilities of archived data.

Retrieval process:
The retrieval process is initiated by an operator request to the local / central archive or directly to the Global LTA. Historical Monitor and Control data may be retrieved directly from local or central archives, depending on how old are the data:

- Historical monitor and control data can be directly retrieved from a Local archive when the relevant data are typically not older than a few days (up to one week for the TTCF).

- Historical monitor and control data can be retrieved from the Central archives when data are not older than three years.
- In case no data are available in the local or Central archive the operator has to request the data to the GMS Global LTA via a Retrieval Query request.

As the Global LTA is implemented by the GMS, the GCS / GMS interface provides the means to allow the transfer of data to the GCS. The retrieval process between the GMS and the GCS foresees the following steps:

1. An Archive retrieval Query request to the GMS LTA is placed by an operator from GCS to GMS
2. A response to the Query request is provided by GMS back to GCS
3. Based on the Query response, GCS issue a request to GMS specifying the files to be retrieved
4. The required data files are retrieved by GMS from the Global LTA and sent to the GCS together with a separate metadata file detailing the result of the Global LTA retrieval operation

- At end of the process, an Archive response on the retrieval operation is provided by GMS to GCS.

The Archive retrieval query in the Global archive storage is based on signal names using a maximum of 50 characters.
All GCS Security Data are readily retrievable from the GCS CLA archive and useable within 10 seconds from user request.

4.7.2.2 SFC Functional Block Diagram

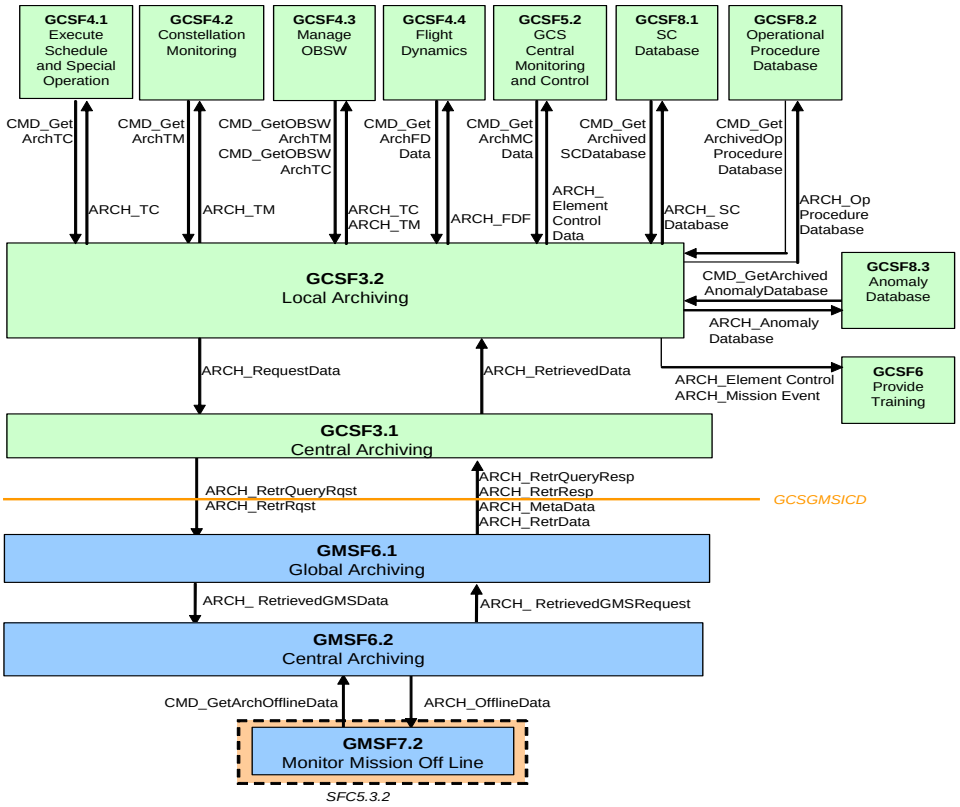


Figure 72 : SFC6.2 Functional Block Diagram

4.8 SFC7 Support Commercial Service

The CS Service Support functional chain is in charge of providing all required functionality to support the dissemination of Commercial Service data received from Commercial Service Providers to the end user.

The first level functional tree for the “Support to CS Services” SFC is shown below.

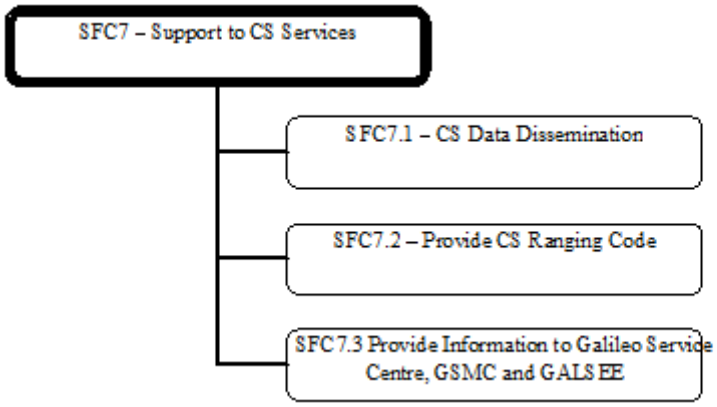


Figure 73 : SFC7 Functional Tree

4.8.1 SFC7.1 CS Data Dissemination

4.8.1.1 SFC description

This SFC is responsible for broadcasting Commercial Service (CS) data provided by external service centres via the Galileo satellites, and in generating dissemination reports to these service centres.

4.8.1.1.1 Global vs. Regional Dissemination

The CS data can be broadcast globally or regionally to the following regions:

1. Global
2. Europe
3. Africa
4. Asia
5. North America
6. South America
7. Oceania

The global dissemination will include all connected satellites and will be exclusive to other regional dissemination (i.e. data can be provided to either the non-overlapping regions (2-7) or to the global region.

4.8.1.1.2 Dissemination process

This dissemination process is composed of the following steps:

a) CS Dissemination grid generation by GMS

The GMS functions in charge of managing uplink schedules (GMSF3.5) is responsible to produce CS dissemination grids (CS-grids) to be provided to the GSC. These grids indicate the list of connected satellites in view of the predefined commercial regions.

Note that the CS dissemination does not constrain the ULS to satellite allocation plan needed to provide the OS I/Nav and PRS mission services.

These grids are generated in advance for a predefined period. In addition, contingency dissemination grids can be generated in case of contingencies (loss of a satellite, loss of an uplink site, loss of an antenna station.

GMS provides Satellite Mask configuration file to GSC in order to allow the identification between Galileo satellites SVID and CCSDS Identifier

b) CS Data routing allocation by the GSC

The allocation of the individual CS data to the corresponding satellites will be performed by the GSC, based on

- the information of the region for dissemination
- the list of connected satellites in view of the target region, indicated in the CS grid

This allocation allows a tailored and optimized dissemination strategy adapted to the individual needs and priorities of the commercial service provider.

c) Nominal CS data dissemination

- The GMS receives commercial service data in real time streams from the Commercial Service providers as specified in GMS to SC ICD.

- The GMS multiplexes the CS data into the Navigation up-link message subframe according to the satellite allocation plan indicated in the CS dissemination grid and up-links them to the satellites as specified in GMS to SS ICD.

Note that there is no start or end dissemination time provided for the CS data. The GMS will start transmission of the CS data just after reception of request on a Satellite basis from the GSC via the MDDN ENE-RT GSC MGF connection. GMS will only uplink data received within the reception window frame internally defined. No cyclic repetition is designed inside GMS.

GMS shall accept Commercial Data received between GST(T)-1 sec and GST(T)-700 msec for uplink that will start in the slot [T; T+1] sec and will end in the slot [T+1; T+2] sec, where T =GST (T).

After reception of the uplink, the satellite builds the navigation message and modulates it with the CS ranging codes.

The considered CS data batch will start leaving the on board antenna in the slot [T+3; T+4] sec and the last bit of the relevant CS data batch will be received by the user receivers in visibility before T+6 sec.

Note:

- The CS ranging code generation and management process are detailed in dedicated SFC7.2 (in particular, process to support the capability to enable or disable the encryption of the CS ranging codes).
- The resulting CS message (CS_CsMessage) is modulated with the E6 carrier. Later on the whole signal is amplified and transmitted in the SIS as specified in SIS-ICD.
- Timing constraint at satellite level for CS data dissemination: each satellite that receives a valid Uplink at the date T(k) including a commercial Service data as specified in GMS to SS ICD shall relay this data in the L-band downlink by the date T(k)+2s corresponding to the last bit transmission in accordance with SIS ICD.

- Time elapsed since the last bit of commercial data is received by the Galileo Core Infrastructure to the time when the last bit of commercial data leaves the satellite antenna output shall be 6 seconds (95%).

#STRB2.0_change: (corresponding to the regularisation of 6s budget): GAL-DCN-TASI-SYST-X-0858, GAL-DCN-ASTG-SYST-X-13497 and GAL-DCN-ASTG-SYST-X-13484

The transmission of the valuable CS-data to the users will be stopped by either:

- GSC stops sending CS for the specific satellite, which are used by GMS to fill the corresponding parts of the uplink message, or
- The corresponding satellite is no longer connected to a ULS and dummy data are filled in by the satellite.

d) Contingency CS data dissemination

When no commercial data is available on ground Dummy packets are uplinked in the Uplink 'Message' sub-frame replacing the missing CS data.
Then the satellite generates and disseminate CS dummy message (C/NAV-0).

e) Monitoring of CS dissemination

The GMS regularly provides the service centres with up-link status reports. These reports contain the status of the commercial data stream and are based on the monitoring of the successful uplink to the satellites. The message is sent on a regular basis (i.e. not on demand or occurrence of a status change) and reports the status of CS data messages scheduled for uplink in the preceding 30 minutes. No monitoring at user level is provided

4.8.1.2 SFC Functional Block Diagram

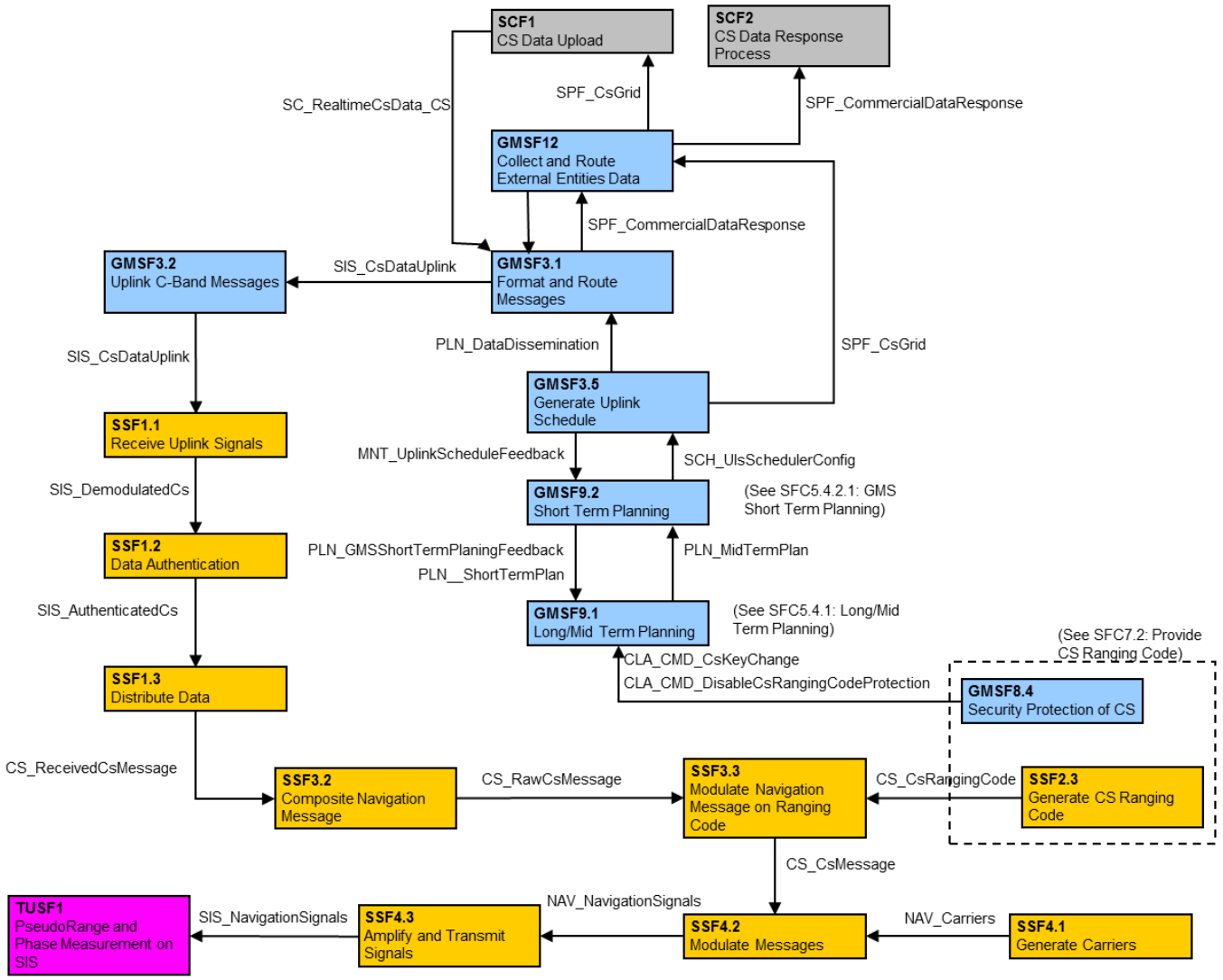


Figure 74 : SFC7.1 Data provision Functional Block Diagram

4.8.2 SFC7.2 Provide CS Ranging Code

Please note that this section is pending the conclusions on CR18 and CR 19 related to CS implementation and E6 encryption.

4.8.2.1 SFC description

Galileo provide means to perform ranging measurements on a so-called third carrier (TCAR).

The provision of the CS ranging code is protected to control the user access. The related functions can be categorised as follows:
1. Protection of CS Ranging Code
2. Remote NSGU Management
3. Key Management for CS

4.8.2.1.1 Protection of CS Ranging Code

The Protection of the CS Ranging Code includes the following functionalities:

Encryption of Ranging Code:

Controlled access to the “E6” ranging code by means of encryption of the ranging code by the satellite.

Enabling/Disabling of Protection:

This protection can be (activated or) deactivated by command from GMS on request of the GSC.

Dissemination of Encryption Status:

The actual protection status is signaled over L-band via a status flag in the I/NAV message which is set by the satellite.

Monitoring of Ranging Code:

- The CS third carrier protection status is verified by GMS, which is capable to access the signals for TCAR on L-Band.
- CS receivers shall be able to acquire and track the encrypted Commercial Service ranging code on the "E6" carrier.

4.8.2.1.2 Remote NSGU Management

The CS functionalities of the NSGU are managed according to the following capabilities:

- The CS ranging code encryption key can be distributed via telecommand to the NSGU (via GCS).
- Failure of key integrity or authentication of the CS ranging code encryption key detected in the satellite are reported via telemetry (standard TM) to GMS.

4.8.2.1.3 Key Management for CS

Key Management:

- The CS ranging code encryption key can be changed in the satellite by command from the GMS.
- The CS ranging code encryption key can be distributed manually (i.e. using removable media) from GMS to CS Test User Receivers, Ground Segment CS Receivers and the CS Service Provider
- The CS ranging code encryption key can be distributed remotely (i.e. via network links) from GMS to Ground Segment CS Receivers and to the CS Service Provider.
- GMS informs the service centres 7 days in advance when the CS ranging code encryption keys are going to be changed (in accordance with GMS to SC ICD).
- The GSC may request GMS to change the CS ranging code encryption key. When requested to change this key, this change will be implemented within 30 days time, in the ground assets and in the satellites.
- A key change for the CS ranging code encryption in a satellite shall take place without any interruption to the signal output from the satellite.

#STRB2.1_change: GAL-DCN-TASI-SYST-X-0738: On Line NSGU CS Key status Report and CS Encryption mode planning, GAL-DCN-TASI-SYST-X-0970 CS KEY TM DATA PROVISION

GMS receives from GCS the CS NSGU Key Status Report with information transmitted by satellite to GCS in the S-Band telemetry. This CS NSGU Key Status Report shall consist of:

- NSGU status (NSGU A ON, NSGU B ON or both NSGU OFF).
- CS Key Type (Base, Traffic E6b, Traffic E6c, Protection)
- List of CS Keys on-board:
- CS Active Key Index

- CS Key CRC

4.8.2.2 SFC Functional Block Diagram

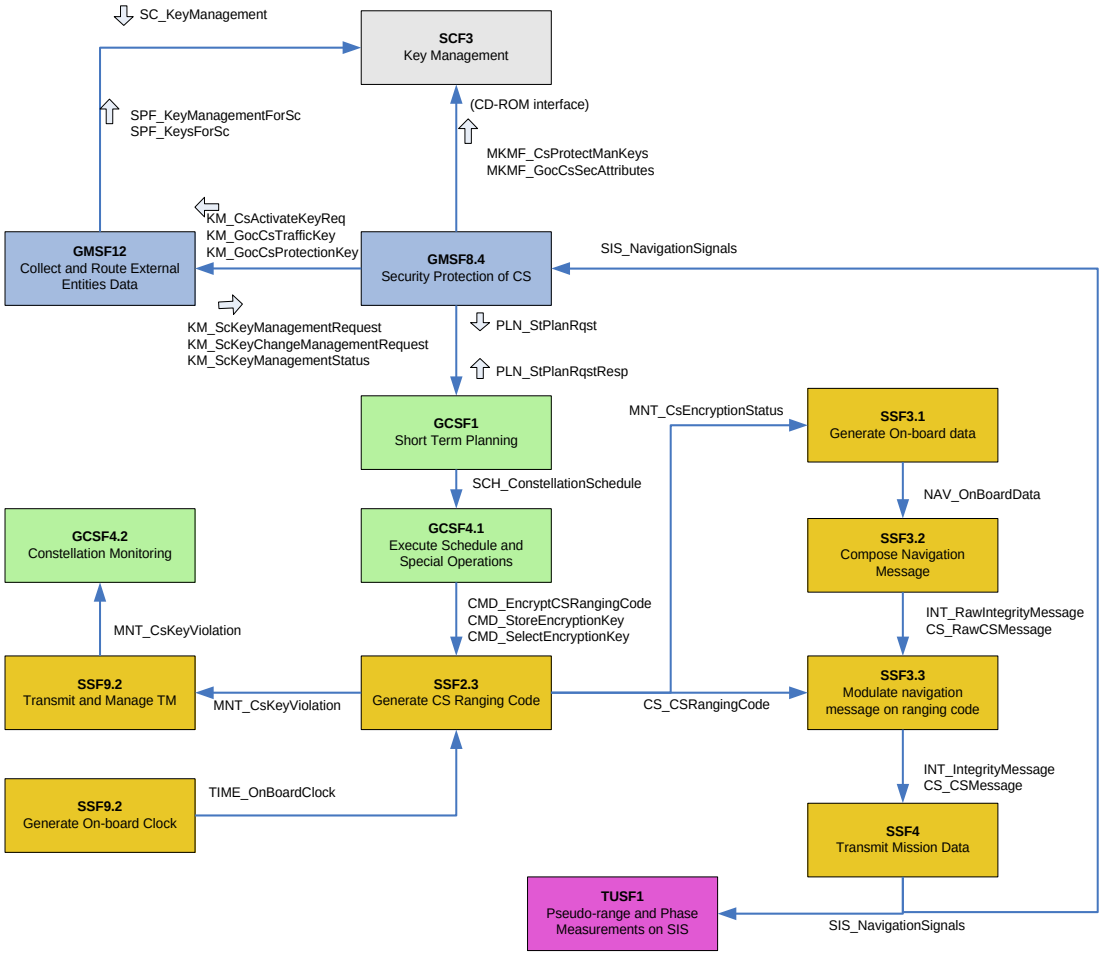


Figure 75 : SFC7.2 Functional Block Diagram

4.8.3 SFC7.3 Provide Information to GSC, GSMC and GALSEE

4.8.3.1 SFC description

This SFC is responsible for providing information to the GSC, the GSMC and GALSEE

4.8.3.1.1 Information provided to the GSC

The Galileo System is responsible for providing to the GSC information related to different products obtained from Galileo Core Infrastructure.
This includes:

- service and system performance reports
- Galileo component overall status
- maintenance and forecast bulletins
- Mission data generated by OD&TS algorithms: almanac, clock corrections, ephemeris, ionospheric data…,
- archived information

- EDBS status.
- Real-time Raw data related to GPS and Galileo receivers, with a maximum latency of 5 seconds (95%)
- Navigation data (F/NAV, I/NAV, C/NAV) included in the C band uplink messge **(#STRB2.1_change: GAL-DCN-TASI-SYST-X-1021, GAL-DCN-TASI-SYST-X-1023)**

The Galileo Core Infrastructure will also be able to receive anomaly notifications, report request and new product request form GSC.

The detail of the data exchanged between the Galileo system and the Service centres is given below:

a) Reports (SPF_ReportToSc) generated by the GMS including:

- Service (INF_DailySrvcPerfoRpt) and System (INF_DailySystemPerfoRpt) statistical performance reports computed on a daily basis. These reports contain service-level and system-level data computed and collected every day by the GMS Mission Monitoring functions (GMSF7). They enable to indicate daily the level of performance of the Galileo system and services to the SC.
- Long-term Service (INF_LTSrvcAssessmtRpt) and System (INF_LTSystemAssessmtRpt) performance assessment reports computed on a monthly, quarterly, and yearly basis. This report contains service and system performance data computed by the GMS Mission Monitoring functions (GMSF7). They enable to indicate the performance level of the Galileo System and Services over a long past period to the SC.
- Service Performance Prediction reports generated 3 days in advance for a 24-hour period (INF_SrvcPerfoPredictions). These prediction reports include for global and designated areas, and for each service, what would be the service performance for the area of interest during these 24 hours. They are computed by the GMS Mission Monitoring functions (GMSF7)
- Specific information on Galileo satellites including:
  - Galileo satellite almanacs (INF_AlmanacData): contains reduced-precision subset of the clock and ephemeris parameters, including the associated time of applicability, for the active SVs in orbit. Parameters list as almanac in SISICD.
  - Galileo Satellite event bulletins computed by the Mission Monitoring function (GMSF7) (INF_NoticeAdvisoryToUsers). This includes information on satellite planned events (such as scheduled maintenance events, notification that a satellite is declared healthy for the first time, etc.) and unplanned events (like unscheduled outages) having impact on the quality of service or on the execution of the mission. They are also known as NAGUs (Notice Advisory to GALILEO Users). NAGU provision can be activate/de-activate by GMS operator.

    Remark: *the automated NAGU generation concept is currently not consolidated in the STRB2.0 baseline. It is expected that the content of the signal INF_NoticeAdvisoryToUsers will be refined in close coordination with GSC in the frame of GSC operational concept consolidation. In the meanwhile, the signal INF_NoticeAdvisoryToUsers will not be disseminated to the external entity and the NAGU will be generated manually and sent to the GSC. In both cases (automated and manual NAGU processes) the authorization for dissemination to the public is granted by the corresponding formal process at GSC level.*

- Specific information on Galileo components including the Galileo Overall Health status (INF_GalileoOverallStatus) computed by the Mission Monitoring function (GMSF7). It contains the summary of health status of constellation spacecraft and other major system elements
- Specific information from the Mission data determination chain including:
- OD&TS Data broadcast in the navigation message (INF_ClkEphemIonoPredData) as computed by the OD&TS functions (GMSF5).

- OD&TS precised restituted Data (INF_ClkEphemIonoRestData) computed by OD&TS (GMSF5) and Mission monitoring (GMSF7) functions. It contains the high precision Clock, Ephemeris and Iono restituted data.

- Additional Specific observable information including:
- RINEX3 Observation files (INF_RinexGalData) based on archived observables
  - RINEX3 Navigation files (INF_RinexGalNavMsg) based on archived navigation data
- Estimated (Restituted) orbits (INF_OrbitFromMsf) and Clocks (INF_ClkFromMsf) from MSF for the Galileo SVs
- ULS Pointing Data for Svalbard Antennas (INF_UlsPointingData) :
- Pointing data file for each ULS chain installed in Svalbard. It contains Antenna pointing data samples (max. 1 sample per minute) including Azimuth, Elevation and Satellite ID. Data to be provided only to the Norwegian NSA **STRB2.0_change: GAL-DCN-ASTG-SYST-X-13900#**
- Galileo orbital data and clocks for SAR User community. The Galileo System is responsible for providing to the GSC with orbital data and clocks to be then distributed by GSC to the SAR User community including :
  - Os and SoL predicted clocks for all Galileo satellites in RINEX Clock Format, containing 7 days of prediction
  - Os and SoL predicted orbits for all Galileo satellites in SP3 Format, containing 7 days of prediction with an accuracy better than 50 meters 3D RMS.
  - Spacecraft Ephemeris data for all Galileo satellites calculated by the GMS with an accuracy better than 50 meters 3D RMS for the validity time span of each Ephemeris

#STRB2.0_change: GAL-DCN-TASI-SYST-X-14165, GAL-DCN-ASTG-SYST-X-14198, GAL-DCN-TASI-SYST-X-1055: Provision of Galileo orbital data for SAR user community

b) Bulletins (SPF_BulletinToSc) generated by the GMS including:

- Planned maintenance bulletin (INF_MaintenanceBulletin): containing information on all maintenance activities on the system resources as planned in the mission planning generated by the mission planning functions (GMSF9). This includes the scope of maintenance, duration and concerned elements.
- Performance Prediction for planning validation (INF_ForecastPerfoBulletin): containing the results from the comparison between the required performances and the system performances computed by the mission planning validation function (GMSF9).

c) Report Requests (SC_ReportRqst) from the GSC including:

- Request from GSC to obtain particular data/reports from Galileo system (INF_ReportInfoRqst): request for a standard report to be generated and transmitted at an instant which is not prescheduled.
- Request for specific data, reports or information which are not included in the standard product catalogue (INF_NewProductRqst).

d) Standard Anomaly Notification from the GSC

Standard Anomaly Notifications generated from the collection and filtering of User Anomaly Reports by the operator of GSC (INF_AnomalyNotif). Additional Specific observable information including:

- RINEX3 Observation files (INF_RinexGalData) based on archived observables
- RINEX3 Navigation files (INF_RinexGalNavMsg) based on archived navigation data

e) Real Time raw data related to GPS and Galileo receivers from GMS including:

Galileo SIS data messages and Galileo pseudorange Measurements from the non-PRS receiver of each GSS at 1 second GST epoch rate

- GPS Navigation and Measurement data from E1 /L2 received from the GPS receiver of each GSS at rate of 1 second.

f) Navigation data (F/NAV, I/NAV, C/NAV) selected for uplink to space segment including

4 batches (F/NAV, I/NAV) of navigation data per satellite, as per F/NAV I/NAV message description in GMS-SS ICD, in SPF_DataForUplink with the uplink timestamp of the batches

- Commercial Service data selected for uplink as per C/NAV message definition in GMS-SS ICD and uplink timestamp

4.8.3.1.2 Information provided to the GSMC

The Galileo Core Infrastructure responsible for providing bulletins to the GSMC including information related to status of Galileo Core Infrastructure, service performance, notice advisory to users, maintenance, forecast performance, and security.
This includes:

a) Reports (SPF_ReportToGSMC) generated by the GMS including:

- Summary of health status of constellation spacecraft and other major system elements (INF_GalileoOverallStatus) provided by the Mission Monitoring function (GMSF7).
- Service-level data for reporting performances computed every day (INF_DailySrvcPerfoRpt).
- Reports with predictions on service performances generated 3 days in advance for a 24-hour period (INF_SrvcPerfoPredictions).
- Bulletins on planned events include information regarding the forecasted events (such as scheduled maintenance events, notification that a satellite is set healthy for the first time, etc.) and unplanned events (like unscheduled outages) having impact on the quality of service or on the execution of the mission (INF_NoticeAdvisoryToUsers). NAGU provision can be activate/de-activate by GMS operator.

b) Bulletins (SPF_BulletinToGSMC) generated by GMS including:

- Information on all maintenance activities on all system resources within the current planning time-frame, including scope of maintenance, duration and concerned elements (INF_MaintenanceBulletin).
- Results from the comparison between the required performances and the system performances assessed from the data of the validation function, planning reports from log data, and NAGUs Forecast Performance Bulletin (INF_ForecastPerfoBulletin), this signal is for internal use by the GSMC only.

c) Security information generated by GMS (classified) including:

- PKMF_ClassifiedData. More information can be found in the classified DDF.
- GSMC_ClassifiedData. More information can be found in the classified DDF.

4.8.3.1.3 Information provided to the GALSEE

The Galileo System is responsible for providing to the GALSEE information related to different products obtained from Galileo Core Infrastructure.
The detail of the data exchanged between the Galileo system and the Service centres is given below:
#STRB2.0_change: GAL-DCN-ASTG-SYST-X-13616, GAL-DCN-ASTG-SYST-X-13537, GAL-DCN-ASTG-SYST-X-13912 and GAL-DCN-ASTG-SYST-X-13974 request to provide additional data

a) Reports (SPF_ReportToGalsee) generated by GMS including:

- Galileo satellite almanacs (INF_AlmanacData): contains reduced-precision subset of the clock and ephemeris parameters, including the associated time of applicability, for the active SVs in orbit. Parameters list as almanac in SISICD.
- OD&TS Data broadcast in the navigation message (INF_ClkEphemIonoPredData) as computed by the OD&TS functions (GMSF5).
- RINEX3 Observation files (INF_RinexData30s, TIME_RinexData30s) archived GAL and GPS Pseudorange and phase measurements for non-PRS and PRS receivers collected at each GSS and PTF combined receiver and related to all Galileo and GPS satellites, sampled every 30 seconds.
- RINEX3 Navigation files (INF_RinexGalNavMsg, TIME_RinexGalNavMsg) based on archived GAL navigation messages from non-PRS and PRS receivers collected at each GSS and PTF combined receiver
- GMS archived files in GACF format including among other information the following data:
  - MM_RawData: GMS mission monitoring messages and associated signals
  - NAV_RawData OSPF produced messages and associated signals containing navigation messages sent to MGF for uplink.
  - INF_SIS_RawData UDP streams extracted from GACF that contains, among other information, SIS_RawData, SIS_RawMeasurements, SIS_PrsData and SIS_PrsMeasurements
  - NAV_SBandUplkDataFile SBand uplink date archived at GACF, after data migration from MGF (Corresponds to GMS message MGF_SBandUplk_NAV from MGF to GCS-SCCF).
  - DISS_RawData containing the MGF produced Message and integrity sub-frames.
  - CNF_SvMask Configuration file including a correspondence table between Identification of the Galileo satellites SVID and CCSDS Identifier
- Estimated orbits and clocks for all Galileo SVs (SPF_EstimatedDataToGalsee) including:
- File in Clock RINEX standard format (INF_ClkFromMsf) containing precise estimates for satellites and station clocks generated by GMS, sent to the SC for validation. The clock data provided by MSF will be the ones calculated for OS (I/NAV) service
- File in SP3 standard format (INF_OrbitFromMsf) containing precise estimates for orbits. The orbit provided by MSF will be the ones calculated for the OS (I/NAV) service.
- Delta Time (SPF_DTimeToGalsee) with measurements files containing the ΔTi=Ti-GST information (where Ti is the i-th reference, i.e. UTC(i) laboratories) as available from PTF and measurements files from TWSTFT containing the ΔT= GST-UTC(USNO) as available from PTF including:
  - TIME_GstTaiOffset
  - TIME_GalClkTwstftMeas
  - TIME_GalClkCvMeas
  - TIME_GalGpsGgto (from PTF only)
  - TIME_GpsClkTwstftMeas
  - TIME_GpsClkCvMeas
- On-board clocks phase difference (SPF_OnBoardClockToGalsee) containing:
- MM_SatSumStatus
- Orbits data (SPF_FdfOrbitDataToGalsee) provided by FDF, containing:
  - vector defining the orbit immediately before and after a manoeuvre (MISC_OrbitPred)
- information on satellite manoeuvre announcements (MISC_ManvrAnnounce)
  - S-Band ranging data (SPF_TtcSBandToGalsee) provided by TTC Stations, including:
  - Ranging data, including calibration (MISC_SBandSatRngData)
  - Doppler data (MISC_SBandSatDopData)
  - Antenna azimuth and elevation angles (MISC_SBandSatAngData)
  - Meteo data (MISC_SBandSatMetData)

- Galileo Satellites configuration data and default static parameters for the satellite configuration (SPF_SVCfgDataToGalsee), including:
  - CNF_SatCfgData
  - CNF_SatCfgDataStatic
- 3D coordinates and velocities of the GSS tracking antennas to the GCC for enabling ODTS functions (SPF_GssCoordDataToGalsee), including:
- CNF_GssCoordinatesVal
- Data from GSS external site environment sensors (SPF_GSSEnvDataToGalsee), including:
- Meteo monitoring data on GSS site (MNT_SiteData)
- Short Term planning for both ULS and TTC antennas (SPF_ContactPlan), including:
  - SCH_OverallUlsContactPlan : Containing the ULS to Satellite connection plan used for dissemination of the navigation messages
  - SCH_UlsTrackPlan : Containing specific ULS to Satellite connection plan used for dissemination of the navigation messages and almanacs for the satellites included in the plan
  - PLN_TtcsContactSched: Containing Short Term TTCs contact schedule.
- S-Band navigation uplink report (MM_NavSBandUplkRpt), containing the report provided by GCS regarding the uplinking of navigation data in S-Band using the TTC stations
- Notifications of GMS-GALSEE interface problem through operational interface (direct communication not using the M-EDDN interface but mainly email exchange). The following communications are envisaged:
  - GSS Data Missing: Report from GALSEE of Galileo/GPS RINEX files missed.
  - GACF Extracted Data Missing: Report from GALSEE of GACF archived data missed.
  - Interface Failure ; GMS-GALSEE Interface report problem from GALSEE
  - Interface Failure ; GMS-GALSEE Interface report problem report from GACF

4.8.3.2 SFC Functional Block Diagram

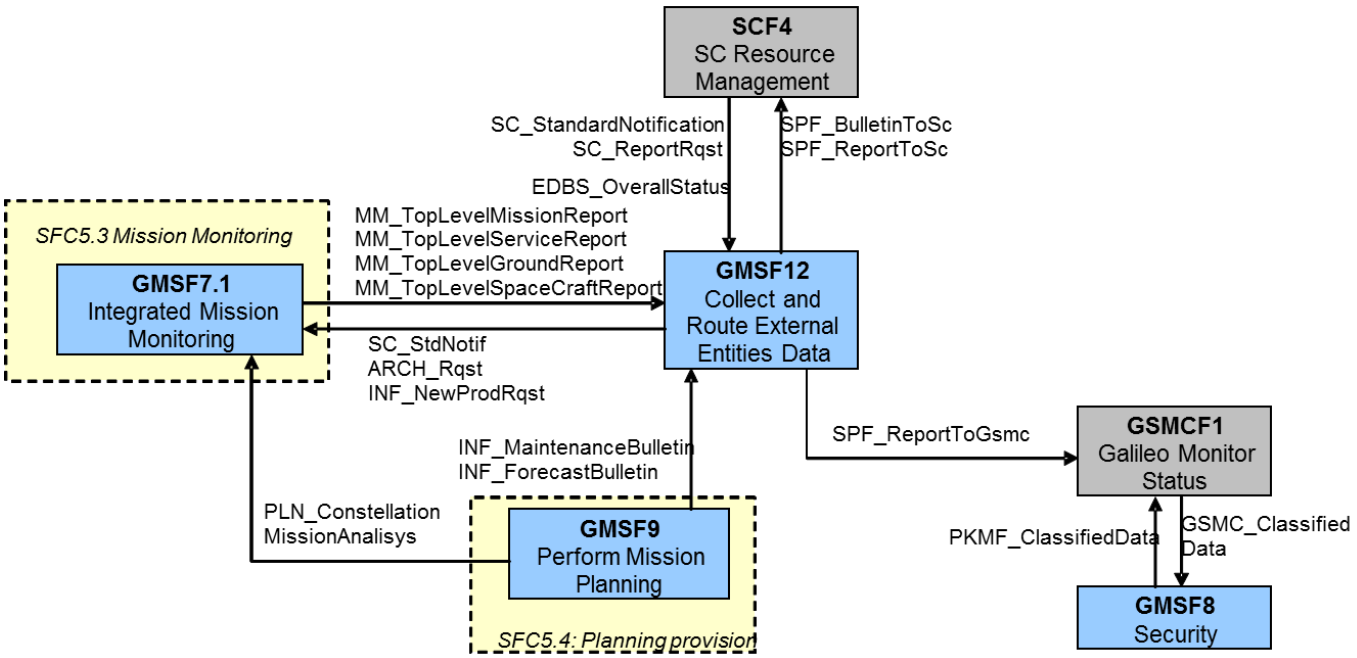


Figure 76 : SFC7.3 Functional Block Diagram for GSC and GSMC

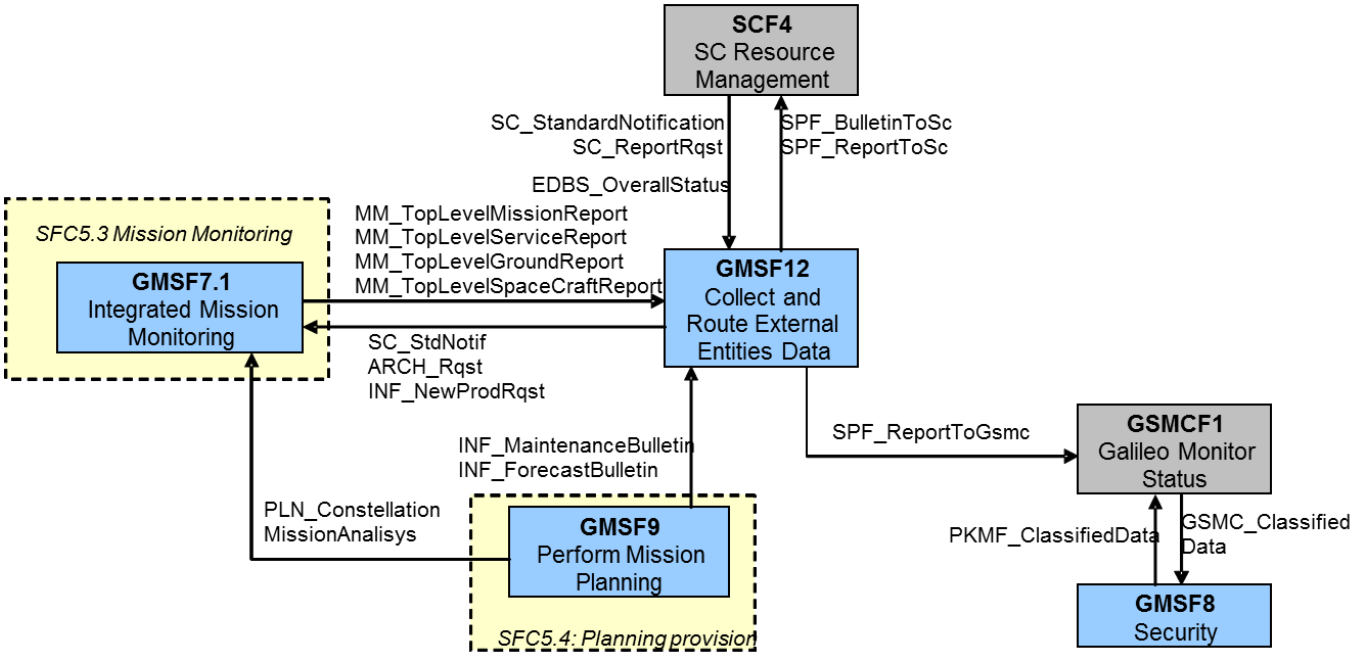


Figure 77 : SFC7.3 Functional Block Diagram for GALSEE

4.9 SFC8 Provide SAR Service

4.9.1 SFC8 Presentation

4.9.1.1 SFC8 High Level Description

The SAR Services functional chain shall provide all required functionality to ensure the Galileo system can provide the SAR forward (Alert) link and Return Link services for SAR users.
The sub SFC's are identified by the different functionality requirements of the SAR service:

- SFC8.1: SAR Forward Link Service
- SFC8.2: SAR Return Link Service, Including the following further sub SFC's:
  - SFC8.2.1 SAR Return Link Resource Management
  - SFC8.2.2 RLM Dissemination and Monitoring
  - SFC8.2.3 SAR Return Link Interface to Cospas/Sarsat

The first level functional tree for the "SAR Service" SFC is shown below.

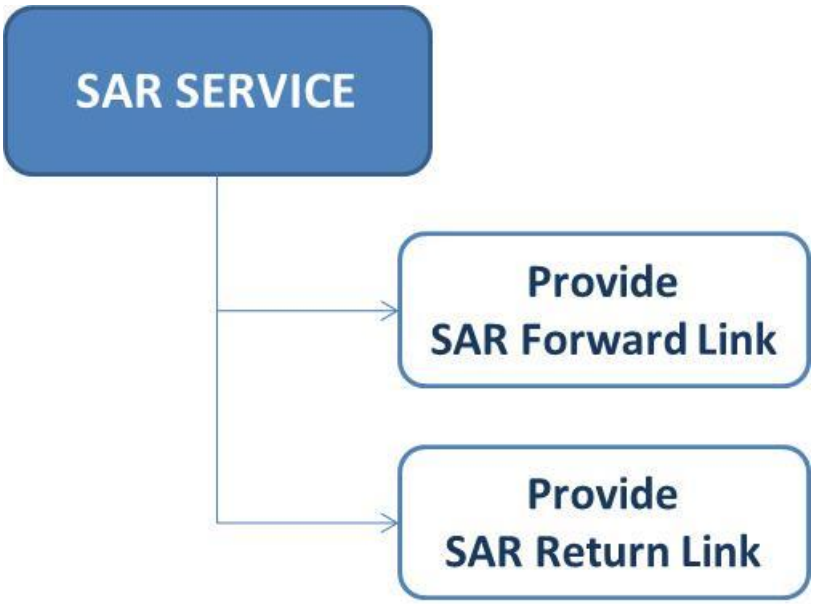


Figure 78 : SFC8 Functional Tree (1st level)

4.9.1.2 Definitions

COSPAS-SARSAT definition: The COSPAS-SARSAT system is designed to assist Search and Rescue activities on a worldwide basis by providing accurate, timely, and reliable alert and location data to the Rescue services at national level.
Forward Link Service Definition: The Forward link service is the service which provides the relay of SAR distress signals from Beacons to the MEOLUT and distribute resultant alerts to a Mission Control Center (MCC).

The following user communities are considered:
- Aviation (using Emergency Locator Transmitters - ELT)
- Maritime (using Emergency Position-Indicating Radiobeacons - EPIRBs)
- Other distress scenarios (using Personal Locator Beacons - PLBs)

Forward Link coverage area definition: coverage area defined by the following points (latitude/longitude), connected by geodetic lines:

SGC-1: 84.00° N, 41.20° E
SGC-2: 29.18° N, 36.07° E
SGC-3: 05.00° N, 38.00° W
SGC-4: 75.76° N, 77.87° W

Return Link Service Definition: The return link service is a new service that SAR Galileo system will provide. Return Link service provides SAR beacons equipped with Galileo receivers a feedback confirmation (Acknowledgement Return Link Message) of the successful transmission of their distress message sent as part of the Forward link service. Other possibilities to provide different types of relevant information to the distress beacon are under study outside of the current contractual perimeter.
The following categories of users are envisaged:
Users of the SAR forward link Service with a Galileo receiver integrated as part of (or an add-on) to their Distress beacon.
SAR Ground Segment for Forward Link Service Definition

The SAR Ground segment is responsible for ensuring the COSPAS/SARSAT forward link service for the European SAR Coverage. It is composed by the following elements: European Meoluts, MTCF, SAR network and 10 RLS Emergency Beacons for test purposes

MEOLUT definition: The Medium Earth Orbit Local User Terminal receives from the SAR Beacon the distress signal/s relayed by GNSS MEOSAR satellites (Galileo, GLONASS and GPS) over a specific European SAR/Galileo Coverage Area
MEOLUT functions:

- to receive alerts from beacons, demodulate and decode messages,
- perform localization using demodulating the beacon message or tri-lateration methods based on differences of time and frequency of arrival from several satellites
- to send alert to MCC

MTCF definition :
The MTCF (MEOLUT Tracking Coordination Facility) is responsible for the coordination of the tracking schedule of the European MEOLUTs and the External MEOLUTs in order to maximize the SAR forward link coverage capability. This results in the generation and dissemination of the European MEOSAR Satellite Tracking Plan. MTCF contribute to the TOA/FOA exchanges between MEOLUTs to support the coordinated behavior of the MEOLUTs.
MTCF is also involved in the centralized functions such as the Monitoring and Control functions, Archive Data Management functions and KPI collection, computation and reporting.

MEOSAR definition: Medium Earth Orbit SAR system for the relaying of 406 MHz SAR Beacon alerts, encompassing the use of MEO satellites for the retransmission of SAR Distress beacon alerts.
Space segment Coverage area definition: coverage area is defined by elevation angles > 10° (i.e. +/- 12.25° off-nadir).
SAR Beacon definition: The SAR beacon is an equipment which once activated, transmits distress signals. The Galileo system supports beacons operating at 406MHz. Some future SAR Distress beacons may have Galileo receivers as an integral part of the beacon to allow exploitation of the SAR Return Link service, and also the possibility to embed receiver position in the forward link message.

4.9.2 SFC8.1 Provide SAR Forward Link Service

The SAR Forward Link Service consists in :

- Rely beacon bursts transmitted by SAR beacons
- reception of the beacon distress signals (relayed by Galileo satellites) by the SAR / Galileo MEOLUTs
- processing of beacon distress signals at SAR / Galileo MEOLUTs
- provision of 2D accuracy according to the performance requirements

MEO Local User Terminals (MEOLUT) operate at COSPAS-SARSAT standards.

SAR/Galileo Forward Link service shall be provided at all times and at all places within the specified coverage to all COSPAS-SARSAT approved SAR beacons and under default operational conditions.

SAR/Galileo Forward Link exploits the MEOSAR transponders on board GLONASS (SAR/GLONASS) and GPS(DASS) satellites

4.9.2.1 SFC description

In the Forward Link Service the SAR/Galileo Space Segment will receive the SAR distress beacon signals from transmitters located anywhere on Earth at any time. After receiving the signals, the SAR/Galileo Space Segment will filter, translate in frequency, amplify, filter the received signals and broadcast them to the ground stations (the MEOLUTs).

SAR Forward Link service shall be guaranteed also in presence of external emissions.
The Galileo System supports up to 450 Distress-Active SAR Beacons, uniformly distributed worldwide.
The following tasks are executed as part of the SAR Forward link function onboard the Galileo Satellites:

- "Receive signals containing messages from SAR beacons in the 406 MHz band with a receiving chain as defined in [AD 19].
- Frequency-convert and amplify these signals - Simple non-inverting transparent frequency translation from the 406 MHz band to the 1544.05 MHz band. To allow accurate frequency stability of the SAR distress downlink signal the onboard clock reference is used from the payload Clock Monitoring and Control Unit.
- Transmit these signals immediately towards the COSPAS/SARSAT MEO Local User Terminals (MEOLUT) at a frequency of 1544.05 MHz with the signal characteristics defined in [AD.19].

Out of the Galileo specified coverage area, Galileo space segment contribute to the dissemination of the beacon bursts through the SAR Payloads.
In fact the GALILEO satellites shall be usable for distress beacon signal detection and localization by any COSPAS-SARSAT approved MEOLUT SAR/Galileo Ground segment (SGS) shall support the tasks associated to the SAR/Galileo Forward link:

- Detect the Distress signals relayed by MEOSAR satellites included into the constellations GPS (DASS), GLONASS (SAR/GLONASS) and GALILEO (SAR/Galileo);

1. Demodulate, Decode and Validate the message contained within distress signals;

- Determine the location on Earth of the distress signal emissions through the Location Protocol Message (if available) and Difference of Arrival methods (DOA)

Note: this section is not meant to provide the description of the algorithm used by the MEOLUTs to locate the beacon. It contains the general principles of Difference of Arrivals (DOA) location method based on TOA and FOA

The SAR Ground Segment shall provide localization of the beacon burst through TOA and FOA method.

TOA is the time of arrival of the beacon burst at the satellite, FOA is the frequency of arrival of the beacon burst at the satellite. Each MEOLUT receives a burst at a certain time and at a certain frequency namely TOA_{MEOLUT}/$FOA_{MEOLUT,}$ then it calculates the TOA/FOA using the detailed information on the satellite positioning available (Ephemeris).

If there are N pairs of TOA/FOA observations available it is possible to build a [2Nx1] matrix, where the [Nx1] sub-matrix contains the linearized TDOA observations converted from seconds to meter and the other [Nx1] sub-matrix reports the linearized FDOA observations converted from Hz to m/s

$$\Delta R_i = (TOA_i - TOA_{ref})c$$

$$\Delta \dot{R}_i = (FOA_i - FOA_{ref})\frac{c}{f_b}$$

with f_b, the beacon transmission frequency, and c the speed of light and *ref* refers to the reference satellite, relative to which the DOA observations are built

This [2Nx1] matrix is part of a linearized observation equations system, that allows, starting from approximate position and velocity of beacon and satellites, to converge to an accurate beacon localization.

The system to be solved in order to determine the beacon location is shown below:

$$\begin{bmatrix} \Delta R_i \\ \Delta \dot{R}_i \end{bmatrix} = \begin{bmatrix} H_{TDOA} \\ H_{FDOA} \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \\ \Delta z \end{bmatrix}$$

The H_{TDOA} and *HFDOA* contain differences between beacon and satellite coordinates and velocity along the three directions (being $\{x_b, y_b, z_b\}$ and $\{\dot{x}_b, \dot{y}_b, \dot{z}_b\}$ the approximate position and velocity of the alert beacon to be localized and $\{x_i, y_i, z_i\}$ and $\{\dot{x}_i, \dot{y}_i, \dot{z}_i\}$ the approximate position and velocity of the satellite (i= 1, 2…N))

Finally the matrix at the right contains the 3 beacon position unknowns, where the first two elements represent the horizontal position and the third one the vertical position

2. Transmit all information on detected distress beacons to the COSPAS-SARSAT operational network

- Locate non-SAR 406 MHz in-band signals interfering with the system anywhere in the coverage area
- SAR/Galileo system monitoring and Support Service Validation campaign by collecting and monitoring the Key Performance Indicators and generating the related reports.

The SGS functions are allocated to the following main components:

- **Three European MEOLUTs** responsible for the Tracking of MEOSAR satellites (GALILEO, GPS/GLONASS), receive and process the SAR downlink messages and locate the emergency beacons, as well as providing the relevant SAR distress data to the associated COSPAS-SARSAT Mission Control Centre (MCC). MEOLUTs location shall be chosen properly in order to ensure the coverage of the SGC area.
- Location can be performed either by decoding the position information in the distress beacon message or by trilateration method based on TOA/FOA.TOA/FOA are exchanged both among European MEOLUTs and with the External Meoluts

  #STRB2.0_change: GAL-DCN-TASI-SYST-X-14165, GAL-DCN-ASTG-SYST-X-14198, GAL-DCN-TASI-SYST-X-1055: Provision of Galileo orbital data for SAR user community
- The **MTCF (MEOLUT Tracking Coordination Facility).** The main functions of the MTCF are:
  - Monitoring and control of all SAR GS components including MTCF itself;
  - Building the tracking schedule for the three MEOLUTs in order to maximize the SAR forward link coverage capability using orbital data retrieved through SiS (baseline) or alternatively by GALILEO Service Center;
  - Contribute to TOA/FOA files exchange among MEOLUTs
  - Archiving all data handled by SAR GS system.
  - KPI data collection, KPI computation and reporting
- The **SAR Network (SARN)** corresponding to the necessary network used for inter-connecting all the SAR GS components.
- **RLS emergency beacons** (REBs): 10 RLS emergency beacons for test purposes for Demonstration and Evaluation test phase
- **SAR/Galileo Reference Beacons:** 5 Reference beacons with stable characteristics placed at known locations that allow the monitoring of SAR/Galileo system performance parameters at the MEOLUT sites and the collection of SAR Key Performance Indicators (KPI)

Detailed Design Justification and Description of SAR/Galileo Ground segment is provided in SYS-RD.82, SYS-RD.86 and SYS-RD.98.

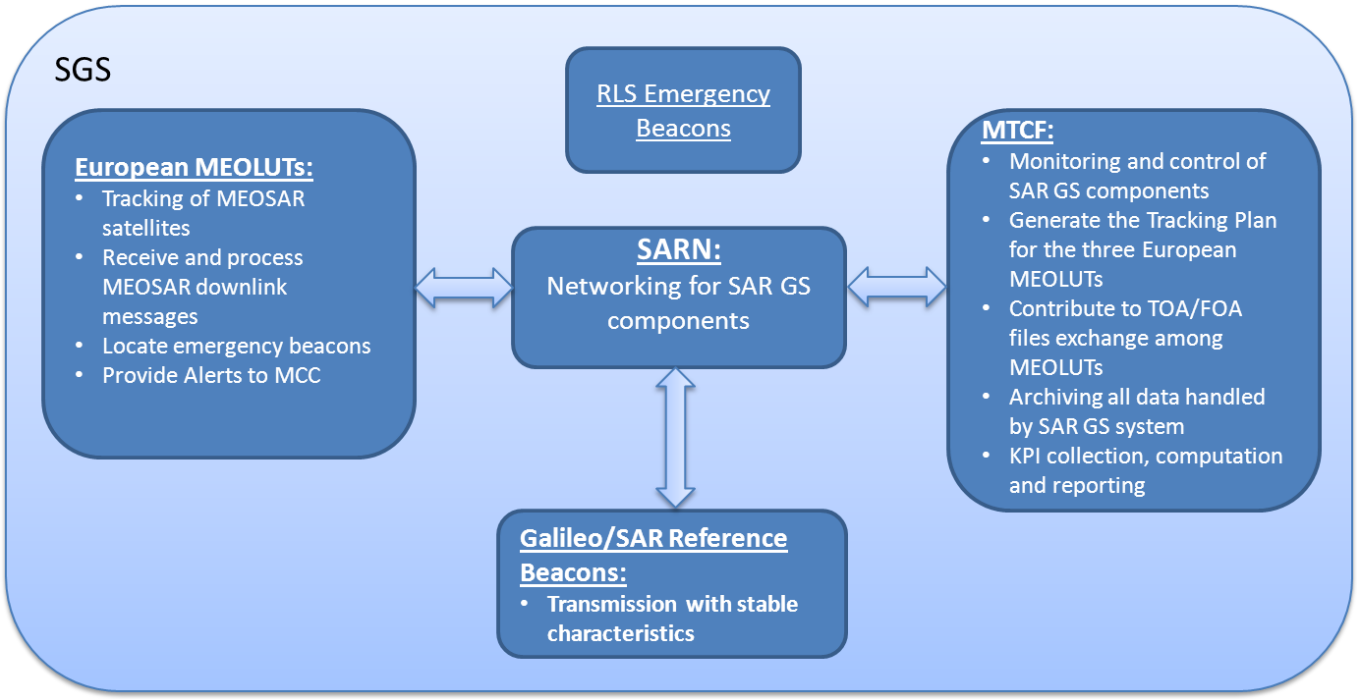


Figure 79 : SGS elements Functional Block diagram

ESA GCI-GSA/SGDSP ES Coordination Scenarios

During Early Services (ES) validation and provision, Galileo Search and Rescue (SAR) Service is managed by a number of Operational Interfaces (OIs) implemented between ESA Galileo Core Infrastructure (GCI) and the European GNSS Agency (GSA) / Search and Rescue Galileo Data Service Provider (SGDSP)

Three types of “Coordination Scenarios” between ESA GCI Operator and GSA/SGDSP during ES SAR provision shall be foreseen:

- Ad-hoc Planning (ES-SAR-SCE-01 - Coordination Scenario 1):
- Ad-hoc Investigation (ES-SAR-SCE-02 - Coordination Scenario 2)
- Reporting (ES-SAR-SCE-03 - Coordination Scenario 3)

All three scenarios are bidirectional. Further details are in SYS-RD.78.
A single SAR transponder is able to retransmit up to 5 simultaneous distress signals in different frequencies, in order to provide SAR Forward Link in its coverage area (SGC).
Each satellite (operating according to the “COSPAS-SARSAT Galileo ICD” document) that receives a SAR distress message, relays the message to the MEOLUT with a probability higher than 99.99% (excluding SAR Payload non-recoverable failures)

The SAR/Galileo Forward Link Service shall be available at least 99.5% of the time over a period of any one year, in the entire Forward Link coverage area, during the nominal operational lifetime.

SAR Beacon Bursts shall be detected and Alert shall be sent to the COSPAS-SARSAT operational network within 10 minutes from the first beacon message transmission, with a probability of at least 0.99 at any time during the system design lifetime

The probability of obtaining a 2D location (latitude/longitude) at the SAR/Galileo MEOLUT, independently of any encoded positional data in the beacon message, within 10 minutes from the first beacon message transmission shall be at least 98%. Location accuracy shall be better than 5 km (with 95% confidence)

The SAR/Galileo system shall not produce more than one processing anomaly for every 10,000 alert (forward link) messages.

4.9.2.2 SFC Functional Block Diagram

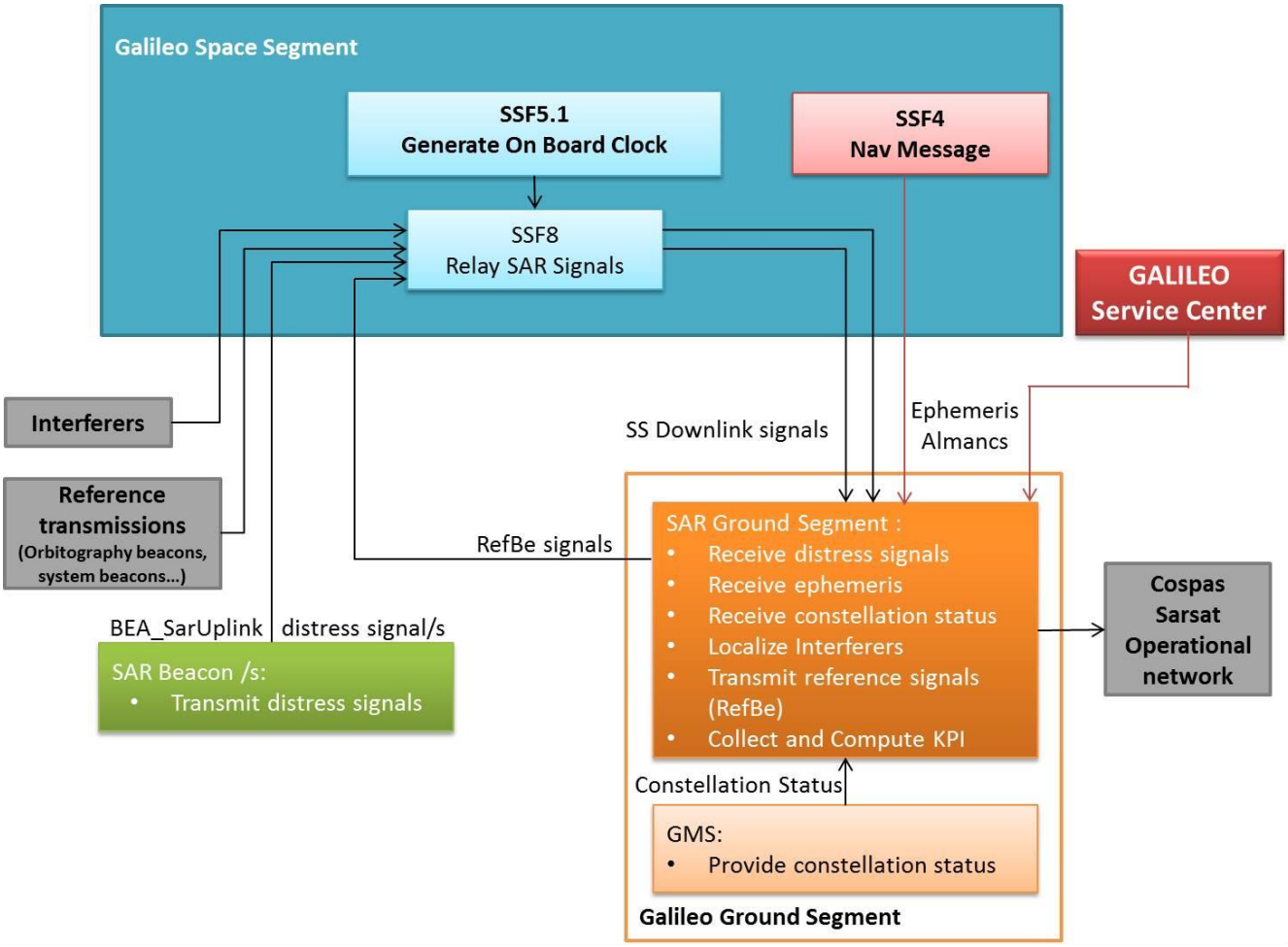


Figure 80 : SFC8.1 SAR Forward Link Service Functional Block Diagram

4.9.3 SFC8.2 Provide SAR Return Link

4.9.3.1 SFC High Level description

This functional chain is in charge of providing the Return Link Service for the Galileo-equipped beacons (Return Link Messages).
Galileo provides support to the SAR Return Link Service by carrying out the following tasks:

1. Interface with Cospas Sarsat
2. Management of the Galileo resources necessary to provide the RLM dissemination

3. RLM dissemination and Monitoring of the RLM dissemination. Including the incorporation of return link messages into the Galileo Navigation uplink message and transmission to the SAR beacons equipped with Galileo receivers

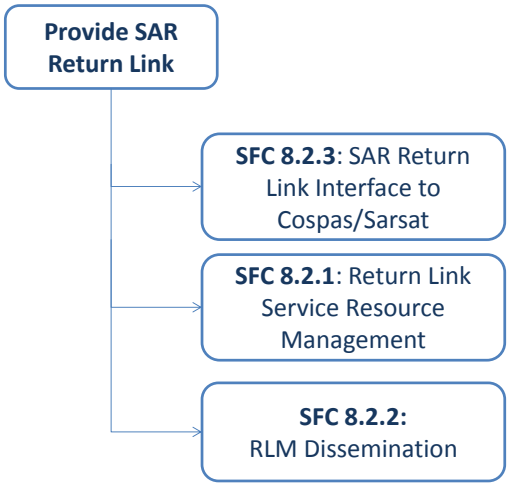


Figure 81 : SFC8.2 Functional tree (2nd Level)

4.9.3.2 SFC8.2.3 SAR Return Link Interface to Cospas/Sarsat

RLSP is the GALILEO interface to Cospas/Sarsat for Return Link Service provision

RLSP interface with FMCC and RCC and

- Receives from FCC the RLM request for Type1 acknowledgment (SIT 134/135)
- Receives from RCC the RLM request for Type2 acknowledgment and RLM cancellation

RLSP sends to FMCC:

- GALILEO satellites ephemeris
- SAR/Galileo and GALILEO status message coming from the RLSP

4.9.3.2.1 SFC Functional Block Diagram

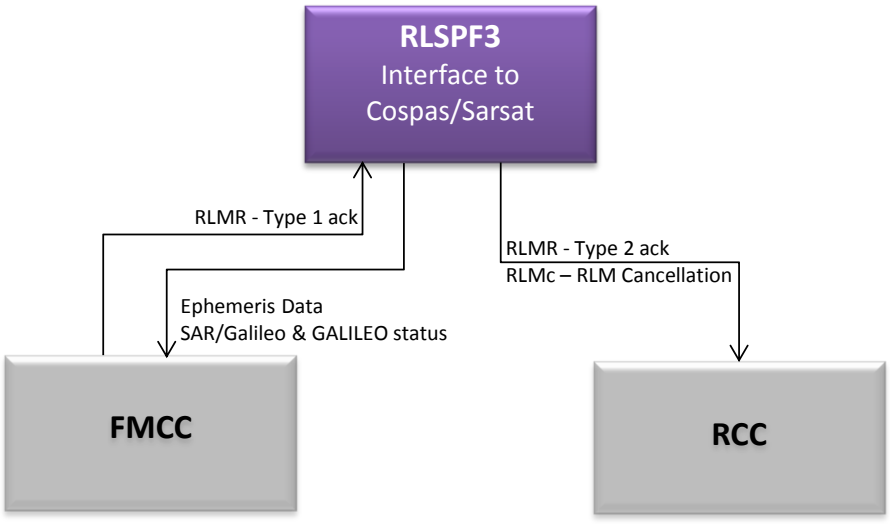


Figure 82: SFC8.2.3 Functional Block Diagram

4.9.3.3 SFC8.2.1 SAR Return Link Resource Management

4.9.3.3.1 SFC Description

Any information of any components of the SAR system is transferred through the GMS-RLSP interface. A single RLSP is interfaced with the GMS
To better coordinate activities in all MEOSAR components the RLSP needs to know particular information about the status of the SAR/Galileo components. This includes the following signals sent by GMS (SPF) to RLSP:

- Current satellite status (for each satellite: Configuration of SAR functions available for Forward Link (bandwidth, gain control mode etc), Return Link Status). This information is contained in *MM_SarSystemStatus* signal included in the SPF_RlmSarStatus dataflow
- Current GMS SAR status (awaiting requests, unable to process requests, degraded modes, estimated RLM transmission delay based on average transmission delay of each new RLM processed within 15 min). This information is contained in *MM_SarSystemStatus* signal included in the SPF_RlmSarStatus dataflow
- Service bulletins (planned satellite Forward Link outages, planned satellite Return Link outages, GMS SAR Service Mode Status, planned RLSP-GMS changes). This information is contained in PLN_SarServiceBulletin signal

In addition the RLSP receives the following data by GMS/SPF:

- Ephemeris data of Galileo Satellites (including satellite clock correction parameters, Ephemeris parameters, Ionospheric correction parameters). This information is contained in the NAV_EphemToRlsp included in the SPF_ReportToRlsp dataflow

- Monitoring information regarding the Return Link Message notification status for each active RLM from the last RLM notification. This information is contained in the INF_RlmNotifMnt included in the SPF_RlmSarStatus dataflow and is sent every 5 minutes

Galileo GMS/SPF sends the required reports to the RLSP either periodically at scheduled instants or upon request.

RLSP can issue a request to GMS(SPF) to obtain particular data/report or to ask for a report to be generated and transmitted at an instant which is not pre-scheduled. This is done through the INF_ReportInfoRqst that is part of the RLSP_ReportRqst dataflow

4.9.3.3.2 SFC Functional Block Diagram

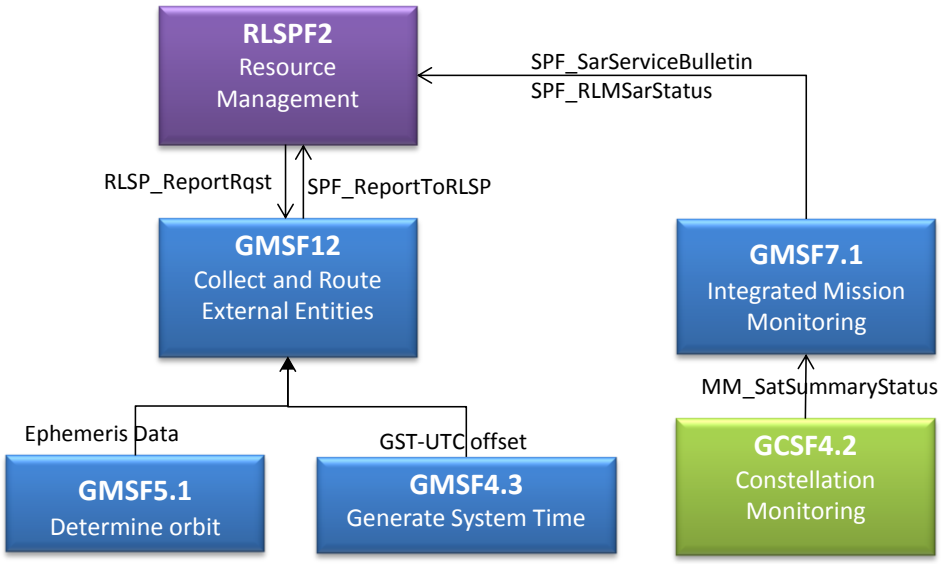


Figure 83: SFC8.2.1 Functional Block Diagram

Note: the data flow between GMS and RLSP is managed by GMSF12 function, nevertheless the RLSP is internal to the Galileo System.

4.9.3.4 SFC8.2.2 RLM Dissemination

4.9.3.4.1 SFC Description

4.9.3.4.1.1 RLM Handling & Dissemination

The Galileo System provides the Return Link Service performing the following tasks

1. RLSP receives from Cospas/Sarsat (FMCC/RCC) a request for a RLM originating from beacons detected by non-Galileo and Galileo MEOLUTs
2. RLSP elaborates the information received by GMS on the Satellites Current status, current GMS SAR status, service bulletins and Ephemeris together with the information contained in the Return Link Message request by FMCC and generate the RLSP request.
3. RLSP sends the RLM request to the GMS (MGF) through the SAR_ReturnLinkAckNotif MSG ID151 signal contained in the RLSP_SarRqst dataflow. The request includes the RLM data and the support information (Estimated location of the beacon (latitude, longitude) and a list of satellites that can be used for the dissemination of the RLM.
4. Galileo GMS receives from RLSP via a dedicated GMS-RLSP interface Return link messages request through the RLSP_SarRqst dataflow.
   - The RLSP-GMS (MGF) interface shall be able to transmit up to 30 RLM requests per minute and up to 30 RLM cancellations per minute.
     The Galileo system itself may not necessarily be able to process this number every minute as it is dependent on the repetition strategy required, the location of the beacon, the distribution of RLMs to satellites, etc etc. If the RLM received cannot be disseminated to an available satellite a saturation notice is sent to RLSP. Each RLM request will be repeated once by the RLSP with a delay of less than 1 minute.
5. Define Satellites to be used for RLM dissemination:
   From the list of satellites provided by RLSP and knowledge of the contact plan, of the online status and the outages of the satellites, the GMS can define the satellites (at least 2) to be used for the RLM dissemination. This list is used to define the uplink scheduling for the SAR RLM.
   #STRB2.0_change: the contact plan will be received by RLSP in GAL-DCN-ASTG-SYST-X-13906

GSM provides the SPF_ContactPlan (SCH_OverallUlsContactPlan) to the RLSP with the lates ULS to satellites contacts.

6. Create Uplink message
   The GMS then creates the uplink message for the selected satellites including the RLM in the navigation message sub-frame.
7. Uplink Message
   - The Galileo system then uplinks the complete navigation uplink message to the satellites (at least 2) according to the following repetition strategy in the C-Band uplink signal:
   - The number of repetitions and time between repetitions of an RLM is a configurable parameter:
     - Value range: 0 to 5 for the number of repetitions and 7 to 15 minutes for the time between them (30 seconds time steps)
       Default value = 5 repetitions and 10 minutes between them.
   - The RLM first transmission to at least 2 satellites is fixed to To+5min (from GMS-REQ-643)
   - Within the 30 RLM Requests per minute requirement the GMS shall always prioritize new RLMs over repetitions of previous RLMs.
   - It is clarified that RLM Requests refer to new RLMs delivered from RLSP to GMS.
   - The GMS shall stop uplink to a satellite upon receiving a cancellation request from RLSP within 5 minutes of reception of the request.
   - The GMS shall ensure that the probability that a received RLM fails to be up-linked to a satellite in the required time or fails to stop a transmission upon cancellation request is less than 2E-4.
8. Satellite processing of RLM
   - From the received message the Galileo satellite prepares the downlink and adds the SAR RLM into the I/NAV (E1-B) downlink channel. Each satellite shall ensure that probability that a received RLM in the uplink is transmitted over T+2minutes is higher than 0.998 (excluding the NAV payload non-recoverable failures
   - The Galileo satellites are capable of disseminating 600bits/minute of SAR RLM data in nominal situations
9. Receive at SAR Beacon
   The SAR beacons equipped with Galileo Receivers receive the Galileo signal with the RLM dedication to the beacon in the Galileo L-Band downlink for confirmation of a distress message being received.

The Galileo System ensures that the RLM is delivered to the SAR beacon within 15 minutes from receiving the RLM from RLSP.

10. Confirmation of RLM processing
    If the GMS is no longer able to process/schedule RLM requests for a particular satellite a saturation notice is sent to RLSP (MGF_SarControlInfo) within 35s. The RLSP shall send an update of the RLM including the new S/C dissemination list. The GMS will update its dissemination data and reroute the following repetitions accordingly.

The SAR/Galileo Return link Service is provided over the whole surface of the Earth.

4.9.3.4.1.2 Monitoring of RLM dissemination

The GMS monitors the successful dissemination of the RLMs and the correct cancellation as requested by RLSP.
The successful dissemination of an individual RLM (and its possible cancellation) is carried out by the monitoring functions of the GMS.
At least one GSS receives the SiS observables with the RLM included. The GMS carries out online monitoring and confirms the RLM was included as requested.
This confirms that the RLM has been correctly scheduled for uplink and received by the GSSs.
A message is sent to RLSP with the details of this monitoring.

4.9.3.4.2 SFC Functional Block Diagram

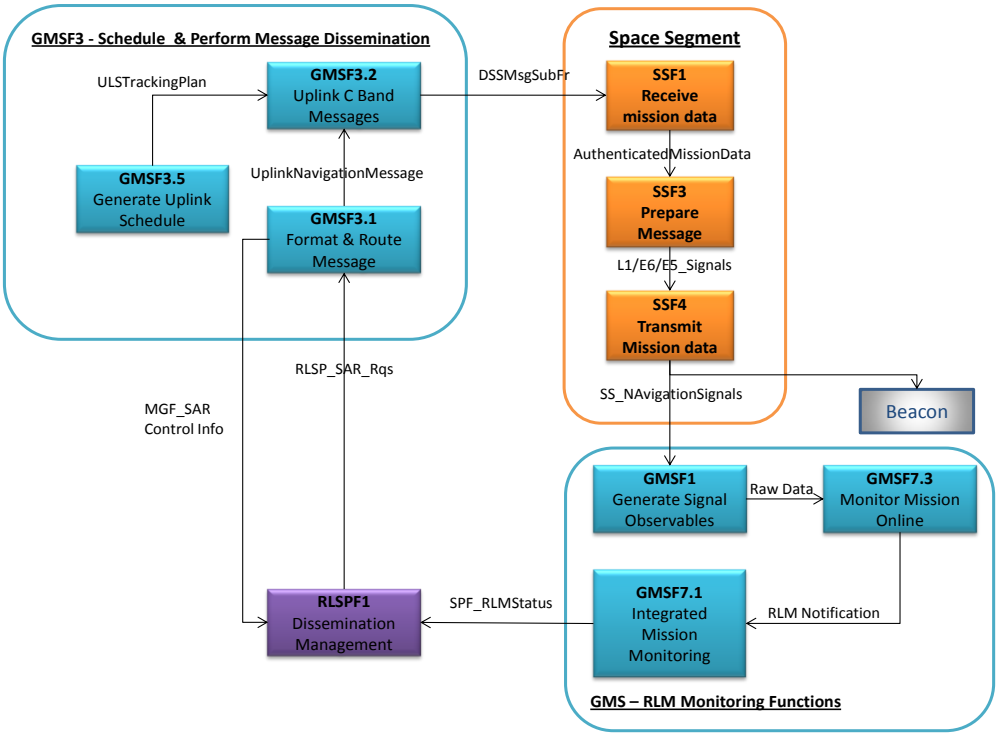


Figure 84: SFC8.2.2 RLM Dissemination Functional Block Diagram

4.10 SFC9 Support EDBS Service

4.10.1 SFC9 Presentation

4.10.1.1 SFC9 High Level Description

The Galileo Core Infrastructure enables the provision of dissemination by Galileo satellites of External Data Broadcast Service generated by Galileo Service Centre.
The Support to EDBS functional chain provides all required functionality to ensure the Galileo system can support the External Data Broadcast Service.
This includes:

- Receiving External Data Broadcast Service for all / a subset of Galileo Satellites and include it in the Galileo C-Band uplink message to the Galileo Satellites
- Providing satellite contact plans to Galileo Service Centre, entity in charge of External Data Broadcast Service generation

  Processing the External Data Broadcast Service onboard the Galileo satellites and transmitting the External Data Broadcast Service data to users in the L-Band Navigation Galileo Signal using old ERIS Bits Allocation.

- Monitoring of the EDBS data transmitted and generating reports to the GSC

The first level functional tree for the “Support to EDBS Services” SFC is shown below.

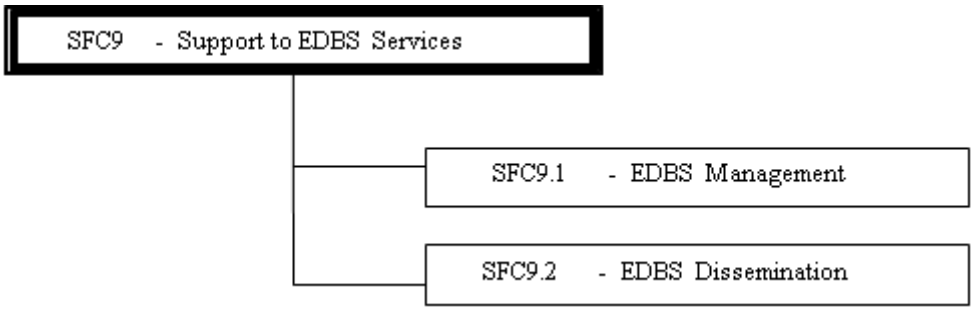


Figure 85 : SFC9 Functional Tree (1st level)

4.10.1.2 Defintions

EDBS: External Data Broadcast Service is an Open Service authentication computed by Galileo Service Centre and provided for inclusion within the Open Service signals of Galileo

EDBS Dissemination: The External Data Broadcast Service dissemination is carried out via the Galileo ground system. This is used for Open Service Authentication.

The Galileo Service Centre generates the Open Service authentication for a subset of satellites and provides it to the Galileo ground system, which then adds this data to the uplink subframe to be uplinked to the specific Galileo satellite. Satellites then add the data in the downlink to be transmitted to the system users.
The design provides an E1b mono frequency authentication service and reserves the E5b dataflow for future uses.

This means EDBS mono-frequency data has to be provided to GMS in those epochs for which then, taking into account the dissemination time, the downlink correspond for Even page in E5b and Odd page in E1b (corresponding to even GST Epochs) as described in SISICD-1180.

4.10.2 SFC9.1 EDBS Management

4.10.2.1 SFC description

The purpose of this system functional chain is:

- to ensure that Galileo Service Centre and GMS have the correct information available to them at the correct time to be able to support the EDBS service.
- to monitor the EDBS service and report to GSC this monitoring

The GSC will receive every 30 minutes from the Galileo system reports related to EDBS data insertion (EDBS_DisseminationReport) .
Galileo periodically sends this information to GSC to allow better co-ordination of the uplinking activities as described in SFC9.2.

Galileo Service Centre sends also periodically (30 min) the status of the EDBS reception performances (MNT_EBDSStatus) to Galileo Mission Segment to better coordinate the up-linking activities

4.10.2.2 SFC Functional Block Diagram

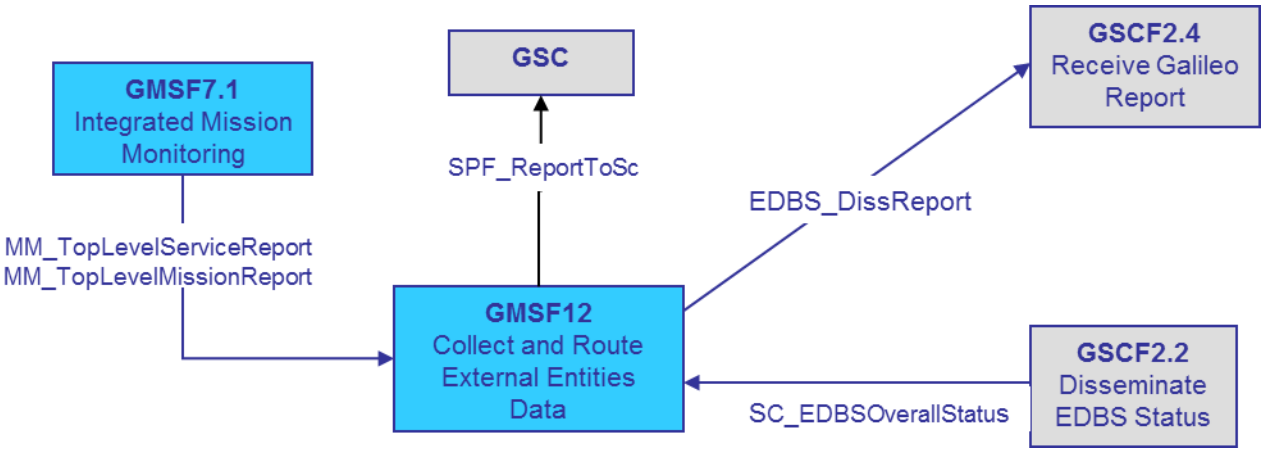


Figure 86 : SFC9.1 General reports Distribution Functional Block Diagram

4.10.3 SFC9.2 EDBS Dissemination

This SFC is in charge of disseminating External Data Broadcast Service messages provided by Galileo Service Centre to the users.

The External Data Broadcast Service messages are disseminated using Galileo Ground Segment to uplink the messages to the satellites (SFC9.2.1). (old indirect ERIS uplink)

The system is able to disseminate EDBS for all satellites in contact.

4.10.3.1 On-Board Processing of EDBS Data

The following block diagram shows the satellite on-board functions to process the EDBS data

- The satellite receives EDBS data uplinked via Galileo uplink stations, and authenticates the uplink to ensure it is from an authorised uplink.
  - the NSGU adds the EDBS data from the Galileo uplink to the appropriate position in the EDBS field in the downlink.
  - If there is no EDBS data received, the NSGU inserts 40 zero bits in the appropriate position in the EDBS field in the downlink message. No EDBS message is downlinked when the satellites are not connected to the Galileo ULS network
- The received messages are modulated and transmitted on the L-band I/NAV downlink message.
- The Galileo user and Galileo Sensor stations receive the EDBS data in the Galileo Signal in Space in the appropriate part of the message (old ERIS bit allocation in E1-b I/NAV pages. Old ERIS bit allocation in E5b is reserved for future uses).

Note related to ERIS direct uplink removal: dissemination of any data formerely allocated to ERIS direct uplink must not occur. Blocking barrier may be set active within the NSGU via TC as such to deny any outbound direct ERIS data transmission.

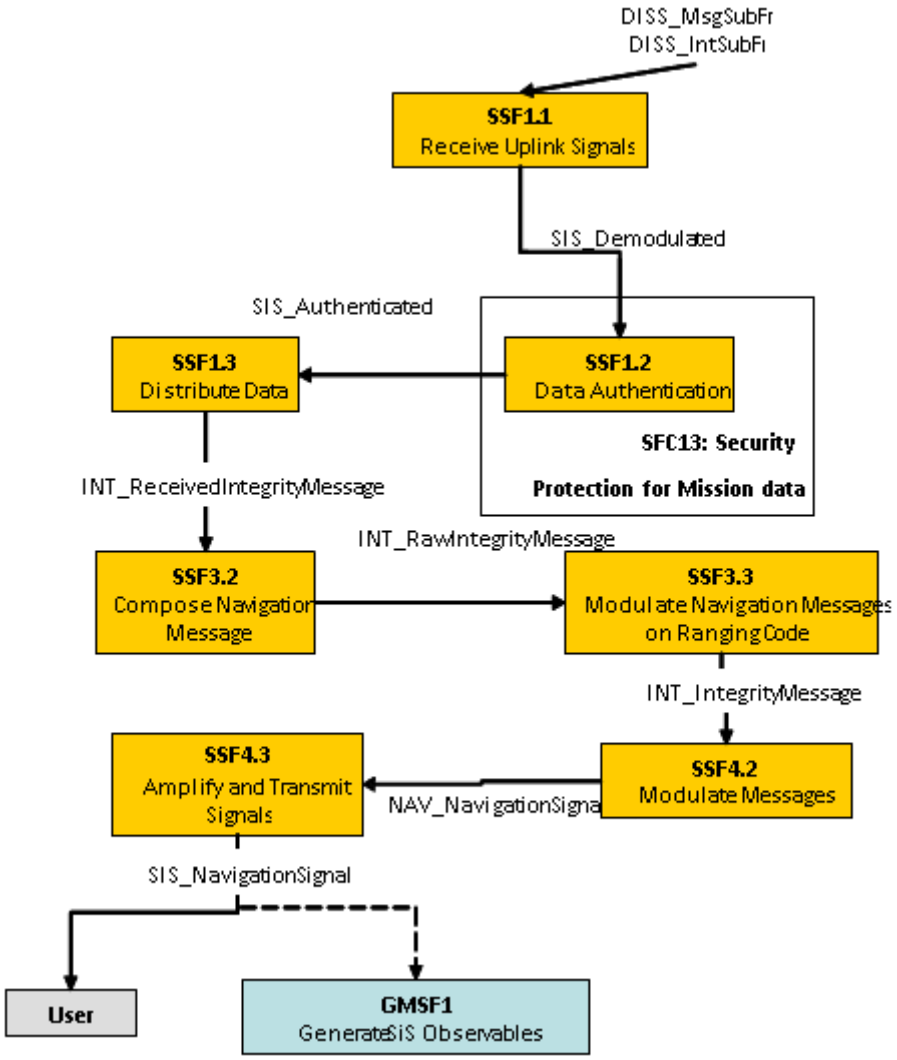


Figure 87 : ERIS data On Board Processing Principle

4.10.3.2 Generation of EDBS Reports: EDBS Monitoring

The Galileo system monitors on-line the activity of the EDBS.
The External Data Broadcast Service monitoring is carried out by GMS, a report of this monitoring is sent to Galileo Service Centre (SPF_DataResponse / EDBS_DisseminationReport). The process of monitoring and dissemination of this report is:

- The Galileo signals sent down by those satellites receiving EDBS data is received by the GMS Sensor stations within the I/NAV message type.
- Within the GMS there will be a display showing for EDBS the monitored activity on each satellite. This will be combined with information on whether the satellite is disseminating or not EDBS data based on input from the MNT_EBDS Status signal.
- This signal is interpreted by the mission monitoring function to allow knowledge that the satellite has EDBS data present as opposed to dummy data. The monitoring function will have the visibility of which satellites are active (for operational use).
- The GMS integrated mission monitoring function creates the EDBS dissemination report (SPF_EDBSDisseminationReport) for further dissemination to Galileo Service

Centre with 30 minutes of statistics. The EDBS dissemination report will show for each spacecraft the times at which activity started and stopped on their channel.

Additionally:

· Galileo will send confirmation of the correct broadcast of the EDBS data to allow the GSC to know that the EDBS data was broadcasted correctly by the Galileo system and by which satellites over which times.

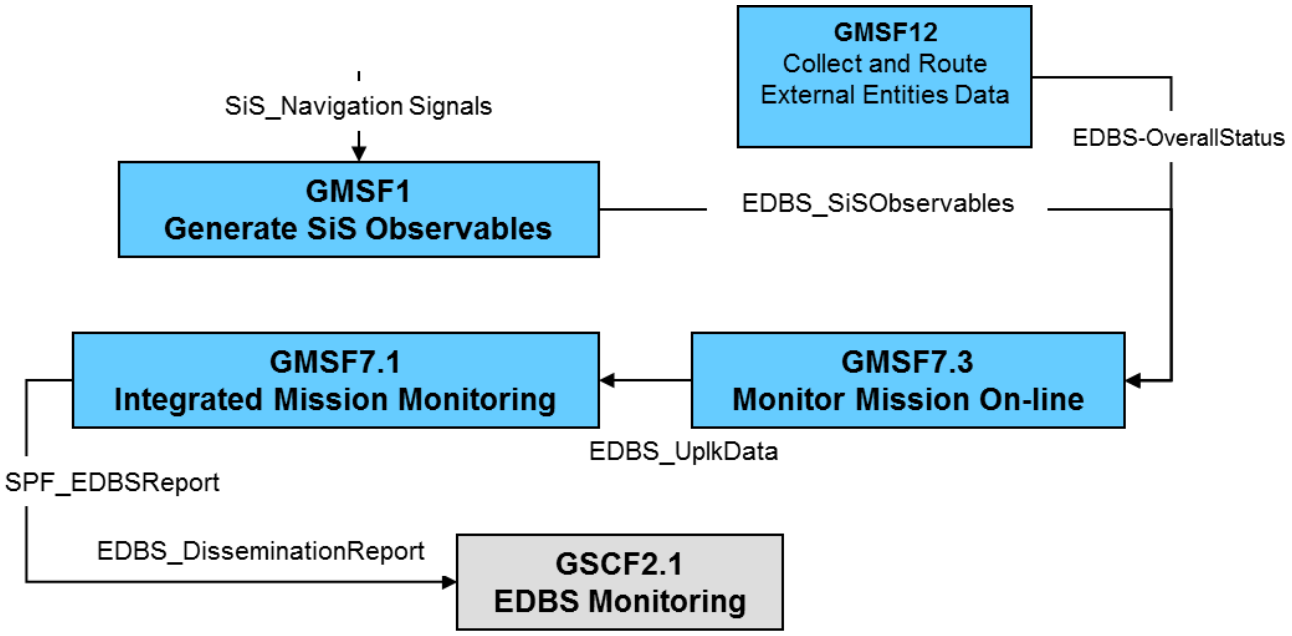


Figure 88 : EDBS Integrity Report Creation Functional Block Diagram

4.10.3.3 SFC9.2.1 EDBS Uplink

4.10.3.3.1 SFC description

The purpose of this system functional chain is to disseminate External Data Broadcast Service received from GSC to the user via the Galileo Ground Segment.

This SFC has the following process:

- GSC is expected to create the External Data Broadcast Service (EDBS) data for Os Authentication for a subset of satellites in contact with ULS and provide it in Real Time to the Galileo Mission Segment within the constraints of the available capacity (40 bit/sec) provided by the Galileo Core system

- The GSC sends each two seconds EDBS data for specific satellites to be transmitted along with auxiliary information, (i.e. Satellite ID, message header, message identifier for monitoring purposes).This data is required to be sent within the specified time period to ensure broadcast in the epoch proceeding the delivered epoch.

EDBS authentication is disseminated through E1b. GSC provides data to be transmitted in those epochs for which then, taking into account the dissemination time, the downlink corresponds for and Odd page in E1b.

- The GMS receives the EDBS data from the selected satellites and creates the Integrity uplink message by including the EDBS data in the integrity uplink subframe (per satellite). If for a satellite in contact with ULS GSC does not send any EDBS data, the GMS fills the EDBS section within the Integrity message subframe with dummy data (0x00000000).

  Note: Even if SoL has been descoped from GMS, Integrity Subframe continues being generated by MGF and uplink to satellite

- EDBS Delivery Time: The Galileo system provides the EDBS data to the users within 3100ms from the epoch of reception of the EDBS data at the Galileo ground segment

4.10.3.3.2 SFC Functional Block Diagram

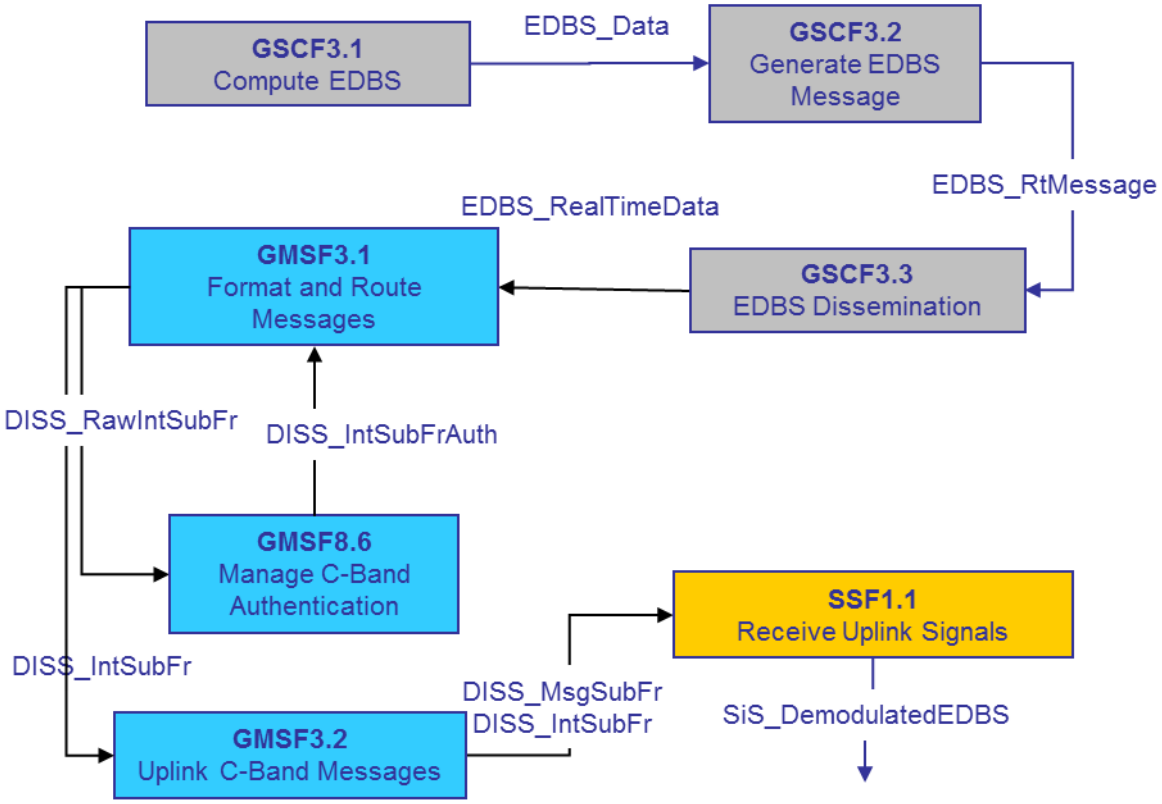


Figure 89 : SFC9.2.1 Functional Block Diagram

4.10.3.4 SFC9.2.2 ERIS Direct Uplink Inhibition

4.10.3.4.1 SFC description

Note related to ERIS direct uplink removal: direct dissemination must not occur. Blocking barriers are to be installed for ensuring direct uplink dissemination blocking.

As described in GAL-DCN-TASI-SYST-X-0797, before EDBS dissemination, each individual Uplinked Direct ERIS will inhibit via Telecommand to the MISREC and NSGU. Then, transmission of the indirect ERIS data shall prevail allowing EDBS dissemination through this channel.

4.11 Security SFCs (Cla)

See Classified Annex.

5. SEGMENT FUNCTION DESCRIPTION

5.1 General Description

The functional architecture replicates the SDL design.
Linking:

- is composed of system scenarios.
- is mapped on organic design
- is allocated to / is constrained by to segment requirements

5.1.1 Core System Functional Block Diagram

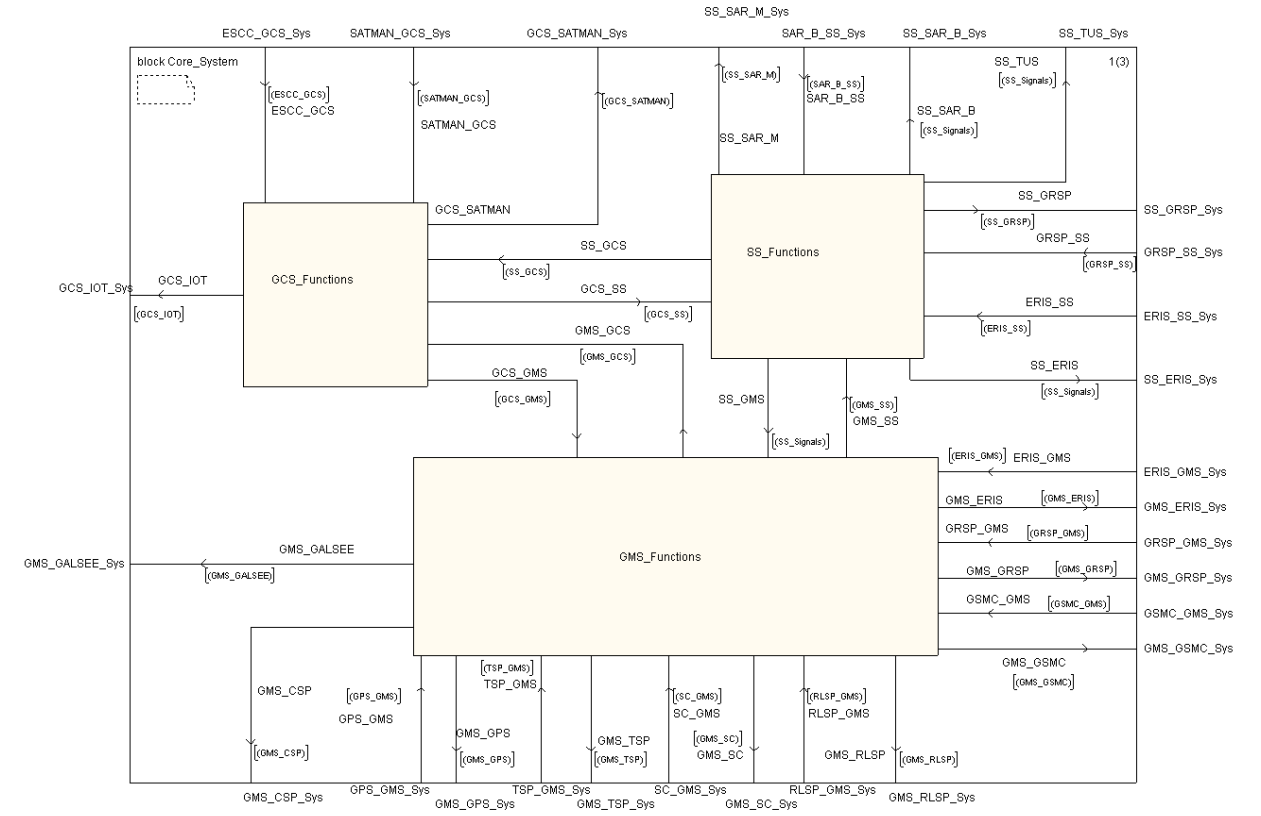


Figure 90: Core System SDL Functional Block Diagram

5.1.2 SDL Functional Tree

5.1.2.1 Core System Functions

- **GCS Functions**

Function ID	Function name	Mapping to Element
GCSF1	**GCS Short Term Planning**	SCPF/FDF/SCCF
GCSF2	**Manage TTC Link**	TTC/SCCF/SDDN
GCSF2.1	TTC Link Modes	TTC/SCCF/SDDN/KMF
GCSF2.2	Two Way Ranging	TTC/SDDN
GCSF3	**Archiving**	OPF/CMCF/FDF/SCCF
GCSF3.1	Central Archiving	OPF/CMCF/FDF/SCCF
GCSF3.2	Local Archiving	Elements
GCSF4	**Constellation Monitoring and Control**	SCCF/FDF/KMF
GCSF4.1	Execute Schedule and Special Operation	SCCF/TTC/FDF/KMF
GCSF4.2	Constellation Monitoring	SCCF/FDF/KMF
GCSF4.3	Manage OBSW	SCCF
GCSF4.4	Flight Dynamics	FDF
GCSF5	**GCS Monitoring and Control**	CMCF
GCSF5.1	GCS Element Monitoring and Control	CMCF/TTC
GCSF5.2	GCS Central Monitoring and Control	CMCF
GCSF5.3	GCS Network Monitoring and Control	SDDN/CMCF
GCSF6	**Provide Training**	CISM
GCSF7	**Perform Simulation and Test**	CSIM
GCSF8	**Manage DataBase**	OPF
GCSF8.1	SC DataBase	OPF/KMF
GCSF8.2	Operational Procedure DataBase	OPF/KMF
GCSF8.3	Anomaly Database	OPF
GCSF8.4	Document Database	OPF
GCSF9	**Security**	KMF/ILS(Secure)
GCSF9.1	Security protection of S-Band	KMF
GCSF9.2	Security protection of ground infrastructure	KMF/ILS(Secure)

Table 11: GCS Function List

- **GMS Functions**

Function ID	Function name	Mapping to Element
GMSF1	**Generate SiS Observables**	GSS
GMSF3	**Schedule and Perform Message Dissemination**	xxx
GMSF3.1	Format and Route Messages	MGF/MDDN/ULS
GMSF3.2	Up Link C Band Messages	ULS
GMSF3.3	Select Navigation Message	MGF/MDDN
GMSF3.5	Generate Uplink Schedule	MUCF
GMSF4	**Generate Time and Geodetic Reference Data**	xxx
GMSF4.1	Generate Coordinates Reference	GACF/GSS/MSF
GMSF4.2	Determine GGTO	PTF/MUCF/GACF/MSF
GMSF4.3	Generate System Time	PTF/GACF/MSF
GMSF4.4	Time Transfer	PTF
GMSF5	**Determine Navigation**	OSPF

GMSF5.1	Determine Orbit	OSPF
GMSF5.2	Determine BGD	OSPF
GMSF5.3	Determine SISA	OSPF/GACF/MSF
GMSF5.4	Determine Satellite Clock Parameters	OSPF
GMSF5.5	Determine Single Frequency Ionospheric Model	OSPF
GMSF5.6	Generate Navigation Batches	OSPF
GMSF6	**Archiving**	GACF
GMSF6.1	Global Archiving	GACF
GMSF6.2	Central Archiving	GACF
GMSF6.3	Local Archiving	All elements
GMSF7	**Monitoring and Control**	xxx
GMSF7.1	Integrated Mission Monitoring	GACF/MUCF
GMSF7.2	Monitor Mission Off Line	MUCF/MSF
GMSF7.3	Monitor Mission On Line	MUCF
GMSF7.4	Ground Asset Monitoring and Control	GACF
GMSF7.4.1	GMS Elements Monitoring and Control	All elements
GMSF7.4.2	GMS Central Monitoring and Control	GACF
GMSF7.4.3	Overall Monitoring and Control	GACF
GMSF7.4.4	Network Monitoring and Control	GACF/MDDN
GMSF8	**Security**	xxx
GMSF8.1	Security Protection of PRS	PKMF
GMSF8.2	Security Protection of SOL	MKMF/SPF
GMSF8.3	Security Protection for Mission Data	MKMF/SPF
GMSF8.4	Security Protection of CS	MKMF/SPF
GMSF9	**Perform Mission Planning**	MUCF
GMSF9.1	Long/Mid Term Planning	MUCF
GMSF9.2	Short Term Planning	MUCF
GMSF10	**Provide Training**	MTPF
GMSF11	**Perform Simulation and Test**	MTPF/MSF/MUCF
GMSF12	**Collect and Route External Entities Data**	SPF/MGF/MUCF
GMSF13	**Manage Database**	GACF/MTPF

Table 12: GMS Function List

- **SS Functions**

Function ID	Function name	Mapping to IOV Element	Mapping to FOC1 Element
SSF1	**Receive Mission Data**	**xxx**	**xxx**
SSF1.1	Receive Uplink Signals	MISREC	MISREC
SSF1.2	Data Authentication	PLSU	CSU
SSF1.3	Distribute Data	NSGU/PLSU	NSGU/CSU
SSF2	**Generate Ranging Code**	**xxx**	**xxx**
SSF2.1	Generate Navigation Ranging Code	NSGU	NSGU
SSF2.2	Generate PRS Ranging Code	PLSU	CSU
SSF2.3	Generate CS Ranging Code	NSGU	NSGU
SSF3	**Prepare Message**	**xxx**	**xxx**
SSF3.1	Generate On-Board Data	NSGU/AVS	NSGU/AVS
SSF3.2	Compose Navigation Message	NSGU	NSGU
SSF3.3	Modulate Navigation Messages On Ranging Code	NSGU	NSGU

SSF4	**Transmit Mission Data**	**xxx**	**xxx**
SSF4.1	Generate Carriers	FGUU	FGUU
SSF4.2	Modulate Messages	NSGU/FGUU	NSGU/FGUU
SSF4.3	Amplify and Transmit Signals	LTX	LTX
SSF5	**Provide Time Reference**	**xxx**	**xxx**
SSF5.1	Generate OnBoardClock	CMCU	CMCU
SSF5.2	Generate OnBoardTime	NSGU	NSGU
SSF6	**Manage Platform**	**AVS**	**AVS**
SSF6.1	FDIR	AVS	AVS
SSF6.2	Health Status	AVS	AVS
SSF6.3	Satellite Mode	AVS/RTU	AVS/RTU
SSF6.4	AOCS	AVS	AVS
SSF6.5	On-Board Data Management	AVS/RTU	AVS/RTU
SSF6.7	Autonomous Action and Control	AVS	AVS
SSF7	**Manage Payload**	**AVS/RTU**	**AVS/RTU**
SSF8	**SAR Payload**	**SAR**	**SAR**
SSF9	**Provide TTC Communication**	**AVS/TTC**	**AVS/TTC**
SSF9.1	Receive and Control TC	AVS/TTC/RTU/ PFSU	AVS/TTC/RTU/CSU
SSF9.2	Transmit and Manage TM	AVS/TTC/RTU/ PFSU	AVS/TTC/RTU/CSU
SSF9.3	Two Way Ranging	AVS/TTC/RTU	AVS/TTC/RTU
SSF10	**Optical Ranging**	**LRR**	**LRR**

Table 13: SSgt Function List

- **TUS Functions**

Function ID	Function name	Mapping to elements
TUSF1	**PseudoRange and Phase Measurement on SIS**	**Antenna/Core receiver**
TUSF2	**Navigation Message Recovery**	**Core Receiver**
TUSF4	**Position Determination**	**Application Unit**
TUSF5	**Corrected Measurements Determination**	**Application Unit**
TUSF5.1	Clock Correction	
TUSF5.2	Relativistic Effect Computation	
TUSF5.3	Troposphere Compensation Computation	
TUSF5.4	Sagnac Correction Determination	
TUSF5.5	Group Delay Determination	
TUSF5.6	Iono Free PR Measurements Determination	
TUSF6	**Measurement Pre-Processing**	**Application Unit**
TUSF6.1	Code Carrier Phase Filter	
TUSF6.4	Valid Measurement Determination	
TUSF7	**Timing Determination Algorithm**	**Application Unit**
TUSF7.1	GST Determination Algorithm	
TUSF7.2	GST-UTC Conversion Algorithm	
TUSF8	**Analyses Tool**	**Application Unit**
TUSF9	**Autonomous Integrity Determination**	**Application Unit**
TUSF10	**Search and Rescue**	**Application Unit**
TUSF11	**Differential Data**	**Application Unit**
TUSF11.1	Differential Data Generation	
TUSF11.2	Differential Data Reception	

Table 14: TUS Function List

5.1.2.2 External Entity Functions

- **ERIS Functions**

- Remark: ERIS descoped and EDBS concept addressed in [SYS-RD.89].

Function ID	Function name
ERISF1	**Direct Uplink Integrity Message (descoped)**
ERISF2	**ERIS Mission Monitoring**
ERISF2.1	ERIS Monitoring
ERISF2.2	Disseminate ERIS Status
ERISF2.3	Request Report and Plan
ERISF2.4	Receive Galileo Report and Plan
ERISF3	**Determine ERIS Integrity**
ERISF3.1	Compute ERIS Integrity
ERISF3.2	Generate ERIS Message
ERISF3.3	Indirect Integrity Dissemination
ERISF4	**Manage Keys**

Table 15: ERIS Function List

- **ESCC Functions**

Function ID	Function name
ESCCF1	Perform LEOP
ESCCF2	Perform Orbit Determination
ESCCF3	Perform Satellite Hand-Over

Table 16: ESCC Function List

- **GPS Functions**

Function ID	Function name
GPSF1	Provide GPS System Time (USNO)
GPSF2	Generate GPS SIS (CV Time Transfer)

Table 17: GPS Function List

- **GRSP Functions**

Function ID	Function name
GRSPF1	Generate GTRF and Provide related/not Related Products

Table 18: GRSP Function List

- **IOT Functions**

Function ID	Function name
IOTF1	Perform IOT Measurements

Table 19: IOT Function List

- **Launch Service Functions**

Function ID	Function name
LSPF1	Perform Launch

Table 20: LS Function List

- **RLSP Functions**

Function ID	Function name
RLSPF1	Dissemination Management
RLSPF2	Resource Management

Table 21: RLSP Function List

- **SAR Functions**

Function ID	Function name
SARF1	SAR Beacon
SARF2	SAR Meolut

Table 22: SARFunction List

- **SATMAN Functions**

Function ID	Function name
SATMANF1	Satellite Manufacturer

Table 23: SATMAN Function List

- **SC Functions**

Function ID	Function name
SCF1	CS Data Upload
SCF2	CS Data Response Process
SCF3	Key Management
SCF4	CS Resource Management

Table 24: SC Function List

- **TSP Functions**

Function ID	Function name
TSPF1	Metrological Timekeeping of GST
TSPF1.1	Time Transfer
TSPF1.2	GST Steering Generation

Table 25: TSP Function List

5.2 GCS Functions

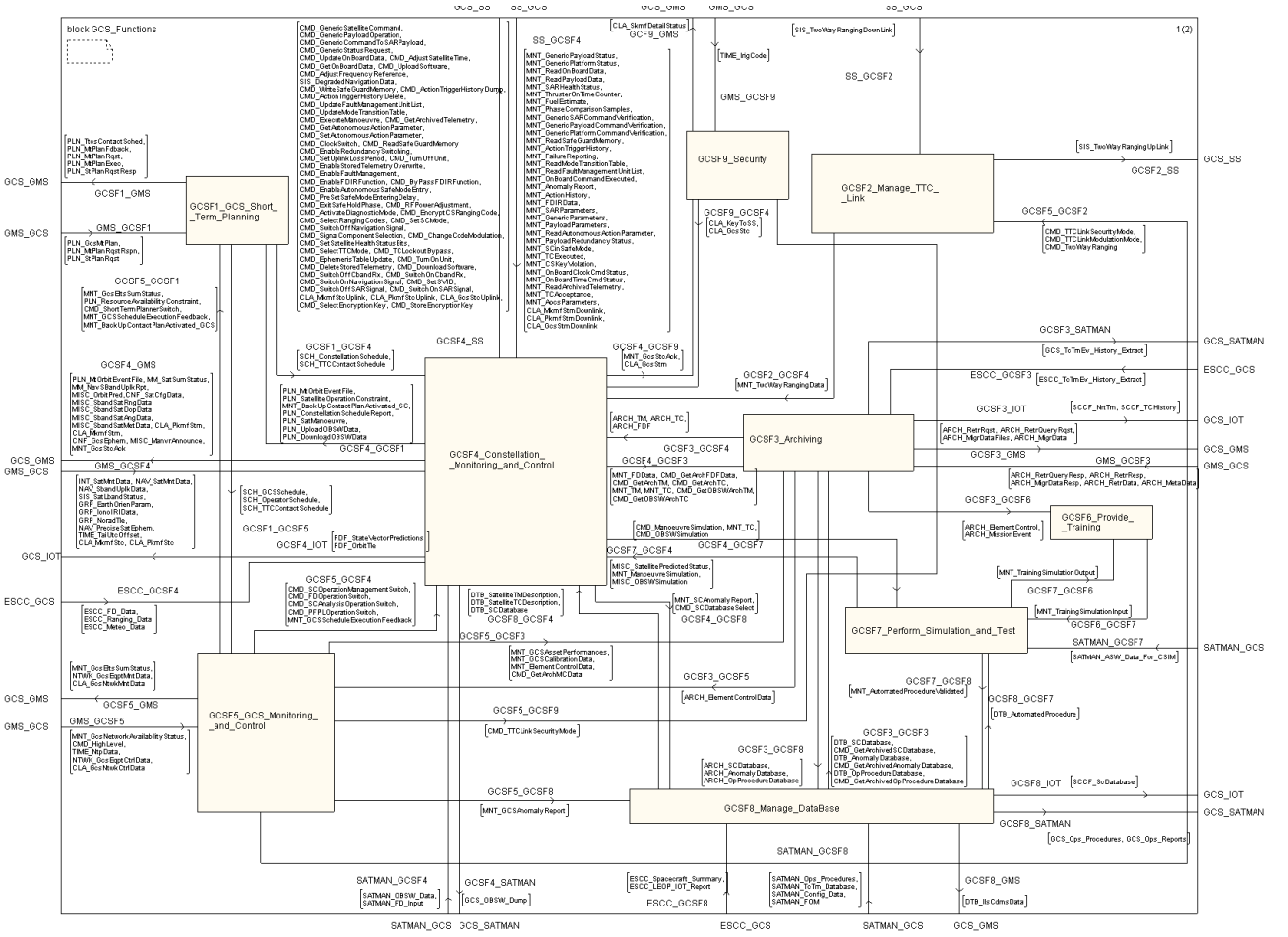


Figure 91: GCS SDL Functional Block Diagram

5.2.1 GCSF1 GCS Short Term Planning

GCSF1 General Description

This function is responsible for performing optimized detailed GCS Short Term planning within the framework provided by the Long/Mid-term Plan for the operations to be performed in GCS allowing the generation of detailed automatically executable schedules (in particular TTC contact schedules), as well as the schedule of the operational tasks to be performed.

This is based on the reception of long and mid term plan and requests for payload operation from GMS (e.g. Satellite Onboard clock Resynchronisation Requests).

This function is also in charge of generating mid-term plan requests to the GMS (basically for any activity which might impact the mission performance).

GCSF1 outputs:

GCSF1 Planning Outputs

The Short Term Planning function produces the following planning outputs:

1 - Planning requests provided to the GMS and feedback to the GMS Long/Mid-Term Plan function,

2 - Planning Reports with the results of the planning execution (provided to GMS) as well as the status of short-term planning requests coming from GMS for CS ranging key management.

GCSF1 Schedule Outputs

The Short Term Planning function produces the following schedule outputs:
1 - Manual Operations: Timelines & Operators Tasks,
2 - Automatic Tasks: Executable Schedule for the Overall, Central and Element M&C functions,
3 - Nominal and backup TTC contact schedules. The nominal contact schedule corresponding to the GCS short-term plan are provided to GMS.

GCSF1 Design Constraints / Performances

The GCS has the capability of:

- producing a validated short-term plan in less than 10mn following the provision of all necessary inputs,
- refining an existing short-term plan in less than 5mn,
- generating executable schedules from a short-term plan shall not take more than 5mn,
- validating an updated schedule in less than 1mn.

The GCS validates that the planned operations fit into the duration of the TTC contact periods. This is also monitored during the contact periods.

GCSF1 Functional Block Diagram

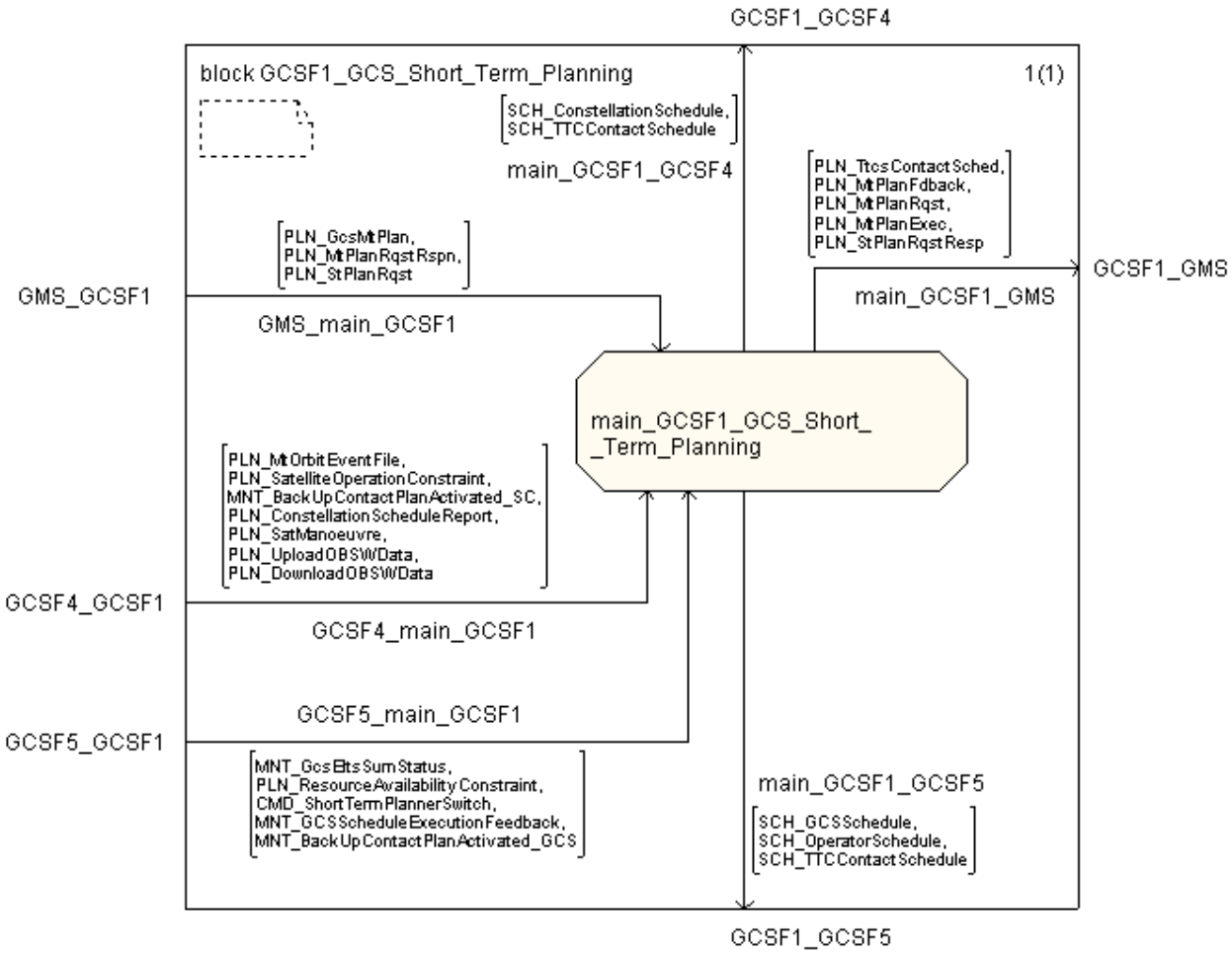


Figure 92: GCSF1 SDL Functional Block Diagram

5.2.2 GCSF2 Manage TTC Link

GCSF2 General Description

This function is to provide RF / baseband and link mode management functionality in support of the operation of the S-band TTC link.
Main functionalities:

- Supporting 2 TTC link RF modulation modes (ie spread spectrum or ESA standard mode) consitent with the characteristics defined in [AD/9] for the Galileo TT&C ground station
- Manage the TTC link security mode (ie CLEAR or PROTECTED).
- Provide two way ranging

GCSF2 Functional Block Diagram

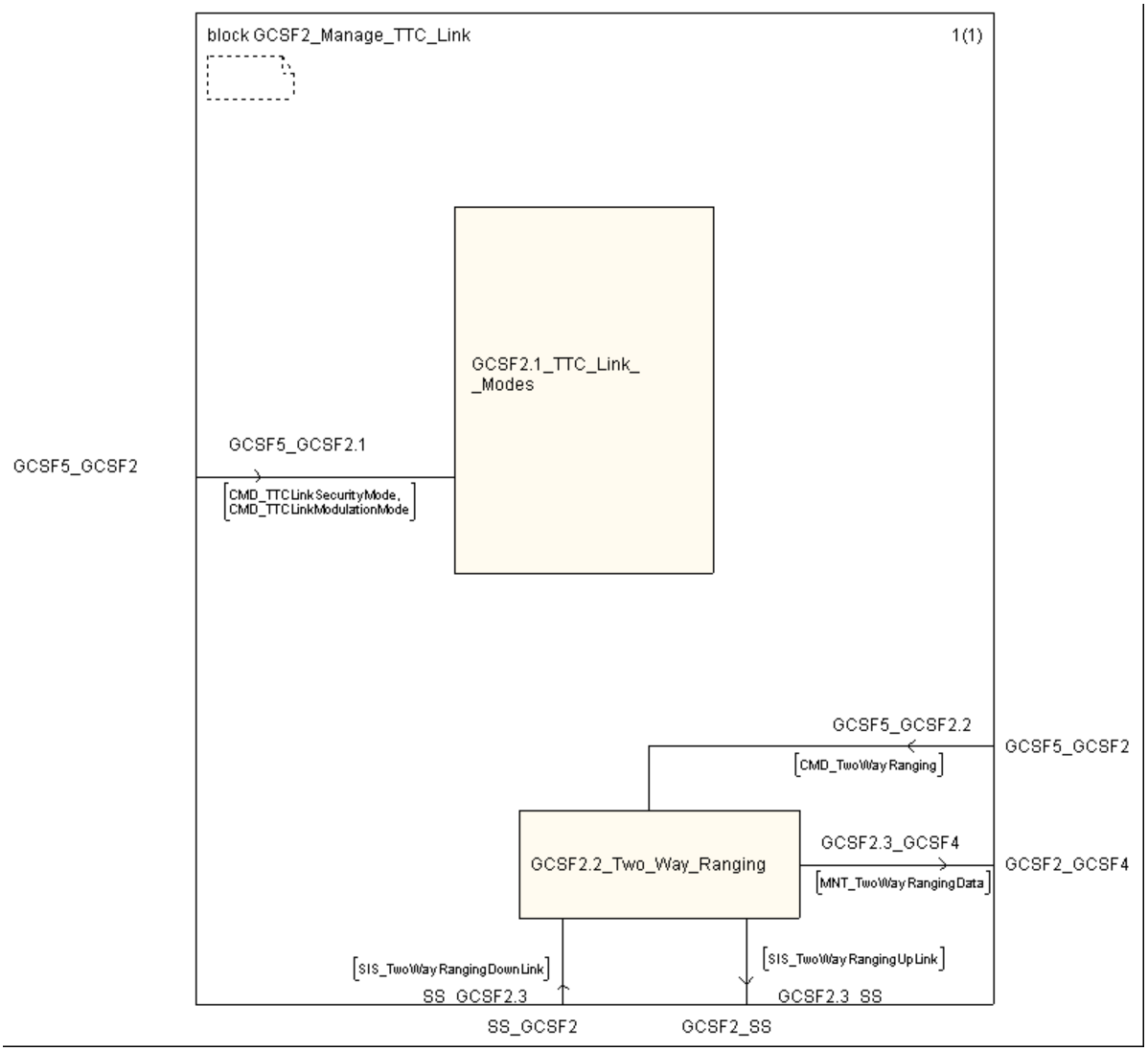


Figure 93: GCSF2 SDL Functional Block Diagram

5.2.2.1 GCSF2.1 TTC Link Modes

GCSF2.1 General Description

This function provides RF / baseband and security functionality in support of the operation of the S-band TTC link. In particular the following functions will be supported:

- Managing the S-band link RF modes (spread spectrum and ESA standard modes)
- Supporting TT&C ground station RF and baseband performances as specified in [AD/9].

GCSF2.1 Design Constraints

This function will support the TTC link according to the following constraints:

- The TTC link will support both spread spectrum and ESA standard modulation modes
- The TTC link will operate according to the RF specification provided in [AD/9].
- Doppler frequency compensation is applied to the TTC links.

GCSF2.1 Functional Block Diagram

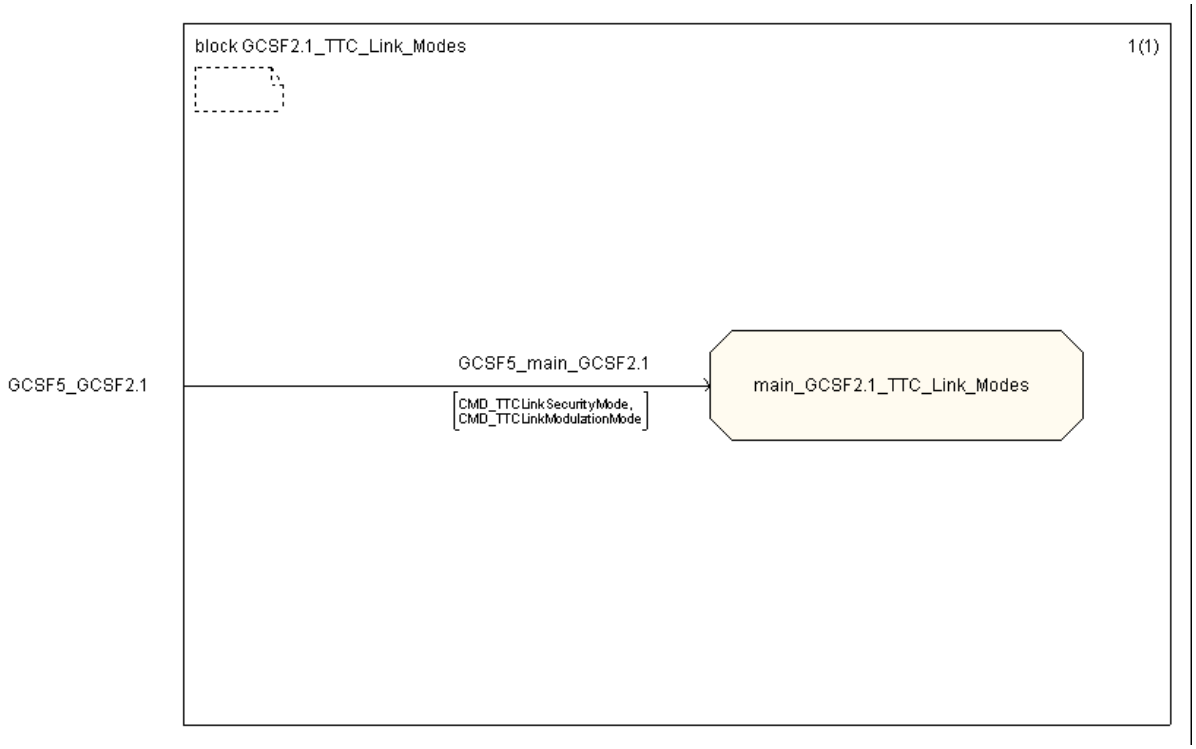


Figure 94: GCSF2.1 SDL Functional Block Diagram

5.2.2.2 GCSF2.2 Two Way Ranging

GCSF2.2 General Description

This function is responsible for providing:

- Two way ranging
- Communication of ranging data to the flight dynamics function

GCSF2.2 Outputs
This function will provide the following outputs:

1. the uplink RF ranging signal (either in ESA standard mode with code ranging, or spread spectrum mode with ranging)
2. two way ranging measurements for input to the flight dynamics functions.

GCSF2.2 Design Constraints
This function will provide the two way ranging capability according to the following constraints:

- It will be possible to perform ranging in parallel with TC/TM
- The following ranging methods will be supported:
  - ESA standard mode with code ranging
  - Spread spectrum ranging
- The two way ranging accuracy achievable will be 10m with 1-sigma uncertainty (including transponder calibration errors)

GCSF2.2 Functional Block Diagram

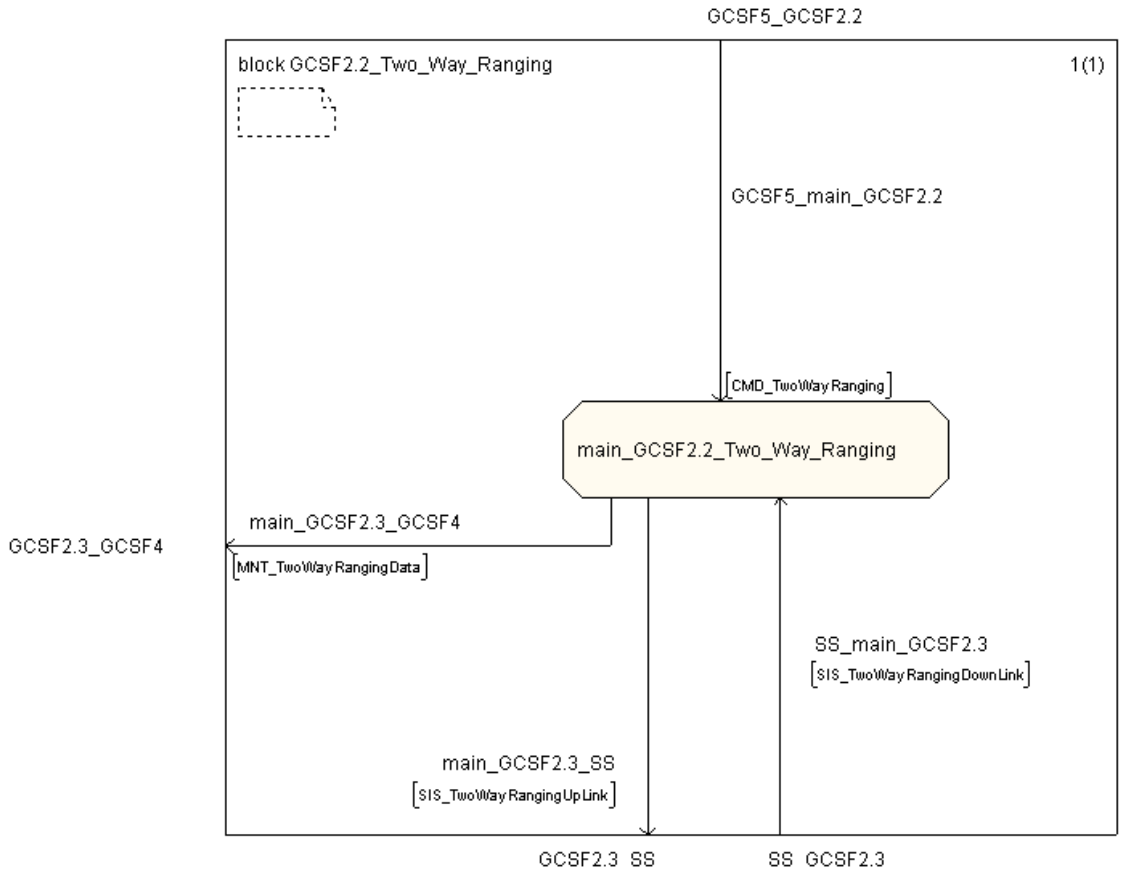


Figure 95: GCSF2.2 SDL Functional Block Diagram

5.2.3 GCSF3 Archiving

GCSF3 Description

This function is in charge to provide archive storage and retrieval capabilities for the following types of data:

- Databases,
- Configuration data,
- TM/TC,
- SC Operations,
- Event Log,
- Monitoring and Control of GCS Assets,
- Flight Dynamics dat,
- Planning data.

In GCS, the archiving is limited to local and central archive:

- Local archive storage and retrieval capabilities (short term archive at element level, notion of circular buffering)
- Central archive and retrieval capabilities (archive of data on a 1 year duration basis)
- The GCS central archive of a GCS element interfaces with the GMS Global archive for permanent archive.

GCSF3 Functional Block Diagram

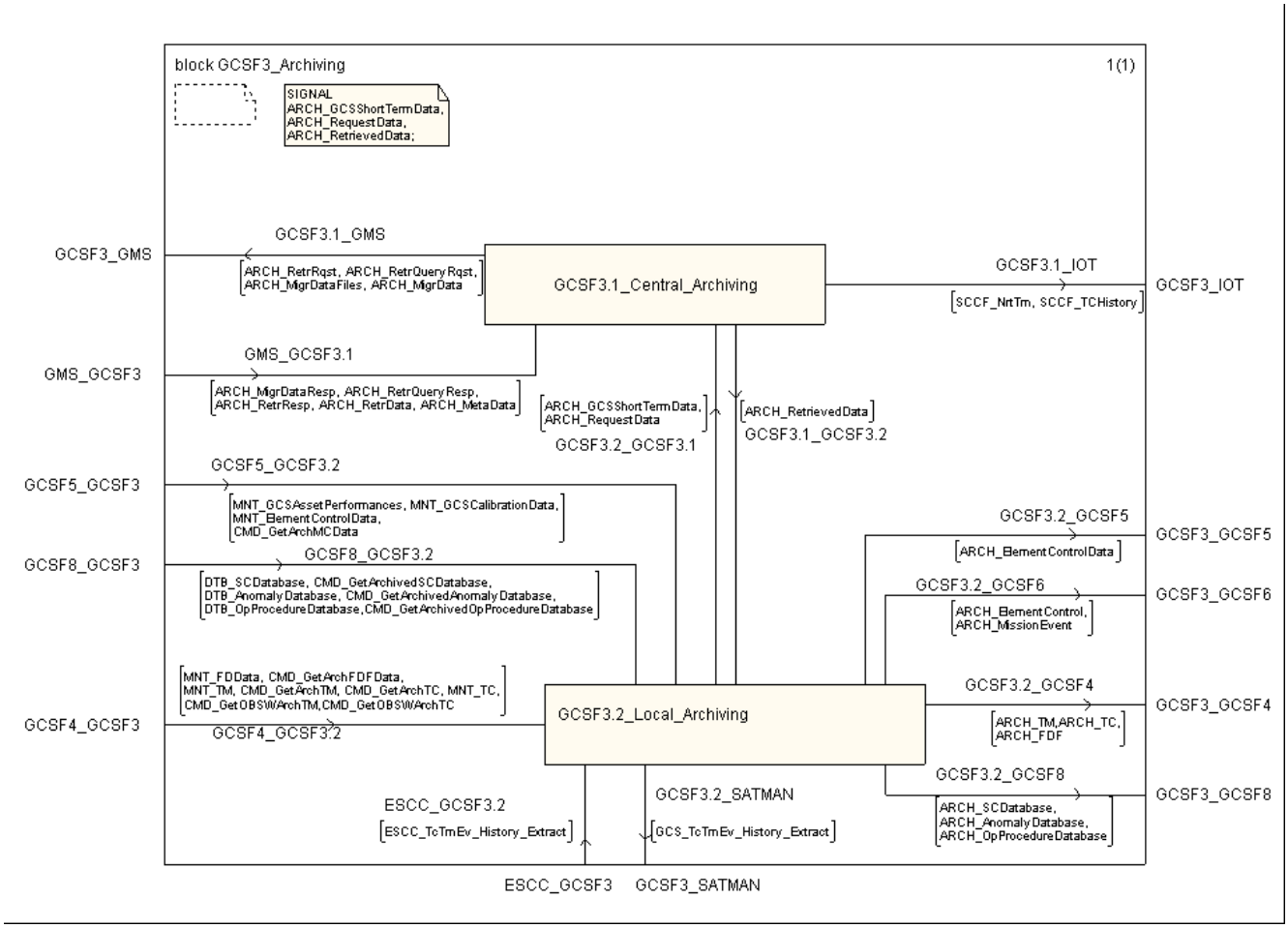


Figure 96: GCSF3 SDL Functional Block Diagram

5.2.3.1 GCSF3.1 Central Archiving

GCSF3.1 General Description
This function is in charge to provide archive storage and retrieval capabilities for the following types of data:

- Databases,
- Configuration data
- TM/TC,
- SC Operations,
- Event Log,
- Monitoring and Control of GCS Assets,
- Flight Dynamics data,
- Planning data.

The function supports:

- a non-permanent high throughput storage media. This function supports a mid-term storage capacity of one year.
- a catalogue allowing organised data archiving and selective retrieval of archived data.

The function is supported by Man Machine Interface function allowing the operator to interact with the Permanent Archive in the GMS.

GCSF3.1 Outputs
The GCS archive functions allows archived data transfer between GMS and GCS. GCS elements perform data archive migrations and retrievals with the GMS Global Archive when needed.

GCSF3.1 Design Constraints/ Performances
The GCS archive functions maintain a record of the source of the archived data to ease the retrieval process of data. Additionally the GCS correlate the on-board (ICDU, PFSU) time with the Galileo System Time with an accuracy of +/- 5 msec to create a correlated S/C time, to be used for archival of TM&TC packets.

This function allows the GCS to retrieve

1. TM & TC data older than 3 years within 5 minutes (from the long-term archive).
2. any GCS element M&C archived data less than 3 years old within 1 minute (from the central archive) in an automated way for further off line analysis (maintenance, troubleshooting, re-play).

GCSF3.1 Functional Block Diagram

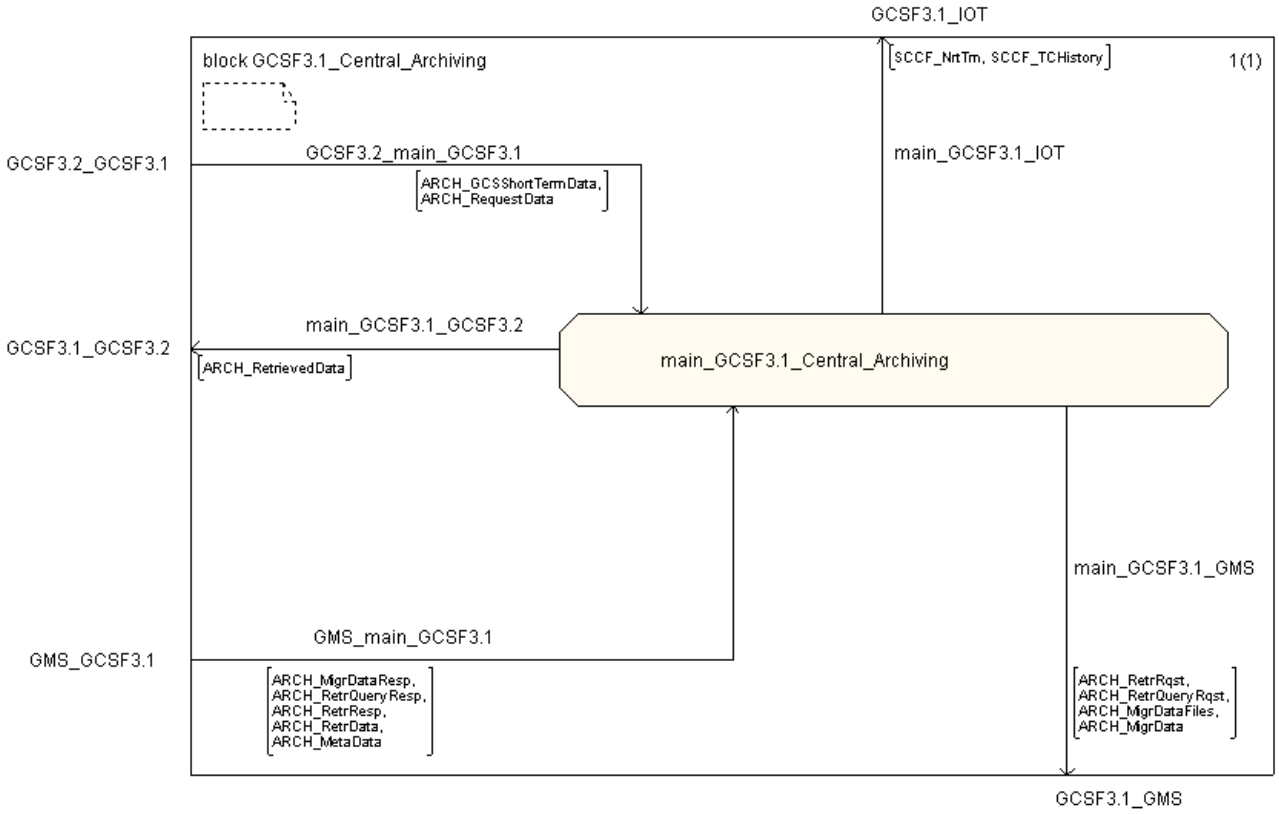


Figure 97: GCSF3.1 SDL Functional Block Diagram

5.2.3.2 GCSF3.2 Local Archiving

GCSF3.2 General Description
This function is in charge of local archiving and retrieval capabilities needed at each element level for the following data:

- Satellite Telemetry and Telecommand
- TTC station Telemetry and Telecommand
- Flight Dynamics Data
- L Band Telemetry
- GCS Assets performances
- GCS Calibration Data
- GCS Element Monitoring and Control Data
- GCS Element Databases
- Event logbook

This function is processed automatically. Local Archive of Galileo Ground elements have a few days of storage capability.

GCSF3.2 Outputs
This function allows archived data transfer with the GCS elements and the central archive for the following data:

- Constellation Management and Satellite Control data,
- Databases,
- TM/TCs,
- Operational Data,
- Flight Dynamics
- Ground assets data,
- M&C data

GCSF3.2 Design Constraints/ Performances
The GCS archive functions maintain a record of the source of the archived data to ease the retrieval process of data. Additionally the GCS correlate the on-board (ICDU, PFSU) time with the Galileo System Time with an accuracy of +/- 5 msec to create a correlated S/C time, to be used for archival of TM&TC packets.
The archiving of TM is processed automatically.

This function allows local data archive in TTCs for data related to the Monitoring and Control with a period of storage limited to one month

GCSF3.2 Functional Block Diagram

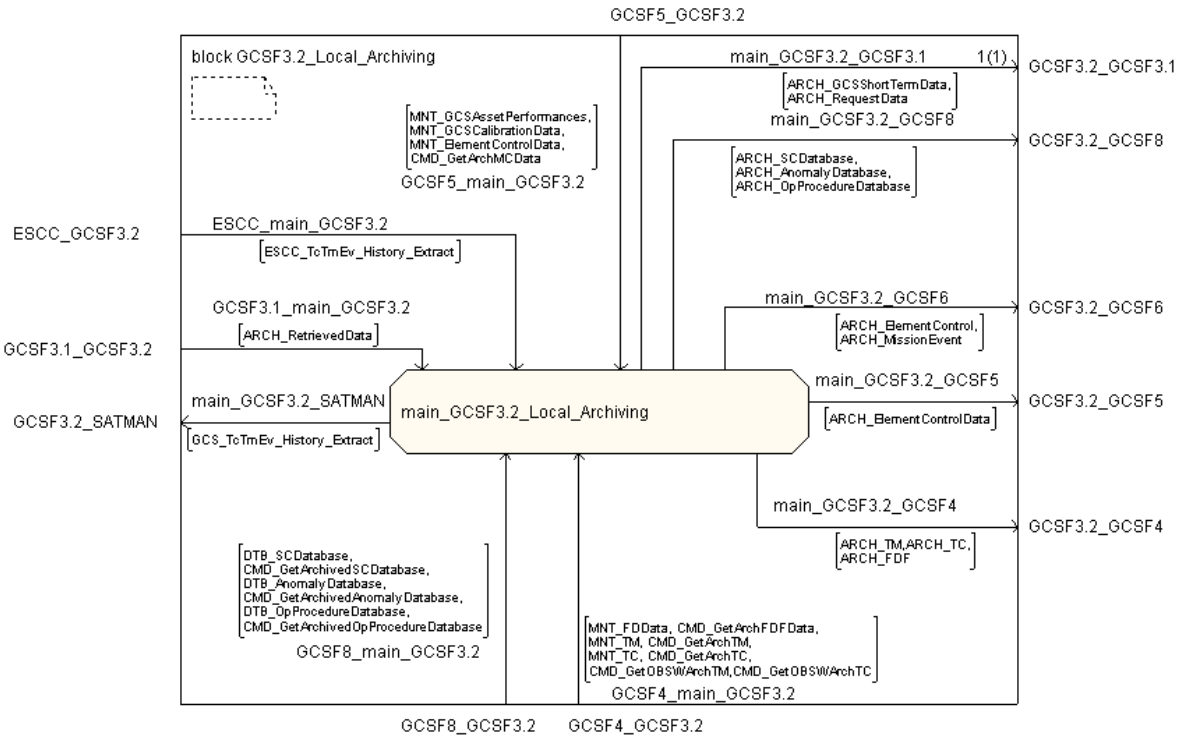


Figure 98: GCSF3.2 SDL Functional Block Diagram

5.2.4 GCSF4 Constellation Monitoring and Control

GCSF4 Description

This function is responsible for monitoring and controlling the operational state of all of the satellites in the Galileo constellation:
Monitoring of the constellation is performed based upon Telemetry received intermittently via S-Band and satellite health status information received continuously via L-band (passed to GCS by GMS).
Control of the constellation is nominally performed through the execution of satellite command schedules defined in the GCS short term plan and uplinked to the satellite via S-Band

The main functionalities are:

- Execute the satellite schedule (including those defined in support of special operations)
- Provide Constellation Status (both current and historical)
- Provide Flights Dynamics functions
- Manage OBSW updates

GCSF4 Functional Block Diagram

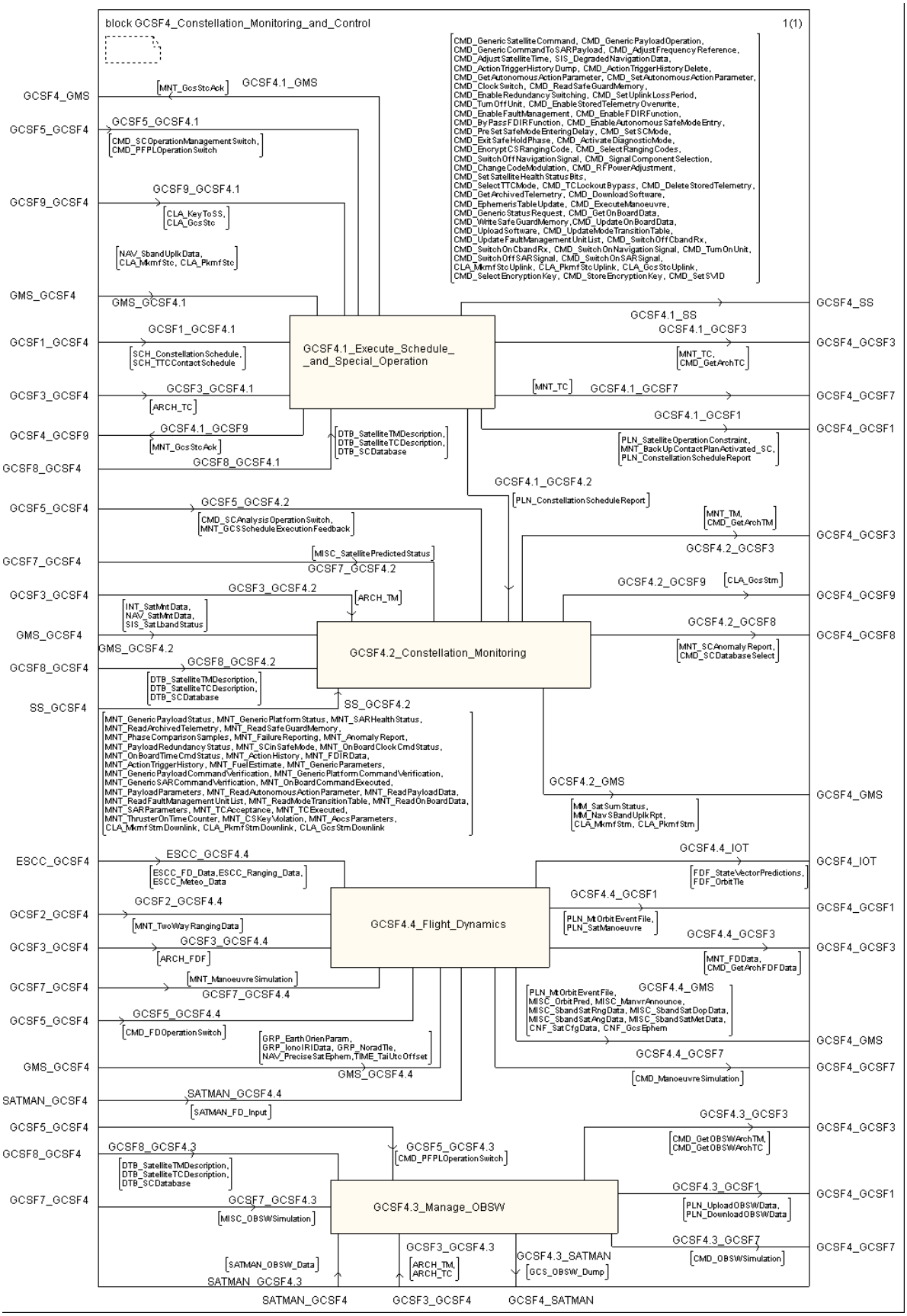


Figure 99: GCSF4 SDL Functional Block Diagram

5.2.4.1 GCSF4.1 Execute Schedule and Special Operation

GCSF4.1 General Description
This function is responsible for controlling the satellite, and in particular:

- Execution of the spacecraft schedule provided by the GCS planning functions (GCSF1) - with capability to execute these automatically, partially automatically or fully manually.
- Automated generation and uplinking of command plans in accordance with the spacecraft schedule
- Support for manual commanding of the spacecraft
- Ability to send any telecommand included in the spacecraft database
- Parallel commanding of up to 6 operational satellites in orbit, plus at least 4 satellites on ground (2 simulated, 2 during LIT or SVT)
- Uplink via telecommand of OS and PRS navigation data provided for degraded mode operation by the GMS
- Management of the satellite on-board schedule (time-tagged telecommands)
- Verification of the acceptance and execution of all telecommands uplinked to the spacecraft
- Supporting special operations with up to 4 satellites in parallel with no impact on routine operations

.
GCSF4.1 Outputs:

GCSF4.1 Platform and Payload Command Outputs
This function is responsible for sending telecommands to the spacecraft platform and payload related to the monitoring and control of:

- On-board FDIR functionality
- Satellite modes
- Management of the on-board telemetry packet stores
- Unit control (on/off etc)
- Management of all on-board data and software parameters
- TT&C link management (RF mode)
- Navigation payload
- SAR payload

GCSF4.1 Satellite Maneuver Outputs
This function is responsible for sending commands to the satellite to execute orbit and attitude control maneuvers.

GCSF4.1 OBSW Management Outputs
This function is responsible for sending commands to the satellite related to the management of the satellite on-board software (patch and dump).
.
GCSF4.1 OS and PRS Navigation Data Outputs
This function is responsible for uplinking telecommands to the satellite containing OS and PRS navigation data provided by the GMS.

GCSF4.1 Secure Telecommand Outputs
This function is responsible for uplinking secure telecommands received from the key management functions within the GCS and GMS and also for providing confirmation that the secure telecommands have been received.

GCSF4.1 CS Service Management Outputs
This function is responsible for sending commands to the satellite related to the management of the commercial service (CS) - including key management.

GCSF4.1 Clock Management Outputs
This function is responsible for sending commands to the satellite related to the management of the commercial service (CS) - including key management.

GCSF4.1 Design Constraints/Performances
This design constraints and performances related to this function can be summarised as follows:
- The provision of the means to control the following worst case
- operational scenario:
    - A constellation with up to 41 satellites (including operational, in-orbit spares and replacements satellites)
    - Perform orbit and attitude control manoeuvres for at least 4 satellites in parallel
    - Parallel commanding of at least 6 satellites in orbit, plus at least 4 satellites on ground (2 simulated, 2 during LIT or SVT)
    - Capability to support special operations for up to 4 satellites in parallel without affecting routine operations.

GCSF4.1 Functional Block Diagram

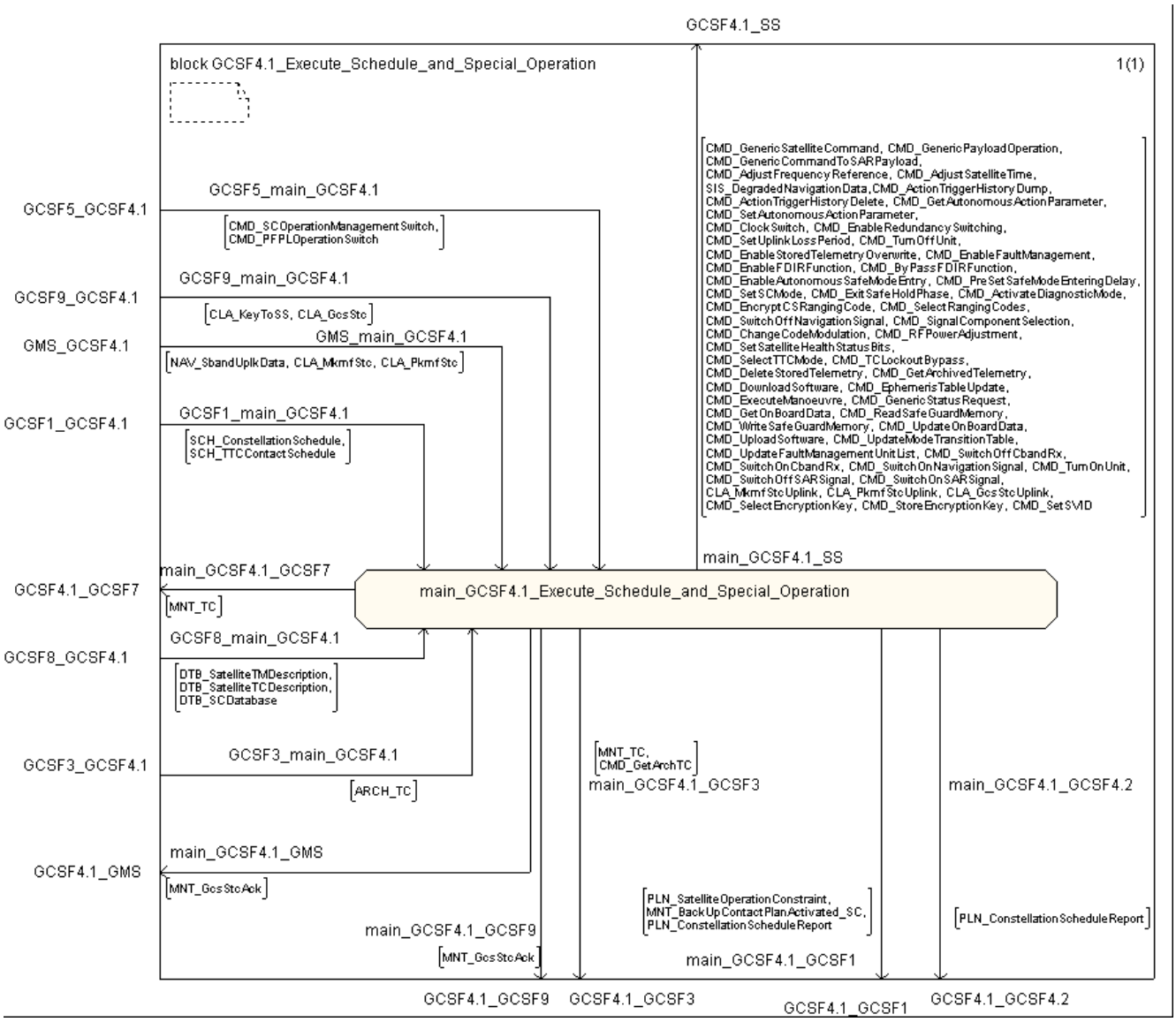


Figure 100: GCSF4.1 SDL Functional Block Diagram

5.2.4.2 GCSF4.2 Constellation Monitoring

GCSF4.2 General Description
This function is responsible for:

- automatically processing telemetry received via S-Band from the satellites.
- automatically processing the satellite health status information received via L-Band at the GMS and forwarded to GCS.
- elaborating the satellite and overall constellation status so that it can be presented to the operators and transmitted to GMS
- verifying telecommand execution status
- providing confirmation to GMS that the OS and PRS navigation data has been successfully uplinked
- detecting spacecraft anomalies (eg OOL), reporting them and reacting autonomously if needed.
- providing an indication to the GMS of the list of satellites available for mission data uplink.

GCSF4.2 Outputs:

GCSF4.2 Secure Telemetry Outputs
This function is responsible for forwarding secure telecommands downlinked from the satellite to the security functions in the GCS and GMS.

GCSF4.2 Archiving Outputs
This function is responsible for sending and retrieving all event and telemetry packets (including calibration data) to and from the local archive.

GCSF4.2 Database Outputs
This function is responsible for populating the GCS anomaly database with anomalies observed in telemetry (from the satellites). It is also responsible for selecting the spacecraft database to be used for telemetry processing.

GCSF4.2 GMS Outputs
This function is responsible for sending the following to the GMS:

- Reports describing the S-band uplink status (ie success or otherwise) of the OS and PRS navigation data originally provided by GMS
- Reports describing the summary status of the satellites.

GCSF4.2 Design Constraints/Performances
The design constraints and performances related to this function provide the capability to:

- Monitor the constellation under the following worst case conditions:
  - 24 operational satellites (contact once per orbit)
  - 6 in-orbit spare satellites
  - 8 replacement satellites (peak replenishment)
  - Perform orbit and attitude control manuvres for at least 6 satellites in parallel
  - Command in parallel at least 5 satellites in orbit, plus 4 satellites on ground (2 simulated, 2 during LIT or SVT)
- Support special operations:
  - Without affecting routine operations
  - With up to 4 satellites in parallel (this does not imply simultaneous contact with all 4 satellites).

- Process and display to the operator in the GCC real time telemetry in not more than 5 seconds after reception of the RF signal at the TT&C station
- Process, in real-time, Telemetry received from any TT&C station, in parallel at both GCCs
- Interpret and display the content of Event and Report telemetry packets in real-time
- Present the real-time status of each spacecraft in the constellation with the capability to update the information not more than 5 second after reception of the pertinent data at the GCC
- Monitor the real-time progress of schedule execution

GCSF4.2 Functional Block Diagram

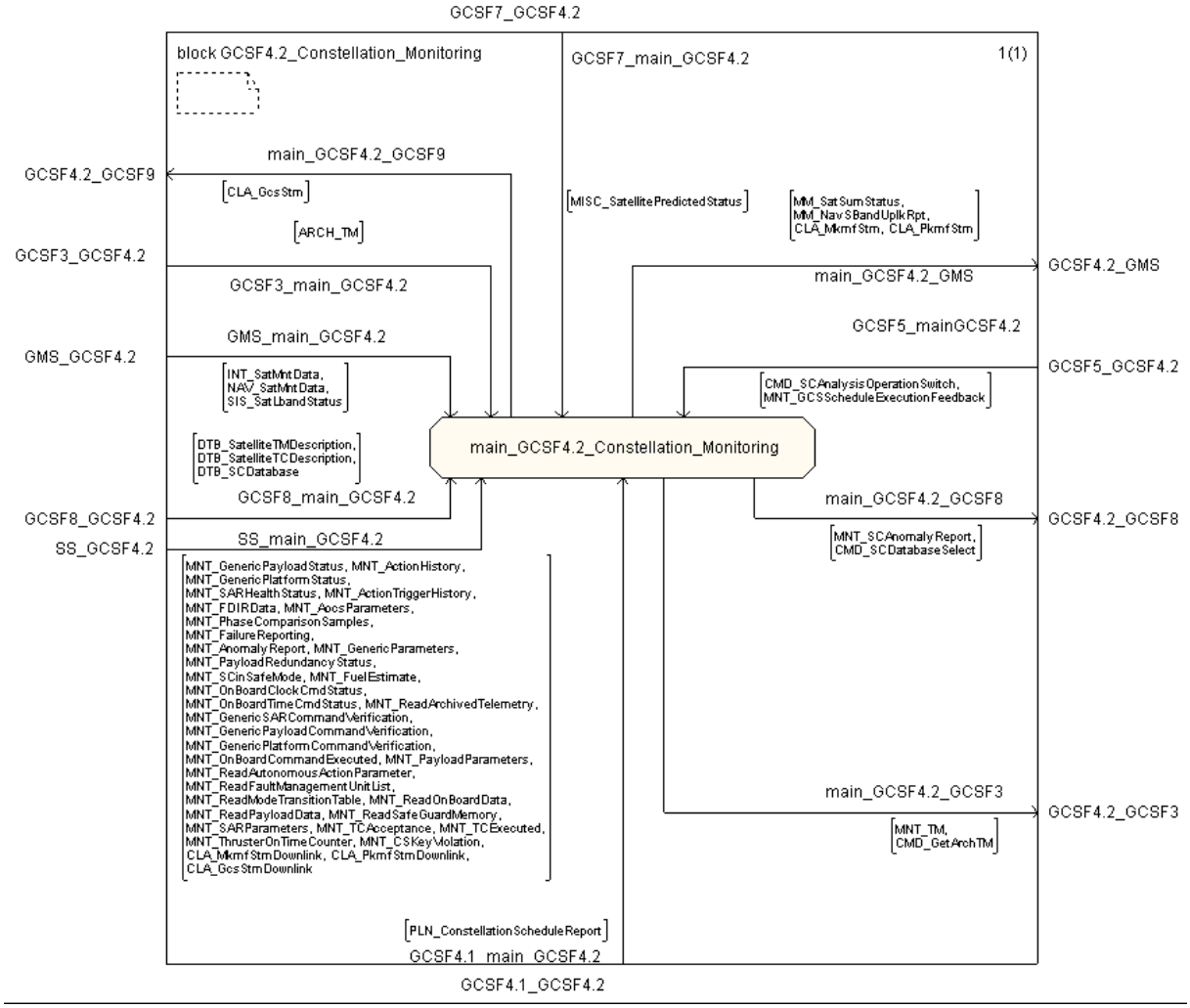


Figure 101: GCSF4.2 SDL Functional Block Diagram

5.2.4.2.1 Expected evolution of GCSF4.2 function

The following DCNs have been generated at system level in the STRB 1.4.1:

- GAL-DCN-TASI-SYST-X-792 v1.0 on GCSGMS ICD (Addition of an extra TTCF)
- GAL-DCN-TASI-SYST-X-788 v1.0 on GCSREQ (6th TTCF in REDU)
- GAL-DCN-TASI-SYST-X 795 v. 1.0 on GCS SSS ICD (6th TTCF in REDU)
- GAL-DCN-TASI-SYST-X-0790 v1.0 on FOC1 GCS implementation requirements (FOC1 Constellation support expansion to 26 satellites)

The main changes introduced by the above DCNs are:

- The increase of the maximum number of supported in orbit satellites by GCS in FOC1 from a 18 to 26 (note that for FOC2 the max number of satellites to be supported by GCS is 41)
- The Addition of a sixth TTCF (addition of Redu on top of the original 5 TTCF located in Kiruna, Kourou, Noumea, Reunion and Papeete)
- The expansion of the capability to support simultaneous parallel TT&C contacts to 6 satellites (vs the previous 5)

Despite all above DCNs were part of the System Technical Requirement Baseline 1.4.1, not all changes have been flown down to GCS in the system build 1.5.0. In particular the GCS presently only foresees:

- a max of 5 TTCF, with Redu replacing Papeete (the 5 TTCF presently supported are hence Kiruna, Kourou, Noumea, Reunion and Redu);
- a max of 5 simultaneous TT&C contacts;
- The capability to support 26 satellites from GCC-I only (a max of 18 satellites can be presently supported by GCC-D).

The implementation of the following missing functionalities (essentially determined by hardware constraints since the GCS is conceived to cope with up to 6 TTCF):

- integration of the 6 TTCF in Papeete;
- capability to support up to 6 simultaneous TT&C contacts ;
- capability to support up to 26 operational satellites from GCC-D,

is only expected for the future system build 1.5.1 (for which design details are not available yet from the GCS). A future GCS evolution is foreseen for FOC, with the necessary upgrade to allow the GCS to support the peak constellation size (41 satellites) specified by current requirements for FOC.

5.2.4.3 GCSF4.3 Manage OBSW

GCSF4.3 General Description

This function is responsible for on-board software management including in particular the following:

- Managing OBSW in the form of memory images (these can either be complete, partial or differential).
- Providing the means to inspect memory images based upon memory maps that define the locations of all variables, constants and executable code.
- Providing the means to import memory images (both full images and patches) from the satellite manufacturer (SATMAN).
- Providing the means to manipulate images (including display, editing, copying, comparing and printing memory images).
- Configuration control of OBSW (images, patches and dumps).
- Providing the means to generate memory load commands for full and partial memory images.
- Providing the means to generate memory dump commands for any area of on-board memory and to generate images based upon the dumped telemetry

GCSF4.3 Outputs

This function is responsible for producing the following outputs:

- Generation of planning requests associated with OBSW maintenance tasks (ie to perform OBSW updates and dumps)
- Requests to verify OBSW updates through off-line simulation
- Request to retrieve archived OBSW TM and TC.
- Delivery of OBSW dumps to SATMAN for anomaly investigations

GCSF4.3 Functional Block Diagram

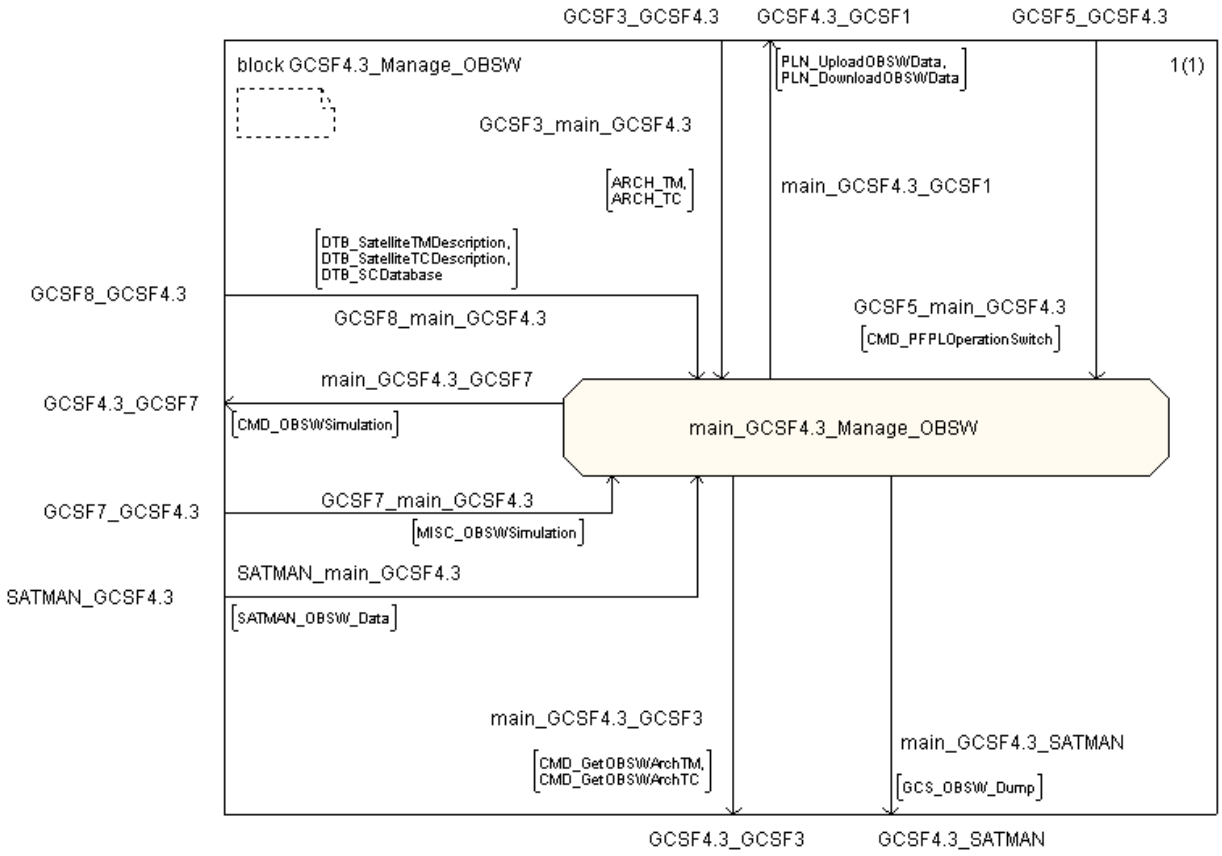


Figure 102: GCSF4.3 SDL Functional Block Diagram

5.2.4.4 GCSF4.4 Flight Dynamics

GCSF4.4 General Description
This function is responsible for all Flight Dynamics related activities including:

- Precise orbit determination based upon:
  - GMS computed ephemeris (nominal case)
  - GCS two-way ranging data (non-nominal case)
- Satellite orbit event predictions for input to planning
- Preparation and validation of satellite orbit manoeuvres
- Flight dynamics analyses (eg for attitude determination, fuel budget estimation, AOCS and thruster performance monitoring, etc).
- Implemention and maintainance of satellite models (including satellite attitude law and centre of mass models).

GCSF4.4 Outputs:

GCSF4.4 Outputs to GMS
This function is responsible for generating the following outputs towards the GMS:

- satellite ephemeris data;
- configuration data for all satellites;
- manoeuvre announcements;
- orbit predictions;
- TT&C measurement data (including ranging, meteo Doppler and angular data);
- orbit event data (including TTC station and spacecraft events).

GCSF4.4 Archiving Outputs
This function is responsible for sending and retrieving flight dynamics data to and from the local archive.

GCSF4.4 Maneuver Preparation Outputs
This function is responsible for producing the following outputs as part of preparing satellite maneuvers:

- requesting maneuver validation through simulation
- providing maneuver data for input to GCS planning functions

GCSF4.4 Design / Performance Constraints
The design constraints and performances related to this function provide the capability to:

- ensure that the absolute out-of-plane position of any spacecraft with respect to the RAAN of the designated reference constellation orbit plane is better than +/- 1 degrees;
- ensure that the absolute along track orbit keeping between any operational satellites and its operational obrit slot in the same orbital plane is better than +/-2 degrees;
- ensure that the absolute out-of-plane position of any spacecraft with respect to the reference constellation inclination of 56 degrees is better than +/-2.0 degrees;
- calculate the orbital position of any Galileo satellite, based on two-way ranging measurements, to within 5 metres accuracy with 1-sigma uncertainty within 24hrs;
- support orbit and attitude control manoeuvres for at least 4 satellites in parallel;
- implement a satellite attitude law model with an accuracy of +/-1 degrees;
- implement a satellite centre-of-mass model with an accuracy of+/-15mm.

GCSF4.4 Functional Block Diagram

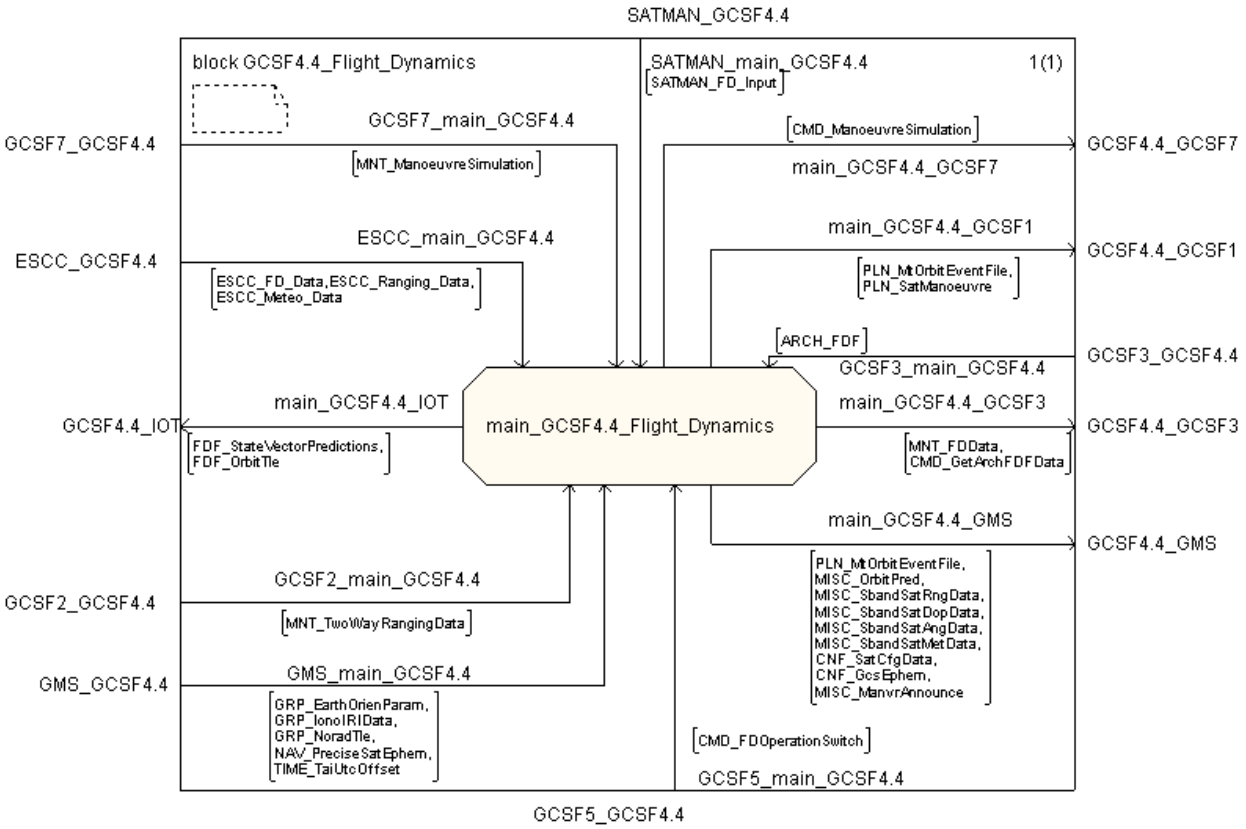


Figure 103: GCSF4.4 SDL Functional Block Diagram

5.2.5 GCSF5 GCS Monitoring and Control

GCSF5 General Description
This function is in charge to monitor and control all the GCS elements.
The main functionalities are:

- GCS elements real-time monitoring and control
- GCS elements maintenance and troubleshooting
- Switching and Handover
- GCS schedule execution
- Reporting
- GCS element time synchronization

This function is also involved in the Galileo ground assets monitoring and control and in the GCS short term planning, providing current status of GCS assets to the relevant engines.

For clarification, this function is derived into three sub-functions:

- GCSF5.1 for element monitoring and control and
- GCSF5.2 for central monitoring and control
- GCSF5.3 for network monitoring and control

GCSF5 outputs
The GCS monitoring and control function produces the following outputs:
1 - summary status of the GCS elements, which is sent to the GMS relevant function for Galileo overall ground asset M&C and sent to the GCS short term planning
2 - database configuration
3 - GCS schedule execution feedback
4 - GCS summary network M&C function for display (collected and provided by the GMS network M&C function responsible of collecting the required data from the network devices)

GCSF5 design constraints/performance
The monitoring and control function of the GCS is performed at three different levels:

- level 0, for local maintenance, performed using element local monitoring and control facility
- level 1, allowing GCS remote maintenance from GCC, for troubleshooting, preventive maintenance
- level 2, allowing GCS central monitoring and control from GCC during operations

This function relies on a database oriented system, based on the SNMP protocol.

The GCS facilities located in the Galileo Control Centre that have a backup mode are able to switch from master to backup mode in less then 1 minute.

This function provides status information that may be used during the GCC hand-over procedure..

GCSF5 Functional Block Diagram

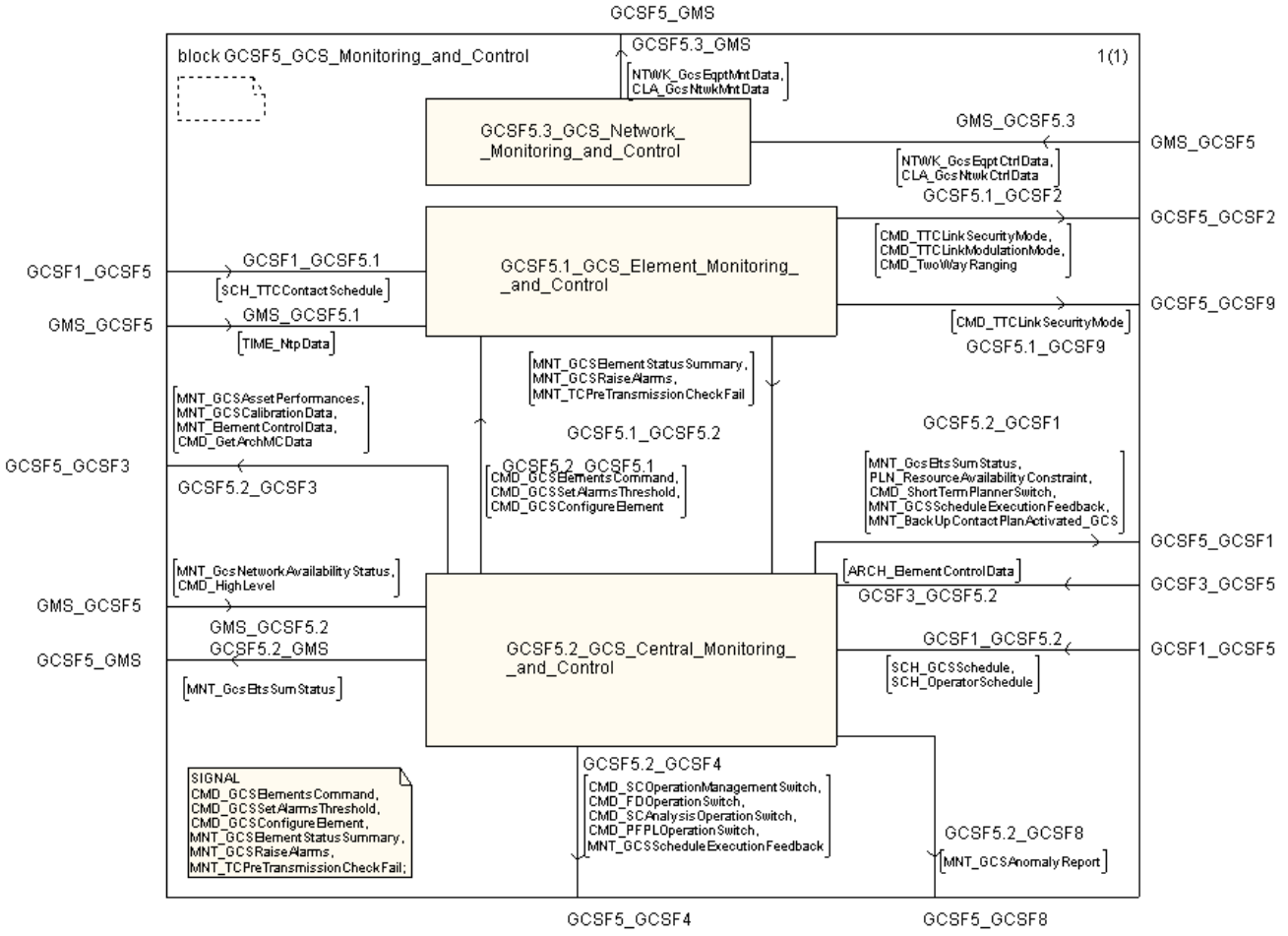


Figure 104: GCSF5 SDL Functional Block Diagram

5.2.5.1 GCSF5.1 GCS Element Monitoring and Control

GCSF5.1 General description

The GCS element M&C function is the element local instance of the GCS monitoring and control function.
It is therefore implemented locally in each element.
Its aim is to:

- monitor the element, i.e. gathering all the element health statuses and alarms (including elements' LANs) and send them to the GCS central M&C function, located in the central GCC location
- execute the configuration commands sent by GCS central M&C function
- execute the TTCF contact plans (contact directives between TTCs and Satellites)
- allow local monitoring and control by an operator for all monitoring and control level-0 activities (i.e. maintenance operations)
- prevent dual control from two different entities (e.g. GCS central M&C function and local operator)
- maintain a local instance of element configuration database

GCSF5.1 outputs

The GCS element monitoring and control function produces the following outputs:
1 - element status of the GCS elements, which is sent to the GCS central M&C function
2 - alarm reporting

GCSF5.1 Functional Block Diagram

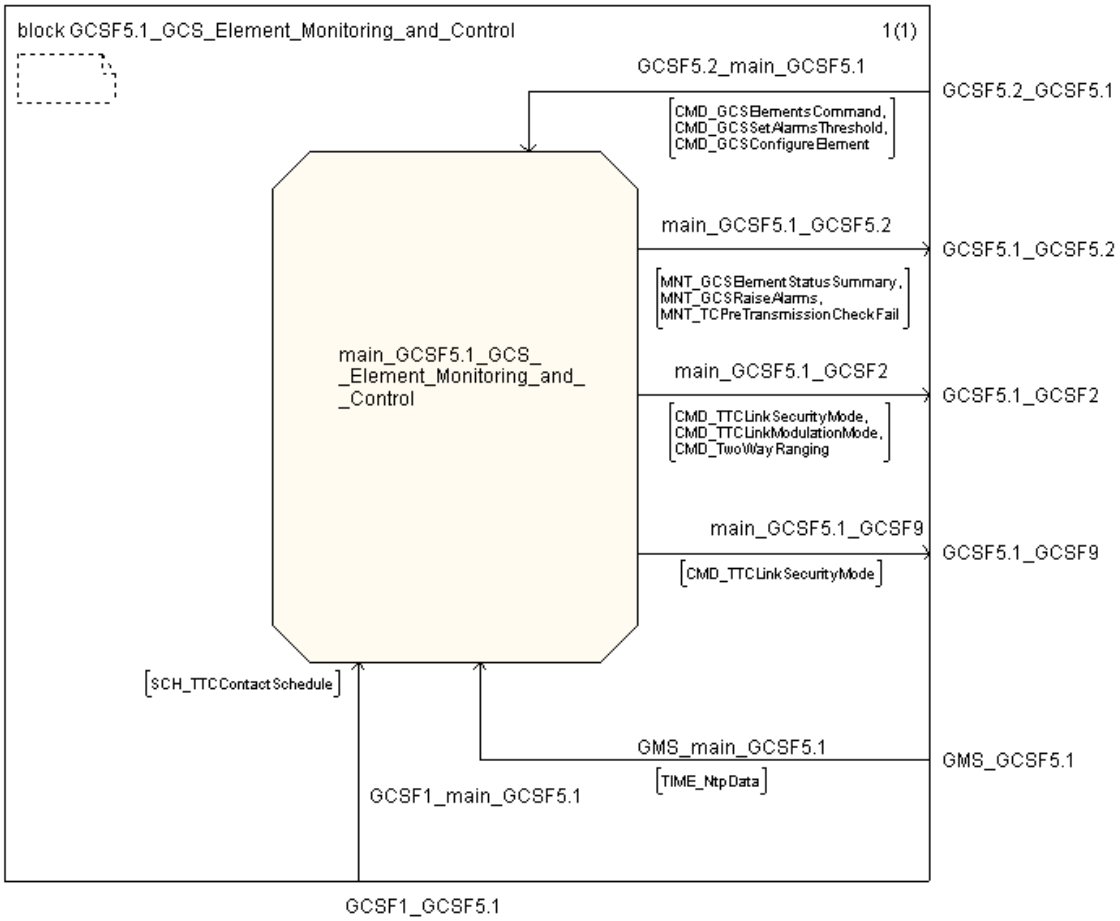


Figure 105: GCSF5.1 SDL Functional Block Diagram

5.2.5.2 GCSF5.2 GCS Central Monitoring and Control

GCSF5.2 General description

The GCS central M&C function is the segment level instance of the GCS monitoring and control function. It is implemented in the GCC.
Its aim is to:

- gather all monitoring health statuses and alarms from all GCS elements (coming from GCS local M&C function)
- send GCS summary status and alarms to the overall M&C function of the GMS
- send GCS summary status to the GCS short term planning function for planning preparation or update if necessary
- execute the configuration macro-commands sent by the GMS overall M&C function
- allow central monitoring and control on all the GCS elements by operators for level-1 activities (i.e. maintenance operations, via central M&C and normal ground operations handling), via adequate displays, MMI, logbooks …
- maintain elements and GCS configuration database

GCSF5.2 outputs

The following outputs are produced by the GCSF5.2 function:
1 - summary status of the GCS elements, sent to the GMS function for ground assets monitoring and to the GCS short term planning in order to be taken into account in the GCS short term planning elaboration
2 - database configuration, sent to the GMS function for ground assets monitoring
3 - configuration commands for all GCS elements
4 - GCS schedule execution feedback to the GCS short term planning function

GCSF5.2 Functional Block Diagram

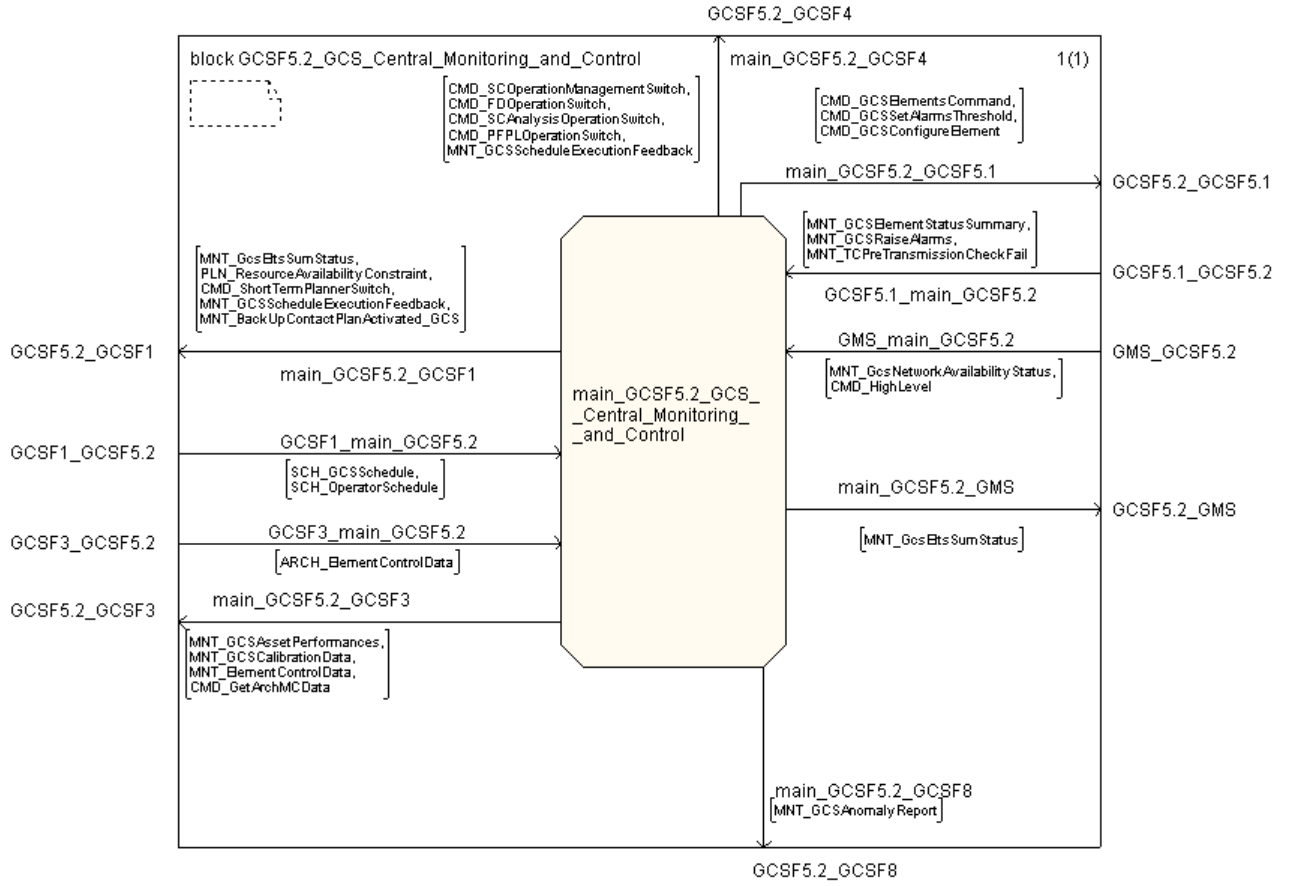


Figure 106: GCSF5.2 SDL Functional Block Diagram

5.2.5.3 GCSF5.3 GCS Network Monitoring and Control

GCSF5.3 General description

The GCS local network M&C function is spread over the GCS part of the Galileo network.
Its aim is to:

- perform control over the GCS network (data switching, data routing, data flow priority setting, …)
- continuously gather GCS network performance measurement of all traffic streams going across the network and generate QoS (quality of service) stamps
- monitor the GCS network equipment
- report to GMS network monitoring and control function

The following performance parameters are measured:

- latency, i.e. delay of packets
- integrity, i.e. loss of packets
- network availability, i.e. possible equipment failure

GCSF5.3 network monitoring and control outputs

The main output of this function is the status of the network equipment and the performance of the GCS network links. The GMS network M&C function is responsible of collecting the required data from the network devices and to forward a summary to the GCS network M&C function which displays and stores this information.

GCSF5.3 Functional Block Diagram

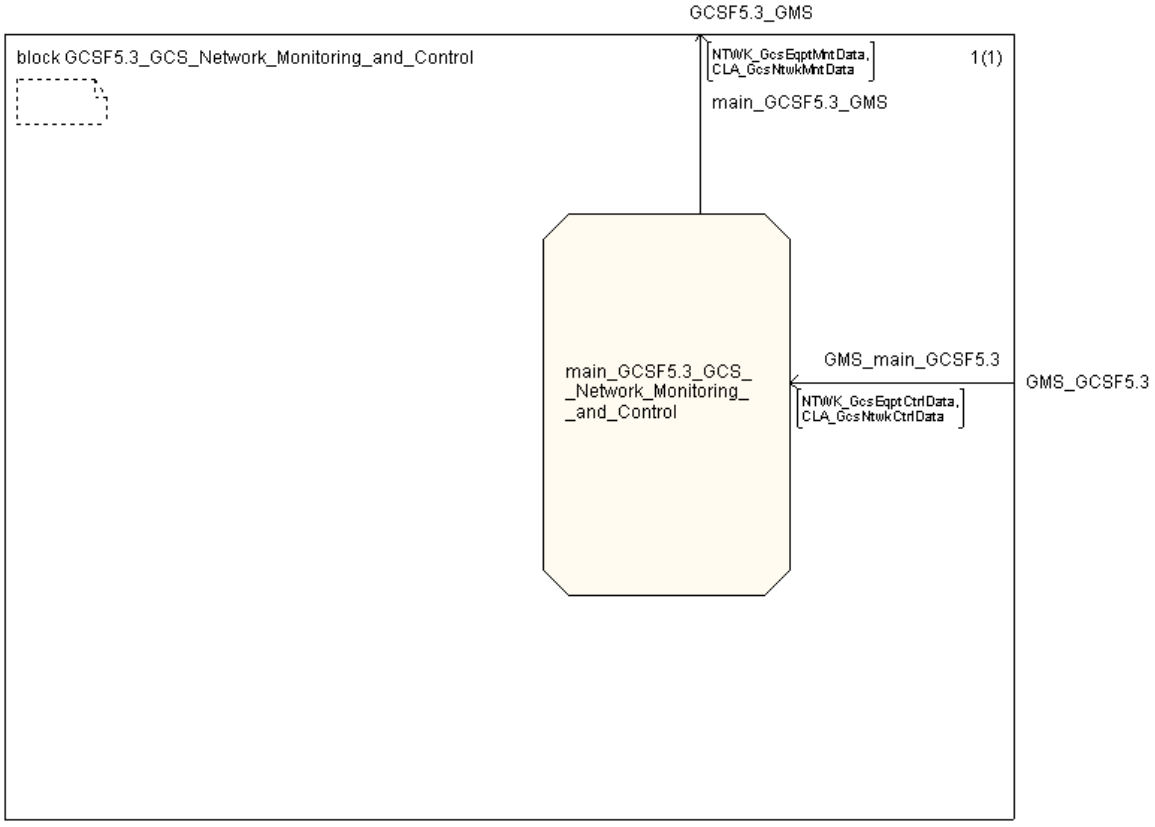


Figure 107: GCSF5.3 SDL Functional Block Diagram

5.2.6 GCSF6 Provide Training

GCSF6 Description

This function is in charge to support operator's training.

GCSF6 Functional Block Diagram

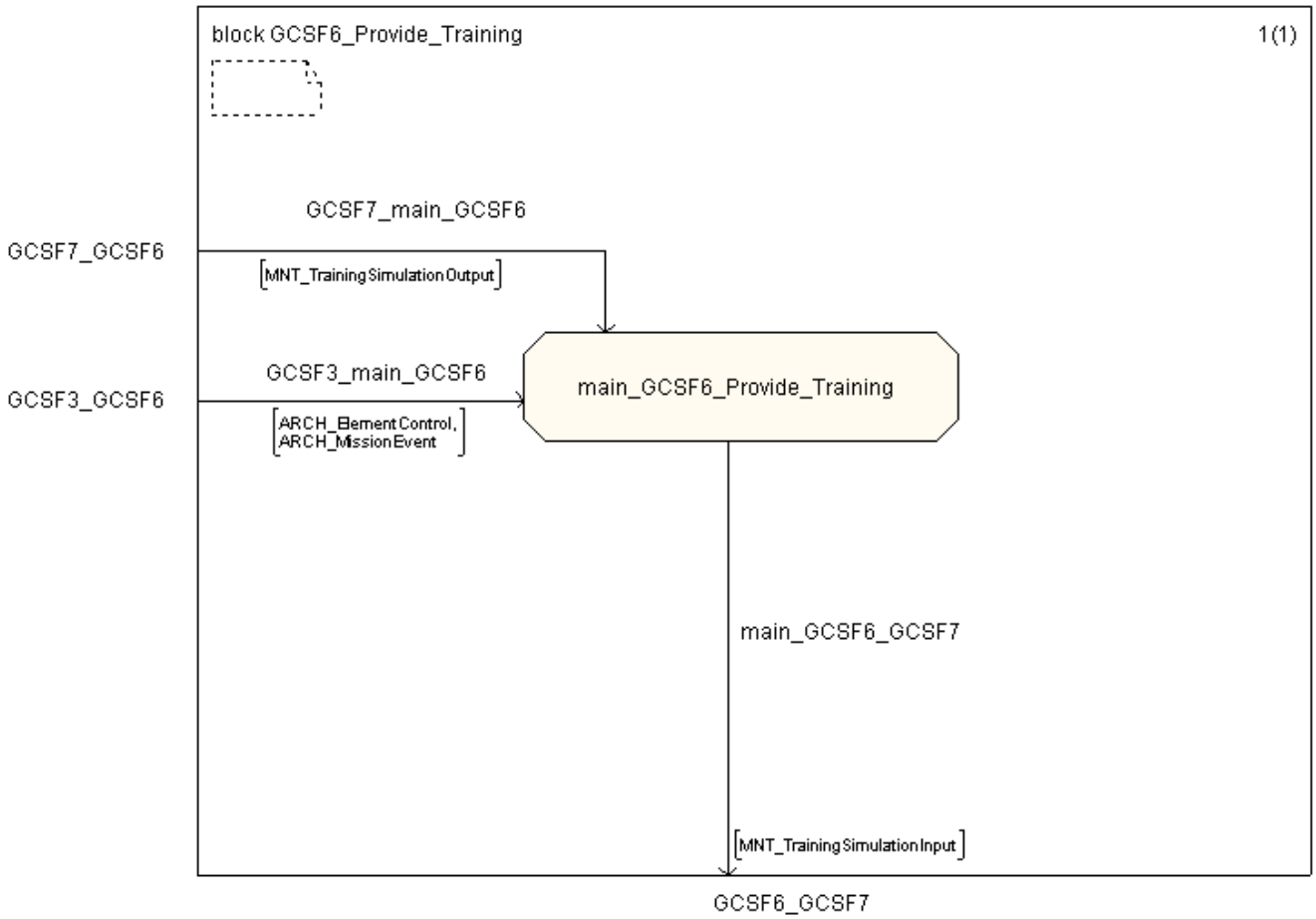


Figure 108: GCSF6 SDL Functional Block Diagram

5.2.7 GCSF7 Perform Simulation and Test

GCSF7 Description

This function is in responsible for providing a simulation and validation environment for the satellite constellation and ground assets.

This includes all the functions in charge of replaying the data retrieved from archive

GCSF7 Functional Block Diagram

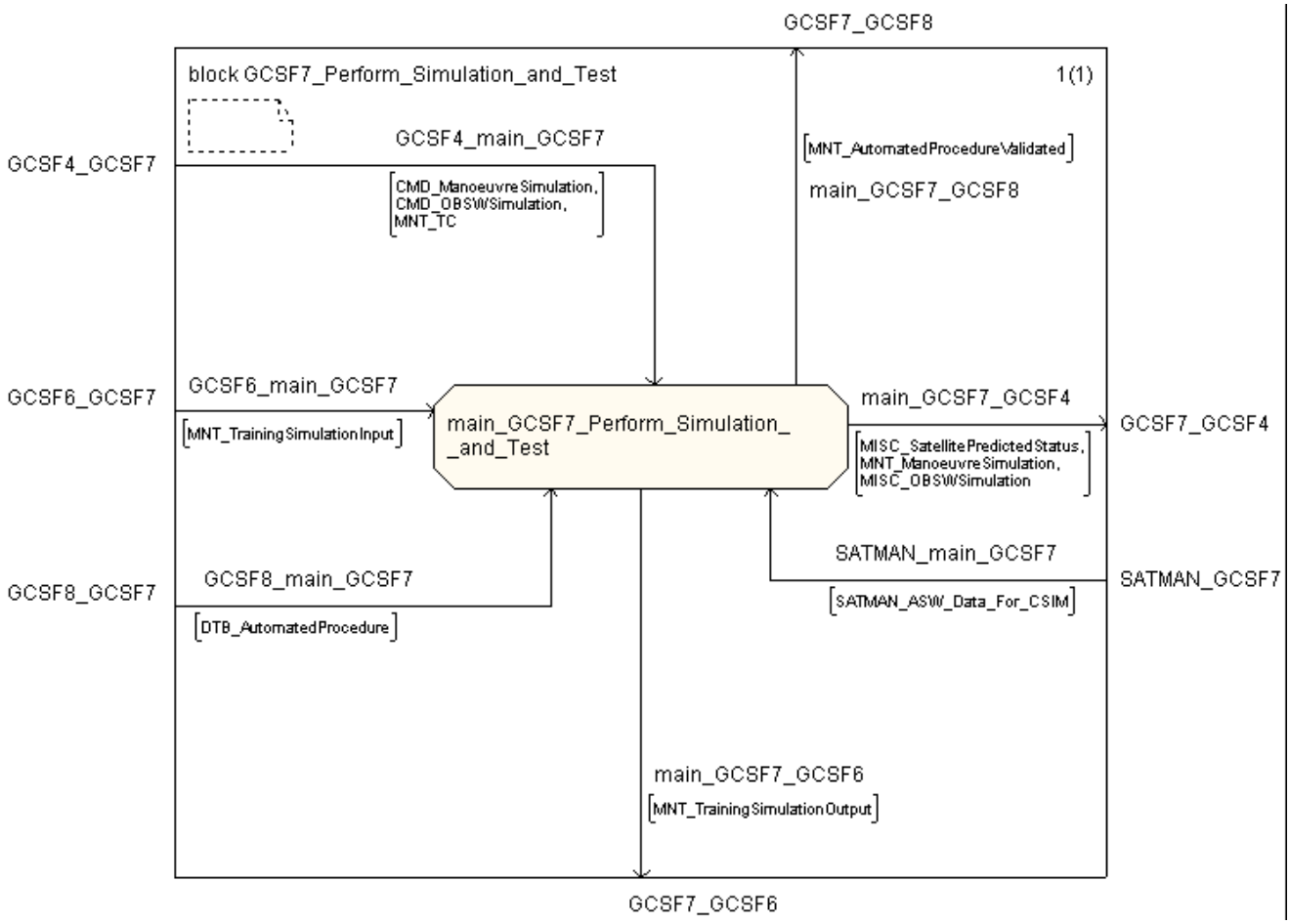


Figure 109: GCSF7 SDL Functional Block Diagram

5.2.8 GCSF8 Manage DataBase

GCSF8 Description
This function is in charge to maintain Databases.
Main functionalities:

- SC TM/TC Database
- Anomaly Database
- Operational procedure Database

GCSF8 Functional Block Diagram

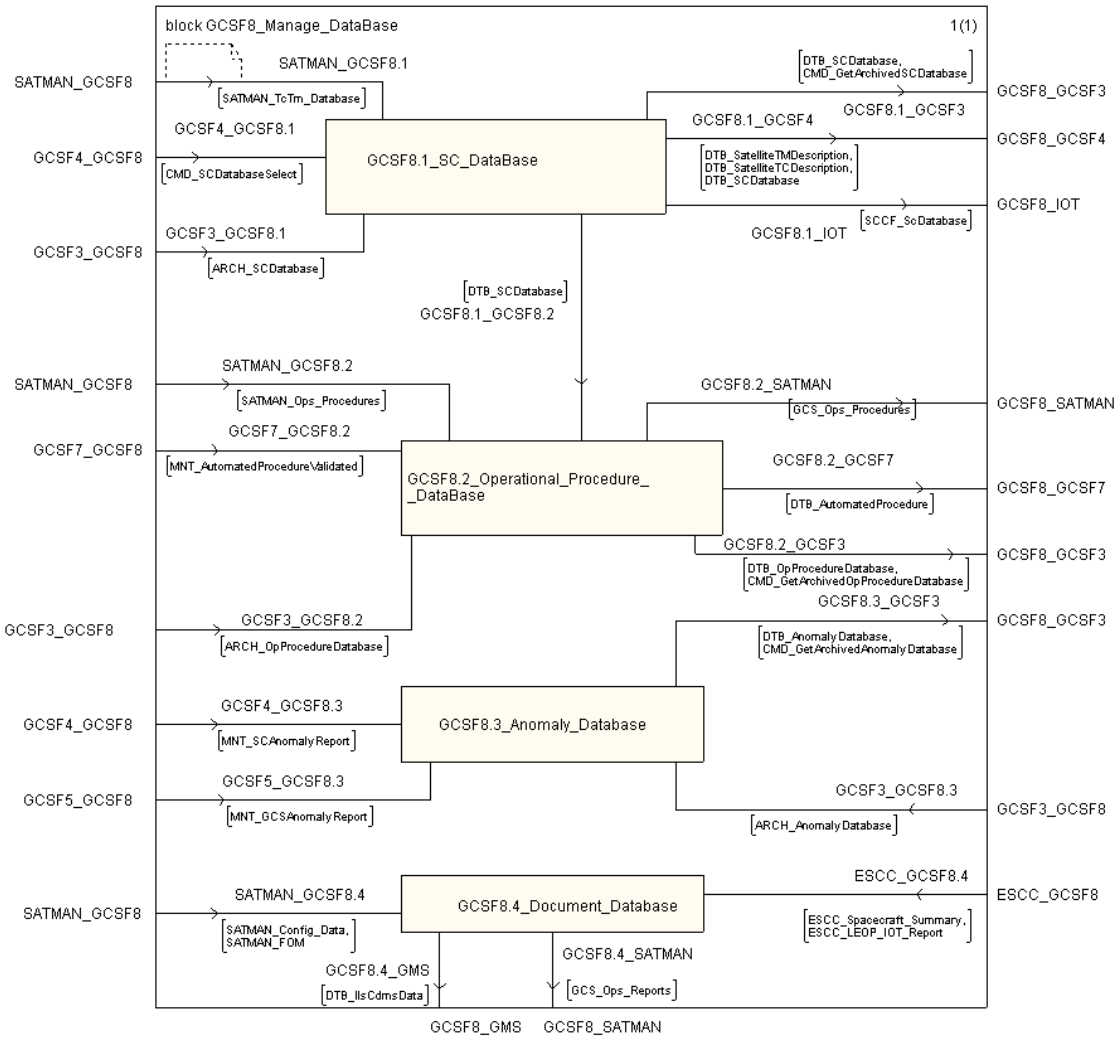


Figure 110: GCSF8 SDL Functional Block Diagram

5.2.8.1 GCSF8.1 SC DataBase

GCSF8.1 Description
Spacecraft configurable database with a common structure for each satellite containing current status information, including at least:

- The actual status and trend analysis of the spacecraft
- Version numbers of any applicable CI
- Operational status of equipment, heater configuration, FDIR context,
- Date/time of last manoeuvres,

This database is fully editable by operators

GCSF8.1 Outputs

This function reports report of differences between the satellite database and an imported EGSE satellite database

This function can export the following data:

- the complete database of an individual satellite or group of satellite
- the telemetry data of an individual satellite or group of satellite
- the telecommand data of an individual satellite or group of satellite

GCSF8.1 Design Constraints

This function provides the means to create, modify and delete the content of the satellite database

The satellite database is design to allow importation of TM/TC from EGSE satellite database

The time to derive and export a complete single spacecraft consistency checked database does not exceed 5 minutes

GCSF8.1 Functional Block Diagram

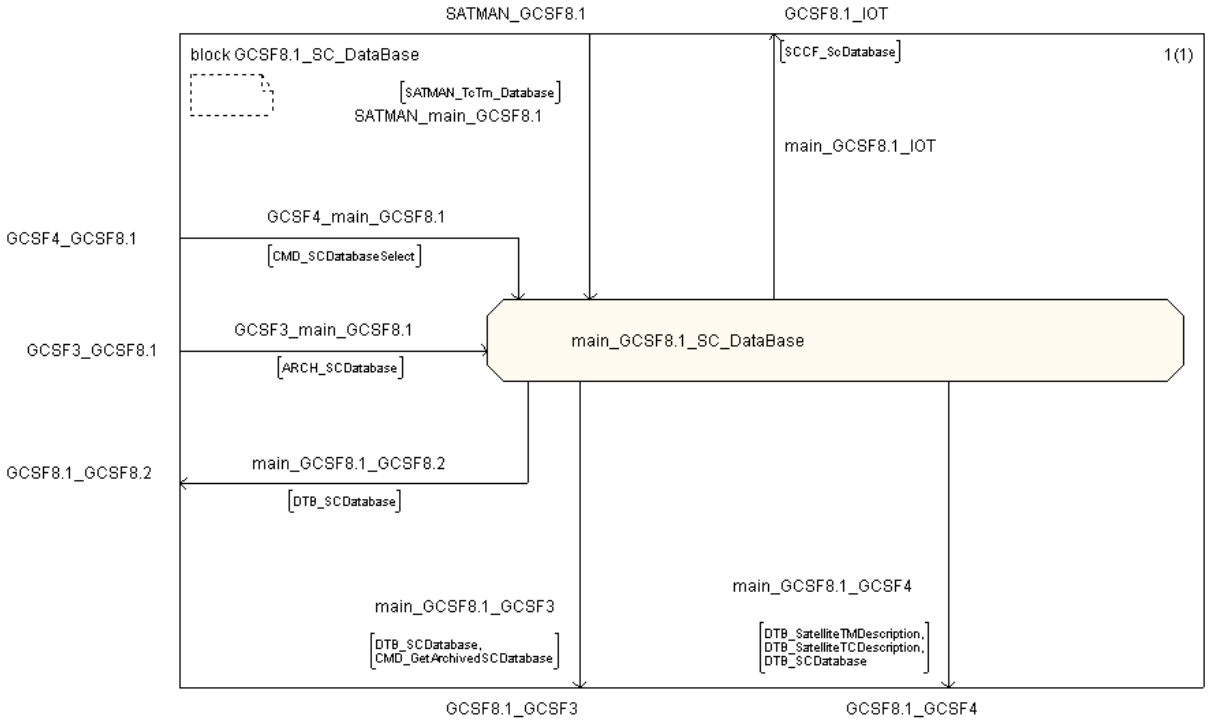


Figure 111: GCSF8.1 SDL Functional Block Diagram

5.2.8.2 GCSF8.2 Operational Procedure Database

GCSF8.2 Description
Manage Database containing:
- Operation procedures for operators
- Flight Control procedures
- Ground Control procedures

GCSF8.2 Outputs
This function provides flight control and ground control procedures directly accessing from the satellite and ground control element database

GCSF8.2 Design Constraints
This function generates flight control and ground control procedures in real time.

GCSF8.2 Functional Block Diagram

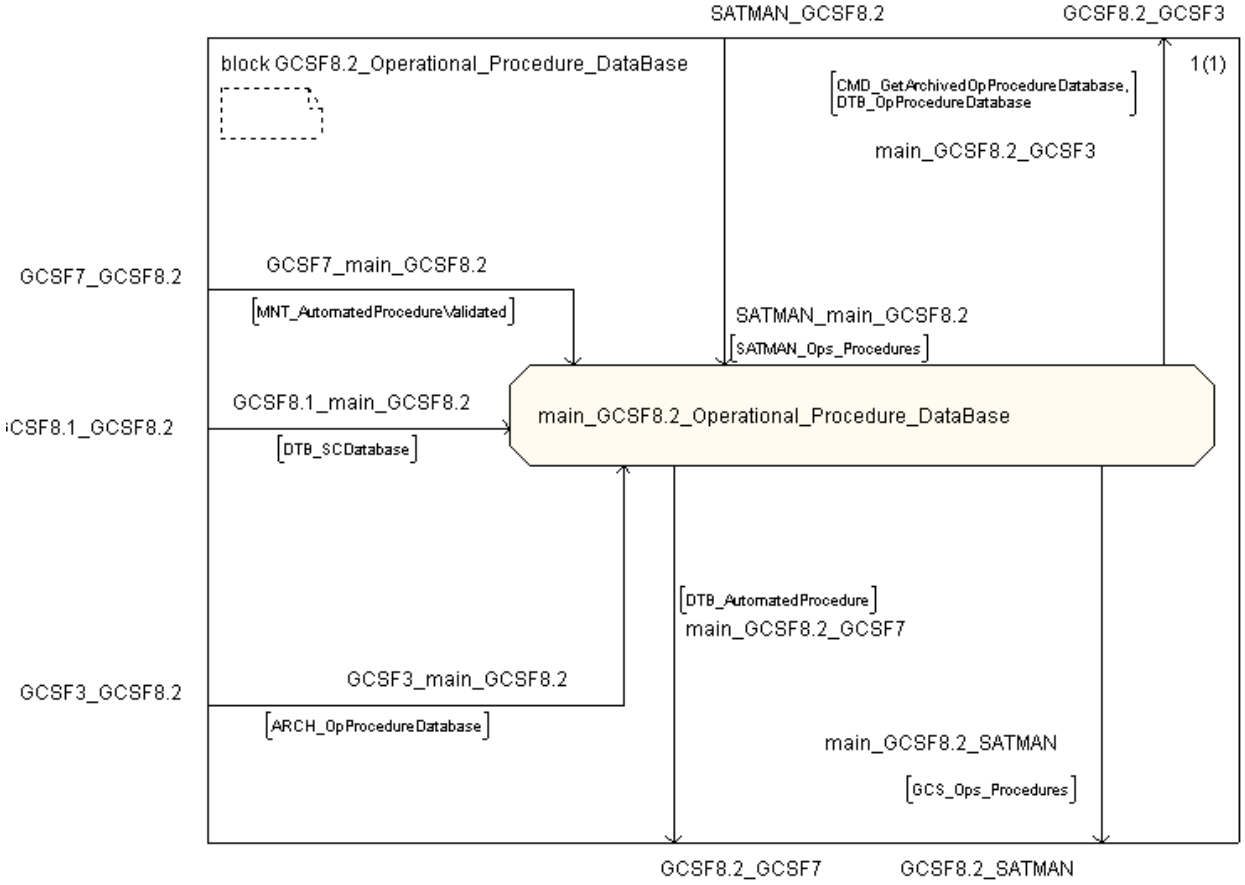


Figure 112: GCSF8.2 SDL Functional Block Diagram

5.2.8.3 GCSF8.3 Anomaly Database

GCSF8.3 Description
Manage Anomaly Database:
- Anomalies observed on the satellites and the related recovery flight control procedures
- Anomalies observed in the GCS (TT&C, computers).

GCSF8.3 Outputs
This function provides reporting for the following space and ground anomaly databases.

GCSF8.3 Design Constraints
This function is manually managed to include all the anomalies observed on the satellites and the related recovery flight control procedures, and the anomalies observed in the GCS (TT&C, computers).

GCSF8.3 Functional Block Diagram

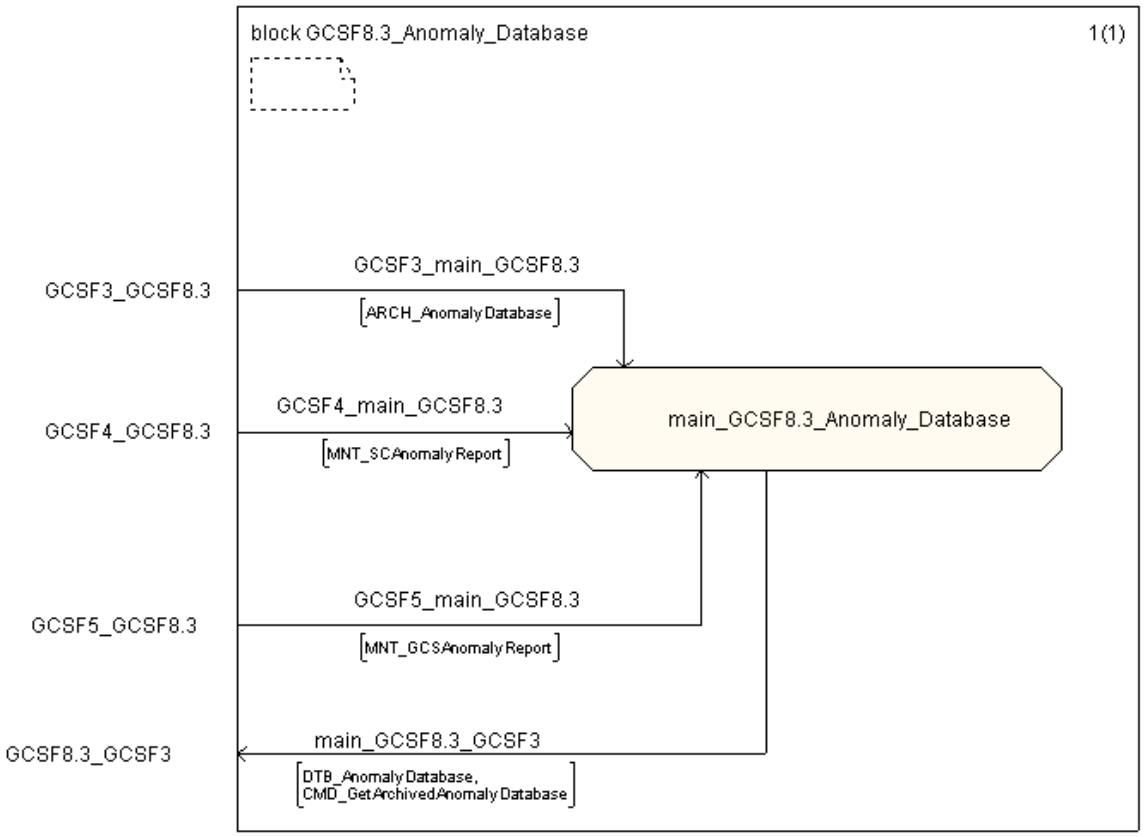


Figure 113: GCSF8.3 SDL Functional Block Diagram

5.2.8.4 GCSF8.4 Document Database

GCSF8.4 Description
This function controls the following as Configurable Items (CI's):

- databases (e.g. Satellite, Flight Dynamics, Mission Planning, ground assets etc.)
- Procedures (Satellite, Ground Station etc.)
- Patches (OBSW etc.)
- Operational Documentation

GCSF8.4 Outputs
This function provides the means to store, display, edit, archive and retrieve Configurable Items.
This function interfaces with the central global archive.

This function provides the means to import and export data directly in electronic form from/to external sources, e.g. the satellite manufacturer.

GCSF8.4 Design Constraints

This function provides the means to import and export data directly in electronic form from/to external sources, e.g. the satellite manufacturer.

GCSF8.4 Functional Block Diagram

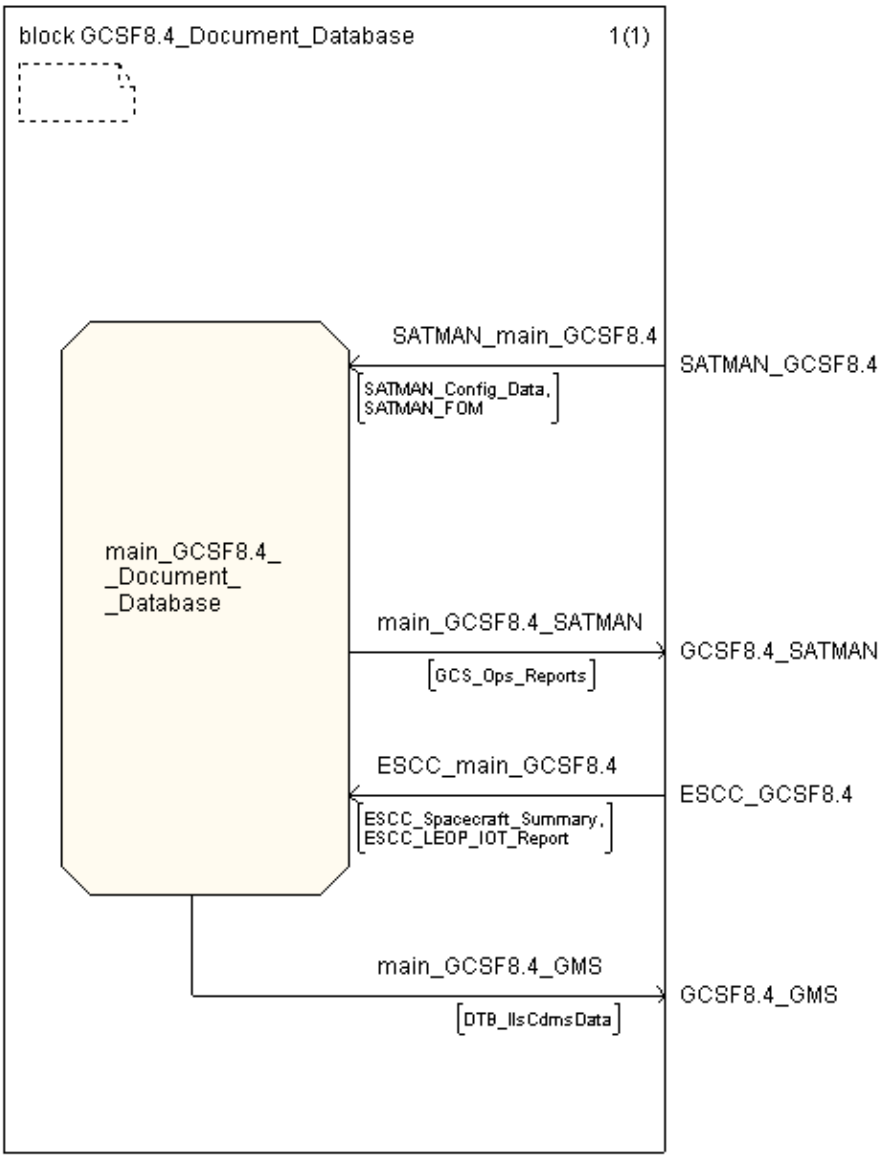


Figure 114: GCSF8.4 SDL Functional Block Diagram

5.2.9 GCSF9 Security

GCSF9 General Description

The GCSF9 includes all functions related to security management at GCS level. These functions are linked to dedicated Security Classified Functions part of Classified SEC-SFCs detailed in the SYSBSLDEF Part I Classified Annex [SYS-RD.07].

- GCSF9.1 (Security protection of S-Band): linked to SEC-GCSF2 and SEC-GCSF3
- GCSF9.2 (Security protection of ground infrastructure): linked to SEC-GCSF2 and SEC-GCSF3

GCSF9 Functional Block Diagram

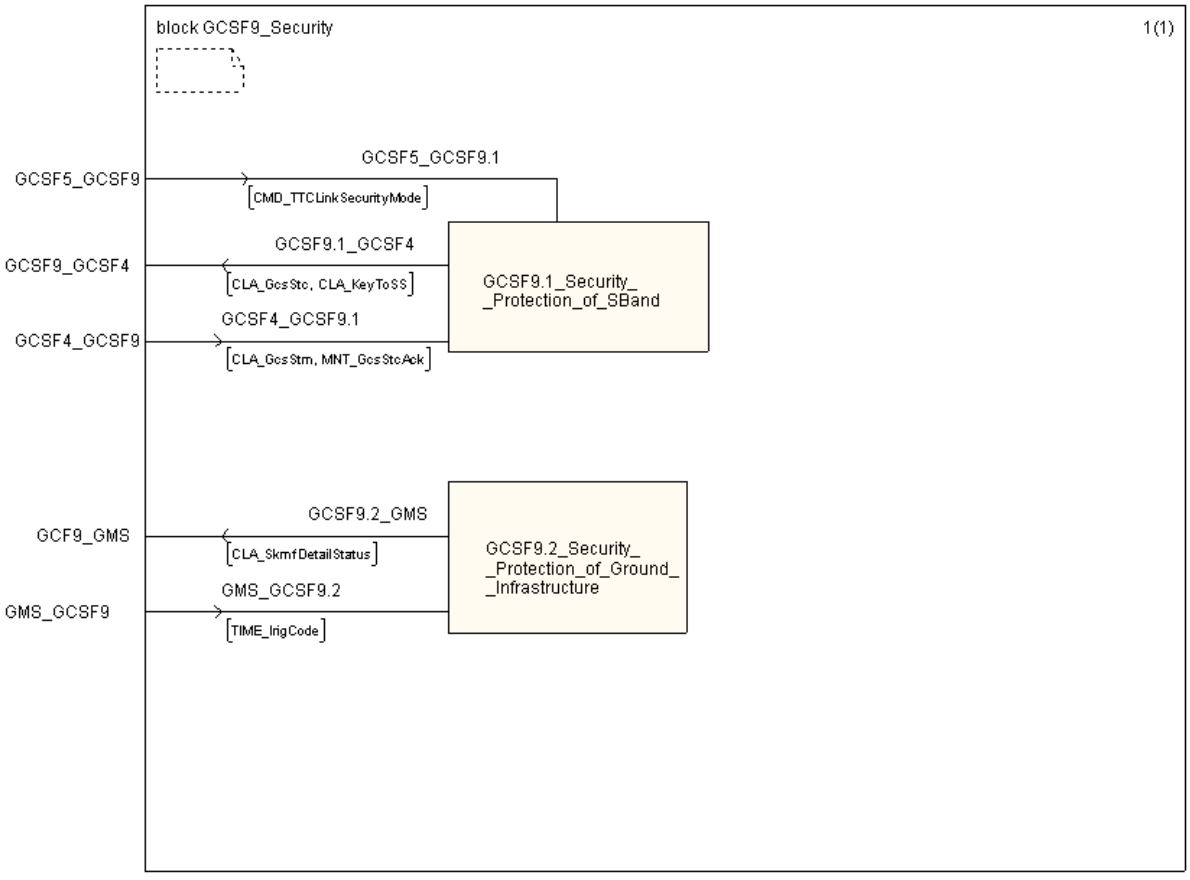


Figure 115: GCSF9 SDL Functional Block Diagram

5.2.9.1 GCSF9.1 Security protection of S-Band

GCSF9.1 General Description
This function is responsible for managing all the security aspects for the protection of the S-Band link. This includes:

- key generation and link mode management for the S-Band protection
- managing the STM/STC related to PRS security unit management dedicated to the satellite PFSU.

This function corresponds to dedicated Classified Security Functions SEC-GCSF2 and SEC-GCSF3. Refer to SYSBSLDEF Part I Classified Annex [SYS-RD.07] for details.

GCSF9.1 Outputs
The outputs of this function are:

- Key for the S-Band protection
- KMF STC dedicated to on-board PFSU

See SEC-GCSF2 and SEC-GCSF3 for further details.

GCSF9.1 Functional Block Diagram

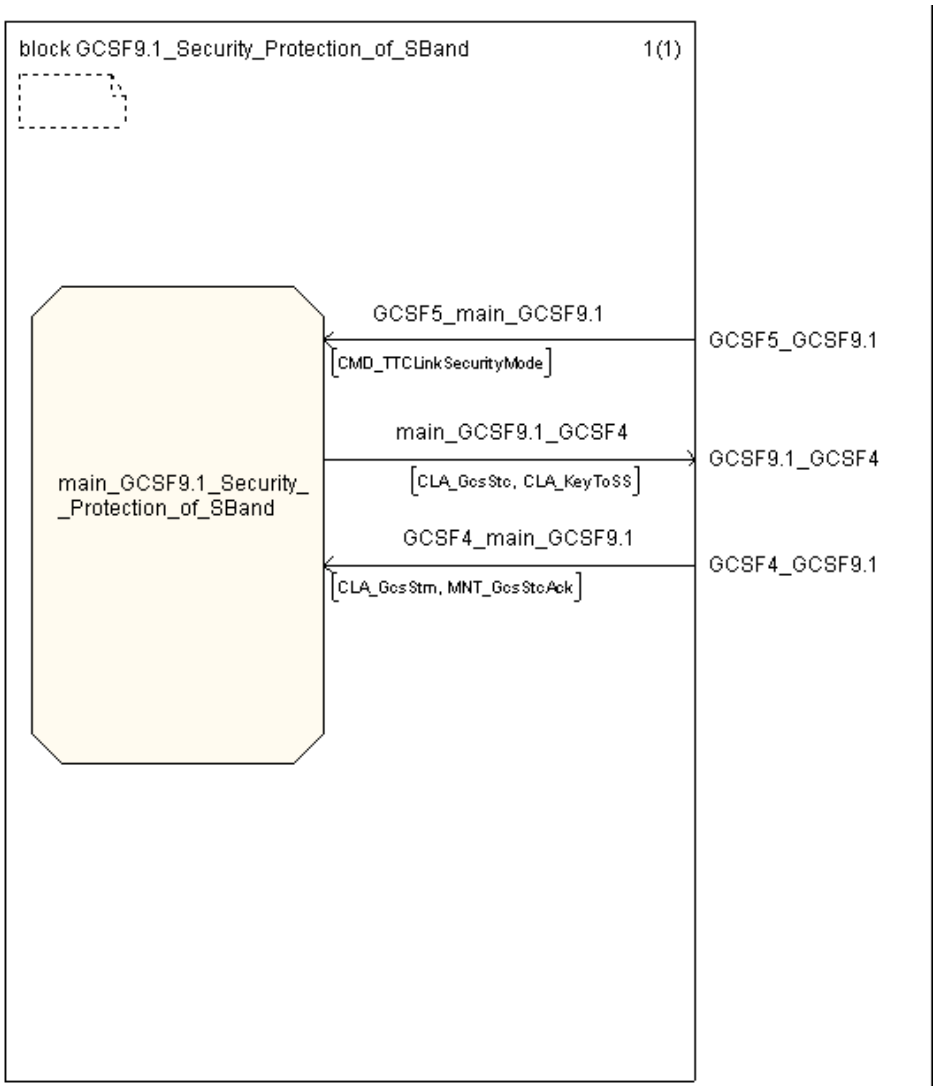


Figure 116: GCSF9.1 SDL Functional Block Diagram

5.2.9.2 GCSF9.2 Security protection of ground infrastructure

GCSF9.2 General Description
This function is responsible for managing all the security aspects for the protection of the ground infrastructure within the GCS. This includes:

- time syncrhonisation of KMFs
- Secured M&C of security elements

This function corresponds to dedicated Classified Security Functions SEC-GCSF2 and SEC-GCSF3. Refer to SYSBSLDEF Part I Classified Annex [SYS-RD.07] for details.

GCSF9.2 Outputs
The outputs of this function are:

- KMF status

See SEC-GCSF2 and SEC-GCSF3 for further details.

GCSF9.2 Functional Block Diagram

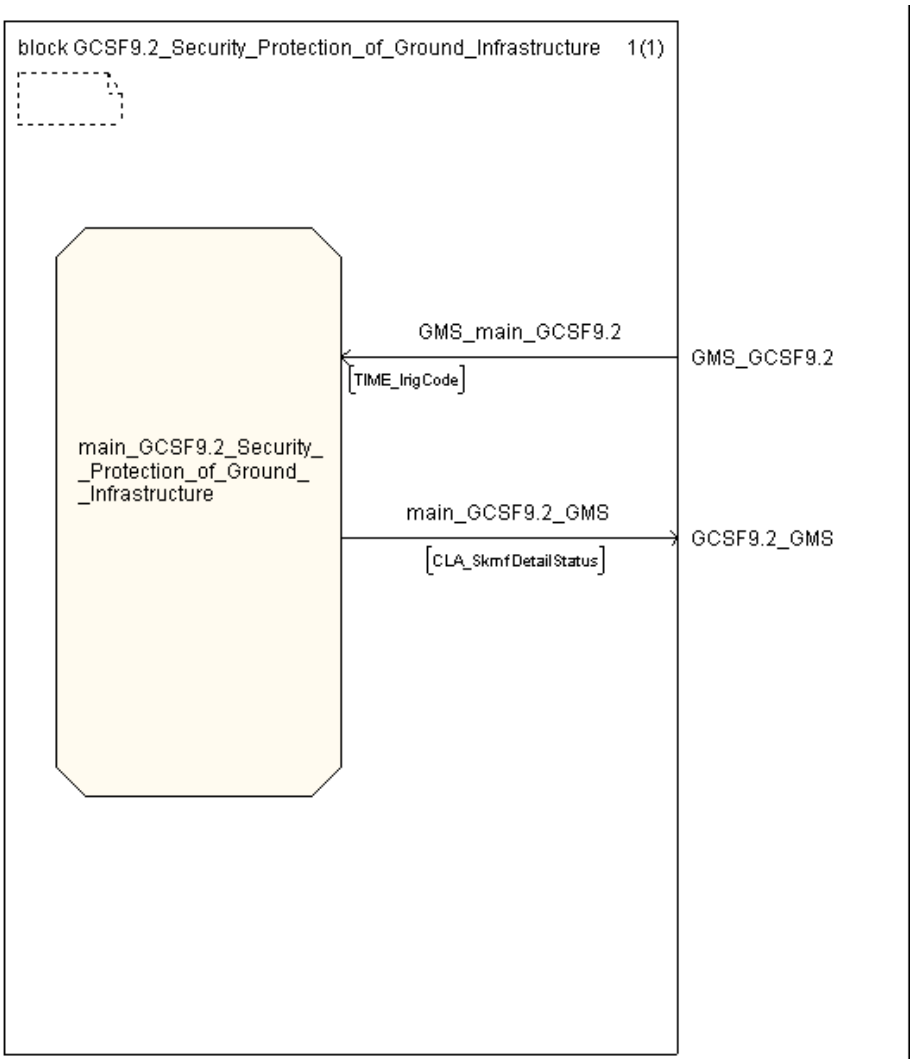


Figure 117: GCSF9.2 SDL Functional Block Diagram

5.3 GMS Functions

General Note applicable to GMS Functions in this section: SoL and integrity have been descoped by MRD 7.1 neverthless design have not changed as per instructions. Therefore in the following SoL and Integrity are still recalled as considered in original design, even if not used anymore

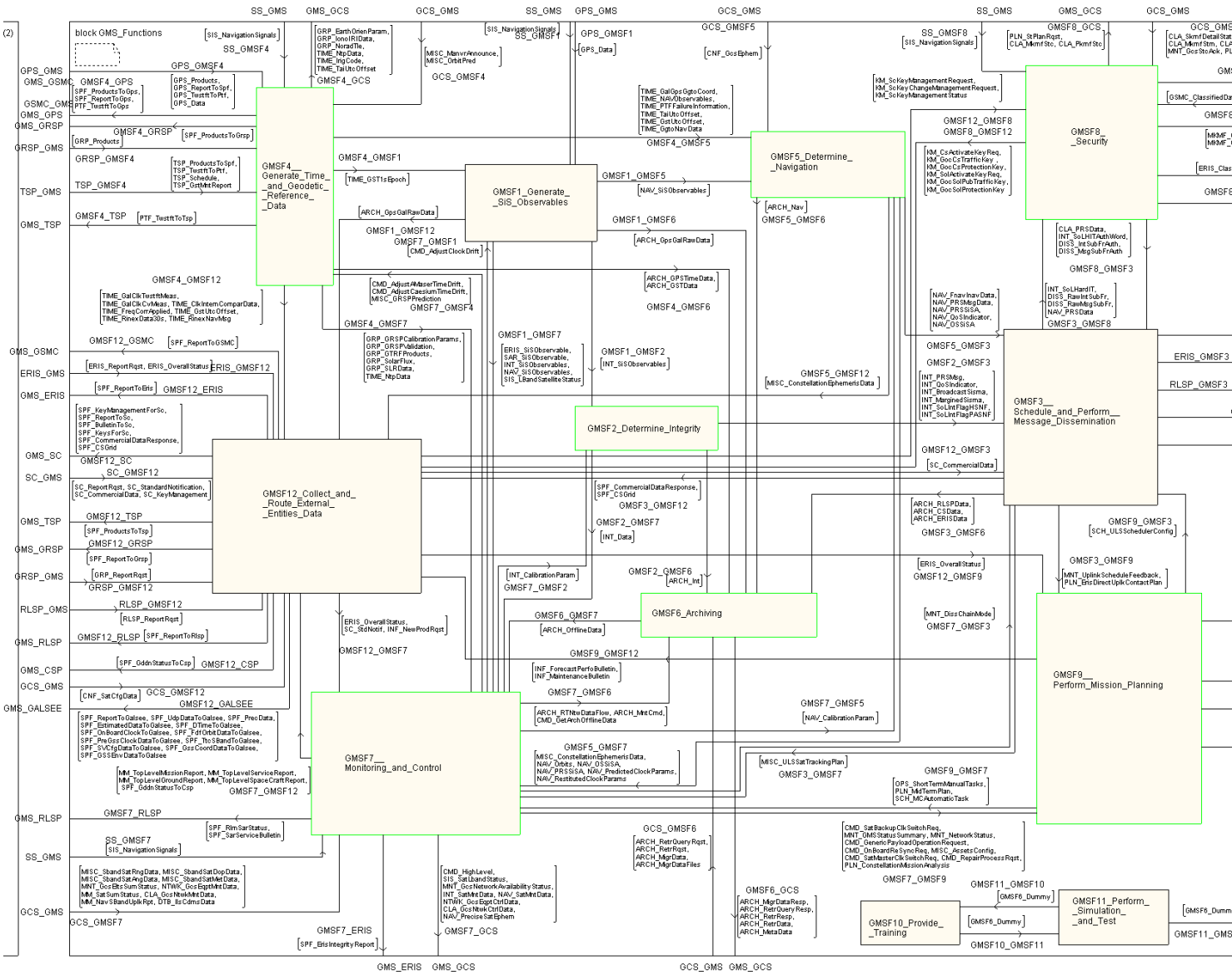


Figure 118: GMS SDL Functional Block Diagram

5.3.1 GMSF1 Generate SiS Observables

GMSF1 General Description
This function is responsiible for generating navigation observables (raw data measurements) derived from SiS measurements performed by the GSS. These observables are generated for each satellite frequency with a 1-second sampling rate from two receiving chains.
The measurements are time-stamped on a local time. In addition this local time is synchronised with the GST one second epoch.

The measurements are then provided to the navigation determination function.

The function includes the following tasks:

- Generate and synchronise GSS local time with the GST,
- Synchronise GSS Measurements to local time,
- Generate Navigation observables,
- Generate SIS observables

The function also provides to the GCC all available GPS observables with a 1-second sampling rate.

In addition, this function is responsible for extracting the satellite health status information (ie limited set of critical satellite telemetry) from the navigation signals and passing this to the GMS monitoring and control functions for onward transmission to the GCS.

Generation and timestamping of the navigation observables, based on the SiS observables and the GSS local time.

Generation of the GSS SiS measurements, including code phase and carrier phase.

Synchronisation of the navigation observables to GST

GMSF1 Outputs
The GMSF1 outputs include:

- The major features of the signals received from the satellites (apparent time of propagation, apparent Doppler, signal to noise ratio …) relative to reference time provided through a local “free-running” clock.
- SIS from all in view satellites including Galileo satellites for all frequencies (non PRS and PRS), and GPS E1/L2 satellites.
- 1 Hz Raw Data Messages which include : Raw measurements and data from the SIS message (navigation non PRS and PRS, commercial services, SAR); with time stamping and parameters concerning the SIS measurement quality,

The results of GSS signal acquisition and "observables" production is sent to the processing facilities at GMS after some local pre-processing of the data.

GMSF1 Design Constraints/Performances

- GSS Code Phase Error
- provide the navigation observables for each satellite frequency with a 1-second sampling rate from two receiving chains
- The 67% error on the carrier phase measurements shall be lower than 1 cm.
- Receive and process observables for all services from up to 36 Galileo satellites

- In a fault-free constellation scenario with GSS Degraded atmospheric environment, the Ranging Accuracy at 95 % obtained by the PRS or OS Navigation Determination shall be better than 1.3 meters
- In a fault-free constellation scenario with GSS Degraded atmospheric environment, the Range Rate Accuracy at 95 % obtained by the PRS or OS Navigation Determination shall be better than 10 mm/s.

GMSF1 Functional Block Diagram

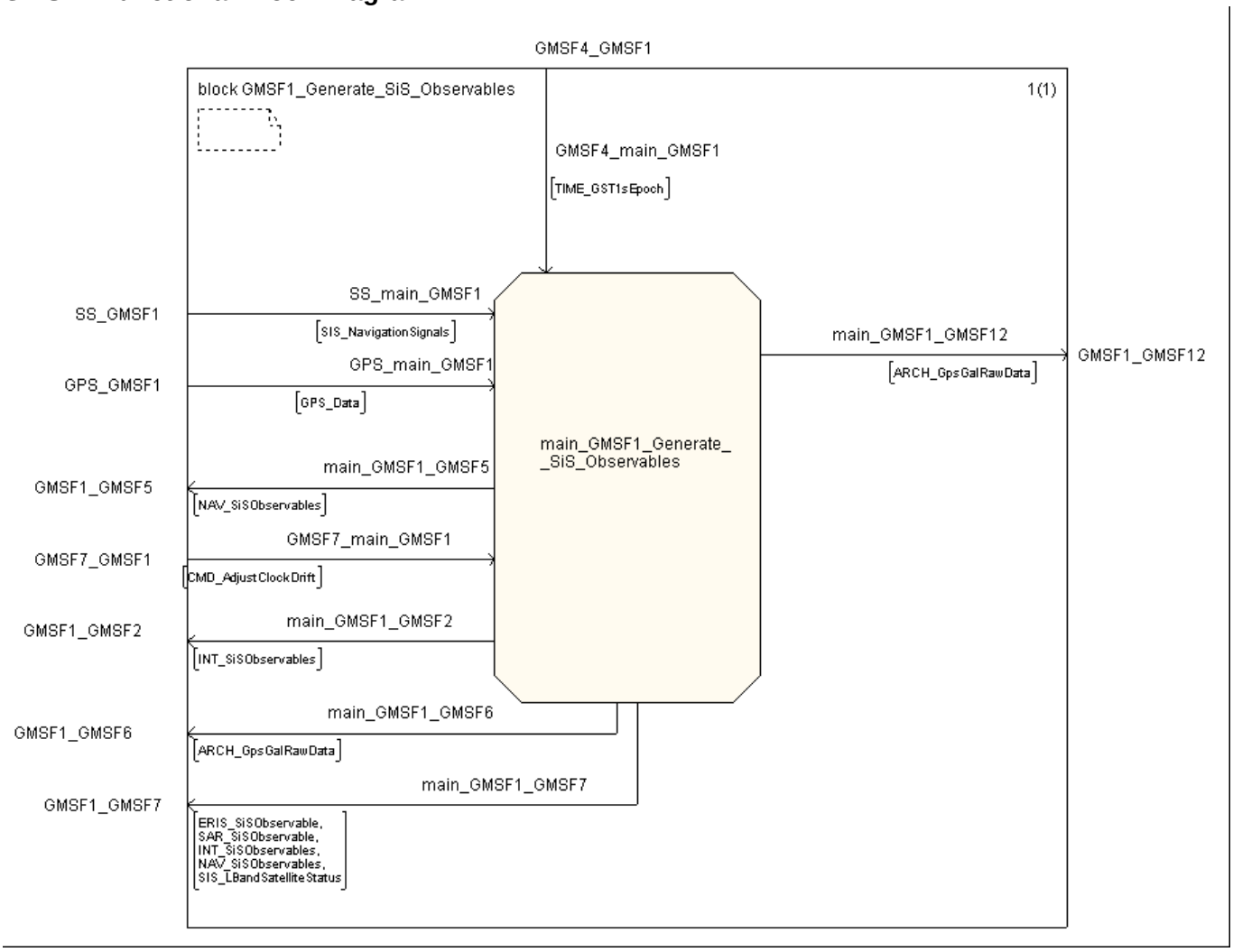


Figure 119: GMSF1 SDL Functional Block Diagram

5.3.2 GMSF2 Determine Integrity

Deleted

5.3.3 GMSF3 Schedule and Perform Message Dissemination

GMSF3 General Description

This function is responsible for managing and disseminating to the Space Segment the mission data for all the Galileo services including: OS, PRS, Commecial Service (CS), EDBS and SAR.

In particular this function performs Selection of:

- OS data
- PRS data (navigation and security)
- Request the encryption of the PRS data.
- Formatting and processing the data from external entities: Commercial data, EDBS data and SAR data.
- Formatting of the data in C-band frames to uplinked to the satellites.
- Schedule the satellite contacts according to the performance and system constraints.
- Handling of the ULS tracking of the constellation.
- Modulate and uplink the data to the satellites.
- Formatting of the data for S-band uplinking.

The complete Navigation dissemination is automated in routine operation.

Note: in this section Integrity and ERIS data is still recalled as design was not changed however ERIS, Integrity and SoL have been descoped from MRD 7.1 (and SoL remapped to Open Service).

GMSF3 Functional Block Diagram

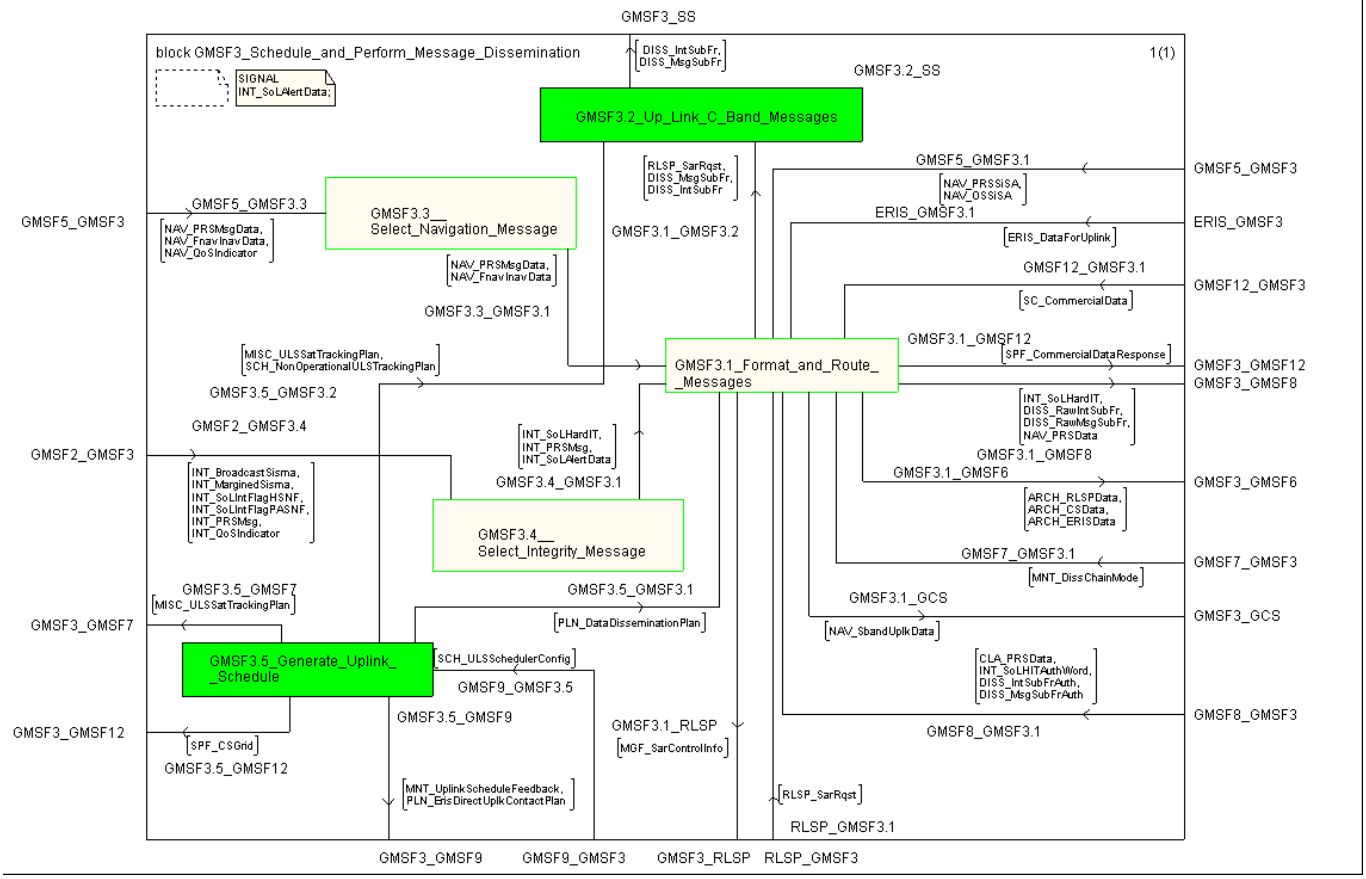


Figure 120: GMSF3 SDL Functional Block Diagram

5.3.3.1 GMSF3.1 Format and Route Messages

GMSF3.1 General Description

This function formats the message to be transmitted to the satellites via:

- C-band (as specified in GMSSSICD [AD.07])
- S-band (as specified in GCSGMSICD [AD.08])

In particular, this function is in charge of:

- Request for the encryption of the PRS navigation message.
- Processing of the SAR return link messages as requested by the RLSP. This function also monitors the RLM use and notifies the RLSP of capacity saturation.
- Processing of the Commercial service (CS) messages (C/NAV) as requested by the Commercial Service provider.
- Generation of dummy and spare data when not available.
- Formatting the C-band frames to be uplinked to the satellite through the C-band in:
  - C-band Integrity sub-frames where it is multiplexed:
    - Indirect ERIS data (to be reused by EDBS)
    - Integrity sub-frame authentication
  - C-band Message sub-frames where it is multiplexed:
    - C-band protocol parameters.
    - Time-of-Applicability (ToA) for:
      - OS navigation messages for both F/NAV and I/NAV (excluding almanac messages)
      - PRS navigation messages (including almanac messages)
      - PRS security data
    - OS navigation messages for both F/NAV and I/NAV
    - PRS navigation encrypted messages
    - PRS Region Status
    - PRS Security messages
    - Commercial messages
    - SAR return link data
    - Message sub-frame authentication
- Request the authentication words for each C-band Integrity and Message sub-frames.
- Generation of a QoS indicator to allow the selection of the data generated by several instance functions.
- Ensuring the routing of the authenticated Integrity and Message sub-frames to the ULS antennas in accordance with the ULS to satellite allocation plan.
- Formatting the message to be sent to GCS for uplinking via S-band including:
  - OS Navigation messages for both F/NAV and I/NAV
  - PRS Navigation encrypted messages
  - PRS Security messages

Generation of a QoS indicator to allow the GCS to select which data should be uplinked to the satellite.

No operator control input is needed in routine operation.

No GMS internal switch affects the dissemination continuity.

GMSF3.1 Outputs

The output of this function depend on which band the data would be uplinked:

- **C-band**: once per second, one Integrity sub-frame and one Message sub-frame as defined in the GMSSSICD document are generated per each connected satellite.
  The contents of the Integrity sub-frame allow to uplink the SoL Region Status, SoL integrity data and EDBS data.The contents of the Message sub-frames follow a predefined 30 seconds structure that is repeated periodically. This allows to uplink all the required data on board the satellite.
- **S-band:** Periodically is sent to GCS for uplinking through the S-band the navigation data as specified in the GCSSSICD document.

GMSF3.1 Design Constraints/Performances

This function contributes to a number of GMS performance budgets:

- Dissemination availability and continuity of navigation data
- Dissemination time of EDBS and CS data
- SAR Performance

GMSF3.1 Functional Block Diagram

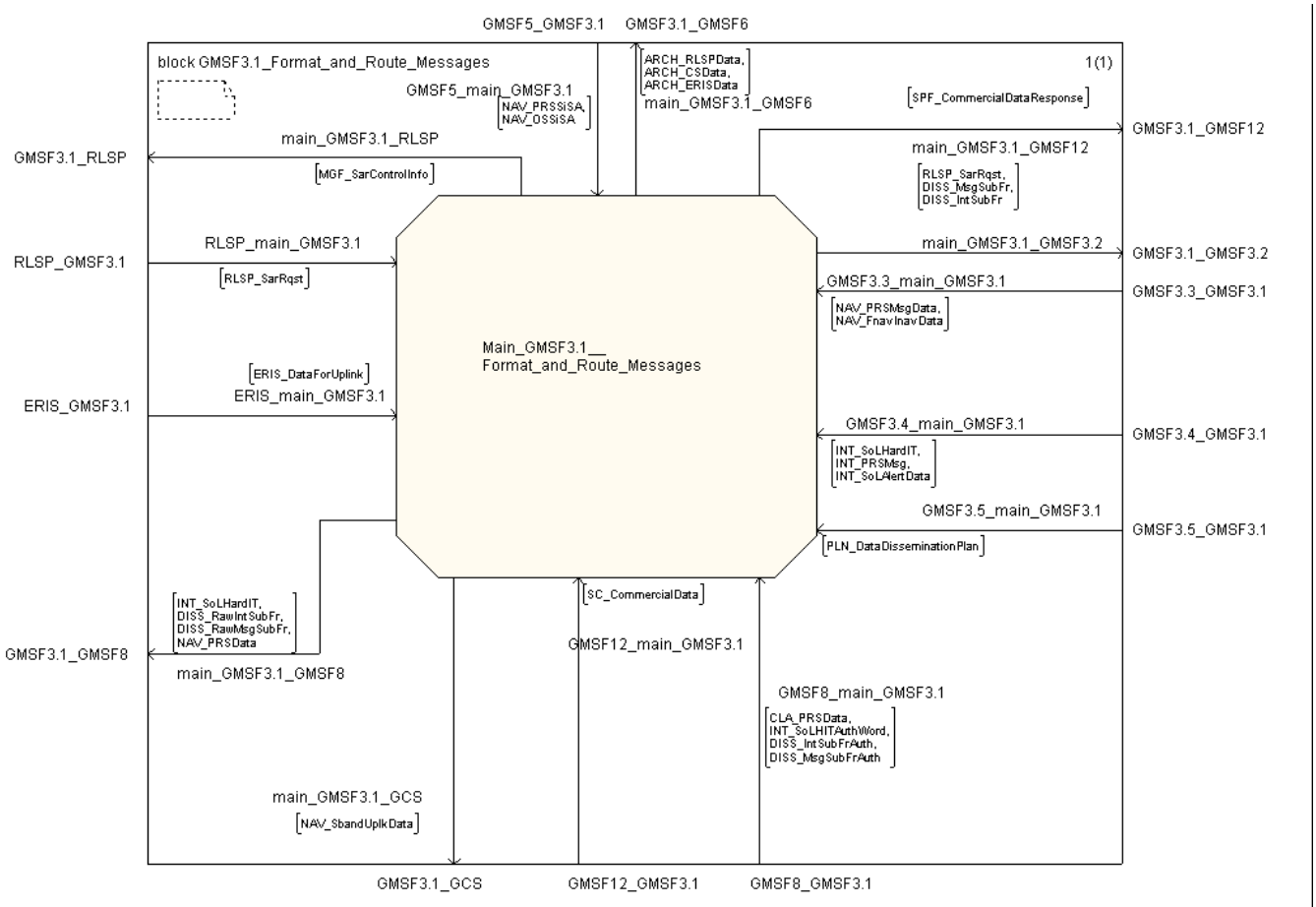


Figure 121: GMSF3.1 SDL Functional Block Diagram

5.3.3.2 GMSF3.2 Up Link C Band Messages

GMSF3.2 General Description
This function is in charge of uplinking the mission data for all the services (OS, PRS, CS, SAR and EDBS). It must then:

- Steer the ULS antennas to the satellites following the ULS to satellite tracking plan generated by the up-link scheduling function.
- Modulate the messages to be up-linked to the satellites in accordance with the GMSSS-ICD [AD.08].

Note that no operator control input is needed in routine operation.

GMSF3.2 Outputs
The output is a continuous modulated signal containing a frame message structure with a duration of 1 second.
The signal is uplinked to the satellite indicated by the satellite tracking plan.
In the scenarios where the data to be uplinked has been corrupted or did not arrive in time, to allow a continuous modulated signal, the output will be a modulated with a dummy frame.

GMSF3.2 Design Constraints/Performances
This function contributes to a number of GMS performance budgets:

- Dissemination availability and continuity of navigation data
- Dissemination time of EDBS and CS data
- SAR Performance

In case of ULS to satellite tracking plan update, the spacecraft is acquired by the ULS in less than 12minutes.

GMSF3.2 Functional Block Diagram

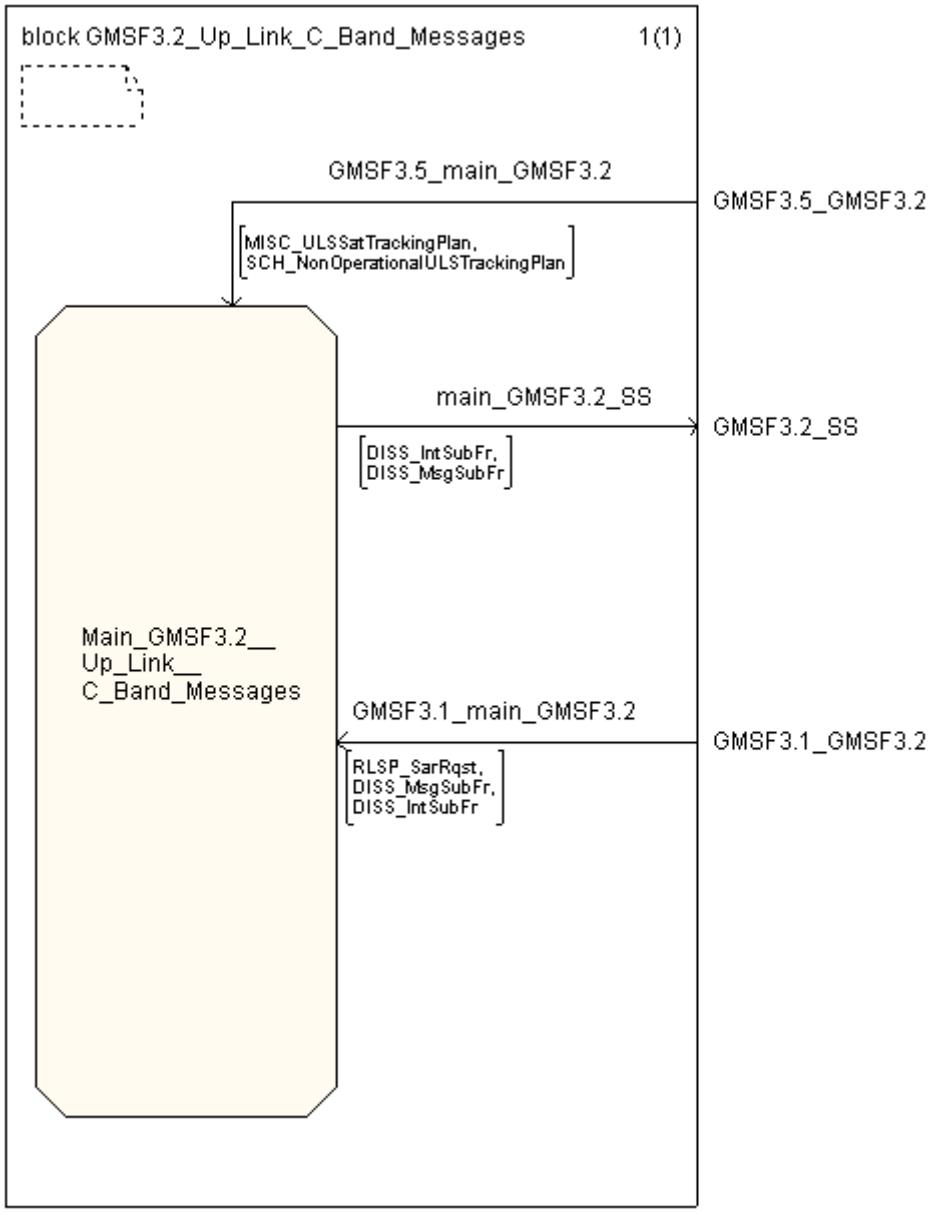


Figure 122: GMSF3.2 SDL Functional Block Diagram

5.3.3.3 GMSF3.3 Select Navigation Message

GMSF3.3 General Description
Automated selection of a single navigation data to be uplinked to the satellites (via C-band and S-band) based on the QoS indicators provided.

GMSF3.3 Outputs
The output is one instances formed by 3 different kinds of messages: OS F/NAV, OS I/NAV and PRS G/NAV provided for each possible satellite (36):

- 8 batches of data for OS F/NAV
- 1 almanac batch for OS F/NAV
- 8 batches of data for OS I/NAV
- 1 almanac batch for OS I/NAV
- 8 batches of data for PRS G/NAV
- 1 almanac batch for PRS G/NAV

Each of those batches has an associated Time-of-Applicability (ToA) to indicate the satellite the starting epoch of validity of each batch and an Issue-of-Data (IOD) to indicate the users about changes and consistency of the disseminated navigation data.

GMSF3.3 Design Constraints/Performances
The selected input must be the best among all received navigation data.

GMSF3.3 Functional Block Diagram

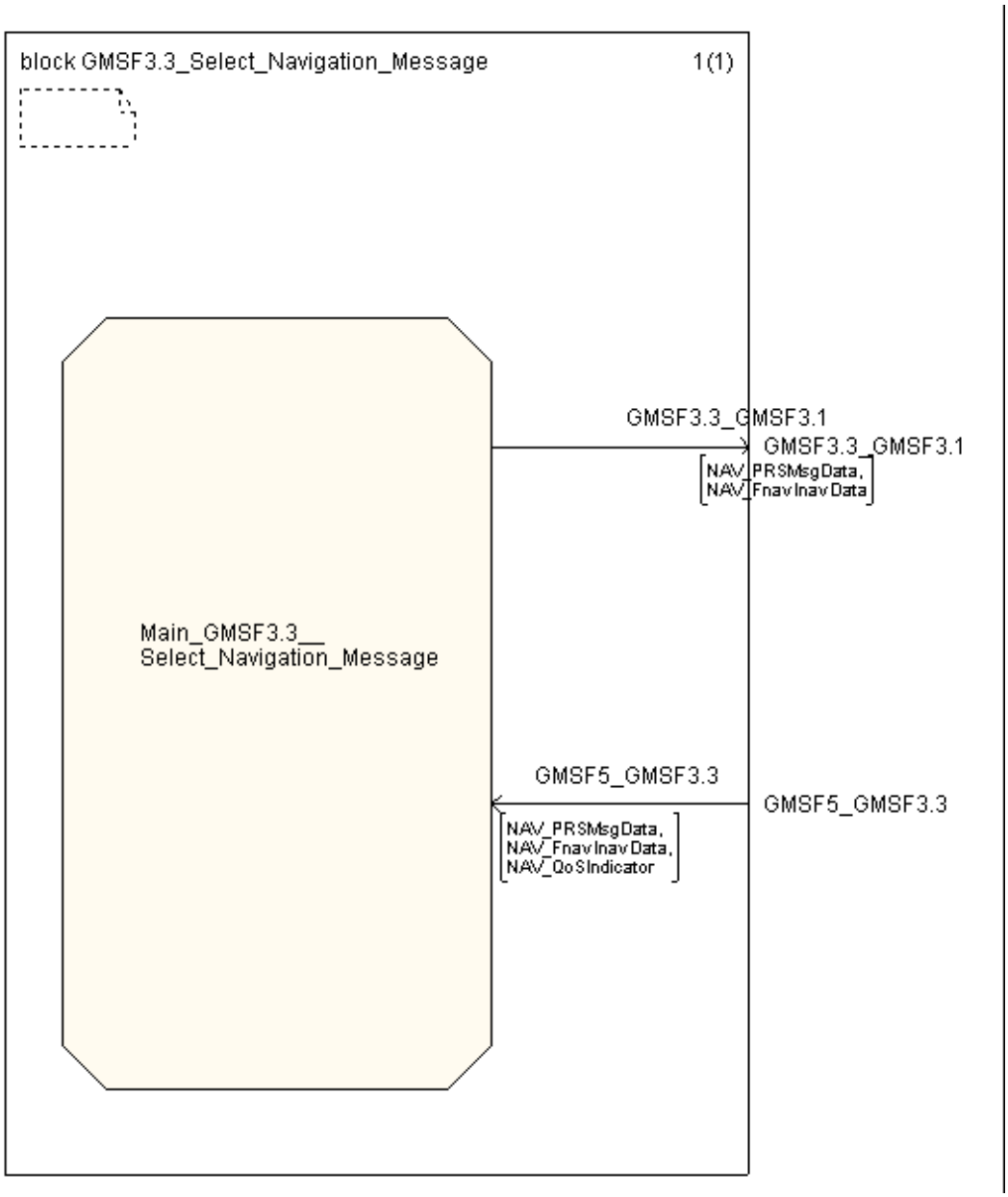


Figure 123: GMSF3.3 SDL Functional Block Diagram

5.3.3.4 GMSF3.4 Select Integrity Message

Deleted

5.3.3.5 GMSF3.5 Generate Uplink Schedule

GMSF3.5 General Description

This function consist in generating:

- ULS to Satellite Tracking Plan
- ULS to Satellite Allocation Plan
- Dissemination Grids for CS
- Up-link contact plan for direct ERIS service providers
- Non operational ULS Tracking Plan for maintenance and test activities.

This function is provided in nominal operational scenario, but the outputs are also provided in case of the following contingencies:

- loss of satellites
- loss of uplink antennas
- loss of uplink stations or associated networks links

Up-Link Scheduling General Constraints

The ULS to satellite tracking plan and allocation plan ensure that the following general physical and performance constraints are met:

- Global coverage for all services
- Use of a maximum of 4 antennas per site.

- In case of tracking plan change, the ULS need a maximum of 12min to reacquire a satellite. This maximum time includes the complete process to change the tracking plan, accept it, start the signal transmission and signal acquisition by the satellite
- The ULS to satellite tracking plan considers that the ULS start tracking the satellites from 5deg elevation angle but are allowed to transmit at the following elevations depending on the station:

Location	Minimum elevation angle at which the ULS can transmit
Svalbard	5°
Kourou	8.5°
Noumea	5°
Papeete	7.5°
Reunion	5.5°

Table 26: ULS minimum elevation angle per station

Up-Link Scheduling Navigation Constraints

The ULS to satellite tracking plan and allocation plan ensure that the following constraints on the navigation data dissemination are met:

- For Navigation data (for OS, SoL, PRS): the scheduling algorithm ensures that the navigation data broadcast by the satellites are refreshed as often as needed to cope with the satellites master clock (RAFS or PHM) performance. Satellites for which the master clock is the RAFS have their navigation data refreshed at least every 100mn.

PRS Security Data Up-link Scheduling
The ULS to satellite tracking plan and allocation plan are not further constrained by the dissemination of PRS security data.
CS Up-link Scheduling
The up-link scheduling functions generates 7 CS dissemination grids:

- 6 regional: Africa, Asia, North America, South America, Oceania, Europe
- 1 global.

The CS dissemination grids ensure that a throughput of 500bits/s on aggregate is provided to the CS users, without constraining the ULS to satellite tracking plan needed to satisfy the navigation and integrity constraints defined above.
Note: SFC7.1 on CS data dissemination explains how the CS dissemination grid are used by the CS service provider.

SAR Up-link Scheduling
The SAR return link dissemination does not further constrain the up-link scheduling function.
Note: SFC8.2 on SAR return link dissemination shows that the SAR service provider determines a list of preferred satellites for the dissemination of each RLM from the constellation ephemeris.

EDBS up-link Scheduling
The ULS to satellite tracking plan and allocation plan are not further constrained by EDBS dissemination.

GMSF3.5 Functional Block Diagram

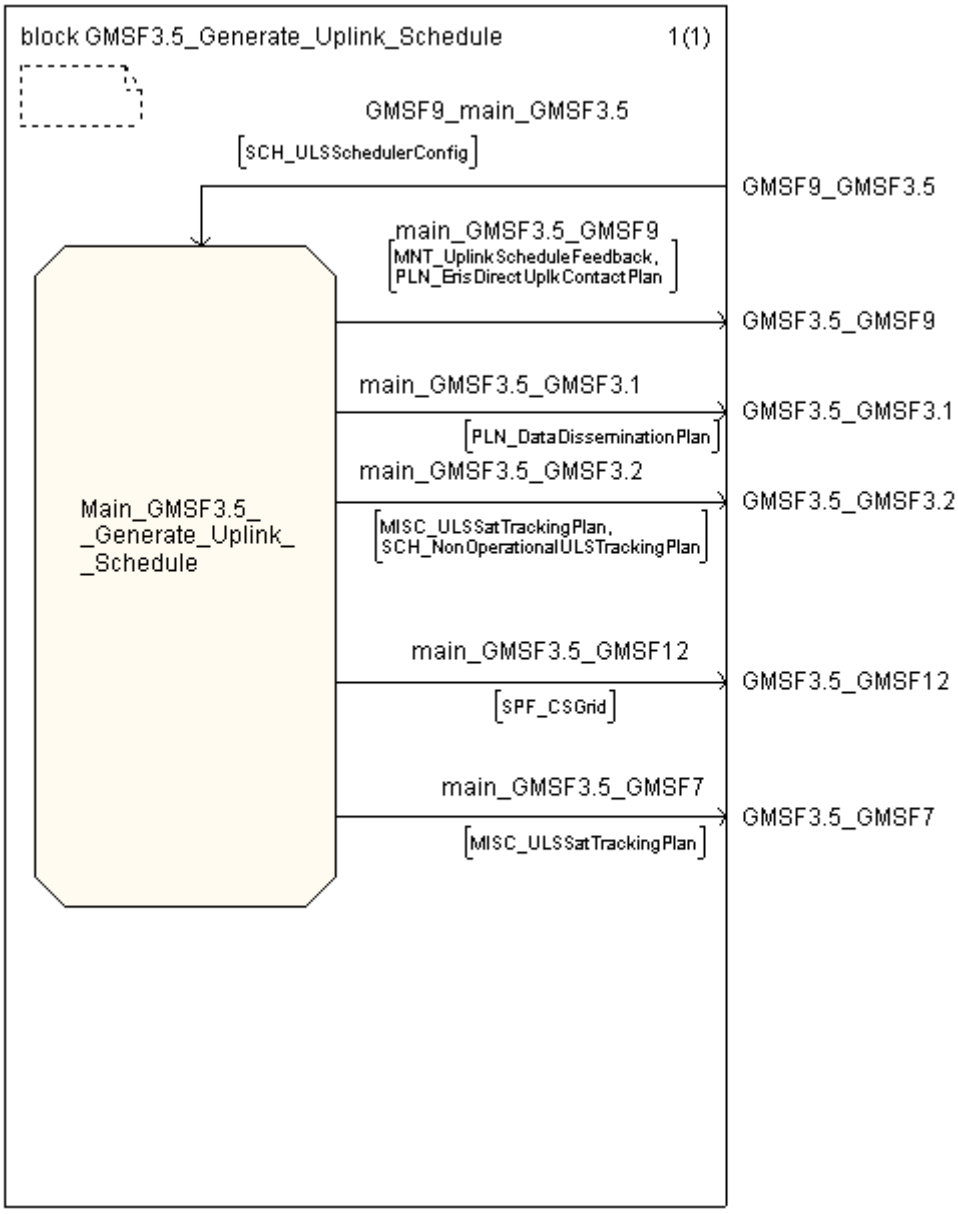


Figure 124: GMSF3.5 SDL Functional Block Diagram

5.3.4 GMSF4 Generate Time and Geodetic Reference Data

GMSF4 Description

This function provides all the non-navigation related mission data to the other GMS and GCS elements and also outside the core system.

This function caters for the following tasks :

- generation of the Galileo System Time (includes the PTF function), interfaced with the external TSP,
- generation of the GGTO, interfaced with the GPS system,
- generation of the Galileo Coordinates Reference, interfaces with the external GSRP

GMSF4 Functional Block Diagram

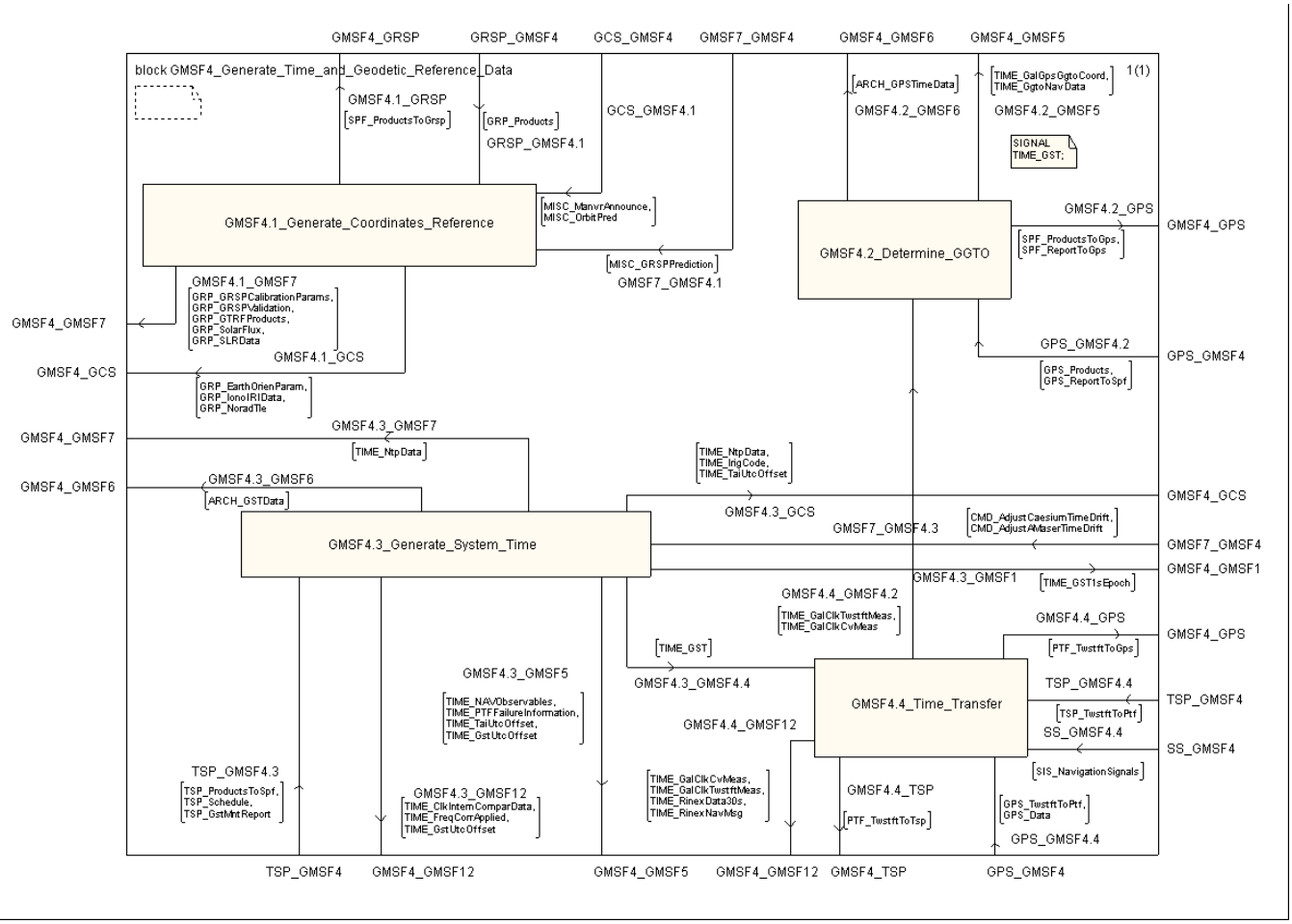


Figure 125: GMSF4 SDL Functional Block Diagram

5.3.4.1 GMSF4.1 Generate Coordinates Reference

GMSF4.1 Description

This funciton offers support to the GSRP for the elaboration of the GTRF including:

- direct data exchange between GMS and GRSP
- playing a role of the interface for the exchange of the data between the GCS and GRSP (for both data consumed and generated by FDF)
- distribution of the GRSP data to the GMS and GCS elements

In addition the function provides the calibration of the GSS-GPS observables needed for the proper calculation of the GTRF products

GMSF4.1 Functional Block Diagram

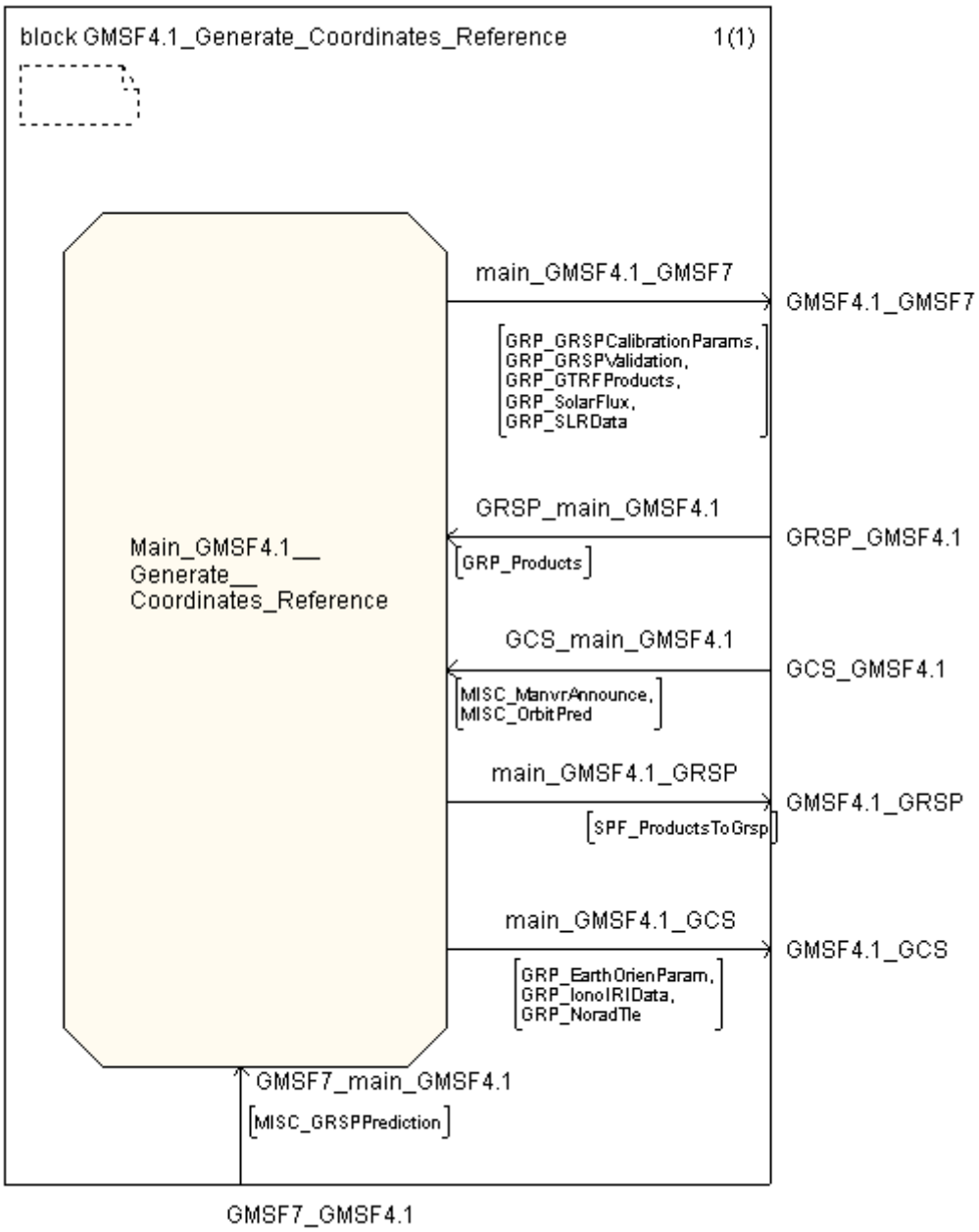


Figure 126: GMSF4.1 SDL Functional Block Diagram

5.3.4.2 GMSF4.2 Determine GGTO

GMSF4.2 General Description

The function is in charge of GGTO computation/determination and the coordination of the GGTO results with the GPS. Function is also responsible for the provision of GGTO to the archive and the navigation generation/determination function for inclusion in the navigation message.
The GGTO determination is done using the GMS time transfer function which allows though the time transfer measurements obtained by employing TWSTFT and GPS-CV techniques for difference between the GPS and GST timescales to be measured. The measured difference is then directly translated into the GGTO.

GMSF4.2 Functional Block Diagram

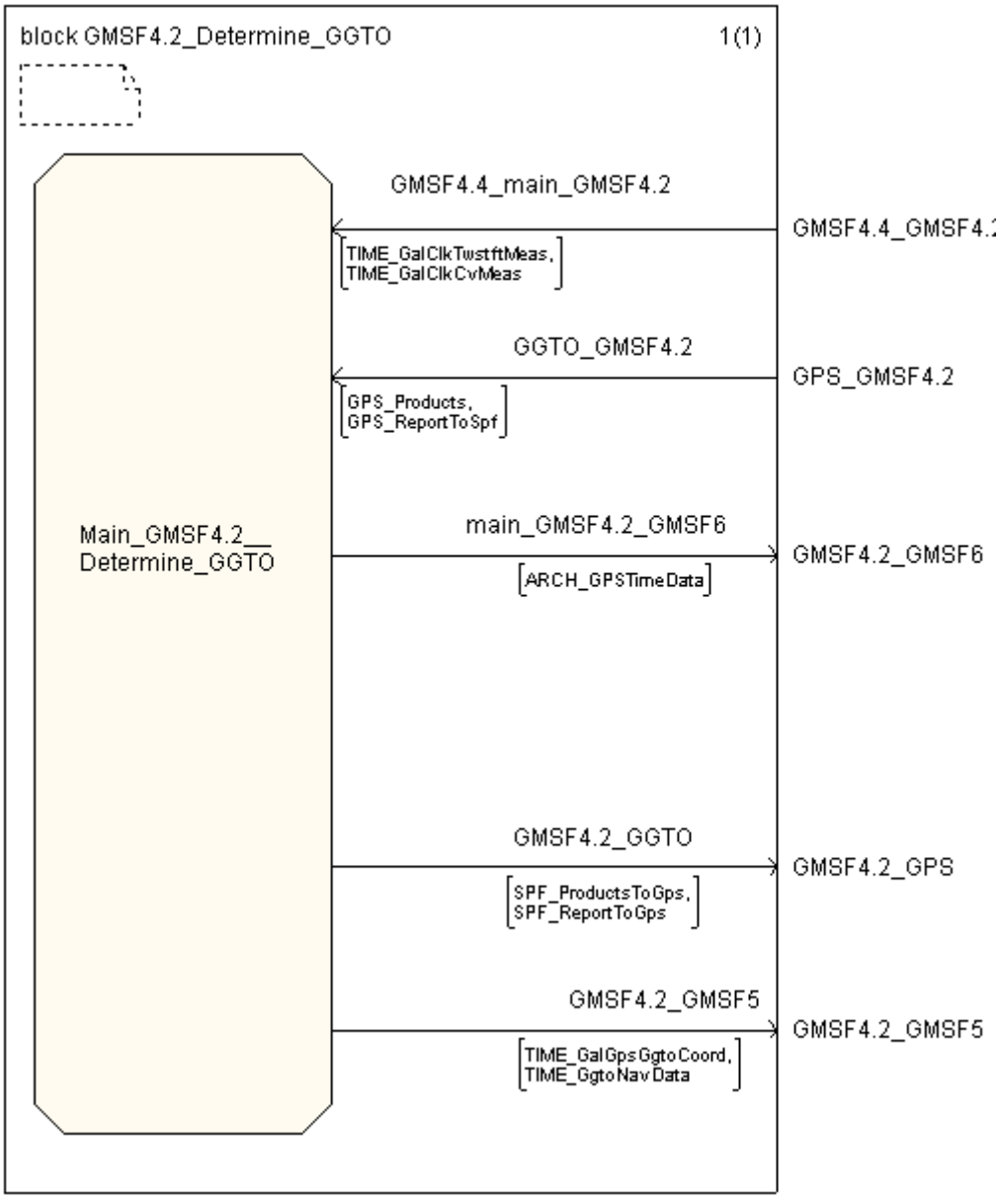


Figure 127: GMSF4.2 SDL Functional Block Diagram

5.3.4.3 GMSF4.3 Generate System Time

GMSF4.3 General Description

This function is in charge of autonomous generation of the Galileo System Time (GST) and its subsequent steering to the TAI using the TSP provided steering corrections.

This function engages Navigation and Metrological timekeeping to generate GST. The metrological timekeeping can be utilized only when the GMS to TSF interface is working according to its specification.

Function embeds following tasks:

- Autonomous generation of the GST and its maintenance Navigation timekeeping (generation of the stable timescale)
- GST to TAI steering (with the TSP provided products) Metrological timekeeping
- Distribution of the timing signals derived from GST, such as 1pps, ntp and IRIG-B to GMS and GCS elements (e.g. for the purpose of GGTO determination, navigation determination etc.)
- Synchronization of the PTF timescales
- Provision of the timing information supporting variety of GMS and GCS functions/services including the UTC dissemination service, function generating navigation message, FDF functionality, orbit determination etc.

GMSF4.3 Functional Block Diagram

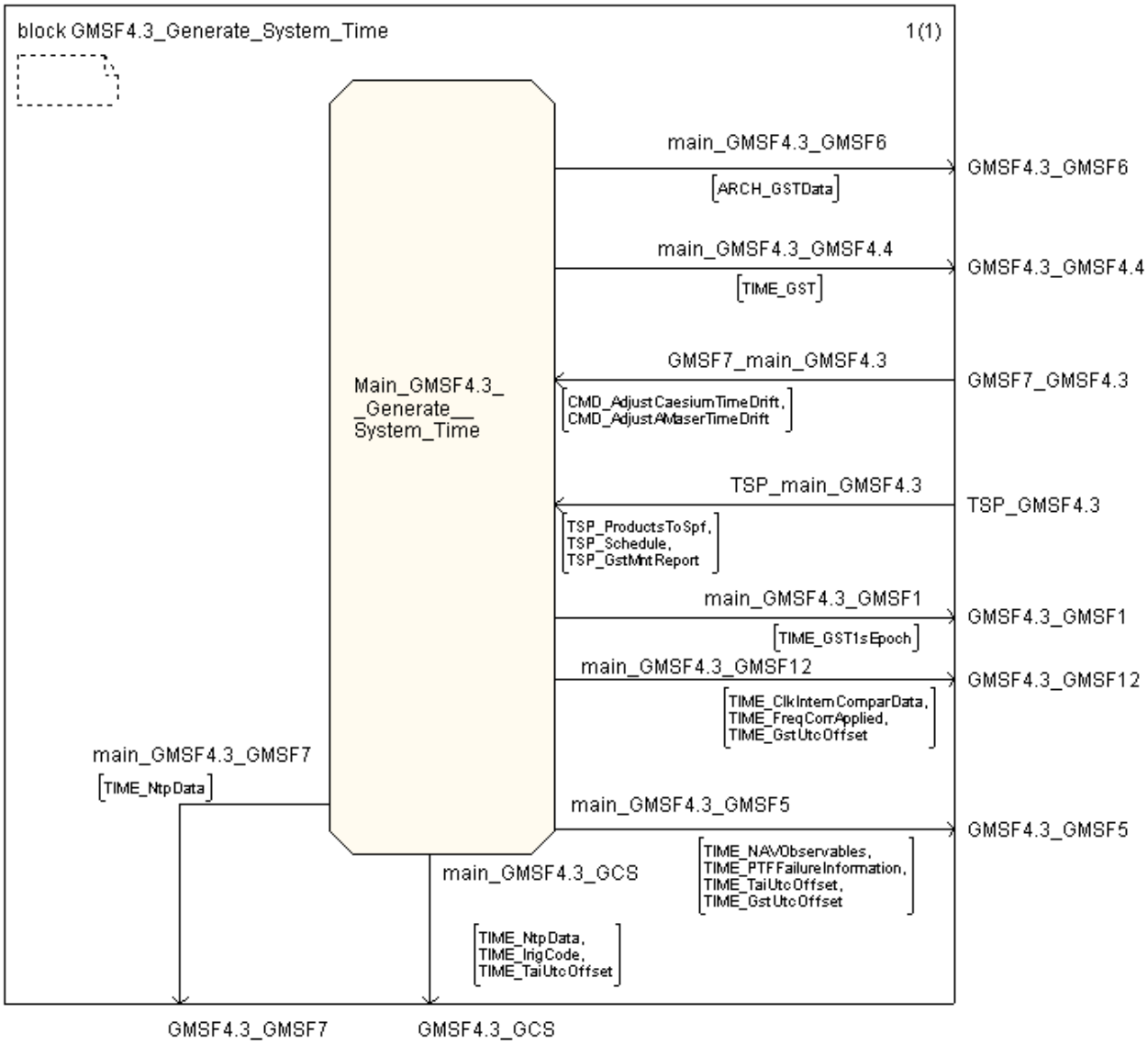


Figure 128: GMSF4.3 SDL Functional Block Diagram

5.3.4.4 GMSF4.4 Time Transfer

GMSF4.4 General Description

This function is in charge of:

- time transfer interfaces such as TWSTFT, GPS/Galileo CV and the interface based on the implementation of the GPS-Galileo combined receiver to external entities e.g. USNO, TSP UTC(k) and also internally within GMS for the PTF sychronisation
- configuration, operation and control of time transfer equipment used for the above
- processing of time transfer results

This function is used by the GMSF4.3 for the purpose of the GST to TAI steering including the PTF synchronisation and also by GMSF4.2 for the purpose of GGTO determination.

GMSF4.4 Functional Block Diagram

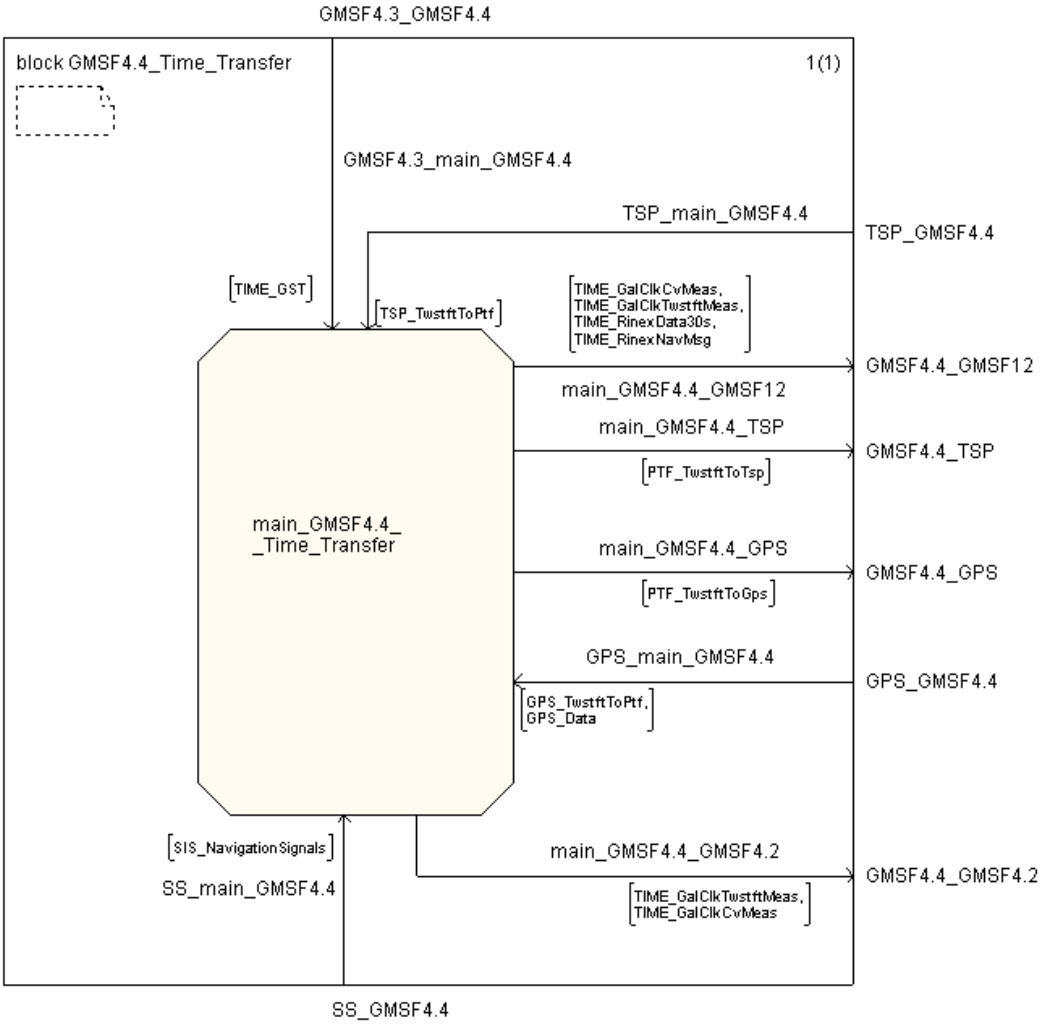


Figure 129: GMSF4.4 SDL Functional Block Diagram

5.3.5 GMSF5 Determine Navigation

GMSF5 Description

This function determines the data related to positioning service for both dual- and single-frequency users. Specifically with respect to satellite orbital parameters, time and satellite clock corrections. Those are computed for both OS and PRS services.

The satellite almanacs (degraded orbital parameters) are also computed at this stage.

The computed data is used for nominal and degraded scenarios for both C-band and S-band links.

All the computed data relies on the GSS navigation observables, the GST and some calibration parameters computed off-line.

A QoS indicator is elaborated for each data flow instance and transmitted to the dissemination functions.

This function is fully automated in routine operations. It also supports off-line mission monitoring.

GMSF5 Functional Block Diagram

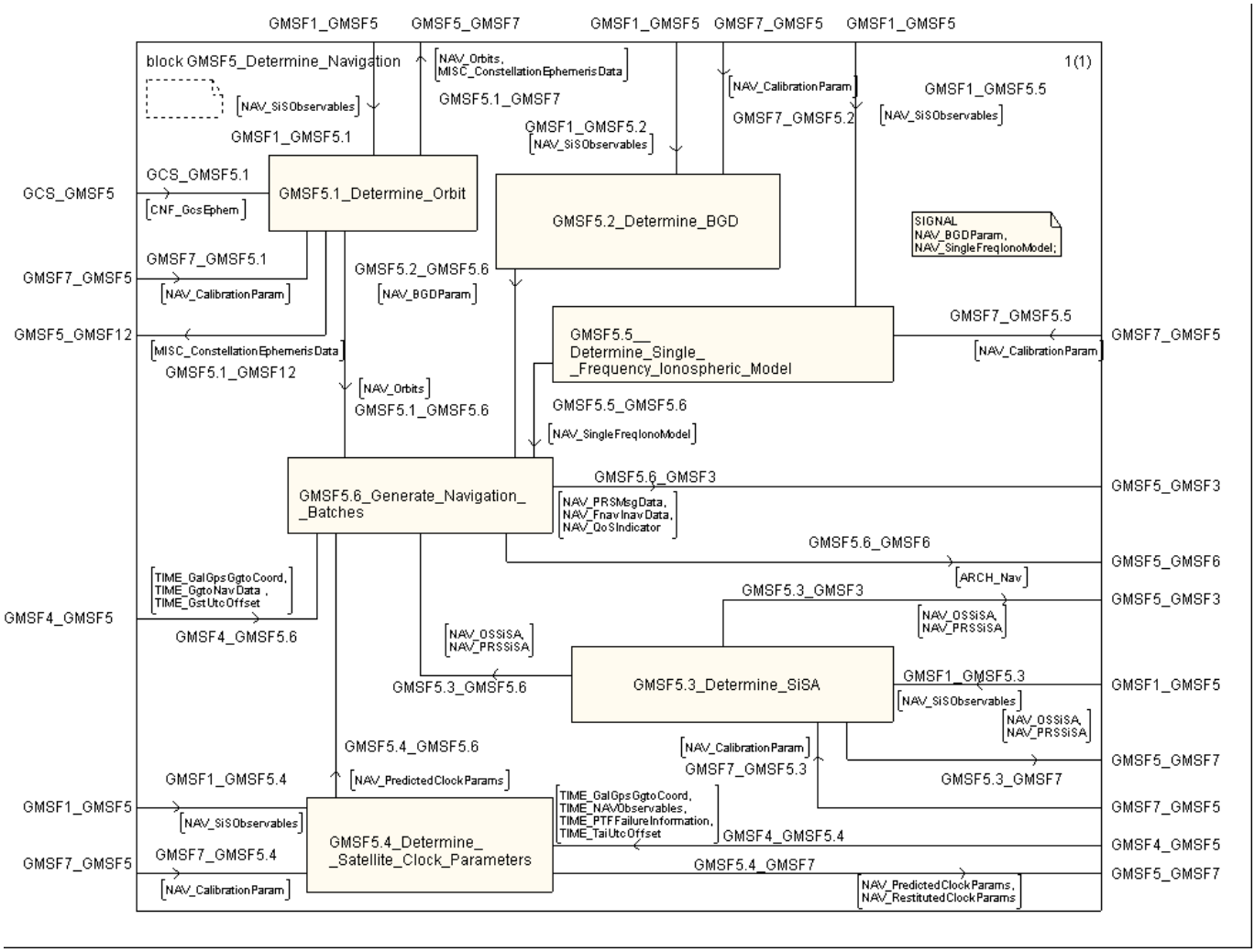


Figure 130: GMSF5 SDL Functional Block Diagram

5.3.5.1 GMSF5.1 Determine Orbit

GMSF5.1 Description
This function computes the satellite orbital parameters prediction estimation (ephemeredes) for all the satellites in orbit (up to a maximum of 36 satellites) for the OS and PRS positioning services.

The ephemeredes are computed based on the raw data (SIS observables) received at the GSS (GMSF1) and their computation is referenced to an initial satellite position at a certain time epoch:

- The initial satellite position is fixed referenced to the Earth through the Greenwich meridian.
- The initial time epoch is referenced to the initial validity time of the navigation batch transmitted.

A total of 8 navigation data batches are generated:

- 4 batches to be uplinked on the C-band and S-band. Each batch has a validity of 4 hours but each batch has an hour overlap between them. In total they cover up to 12 hours of prediction.
- 4 extra batches to be uplinked exclusively on the S-band that complement the last 4 batches to cover up to 24 hours of prediction.

This function also computes, based on the precise satellite ephemerides, the low precision orbit parameters used to generate the almanacs.

Other functionalities include:

- Provide to the GCS constellation ephemeris data
- A posteriori processing to support mission monitoring

GMSF5.1 Outputs
The outputs include the navigation data computed to generate the navigation messages for each of the satellites in orbit.

GMSF5.1 Design Constraints/Performances
Capability to process satellite observables for up to 36 satellites

GMSF5.1 Functional Block Diagram

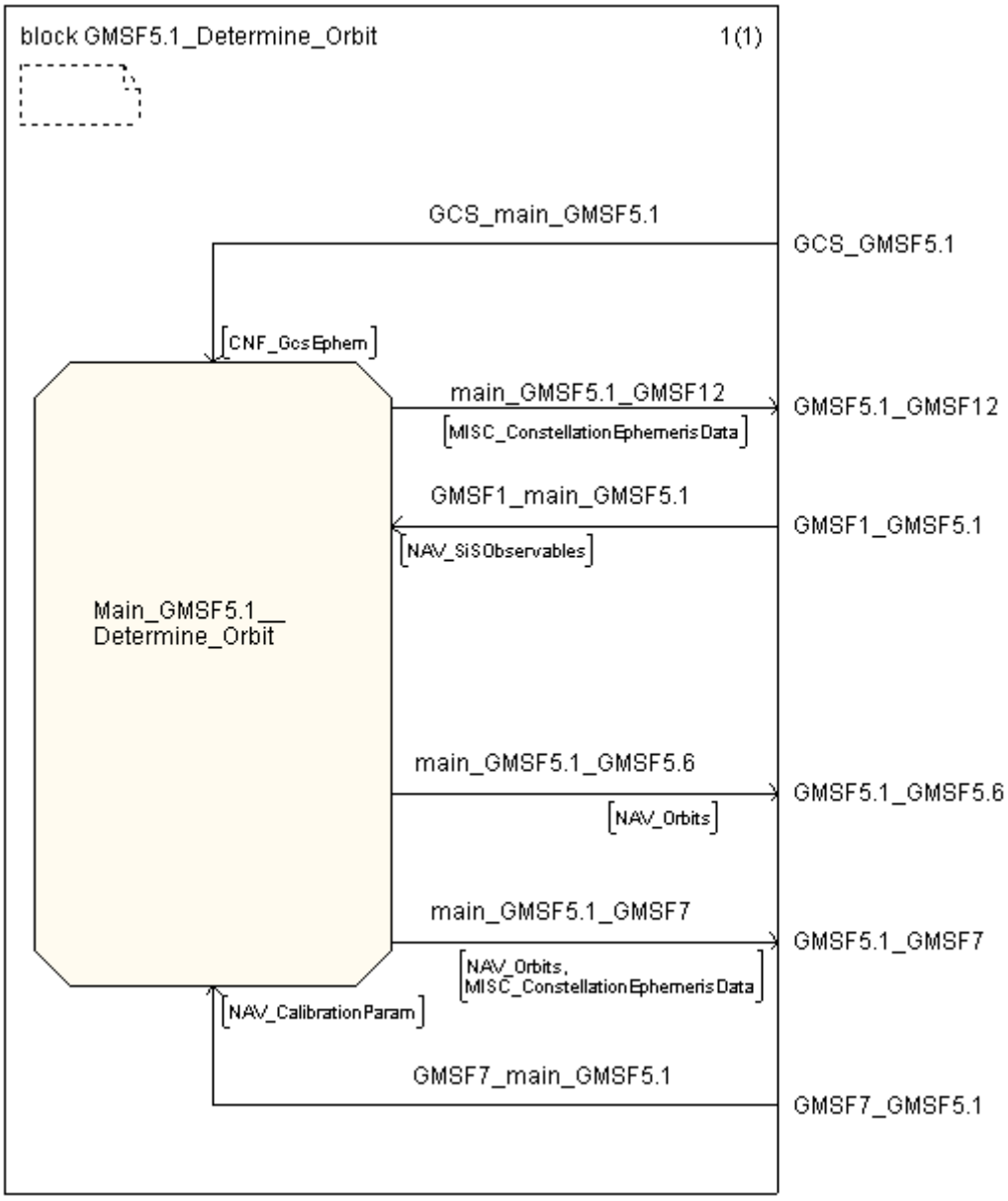


Figure 131: GMSF5.1 SDL Functional Block Diagram

5.3.5.2 GMSF5.2 Determine BGD

GMSF5.2 General Description
This function is in charge of computing the Broadcast Group Delay (BGD) exclusively for single frequency users.

The BGD compensates the errors introduced by the use of the satellite clock correction parameters, based on dual frequency clock models, by single frequency users.

Computation of the Broadcast Group Delay parameters :

- E1-E5b Broadcast Group Delay parameter to be used by E1 and E5b single frequency users.
- E1-E5a Broadcast Group Delay parameters to be used by E5a single frequency users.
- E1A-E6A Broadcast Group Delay parameters to be used by E1A and E6A single frequency users.

GMSF5.2 Outputs
The outputs include the Broadcast Group Delay computed to generate the navigation messages for each of the satellites in orbit:

- E1-E5b Broadcast Group Delay parameter.
- E1-E5a Broadcast Group Delay parameter.
- E1A-E6A Broadcast Group Delay parameter.

GMSF5.2 Design Constraints/Performances

- Capability to process BGD parameters for up to 36 satellites.
- In a fault-free constellation scenario with GSS Reference atmospheric environment, the GMS shall provide the “E1-E5b Broadcast Group Delay” parameter to be used by E1 or E5b single frequency users with an accuracy better than 50 cm (95%).
- In a fault-free constellation scenario with GSS Reference atmospheric environment, the GMS shall provide the “E1-E5a Broadcast Group Delay” parameters to be used by E5a single frequency users with an accuracy better than 50 cm (95%).

In a fault-free constellation scenario with GSS Reference atmospheric environment, the GMS shall provide the “E1A-E6A Broadcast Group Delay” parameters to be used by E1A or E6A single frequency users with an accuracy better than 50 cm (95%).

GMSF5.2 Functional Block Diagram

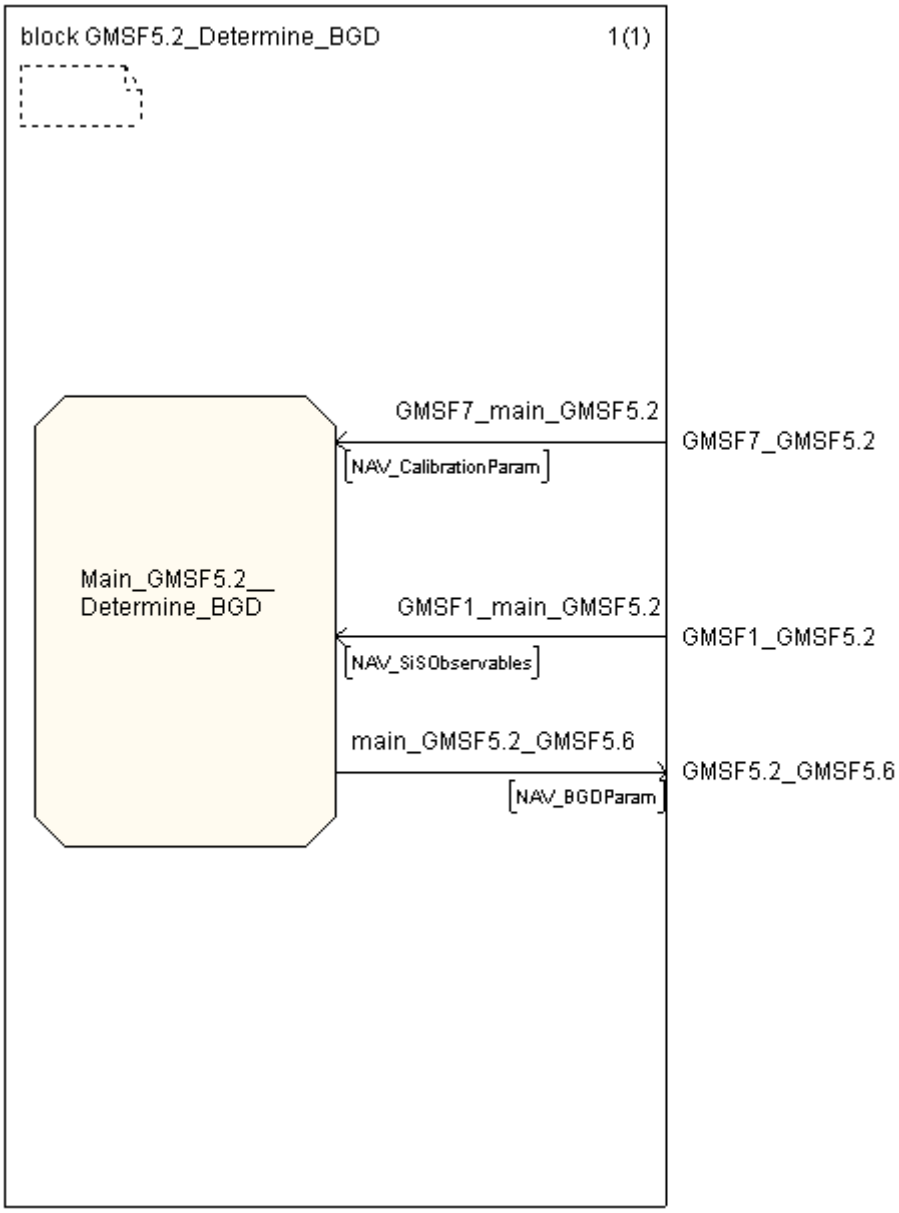


Figure 132: GMSF5.2 SDL Functional Block Diagram

5.3.5.3 GMSF5.3 Determine SiSA

GMSF5.3 General Description
This function is in charge on computing the Signal-In-Space Accuracy (SISA).

This computation is based on the SIS observables received from the GSS (GMSF1) and calibration parameters so called Historical SISA (HSISA).

Determination of SISA parameters is based on the following pairs of frequencies. They are the same as per the satellite clock correction parameters:

- E1-E5a SISA parameter to be used by E1-E5a dual-frequency users.
- E1-E5b SISA parameter to be used by E1-E5b dual-frequency users.
- E1A-E6A SISA parameter to be used by E1A-E6A dual-frequency users.

GMSF5.3 Outputs

- E1-E5a SISA parameter
- E1-E5b SISA parameter
- E1A-E6A SISA parameter

GMSF5.3 Design Constraints/Performances

- In a fault-free constellation scenario with GSS Degraded atmospheric environment, the SISA values for each SOL and PRS service shall be lower than 85 cm.

In a fault-free constellation scenario with GSS Degraded atmospheric environment, the probability that the SoL and PRS SISE 1 sigma bound exceeds broadcast PRS SISA shall be lower than 1E-4 in any 150s.

GMSF5.3 Functional Block Diagram

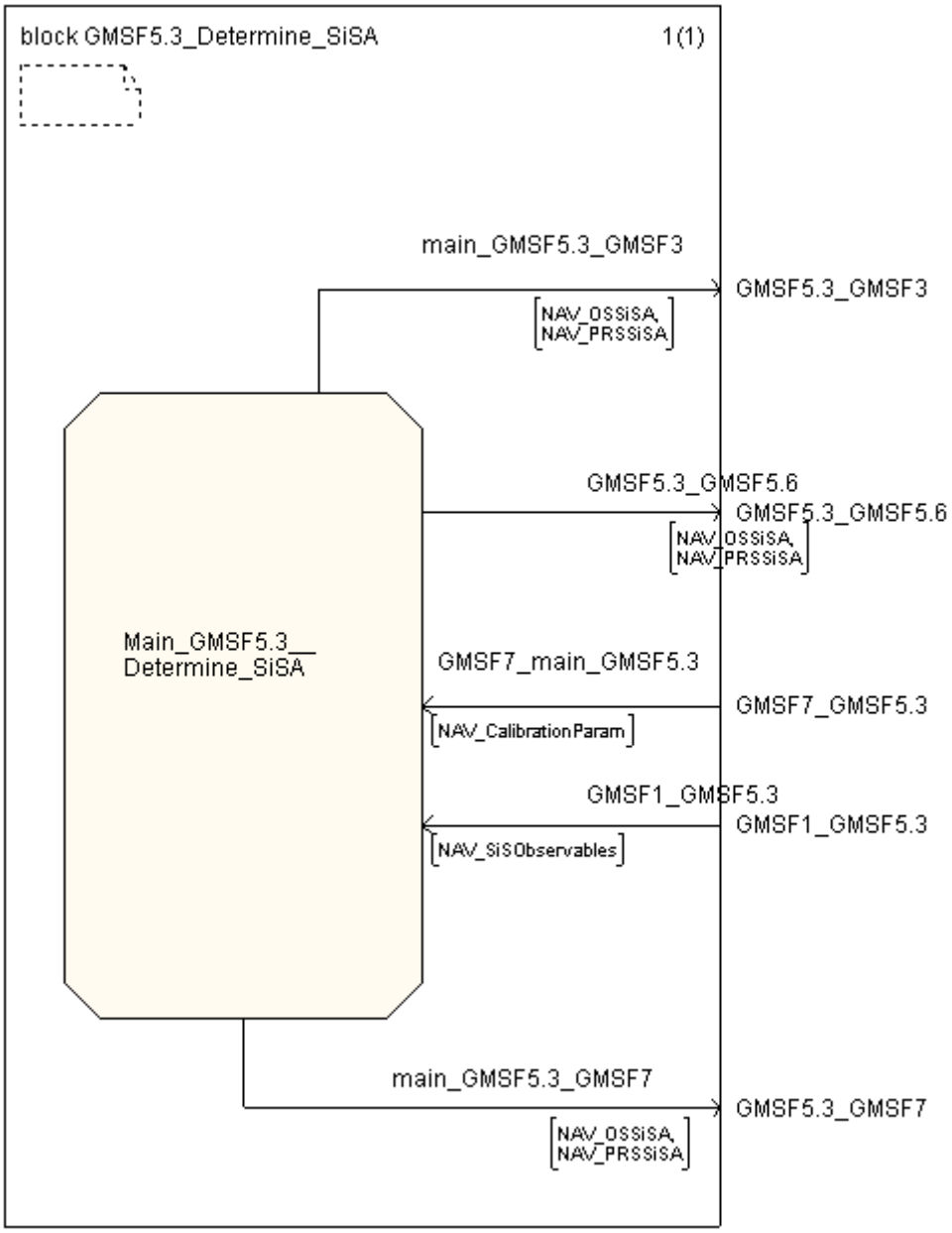


Figure 133: GMSF5.3 SDL Functional Block Diagram

5.3.5.4 GMSF5.4 Determine Satellite Clock Parameters

GMSF5.4 General Description
This function generates the clock correction estimation parameters that dual-frequency users need to apply to align the satellite onboard clock (LGST) measured by the user with the Galileo System Time (GMSF4.3).

The determination of the specific dual frequency clock parameters is made for the following pairs of frequencies:

- E1 / E5a clock model parameters using E5a-E1 ch B/C frequency pair.
- E1 / E5b clock model parameters using E5b-E1 ch B/C frequency pair.
- E1A / E6A clock model parameters (E1A and E6A stands for PRS E1/E6 channels A)

The clock correction estimation compensates the errors introduced on the user measurement by the Broadcast Group Delay after ionospheric compensation by the user.

The estimated clock correction parameters are computed based on the satellite clock models and the SIS observables received from the GSS (GMSF1)

This function also computes, based on the precise satellite clock correction parameters, the low precision parameters used to generate the almanacs.

GMSF5.4 Outputs
The outputs include the clock corrections computed to generate the navigation messages for each of the satellites.

GMSF5.4 Design Constraints/Performances
Capability to process satellite clock parameters for up to 36 satellites.

GMSF5.4 Functional Block Diagram

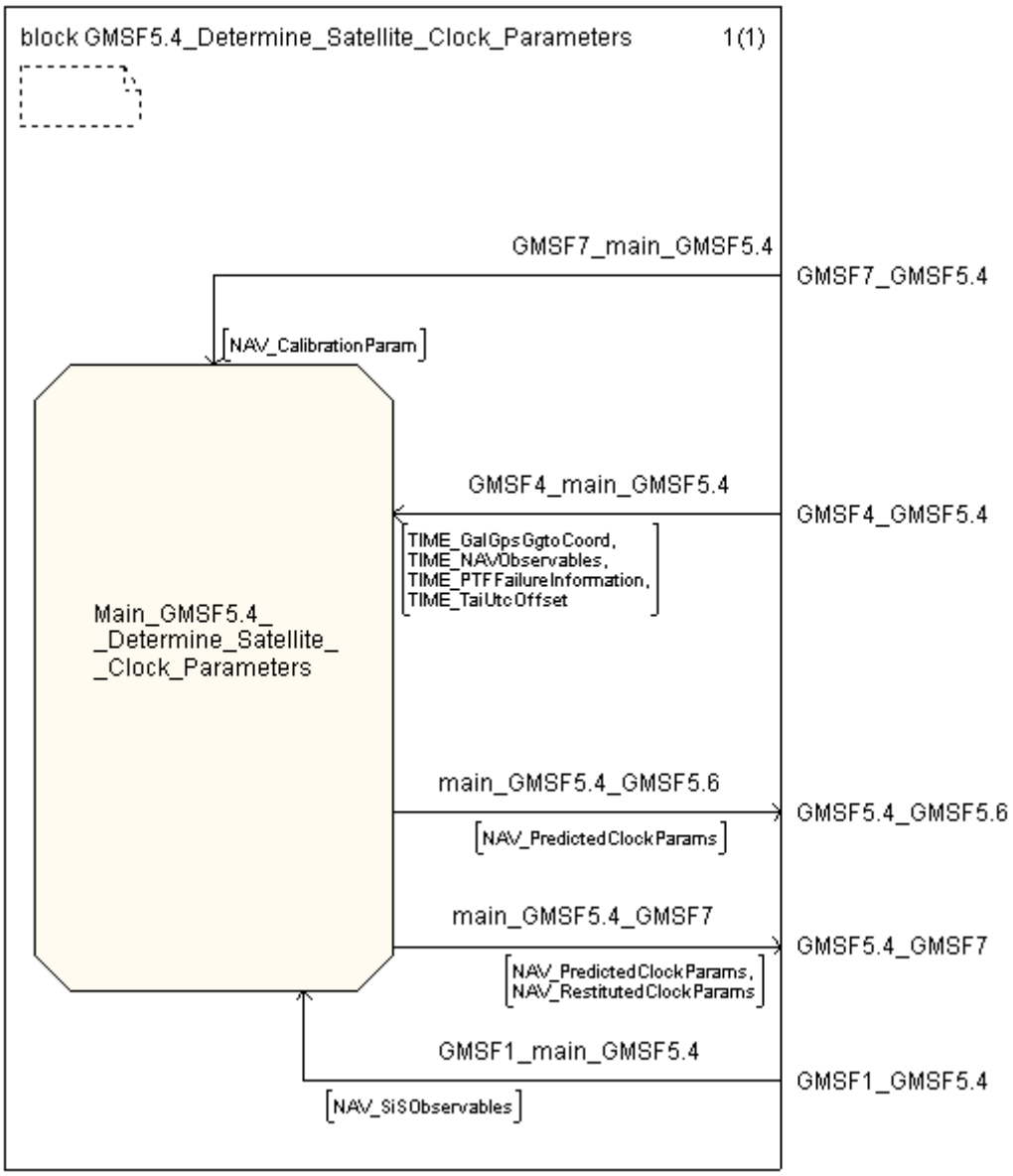


Figure 134: GMSF5.4 SDL Functional Block Diagram

5.3.5.5 GMSF5.5 Determine Single Frequency Ionospheric Model

GMSF5.5 General Description
This function is in charge of the computation of:

- the ionospheric model messages based on GSS measurements and
- Ionospheric disturbance flag or flag storm for 5 predefined regions.

The ionospheric model messages allow to compute the effective ionization level of the atmosphere. This will allow to single frequency users to compensate the delay introduced by the ionosphere on the measured onboard satellite time.

It is noted that dual frequency users are able to automatically compensate the ionospheric effects.

GMSF5.5 Outputs

- Single frequency ionospheric model as Nequick model
- Ionospheric storm flags

GMSF5.5 Design Constraints/Performances
The Single frequency Iono. Model messages shall be produced from a fit of the simplified Nequick model parameters to GSS measurements. These parameters include the ionospheric storm bit under relevant conditions.
In a fault-free constellation scenario with GSS Reference atmospheric environment (“storm flag” not set), the Single frequency Ionospheric Correction Algorithm, initialised by the broadcast correction message and applied to geometries with an elevation angle of more than 10 degrees, shall have a residual error of no more than 20 TEC Units or 30 % r.m.s. of the actual Slant Total Electron Content, whichever is larger. The resulting error in single frequency Ionospheric excess path length ("delay") on E5a, E5b, E1 and E6, is to be calculated using the equation: Ds = 40.3 x 1016 TEC / f 2 (m) where TEC is in TEC-units and f is the frequency in Hz.

GMSF5.5 Functional Block Diagram

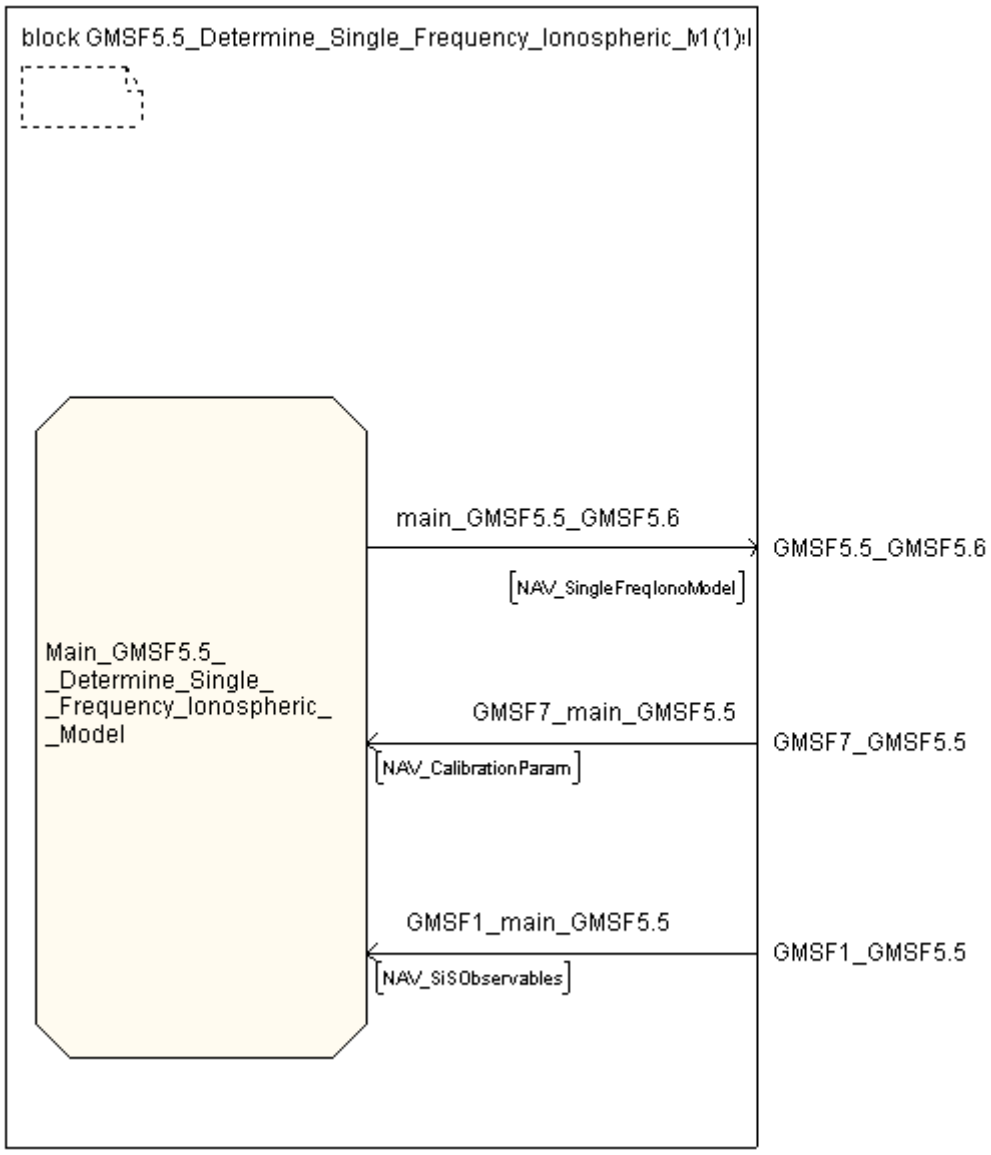


Figure 135: GMSF5.5 SDL Functional Block Diagram

5.3.5.6 GMSF5.6 Generate Navigation Batches

GMSF5.6 General Description

This function is in charge of collecting all OS and PRS (positioning and timing provision) data. This data is formatted and structured in F/NAV, I/NAV and G/NAV messages to be processed by the end users.

The output messages will be part of the data that will be uplinked to the satellite (formatted in GMSF3.1) either on C-band or on S-band.

The data collected by this function contains for each satellite (36 in total):

- 8 batches of ephemeris data for OS.
- 8 batches of ephemeris data for PRS.
- Satellite almanacs for OS with an associated IODa.
- Satellite almanacs for PRS with an associated IODa.
- 3 sets of satellite clock correction parameters (one per pair of frequecies).
- 3 sets of Broadcast Group Delay parameters (one per pair of frequencies).
- Ionospheric model parameters (common to all satellites).
- GST-UTC offset (common to all satellites).
- GPS-Galileo Time Offset (common to all satellites).

All this data is formatted on 3 different kinds of messages: OS F/NAV, OS I/NAV and PRS G/NAV providing for each possible satellite (36):

- 8 batches of data for OS F/NAV
- 1 almanac batch for OS F/NAV
- 8 batches of data for OS I/NAV
- 1 almanac batch for OS I/NAV
- 8 batches of data for PRS G/NAV
- 1 almanac batch for PRS G/NAV

Each batch of data has an associated Issue-of-Data (IOD), incremented every 10 minutes for identification of the batch, and Time-of-applicability (ToA). The IOD repeats unambiguously every 127 OSPF 10 minutes navigation message generation intervals (at 1270 min) for the same batch. (**#STRB2.0_change: GAL-DCN-ASTG-SYST-X-13684)**

OSPF computation time	To	To+10min	To+20min		To+1250min	To+1260min	To+1270min	To+1280min
IOD batch 1	0	1	2		125	126	0	1
IOD batch 2	127	128	129		252	253	127	128
IOD batch 3	254	255	256		379	380	254	255
IOD batch 4	381	382	383		506	507	381	382
IOD batch 5	508	509	510	...	633	634	508	509
IOD batch 6	635	636	637		760	761	635	636
IOD batch 7	762	763	764		887	888	762	763
IOD batch 8	889	890	891		1014	1015	889	890

It is noted that the Signal Health Status (SHS) flag is set by GMSF3.1. The final F/NAV, I/NAV and G/NAV messages will be provided by the satellite after onboard processing (SSF3.2).

This function generates several data instances, each with an associated Quality-of-Service (QoS).

GMSF5.6 Outputs
The outputs are several redundant instances each formatted by 3 different kinds of messages: OS F/NAV, OS I/NAV and PRS G/NAV provided for each possible satellite (36):

- 8 batches of data for OS F/NAV
- 1 almanac batch for OS F/NAV
- 8 batches of data for OS I/NAV
- 1 almanac batch for OS I/NAV
- 8 batches of data for PRS G/NAV
- 1 almanac batch for PRS G/NAV

A Quality-of-Service (QoS) indicator is associated to each redundant instance for OS F/NAV and OS I/NAV data and another one is associated to PRS G/NAV data.

Each of those batches has an associated Time-of-Applicability (ToA) to indicate the satellite the starting epoch of validity of each batch and an Issue-of-Data (IOD) to indicate the users about changes and consistency of the disseminated navigation data

GMSF5.6 Design Constraints/Performances
The output must be provided once every 10 minutes as a maximum

GMSF5.6 Functional Block Diagram

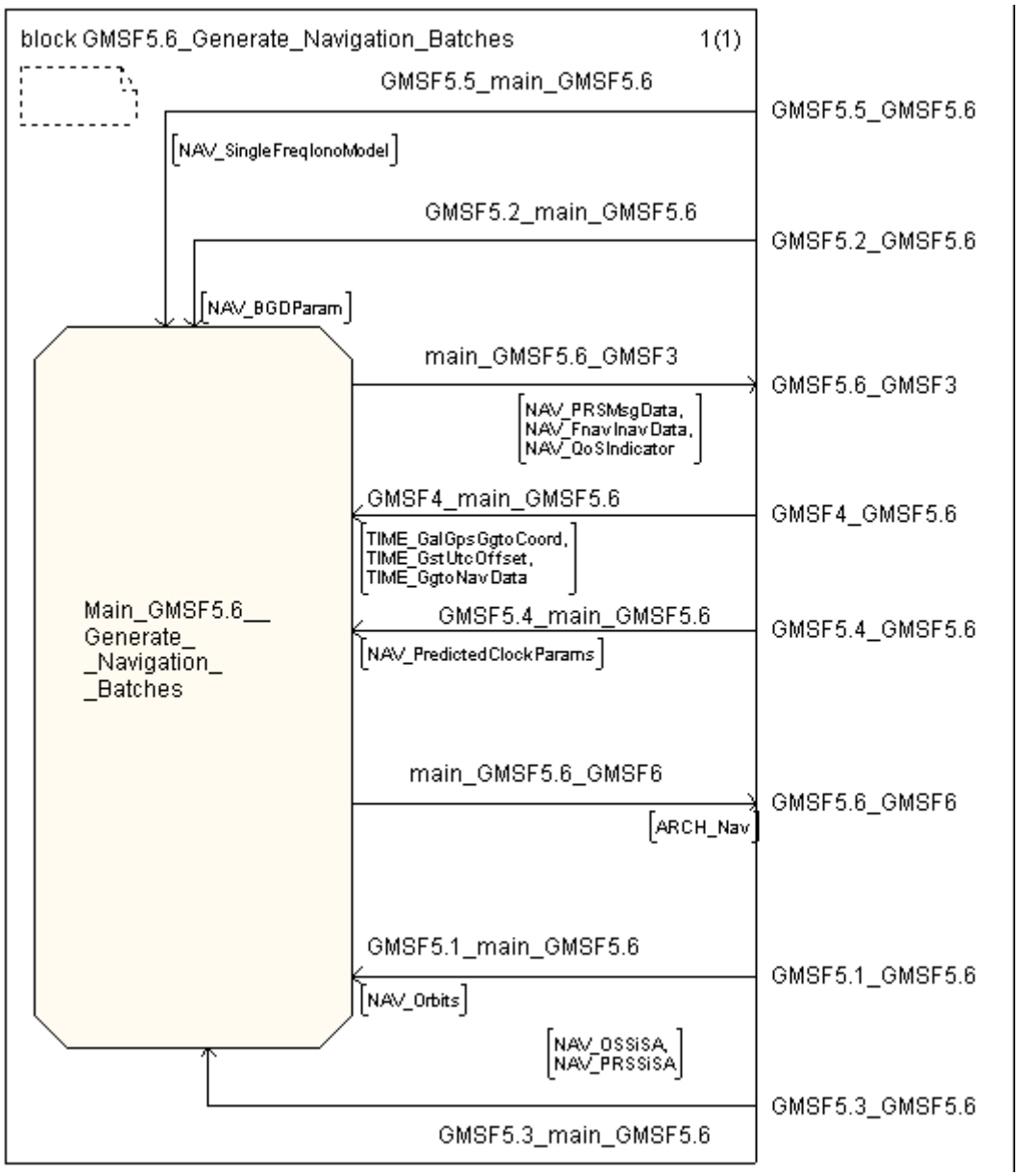


Figure 136: GMSF5.6 SDL Functional Block Diagram

5.3.6 GMSF6 Archiving

GMSF6 Description
This function provides the data archive and retrieval function

Data management includes local and the central archive (storing also GCS data), configuration management.

GMSF6 Functional Block Diagram

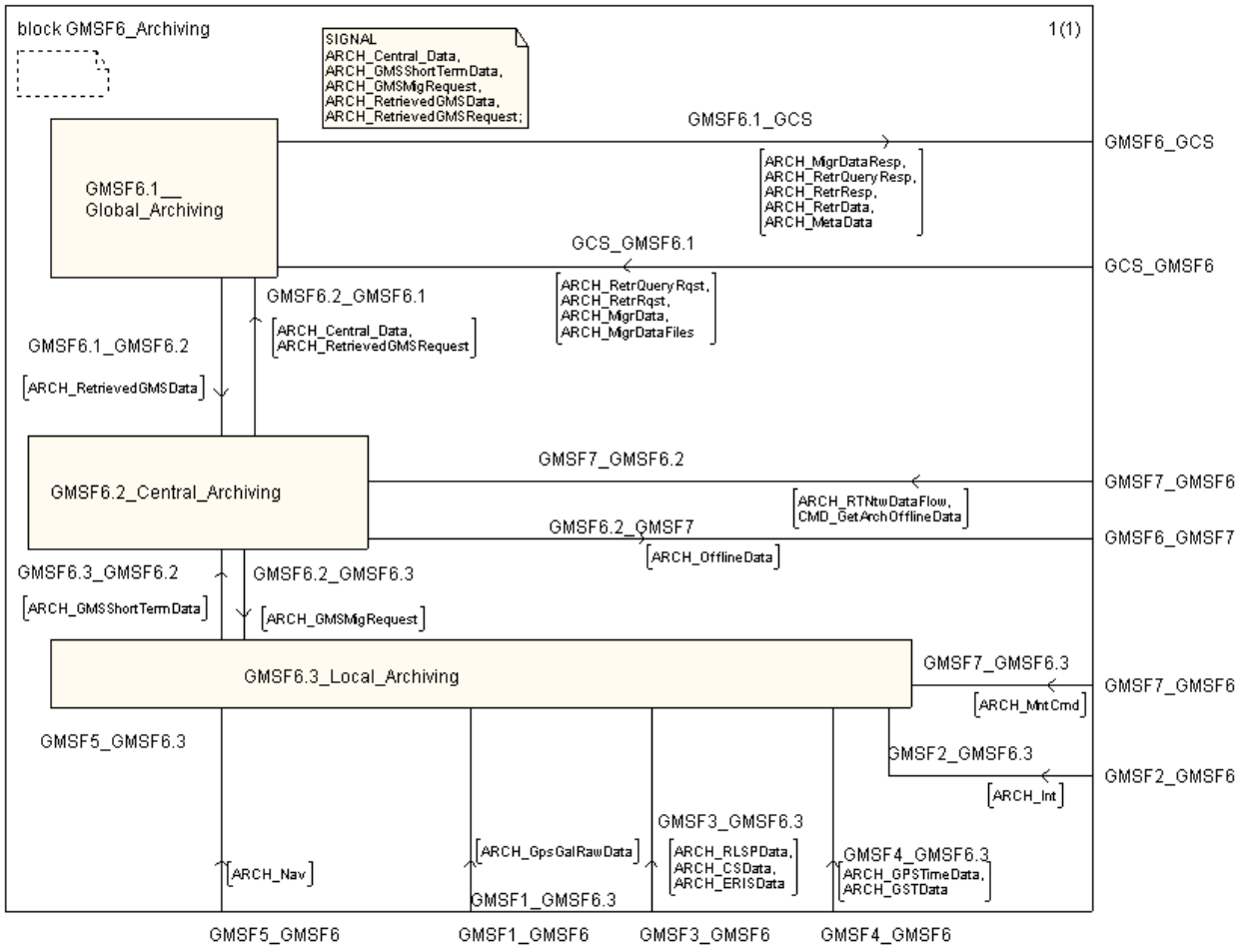


Figure 137: GMSF6 SDL Functional Block Diagram

5.3.6.1 GMSF6.1 Global Archiving

GMSF6.1 General Description
The Global archive function inside the GMS is responsible to migrate the central archived data from GMS and GCS..
This function allow the Central Archives from GMS and GCS to retrieve data which have been previously migrated.

GMSF6.1 Outputs

The GMS record and retrieve the following data of the mission processes, mission elements, M&C as well as the operators actions:

- All the data exchanged between GMS Elements,
- All the data exchanged with other external systems,
- All Monitoring and Control information (events-messages logs),
- Services Performance Monitoring [+ Prediction] data & results,
- Reports and audit files,
- Operator Actions log-files,
- All main processes related data.

GMSF6.1 Design Constraints/Performances

The data are stored on the global archive for the entire system lifetime + 5 years.
The global permanent archive function interfaces with the GCS central function to allow permanent archive of GCS data.

GMSF6.1 Functional Block Diagram

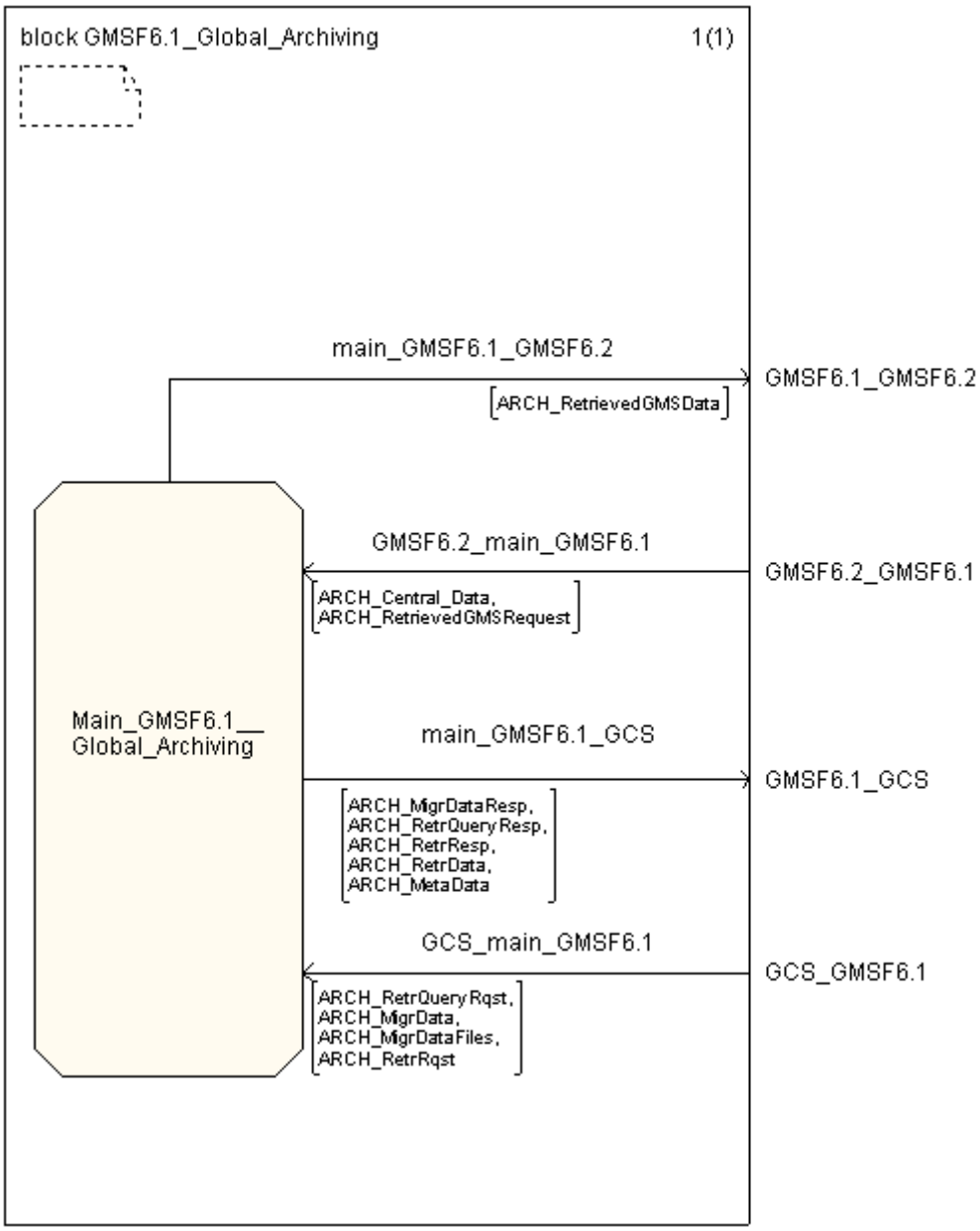


Figure 138: GMSF6.1 SDL Functional Block Diagram

5.3.6.2 GMSF6.2 Central Archiving

GMSF6.2 General Description
The Central archive function inside the GMS is responsible to migrate periodically and to store for a medium duration period (one year):

- the local archived data from GMS elements
- the external data gathered from the external entities (TSP, GPS, GRSP, CS, ERIS, RLSP)
- the Real Time Network dataflow capture inside the GMS

This function interface with the Off Line Mission Monitoring function inside GMS for retrieval of data which have been previously migrated.

GMSF6.2 Outputs
The GMS record and retrieve the following data of the mission processes, mission elements, M&C as well as the operators actions:

- All the data exchanged between GMS Elements,
- All the data exchanged with other external systems,
- All Monitoring and Control information (events-messages logs),
- Services Performance Monitoring [+ Prediction] data & results,
- Reports and audit files,
- Operator Actions log-files,
- All main processes related data.

GMSF6.2 Design Constraints/Performances
The central archive function interfaces with the local archive function at element level.
The archive function allow clean ups, i.e. deletion of files from the archive without impacting other operations

GMSF6.2 Functional Block Diagram

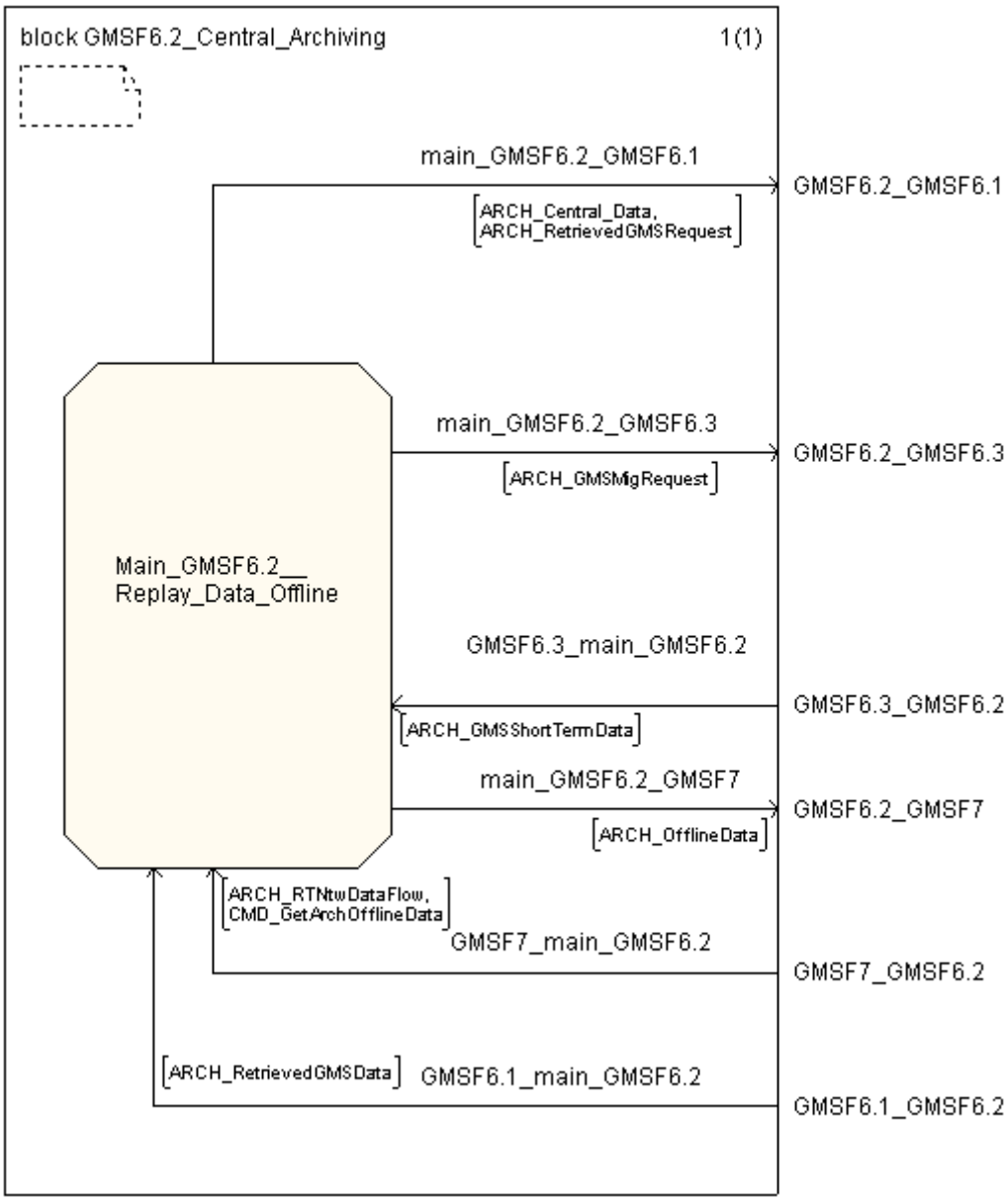


Figure 139: GMSF6.2 SDL Functional Block Diagram

5.3.6.3 GMSF6.3 Local Archiving

GMSF6.3 General Description

The Local Archive inside each GMS element is responsible to store all data processed and received by the element as well as the event log file (for manned elements) with a limited period of storage (one week).

GMSF6.3 Outputs

The GMS record and retrieve the following data of the mission processes, mission elements, M&C as well as the operators actions:

- All the data exchanged between GMS Elements,
- All the data exchanged with other external systems,
- All Monitoring and Control information (events-messages logs),
- Services Performance Monitoring [+ Prediction] data & results,
- Reports and audit files,
- Operator Actions log-files,
- All main processes related data.

GMSF6.3 Design Constraints/Performances

The same archiving and playback principle (format, procedure) are used for every element.

GMSF6.3 Functional Block Diagram

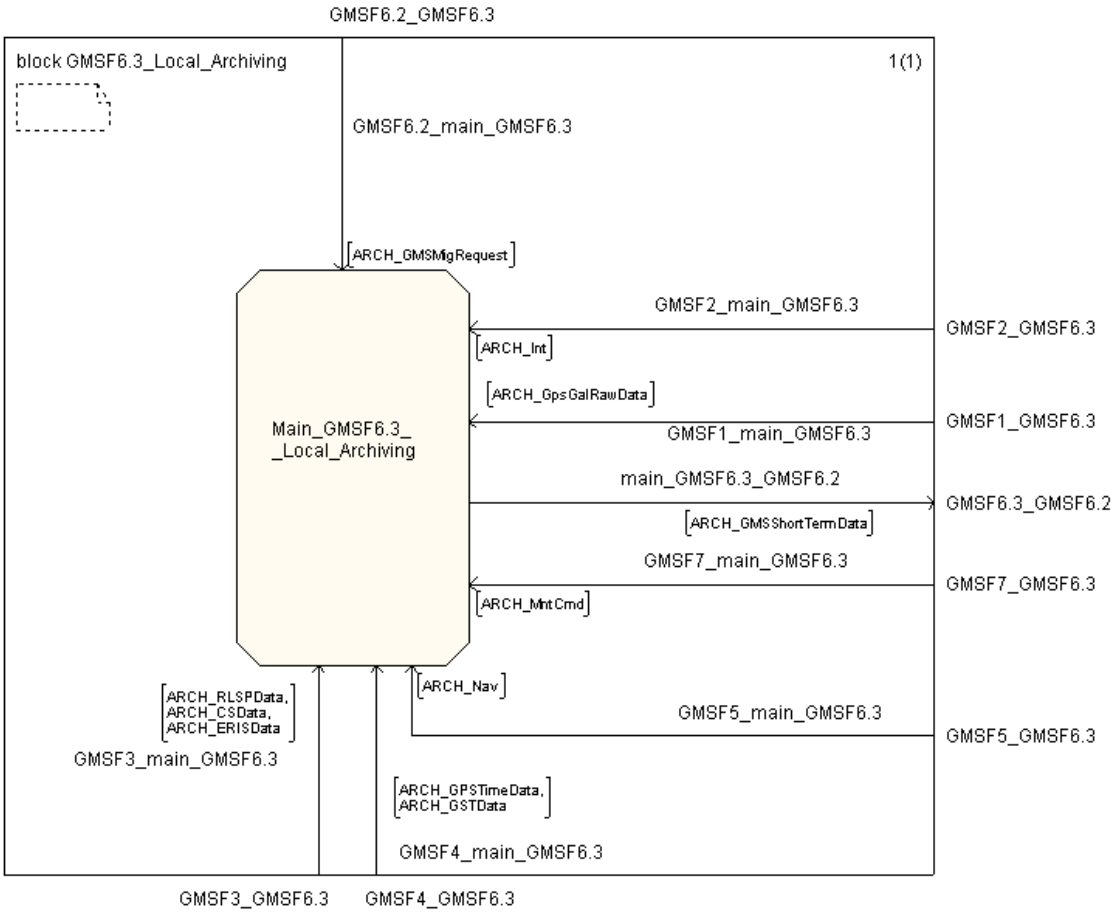


Figure 140: GMSF6.3 SDL Functional Block Diagram

5.3.7 GMSF7 Monitoring and Control

GMSF7 Description

The Mission Monitoring and Control is in charge of:

- allowing a proper display and reporting of mission monitoring parameters, with associated alarms when required: GMSF7.1 Integrated mission monitoring
- on-line mission monitoring to provide inputs to mission controllers in real time: GMSF7.2
- off-line mission monitoring to provide inputs to mission analysts: GMSF7.3
- monitoring and controlling all the ground assets: GMSF7.4 (GMS and GCS elements, controller and operator inputs/outputs)

GMSF7 Functional Block Diagram

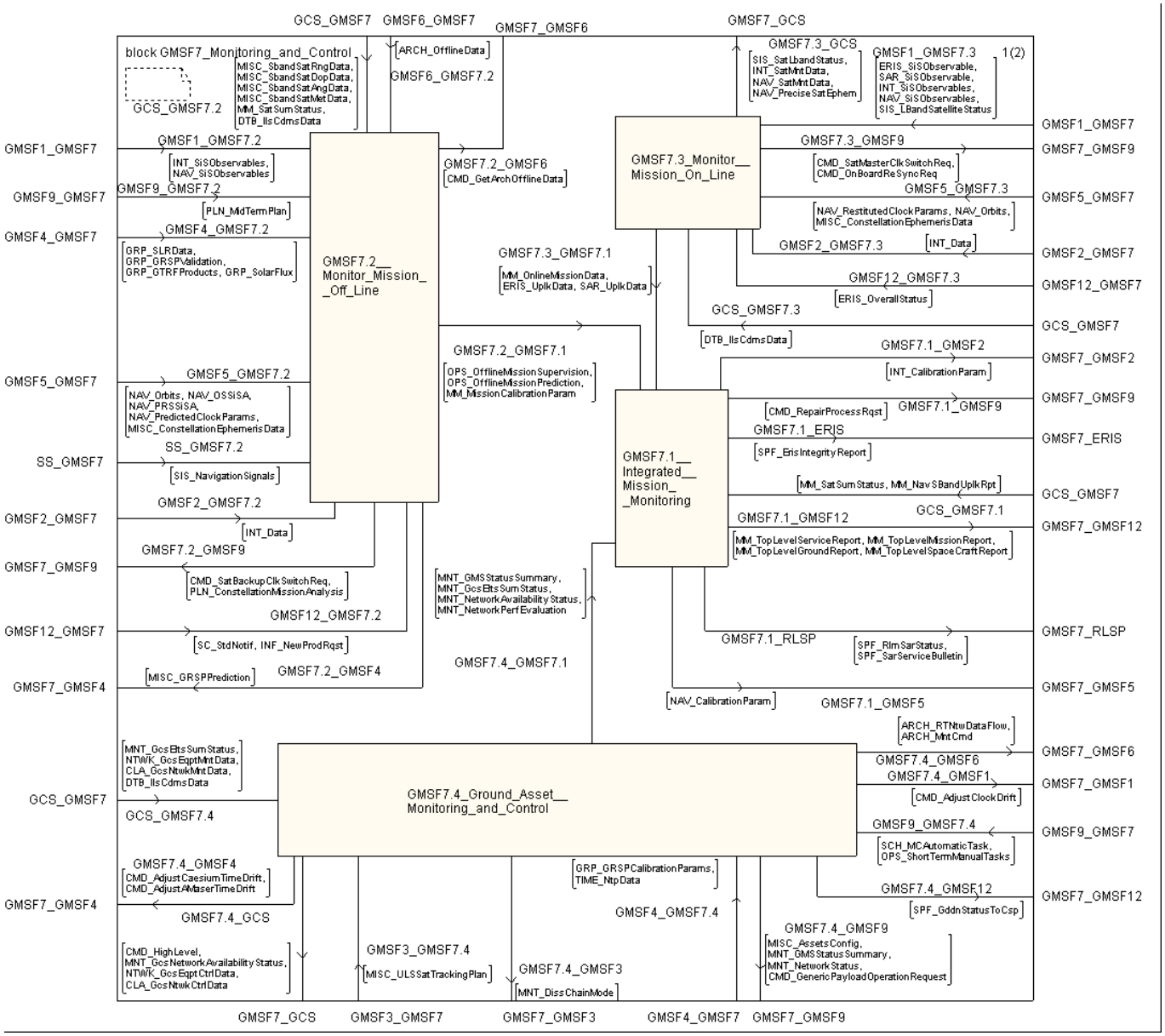


Figure 141: GMSF7 SDL Functional Block Diagram

5.3.7.1 GMSF7.1 Integrated Mission Monitoring

GMSF7.1 General Description

This Integrated Mission Monitoring is responsible for implementing all the needed functions and associated interface for:

- Displaying in real-time the following information provided by mission on-line to the mission controllers:
  - Service performances parameters
  - Ground assets functional and performance status including:
    - QoS of mission provision chain
    - Mission data provision chain configuration display
    - Satellite functional and performance status (including on-board clocks)
- Displaying periodically or on request the information provided by mission off-line monitoring functions, and needed by the mission analysts to support:
  - the calibration of OD&TS algorithms.
  - the verification of SIS RF characteristics
  - the past performances statistical analyses
  - the performances trend analyses
  - the performance prediction analyses
  - the on-bard clock monitoring
- Generating various reports, related to mission, operation or equipment for the external world, including:
  - Quality of uplink service and summary status for ERIS
  - SAR Return link message dissemination summary status for RLSP
  - Operation, Ground assets and Mission performance reports for the Service Centers.
  - Generating alarms in case of non nominal service performance
  - Generating command request to the planning function to repair abnormal identified process.

GMSF7.1 Outputs

This Integrated Mission Monitoring function produces the following outputs:

1. The products provided by the on-line mission monitoring function (mission alarms, QoS, mission chain configuration, satellite and ground assets status)
2. Mission service performance report
3. SAR Service performance report and RLM status
4. ERIS Service performance report
5. SISA check for SISA computation
6. Calibration parameters

GMSF7.1 Design Constraints / Performances

The Integrated Mission Monitoring shall implement an MMI allowing to manage the parameters monitored in an efficient way. This means for instance the implementation of:

- zooming capabilities.
- Alarm cascading barrier

In addition the operator shall be able to have access and modify the alarm threshold.

GMSF7.1 Functional Block Diagram

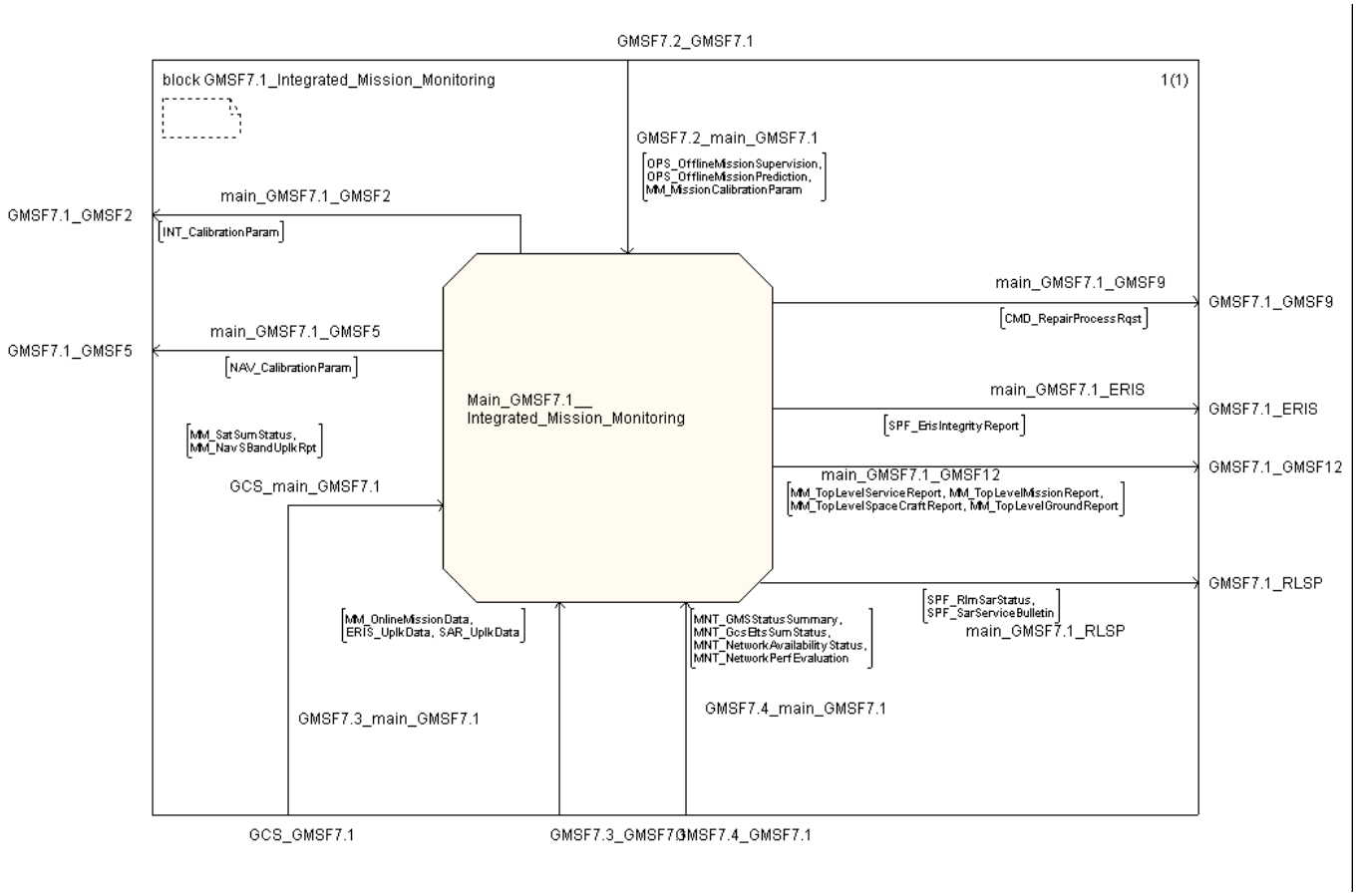


Figure 142: GMSF7.1 SDL Functional Block Diagram

5.3.7.2 GMSF7.2 Monitor Mission Off Line

GMSF7.2 General Description

The Off-Line Mission Monitoring function is responsible for providing all required functionality to monitor and analyse the parameters needed to guarantee a sufficient level of performance of the system with off-line processes.
In particular, it shall provide the needed data for:

- System calibration: support to algorithm and element parameters calibration (including OD&TS, GRSP, GST)
- Statistical and trend mission performance analyses: support to report generation
- Satellite on-board hot backup clock monitoring and control
- Mission performance prediction: support to mission planning validation

GMSF7.2 Outputs

The Off-line mission monitoring function produces the following outputs to the mission analysts operators via the integrated mission monitoring function:

1. Calibrated OD&TS mission data (orbits, clocks, ionospheric model
2. SISA check for SISA computation
3. Calibrated SIS RF characteristics (from Signal spectrum analyses)
4. GST Stability Check (check to monitor the TSP performance)
5. GRSP products calibration
6. Statistical service and system performances (accuracy, availability) for report generation
7. Performance trend analyses to potentially anticipate performance impact
8. Satellite Hot backup Clock switch request if needed (based on on-board hot backup clock phase measurement sampling received via telemetry)

GMSF7.2 Design Constraints / Performances

The design shall provide 6 positions for the off-line mission analyst operators:

- 3 for off-line support to ground asset performance and trouble shooting operations
- 1 for off-line mission performance calibration and configuration
- 2 for off-line mission performance analyses and prediction

The off-line mission monitoring function shall be capable to integrate the AIV tools for mission performance analyses.

GMSF7.2 Functional Block Diagram

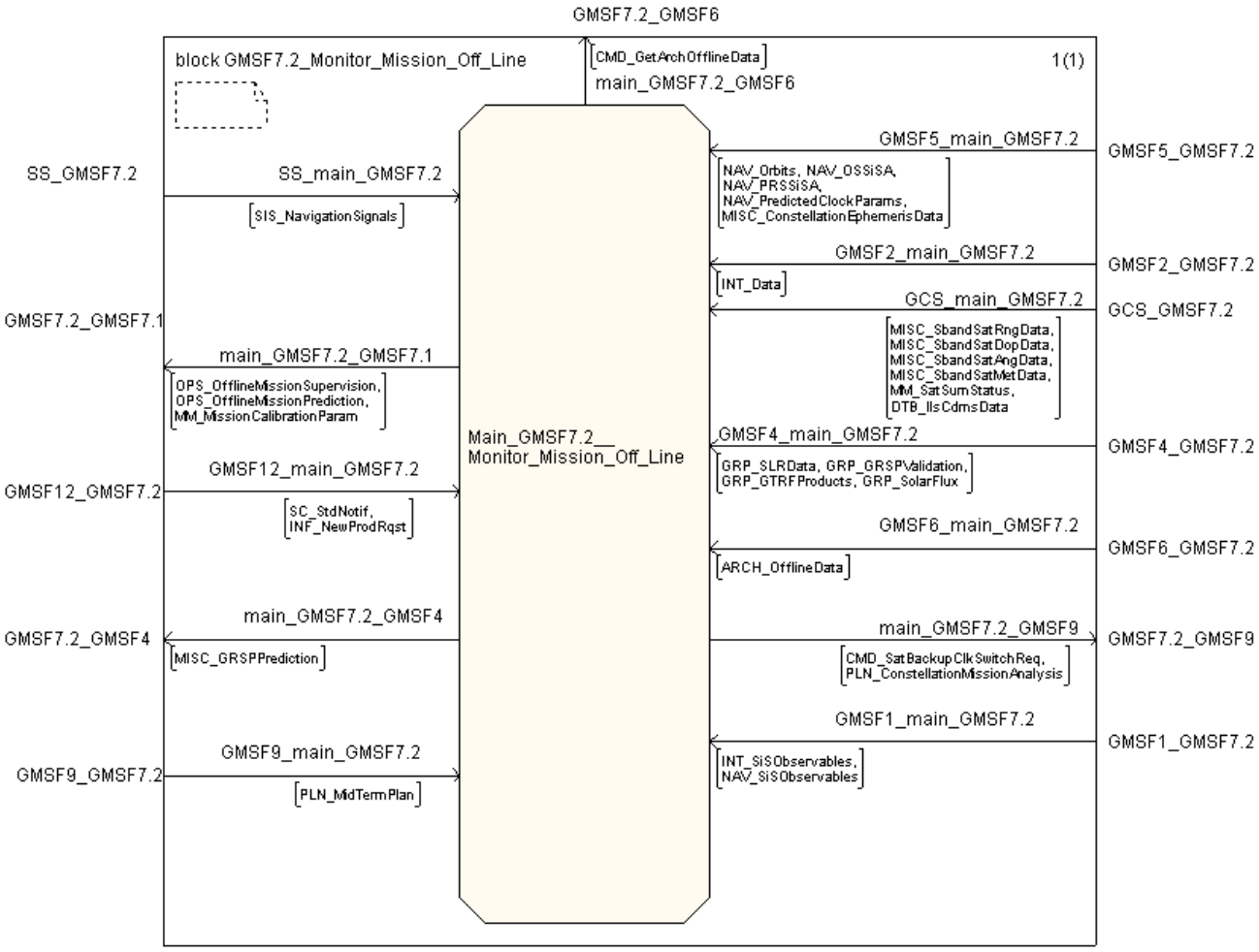


Figure 143: GMSF7.2 SDL Functional Block Diagram

5.3.7.3 GMSF7.3 Monitor Mission On Line

GMSF7.3 General Description

The On-Line Mission Monitoring funciton is responsible for providing all required functionality to monitor and analyse in real-time the parameters that are critical for the mission performance.
In particular, it shall provide the needed data for:

- On-line verification of the service performances (OS, SoL and PRS) at user level including:
  - OS and PRS Positioning and Timing accuracy and availability
  - On-line verification of the mission data provision chain quality (mission data determination products, Ground Segment Ressources monitoring)
- On-line satellite status monitoring including:
  - On-board master clock frequency stability, comparing the on-board restituted clock parameters (in real time) computed by the OD&TS to on-ground clock models
  - satellite general status

In addition the on-line monitoring function is responsible for providing the GCS with:

- Satellite status
- Precise ephemeris computed by the GMS OD&TS function

GMSF7.3 Outputs

The On-line mission monitoring function produces the following outputs to the mission controller operators via the integrated mission monitoring function:

1. Mission alarms in case service performances are not met at user level
2. OD&TS chain QoS and configuration
3. Request for commands to the GCS to steer the on-board frequency reference or time of the satellite master clock, or to switch it to the hot backup clock

In addition the on-line mission monitoring transmits the following information to the GCS in real time:

- the Satellite health status information retrieved from the L-Band Navigation messages (L-Band Telemetry Status)
- a satellite status information obtained from the internal processes (OD&TS, signal analysis).
- Precise Ephemeris computed by the GMS OD&TS algorithm

GMSF7.3 Design Constraints / Performances

The design shall provide 2 positions for the on-line mission controller operators:

- 2 for on-line mission performance monitoring and schedule support

GMSF7.3 Functional Block Diagram

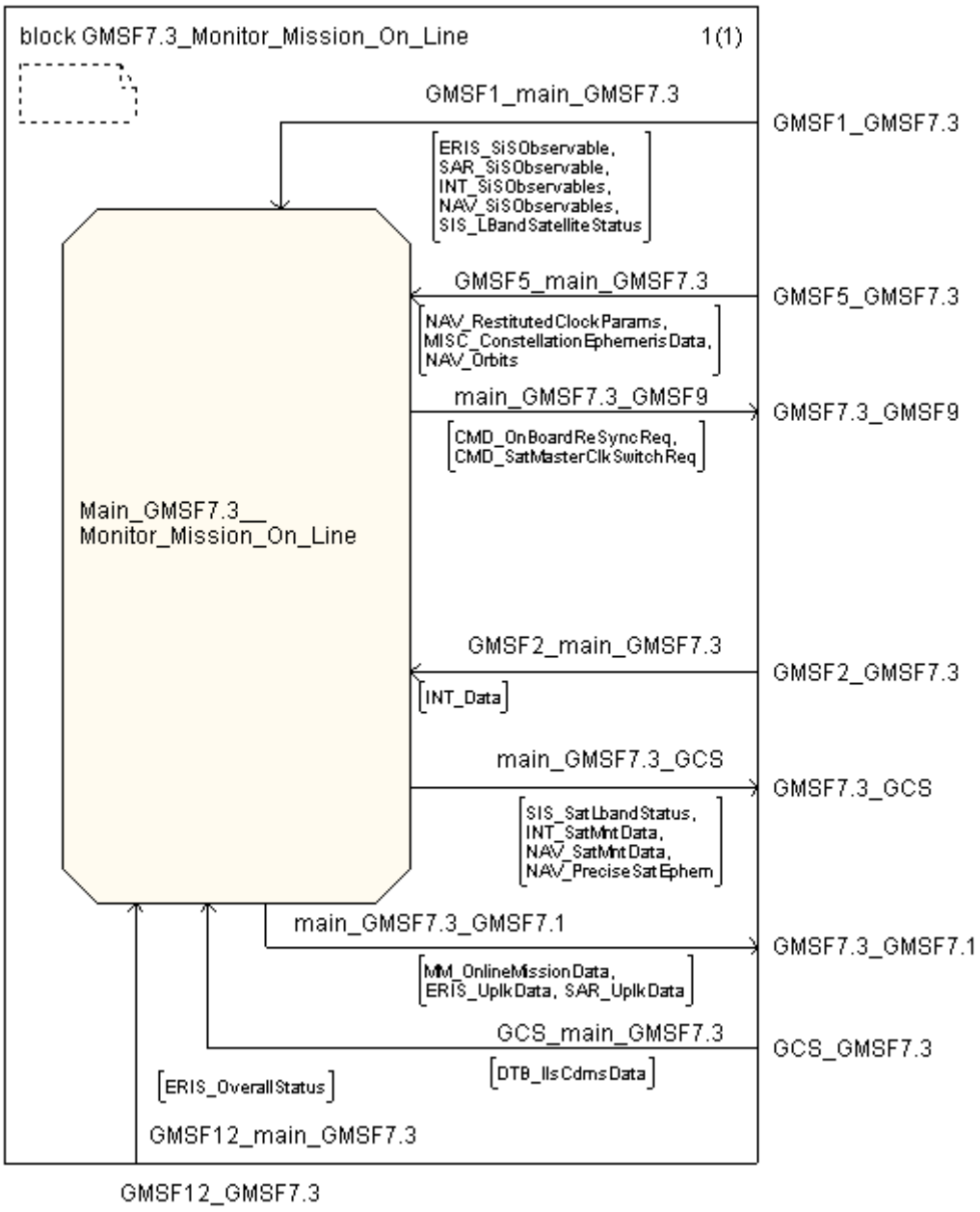


Figure 144: GMSF7.3 SDL Functional Block Diagram

5.3.7.4 GMSF7.4 Ground Asset Monitoring and Control

GMSF7.4 General description

This function is in charge to monitor and control all the Galileo ground assets.
The main functionalities are:

- Ground assets real time monitoring and control and report to the mission monitoring and short term planning functions
- Ground assets maintenance and troubleshooting, alarm reporting
- Ground schedule execution
- Reporting and interface to operators
- network performance assessment
- Interface with operators

For clarification, this function is derived into the following sub-functions:

- GCSF7.4.1 for element monitoring and control and
- GCSF7.4.2 for central monitoring and control
- GCSF7.4.3 for ground assets overall monitoring and control
- GMSF7.4.4 for ground network monitoring and control

GMSF7.4 outputs

The GMS monitoring and control function produces the following outputs:
1 - summary status of the GMS and GCS elements, sent to the mission monitoring function
2 - summary status of the GMS segment to the planning functions
3 - on operator request, some GCS high level commands
4 - summary performance evaluation of the ground networks (GMS and GCS networks) sent to the mission monitoring function and to the planning functions
5 - GCS network summary status, sent to the GCS central monitoring and control function, since this latter has no capability to monitor the GCS network
6 - GDDN performance assessment, sent to the CSP through GMS external interface management function
7 - GCS network management commands

GMSF7.4 design constraints

The monitoring and control function of the GMS is performed at the following different levels:

- level 0, for local maintenance, performed using element local monitoring and control facility
- level 1, allowing GMS remote maintenance from GCC, for troubleshooting, preventive maintenance
- level 2, allowing GMS central monitoring and control from GCC during normal operations

This function relies on a database oriented system, based on the SNMP protocol.

GMSF7.4 Functional Block Diagram

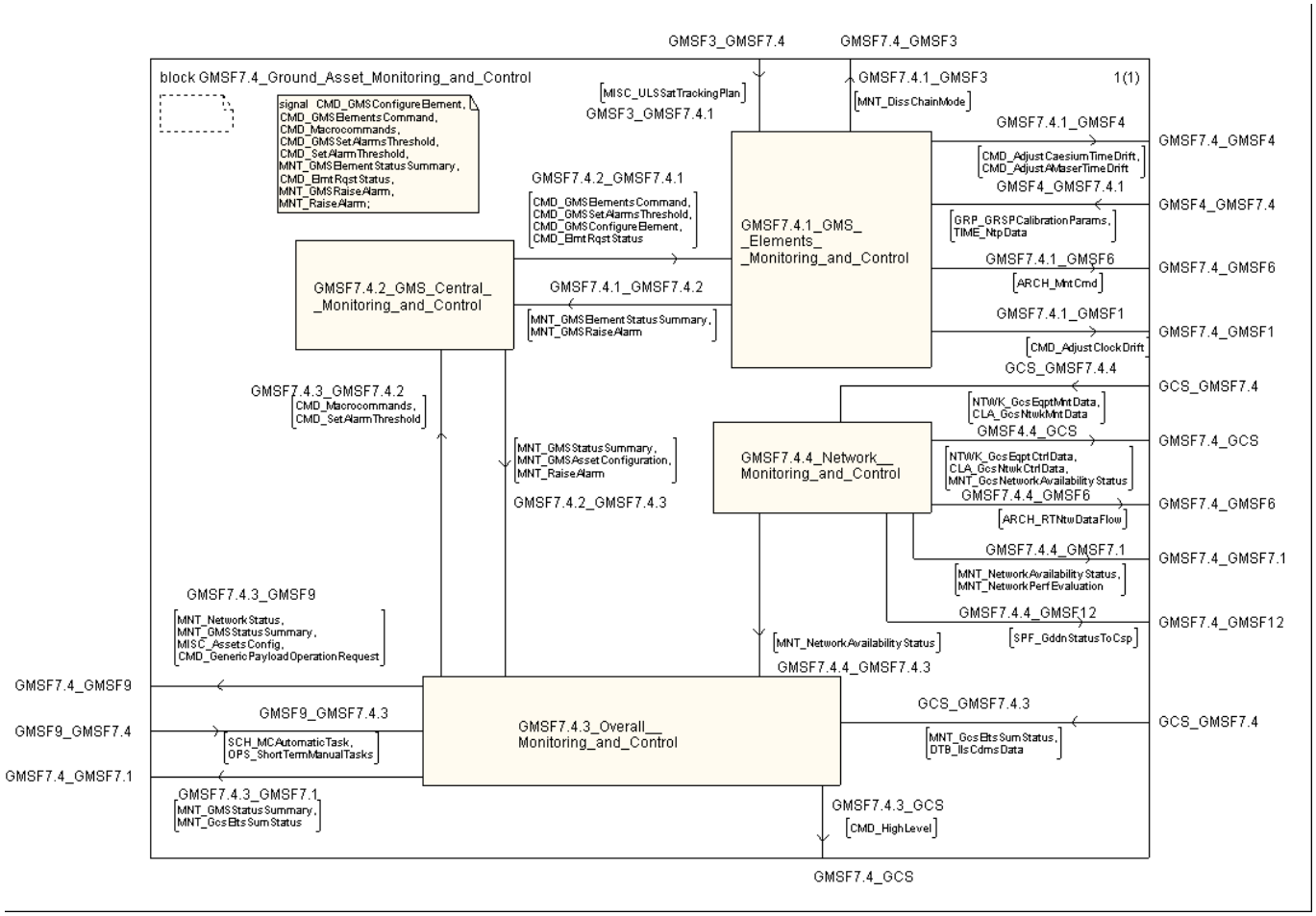


Figure 145: GMSF7.4 SDL Functional Block Diagram

5.3.7.4.1 GMSF7.4.1 GMS Elements Monitoring and Control

GMSF7.4.1 General description

The GMS element M&C function is the element local instance of the GMS monitoring and control function.
It is therefore implemented locally in each element.
Its aim is to:

- monitor the element, i.e. gathering all the element health statuses and alarms (including elements' LANs) and send them to the GMS central M&C function, located in the central GCC location
- execute the configuration commands sent by GMS central M&C function
- allow local monitoring and control by an operator for all monitoring and control level-0 activities (i.e. maintenance operations)
- prevent dual control from two different entities (e.g. GMS central M&C function and local operator)

GMSF7.4.1 outputs

The GMS element monitoring and control function produces the following outputs:
1 - element status of the GCS elements, which is sent to the GCS central M&C function
2 - alarm reporting

GMSF7.4.1 Functional Block Diagram

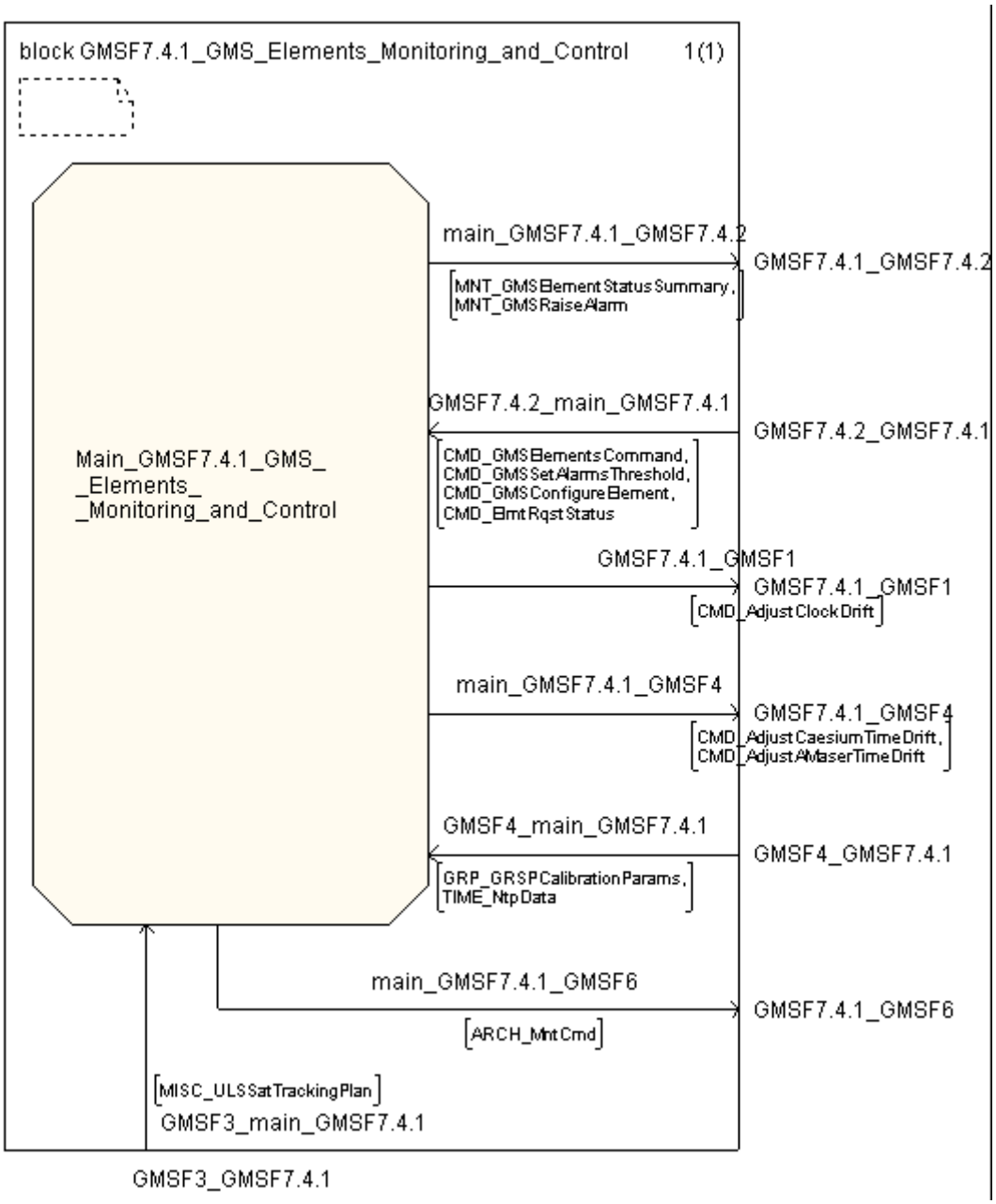


Figure 146: GMSF7.4.1 SDL Functional Block Diagram

5.3.7.4.2 GMSF7.4.2 GMS Central Monitoring and Control

GMSF7.4.2 General description

The GMS central M&C function is the segment level instance of the GMS monitoring and control function.
It is implemented in the GCC.
Its aim is to:

- gather all monitoring health statuses and alarms from all GMS elements (coming from GMS local M&C function)
- send GMS summary status and alarms to the overall M&C function of the GMS
- develop and execute the configuration macro-commands sent by the GMS overall M&C function
- allow central monitoring and control on all the GMS elements by operators for level-1 activities (i.e. maintenance or troubleshooting operations, via central M&C located at GCC and ground normal operations handling) through displays, MMI, logbooks, …

GMSF7.4.2 outputs

The GMS central monitoring and control function produces the following outputs:
1 - GMS elements control command to all the GMS elements
2 - Summary of the GMS element status and alarms, for reporting to the ground overall M&C function
3 - GMS element specific status request for troubleshooting purposes

GMSF7.4.2 Functional Block Diagram

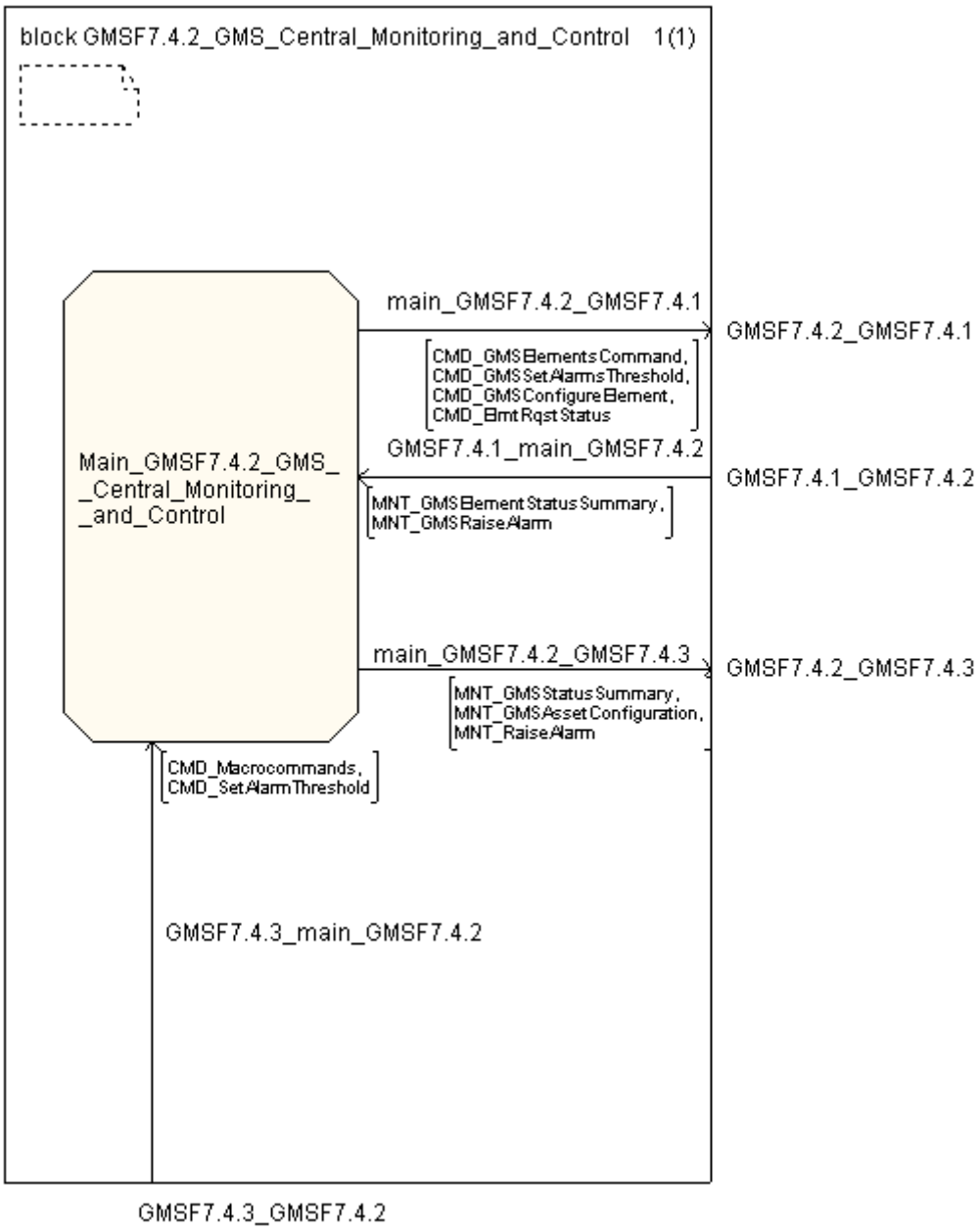


Figure 147: GMSF7.4.2 SDL Functional Block Diagram

5.3.7.4.3 GMSF7.4.3 Overall Monitoring and Control

GMSF7.4.3 General description

The GMS overall monitoring and control function is in charge of the monitoring and control of all the Galileo ground assets (GMS, GCS or networks).
Its aim is to:

- gather the GMS and GCS summary statuses and alarms from the relevant central M&C functions and present them in a synthetic way to the operators, through displays, MMI, logbooks, …
- send ground assets summary statuses and alarms to the integrated mission monitoring function and to the GMS short term planning for planning preparation or update if necessary (level-3 operations)
- execute schedule tasks or operator requests by sending macro-commands to the relevant central M&C functions
- Define the alarm thresholds for each element to be monitored
- Define the alarm thresholds for each element to be monitored

GMSF7.4.3 outputs

The GMS overall monitoring and control function produces the following outputs:

1. summary of the ground segments status (GMS and GCS), sent to the GMS planning function and to the mission monitoring function
2. macro-commands, sent to the GMS and GCS segment central monitoring functions
3. Network configuration commands

GMSF7.4.3 Functional Block Diagram

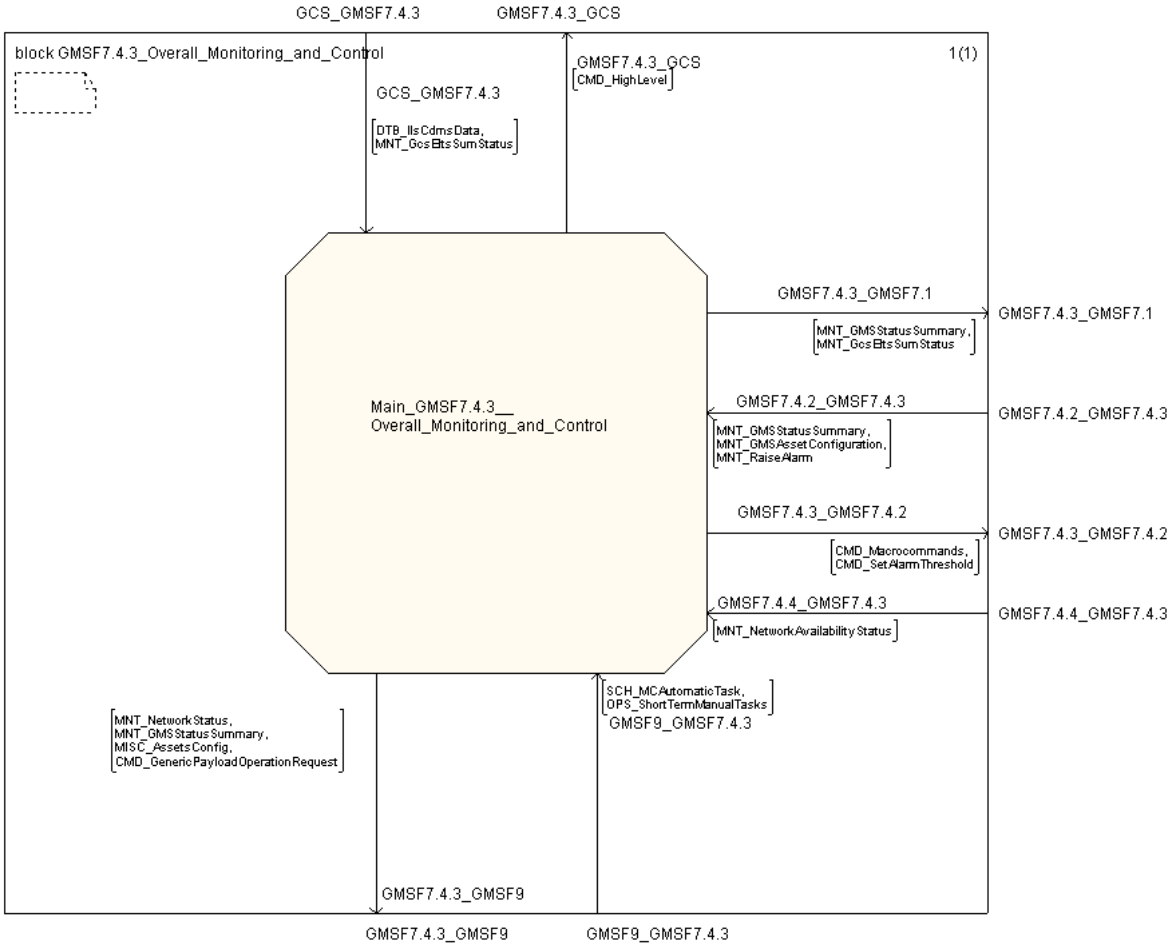


Figure 148: GMSF7.4.3 SDL Functional Block Diagram

5.3.7.4.4 GMSF7.4.4 Network Monitoring and Control

GMSF7.4.4 General description

The network M&C function is included in the GMS. Its aim is to monitor and control the entire Galileo network.
In more detail, this function:

- performs control over the GMS network (data switching, data routing, data flow priority setting, …)
- continuously gather GMS network performance measurement of all traffic streams going across the network and generate QoS (quality of service) stamps, based on data latency, integrity and availability characteristics
- perform the overall network performance assessment by processing the performance measurement performed on GMS network and received from GCS network M&C function
- send GCS network summary status to GCS central M&C function in case any failure or performance issue arises
- handle fault and configuration management of the Galileo network equipment
- interface with overall M&C function for monitoring and control of the overall Galileo ground network
- perform GDDN network performance assessment and transfer the result of this assessment to CSP. Note that this assessment is the basis for CSP service fulfillment check

GMSF7.4.4 outputs

The ground network management function produces the following outputs:
1 - commands for GCS network management
2 - network availability and performance measurement, sent to mission monitoring and planning functions
3 - GCS network status, sent to the GCS central monitoring function, since this latter does not have this capability
4 - GDDN performance assessment, to be transferred to the CSP for service level fulfillment assessment

GMSF7.4.4 Functional Block Diagram

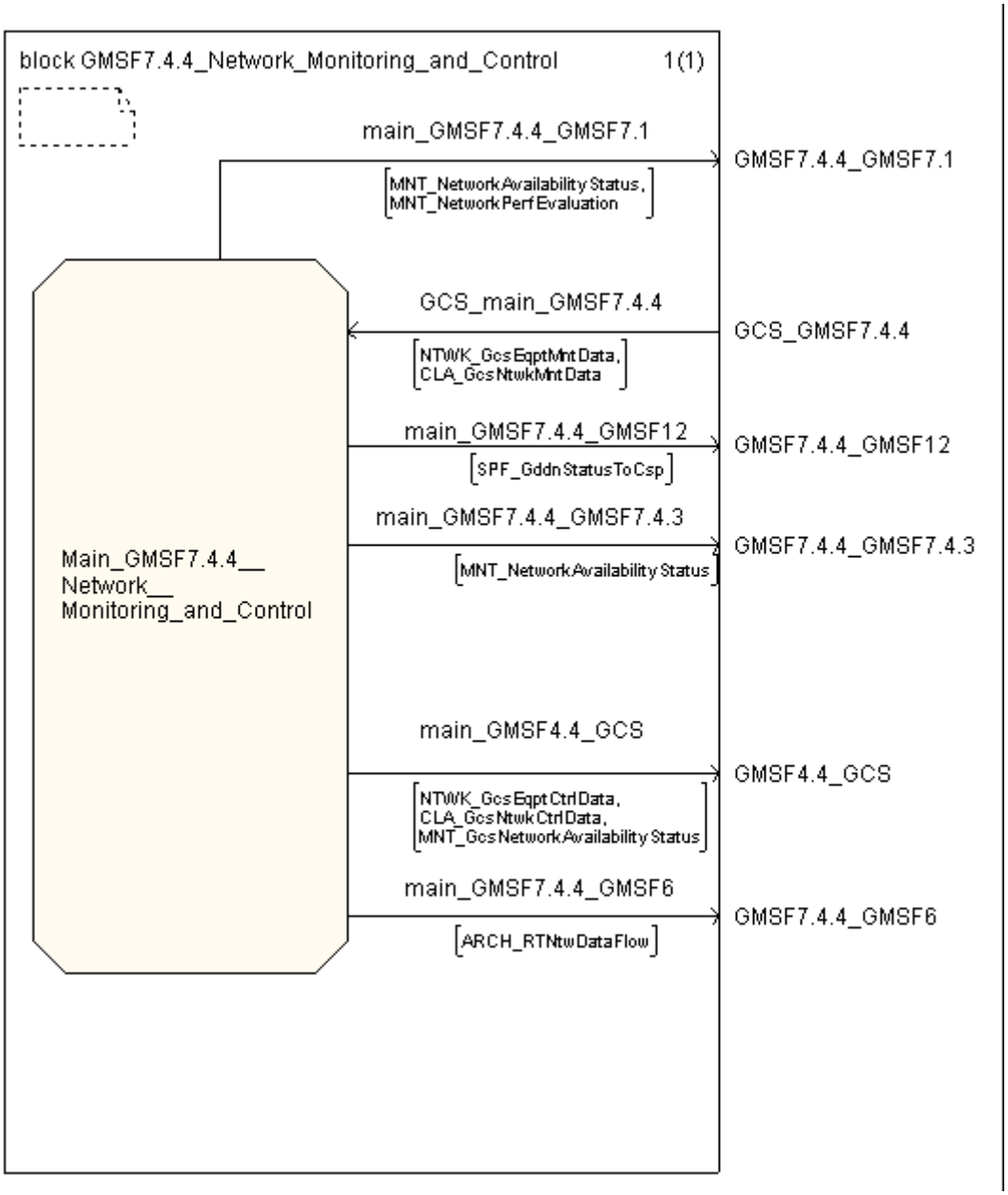


Figure 149: GMSF7.4.4 SDL Functional Block Diagram

5.3.8 GMSF8 Security

GMSF8 Description

The GMSF8 includes all functions related to security management at GMS level. These functions are linked to dedicated Security Classified Functions part of Classified SEC-SFCs detailed in the SYSBSLDEF Part I Classified Annex [SYS-RD.07].

- GMSF8.1 (Security protection of PRS): linked to SEC-GMSF1
- GMSF8.2 (Security protection of SoL): descoped
- GMSF8.3 (Security protection for mission data): linked to SEC-GMSF1
- GMSF8.4 (Security protection of CS): not classified (not linked to any SEC Function)

GMSF8 Functional Block Diagram

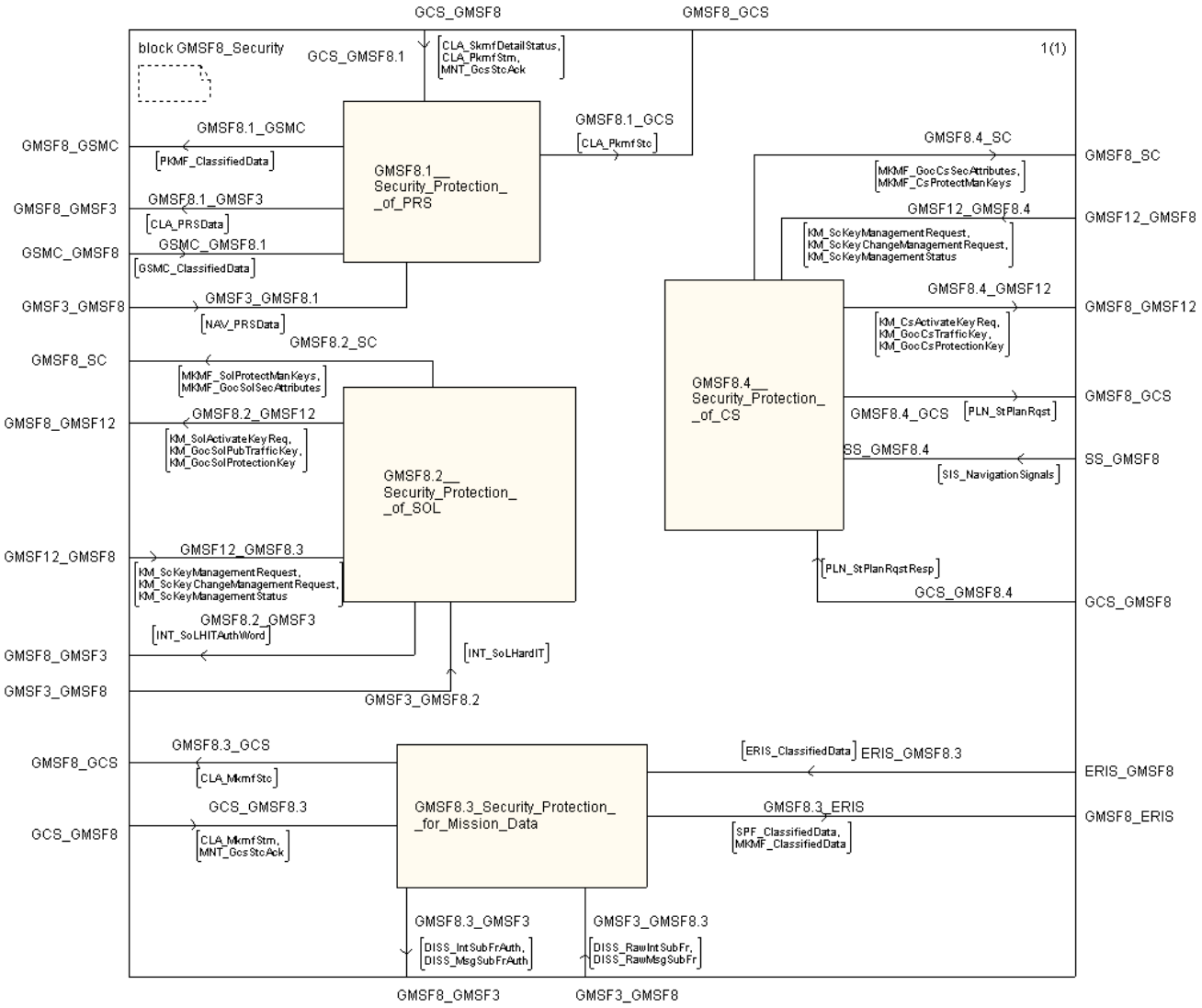


Figure 150: GMSF8 SDL Functional Block Diagram

5.3.8.1 GMSF8.1 Security Protection of PRS

GMSF8.1 General Description
This function is responsible for managing all the security aspects for the protection of the PRS service at GMS level. This includes:

- managing all the security aspects for the protection of the PRS data and service, namely generation and encryption of the PRS security messages. This function relies on the support of GSMC via a dedicated interface.
- managing the STM/STC related to PRS security unit management dedicated to the satellite PLSU

This function corresponds to dedicated Classified Security Function SEC-GMSF1. Refer to SYSBSLDEF Part I Classified Annex [SYS-RD.07] for details.

Note: this function is linked to SSF2.2 related to security protection for PRS data at Satellite level.

GMSF8.1 Outputs
The outputs of this function are:

- Protected PRS Message to be disseminated in the Message uplink subframe
- Classified data dedicated to GSMC
- PKMF STC dedicated to on-board PLSU

See SEC-GMSF1 for further details.

GMSF8.1 Functional Block Diagram

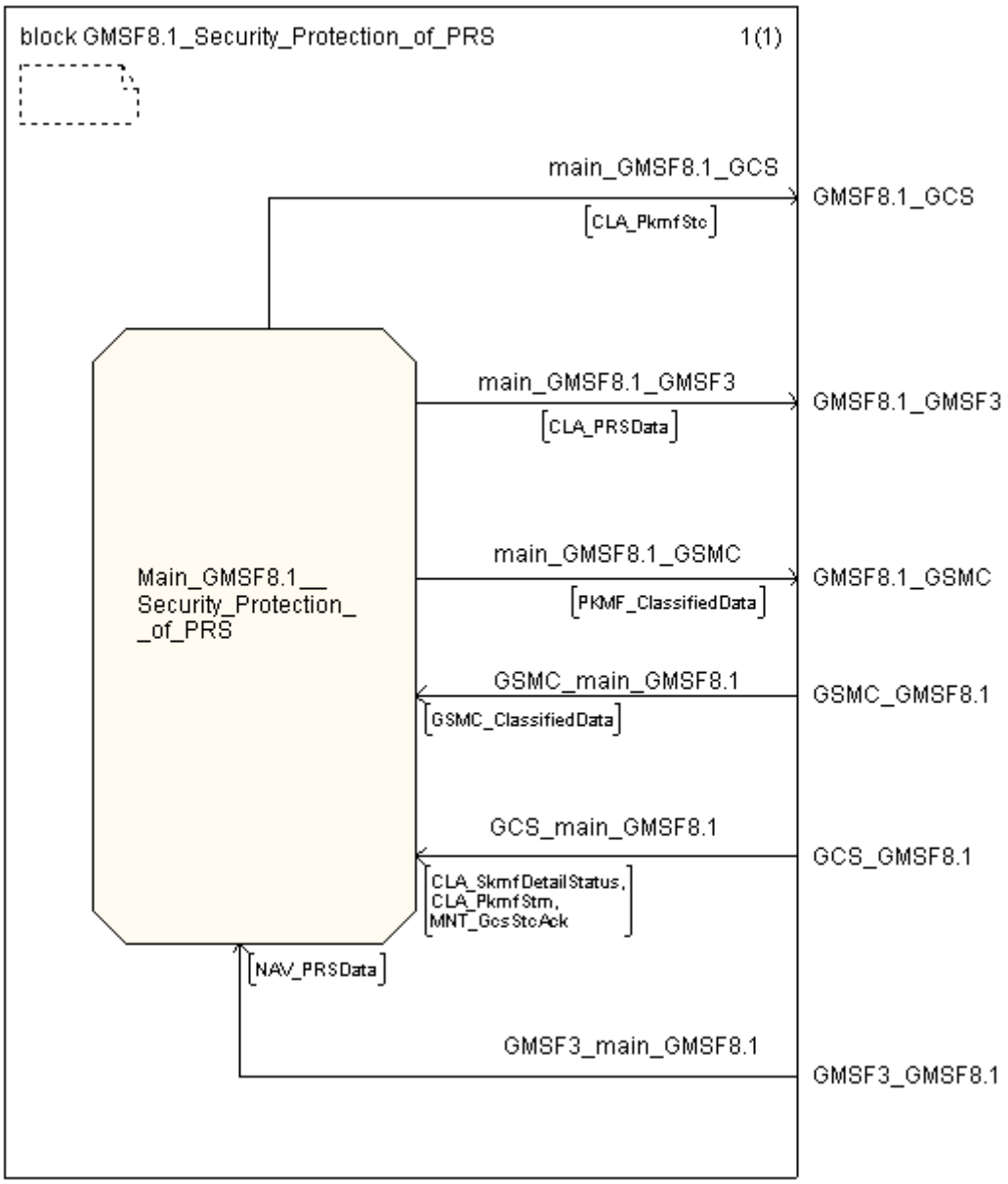


Figure 151: GMSF8.1 SDL Functional Block Diagram

5.3.8.2 GMSF8.2 Security Protection of SOL

Descoped

5.3.8.3 GMSF8.3 Security Protection for Mission Data

GMSF8.3 General Description
This function is in charge of managing all the protection aspects for the mission data at GMS level. This includes:

- Protection of the C-Band Uplink including authentication for the Message and Integrity uplink subframes
- Managing the ERIS Key distribution for ERIS direct access to the satellites (ERIS direct uplink)
- Managing the STM/STC related to C-Band authentication management dedicated to the satellite PLSU (see classified DDF)

This function corresponds to dedicated Classified Security Function SEC-GMSF1. Refer to SYSBSLDEF Part I Classified Annex [SYS-RD.07] for details.

Note: this function is linked to SSF1.2 related to security protection for mission data at Satellite level

GMSF8.3 Outputs
The main outputs of this function are:
1. Protected Message and Integrity Uplink subframes
2. ERIS key for direct access to satellites
3. MKMF STC dedicated to on-board PLSU

See SEC-GMSF1 for further details.

GMSF8.3 Functional Block Diagram

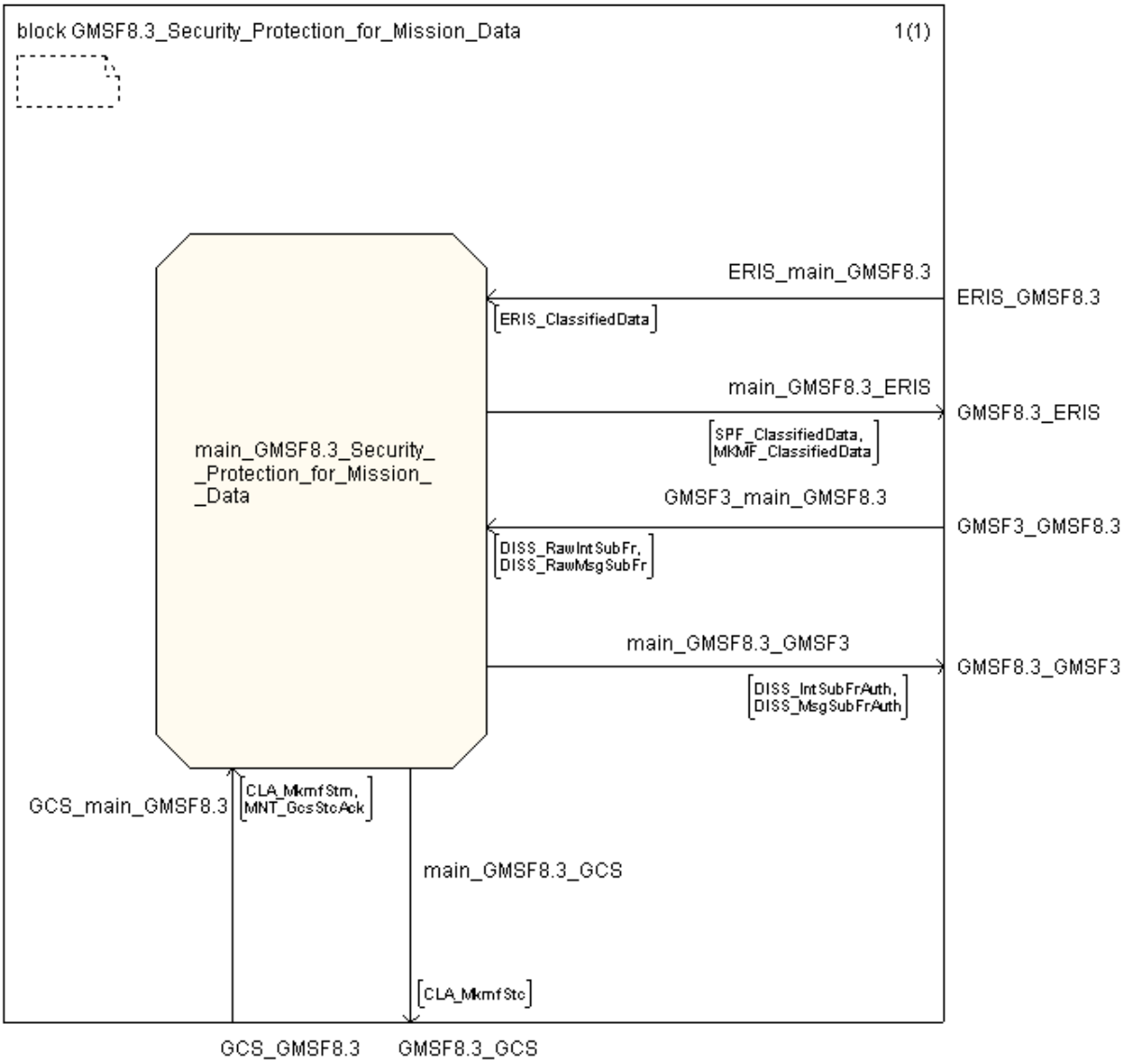


Figure 152: GMSF8.3 SDL Functional Block Diagram

5.3.8.4 GMSF8.4 Security Protection of CS

GMSF8.4 General Description

This function is responsible for:

- Verifying third carrier CS protection
- Interface with GCS to enable or disable the CS ranging code encryption.
- Managing the CS key with the support of SC via a dedicated interface. In particular, distribute the CS-related keys upon request, including manual and network distribution

GMSF8.4 outputs

- The output in response to the CS protection verification is a status indication.
- The output to enable or disable the CS ranging code encryption of a satellite is in the form of a short term planning request to GCS.
- The output to a key request is a CS-related key (with an activation date in case of a CS traffic key) either sent via a network message or in CD-ROM format. The key material is protected (encrypted and authenticated) using previously exchanged protection keys.

GMSF8.4 Functional Block Diagram

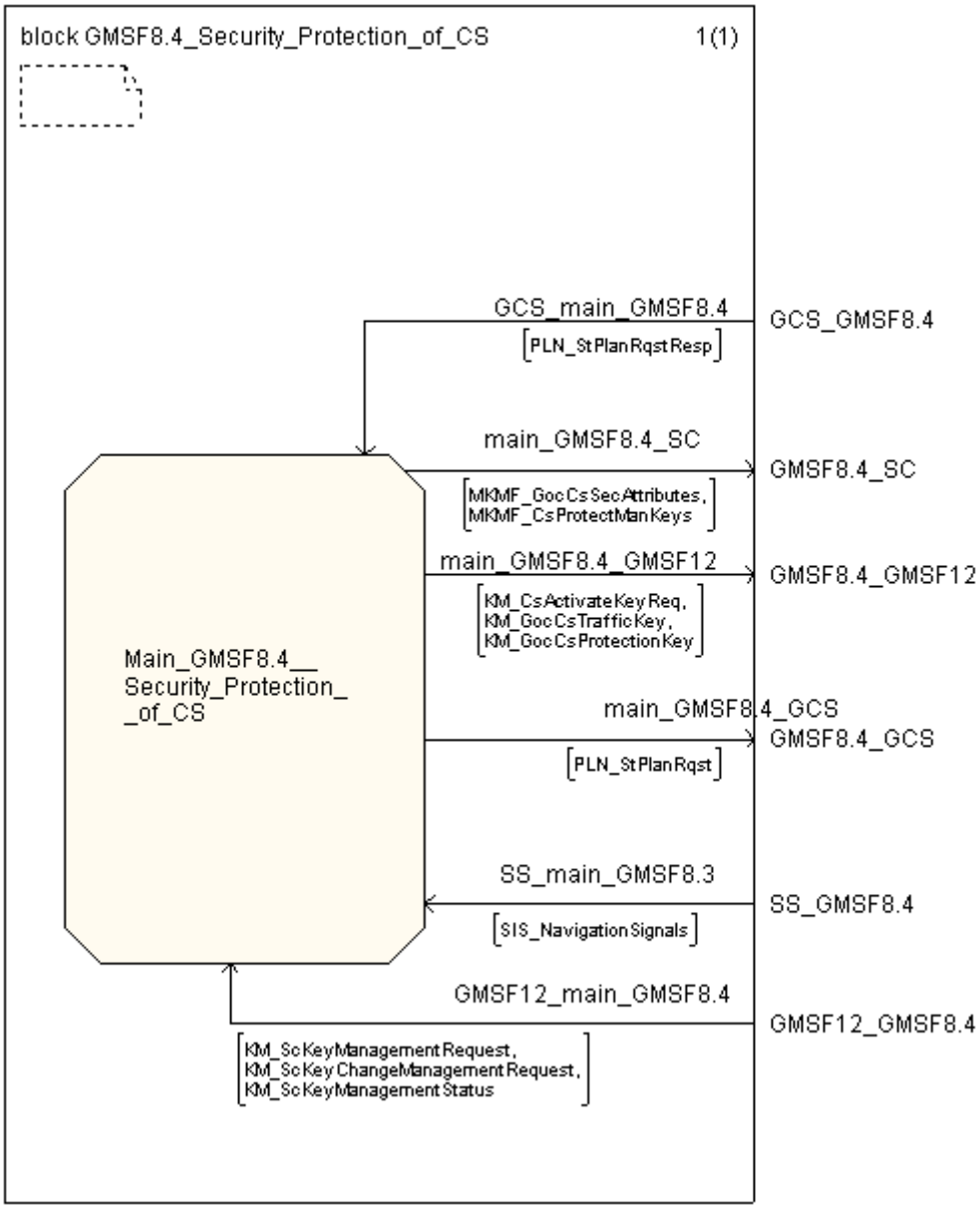


Figure 153: GMSF8.4 SDL Functional Block Diagram

5.3.9 GMSF9 Perform Mission Planning

GGMSF9 Description

The GMS planning function prepares the mission routine & contingency planning for the whole system for long and mid term, and for short term for the GMS only.

GMSF9 Functional Block Diagram

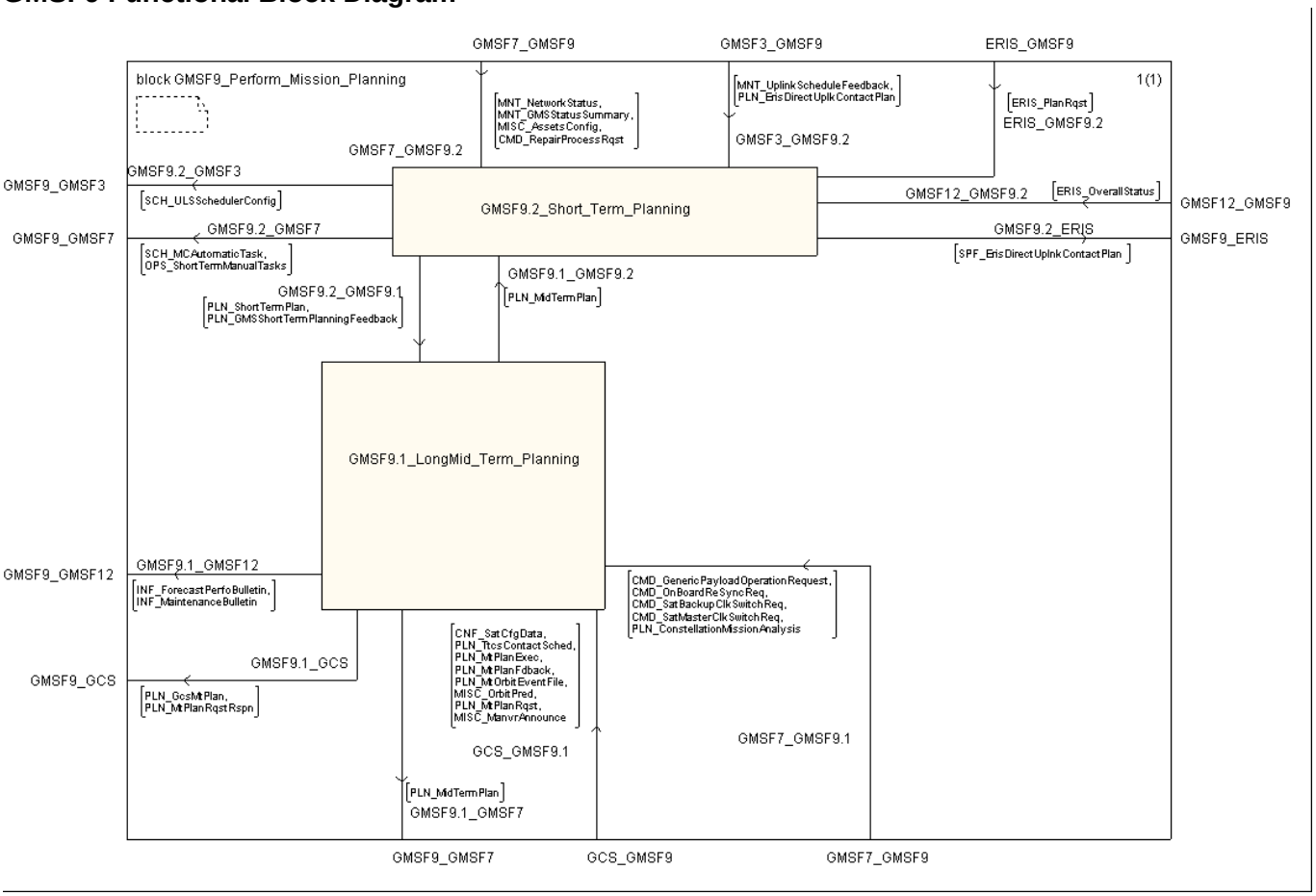


Figure 154: GMSF9 SDL Functional Block Diagram

5.3.9.1 GMSF9.1 Long/Mid Term Planning

GMSF9.1 General Description

The Long/Mid-Term Planning Function is in charge of the planning of high level activities for the entire system, covering events such as maintenance activities, specific manoeuvres, onboard software maintenance, satellites replenishment or phasing out, new launch, etc. The plan shall be refined and optimized on operator requests to resolve resource conflict and ensure that the Galileo Services performances are met. The resulting plan shall state dates, time and duration for activities in terms of windows where it is expected they will be performed.

Handling of GCS Mid-Term Plan Requests and Feedbacks

The Long/Mid-Term Planning Function is able to process the following GCS information to generate the long/mid-term plans:

- Planning requests coming from the GCS (e.g. satellite replenishment, on-board software management...). The planning request shall encompass maintenance and testing activities.
- Feedback from the GCS short-term planning functions (including TTC contact schedule from GCS).
- Feedback on execution of the plan from GCS.
- Flight dynamic and satellite configuration parameters coming from GCS.

Handling of Satellite Clocks Commands in the Long/Mid-Term Plan

The Long/Mid-Term Planning Function enables to plan the following tasks relative to satellites clock control:

- Satellite clock synchronization
- Satellite clocks switching

Handling of CS Codes in the Long/Mid-Term Plan

The Long/Mid-Term Planning Function enables to plan the tasks relative to satellite CS key management.

GMSF9.1 Outputs

The GMS shall produce at least the following outputs:

- The time ordered activities to be performed. This information is used as input by the GMS and GCS short-term planning functions.
- Planning reports with the result of the planned activities
- Bulletins to announce the unavailability of resources and/or periods when the services performance is not met. Such bulletins shall be delivered to the external service centres more than 48hrs in advance of the planned unavailability periods.

GMSF9.1 Design Constraint

The following constraints are considered in the Long/Mid-Term Function:

- The Long-Term Plan is considered to cover at least a 6 months period, while the Mid-Term Plan is considered to be a sliding window of 1-4 weeks in which optimization is performed.
- A planning configuration database is maintained including at least all the possible tasks as well as the involved resources for each task.
- Except in contingency cases the generation of plan and schedules as well as their verification is automated.

GMSF9.1 Functional Block Diagram

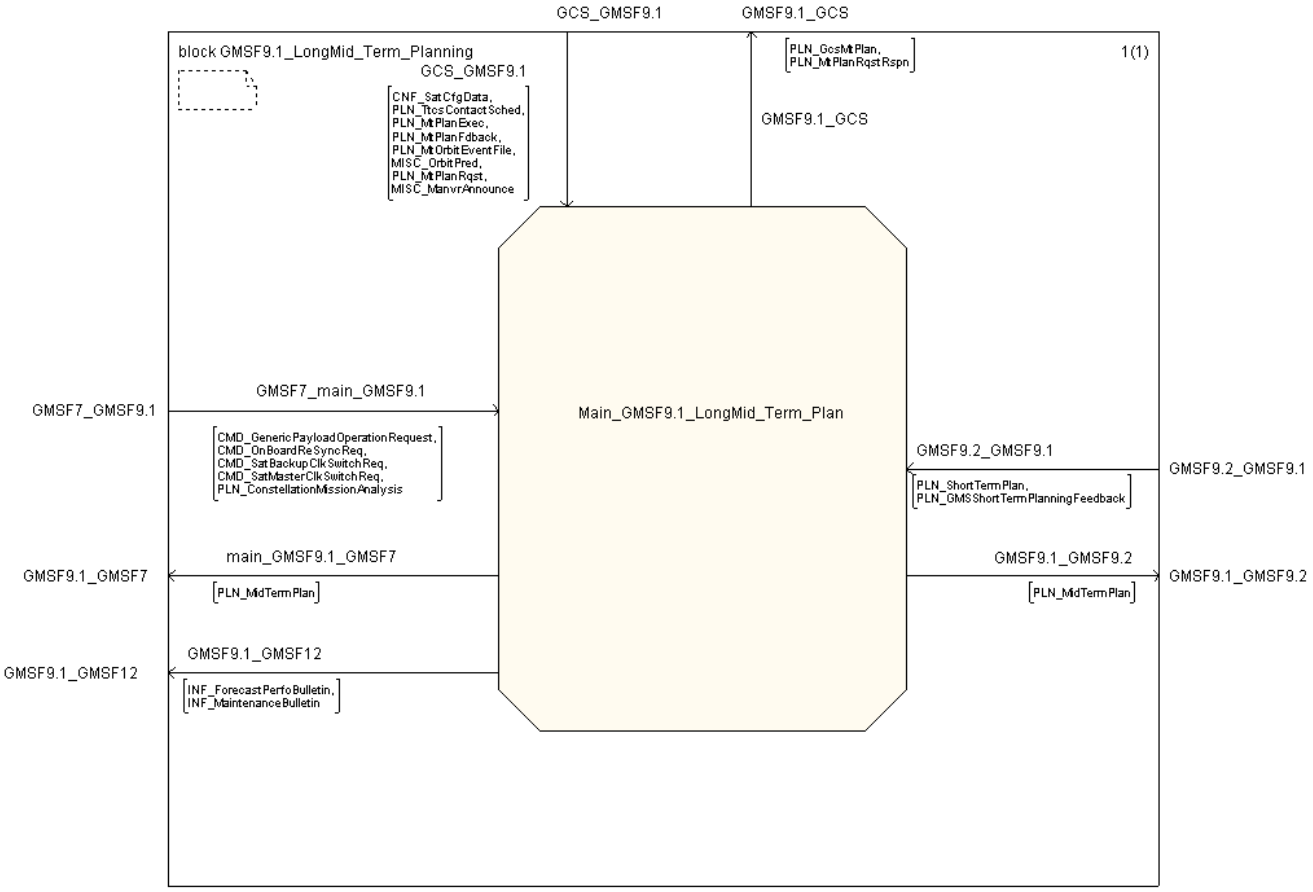


Figure 155: GMSF9.1 SDL Functional Block Diagram

5.3.9.2 GMSF9.2 Short Term Planning

GMSF9.2 General Description
The GMS Short Term Planning function derives a detailed and conflict-free planning including executable schedule for maintenance and test tasks, within the framework provided by the Mid-term Plan for the operations to be performed inside the GMS.

This function is also in charge of configuring the uplink scheduling function (GMSF3.5).

This function takes, as a minimum, the following inputs into account:

- Mid-term plan constraints,
- Current status of GMS ground-assets,
- Configuration parameters of each involved element (masking angles, etc) and provided by operators.

GMSF9.2 Outputs:

GMSF9.2 Planning Outputs
The GMS Short Term Planning function produces at least the following outputs:

- The GMS Short-Term Plan and feedback to the GMS Long/Mid-Term Plan function including planning reports with the result of the planning execution,

GMSF9.2 Schedule Outputs
The GMS Short Term Planning function produces at least the following schedule outputs:

- Manual Operations: Timelines & Operators Tasks,
- Automatic Tasks: Executable Schedule for the Overall, Central and Element M&C functions,

GMSF9.2 Configuration of Uplink Scheduler Outputs
The GMS Short Term Planning function produces at least the following outputs relative to the uplink schedule:

- Configuration parameters for the uplink schedule function,

GMSF9.2 Design Constraints / Performances
The GMS short-term planning function must take into account the following constraints:

- automatic execution of all mission data determination and dissemination functions.

GMSF9.2 Functional Block Diagram

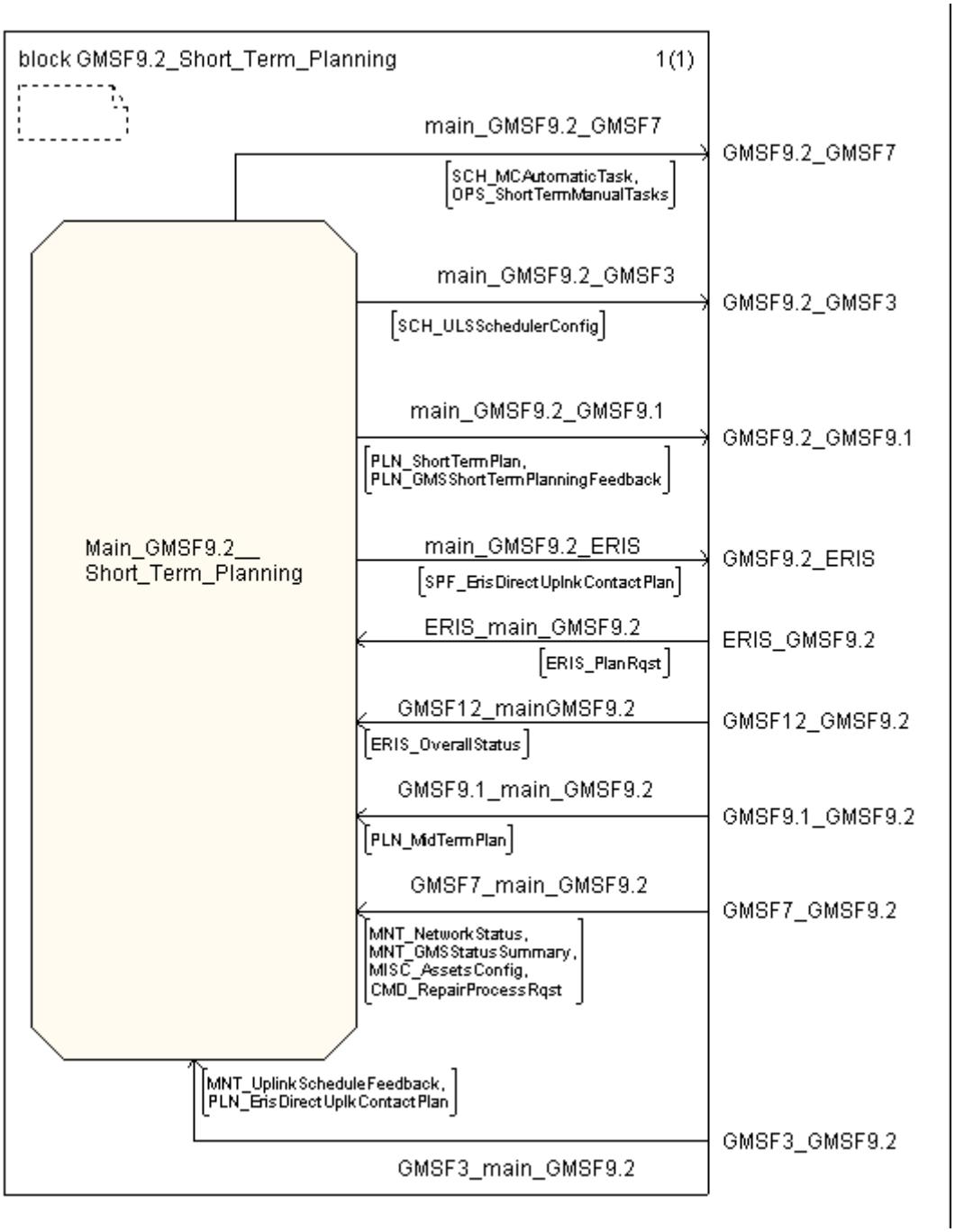


Figure 156: GMSF9.2 SDL Functional Block Diagram

5.3.10 GMSF10 Provide Training

GMSF10 Description
This function is in charge to support operator's training in GMS.

GMSF10 Functional Block Diagram

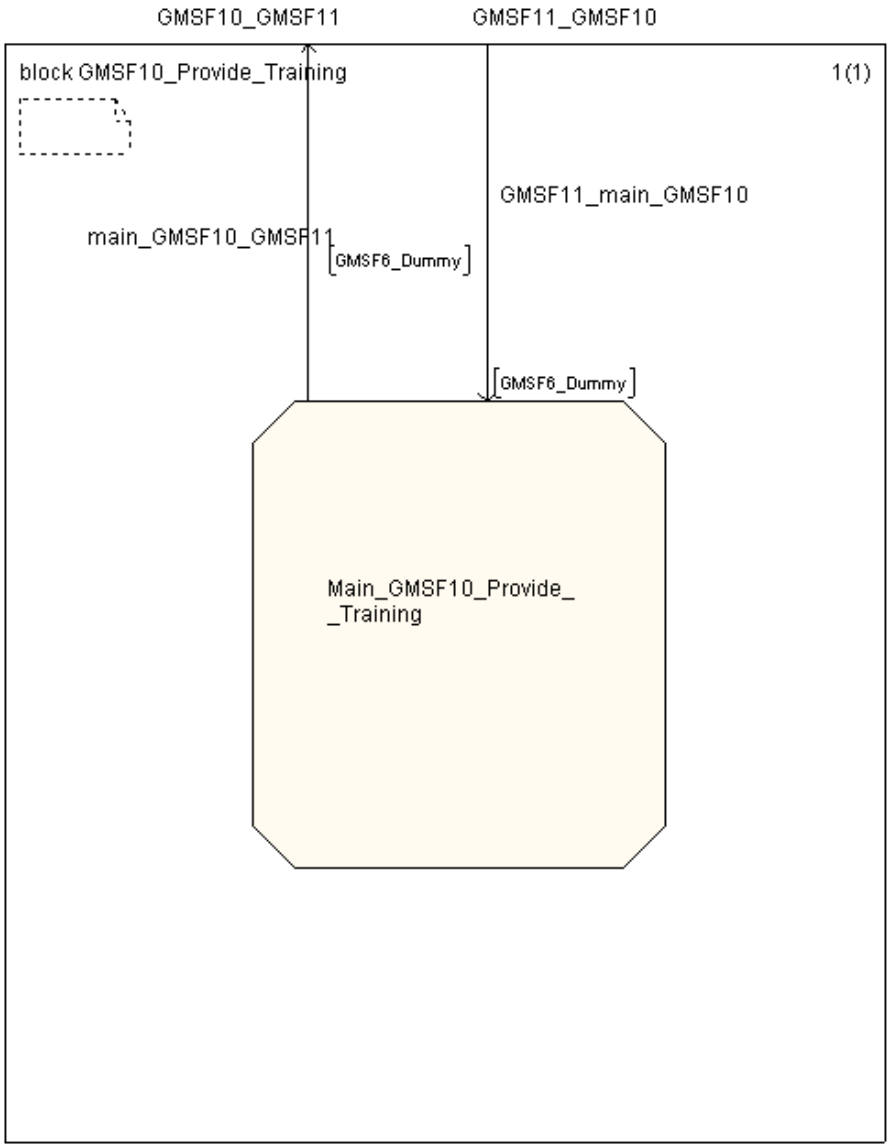


Figure 157: GMSF10 SDL Functional Block Diagram

5.3.11 GMSF11 Perform Simulation and Test

GMSF11 Description
This function is in charge to support all the simulation and test needed

GMSF11 Functional Block Diagram

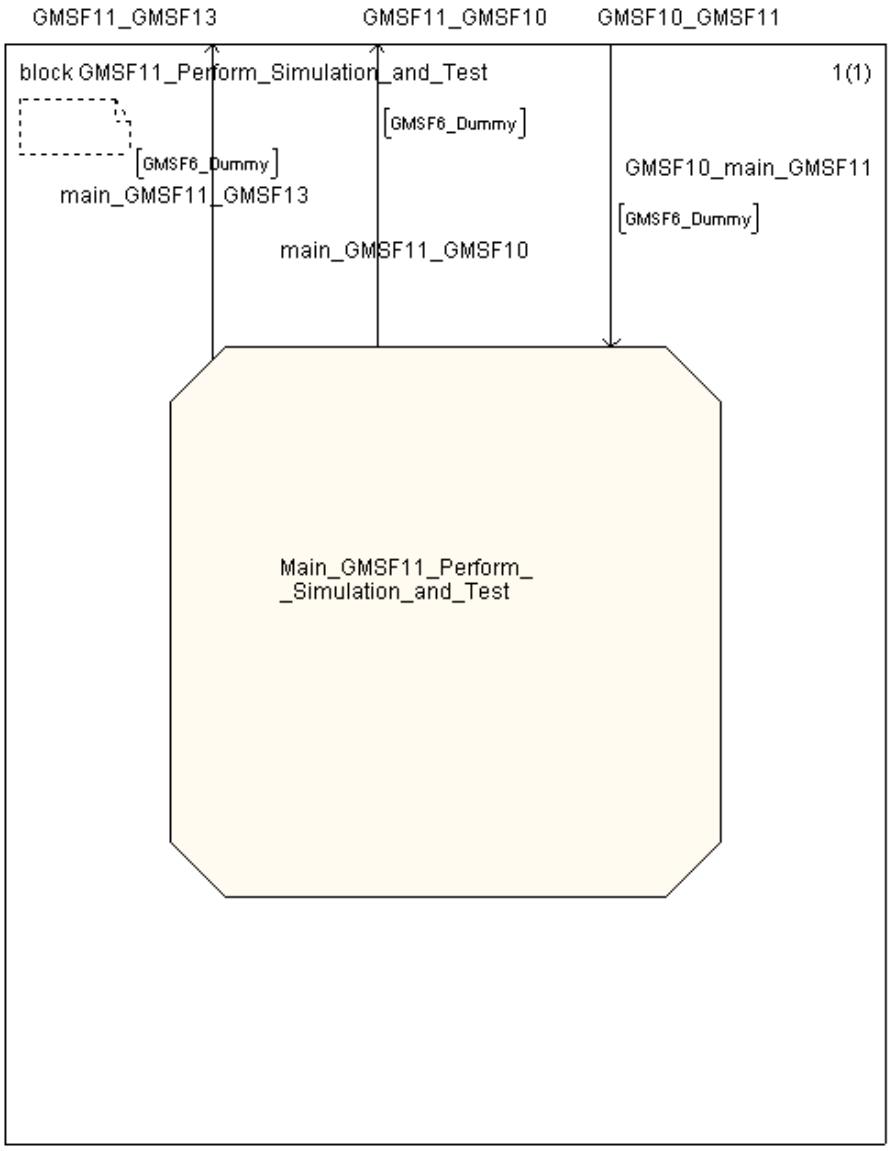


Figure 158: GMSF11 SDL Functional Block Diagram

5.3.12 GMSF12 Collect and Route External Entities Data

GMSF12 Description

This function is in charge to collect data (products, reports, keys management) from the GMS functions where they are produced and send these data on an agreed basis to the relevant external entity (SC, RSLP, TSP, CSP, GSMC, GRSP, GGTO and GALSEE). Inside this function there is no process or modification of the information provided, only for the Galileo observables and navigation message are converted into RINEX version 3 files in order to be sent after filtering and compression to relevant external entities.

On the other hand this function also:

- Handle requests from the external entities of data (products, reports, keys management) from GMS functions.

Receive data (products, reports, keys management) from external entities in order to be distributed to GMS functions.

GMSF12 Functional Block Diagram

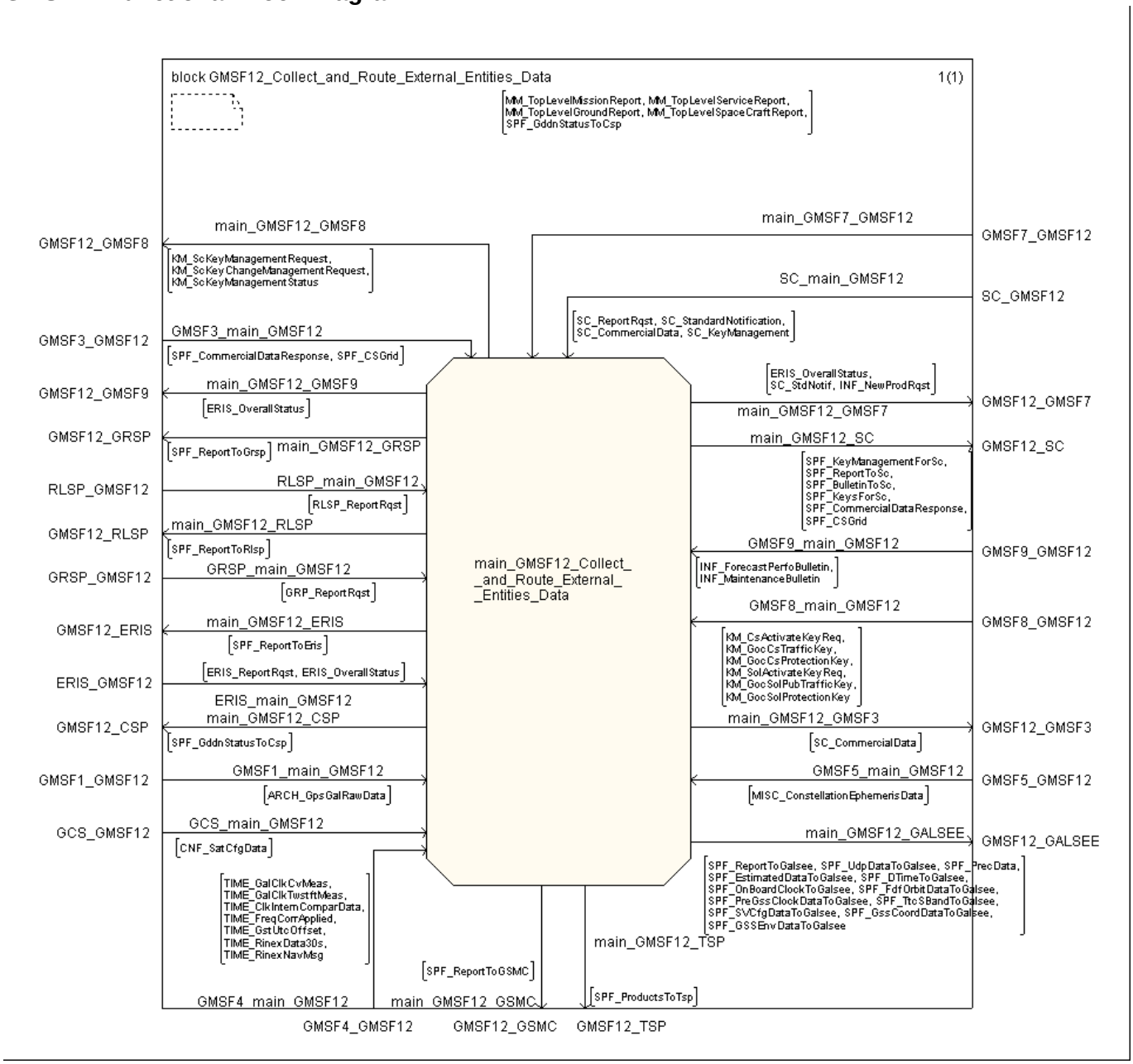


Figure 159: GMSF12 SDL Functional Block Diagram

5.3.13 GMSF13 Manage Database

GMSF13 Description

This function is in charge to manage a common database management system for all GMS functions requiring data-base capability.
This function is in charge to manage the database needed to configure the GMS elements, operational procedures, anomaly events, and configuration control and maintenance database.

GMSF13 Outputs

This function provides exchanges configuration information between each GCC centre.

GMSF13 Design Constraints

Configuration databases have to be kept aligned between each GCC allowing smooth operational hand-over.
The installation and transfer of new and database files should be performed in a controlled manner and should not affect the system real-time operation.
The GMS has to use a common database management system for all GMS functions requiring database capability.
The GMS design has to minimise the number of databases.

GMSF13 Functional Block Diagram

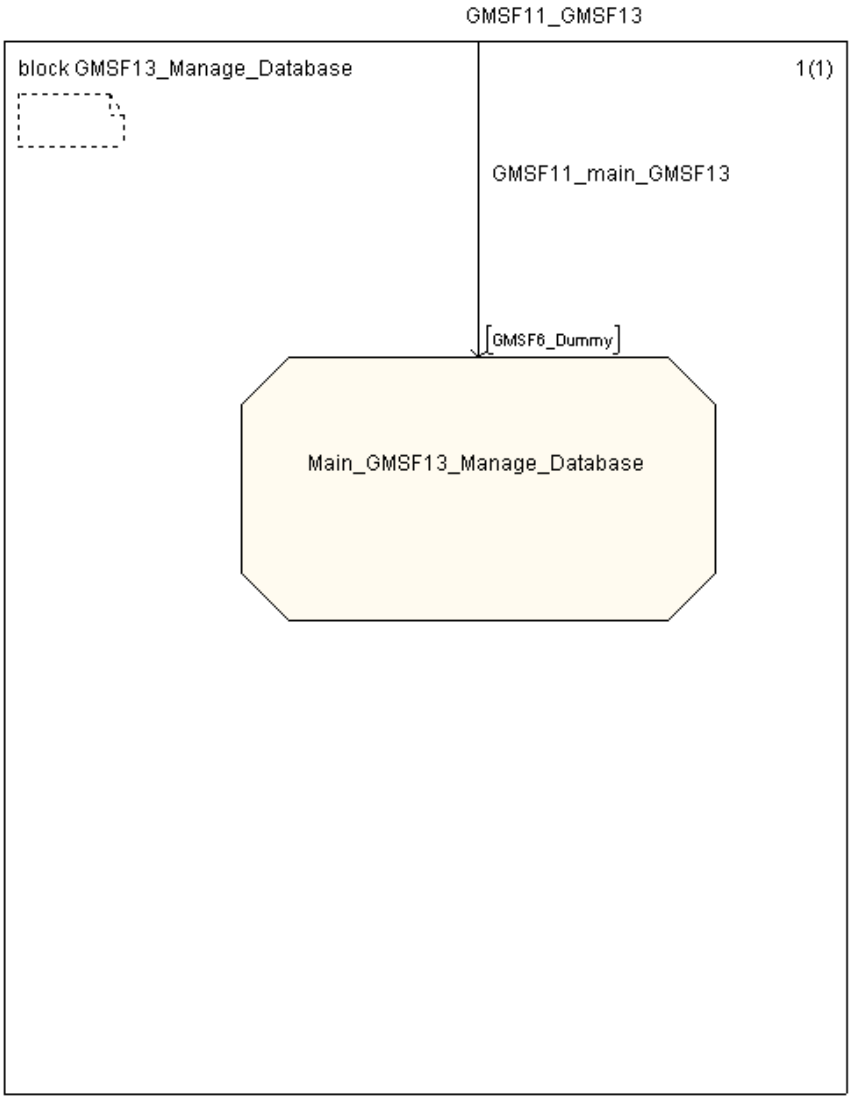


Figure 160: GMSF13 SDL Functional Block Diagram

5.4 SS Functions

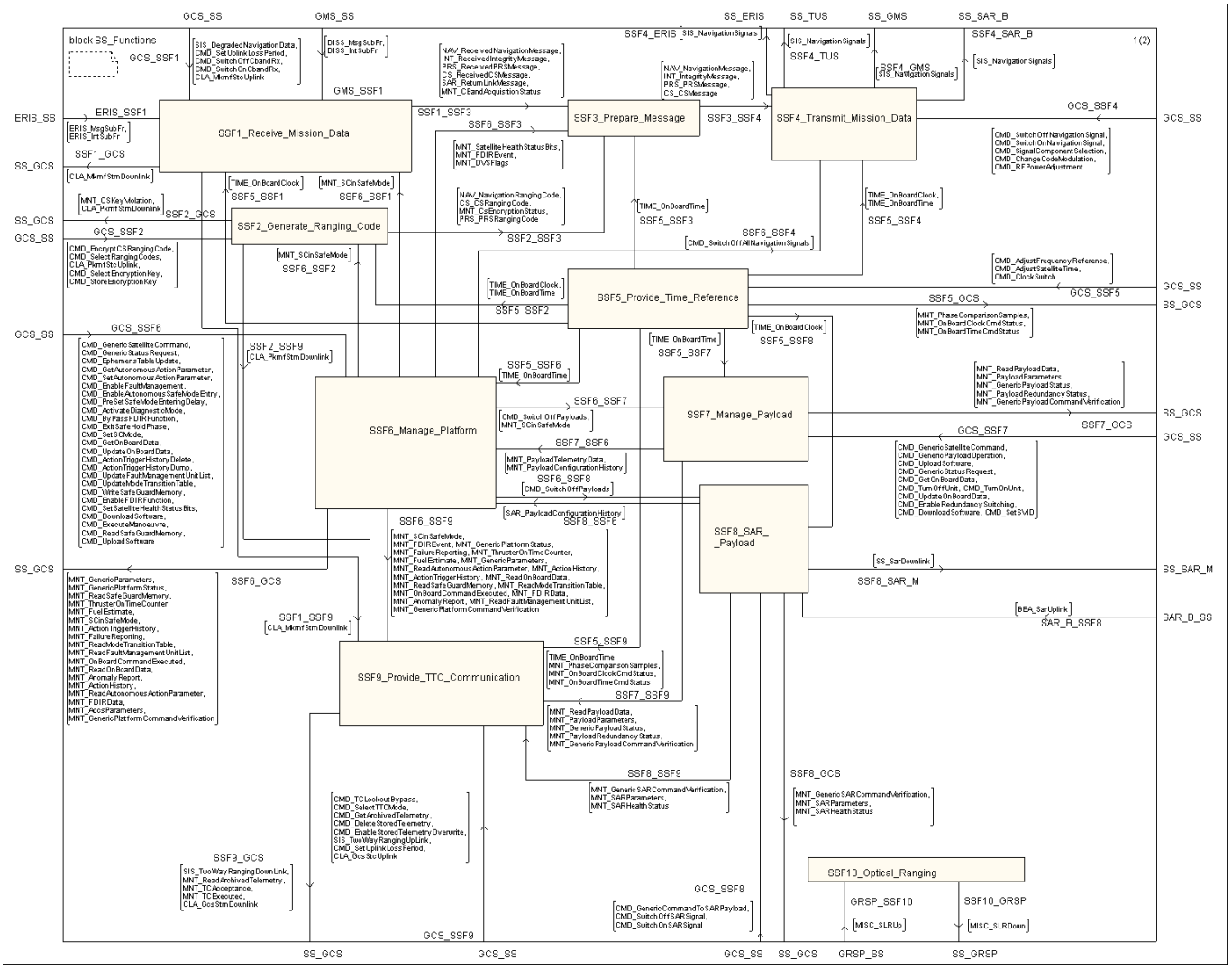


Figure 161: SSgt SDL Functional Block Diagram

5.4.1 SSF1 Receive Mission Data

SSF1 Description

This function is in charge for receiving from C-band all the Mission data for OS, PRS, EDBS, SAR data and Commercial data that must be transmitted by the satellite.

Once demodulated, the C-band data is authenticated and then:

- Stored on board when it is not constrained to Time-to-Alert (TTA) or Real time dissemination (positioning service for OS and PRS and PRS security data).
- Or immediately processed for its immediate dissemination (EDBS, SAR RL and Commercial data)

It also manages the following points:

- Authentication keys management.
- C Band acquisition status for all 6 C-band channels.
- Short/long C band uplink loss flag.
- Degraded Navigation Data.

SSF1 Functional Block Diagram

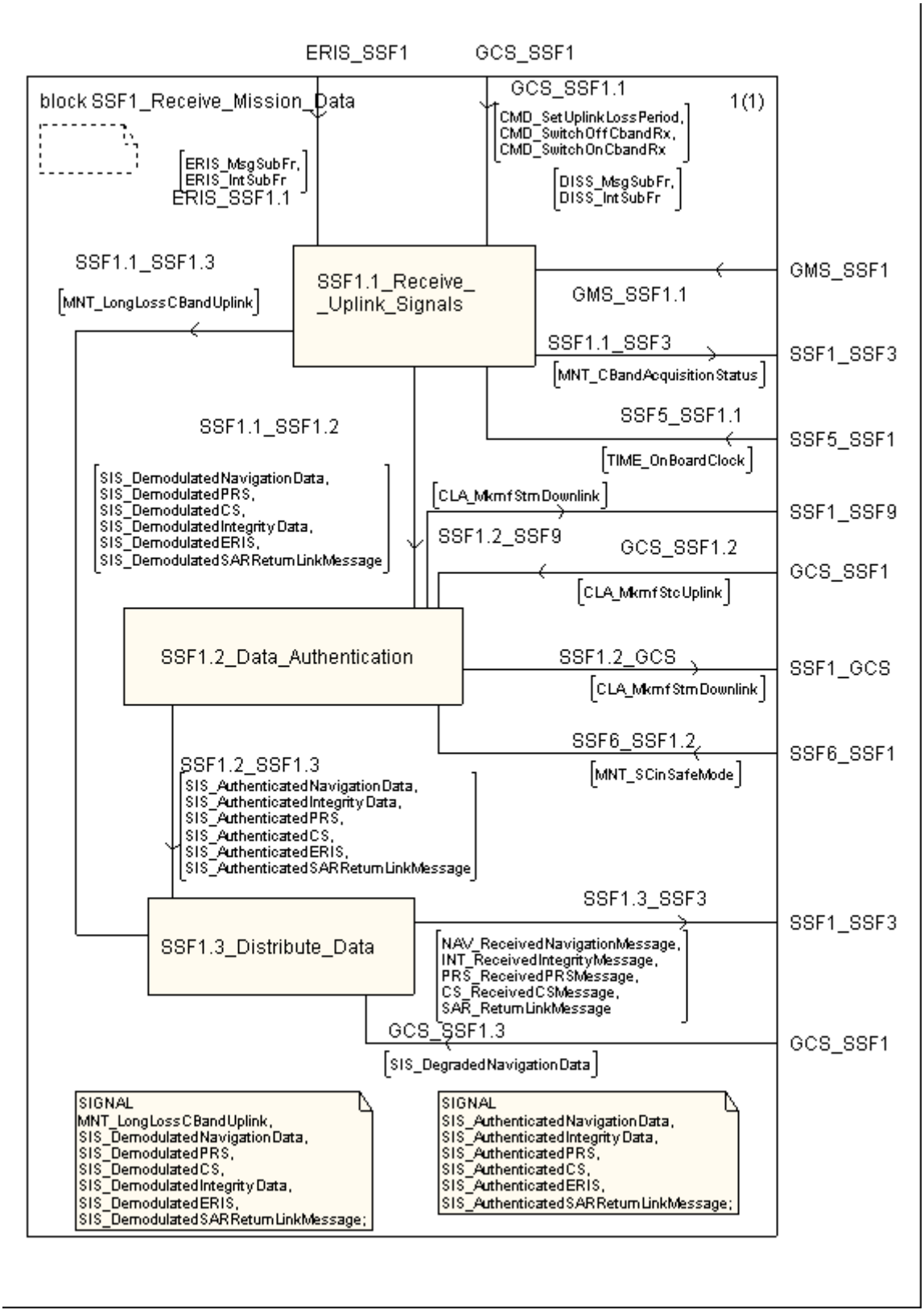


Figure 162: SSF1 SDL Functional Block Diagram

5.4.1.1 SSF1.1 Receive Uplink Signals

SSF1.1 General Description
This function is in charge to receive and demodulate up to 6 simultaneous signals:

- 1 direct signals from GMS
- Up to 5 direct ERIS signals each one containing ERIS data from one specific region.

All 6 signals are modulated on CDMA.

The GMS signals contains:

- Navigation Data for the OS (4 batches) and PRS (up to 5 batches) services.
- PRS Data (Integrity tables and alerts and security data)
- CS Data
- SAR Return Link Message
- indirect ERIS data

The integrity for both SoL and PRS services and ERIS data was considered time critical and designed to be disseminated on real time. They are contained in specific sub-frames.
The other data (navigation, CS, SAR…) is included in a general message sub-frame.
The signals must be acquired within a specified time to meet ULS access plan.
This function was critical for SoL and therefore originally designed to meet output feared event requirements.
This function is subject to timing requirements to meet end-to-end dissemination requirements (TTA, SAR and CS).
The outputs are authenticated data packets and C-Band reception status TM.
The outputs are data packets to be authenticated by SSF1.2 and C-Band reception status TM.

SSF1.1 Outputs
The possible outputs per second are:

- Authenticated data packets (when correctly received) as follows:
  - GMS Integrity sub-frame
  - GMS Message sub-frame
  - 5 ERIS Integrity sub-frames (one per region)
  - 5 ERIS Message sub-frames (one per region)
- C-Band reception status TM.

SSF1.1 Design Constraints/Performances
This function is subject to timing requirements to meet end-to-end dissemination requirements (TTA, SAR and CS).
This function is critical for SoL and therefore must meet output feared event requirements.
The signals must be acquired within a specified time to meet ULS access plan.

SSF1.1 Functional Block Diagram

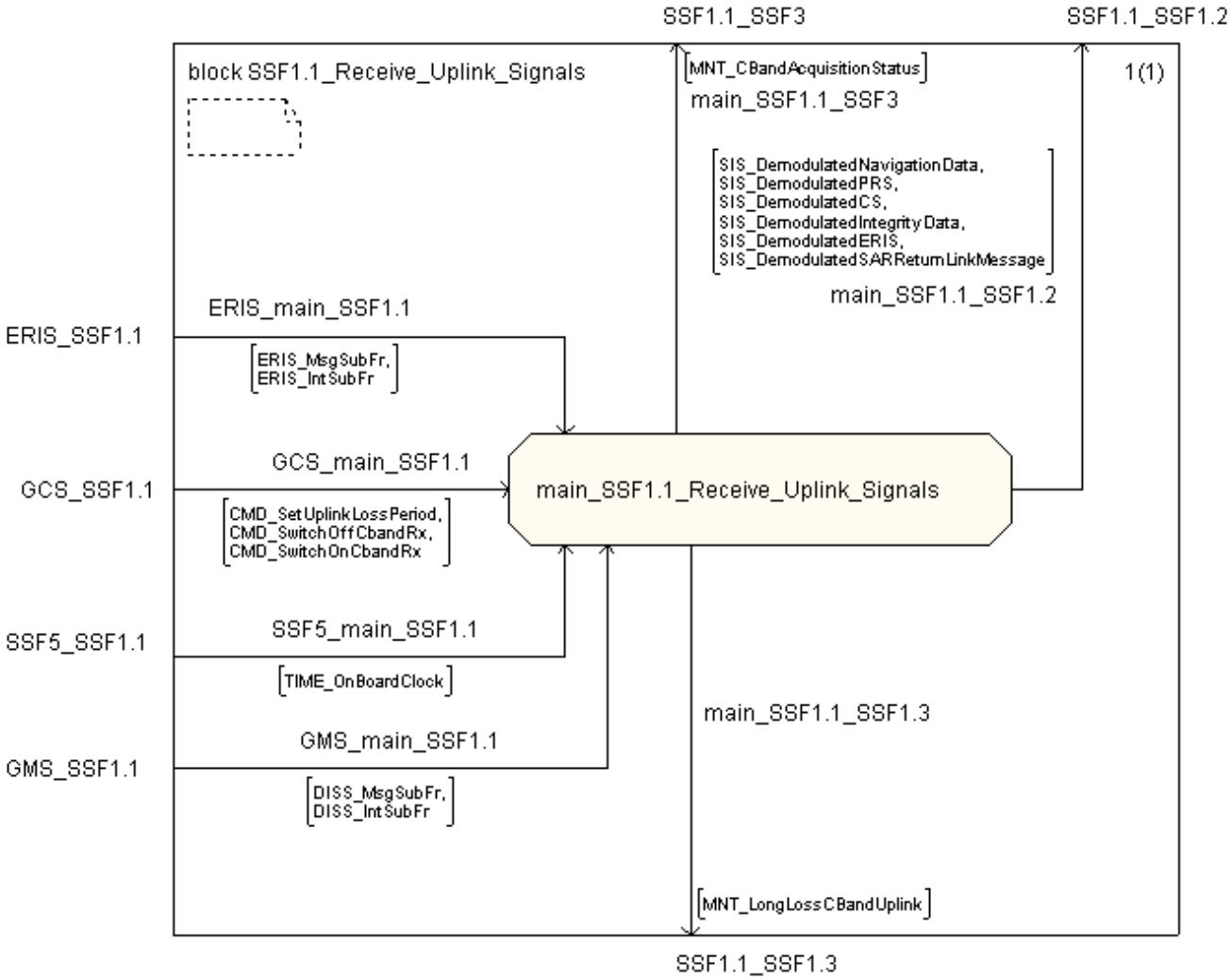


Figure 163: SSF1.1 SDL Functional Block Diagram

5.4.1.2 SSF1.2 Data Authentication

SSF1.2 General Description
This function is in charge of managing all the protection aspects for the mission data at Satellite level. This includes:
- Protection of the C-Band Uplink including authorization check for the Galileo C-Band Message and Integrity uplink subframes and ERIS direct uplinks (authentication check, anti-replay,…)
- Managing the ERIS Key reception from ground for ERIS direct access to the satellites (ERIS direct uplink)

This function corresponds to dedicated Classified Security Function SEC-SSF4. Refer to SYSBSLDEF Part I Classified Annex [SYS-RD.07] for details.

Note: this function is linked to GMSF8.3 related to security protection for mission data at GMS level

SSF1.2 Outputs
The possible outputs per second are:

- Authenticated data packets (when correctly received) as follows:
  - GMS Integrity sub-frame
  - GMS Message sub-frame
  - 5 ERIS Integrity sub-frames (one per region)
- 5 ERIS Message sub-frames (one per region)

Note: The input and output of this function are identical (have the same contents and structure) to allow the satellite to work in clear mode (i.e. without authentication validation). In that scenario the data is passed transparently from SSF1.1 to SSF1.3 without any processing on SSF1.2.

See SEC-SSF4 for details.

SSF1.2 Design Constraints/Performances
This function is subject to the following constraints:

- timing requirements to meet end-to-end dissemination requirements (TTA, SAR and CS).
- Feared event requirements as this function is critical for SoL
- In case the authentication fails a specific counter is increased. This counter is accessible on ground through TM. If the authentication is validated the data is passed to SSF1.3 for its processing.

SSF1.2 Functional Block Diagram

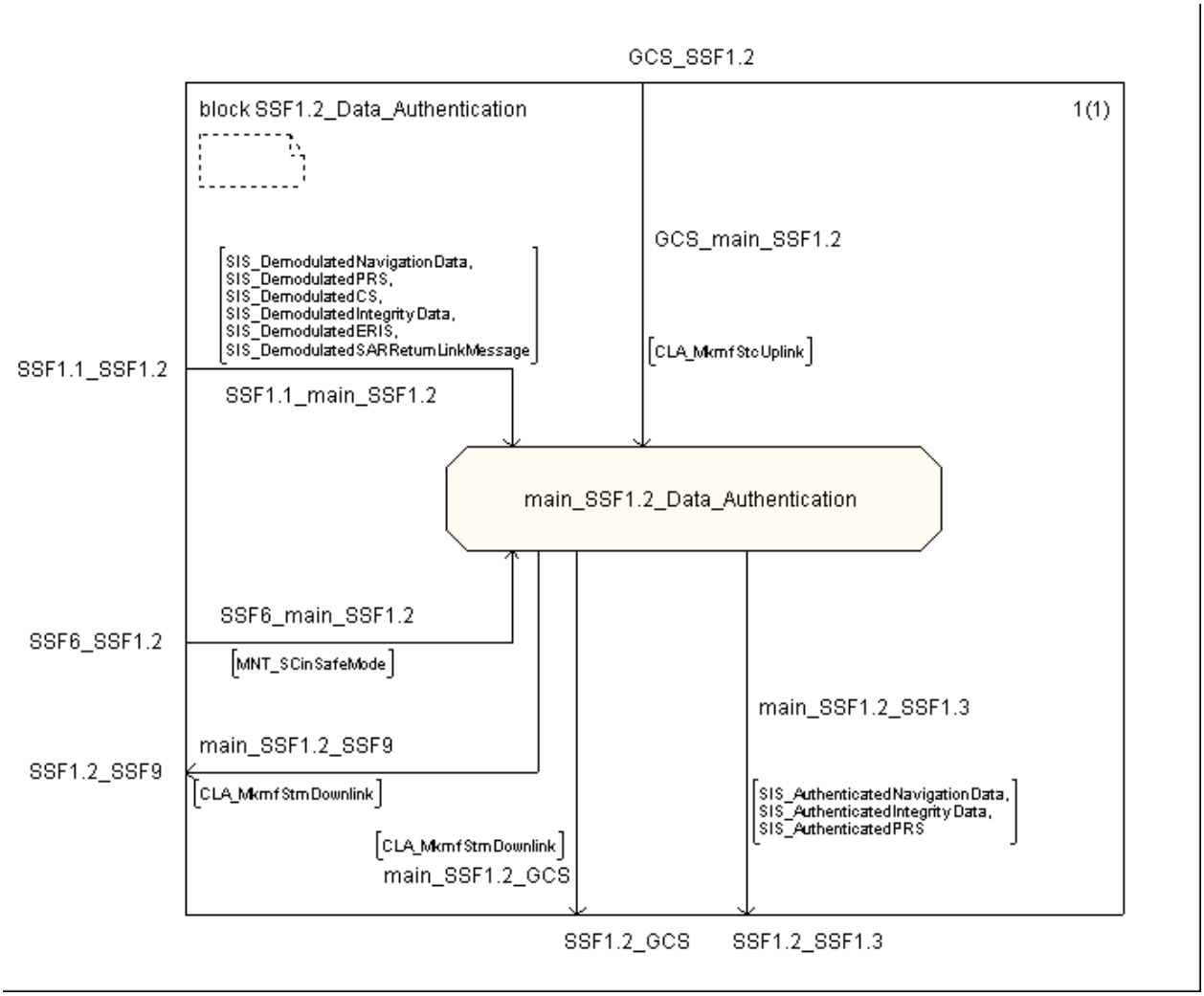


Figure 164: SSF1.2 SDL Functional Block Diagram

5.4.1.3 SSF1.3 Distribute Data

SSF1.3 General Description

This function is responsible for receiving all uplinked data from both C-band and S-band and generate every second the message (F/NAV, I/NAV, G/NAV and C/NAV) that must be transmitted on each navigation frequency band (E5a, E5b, E1A, E1B, E6A, E6B):

- 4 OS batches for F/NAV received from C-band
- 1 batch F/NAV almanacs received from C-band
- 4 OS batches I/NAV received from C-band
- 1 batch I/NAV almanacs received from C-band
- 4 or 5 batches for G/NAV received from C-band
- 2 batches G/NAV almanacs received from C-band
- SoL integrity data: Integrity tables, Integrity alerts and SoL Region Status (designed when SoL and integrity was still in)
- PRS integrity data: Integrity tables, Integrity alerts and PRS Region Status (designed when SoL and integrity was still in)
- 8 OS batches for F/NAV received from S-band
- 1 batch F/NAV almanacs received from S-band
- 8 OS batches I/NAV received from S-band
- 1 batch I/NAV almanacs received from S-band
- 9 batches for G/NAV received from S-band
- 2 batches G/NAV almanacs received from S-band
- 2 batch of security data (G/NAV) from C-band and S-band
- Direct and indirect ERIS data
- SAR return link data
- Commercial Service data

All the above data is used according to the SISICD considering:

- The use of OS and PRS data uplinked via S-band is transmitted in contingency scenarios when the C-band data refreshment fails.
- Interruption of I/NAV and G/NAV messages nominal transmission by Integrity Alerts (designed when SoL and integrity was still in)
- Transmission of the proper SoL and PRS Region Status depending on the C-band link loss.
- Generation of dummy messages in case of lack of valid data for: Integrity tables, Commercial data, ERIS data, SAR return link data.

SSF1.3 Outputs

The output selection is based on:

- The use of OS and PRS data uplinked via C-band or S-band depending on the contingency scenarios when the C-band data refreshment fails.
- Interruption of I/NAV and G/NAV messages nominal transmission by Integrity Alerts (designed when SoL and integrity was still in)
- Transmission of the proper SoL and PRS Region Status depending on the C-band link loss.

Generation of dummy messages in case of lack of valid data for: Integrity tables, Commercial data, ERIS data, SAR return link data.

SSF1.3 Design Constraints/Performances
This function is subject to timing requirements to meet end-to-end dissemination requirements (TTA, SAR and CS).
This function was critical for SoL and therefore originally designed to meet output feared event requirements.

SSF1.3 Functional Block Diagram

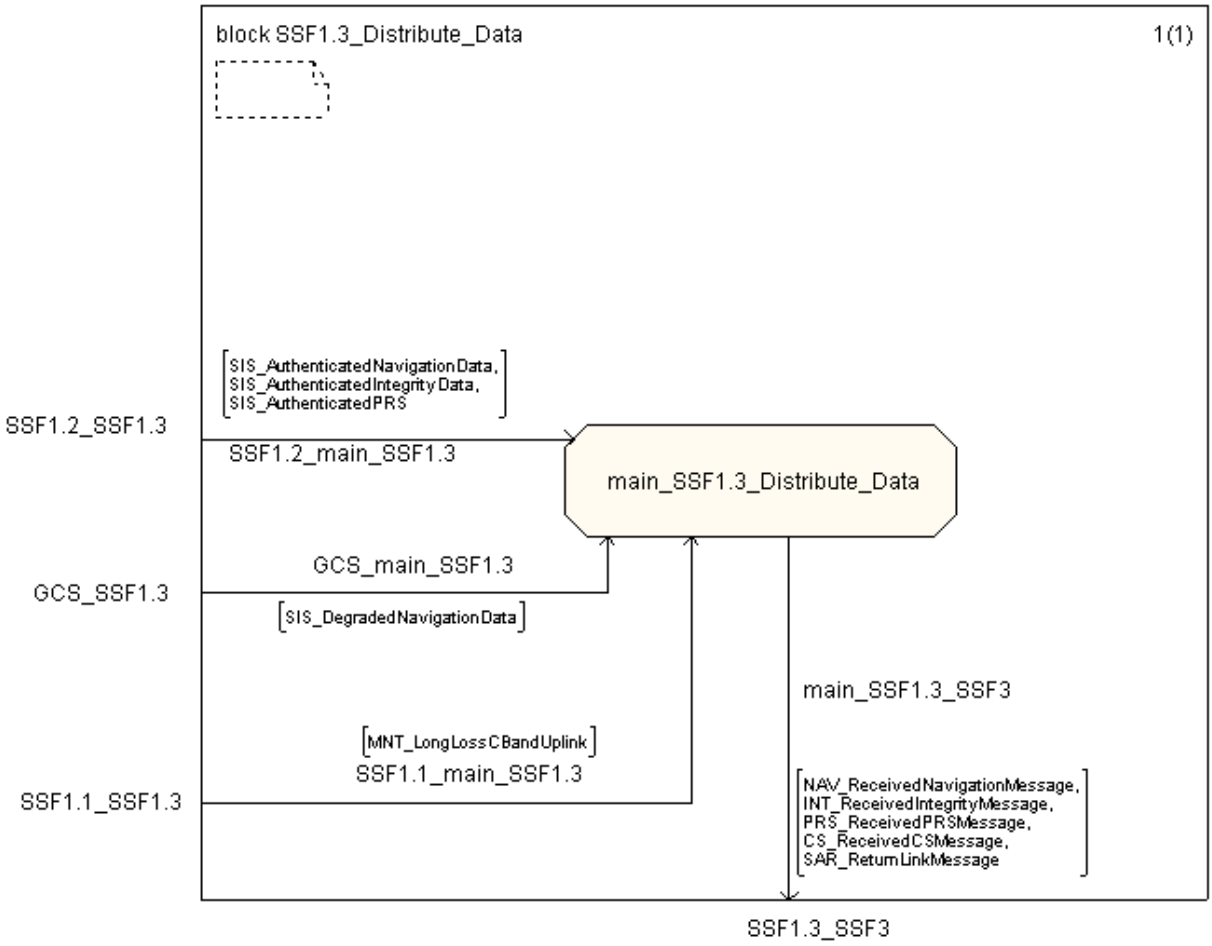


Figure 165: SSF1.3 SDL Functional Block Diagram

5.4.2 SSF2 Generate Ranging Code

SSF2 Description
This function is in charge of generating the ranging codes for all the navigation signals (including pilot signals): E5a, E5b, E6A, E6B, E6C, E1A, E1B, E1C.

It manages the following points:

- Navigation Ranging Code selection
- PRS and CS Ranging Code encryption key management

- SSF2 Functional Block Diagram

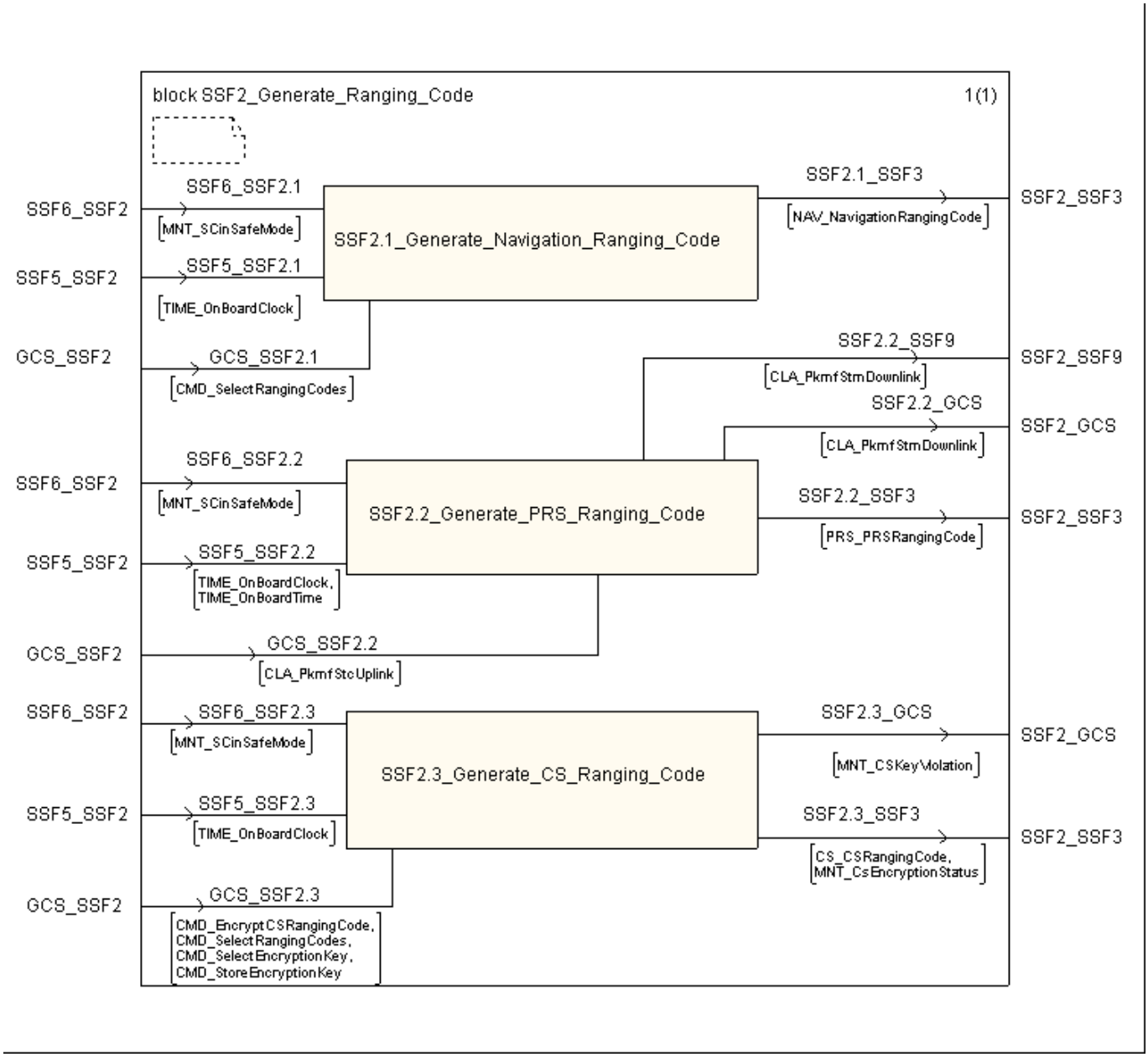


Figure 166: SSF2 SDL Functional Block Diagram

5.4.2.1 SSF2.1 Generate Navigation Ranging Code

SSF2.1 General Description
This function provides navigation ranging codes for the signals transmitted on E5a-I, E5a-Q, E5b-I, E5b-Q, E1B and E1C.

It also provides the ways to select the ranging codes used.

SSF2.1 Outputs
The outputs are the chip codes to be modulated on each frequency carrier carrying either pilot or navigation data streams as specified in the SISICD document.

SSF2.1 Design Constraints / Performances
This function is critical for SoL and therefore must meet output feared event requirements.

SSF2.1 Functional Block Diagram

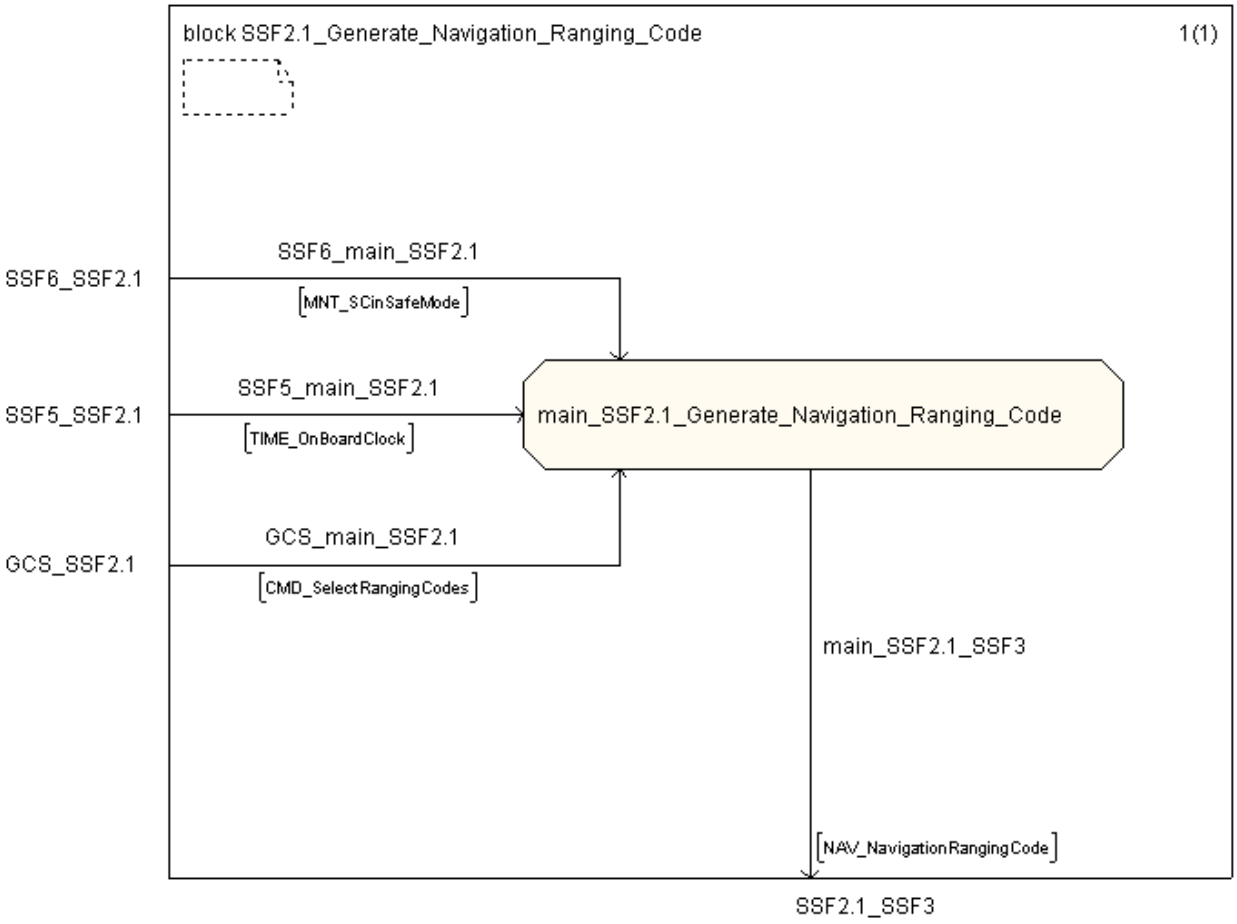


Figure 167: SSF2.1 SDL Functional Block Diagram

5.4.2.2 SSF2.2 Generate PRS Ranging Code

SSF2.2 General Description

This function is in charge of managing all the protection aspects for the PRS data at Satellite level. This includes:

- Generation of PRS encrypted ranging codes for the signals transmitted on the frequency bands E6A and E1A.
- Management of the PRS encryption key via STC/STM

This function corresponds to dedicated Classified Security Function SEC-SSF1.1. Refer to SYSBSLDEF Part I Classified Annex [SYS-RD.07] for details.

Note: this function is linked to GMSF8.1 related to security protection for PRS data at GMS level.

SSF2.2 Outputs

The output for this function is the ranging code to modulate the G/NAV message that must be transmitted by the satellite.

SSF2.2 Functional Block Diagram

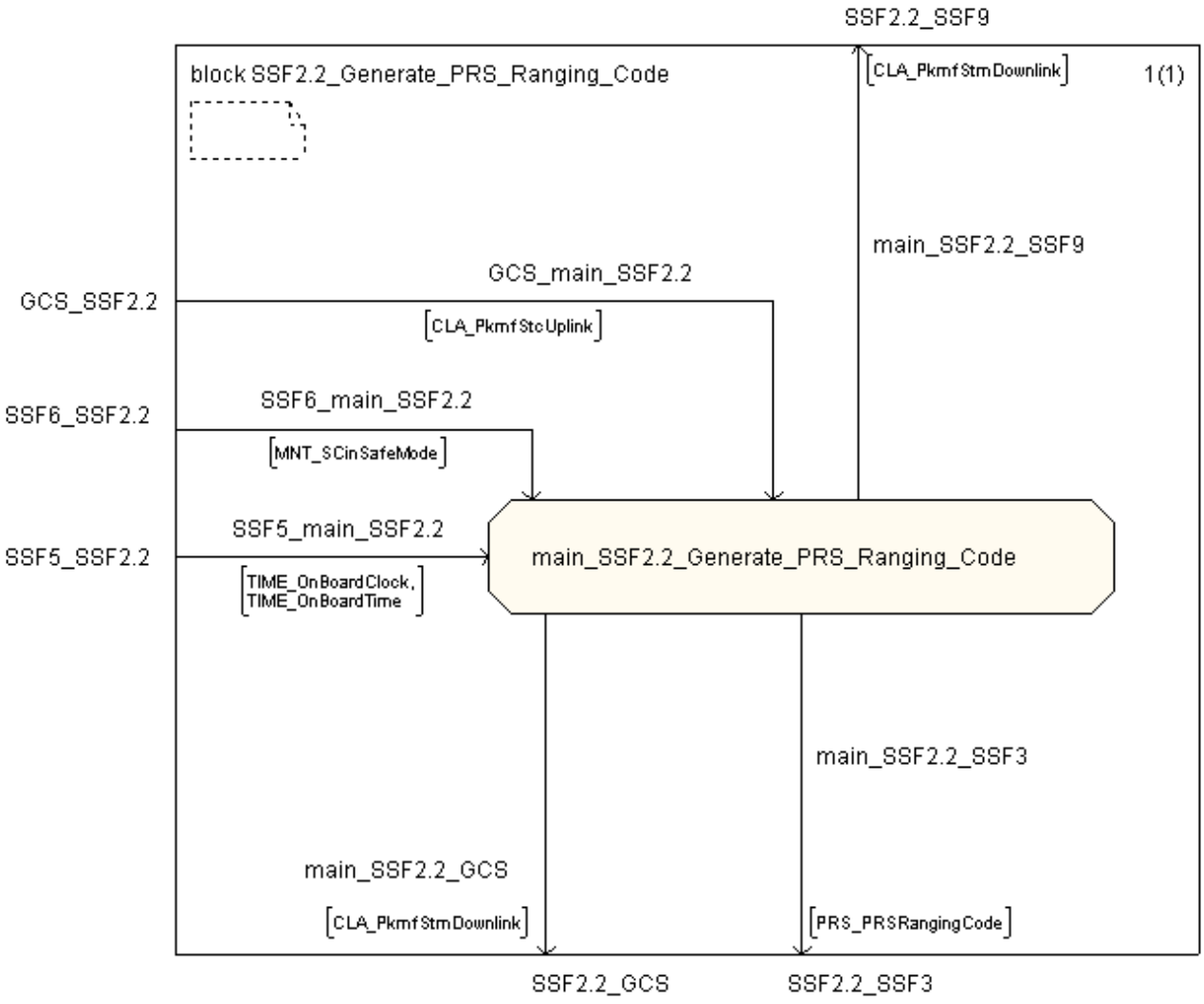


Figure 168: SSF2.2 SDL Functional Block Diagram

5.4.2.3 SSF2.3 Generate CS Ranging Code

SSF2.3 General Description

This function is responsible for the generation of CS ranging codes for the signals transmitted on the frequency bands E6B and E6C.

The CS ranging codes can be either encrypted or unencrypted by telecommand selection.

It also provides functionalities to manage the CS encryption keys.

SSF2.3 Outputs

The output of this function is the ranging code (either encrypted or not) to modulate the C/NAV message.

SSF2.3 Functional Block Diagram

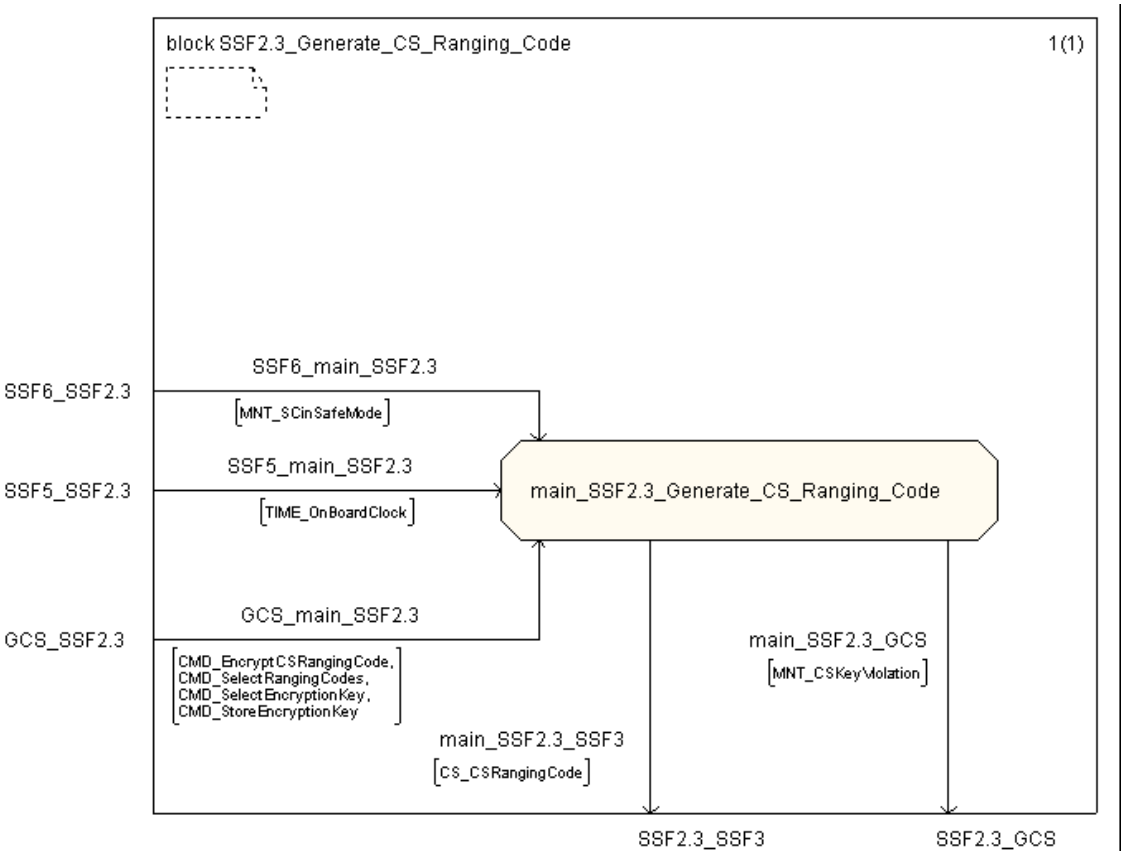


Figure 169: SSF2.3 SDL Functional Block Diagram

5.4.3 SSF3 Prepare Message

SSF3 Description

This function is responsible for composing the navigation message that will be transmitted and modulating it on the appropriate ranging code. The output will be modulated onto the specific carrier frequencies.

The messages (F/NAV, I/NAV, G/NAV and C/NAV) are composed by the data received from ground but also data generated on board by this functionality.

SSF3 Functional Block Diagram

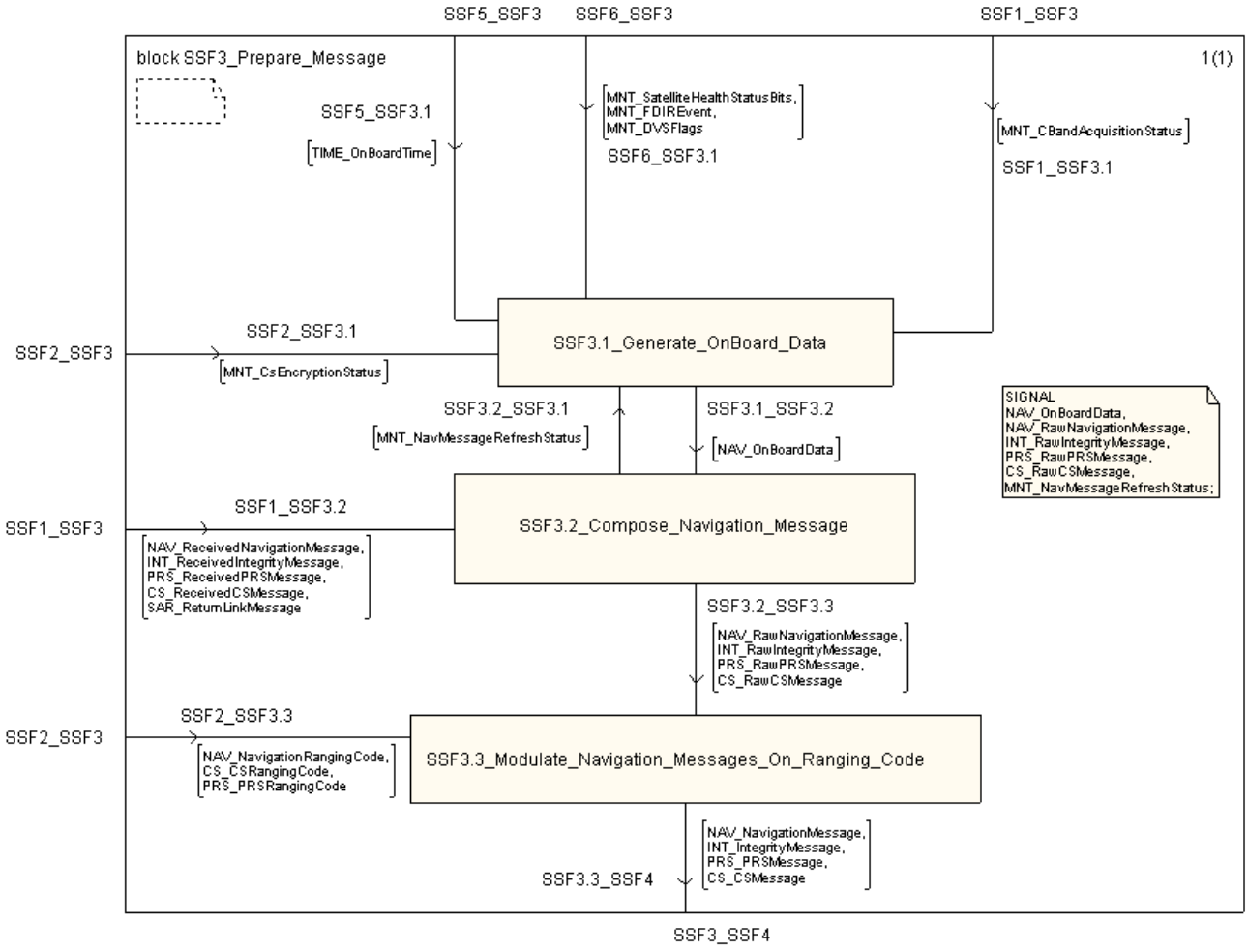


Figure 170: SSF3 SDL Functional Block Diagram

5.4.3.1 SSF3.1 Generate OnBoard Data

SSF3.1 General Description

This function collect all on board data which will be included in the broadcasted F/NAV, I/NAV and G/NAV messages:

- Health Status Alarm Bit

- Data validity status flag (DVS). Specific for each service: OS, SoL and PRS. It is also defined for the Commercial Service (CS) but it is not currently foreseen its transmission within the C/NAV messages.
- C/NAV encryption status parameter
- Housekeeping TM summary
- FDIR events
- C-Band link status for all 6 C-band channels (GMS and 5 ERIS)
- On-Board Time parameters (ToW and WN)

All those parameters are described in the SISICD.

SSF3.1 Outputs
The outputs for this function are:

- Health Status Alarm Bit. It is transmitted within the G/NAV messages.
- Data validity status flag (DVS)
- C/NAV encryption status parameter. It is transmitted within the I/NAV message.
- Housekeeping TM summary. It is transmitted within the G/NAV messages.
- C-Band link status for all 6 C-band channels (GMS and 5 ERIS). It is transmitted within the G/NAV message.
- On-Board Time parameters (ToW and WN) are transmitted within F/NAV and I/NAV messages.

SSF3.1 Design Constraints / Performances
This function, when generating the Data Validity Status flag is critical for SoL and therefore must meet output feared event requirements.
The generation of the On-board time parameters are also critical for navigation and SoL operations.
The C/NAV encryption status parameter is used as indication to the Commercial Service users of the encryption signal status. However it is not considered critical since the Commercial Service users would receive this information directly from the Commercial Service provider.
The rest of data generated is not used for any critical operation but for information only to the on ground system operators.

SSF3.1 Functional Block Diagram

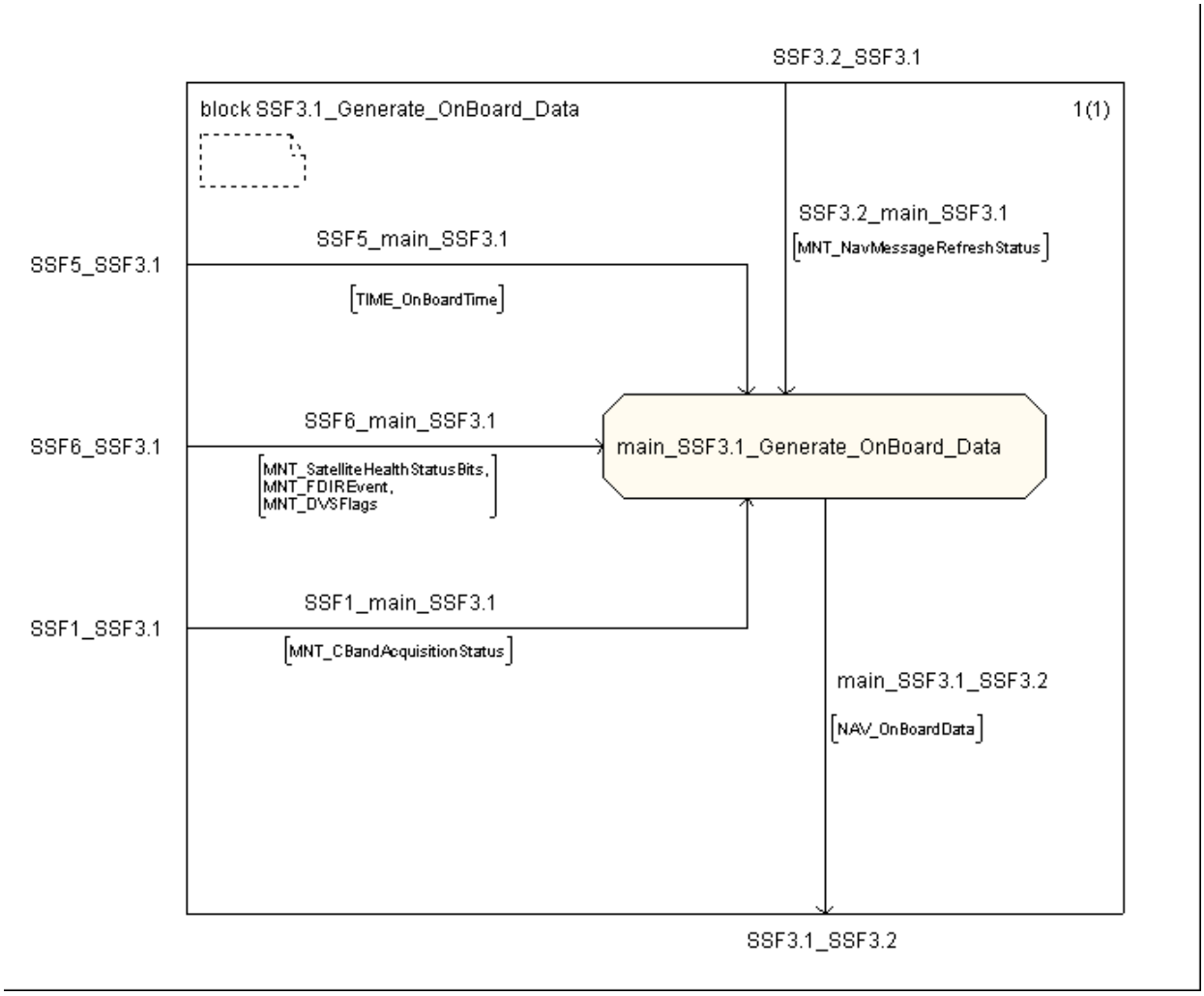


Figure 171: SSF3.1 SDL Functional Block Diagram

5.4.3.2 SSF3.2 Compose Navigation Message

SSF3.2 General Description
This function receives the data to be transmitted from the function SSF1.3 (uplinked data) and SSF3.1 (on board generated data) and composes the F/NAV, I/NAV, G/NAV and C/NAV messages as described in the SISICD to be modulated on dedicated ranging codes.

It also computes and inserts into the messages:

- the CRC on 24 bits for the F/NAV, I/NAV and C/NAV
- the parity bit and spare bits inserted in the G/NAV messages

This function also allows ERIS direct uplink data to be blocked in the downlink when requested by Telecommand.

SSF3.2 Outputs
The outputs are the data to be transmitted for all the services on the frequencies E5a, E5b, E6A, E6B, E1A, E1B, E1C as specified in the SISICD document.

SSF3.2 Design Constraints / Performances
This function is subject to timing requirements to meet end-to-end dissemination requirements (TTA, SAR and CS).
This function is critical for SoL and therefore must meet output feared event requirements.

SSF3.2 Functional Block Diagram



Figure 172: SSF3.2 SDL Functional Block Diagram

5.4.3.3 SSF3.3 Modulate Navigation Messages On Ranging Code

SSF3.3 General Description
This function is in charge to modulate messages(PRS, CS, Navigation) on appropriate ranging codes.

SSF3.3 Outputs
The output is the signals to be transmitted on each frequency modulated with the appropriate ranging codes as described in the SISICD document.

SSF3.3 Design Constraints / Performances
This function is subject to timing requirements to meet end-to-end dissemination requirements (TTA, SAR and CS).
This function is critical for SoL and therefore must meet output feared event requirements.

SSF3.3 Functional Block Diagram

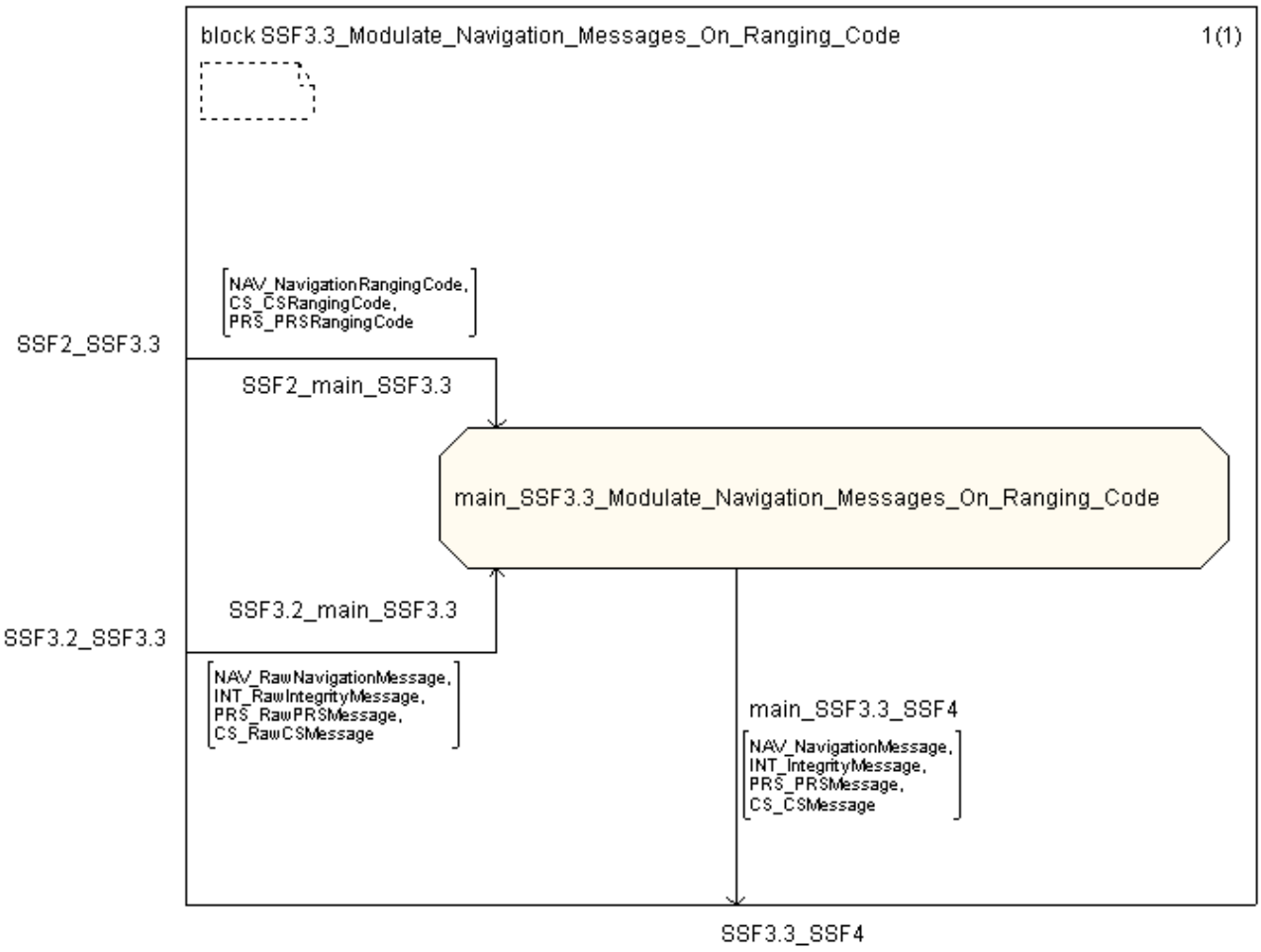


Figure 173: SSF3.3 SDL Functional Block Diagram

5.4.4 SSF4 Transmit Mission Data

SSF4 Description

This function is in charge to modulate Navigation Messages onto specific carriers and transmit L-Band navigation signals to ground.
Main functionalities:

- Generate carriers
- Modulate Messages onto specific carrier
- Amplify and Transmit Signals

SSF4 Functional Block Diagram

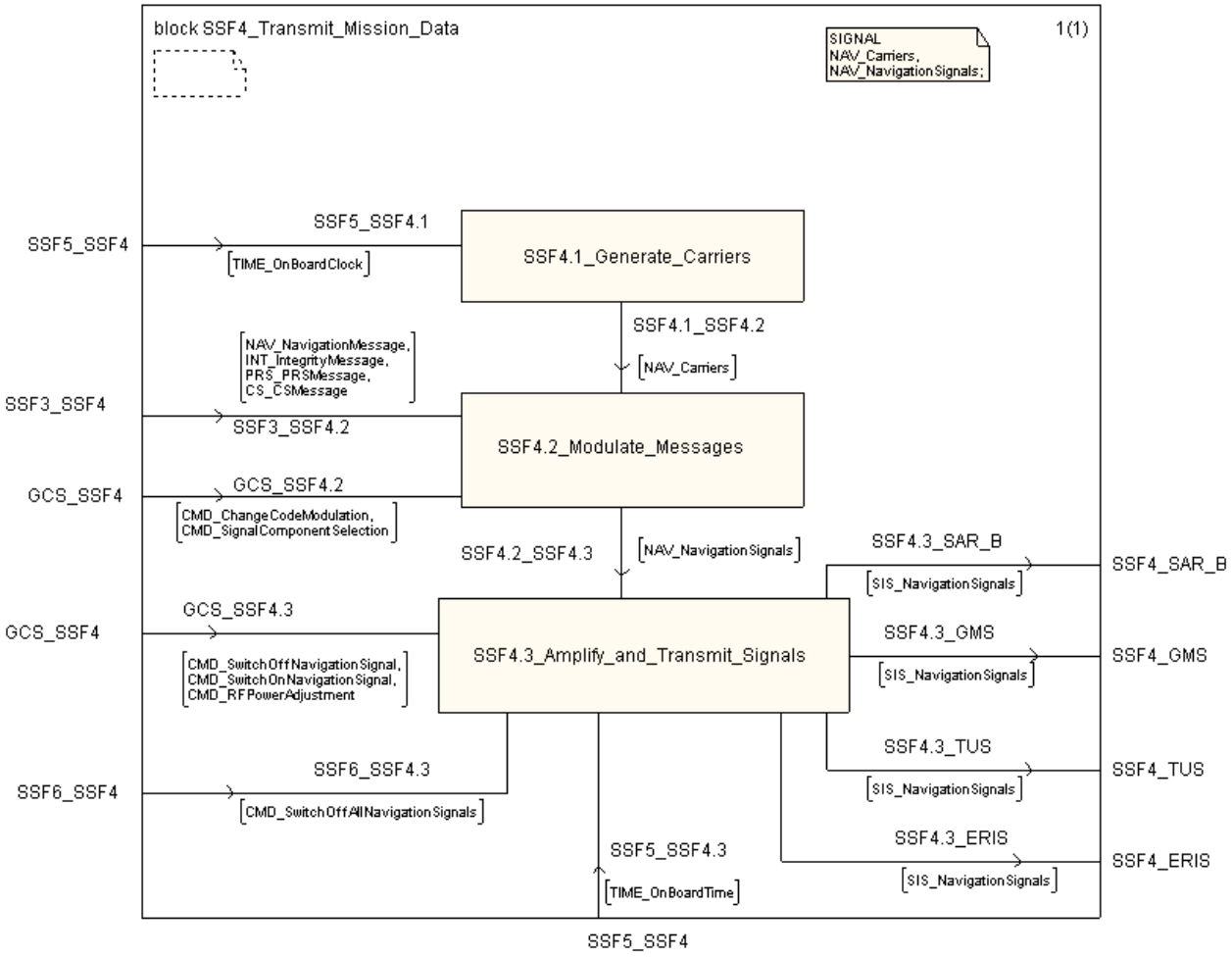


Figure 174: SSF4 SDL Functional Block Diagram

5.4.4.1 SSF4.1 Generate Carriers

SSF4.1 General Description

This function generates carrier frequencies for the frequency bands E5, E6 and E1 for signals modulation.

SSF4.1 Outputs

This function generates the carrier frequencies as specified in the SISICD document.

SSF4.1 Design Constraints/Performances

- This function is critical for SoL and therefore must meet output feared event requirements.
- The phase noise spectral density of each non-modulated carrier shall be as specified in SIS ICD.

SSF4.1 Functional Block Diagram

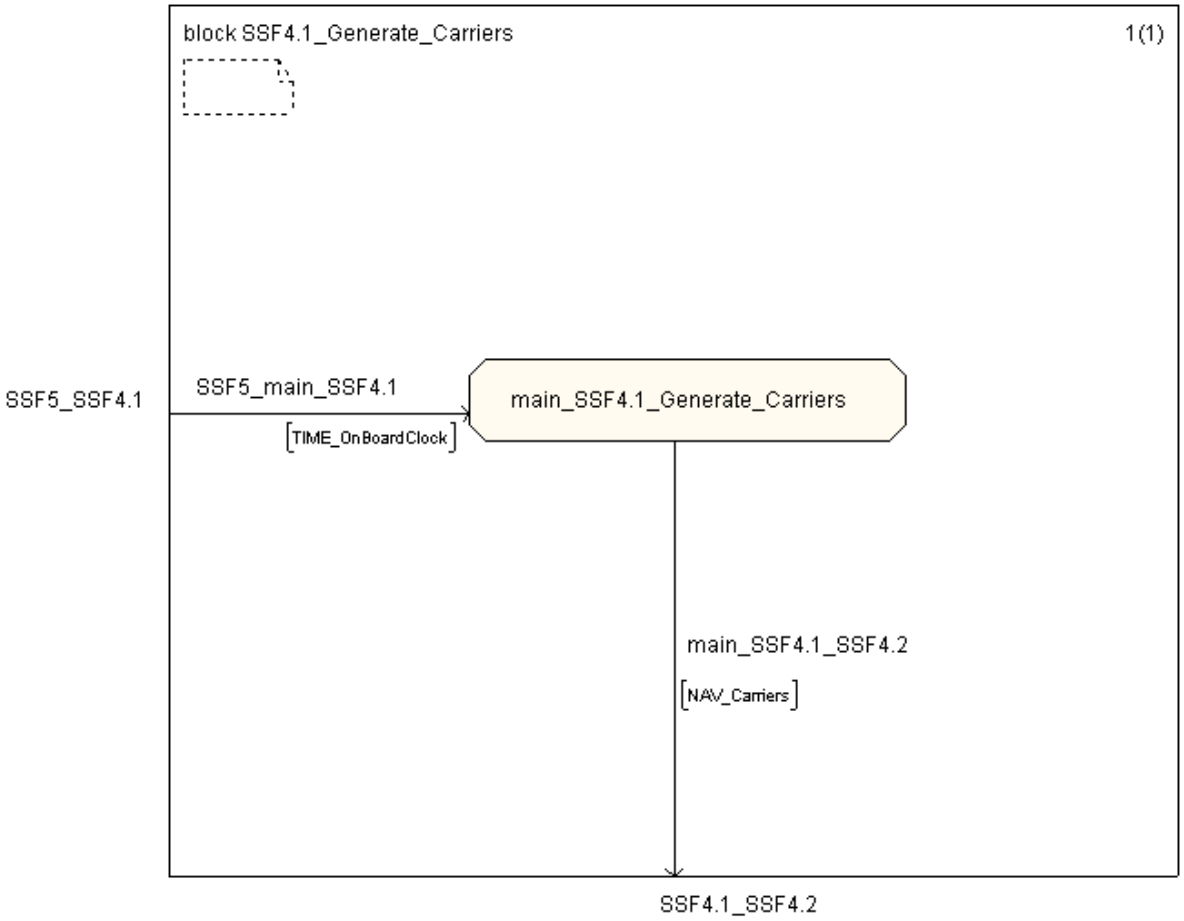


Figure 175: SSF4.1 SDL Functional Block Diagram

5.4.4.2 SSF4.2 Modulate Messages

SSF4.2 General Description
This function modulates messages (F/NAV, I/NAV, G/NAV, and C/NAV) onto dedicated carriers.
SSF4.2 Outputs
The outputs of this function are all the modulated signals to be transmitted by the satellite according to the SISICD document.

SSF4.2 Design Constraints/Performances
This function is critical for SoL and therefore must meet output feared event requirements.

SSF4.2 Functional Block Diagram

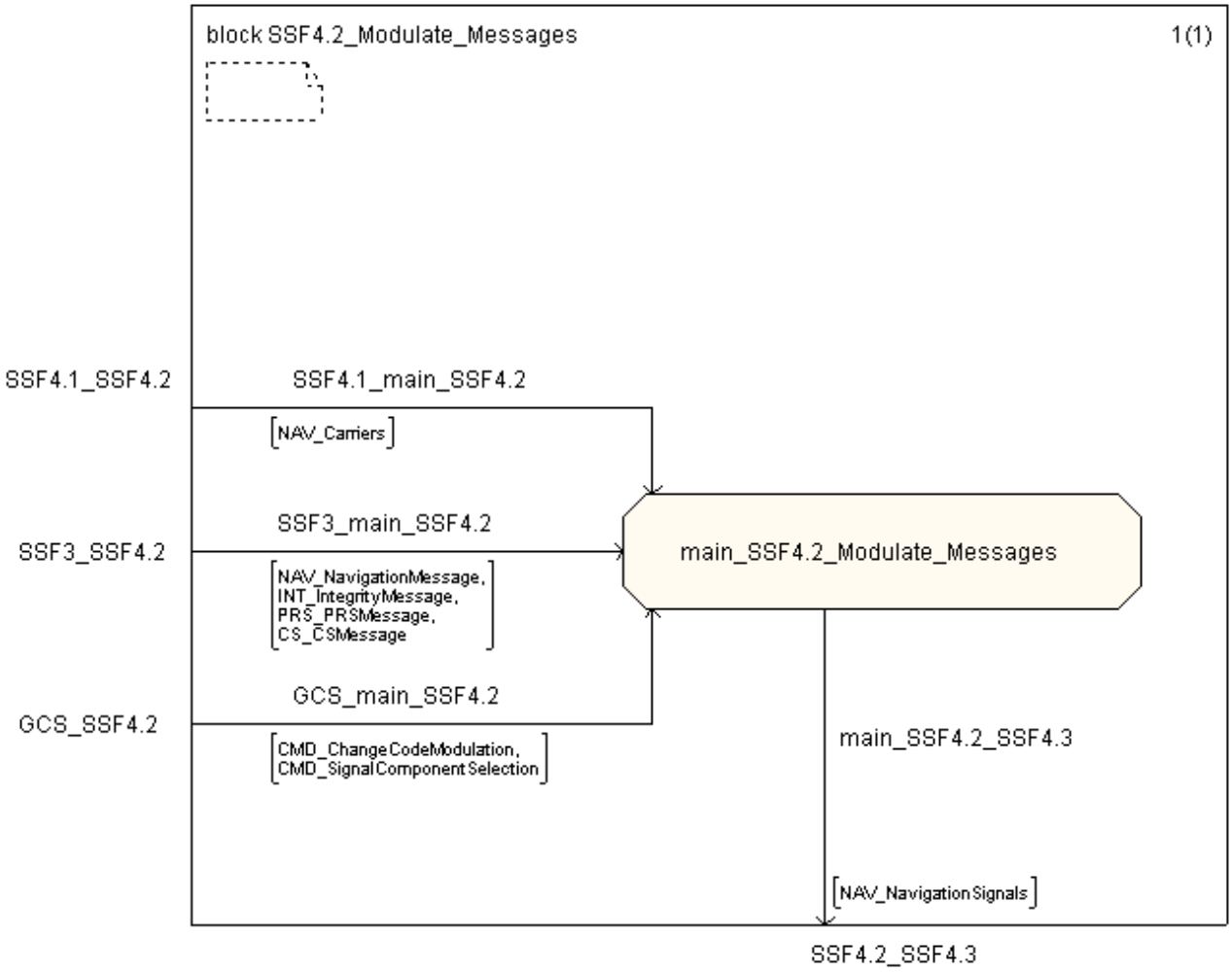


Figure 176: SSF4.2 SDL Functional Block Diagram

5.4.4.3 SSF4.3 Amplify and Transmit Signals

SSF4.3 General Description

This funtion is in charge to amplify and transmit the modulated signals. It manage all related aspects to this functionnalities like: RF power adjustement, frequency band switch off.

SSF4.3 Outputs

The outputs of this function are all the modulated signals to be transmitted by the satellite according to the SISICD document

SSF4.3 Design Constraints/Performances

- This function is critical for SoL and therefore must meet output feared event requirements.
- The minimum EIRP and EIRP stability requirements for Navigation Payload shall be met including all contributors to the satellite antenna pointing error.

The minimum and maximum transmitted EIRP in E5, E6 and E1 bands to any point inside the coverage zone (defined by a 10 masking angle at earth surface) shall be as specified in SIS ICD.

SSF4.3 Functional Block Diagram

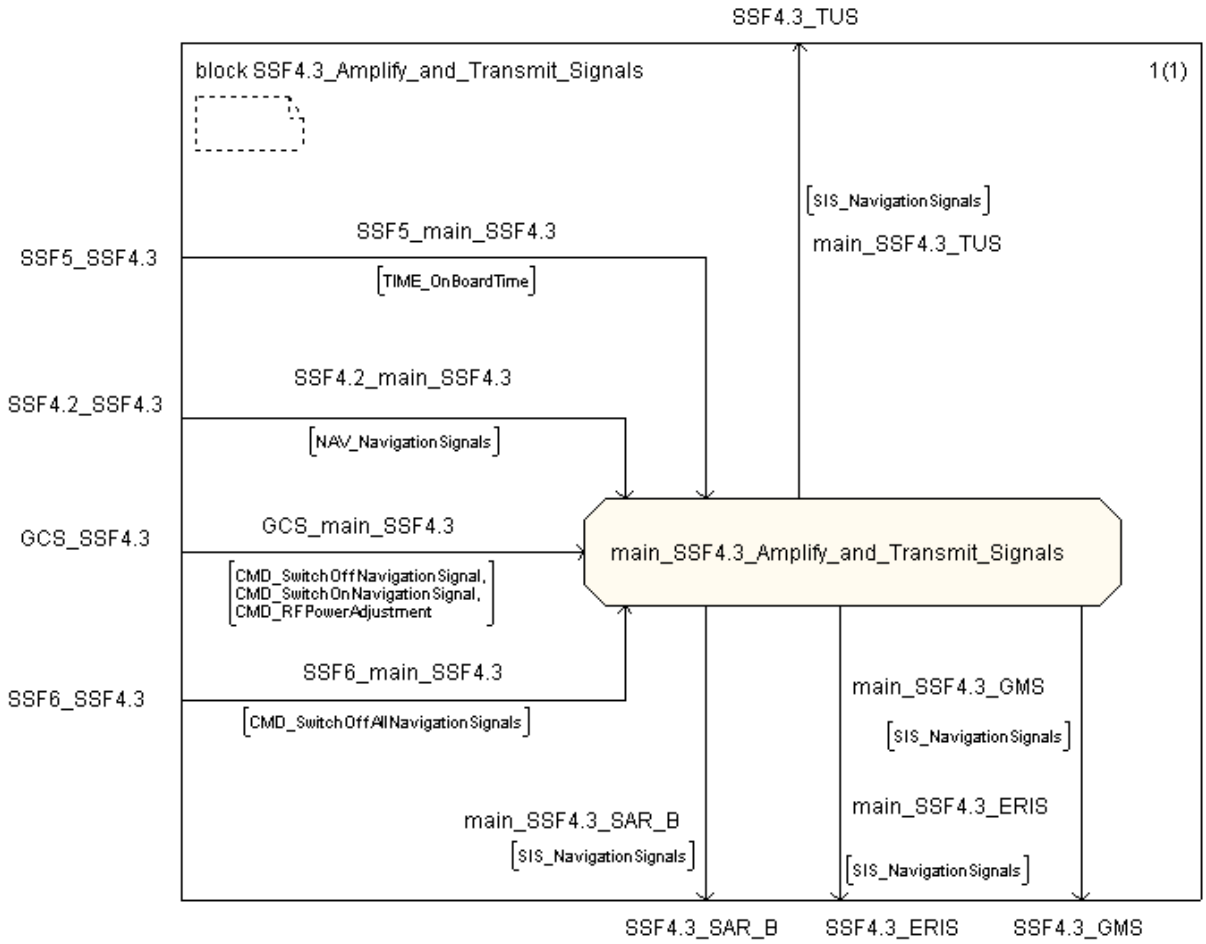


Figure 177: SSF4.3 SDL Functional Block Diagram

5.4.5 SSF5 Provide Time Reference

SSF5 Description

This spacecraft function is in charge of :

- Generating and maintaining a satellite frequency reference (SSF5.1)
- Generating and maintaining satellite time reference (LGST) derived from this frequency (SSF5.2)
- Adjust on board frequency reference or LGST on request from ground commands (SSF5..1 and SSF5.2)
- Manage the clocks redundancy on request from ground commands (SSF5.1)

SSF5 Functional Block Diagram

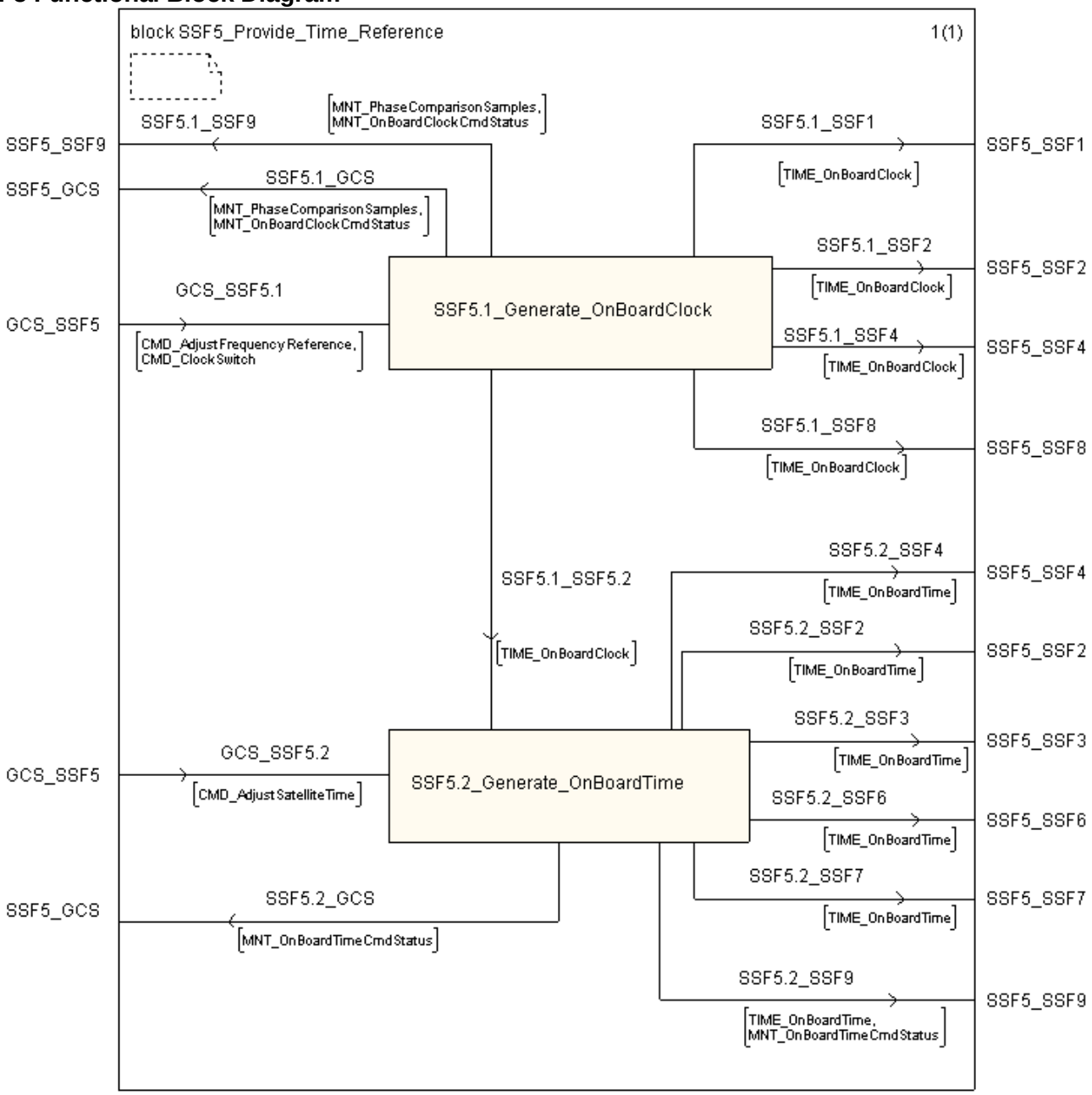


Figure 178: SSF5 SDL Functional Block Diagram

5.4.5.1 SSF5.1 Generate OnBoardClock

SSF5.1 General Description
This function is responsible for generating and maintaining the payload frequency reference. .

The following functionalities are handled:

- Generate a precise frequency reference equal to 10.23MHz derived from the very stable 10MHz frequency of the master on-board atomic clocks.
- Distribute this frequency reference to:
  - the on-board time generation and maintenance functions
  - the navigation signals (ranging codes and carriers) generation functions
  - the SAR signal generation functions
- Adjust this frequency reference applying frequency steps when requested by the Ground Segment via telecommand
- Generate a backup precise frequency reference equal to 10.23MHz derived from the 10MHz frequency of the hot backup atomic clocks.
- Monitor phase comparison samples between the frequency references derived from primary and back-up clock
- Handling any atomic clock switch when requested by the Ground Segment via telecommand.

Note: the switch process is managed by the SSF7 responsible to guarantee the redundancy configuration at any time

SSF5.1 Outputs
The On-Board Clock Generation function produces the following outputs:
1 - Payload frequency reference: Very stable frequency reference produced on-board at Payload level. This frequency is characterized by its long term frequency drift, its frequency stability and its Allan Deviation

2 - Phase Comparison Samples: collection of phase comparison samples between frequency references derived from master and hot backup clocks, transmitted to the Ground Segment via Telemetry.

3- Command execution Status: status of the execution of the commands such as clock switch or frequency adjustment, transmitted to the Ground Segment via Telemetry.

SSF5.1 Design Constraints / Performances
1 - Performance constraints of the frequency reference:
Long term frequency drift, Frequency Drift Stability, Allan deviation

2- Accuracy of the frequency step adjustment:
Adjustment Size capability: relative step size of 5.44 x 10-15.

2 - Performance constraints of the phase measurement sampling:

- The sampling rate shall be programmable between 0.1 second and 1000 seconds with a resolution of 0.1 seconds or less and shall be selectable by telecommand
- The phase measurement has a resolution of 88 millidegrees for a 360 degrees range

3- Clock redundancy configuration: the atomic clocks shall operate in a hot redundant, parallel configuration. Nominally the S-PHM shall be the master reference clock and the RAFS shall be in hot redundancy

SSF5.1 Functional Block Diagram

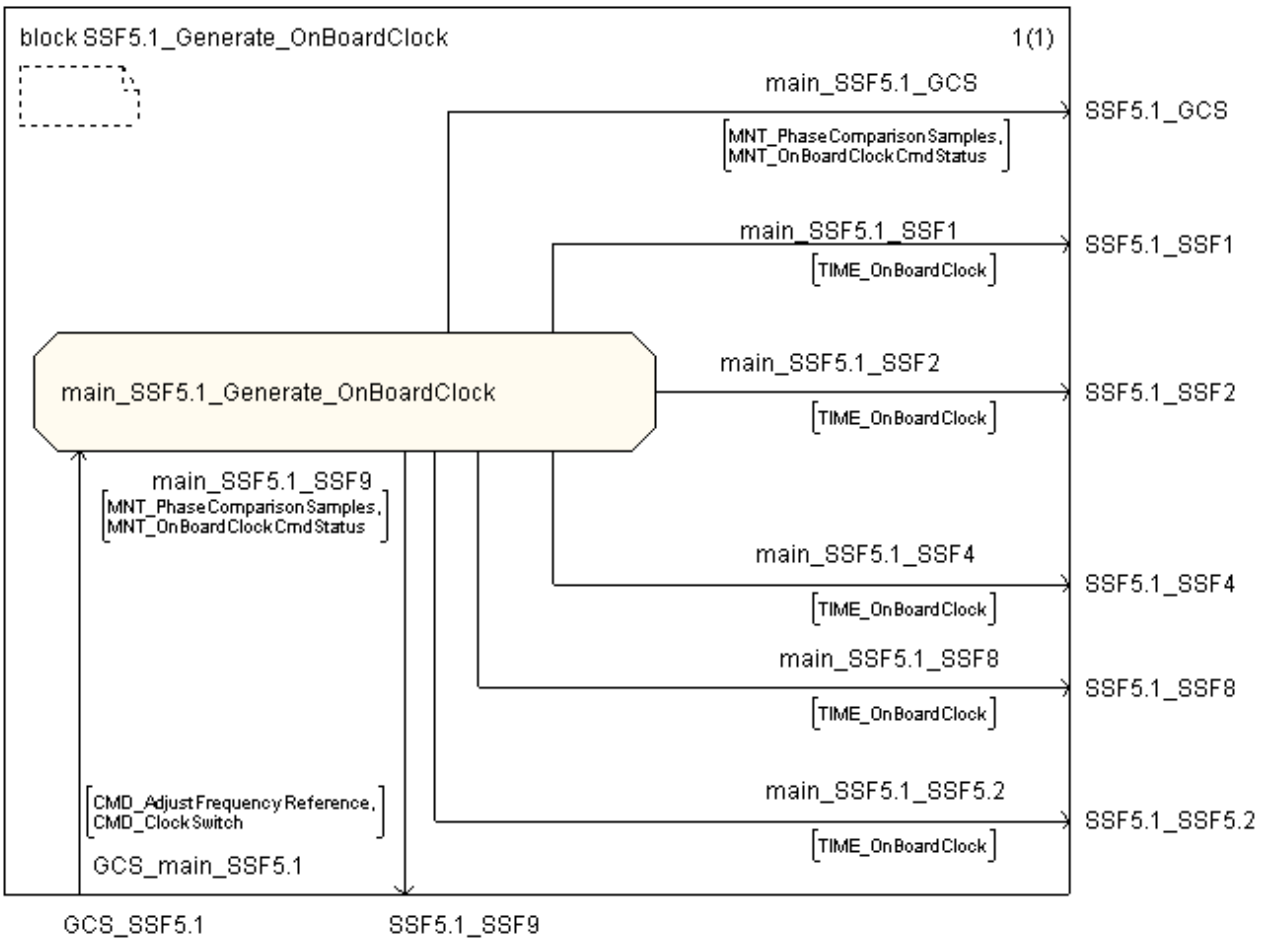


Figure 179: SSF5.1 SDL Functional Block Diagram

5.4.5.2 SSF5.2 Generate OnBoardTime

SSF5.2 General Description
This function is responsible for generating and maintaining the satellite Local GST (or on-board time).

The following functionalities are handled:

- Generate the satellite LGST derived from the 10.23MHz frequency reference generated by SSF5.1
- Generate the timing parameters to be inserted in the L-Band navigation message:
  - the Time Of Week (TOW): counter (time register) incremented every second, equal to the on-ground GST 1 second value: this is steered during initialization phase and shall not be steered along satellite life time
  - the Week Number (WN)
- Synchronise the start of transmission of the each downlink page with the LGST 1-second epoch (NSGU Time-of-Transmission)
- Distribute this LGST to:
- The platform for platform time adjustment
- Adjust this LGST applying time steps when requested by the Ground Segment via telecommand

SSF5.2 Outputs
The On-Board Time Generation function produces the following outputs:
1 - Satellite LGST: Local realization of the GST derived from the payload frequency reference

2- Command execution Status: status of the execution of the commands such as time adjustment, transmitted to the Ground Segment via Telemetry.

SSF5.2 Design Constraints / Performances
1 - Performance constraints of the LGST: the LGST-GST offset is specified at system level and shall remain within [-18ms,+80ms]

2- Accuracy of the time step adjustment:

- The Satellite Time of Week (resp. Week Number) shall be adjustable with a 20 bits (resp. 12bits) field
- The Satellite Time of Transmission (start of page) in the NSGU shall be adjustable over a range of +/- 0.5 s, with a resolution of 100 ns

SSF5.2 Functional Block Diagram

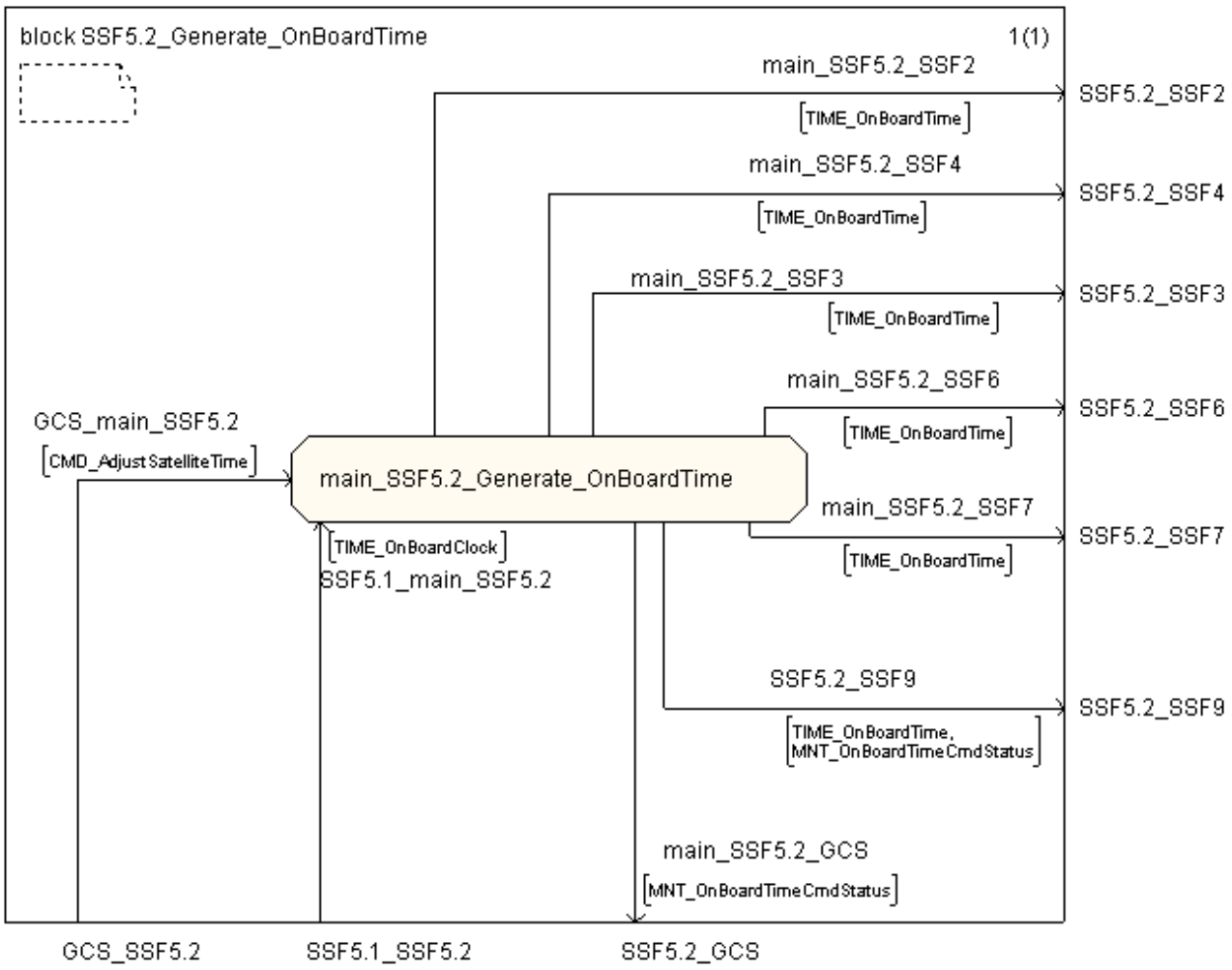


Figure 180: SSF5.2 SDL Functional Block Diagram

5.4.6 SSF6 Manage Platform

SSF6 Description

This function is responsible for providing satellite monitoring and control. It also manages all autonomous S/C operation and FDIR facilities.

Main functionalities:

- Provision of satellite FDIR functionality
- Management of satellite health status for all S/C units
- Generation of satellite health status reports for downlinking via L-band (summary only) and S-band telemetry
- Management of satellite modes
- Management of AOCS functionality
- On Board Data Management (all on-board parameters, software etc)
- Autonomous monitoring and control of the S/C (platform and payload)

SSF6 Functional Block Diagram

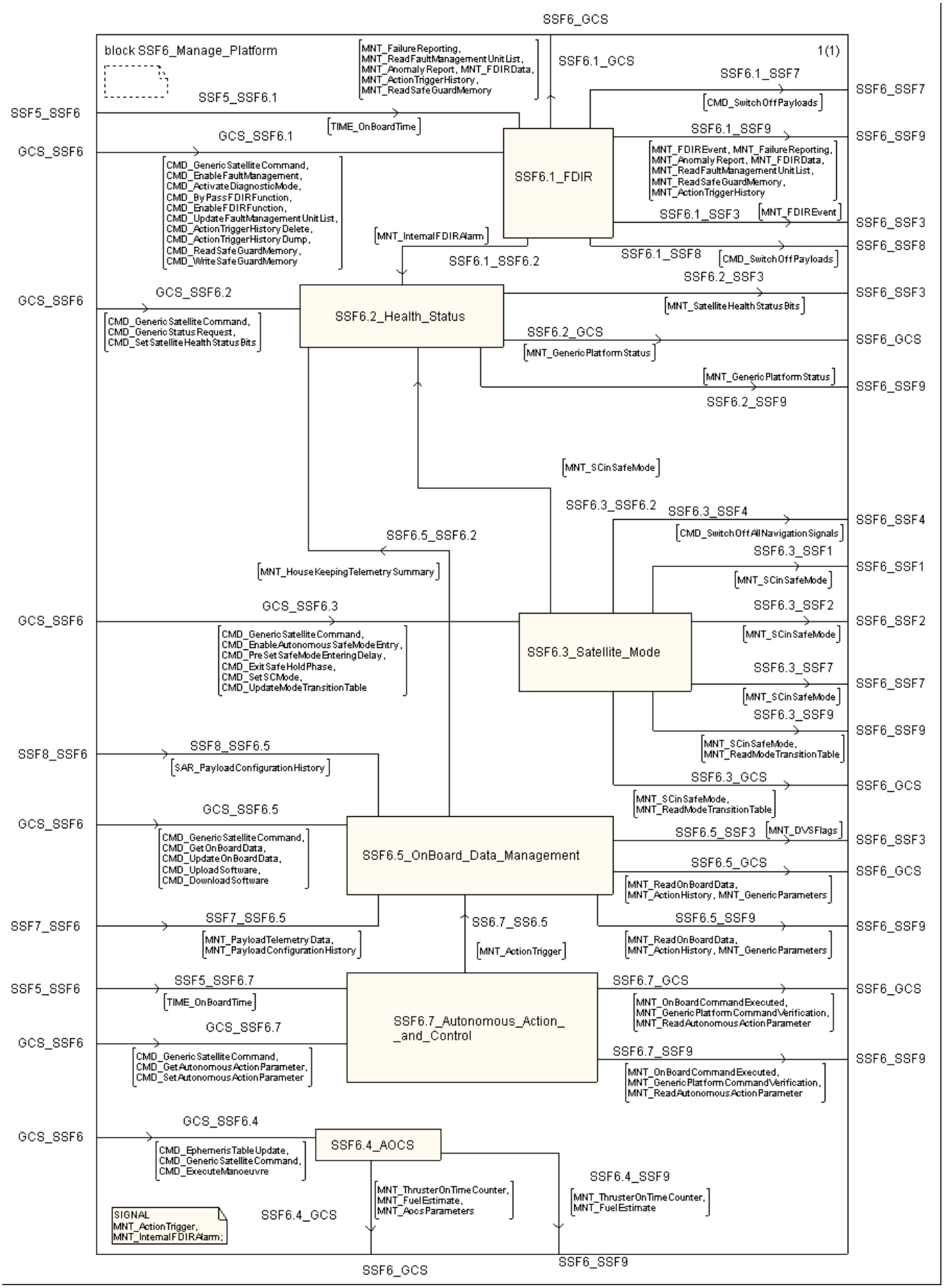

block SSF6_Manage_Platform
1(1)
SSF6_GCS
SSF6.1_GCS
SSF6.1_SSF7
SSF6_SSF7
CMD_Switch Off Payloads
SSF5_SSF6
SSF5_SSF6.1
TIME_On Board Time
GCS_SSF6
GCS_SSF6.1
SSF6.1_SSF9
SSF6_SSF9
SSF6.1_FDIR
SSF6.1_SSF3
SSF6_SSF3
MNT_FDIR Event
MNT_Internal FDIR Alarm
SSF6.1_SSF8
SSF6_SSF8
CMD_Switch Off Payloads
SSF6.1_SSF6.2
SSF6.2_SSF3
SSF6_SSF3
GCS_SSF6
GCS_SSF6.2
SSF6.2_Health_Status
MNT_Satellite Health Status Bits
SSF6.2_GCS
SSF6_GCS
MNT_Generic Platform Status
MNT_Generic Platform Status
SSF6_SSF9
SSF6.2_SSF9
MNT_SCin SafeMode
SSF6.3_SSF6.2
SSF6.3_SSF4
SSF6_SSF4
CMD_Switch Off All Navigation Signals
SSF6.5_SSF6.2
SSF6.3_SSF1
SSF6_SSF1
MNT_House Keeping Telemetry Summary
MNT_SCin SafeMode
GCS_SSF6.3
SSF6.3_SSF2
SSF6_SSF2
GCS_SSF6
SSF6.3_Satellite_Mode
MNT_SCin SafeMode
SSF6.3_SSF7
SSF6_SSF7
MNT_SCin SafeMode
SSF6.3_SSF9
SSF6_SSF9
MNT_SCin SafeMode,
MNT_ReadMode Transition Table
SSF8_SSF6
SSF8_SSF6.5
SAR_Payload Configuration History
SSF6.3_GCS
SSF6_GCS
MNT_SCin SafeMode,
MNT_ReadMode Transition Table
GCS_SSF6.5
GCS_SSF6
SSF6.5_SSF3
MNT_DVS Flags
SSF6_SSF3
CMD_Generic Satellite Command,
CMD_Get On Board Data,
CMD_Update On Board Data,
CMD_Upload Software,
CMD_Download Software
SSF6.5_OnBoard_Data_Management
SSF6.5_GCS
SSF6_GCS
MNT_Read On Board Data,
MNT_Action History, MNT_Generic Parameters
SSF7_SSF6
SSF7_SSF6.5
SS6.7_SS6.5
SSF6.5_SSF9
SSF6_SSF9
MNT_Payload Telemetry Data,
MNT_Payload Configuration History
MNT_Action Trigger
MNT_Read On Board Data,
MNT_Action History, MNT_Generic Parameters
SSF5_SSF6
SSF5_SSF6.7
SSF6.7_GCS
SSF6_GCS
TIME_On Board Time
MNT_On Board Command Executed,
MNT_Generic Platform Command Verification,
MNT_Read Autonomous Action Parameter
GCS_SSF6.7
GCS_SSF6
SSF6.7_Autonomous_Action_
_and_Control
SSF6.7_SSF9
SSF6_SSF9
CMD_Generic Satellite Command,
CMD_Get Autonomous Action Parameter,
CMD_Set Autonomous Action Parameter
MNT_On Board Command Executed,
MNT_Generic Platform Command Verification,
MNT_Read Autonomous Action Parameter
GCS_SSF6.4
GCS_SSF6
SSF6.4_AOCS
CMD_Ephemeris Table Update,
CMD_Generic Satellite Command,
CMD_Execute Manoeuvre
SSF6.4_SSF9
MNT_Thruster On Time Counter,
MNT_Fuel Estimate,
MNT_Aocs Parameters
MNT_Thruster On Time Counter,
MNT_Fuel Estimate
SIGNAL
MNT_Action Trigger,
MNT_Internal FDIR Alarm;
SSF6.4_GCS
SSF6_GCS
SSF6_SSF9

Figure 181: SSF6 SDL Functional Block Diagram

5.4.6.1 SSF6.1 FDIR

SSF6.1 General Description
This function provides the on-board Failure Detection, Isolation and Recovery (FDIR) capability. In particular, it provides the following functions:

- Ensures spacecraft survival ensured for any single on-board failure
- Detection and categorisation of failures
- Automatic failure response with minimum impact on spacecraft
- Implements a hierarchical FDIR structure such that faults, failures or errors are detected, isolated and recovered at the lowest possible level.
- Maintains overall mission for any failures which do not require ground intervention
- Manages spacecraft outages according to OFE requirements
- Preserves spacecraft life in terms of fuel use or unit/equipment loss
- Allows all automated FDIR actions to be enabled / disabled / overriden by TC from ground.
- Provides a non-volatile safeguard memory (fully accessible from ground) capable of.storing all satellite context data necessary to restore a proper configuration following a failure on-board.

SSF6.1 outputs
This function is responsible for producing the following output signals:

- telemetry reporting all parameters associated with a failure event and all mode and configuration changes executed in response to it
- telemetry reporting all satellite anomalies and actions taken to recover from them
- telemetry reporting of FDIR related events that do not adversely affect mission performance for inclusion in the satellite health status information transmitted via L-band
- telemetry providing a history log of all FDIR Data to provide a full satellite observability for ground investigation
- telemetry providing the contents of the non-volatile safeguard memory
- telemetry reporting the list of equipment required to start a mode
- telemetry reporting the history of events occuring before and after detection of a failure
- commands to switch off the payload according to a pre-defined operational sequence.

SSF6.1 Design Constraints
This function supports the following design / performance constraints:

- Ensuring satellite survival, without irrevocable consequence, while receiving no valid TC for at least 24 hours, assuming any single failure on-board (excluding single point failures).
- Setting the Data Validity Status flags to "Working Without Guarantee” in less than 4 seconds in response to events which adversely affects the navigation service performance
- Ensuring compliance with the FDIR mission availability OFE requirements.

SSF6.1 Functional Block Diagram

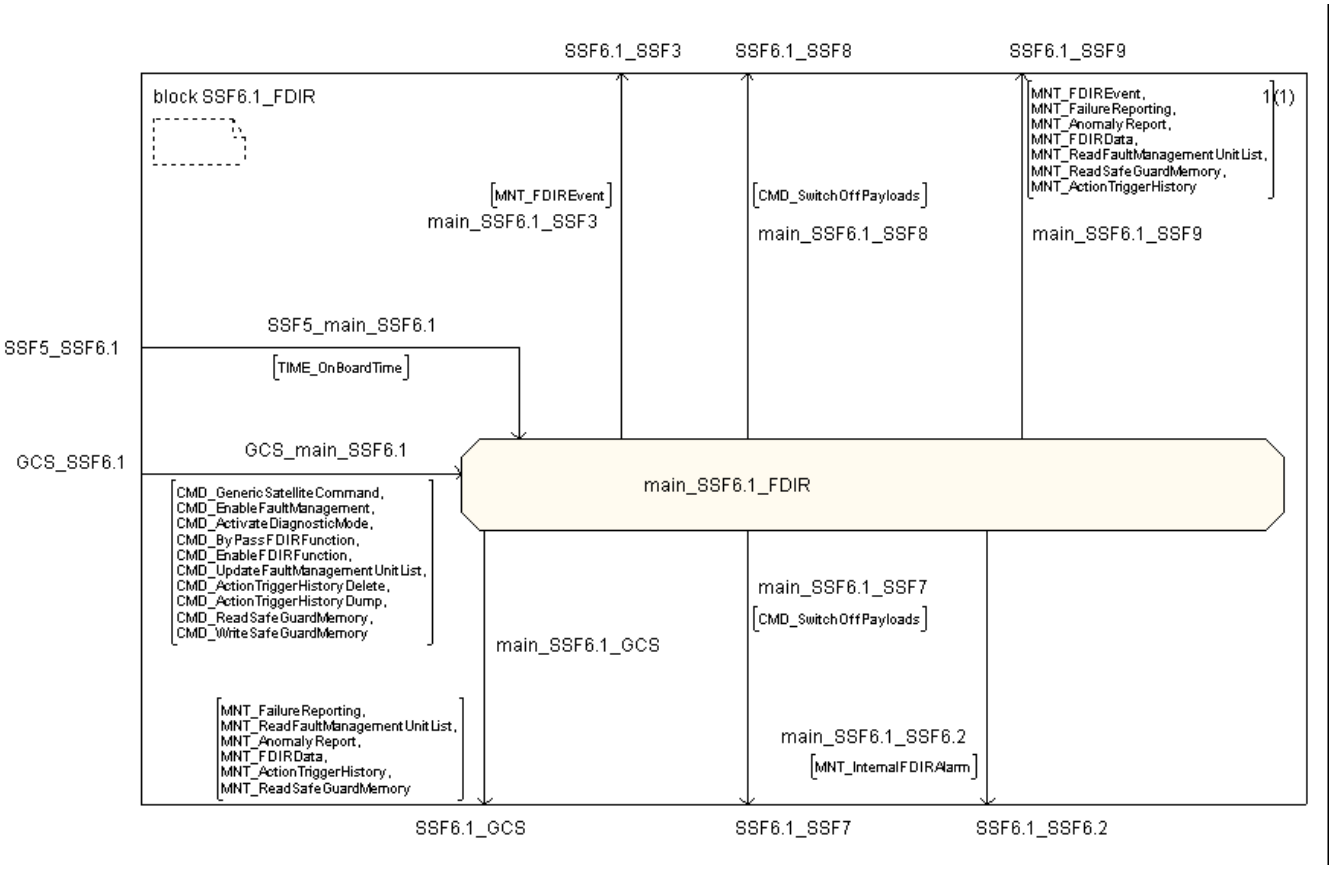


Figure 182: SSF6.1 SDL Functional Block Diagram

5.4.6.2 SSF6.2 Health Status

SSF6.2 General Description

This function monitors the health status of all spacecraft units and reports this to ground through:

- generation of nominal telemetry reported via S-band TT&C link. The data provided in this way will, for nominal operations, allow assessment of spacecraft status and performance without the need for reference to the TC history.
- generation of the summary satellite health status data bits and providing these to payload for transmission via L-band. Note that these bits can be reset by ground TC.

SSF6.2 outputs

This function is responsible for producing the following output signals:

- Spacecraft health status telemetry to be reported via the S-band TT&C link
- A summary health status report based on critical housekeeping TM data and internal FDIR alarm signals to be included in the L-band PRS navigation service

SSF6.2 Design Constraints

This function provides the health status information according to the following constraints:

- The health status information will be derived from direct measurements from operating units and not from secondary effects
- The health status of critical satellite functions (as defined by RAMS analysis) will be provided by at least 2 independently obtained measurements.

SSF6.2 Functional Block Diagram

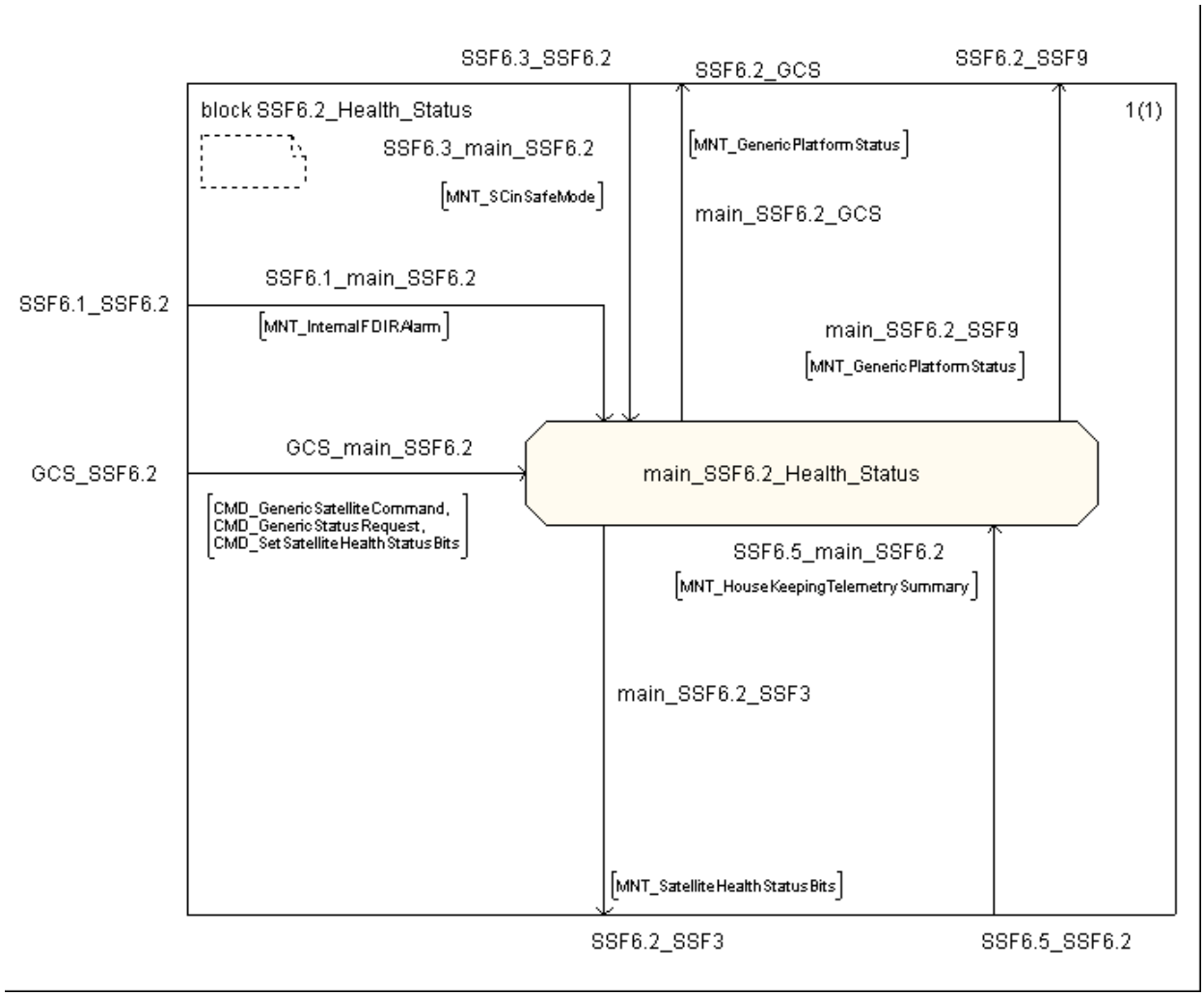


Figure 183: SSF6.2 SDL Functional Block Diagram

5.4.6.3 SSF6.3 Satellite Mode

SSF6.3 General Description

This function is responsible for the management of all satellite modes (including safe mode):

- Managing the conditions for transition between satellite modes including:
  - Automatic mode entry (eg entry into safe mode if no valid TC is received for a pre-set period from 1 hour to 90 days)
  - By TC from ground (entry into each spacecraft mode possible using a single TC)
- Providing and managing all functionality required to enter a each mode and also the functionality allowed withinin each mode (eg essential functions such as fault management, telemetry reporting etc).
- Providing safety logic to ensure that only allowed mode transitions can take place.

SSF6.3 outputs

This function is responsible for producing the following output signals:

- Telemetry defining the allowed and forbidden transitions between all modes
- Telemetry allowing unambiguous identification of the entered mode
- Provision of telemetry to the on-board security unit indicating that the S/C has entered into safe mode
- A signal to ensure that the navigation signals are turned off when the S/C enters safe mode.

SSF6.3 Design Constraints

This function will meet the following constraints and performances with regard to safe mode:

- Ensuring S/C survival and subsequent re-commissioning with no permanent degradation of performance while receiving no signals of any kind from the ground, for at least 90 days [TBC / Pending on the operational concept consolidation]
- Ensuring S/C survival without irrevocable consequence while receiving no valid TC for at least 24 hours, assuming any single failure on-board (excluding single point failures).
- Achieving nominal S/C operational performance within 15 days of recovery from safe mode. This period does not include any time necessary for failure analysis and corrective action.
- Ensuring that exit from Safe Mode is only possible by ground TC
- Ensuring the availability of the following functions in safe mode:
  - S-Band TT&C communications and two way ranging with the Ground
  - Positive power balance
  - Controlled stable attitude
  - Thermal safety

SSF6.3 Functional Block Diagram

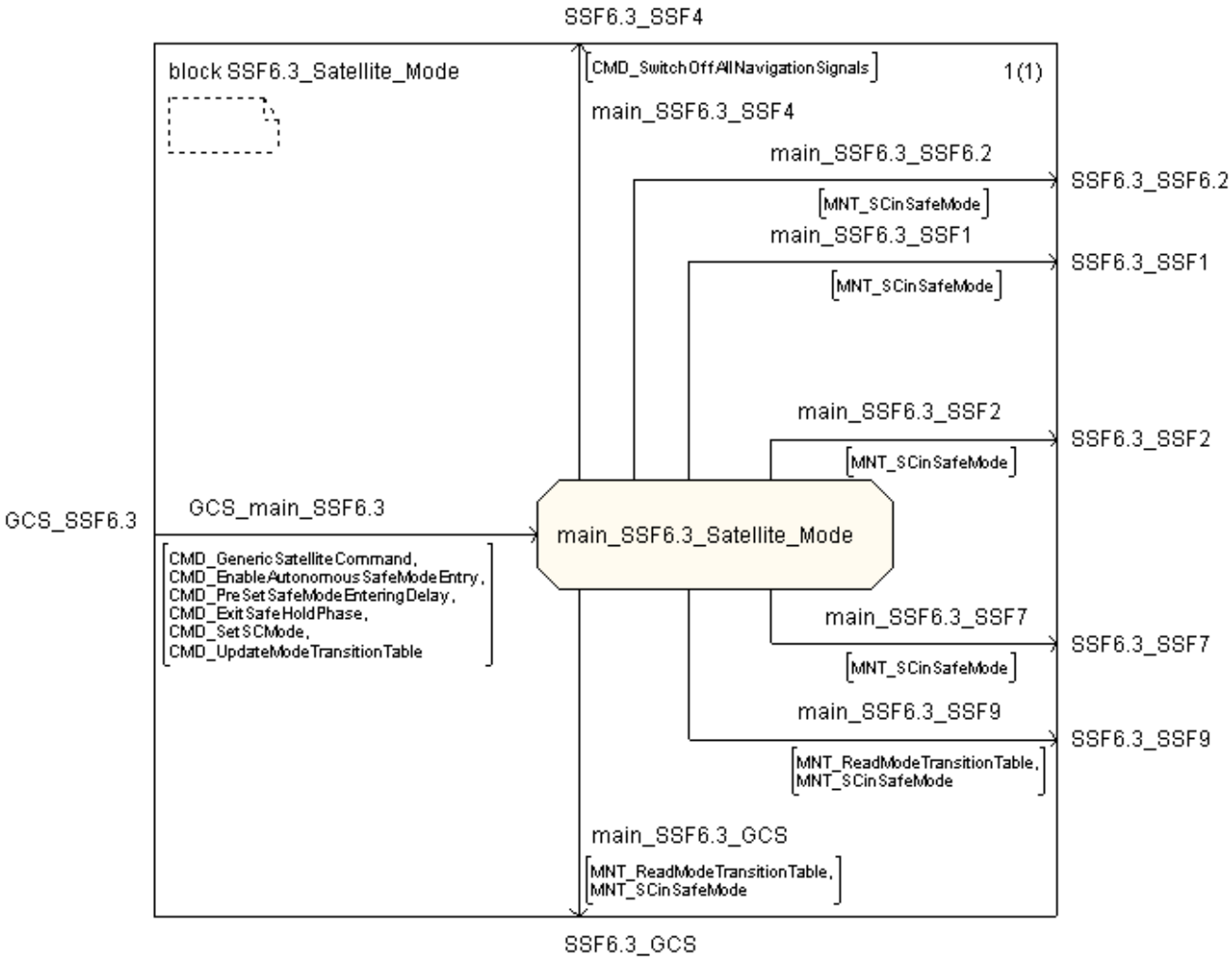


Figure 184: SSF6.3 SDL Functional Block Diagram

5.4.6.4 SSF6.4 AOCS

SSF6.4 General Description
This function is responsible for the following:

- Provision of an automated Attitude and Orbit Control Sub-system (AOCS) that is the to the maximum extent autonomous from ground;
- Ground intervention to AOCS will be limited to special operations (eg LEOP, or recovery operations after multiple failures).
- AOCS autonomy includes the capability, during all missions phases, to determine and maintain the required satellite attitude pointing, rate and acceleration.
- Autonomous control and execution of all Attitude and Orbit Control (AOCS) functions (eg thrusters, actuators etc) for nominal orbit manoeuvres
- Management of the S/C mass, power and fuel budgets
- Provision of the following S/C models:
  - Satellite orientation model
  - Satellite centre of mass model

SSF6.4 outputs
This function is responsible for producing the following output signals:

- Telemetry providing a counter for the accumulated commanded thruster "on times" for each thruster independently.
- Telemetry to allow fuel estimation performed on ground with an accuracy of better than 3%
- Telemetry to provide the ground with sufficient measurement information to permit the satellite attitude and position reconstitution with the accuracy required by the mission, system and operational requirements, throughout all mission phases.

SSF6.4 Design Constraints
This function will meet the following constraints and performances:

- Ensuring that the semi-major axis of the satellite orbit can be corrected to within +/- 5 metres.
- Providing an absolute pointing and rate stability consistent with the payload stability requirements
- Support the capability to:
  - re-position the satellite twice (once, if the satellite was a spare satellite re-positioned to a new orbital position) during its operational lifetime from one operational orbit location to a new operational orbit location up to 30 degrees away on the same orbit plane within a period of 5 days.
  - Move an in-orbit spare to its operational position in the same orbit plane within an average of 8.9 days from call-up
- Resumption of all AOCS functions with full performance after a single failure and following reconfiguration
- The on-board ephemeris tables will only require an update from ground once every 48 hours in all critical mission phases. In routine phases this frequency is reduced to once per week
- Proide sufficient telemetry to allow fuel estimation to be performed on ground with an accuracy of better than 3%.
- Ensure that the centre-of-mass (CoM) stability relative to the internal SV reference frame is be better than 20mm (1 sigma) in any direction over 56.3 hours (including any operation of actuators on devices on board).
- Provide a CoM model capable of predicting the satellite CoM (with respect to satellite reference frame) with an accuracy better than 30mm (1 sigma) in any direction (including any

operation of actuators on devices on board) with all predicted parameters valid for at least the next 8.3h.

- Provide an orientation model with:
  - a maximum yaw error of 2.1
  - antenna NADIR axis (pitch & roll) of 0.55 (half-cone, already covering antenna pointing value; overall antenna budget).
  - an accuracy of 2 in the modelling of the rotation of the solar arrays around the y-axis

SSF6.4 Functional Block Diagram

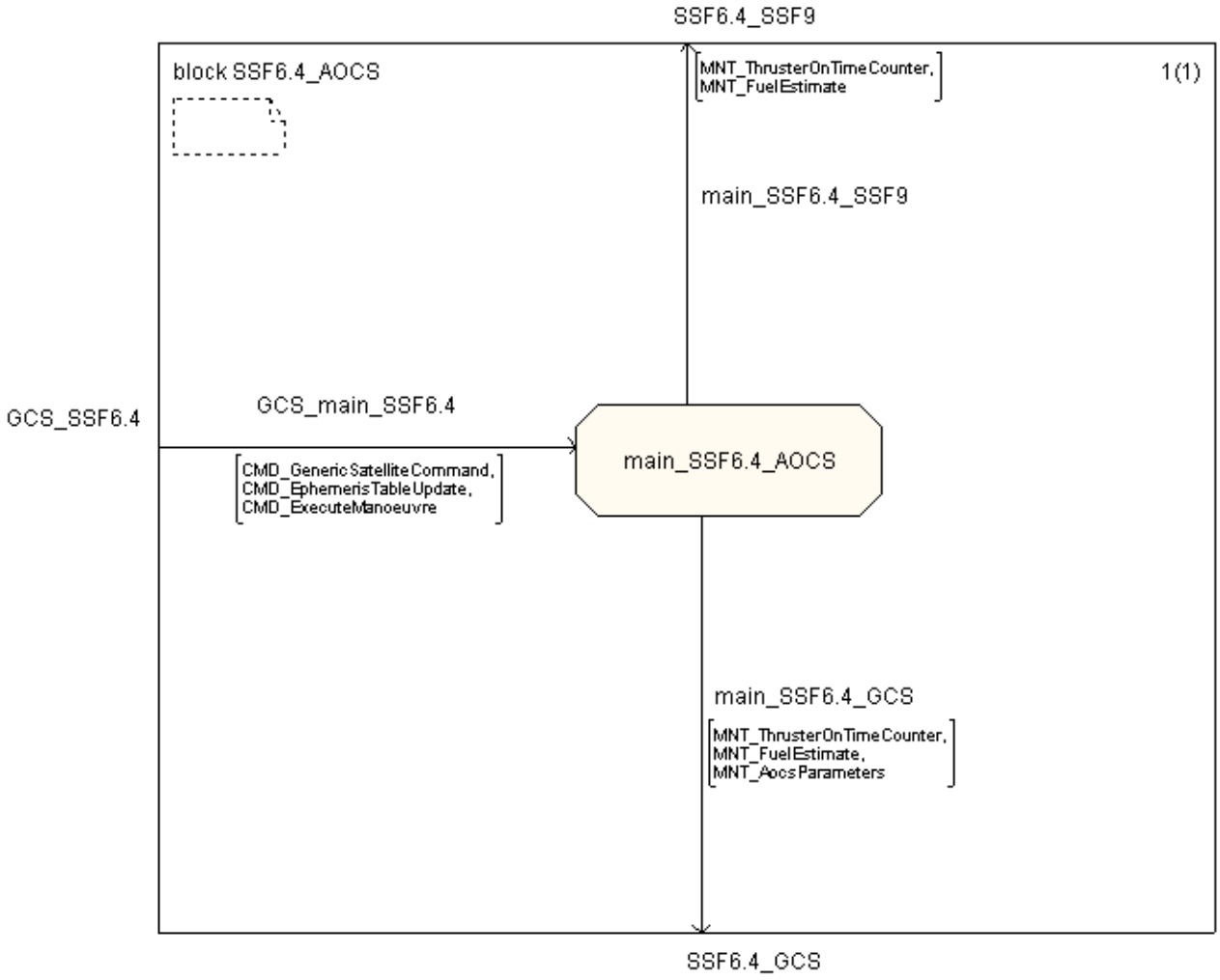


Figure 185: SSF6.4 SDL Functional Block Diagram

5.4.6.5 SSF6.5 OnBoard Data Management

SSF6.5 SGeneral Description
This function is responsible for the management of all on-board software and associated data parameters. In particular, it provides the following:

- Provides maintainability of all on-board software through:
- in-flight modification (through the application of software patches from ground)
- in-flight inspection (through software dumping to ground).
- Manages all parameters used by the on-board software(eg thresholds for limit checking or thresholds and biases for attitude control (updatable by TC and available via TM).

SSF6.5 outputs
This function is responsible for producing the following output signals:

- Telemetry listing all actions taken by on-board software which can affect the satellite performance
- Telemetry providing all commandable parameters such as, for example, monitoring and control thresholds, software tables, flags, global variables used by FDIR
- Telemetry providing all data or parameters used by the on-board software.

SSF6.5 Design Constraints
This function will meet the following constraints and performances:

- All on-board EEPROM and RAM memory will patchable by ground TC.
- All on-board application processes will ensure the integrity of the memory area in order to minimize possible inhibits and constraints

SSF6.5 Functional Block Diagram

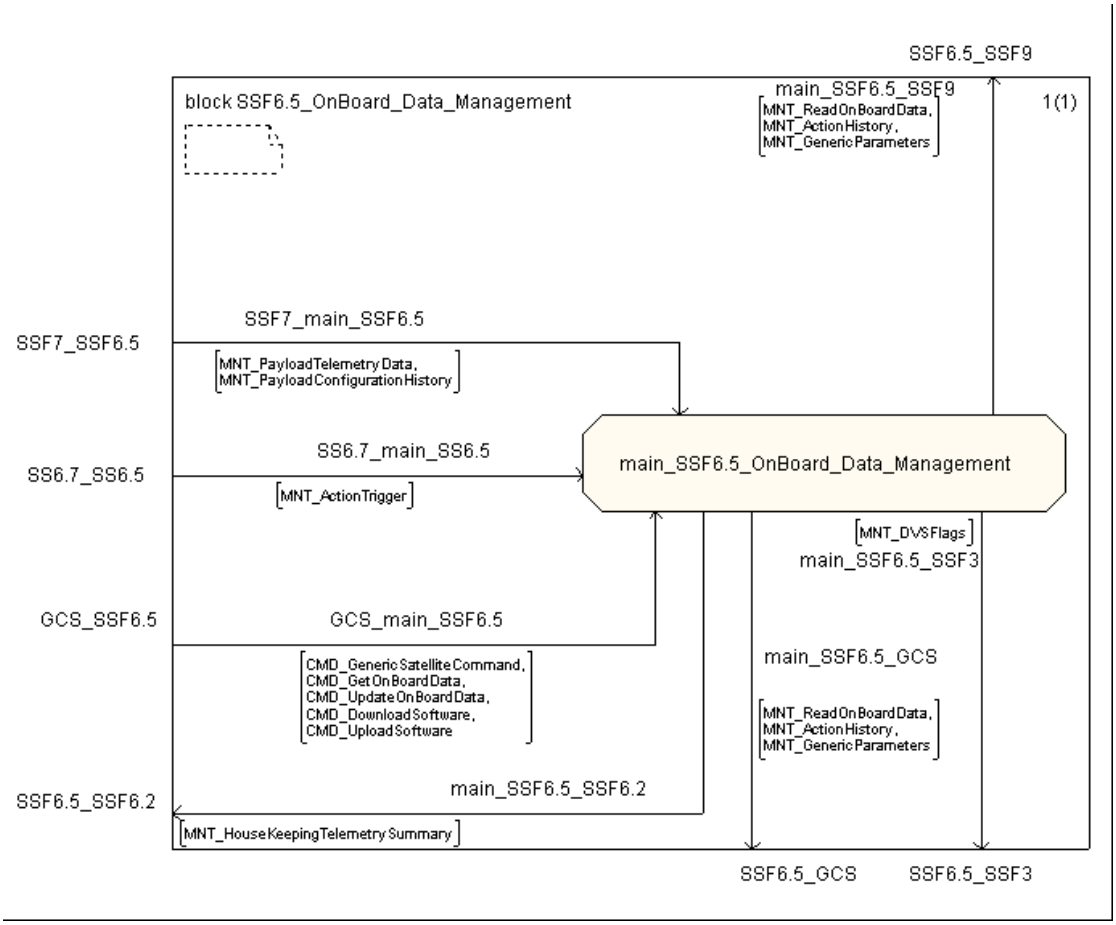


Figure 186: SSF6.5 SDL Functional Block Diagram

5.4.6.6 SSF6.6 Deleted

5.4.6.7 SSF6.7 Autonomous Action and Control

SSF6.7 General Description
This function is responsible for the following aspects:

- Initial switch-on sequence without the need for ground TCs
- Routine autonomous operation of the satellite
- Autonomous restart of the satellite following power unavailability
- Automatic verification of on-board command execution
- Autonomous house-keeping functions (including battery charge, thermal control etc).

SSF6.7 outputs
This function is responsible for producing the following output signals:

- Telemetry regarding all parameters input to or used by the on-board autonomy functions (e.g. thresholds for limit checking or thresholds and biases for attitude control)
- Telemetry to verify the execution of all commands generated on-board

SSF6.7 Design Constraints
This function will meet the following constraints and performances:

- Full autonomous operation for at least two orbital periods in nominal or contingency cases (with the exception of well identified scenarios)
- Full autonomous operation with no degradation below nominal performance while receiving no valid TC for two consecutive orbital periods (assuming no failure on-board).
- Operating within specification without ground intervention for at least:
  - 100 minutes when the Rubidium Clock is providing the satellite timing source
  - 8 hours when the Passive Hydrogen Maser Clock is providing the satellite timing source
- Ensuring new satellites finish IOT within 5 days of arrival at their operational orbital position.
- Ensuring in-orbit spare satellites achieve nominal operational performance within 1 day of arrival at their operational orbital position.

SSF6.7 Functional Block Diagram

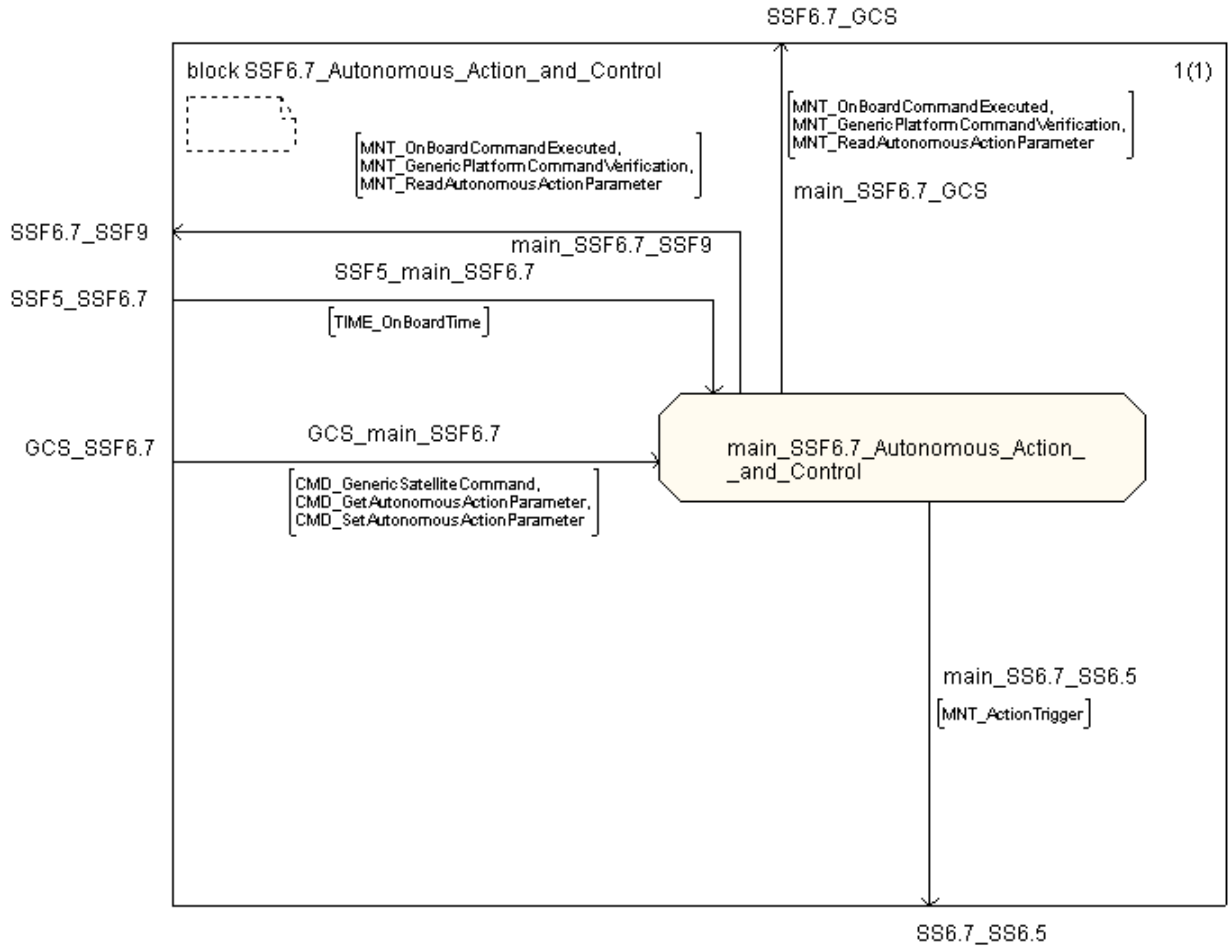


Figure 187: SSF6.7 SDL Functional Block Diagram

5.4.7 SSF7 Manage Payload

SSF7 General Description

This function is responsible for the management, monitoring and control of all units within the navigation payload.
This includes:

- Redundancy and Switch control
- Unit power on/off
- Payload mode control
- Collecting of payload TM to be transferred to the platform for S-Band downlink
- Managing TC dedicated to payload and verification status
- Payload software management

SSF7 Outputs

The Manage Payload function produces the following TM as outputs:

1 - Several status types of the payload including:
- Redundancy status
- Configuration history
- TC execution

2 - All the TM data collected from the payload units
3- Addional paramanters of the payload

SSF7 Functional Block Diagram

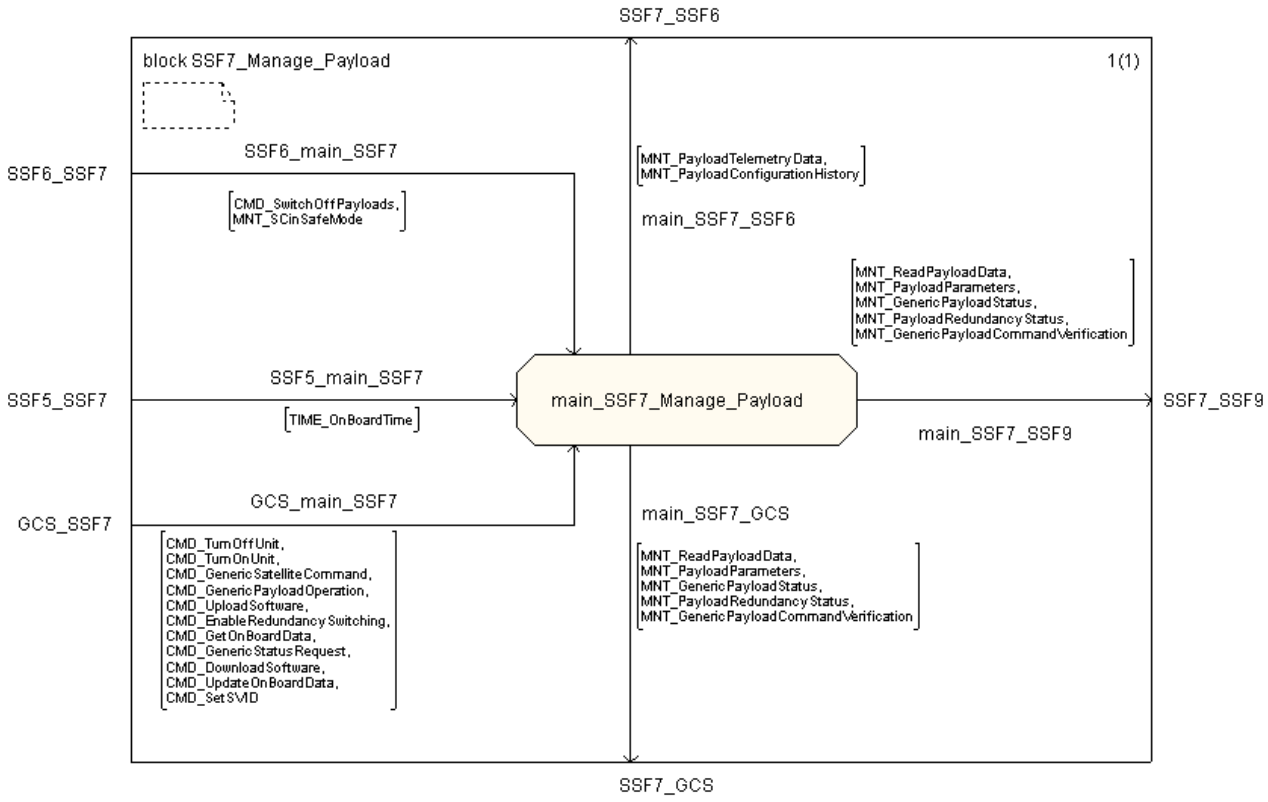


Figure 188: SSF7 SDL Functional Block Diagram

5.4.8 SSF8 SAR Payload

SSF8 General Description
This function is responsible for the relay of SAR Beacon Distress Signals to the SAR MEOLUT ground station which consist of::

1. Receiving signals from UHF beacons in the 406MHz - 406.1MHz band
2. Frequency convert and retransmit to MEOLUT in the 1544.05MHz - 1544.15MHz band.

SAR Transponder Modes:

1. Two frequency band modes: Normalband (90kHz) and Narrowband (50kHz)
2. Two gain control modes: Fixed gain and Automatic Gain Control (AGC)

The SAR payload consists of a UHF receive chain a SAR transponder, and a L-Band receive antenna which is isolated from the Rx antennas to guarantee RF compatibility requirement

The Payload uses the Navigation payload frequency from CMCU to aid the frequency conversion accuracy

The SAR Payload is remotely monitored and control from ground:

The SAR Payload is considered to be used throughout the satellite lifetime

SSF8 Outputs
The SAR Payload function produces the following outputs:

1. SAR Forward link downlink to SAR MEOLUT
2. Various telemetry outputs transmitted to the Ground Segment:

- Status of the execution of the commands,
- SAR Payload Parameters
- SAR Payload Health Status
- SAR Payload Configuration History

SSF8 Design Constraints
The SAR Payload receives the uplink signal from the SAR Beacons with the following basic parameters: This includes the following:

1. Receive G/T >13.7dB/K at EOC, and -12.4dB/K at COC.
2. Polarisation: Right Hand Circular (RHCP)
3. Antenna coverage: Whole earth
4. Axial ratio <2.5dB
5. Up to 5 simultaneous bursts at different frequencies arriving at satellite.

The SAR Payload shall downlink the SAR distress message with the follow signal characteristics:

1. Uninverted frequency conversion
2. EIRP
    - .>18.0dBW over entire earth coverage defined by elevation angles >10deg;
    - >16.8 dBW over entire earth coverage
3. Accuracy of frequency translation: +/-2E-11

4. Stability (short term) 1E-11 (over 100ms)
5. Group delay variation <10microsec, over any 4kHz anywhere in the band.
6. Linearity > 30dBc
7. Dynamic range > 30dB
8. Polarisation: Left Hand circular (LHCP)

Other general design constraints of the SAR Payload functions:

- Pointing Error: EIRP, G/T and stability requirements met taking into account all contributions to the satellite antenna pointing error
- Availability: Payload relays message with a probability higher than 0.9999 (excl. SAR payload non-recoverable failures)
- Reliability: (TBC) ~1920 FITs
- Out-of-band emissions in Inmarsat-E band: <-27 dB(W/kHz)
- Out-of-band emissions in Cospas-Sarsat band: < -48 dB(W/kHz), <-58.2dB(W/kHz)
- Nominal Operations from switch on or safe mode: 1 day

SAR Transponder Gain Control: the SAR transponder shall have two operational modes, controllable by telecommand:

- Automatic Gain Control
- Fixed Gain: in that case, the Gain can be varied in 1 dB steps over a 30 dB range.

SSF8 Functional Block Diagram

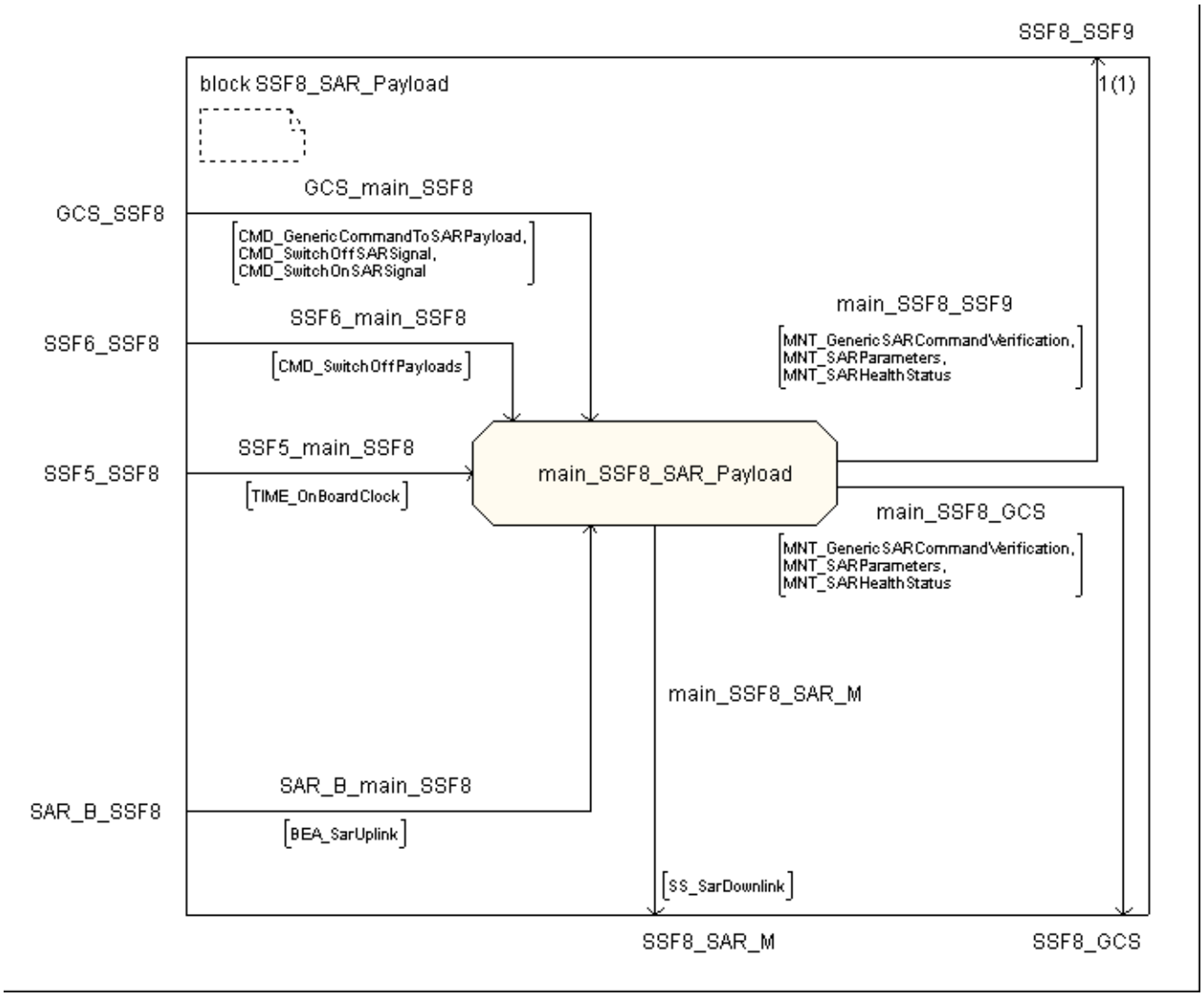


Figure 189: SSF8 SDL Functional Block Diagram

5.4.9 SSF9 Provide TTC Communication

SSF9 Description

This function is provides the spacecraft TC/TM data handling, TTC RF and platform security functions. In particular:

- Support communication with the ground through provision of TC uplink and TM downlink functionality
- Management of time tagged TCs and TC buffer
- Management of RF operational mode (ie Spread Spectrum, ESA Standard, coherent, non-coherent) and RF performance characteristics
- Providing the functions required for the S-band TC/TM security layer (ie COMSEC TC/TM) including support for ESA authentication (in CLEAR mode)
- Management of the S-band security layer through provision of STC/STM functionality (ie COMSEC STC/STM)
- Supporting the TC/TM protocol and associated services in accordance with [AD/9] and [AD/24] Galileo IOV PUS definition, GAL-ICD-ASTD-SS-A-0028
- Supporting S-band 2-way ranging
- Management of on-board packet stores (ie TM archive).

SSF9 Functional Block Diagram

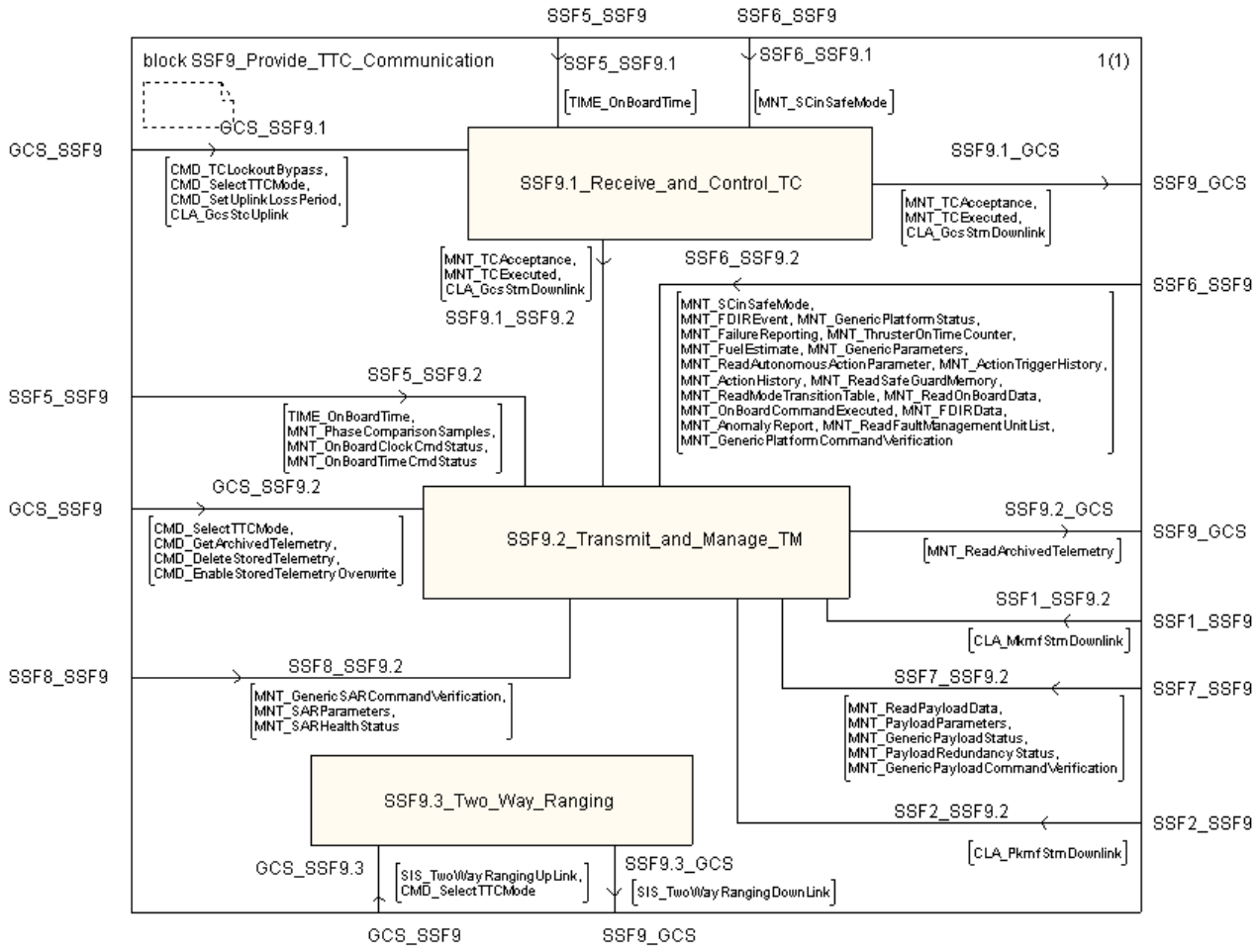


Figure 190: SSF9 SDL Functional Block Diagram

5.4.9.1 SSF9.1 Receive and Control TC

SSF9.1 General Description
This function is responsible for:

- TC Data Handling Functions:
  - Management of time tagged TCs and TC buffer
  - Arm and Fire management for potentially hazardous TCs
  - Supporting the TC protocol and associated services in accordance with [AD.09] (including support for ESA authentication).
- TTC Uplink RF Functions:
  - Reception of all TC uplinked from the ground (ie amplification, demodulation, decoding etc)
  - Management of uplink RF operational modes (ie Spread Spectrum and ESA standard) and associated parameters (eg PN codes).
  - Supporting uplink RF performance in accordance with [AD.09]
- Platform Security Functions:
  - Providing the functions required for the S-band TC/TM security layer - uplink and downlink (ie COMSEC TC/TM)
  - Management of the S-band security layer through provision of STC/STM functionality - uplink and downlink (ie COMSEC STC/STM)

This function covers all the satellite security functions required to fulfill the security SEC-SFC12 part related to space segment S-Band protection. Refer to this SEC-SFC for further details and traceability.

Note: this function is to be linked to GCSF9.1 related to security protection of the S-Band link at GCS level

SSF9.1 outputs
This function is responsible for producing the following outputs:

- Signals to provide unambiguous verification of the acceptance and execution of all TCs (sent from ground or generated on-board)
- Secure telemetry (STM) from the platform security unit.

SSF9.1 Design Constraints
This function will support the following design / performance constraints:

- TC Data Handling Functions:
  - All timing functions will be synchronised to the on-board time with an accuracy better than 1 ms.
  - The TC buffer will be consistent with the worst case assumption of a ground contact only once every two orbits
  - Support routine TT&C communications in a time slot not longer than 60 minutes once every orbit
  - Support special operations TT&C communications (eg SW patch, recovery operations, autonomy reconfiguration) in a time slot not longer than 4 hours excluding ground reaction time
- TTC Uplink RF Functions:
  - The combined radiation pattern of the TT&C antennas will allow the TC link to be maintained for 100% of off-axis angles of each antenna between 0 and 75 degrees

and for 90% of off-axis angles of each antenna between 75 and 90 degrees for all azimuth angles between 0 and 360 degrees.
    - The TC eception chain will support hot-redundancy such that no single failure shall lead to the loss of satellite commanding capability
    - Availability of all TC functions during all mission phases, including pre-launch and LEOP
    - Allow the setting the S-band loss of link period to any value between 100 minutes and ten orbits
- Platform Security Functions:
- The Platform Security functions will support internal hot redundancy and cross-strapping.

SSF9.1 Functional Block Diagram

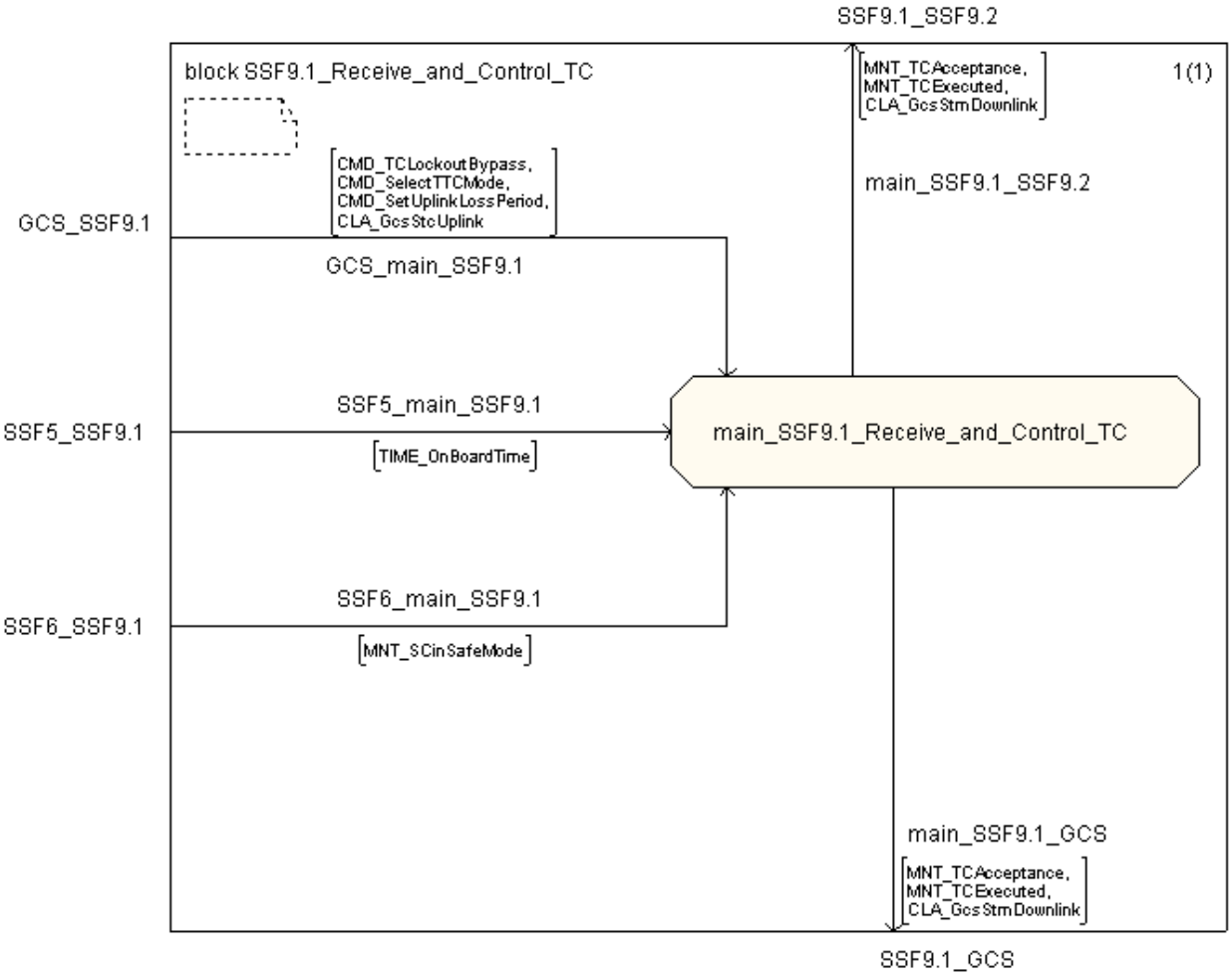


Figure 191: SSF9.1 SDL Functional Block Diagram

5.4.9.2 SSF9.2 Transmit and Manage TM

SSF9.2 General Description
This function is responsible for the following:

- TM Data Handling Functions:
  - Management of TM packet stores (eg TM archiving when out of TT&C contact)
  - Supporting the TC/TM protocol and associated services in accordance with [AD/9] and [AD/24]
- TTC Downlink RF Functions:
  - Supporting downlink RF performance in accordance with [AD/9]
  - Management of downlink RF operational modes (ie Spread Spectrum and ESA standard) and associated parameters (eg PN codes).

SSF9.2 outputs
This function is responsible for producing the following output signals:

- Downlink of telemetry archived in the on-board packet stores.

SSF9.2 Design Constraints
This function will support the following design / performance constraints:

- TM Data Handling Functions:
  - All timing functions (telemetry time-stamping) will be synchronised to the on-board time with an accuracy better than 1 ms.
  - A TM archive capacity sized to be consistent with the worst case assumption of a ground contact frequency of only once every two orbits, at the nominal TM sampling rate.
  - Support routine TT&C communications in a time slot not longer than 60 minutes once every orbit
  - Support special operations TT&C communications in a time slot not longer than 4 hours excluding ground reaction time
- TTC Downlink RF Functions:
  - The combined radiation pattern of the TT&C antennas will allow the TM link to be maintained for 100% of off-axis angles of each antenna between 0 and 75 degrees and for 90% of off-axis angles of each antenna between 75 and 90 degrees for all azimuth angles between 0 and 360 degrees.
  - Availability of all TM functions during all mission phases, including pre-launch and LEOP
  - Supporting the following EIRP spectral density limits:
  - Less than -9.6 dBW in any 1 kHz frequency band in any direction at any time in the GPS bands (1 215.60 - 1 239.60 MHz)
  - Less than -15.1 dBW in any 1 kHz frequency band in any direction at any time in the Glonass bands (1 237.8275 - 1 252.2220 MHz and 1 592.9575 - 1 610.0000 MHz)
  - Less then - 70 dBW in any 1 kHz frequency band at any time, in any direction, at any frequency from 10 MHz up to 32 GHz, except within the frequency range 1129.450 - 1323.75 MHz and 1529.052 - 1621.788 MHz, and within 50 MHz of the TTC downlink frequency.
  - Less than -48 dB(W/kHz) in the 1544.24 - 1544.40 MHz and 1544.60 - 1544.76 MHz frequency bands (Cospas-Sarsat)
  - Less than -58.2 dB(W/kHz) in the 1544.40 - 1544.60 MHz frequency band (Cospas-Sarsat)

- Less then -27.0 dB(W/kHz) in the band 1 544.20 - 1 544.24 MHz (Inmarsat-E service band)
- Limit the EIRP spectral density in radioastronomy and passive service bands according to the table specified in SSREQ-1725

SSF9.2 Functional Block Diagram

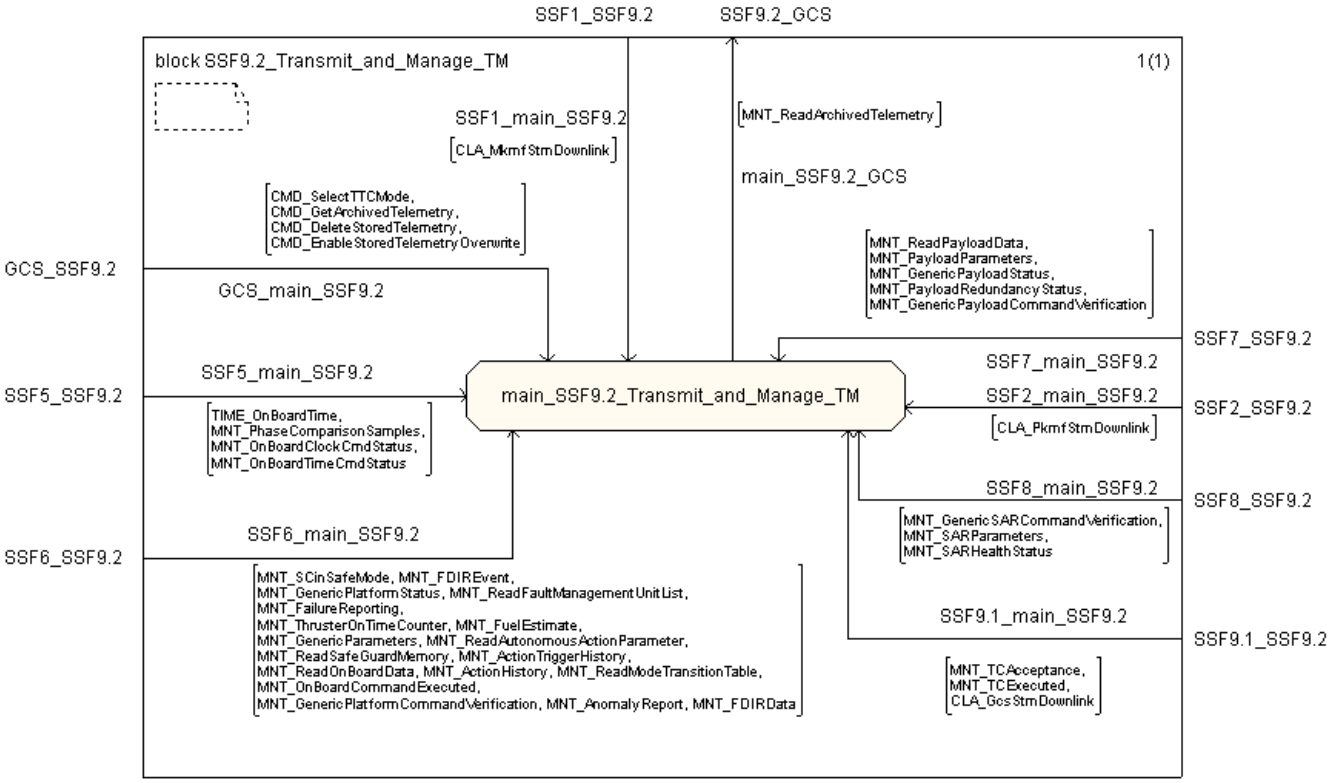


Figure 192: SSF9.2 SDL Functional Block Diagram

5.4.9.3 SSF9.3 Two Way Ranging

SSF9.3 General Description

This function is responisble for providing the spacecraft two-way ranging capability. In particular, this function is responsible for the reception and re-transmission of two-way ranging signals from / to the Galileo Ground Segment and the management of transponder operational mode for ranging (ie coherent transponder mode for ranging). This function enables support for two-way ranging using Standard TTC Mode, compatible with both code and tone ranging techniques, and using spread spectrum mode (as defined in [AD/9]).

SSF9.3 outputs

This function is responsible for producing the following output signals:

- Re-transmission of the two-way ranging signals to the Galileo Ground Segment

SSF9.3 Design Constraints

This function will support the following design / performance constraints:

- The residual uncalibrated delay of the TTC subsystem shall not be greater than 3ns 1-sigma, for a time not longer than 1 hour (TBC). This residual delay is defined between the I/O ports of the diplexer.
- Function availability during all mission phases, including pre-launch and LEOP.

SSF9.3 Functional Block Diagram

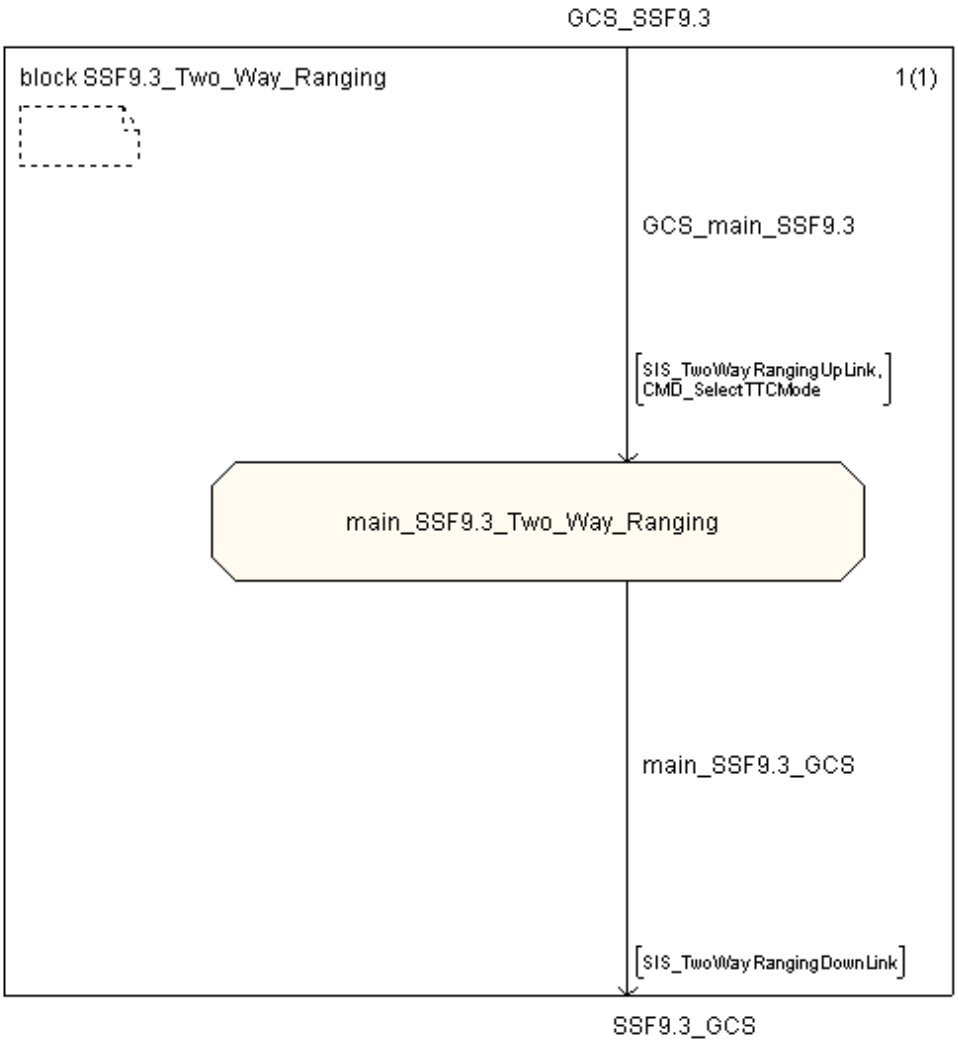


Figure 193: SSF9.3 SDL Functional Block Diagram

5.4.10 SSF10 Optical Ranging

SSF10 General Description
This function is responsible for providing laser retro-reflector for use in precise orbit determination., with a, to be used:In routine (Earth-pointing mode) for precise orbit determination (calibration need).
SSF10 outputs
This function is responsible for producing the following output signals:

- The laser signal reflected from the satellite laser reflector.

SSF10 Design Constraints
This function will support the following design / performance constraints:

- Providing a minimum aggregate effective reflecting surface area, viewed from any point on the Earth, of 660 square centimeters

SSF10 Functional Block Diagram

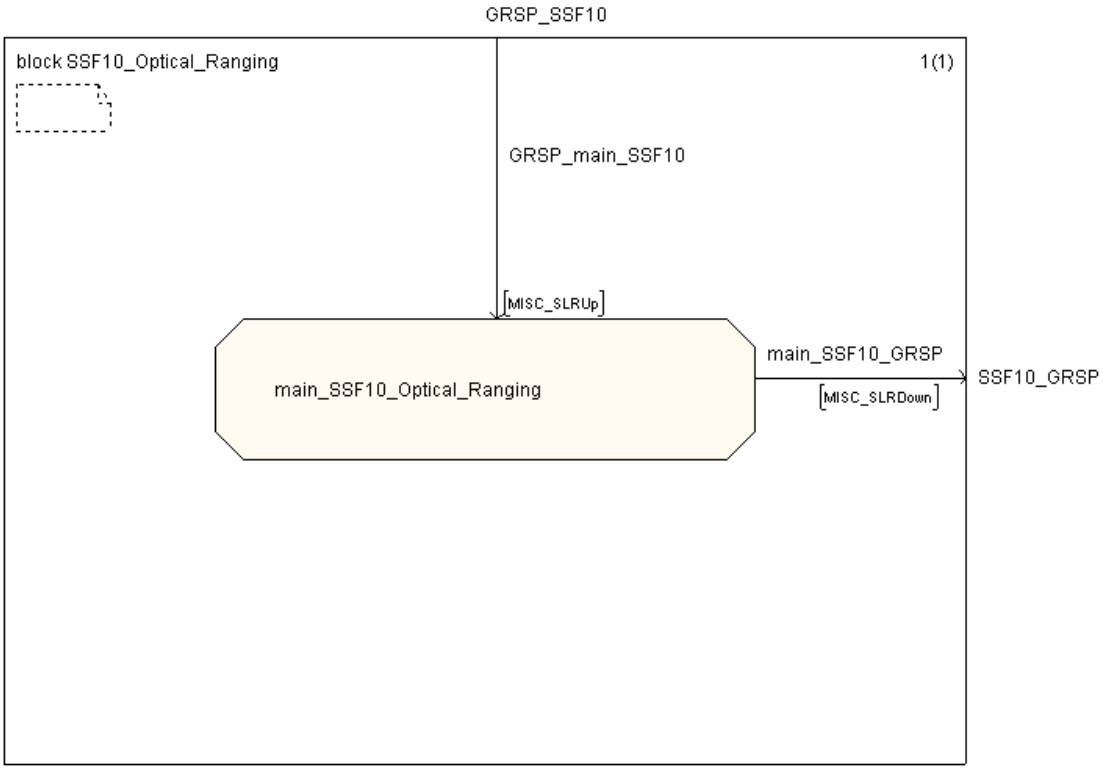


Figure 194: SSF10 SDL Functional Block Diagram

5.5 TUS Functions

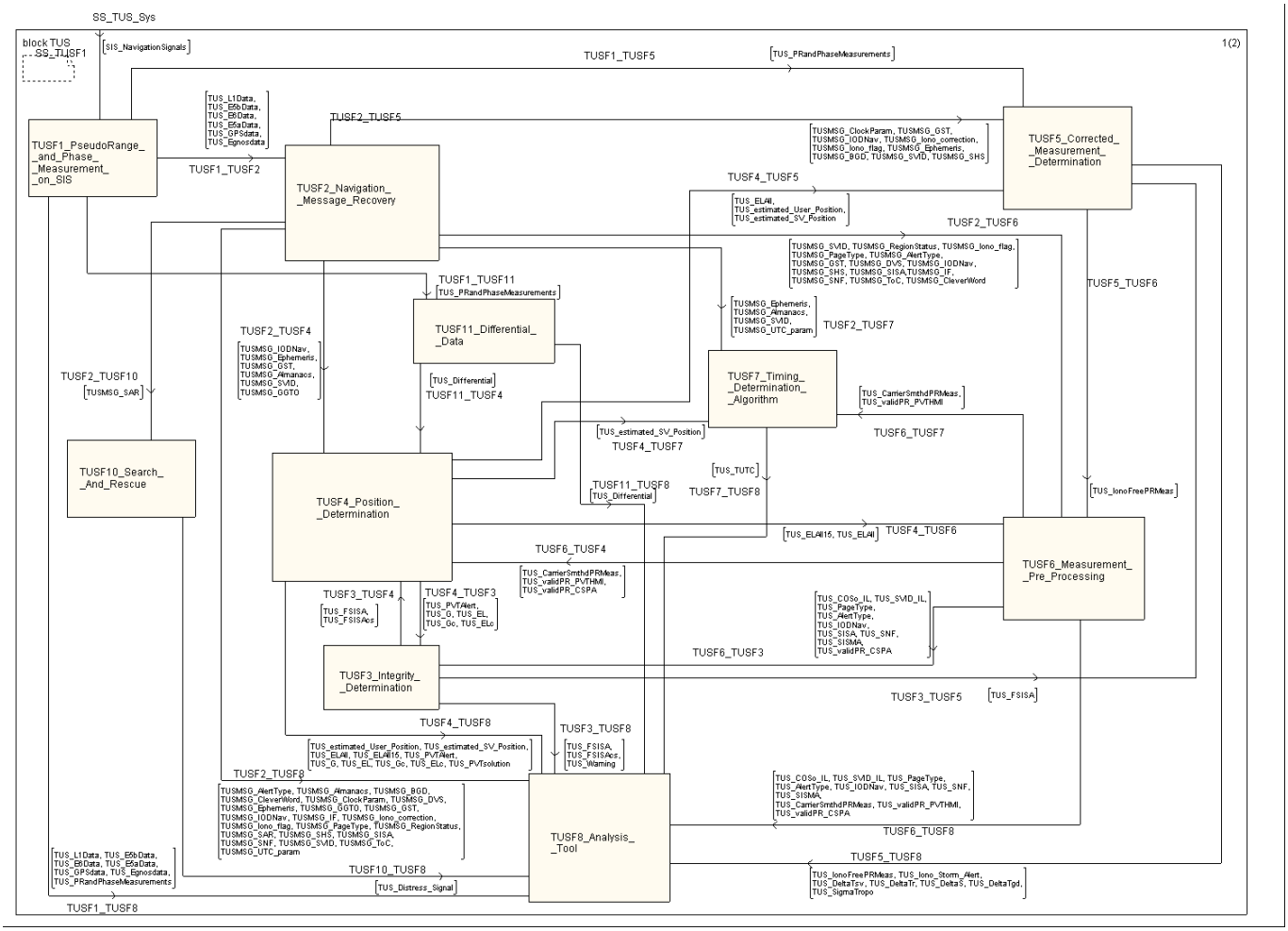


Figure 195: TUS SDL Functional Block Diagram

5.5.1 TUSF1 PseudoRange and Phase Measurement on SIS

TUSF1 General Description

This function is in charge of performing pseudo-range measurements on the SIS ranging codes and pseudo-phase measurements on the SIS carrier.
The following tasks are executed:

- Generation of reference clock
- Acquisition of SIS
- Carrier tracking (including Doppler effect estimation)
- Carrier phase measurement
- Ranging code tracking
- Ranging code measurement
- Multipath mitigation
- Interference mitigation
- Cycle-slip detection and correction

TUSF1 Outputs

- Raw measurements for each tracked signal
- Navigation data from received signal

TUSF1 Design constraints/Performances

This function shall:

- Track all Galileo satellites above an elevation angle of 10 degrees simultaneously meeting all performance specifications
- Generate its own reference timing or use external reference timing
- Be able to acquire any signal without aiding information (eventually with encrypted code)
- Solve for code length ambiguity in the pseudoranges
- Have navigation signal and channels flexibility
- Compute the carrier Doppler data once per second

TUSF1 Functional Block Diagram

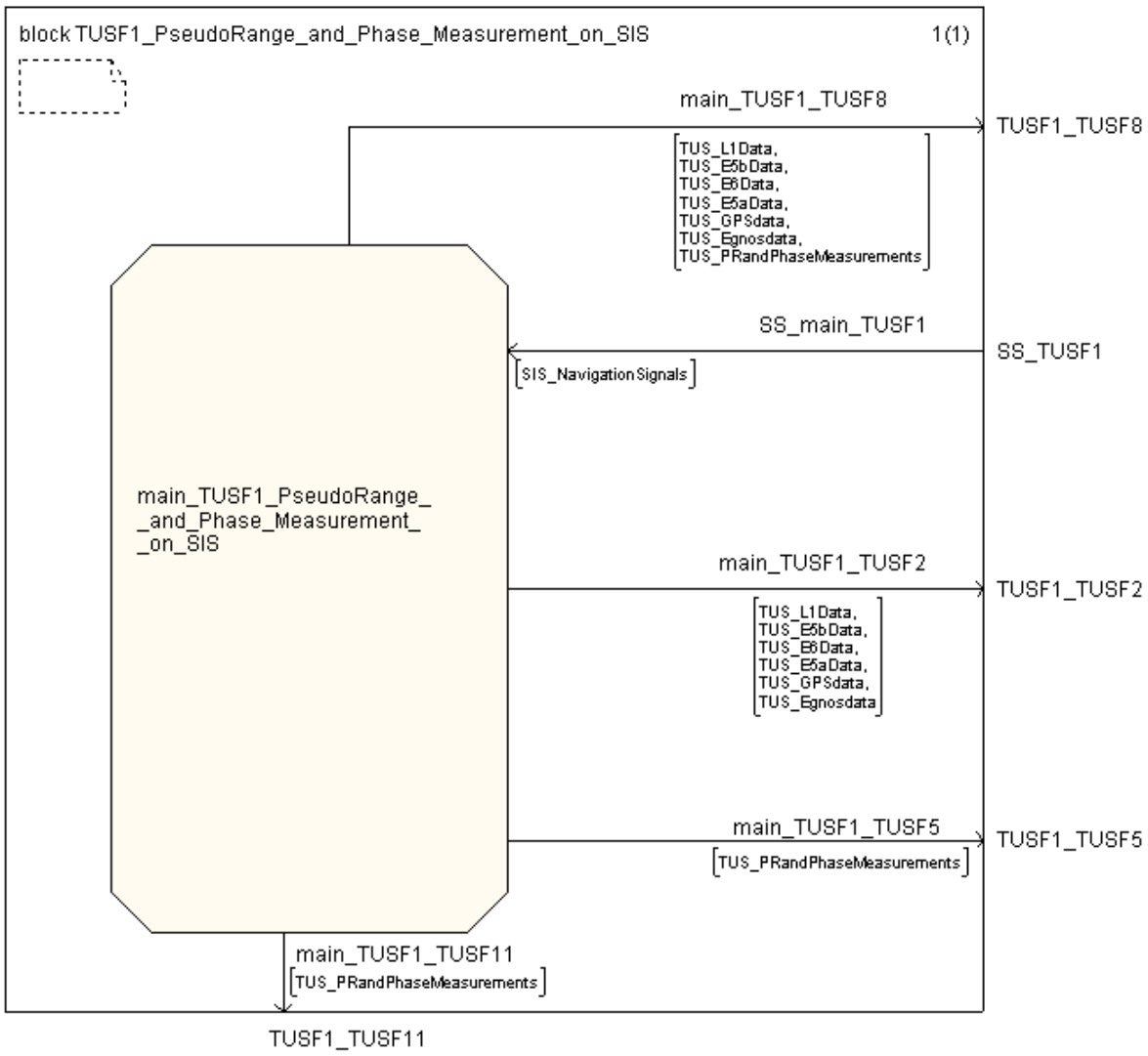


Figure 196: TUSF1 SDL Functional Block Diagram

5.5.2 TUSF2 Navigation Message Recovery

TUSF2 General Description

This function is in charge of recovering the navigation data transmitted by the Navigation Signals (satellite parameters, satellites flags…).
The following tasks are executed:

- Bit synchronisation
- Frame synchronisation
- Frame Demultiplexing
- Symbol recovery
- CRC check
- Viterbi decoding
- De-interleaving
- Bit recovery

TUSF2 Outputs

- Signal Health Status
- Navigation data Validity Status
- Galileo/GPS time offset
- Navigation data (including GPS/Egnos data for adequate configuration)

TUSF2 Design constraints/Performances

This function shall:

- Identify subframes tagged with satellite identifier and frequency designator
- Have message services flexibility
- Decrypt CS data, if present

TUSF2 Functional Block Diagram

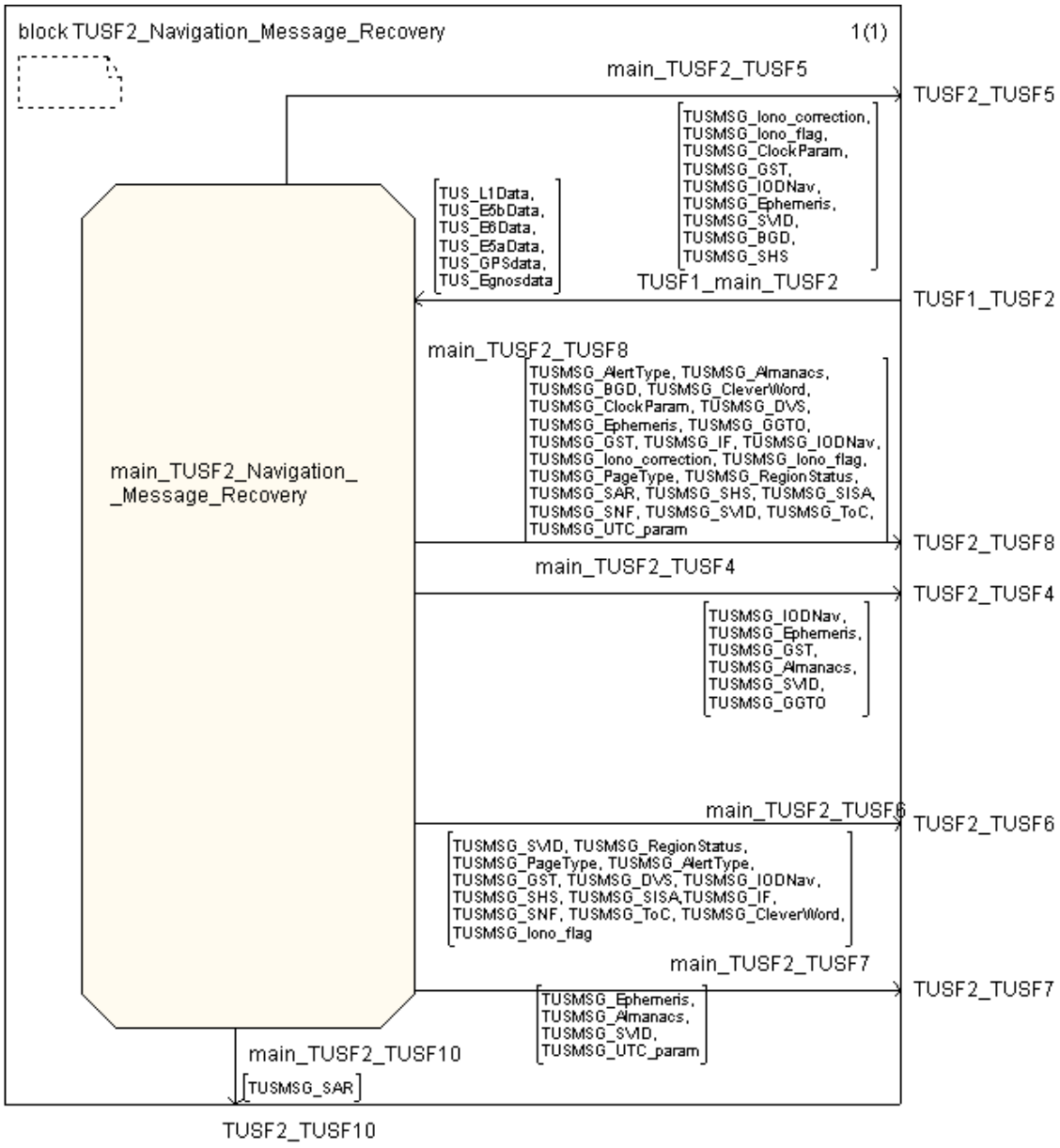


Figure 197: TUSF2 SDL Functional Block Diagram

5.5.3 TUSF3 Integrity Determination

Deleted

5.5.4 TUSF4 Position Determination

TUSF4 General Description
The Position Determination Algorithm is in charge of elaborating the navigation solution (position, velocity, and time) based on valid Pseudorange Measurements.

TUSF4 Outputs

- User Position in GTRF, NED and ellipsoidal, latitude, longitude and altitude coordinates,
- Estimates of Galileo time,
- PVT Solution Algorithm alert indicating form PVT Algorithm fails to compute PVT solution,
- Geometric Matrix at current epoch (cosine directors matrix) in the selected local navigation frame (NED)
- Elevation Angles for the set of satellites
- Elevation Angles for all satellites for troposphere correction function

TUSF4 Design constraints/performances
This function shall:

- Provide PVT solution with a selectable output rate of 1Hz or 10Hz
- Be able to compute navigations solutions using valid measurements from Galileo and/or GPS satellites

TUSF4 Functional Block Diagram

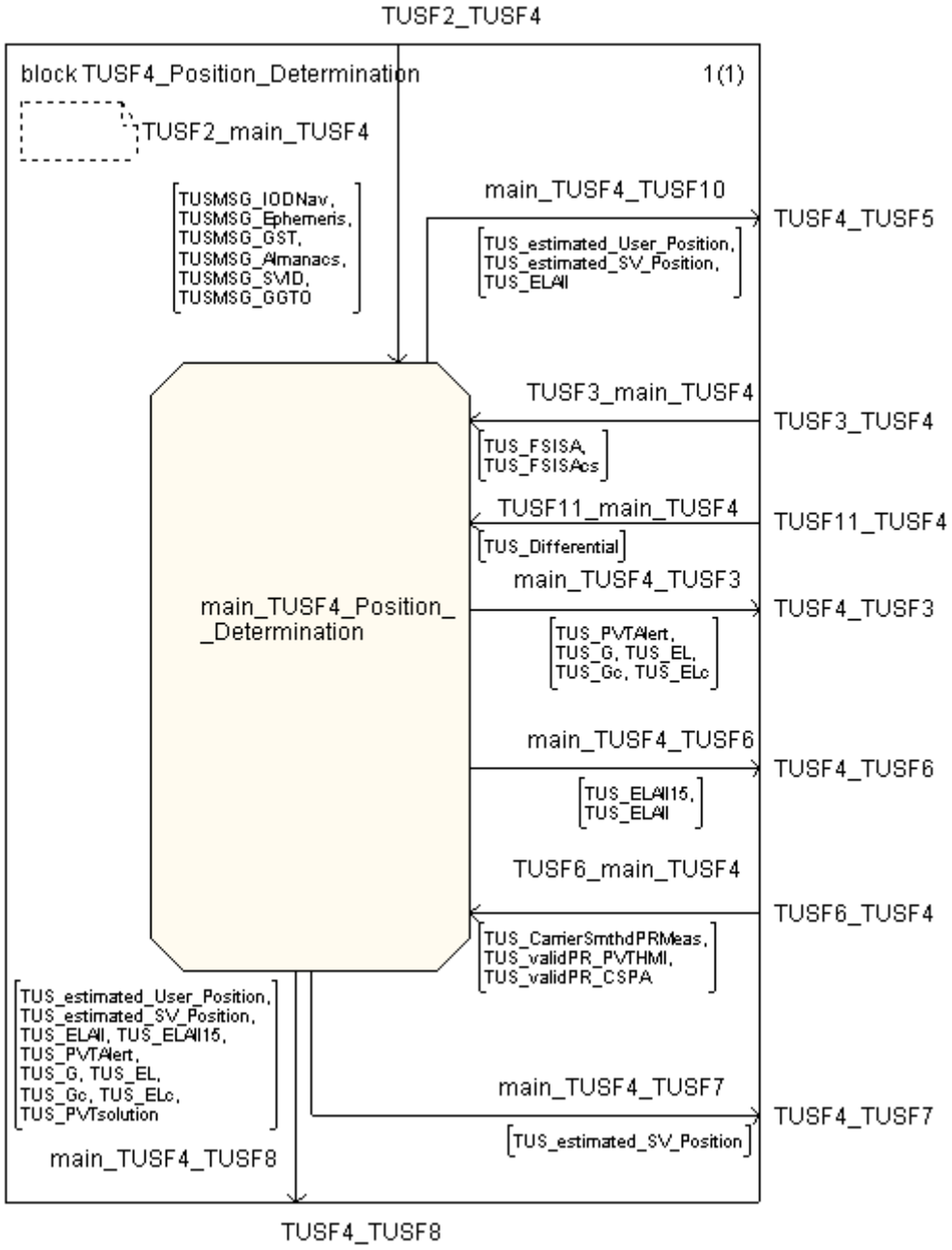


Figure 198: TUSF4 SDL Functional Block Diagram

5.5.5 TUSF5 Corrected Measurements determination

TUSF5 Description

This function shall correct pseudorange measurements from transmission errors, including:

- Satellite clock synchronization error (with Group delay for single frequency configurations)
- Relativistic effects
- Troposphere effects
- Sagnac effect
- Ionosphere effects

TUSF5 Outputs

Unfiltered valid code phase measurement data and unfiltered valid carrier phase measurement data

TUSF5 Functional Block Diagram

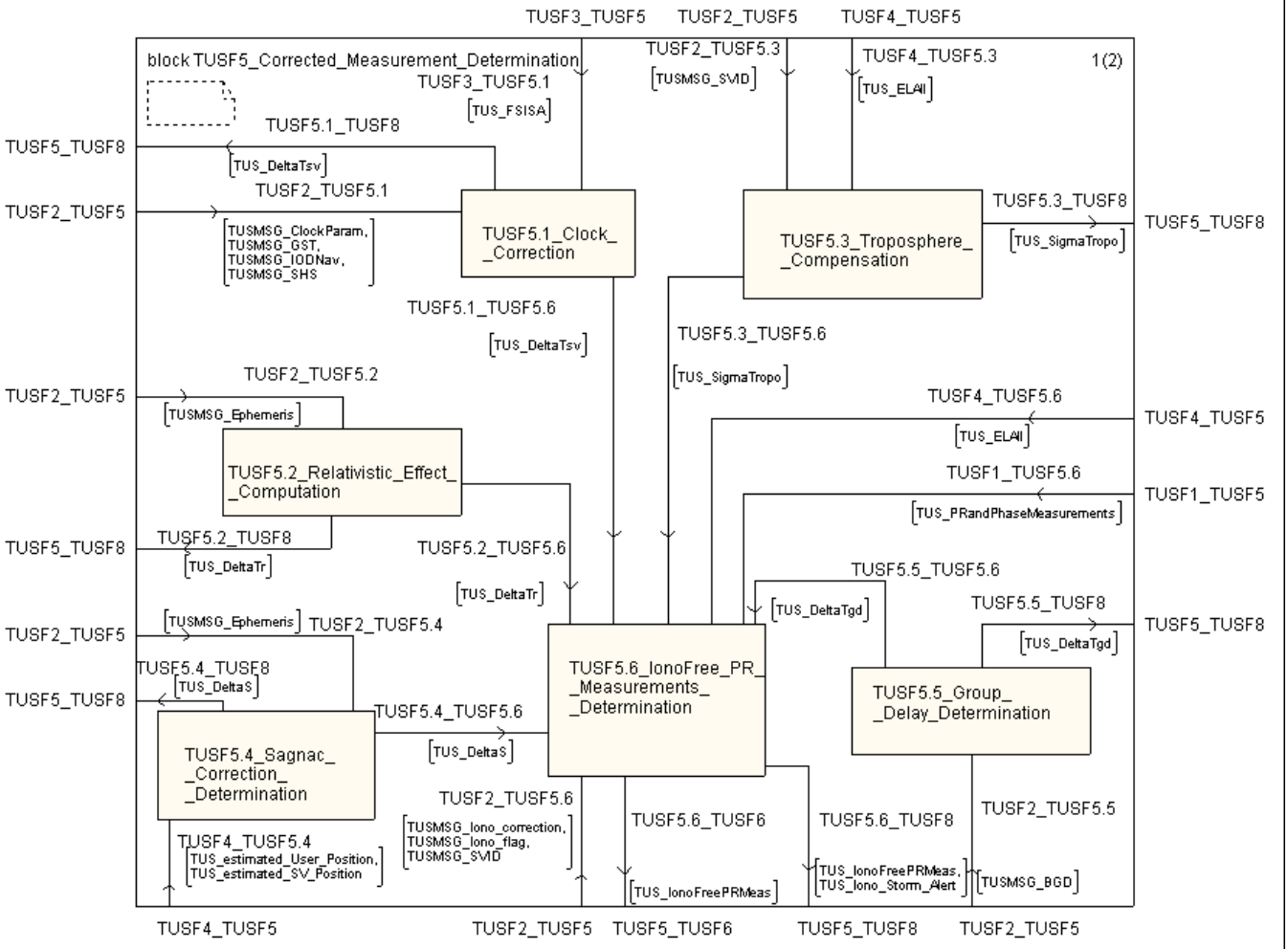


Figure 199: TUSF5 SDL Functional Block Diagram

5.5.5.1 TUSF5.1 Clock Correction

TUSF5.1 General Description

The Satellite Clock Correction Computation Algorithm shall compute, for each visible (healthy) Galileo satellite, the satellite clock offset wrt GST using the relevant clock model parameters broadcast through the navigation message.

TUSF5.1 Outputs

- E5a Satellite clock correction, tSV(E5a), for all visible Galileo satellites
- E5b Satellite clock correction, tSV(E5b), for all visible Galileo satellites
- E1 (Channel B/C) Satellite clock correction, tSV(E1), for all visible Galileo satellites
- E6 (Channel A) Satellite clock correction, tSV(E6A), for all visible Galileo satellites
- E1 (Channel A) Satellite clock correction, tSV(E1A), for all visible Galileo satellites

TUSF5.1 Design constraints/performances

The Satellite Clock corrections with respect to GST shall then be computed using the algorithm specified in Appendix B of TUSREQ

TUSF5.1 Functional Block Diagram

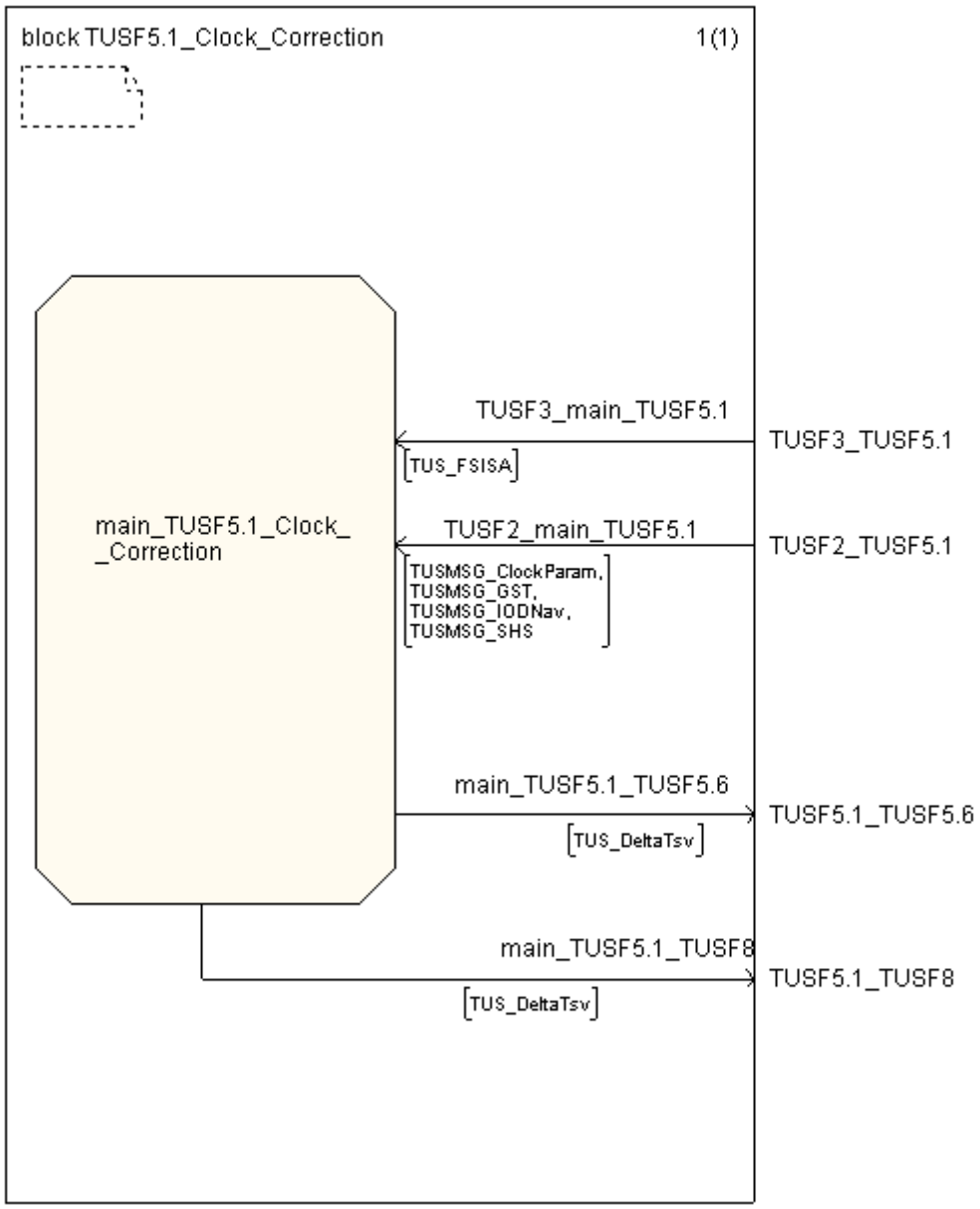


Figure 200: TUSF5.1 SDL Functional Block Diagram

5.5.5.2 TUSF5.2 Relativistic Effect Computation

TUSF5.2 General Description
Test User Receiver computes, for each visible Galileo satellite, the satellite signal time of transmission correction for the relevant clock, as required to remove the relativistic effects due to orbit eccentricity.

TUSF5.2 Outputs
Satellite clock relativistic correction Delta Tr

TUSF5.2 Design constraints/performances
The relativistic effect corrections shall be computed using the algorithm specified in Appendix A of TUSREQ.

TUSF5.2 Functional Block Diagram

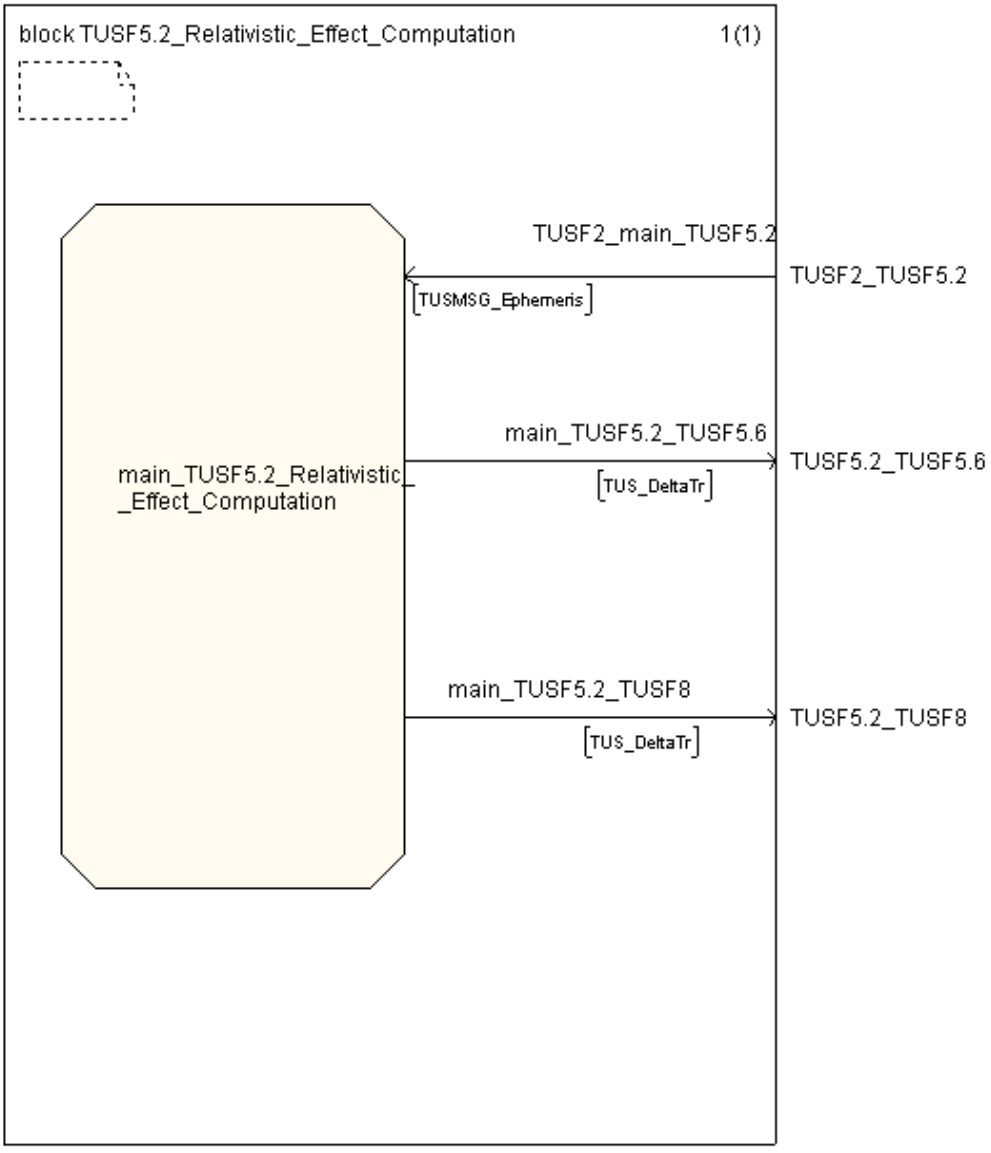


Figure 201: TUSF5.2 SDL Functional Block Diagram

5.5.5.3 TUSF5.3 Troposphere Compensation Computation

TUSF5.3 General Description
Test User Receiver estimates for each visible Galileo satellite the range error component due to troposphere effects. The model to be applied shall be user selectable

TUSF5.3 Outputs
Range error component due to troposphere effects Sigma Tropo

TUSF5.3 Design constraints/performances
The Galileo Test User Receiver Troposphere Correction Algorithm shall compensate for tropospheric delay to within 5% of the actual delay or 2 cm equivalent vertical error, whichever is greater, when operating in the specified environmental conditions (see Appendix N of TUSREQ).

TUSF5.3 Functional Block Diagram

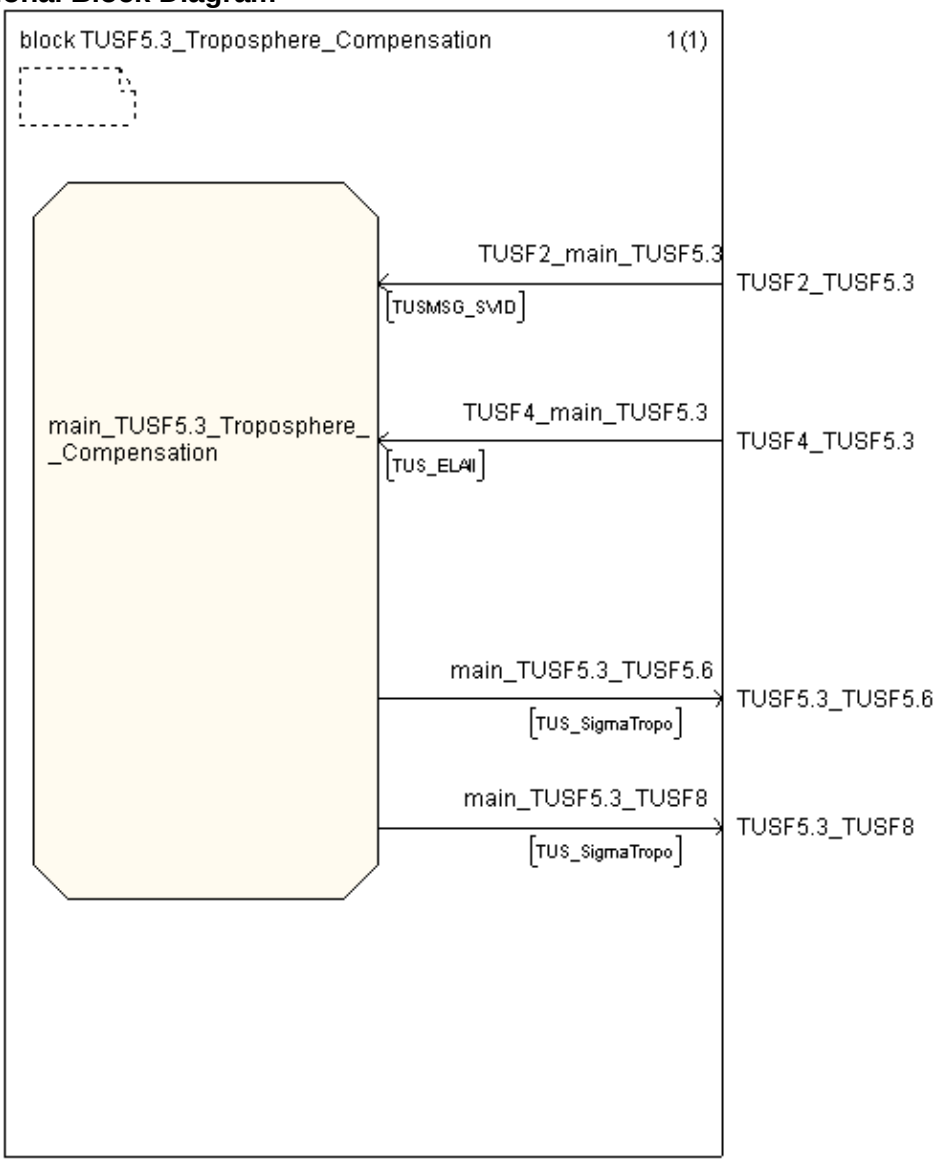


Figure 202: TUSF5.3 SDL Functional Block Diagram

5.5.5.4 TUSF5.4 Sagnac Correction Determination

TUSF5.4 General Description

The Geometrical Range Correction Algorithm shall determine the Geometrical Range (or Sagnac effect) correction to be applied to the received signal to take into account the fact that the receiver moves during the satellite-receiver transit time, as the Earth spins.

TUSF5.4 Outputs

Geometric Range corrections for each visible satellite Delta S

TUSF5.4 Design constraints/performances

Geometrical Range (or Sagnac effect) correction shall be computed using the algorithms specified in Appendix D.

TUSF5.4 Functional Block Diagram

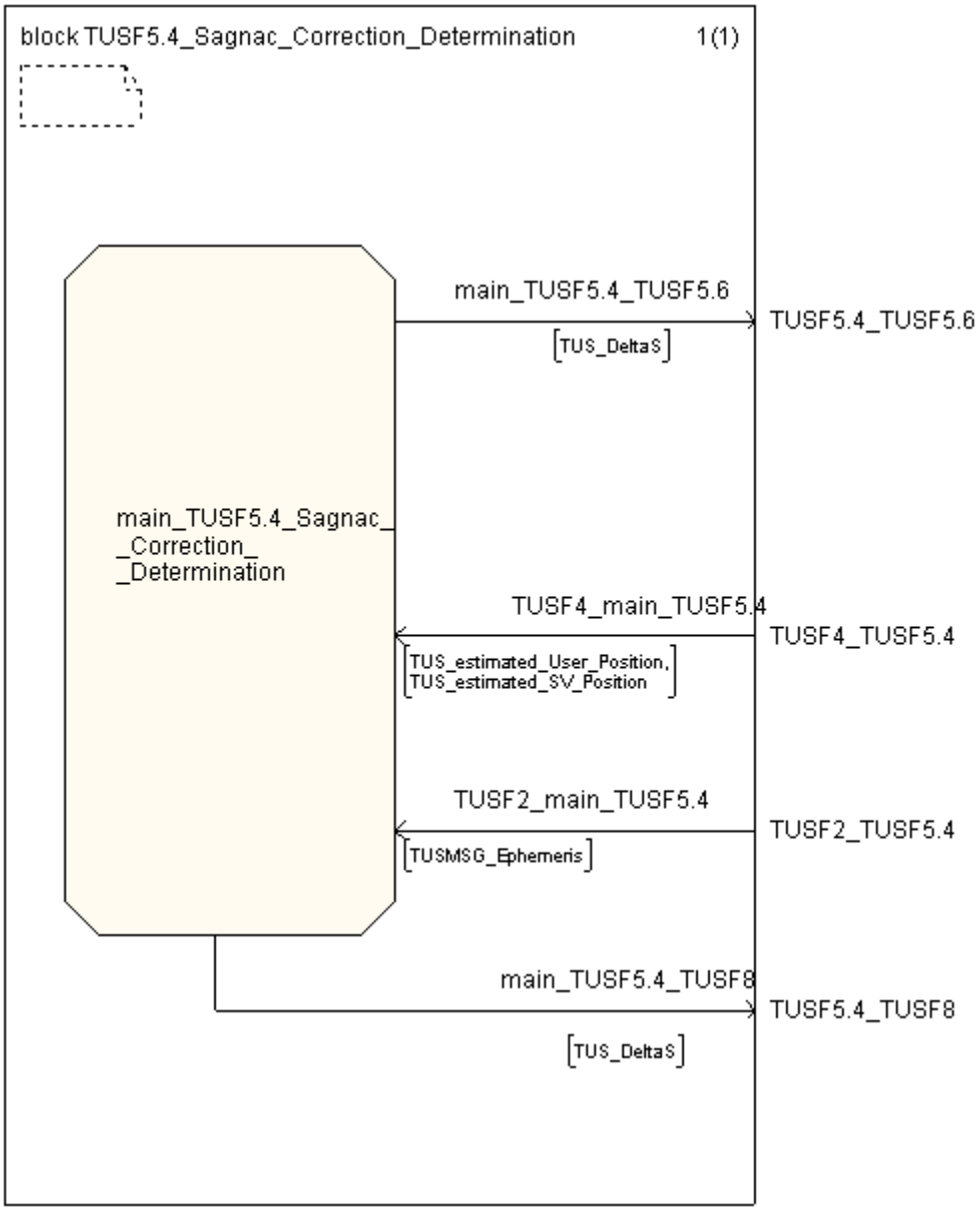


Figure 203: TUSF5.4 SDL Functional Block Diagram

5.5.5.5 TUSF5.5 Group Delay Determination

TUSF5.5 General Description
The Satellite Group Delay Correction Algorithm shall allow the Single Frequency TUR to determine the satellite group delay corrections, for all visible Galileo satellites and relevant Ei frequencies.

TUSF5.5 Outputs
Satellite interfrequency bias correction Delta T_{GD}

TUSF5.5 Design constraints/performances
The satellite interfrequency bias correction shall be computed according to appendix C of TUSREQ.

TUSF5.5 Functional Block Diagram

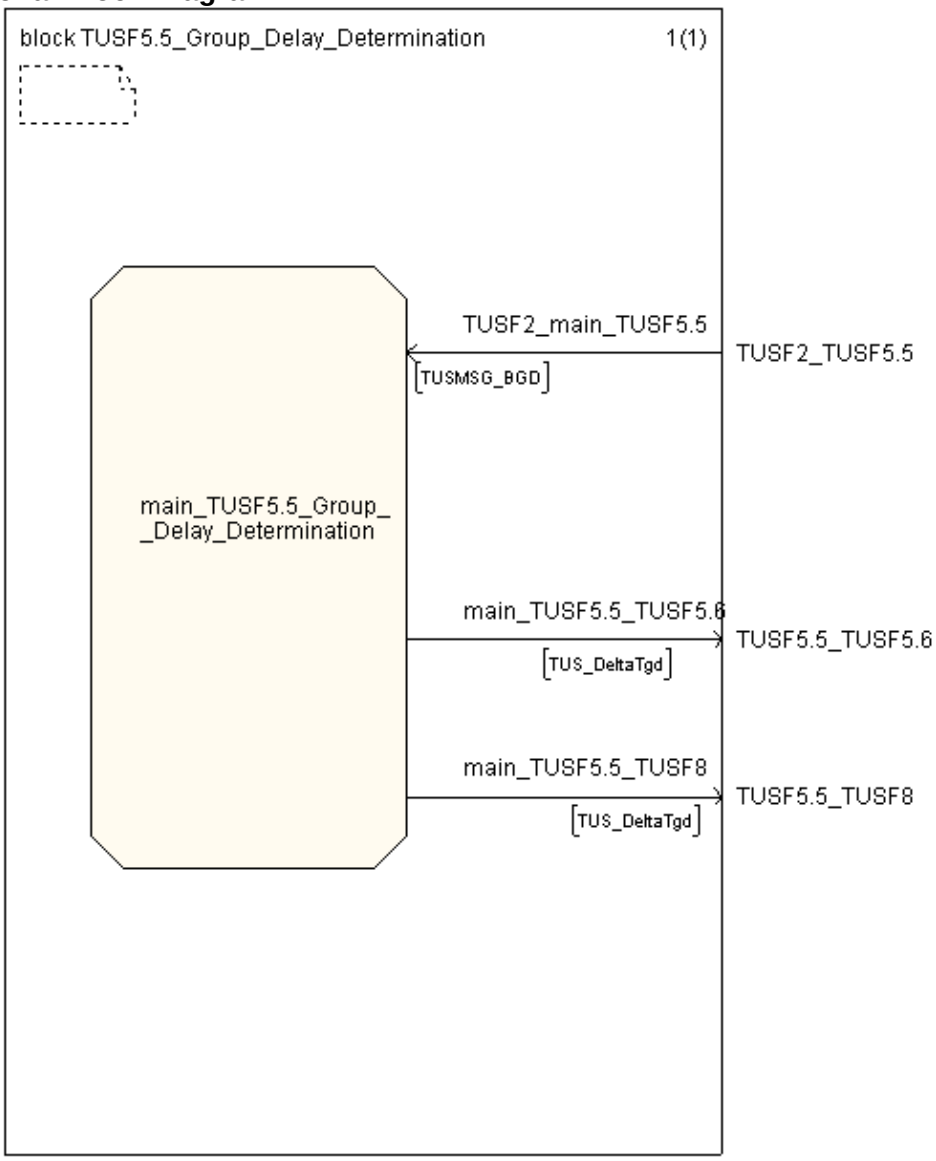


Figure 204: TUSF5.5 SDL Functional Block Diagram

5.5.5.6 TUSF5.6 IonoFree PR Measurements Determination

TUSF5.6 General Description

Test User Receiver estimates for each visible Galileo satellite the range error component due to ionosphere effects.
The Galileo Dual Frequency Test User Receiver shall compensate for ionospheric delays using dual frequency measurements.
The Galileo Single Frequency Test User Receiver shall compensate for ionospheric delays by using the single-frequency model whenever the received storm flag is set to "Normal Ionosphere Condition" status.

TUSF5.6 Outputs

The Single Frequency Iono Correction Algorithm shall generate a Storm Alert, whenever the received storm flag is set to "Storm IONO Condition" status.
This function shall compensate for ionospheric delays.

TUSF5.6 Design constraints/performances

Performances shall be met whenever ionosphere conditions do not exceed limits specified in Appendix M of TUSREQ (240TECU).

TUSF5.6 Functional Block Diagram

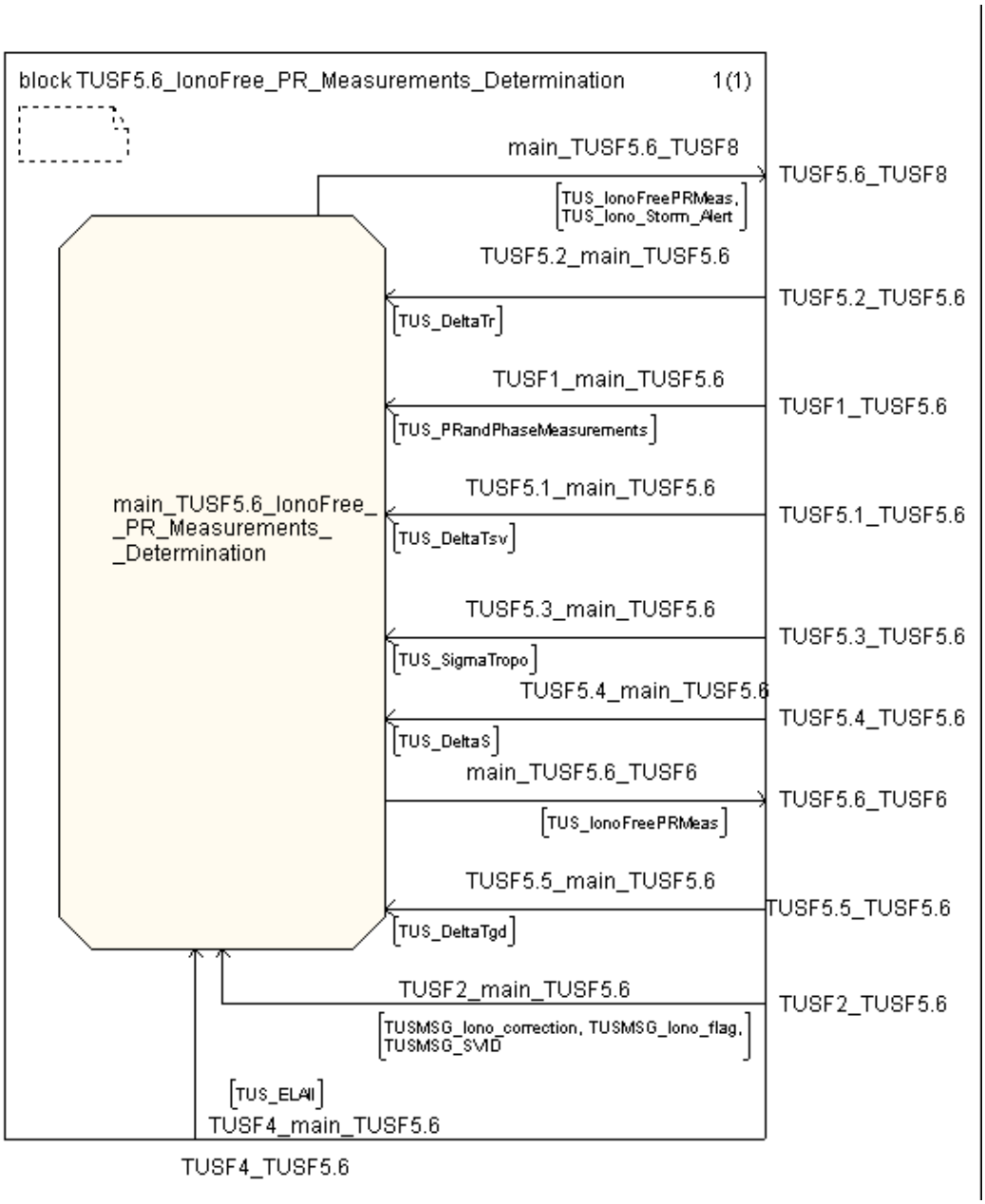


Figure 205: TUSF5.6 SDL Functional Block Diagram

5.5.6 TUSF6 Measurement Pre Processing

TUSF6 Description

This function is responsible to filter the iono-free PR measurements

TUSF6 Functional Block Diagram

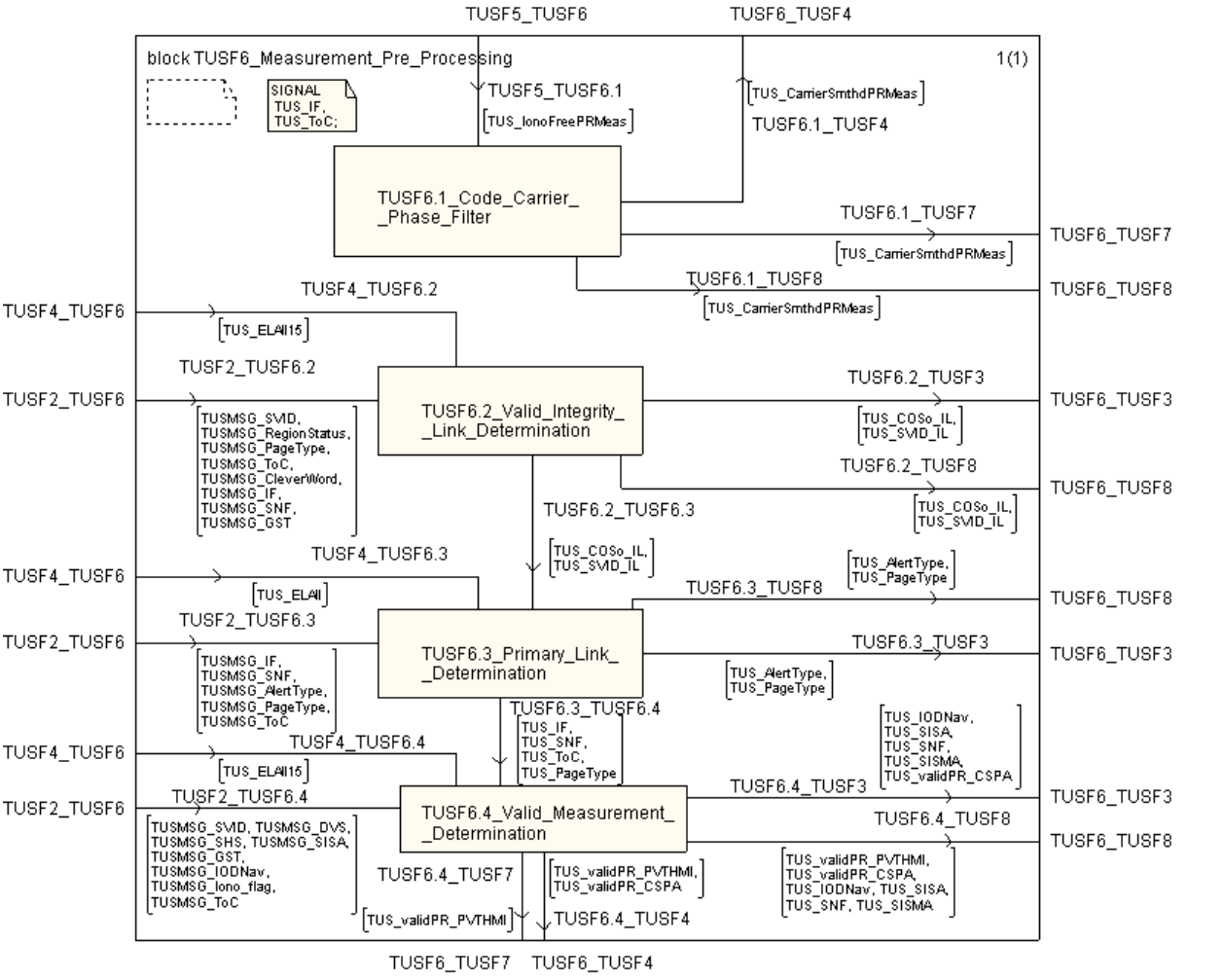


Figure 206: TUSF6 SDL Functional Block Diagram

5.5.6.1 TUSF6.1 Code Carrier Phase Filter

TUSF6.1 General Description
This function performs smoothing of the carrier phase measurements to mitigate the effects of random error sources on satellite pseudorange measurements.

TUSF6.1 Outputs
- Single frequency carrier smoothed code phase PR measurement data or
- E1A-E6A carrier smoothed code phase PR measurement data or
- E1-E5a carrier smoothed code phase PR measurement data or
- E1-E5b carrier smoothed code phase PR measurement data

TUSF6.1 Design constraints/performances
None

TUSF6.1 Functional Block Diagram

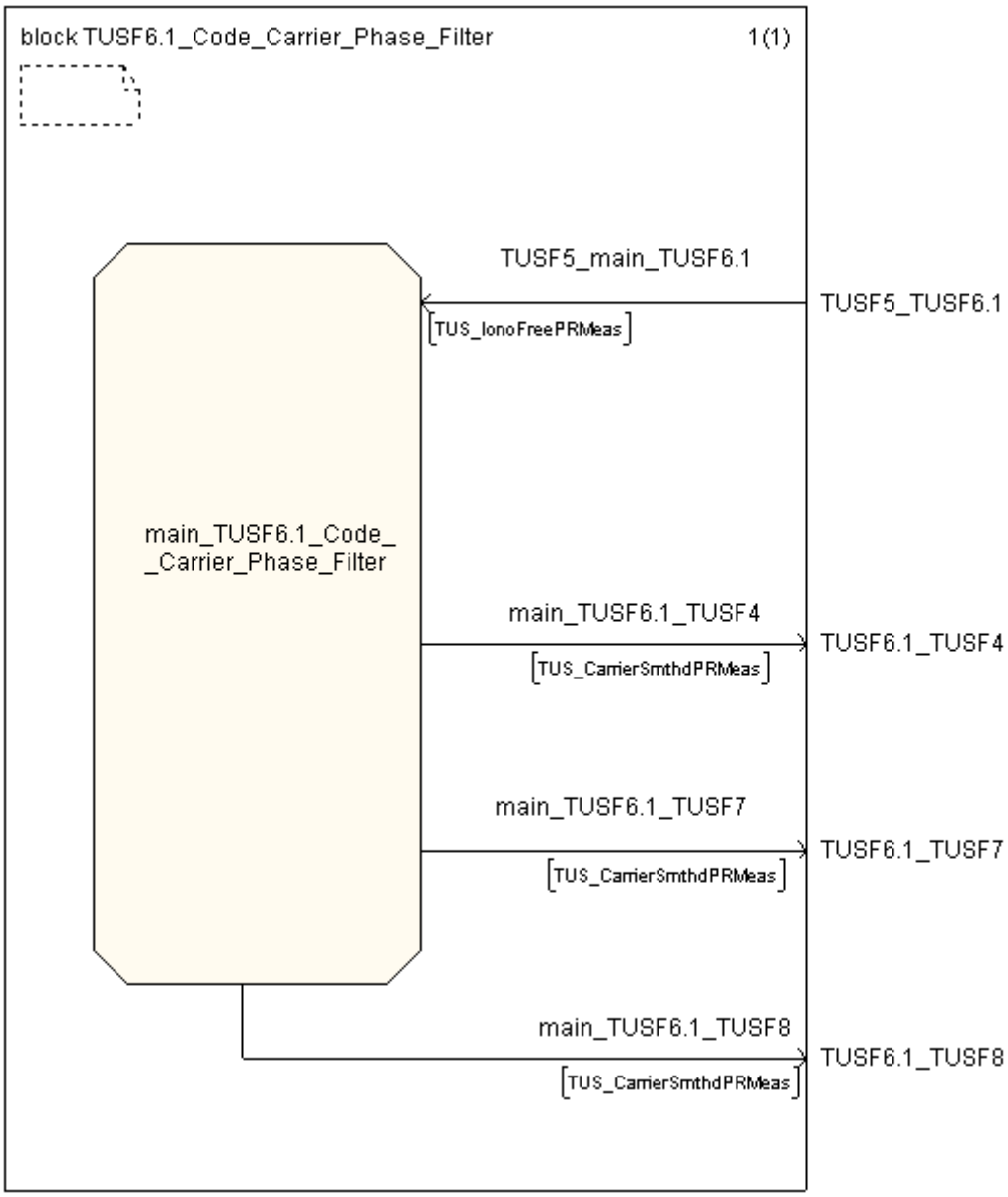


Figure 207: TUSF6.1 SDL Functional Block Diagram

5.5.6.2 TUSF6.2 Valid Integrity Link Determination

Deleted

5.5.6.3 TUSF6.3 Primary Link Determination

Deleted

5.5.6.4 TUSF6.4 Valid measurement Determination

TUSF6.4 General Description
This function determines the Valid Pseudorange Measurements for PVT computation.

TUSF6.4 Outputs
- Valid Pseudorange Measurements for PVT computation

TUSF6.4 Functional Block Diagram

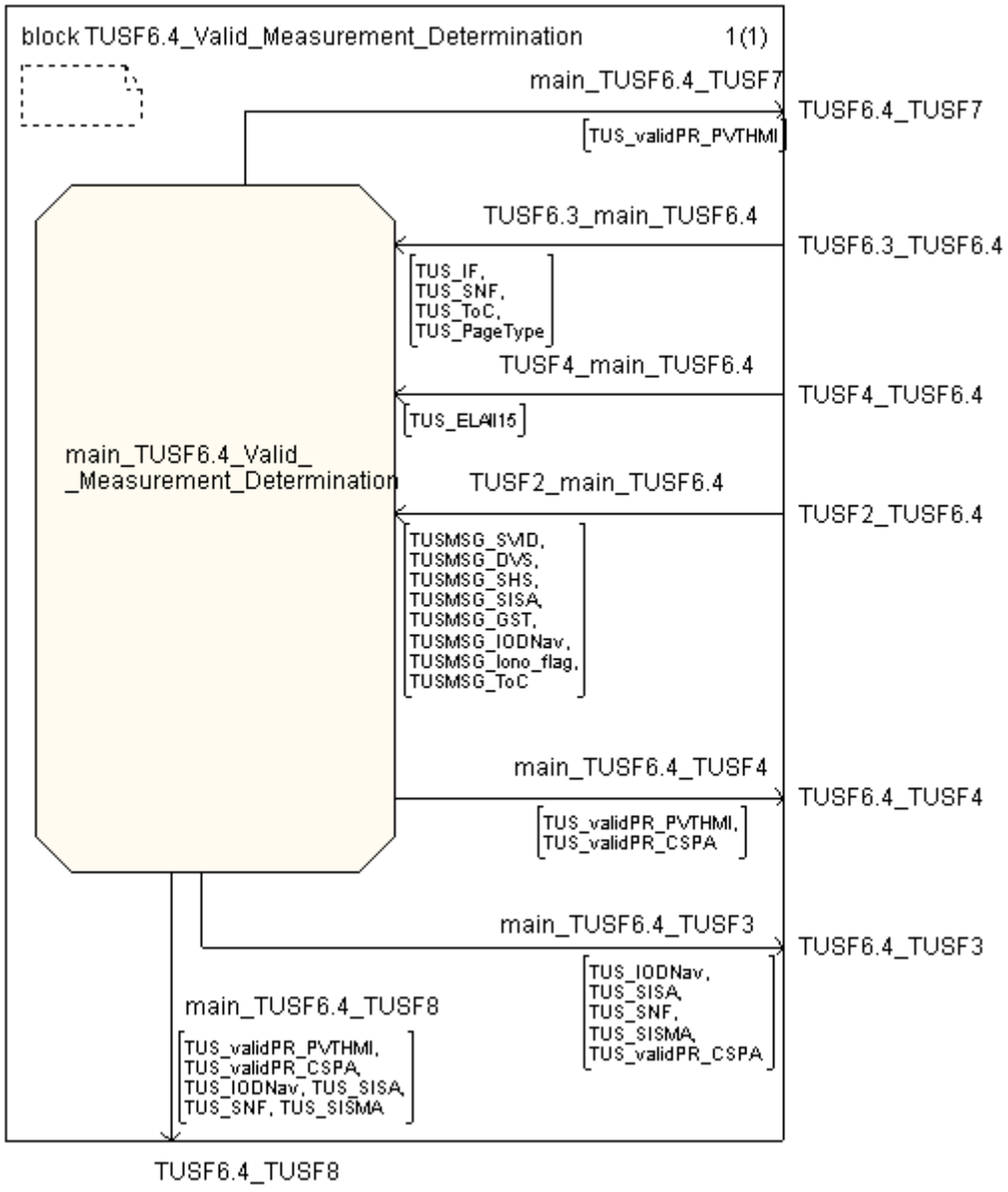


Figure 208: TUSF6.4 SDL Functional Block Diagram

5.5.7 TUSF7 Timing Determination Algorithm

TUSF7 General Description

The Time Determination Algorithm determine the UTC. The TUR will retrieve UTC by operating in stand-alone mode, that is by determining GST through measurements with all visible Galileo satellites and applying the (GST-UTC) difference data included in the navigation data message and produced through Time Service Provider (TSP) interface.
It includes the following Algorithms:
- GST Determination Algorithm
- GST-UTC Conversion Algorithm

TUSF7 Functional Block Diagram

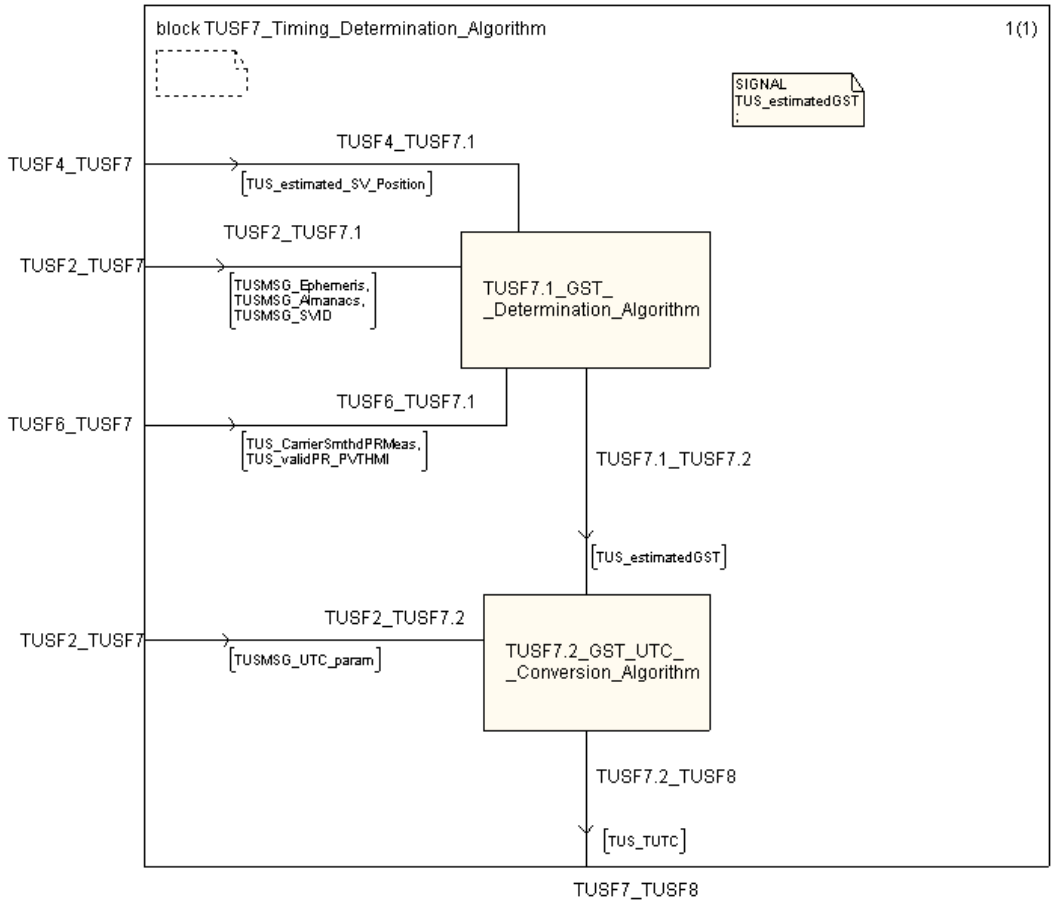


Figure 209: TUSF7 SDL Functional Block Diagram

5.5.7.1 TUSF7.1 GST Determination Algorithm

TUSF7.1 General Description
The GST Determination Algorithm shall determine, at each fixing epoch, the user clock offset with respect to GST.

TUSF7.1 Outputs
Estimate of GST t_E

TUSF7.1 Design constraints/performances
The GST Determination Algorithm shall determine, at each fixing epoch, the user clock offset with respect to GST by means of Weighted Least Mean Square (WLMS) technique by using all valid pseudorange measurements.
This function shall accept from the Navigation Data Message Recovery Function the ephemeris and almanac data, as necessary to determine and predict the position of Galileo Satellites in GTRF.

TUSF7.1 Functional Block Diagram

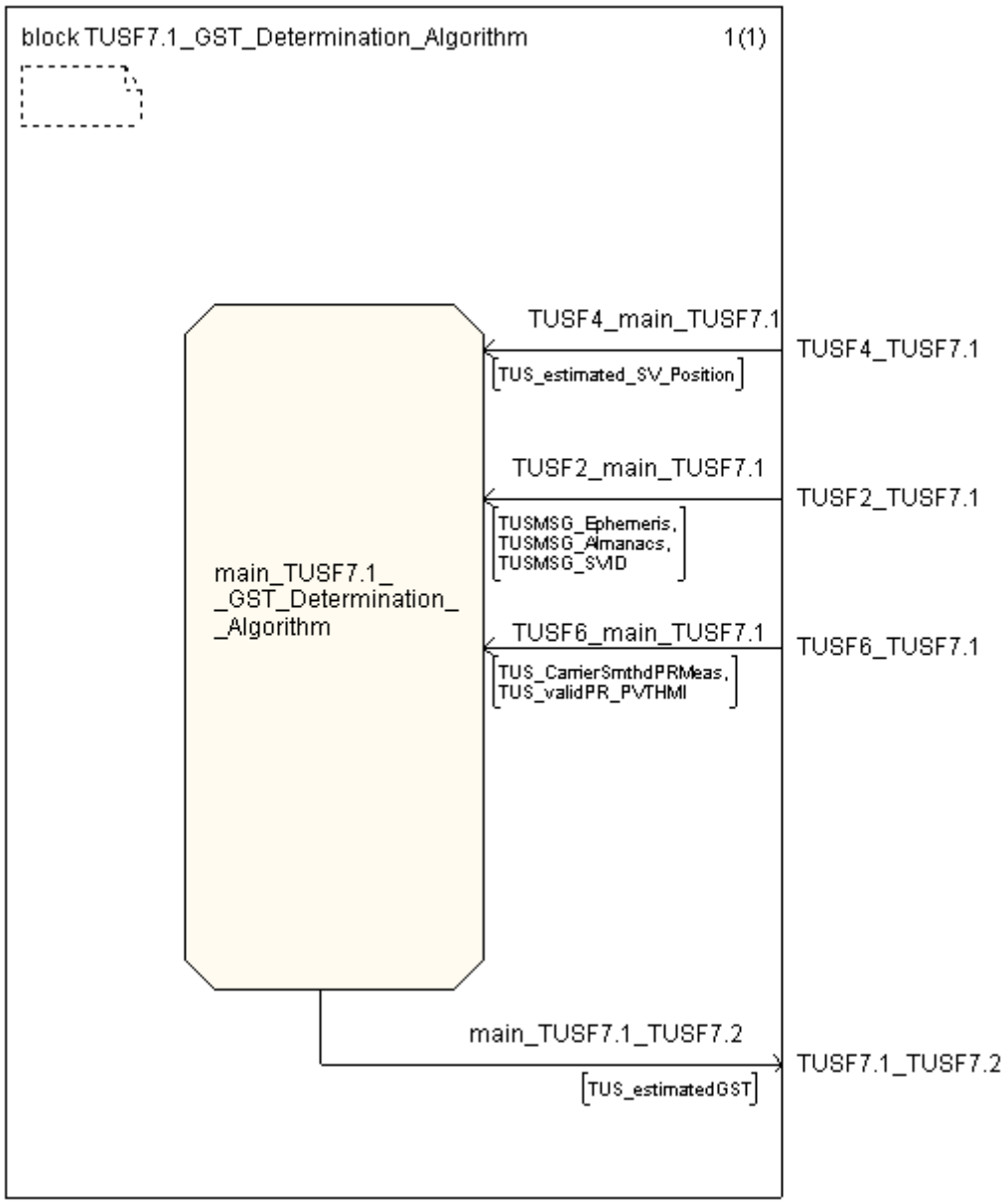


Figure 210: TUSF7.1 SDL Functional Block Diagram

5.5.7.2 TUSF7.2 GST-UTC Conversion Algorithm

TUSF7.2 General Description
The GST-UTC Conversion Algorithm shall, at each fixing epoch, determine the UTC

TUSF7.2 Outputs
UTC time t_{UTC}

TUSF7.2 Design constraints/performances
The GST-UTC Conversion Algorithm shall accept from the Navigation Data Message Recovery Function the Parameters for the following GST - UTC conversion.
The GST-UTC Conversion Algorithm shall accept from the GST Determination Algorithm the current GST estimate relative to the start of the week, as estimated by the user through the GST determination algorithm.

TUSF7.2 Functional Block Diagram

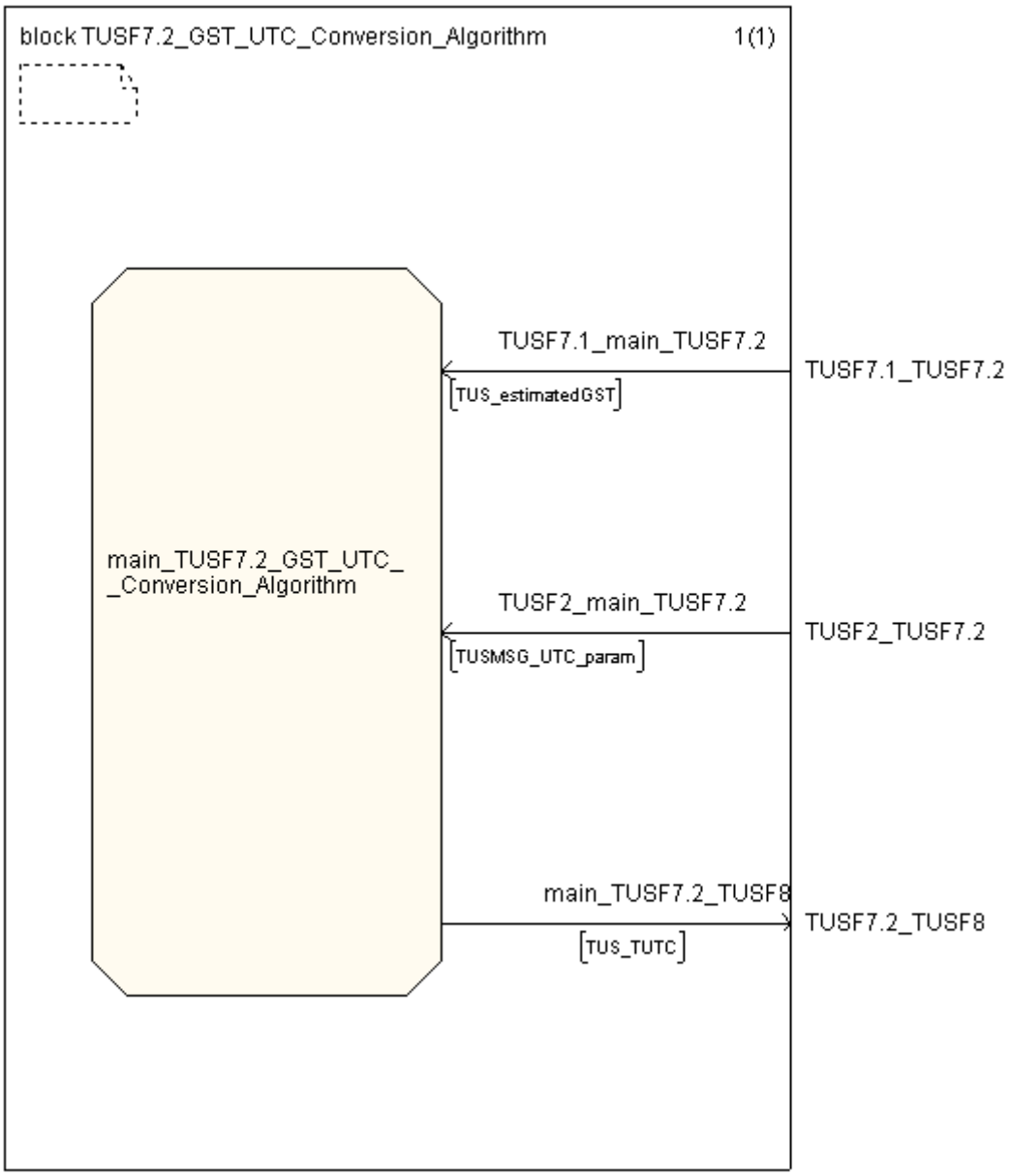


Figure 211: TUSF7.2 SDL Functional Block Diagram

5.5.8 TUSF8 Analysis Tool

TUSF8 General Description

The Test User Receiver will produce, and store on permanent media, all Raw data Outputs. Some of them are processed in real-time by the TUR MMI subsystem, and displayed to the operator for receiver monitoring and control. Some other data have to be post-processed in order to provide charts and statistics on relevant receiver parameters.

The Analysis Tool shall be able to provide data analysis and post-processing functionality for all TUR stored observables, to sustain the verification process for all receiver configurations (Off-line Mode). Moreover, the Analysis Tool shall have the possibility to interface in real-time with the Test User Receivers, to provide some useful processing capabilities during test runs (On-line Mode).

The MMI for this Tool will be the standard computer interface: keyboard, mouse, colour display and printer. Additional massive reception and storage capabilities are required. The operational SW will be provided by menu-driven interfaces.

Both on-line and off-line processing functionalities of the Analysis Tool assume that the reference position of the Test User Receiver antenna is known (this allows for real-time errors computation).

TUSF8 Design constraints/performances

This function shall:
- To store all the data collected during TUS operation, for later off-line processing and/or archiving
- to perform analysis
- Have Geoid undulation model stored in memory
- collect, display and store at least:
    - self-test results
    - Operating parameters
    - Installed HW and running SW ID
    - its status
    - observables and measurements
    - Navigation data
    - Post-processing data
    - historical log file
- time stamp all stored data with receiver internal timing
- storage capacity
    - up to 10 days operation for logging measurement and navigation data at 1Hz;
    - up to 20 SW release
- Display data (geometry, position range error, C/No, etc) with configurable rate, with a separate display for each system (Galileo, GPS, Egnos)
- Be able to receive real-time reference data form the Galileo simulator
- Calculate code and carrier phase jitter, BER, statistics
- Be able to interface with simulator

TUSF8 Functional Block Diagram

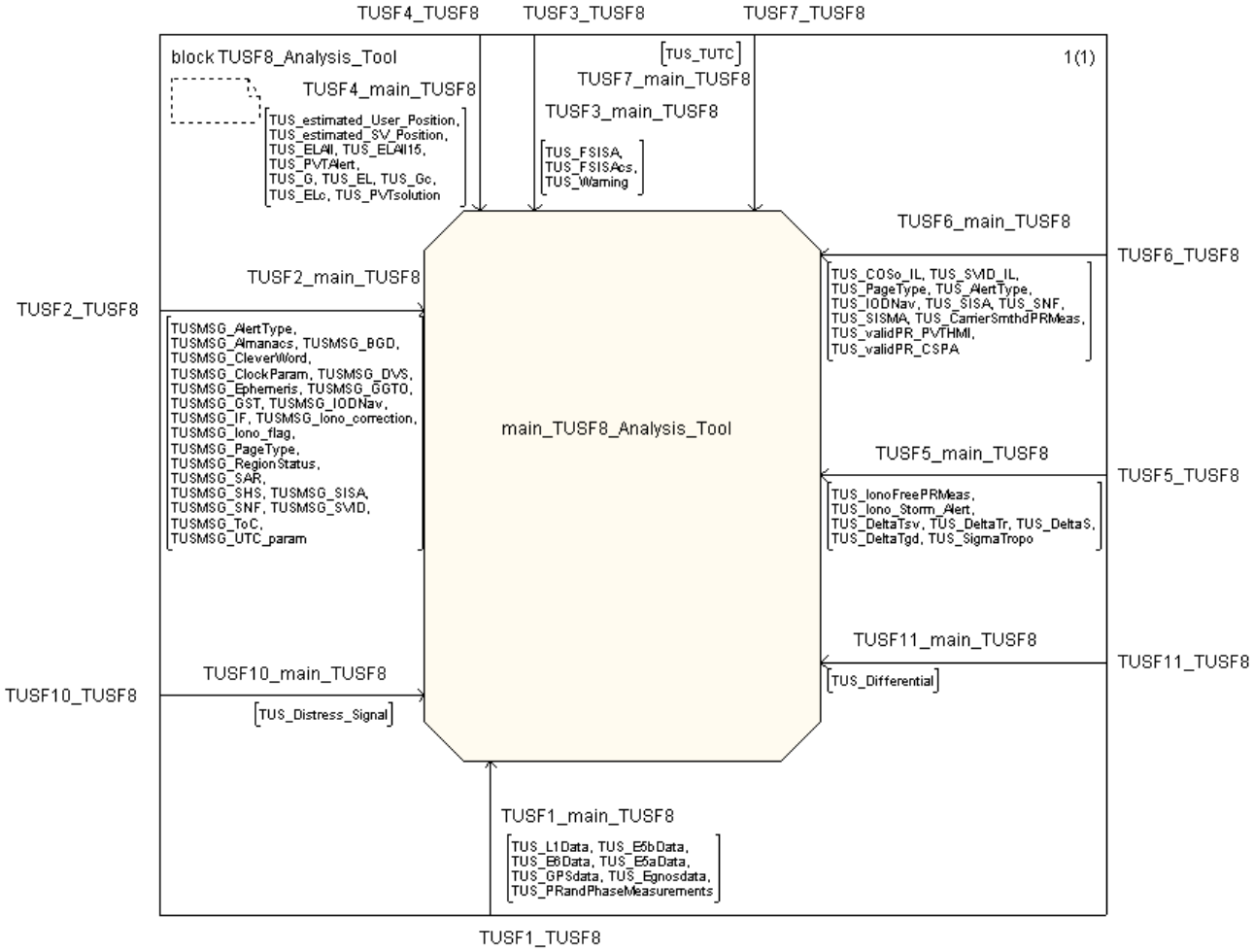


Figure 212: TUSF8 SDL Functional Block Diagram

5.5.9 TUSF9 Autonomous Integrity Determination

TUSF9 General Description

This function is in charge of implementing the RAIM algorithm in order to determine the integrity of the navigation solution. The user shall be allowed to enable or disable this function.
Note: this function is independent from other functions and is thus not represented in the SDL funcional model

5.5.10 TUSF10 Search And Rescue

TUSF10 General Description

This (configuration-specific) function is in charge to support the Search And Rescue Service, generating the Distress Signals and receive the SAR Return Link Messages incorporated into the Navigation Data Messages.

TUSF10 Functional Block Diagram

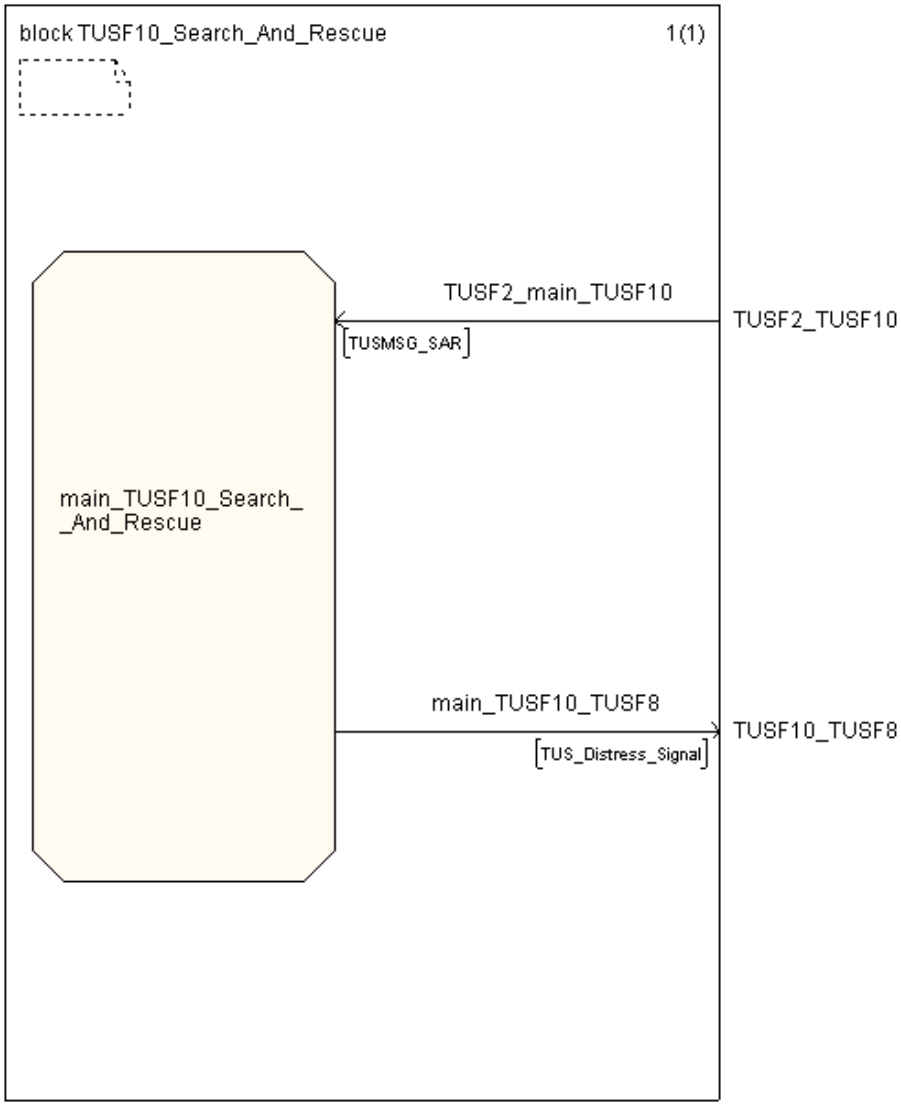


Figure 213: TUSF10 SDL Functional Block Diagram

5.5.11 TUSF11 Differential Data

TUSF11 Description

This (configuration-specific) function is in charge of providing the differential Data Correction for Local Components to support the Galileo Precision and High Precision Navigation Service. It is a combination of Rover Receiver (for Local Precision Services reception) and a Reference Station (for Local Precision Services generation).

TUSF11 Functional Block Diagram

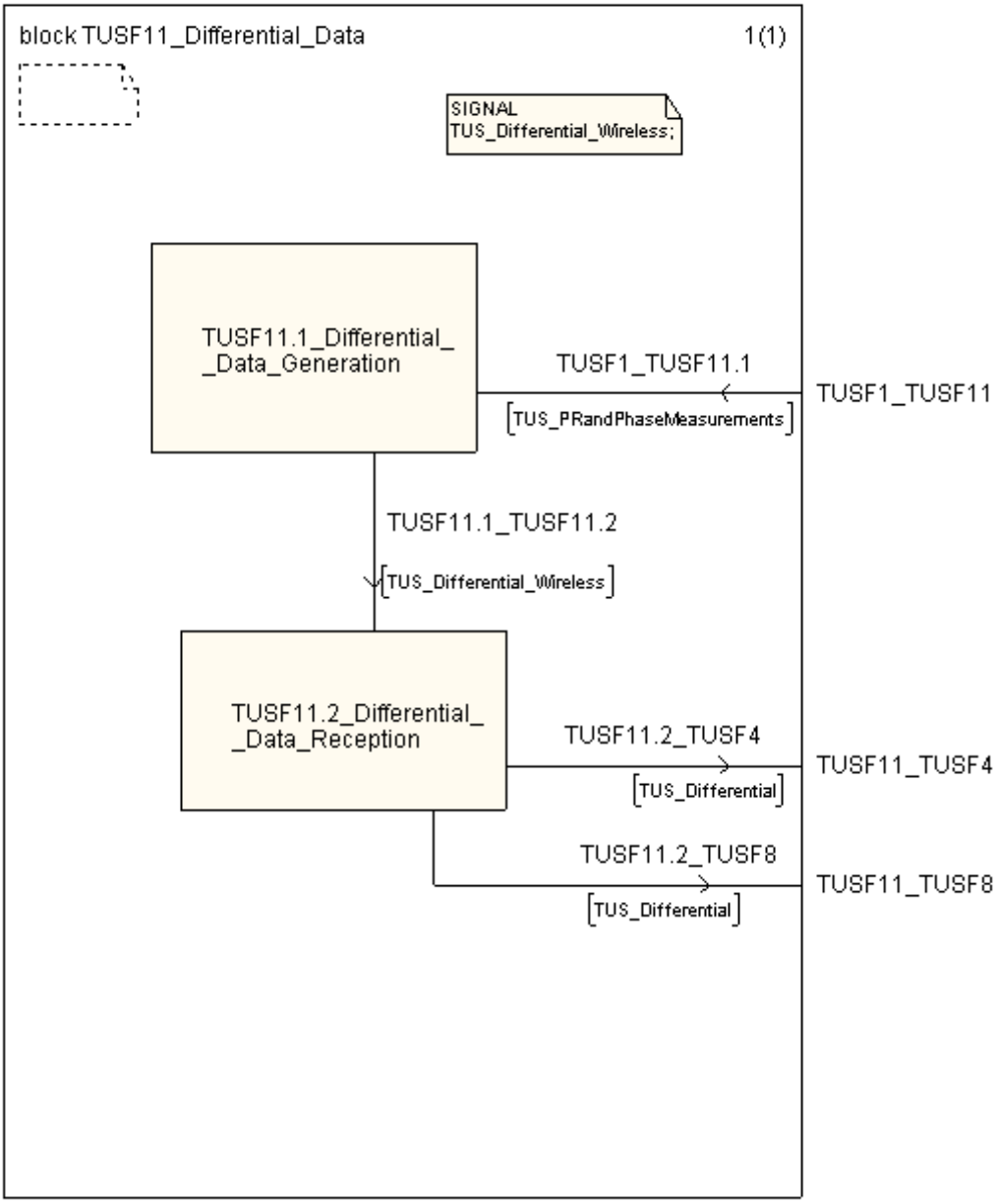


Figure 214: TUSF11 SDL Functional Block Diagram

5.5.11.1 TUSF11.1 Differential Data Generation

TUSF11.1 General Description

This function is in charge of generating the differential Data Correction for Local Components to support the Galileo Precision and High Precision Navigation Service.
It shall generate and broadcast appropriate (high precision) differential corrections for single and dual frequency users, giving estimates of common errors due to satellite clocks and ephemeris propagation, as well as ionospheric (for single frequency) and tropospheric local errors. These corrections will be used by Rovers to correct pseudorange measurements to obtain precision PVT information.

TUSF11.1 Outputs

Precision differential correction messages for all Galileo satellites in view, via wireless link.

TUSF11.1 Design constraints/performances

- Correction messages within a hemispherical area of radius 10 km

TUSF11.1 Functional Block Diagram

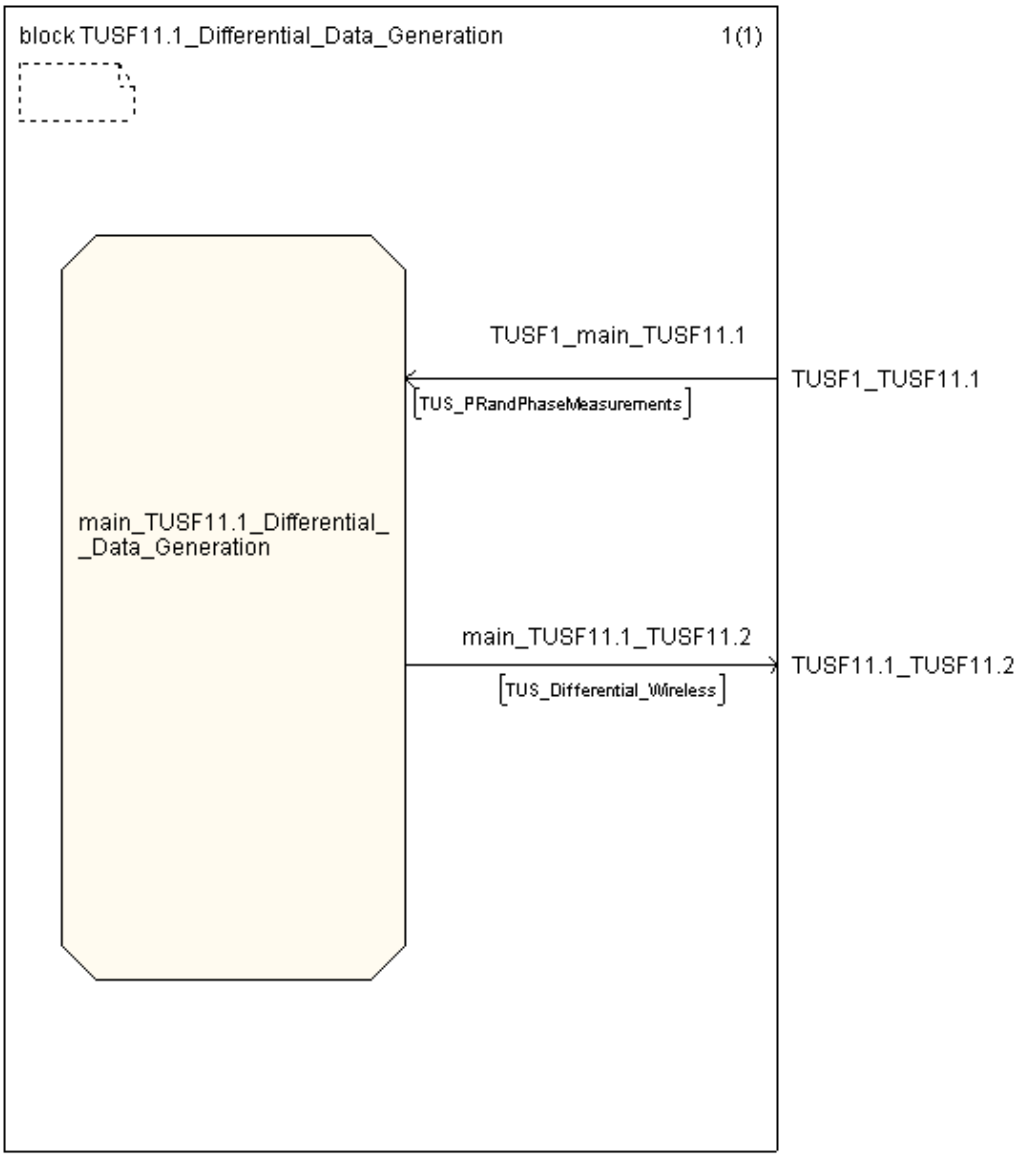


Figure 215: TUSF11.1 SDL Functional Block Diagram

5.5.11.2 TUSF11.2 Differential Data Reception

TUSF11.2 General Description

This function is in charge of receiving the differential Data Correction for Local Components to perform the Galileo High Precision Navigation Service. The received data shall be used in the determination of the Navigation Solution.

It shall receive differential corrections for single/dual frequency users, provided by the Local Reference, giving estimates of common errors due to satellite clocks and ephemeris propagation, as well as ionospheric and tropospheric local errors. These corrections will be used to correct pseudorange measurements to obtain precision PVT information.

TUSF11.2 Functional Block Diagram

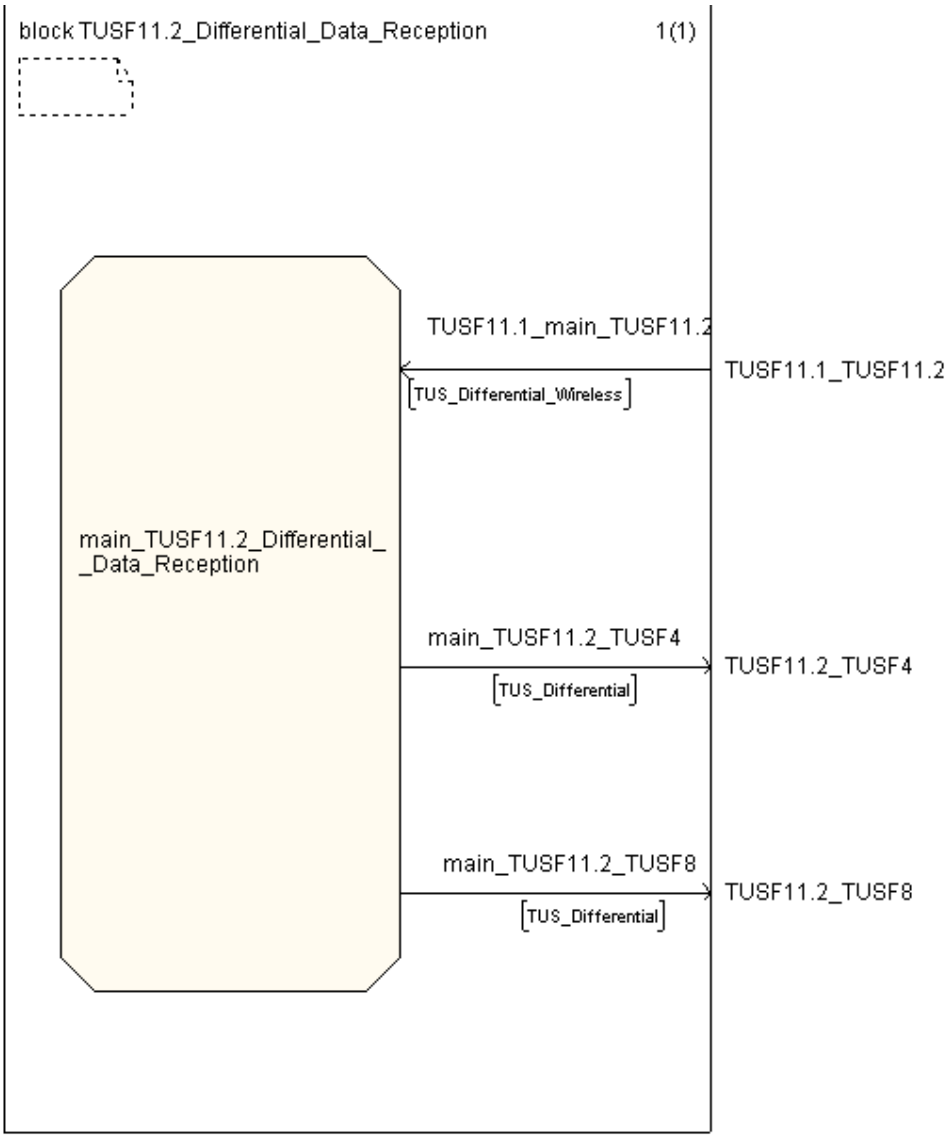


Figure 216: TUSF11.2 SDL Functional Block Diagram

5.6 External Entities Functions

5.6.1 CSP Functions

5.6.1.1 CSPF1 Provide GDDN Connectivity

CSPF1 General Description

This CSP Function consists in providing GDDN connectivity services to Galileo system.
This includes the analsyes of GDDN link data provided by GMS to assess the network performance.

CSPF1 Functional Block Diagram

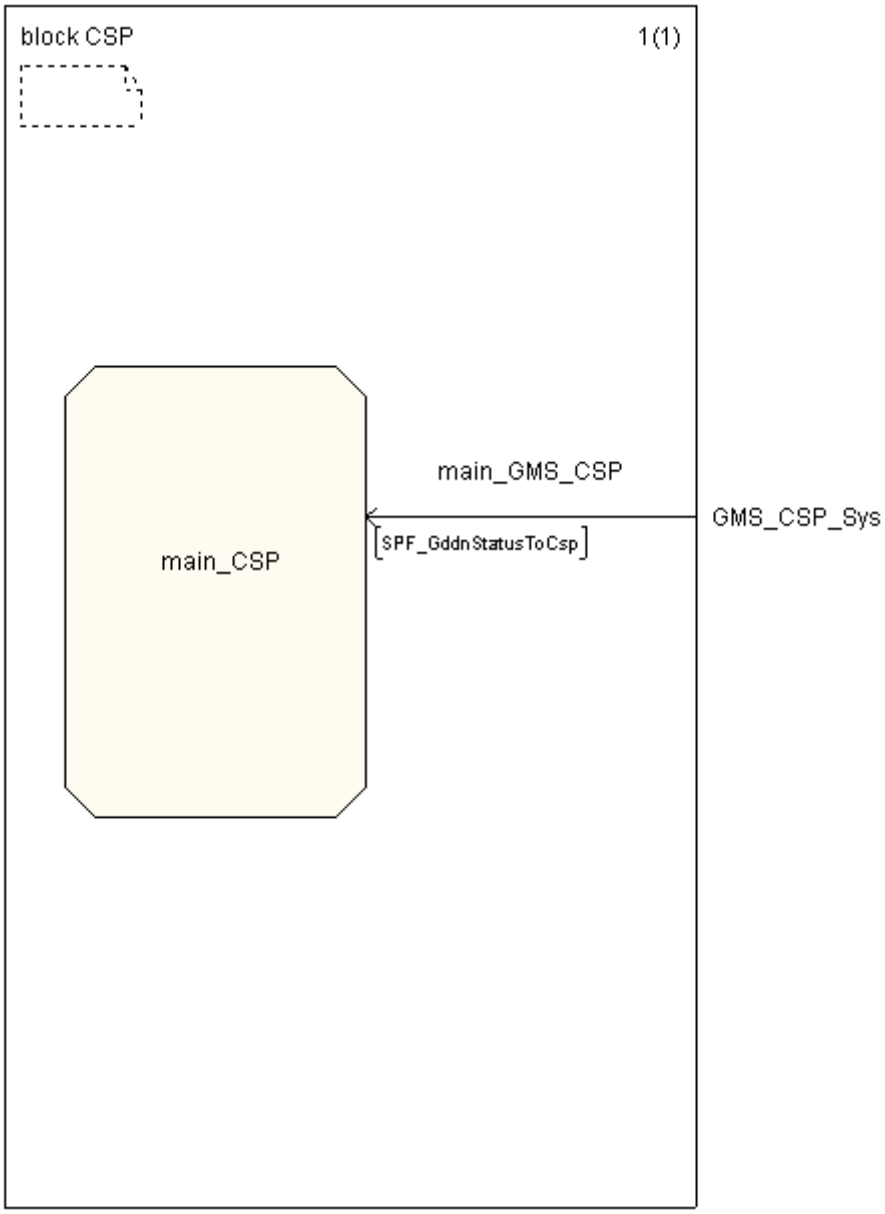


Figure 217: CSP SDL Functional Block Diagram

5.6.2 ESCC Functions

5.6.2.1 ESCCF1: Perform LEOP and Satellite Hand-Over

ESCCF1 General Description

This segment function is responsible for performing the Early Orbit Phase for the Galilieo satellites and the handover of the satellites to the GCC following completion of the LEOP.

ESCCF1 Outputs

This function is responsible for producing the following output signals:

- Flight dynamics data report containing all data required as input to the flight dynmamics functions in the GCS.
- A history of all TC, TM, and Event packet data sent between the ESCC and the spacecraft.
- Records of all meteorological and ranging data recorded by the ESCC for each spacecraft
- Report(s) summarizing the activities performed during LEOP
- Report(s) provided on a per spacecraft which include as a minimum the following contents:
  - Spacecraft ephemeris data
  - Spacecraft propellant consumption, mass and mass properties
  - Spacecraft configuration at handover
  - Preferred spacecraft configuration
  - Spacecraft documentation configuration status
  - Recommendation forms
- A signal indicating that the IOT testing can start

ESCCF1 Design Constraints

This function will meet the following constraints and performances:

- Deploy new satellites for constellation renewal in their nominal orbital positions within 35 days after injection phase including IOT
- Deployment of one spare satellite in its nominal orbital position within 35 days after injection phase, assuming that the maximum phasing angle for station acquisition is reduced to +/- 20 (target location is the nearest gap between two S/C).
- Deployment of two spare satellites in their nominal orbital positions within 35 days after injection phase; assuming that the maximum phasing angle for station acquisition is reduced to +/- 110 (target location is the nearest gap between two S/C and 2 spares are located 180 apart).
- The following launcher performances for all launcher services (ie Ariane and Soyuz
  - 6 months (TBC) delay in case of launch failure,
  - launch in less than 6 months (TBC) from notice, any 2 launches being separated by at least 3 months (TBC)
- Optimising the initial biases of the orbit parameters to satisfy the tolerances of a Walker constellation on a 12 year timeframe with one orbit keeping manoeuvre maximum per satellite life-time.
- Ensuring an initial orbit accuracy is such that the semi-major axis of the satellite orbit is corrected to within +/- 5 metres.

ESCCF1 Functional Block Diagram

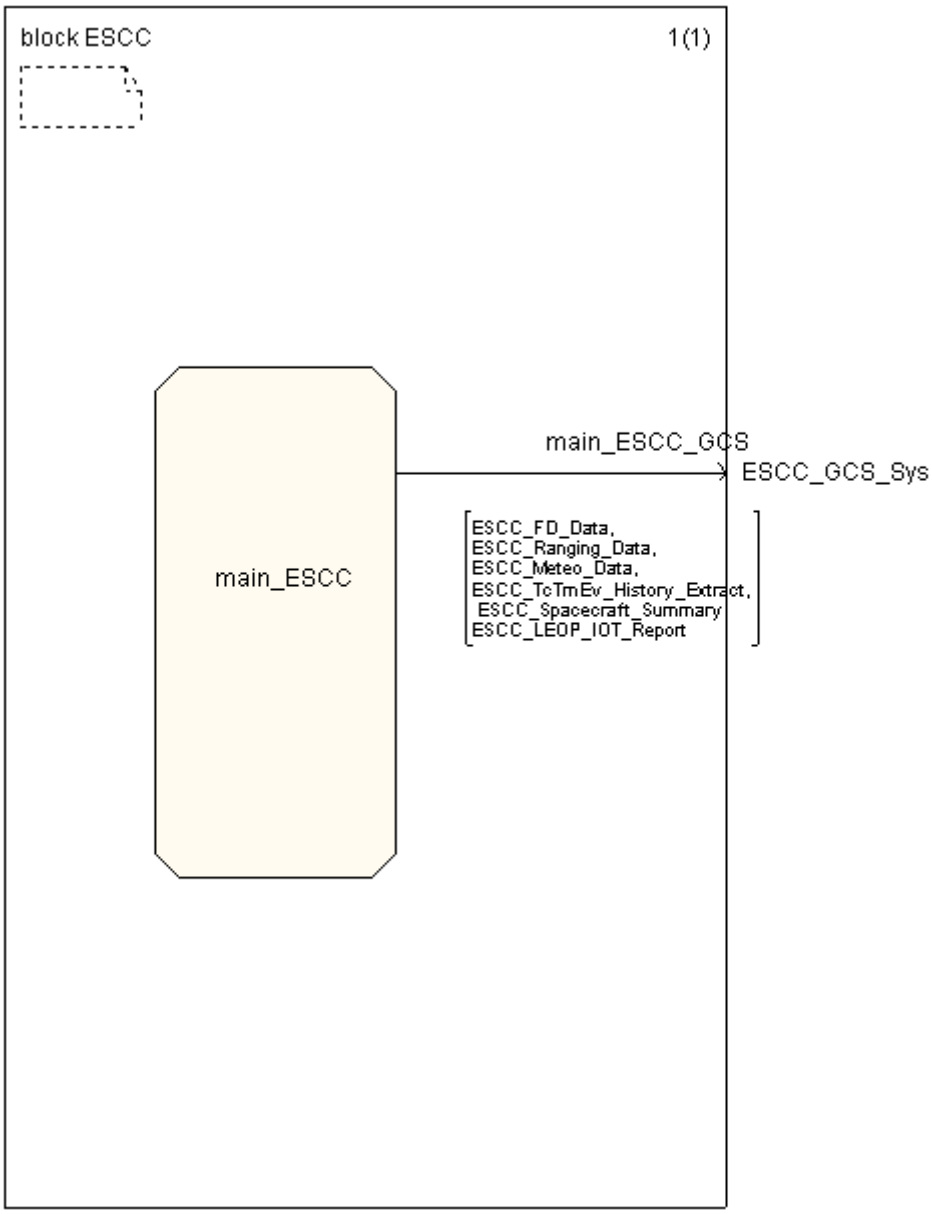


Figure 218: ESCC SDL Functional Block Diagram

5.6.3 GALSEE Functions

5.6.3.1 GALSEEF1 Data Management and Control

GALSEEF1 General Description

This function is responsible for centralizing and storage of all data used as input for test activity, all support data and all the results of the testing activity.
This include the following functionalities:
a. Data Exchange
b. Data Storage
c. Data Archiving
d. Data Configuration
e. Monitoring & Control of all GALSEE elements
f. Connectivity (External I/F and Internal I/F including GALSEE LAN and Internet Access)

GALSEEF1 Outputs

The main outputs of the GALSEE function are the results of the testing activities, including configuration data.

GALSEEF1 Functional Block Diagram

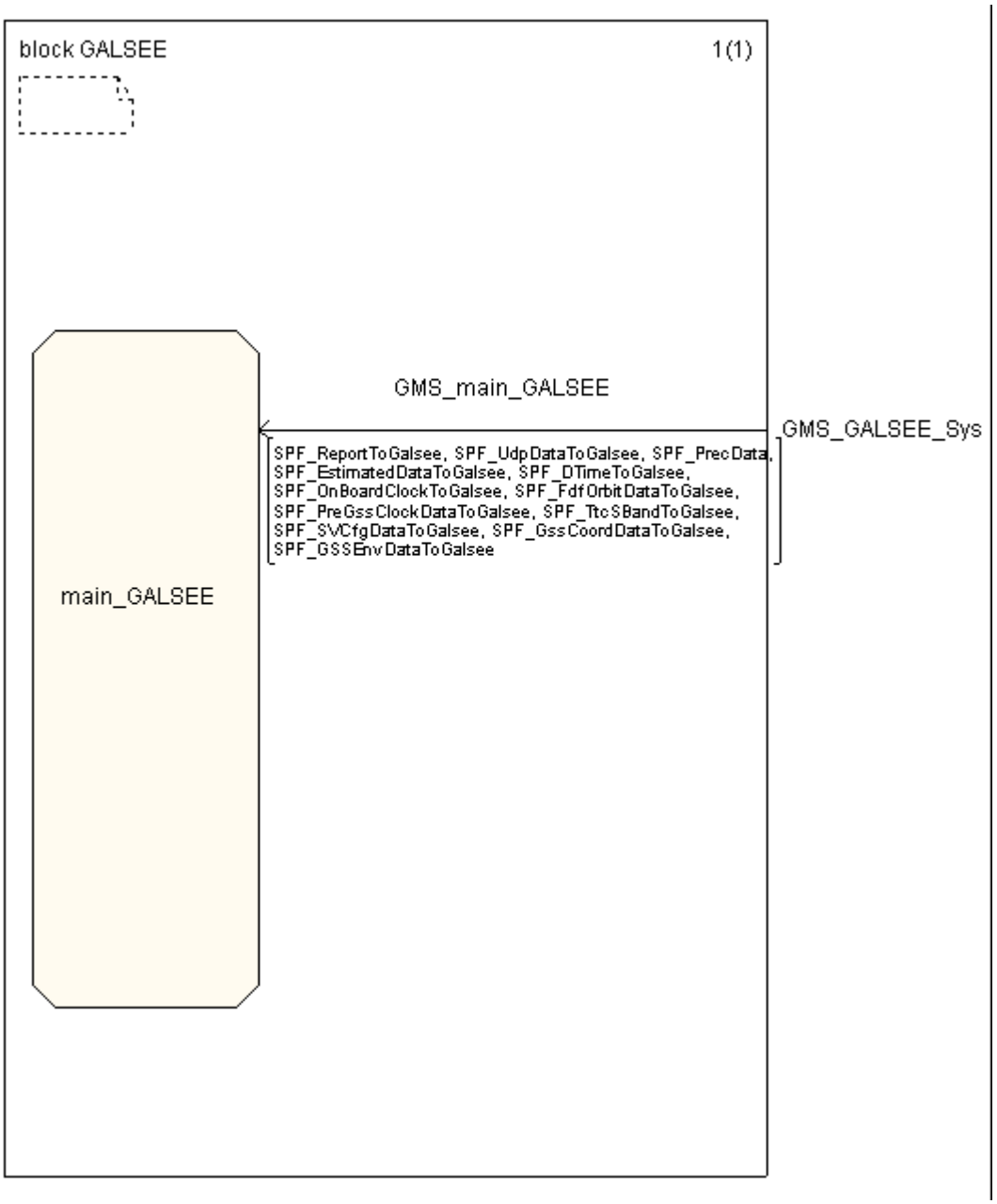


Figure 219: GALSEE SDL Functional Block Diagram

5.6.4 GPS Functions

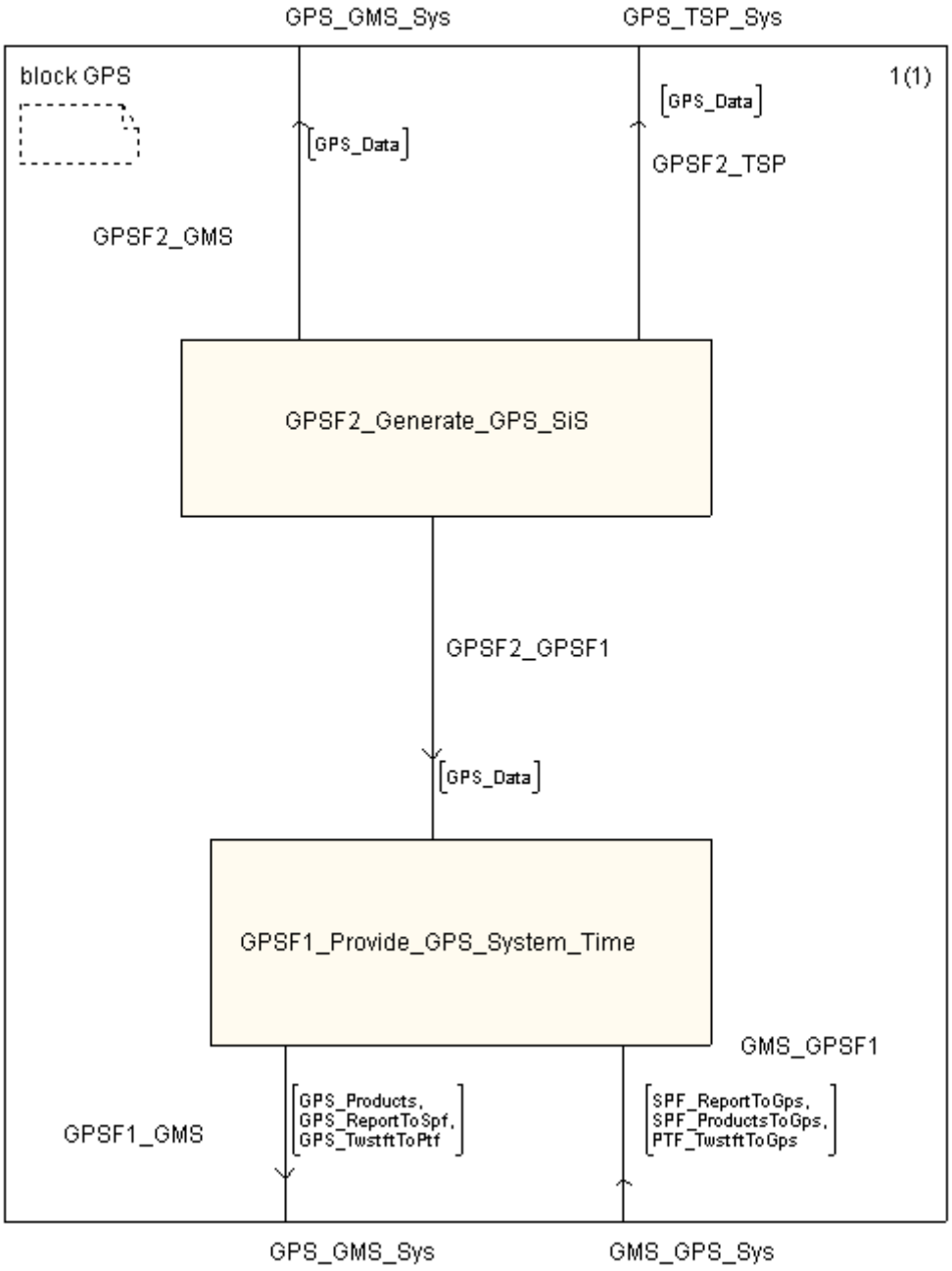


Figure 220: GPS SDL Functional Block Diagram

5.6.4.1 GPSF1 Provide GPS System Time (USNO)

GPSF1 Description

This function is delivered by GPS/USNO an entity external to Galileo core. It provides/allows:
- Maintenance of the GPS System Time
- the access to GPS System Time (via Y-code GPS receiver installed at GPS/USNO premises (GGTO)
- Comparison of the GPS system time with the external timescales, system times such as GST

- Configuration, operation and control of time transfer equipment used for GGTO determination
- GGTO determination and coordination on US side of interface
- Exchange of time transfer results and GGTO coordination
- Prosision of the GGTO to the GPS control centre for its further dissemination to the GPS users.

Function is under direct responsibility of USNO (U.S. Naval Observatory). USNO plays an important role in GPS operation given that GPS System Time is steered to UTC(USNO) on daily basis. USNO role in GPS system time keeping is analogical to the TSP role for the GST to TAI steering

GPSF1 Functional Block Diagram

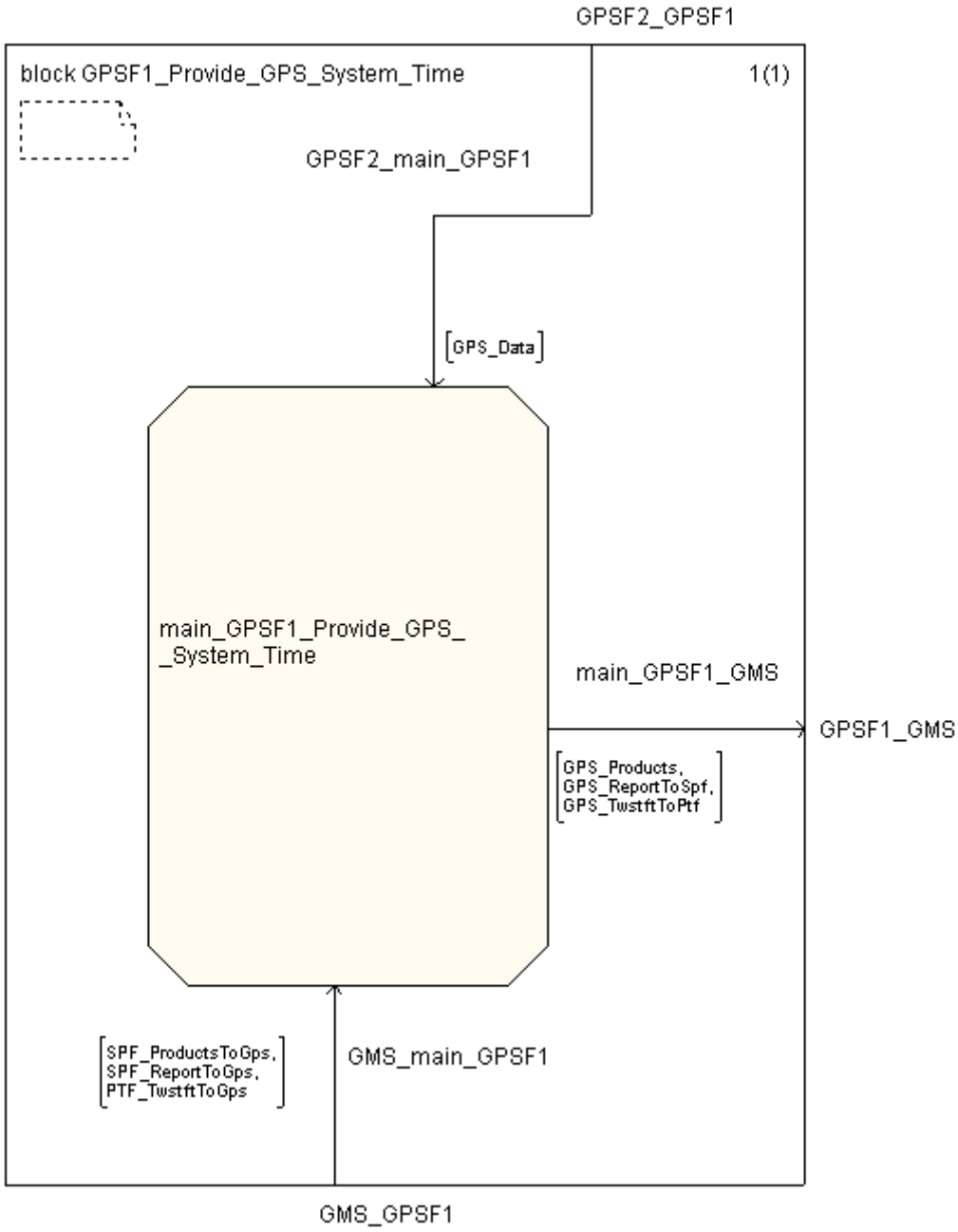


Figure 221: GPSF1 SDL Functional Block Diagram

5.6.4.2 GPSF2 Generate GPS SiS (CV Time Transfer)

GPSF2 Description

Function is under direct responsibility of U.S. Department of Defense and Department of Homeland Security. Function delivers GPS SiS including both pseudo-range observables and navigation data/messages , necessary for the GPS-CV (Common View) time transfer and GPS-Galileo combined receiver measurements and also for the GPS SIS pseudo-range measurements made by GSSs.

The GPS SIS measurements are used for :

- GGTO determination
- GST-TAI steering (GST metrological timekeeping)
- PTF synchronisation
- GTRF determination (GSS raw measurements consists also of the GPS SIS observables including the GPS navigation messages in RINEX3.0 files)

GPSF2 Functional Block Diagram

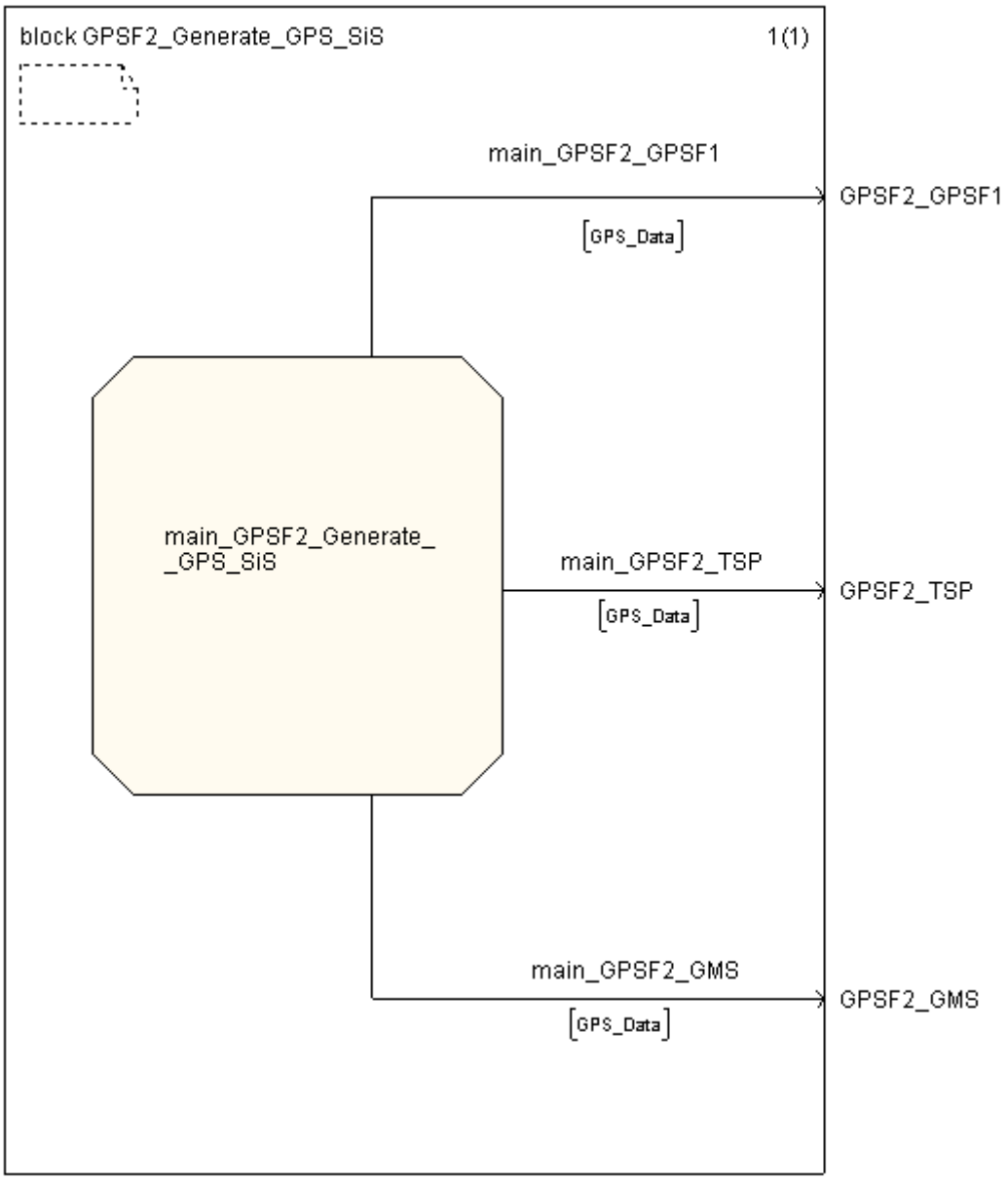


Figure 222: GPSF2 SDL Functional Block Diagram

5.6.5 GRSP Functions

5.6.5.1 GRSPF1: Generate GTRF and Provide related/not Related Products

GRSPF1 General Description

This function is provided by GRSP, an entity external to Galileo Core. This function provides for Galileo Core the following tasks:

- Generation and Maintenance of the GTRF
- Provision of the scientific data to GMS and GCS
- SLR data provision
- Validation orbit, clock and Earth rotation parameters estimates made by MSF

This function needs to deliver the data meeting the manner volume and performance specifications given in the GMSExtIRD and GMS to GRSP ICD.

GRSPF1 Functional Block Diagram

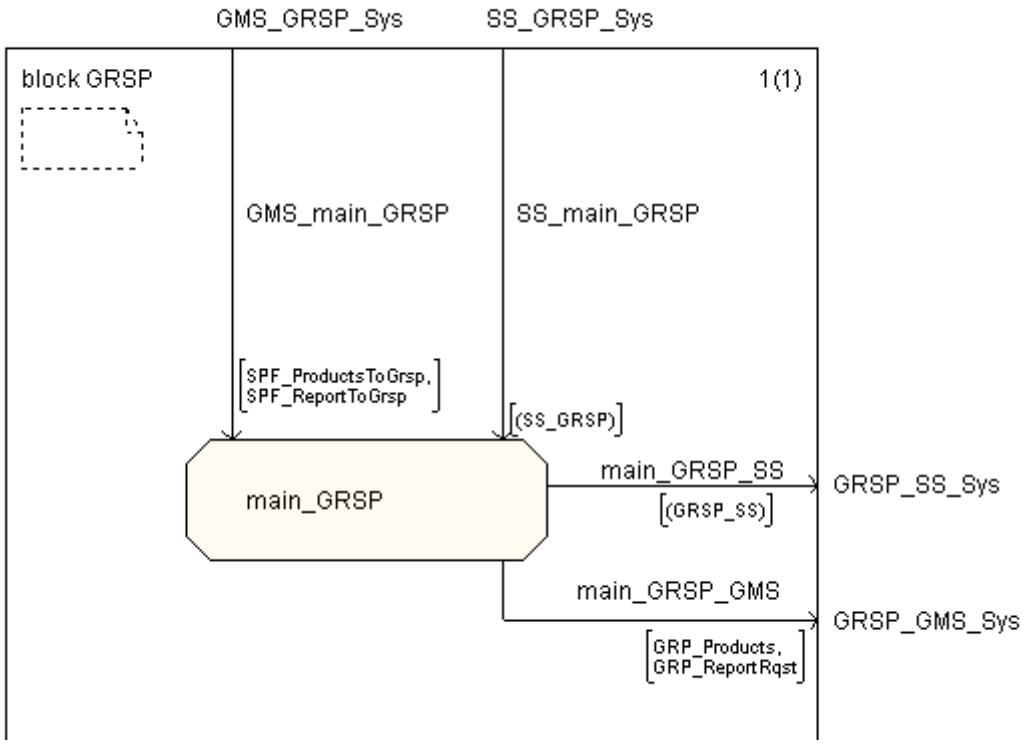


Figure 223: GRSP SDL Functional Block Diagram

5.6.6 GSMC Functions

GSMCF1 General Description

This function is responsible for providing support to core system for the management of all the security aspects for the protection of the PRS service. This includes a dedicated interface with GMSF8.1 (Security protection of PRS) to support the generation and encryption of the PRS security messages.

This function covers all the GSMC security functions required to fulfill the security SEC-SFC15 part related to PRS protection. Refer to this SEC-SFC for further details and traceability.

GSMCF1 Functional Block Diagram

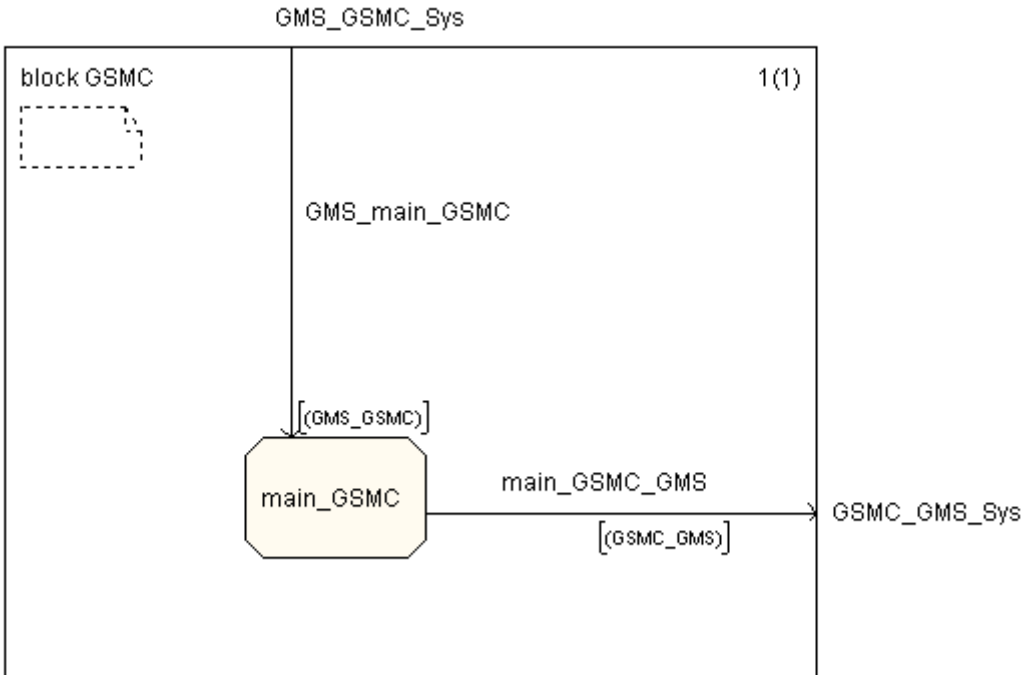


Figure 224: GSMC SDL Functional Block Diagram

5.6.7 IOT Functions

5.6.7.1 IOTF1: Perform IOT Measurements

IOTF1 General Description

This segment function is responsible for the execution of the In-Orbit Test (IOT) campaign to verify that no degradation to the satellite has occurred since on-ground acceptance and confirm that the spacecraft meets its requirements. This includes the capability to start IOT testing during drift phases prior to the satellite arrival on the operational orbit or during re-positioning.

IOTF1 Outputs

This function is responsible for producing the following outputs:

- A signal indicating that the IOT testing is complete

IOTF1 Design Constraints

This function will meet the following constraints and performances:

- Ensure that new satellites finish IOT within 5 days of arrival at their operational orbital position.
- Support the requirement for new satellites to be deployed in their nominal orbital positions within 35 days after injection phase including IOT

IOTF1 Functional Block Diagram

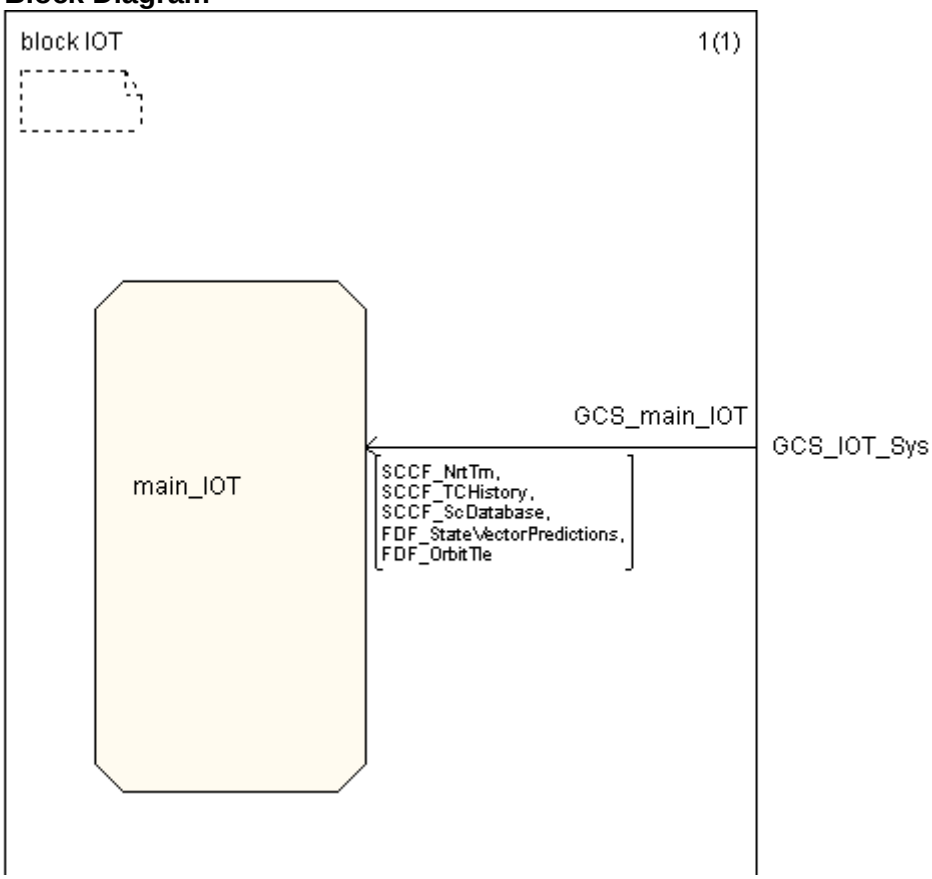


Figure 225: IOT SDL Functional Block Diagram

5.6.8 Launch Service Provider Functions

5.6.8.1 LSPF1: Perform Launch

LSPF1 General Description

This segment function is responsible for performing the Early Orbit Phase for the Galilieo satellites and the handover of the satellites to the GCC following completion of the LEOP and IOT. In particular this function is responsible for providing:

- The following launcher capabilities:
  - Ariane up to 4 satellites per launch;
  - Soyuz one or two satellites per launch;
- Minimising probability of collisions (ie collision avoidance between satellite and the launcher upper stage, between satellites in multiple launch scenario etc)
- Launcher disposal to a graveyard orbit clear of operational orbits
- Optimising the initial biases of the orbit parameters to satisfy the tolerances of a Walker constellation

5.6.9 RLSP Functions

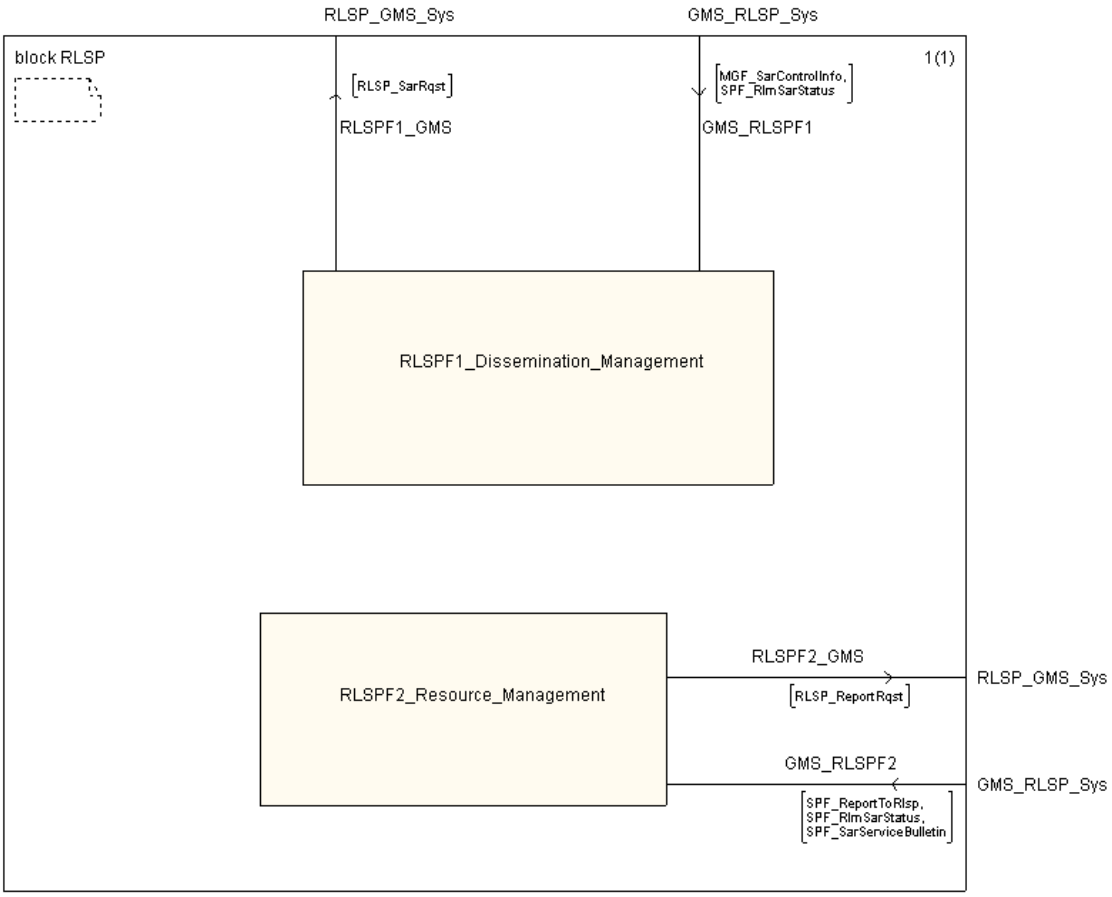


Figure 226: RLSP SDL Functional Block Diagram

5.6.9.1 RLSPF1: Dissemination Management

RLSPF1 General Description

This RLSP function is responsible for the dissemination of Return Link message requests and cancellations to GMS and receiving monitoring information on the RLM dissemination from GMS and adapting the RLM dissemination strategy approporiately. This function creates the return link message information including the list of satellites to be used for dissemination of the message created by the SAR MEOLUT.

RLSPF1 Outputs

The RLSP Dissemination Management function generates the following outputs:

- Return Link Message Requests including:
  - RLM ID
  - RLM Type
  - Return Link Message Content
  - List of satellites for dissemination of the RLM
- Return Link Cancellation Requests

RLSPF1 Design Constraints

The main design constraints of the RLSP Dissemination Management function are considered to be:

- RLM Request Format and content as described above
- RLM Cancellation request format
- RLM Capacity: This function shall be able to send up to 27 RLM requests or cancellations per minute.

RLSPF1 Functional Block Diagram

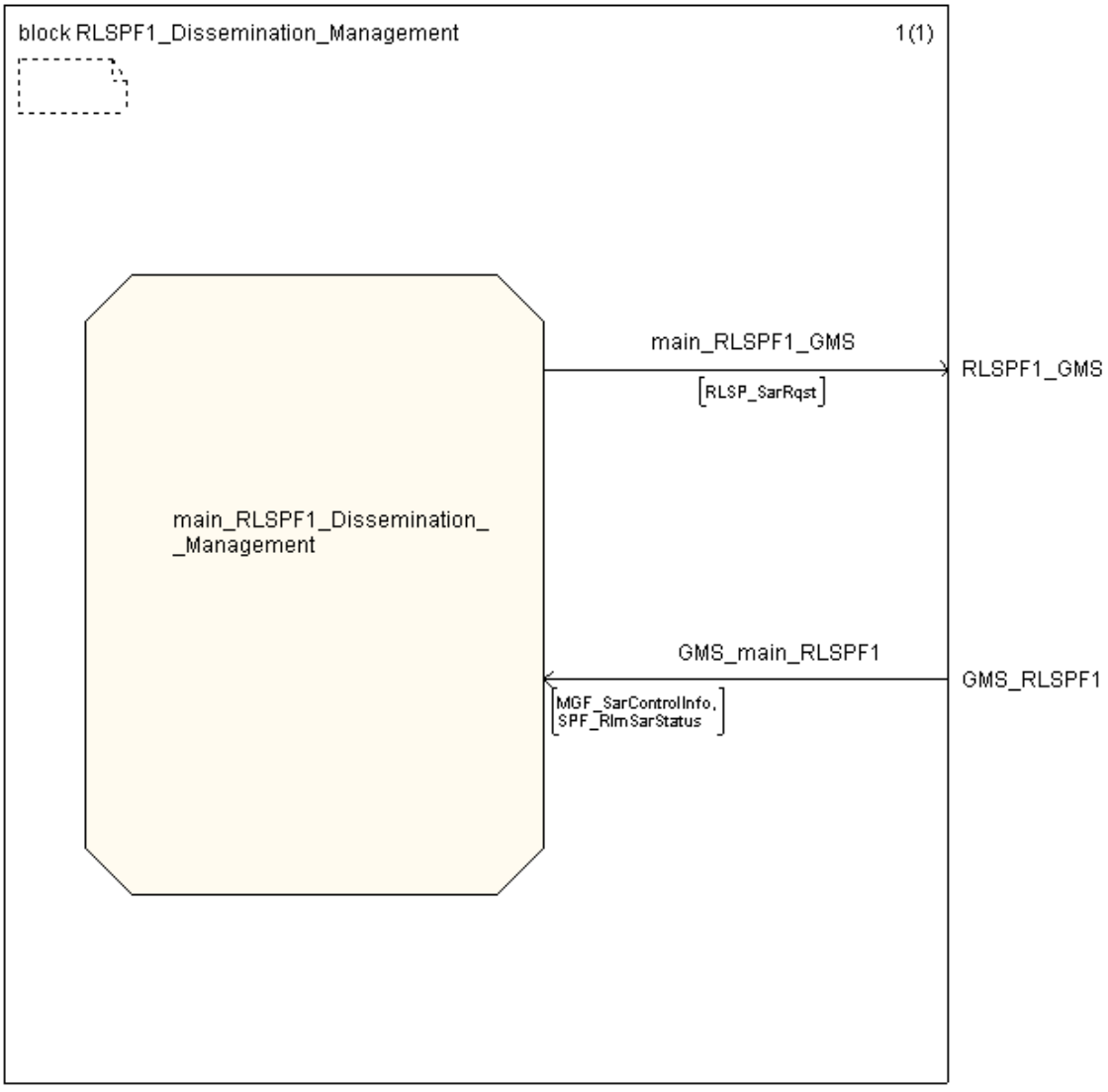


Figure 227: RLSPF1 SDL Functional Block Diagram

5.6.9.2 RLSPF2: Resource Management

RLSPF2 General Description

This RLSP function is responsible for the management of information received from GMS to allow the RLSP to carry out Return link service provision.

The RLSP receives the following data from Galileo:

- Current Satellite status related to SAR functions
- Current GMS SAR Status
- Service Bulletins
- Ephemeris data of Galileo Satellites

This function is also capable of sending a request for unscheduled reports in case these reports are required again by RLSP.

RLSPF2 Outputs

The RLSP Resource Management function generates the following outputs:

- Report requests to receive the Ephemeris data at a unscheduled time

RLSPF2 Design Constraints

The main design constraints of the RLSP Resource Management function are considered to be:

- RLSPF2 shall be able to handle up to 1 SAR/Galileo Status information from Galileo every 15 minutes.

RLSPF2 Functional Block Diagram

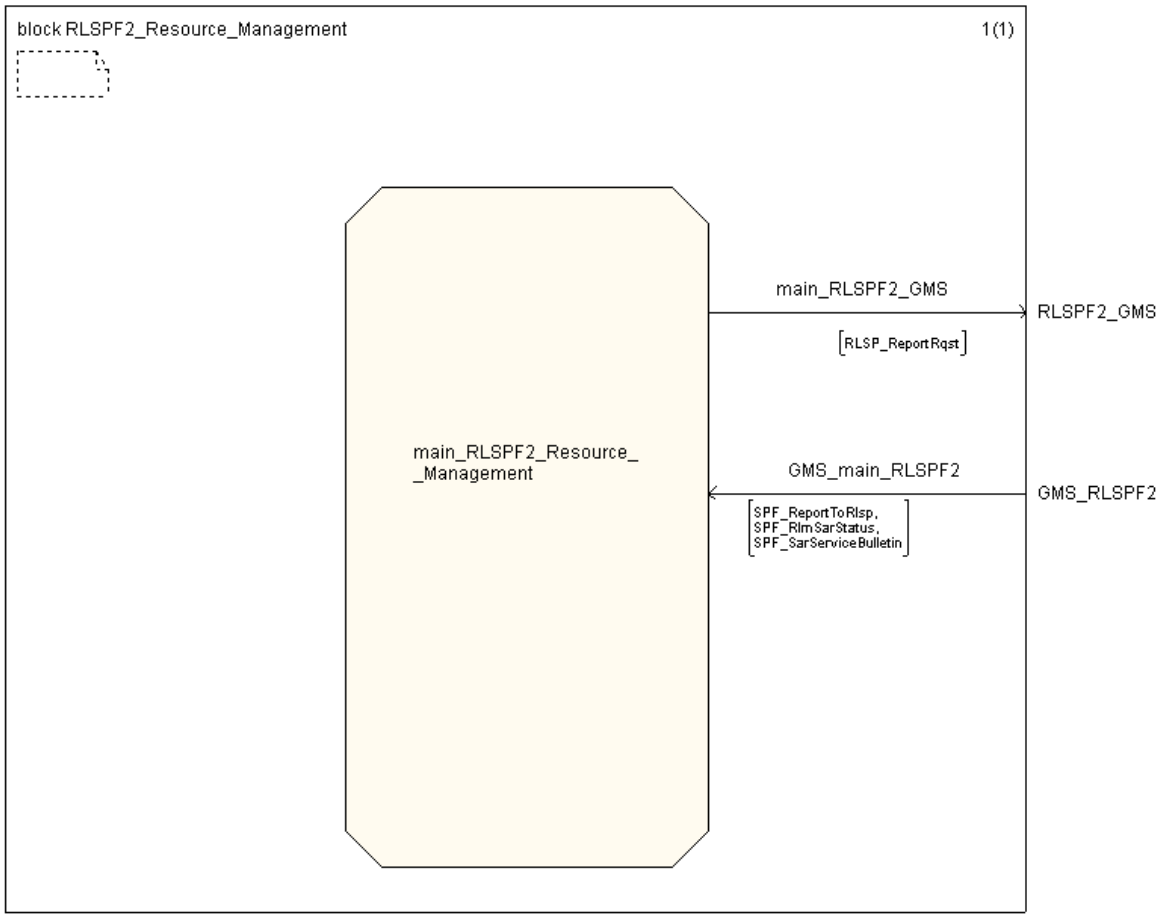


Figure 228: RLSPF2 SDL Functional Block Diagram

5.6.10 SAR Functions

5.6.10.1 SARF1: SAR Beacon

SARF1 General Description

This SAR Beacon function is responsible for the transmission of a SAR distress message upon activation to the Galileo satellite constellation (or other MEOSAR System (GPS, GLONASS). This distress message shall contain the Beacon identification and location data (when available).

SARF1 Outputs

The SAR Beacon function generates the following outputs:
- Short or Long SAR Forward Link Distress Message containing the Beacon identification and the location data (when available)

SARF1 Design Constraints

The main design constraints of the SAR Beacon function are considered to be:

- Short and Long SAR distress message shall contain at least the Beacon ID and location information when available
- Output frequency shall be 406.0 - 406.1 MHz
- Transmitted Power output is 5 W +/-2dB.
- Transmitter Antenna
  - Polarization: circular or linear
  - Gain: -3dBi to 4dB
  - Gain Variation: <3dB

SARF1 Functional Block Diagram

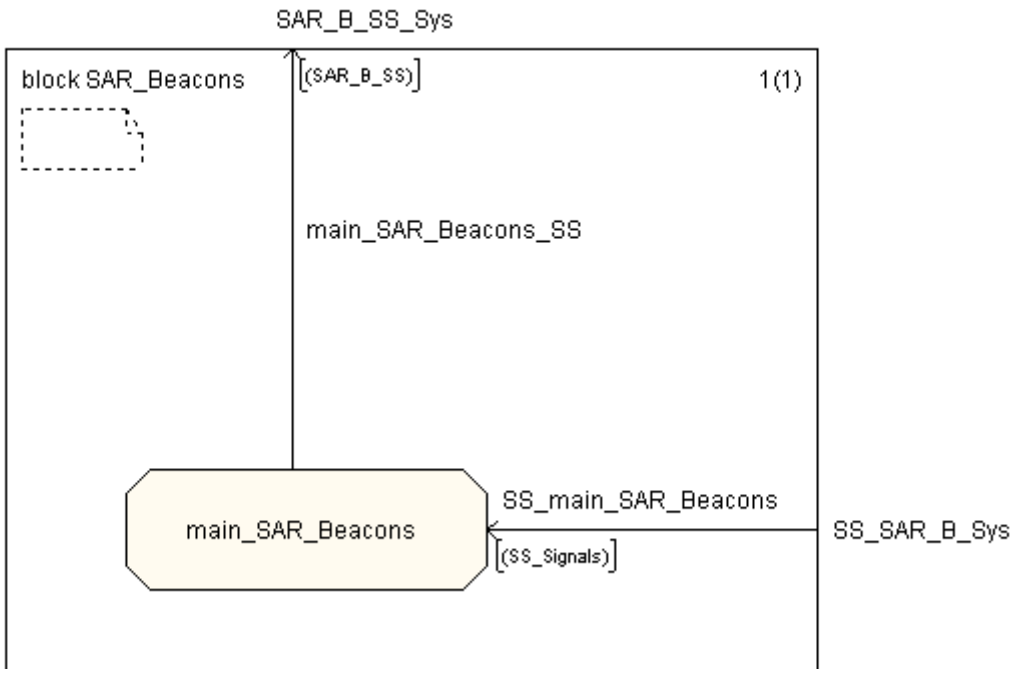


Figure 229: SARF1 SDL Functional Block Diagram

5.6.10.2 SARF2: SAR Meolut

SARF2 General Description

This SAR MEOLUT function for what concerns the Galileo system is responsible for:

- Reception of SAR Distress Messages from up to 150 active beacons relayed by Galileo Satellites from SAR Beacons to start rescue response
- Creation of list of satellites to be used for Return link service dissemination processed through the RLSP.

SARF2 Outputs

The SAR MEOLUT function generates no outputs directly to the Galileo System.

SARF2 Design Constraints

The main design constraints of the SAR MEOLUT function are considered to be:

- BER < 5E-5 within 5minutes
- Be able to receive SAR Forward link downlink in frequency band of 1544.05 - 1544.15MHz
- Masking angle 5 deg

SARF2 Functional Block Diagram

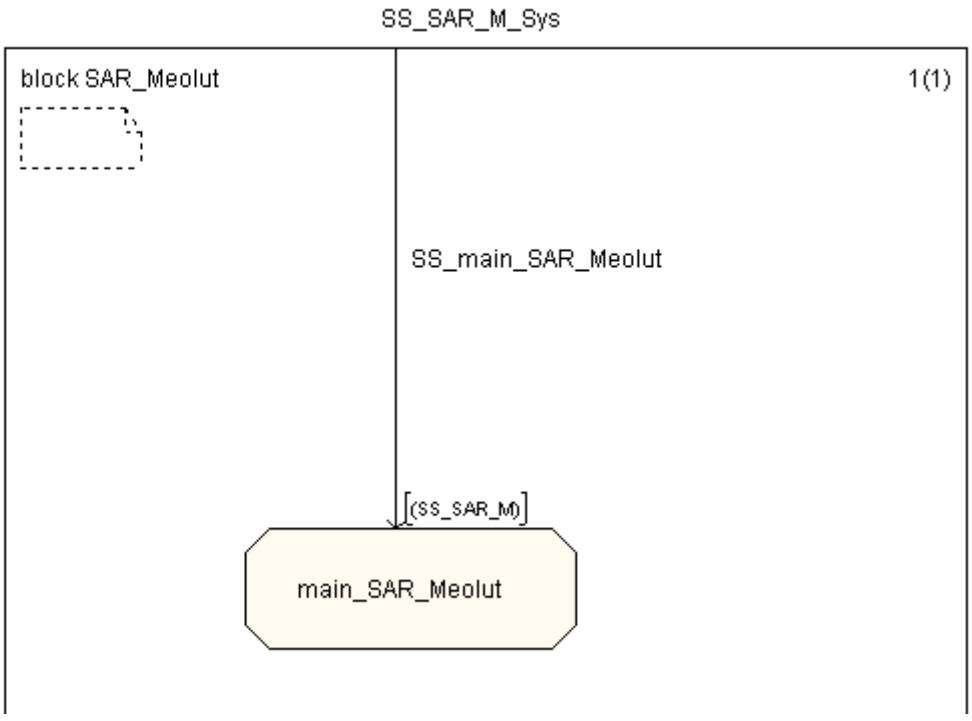


Figure 230: SARF2 SDL Functional Block Diagram

5.6.11 SATMAN Functions

5.6.11.1 SATMANF1: Satellite Manufacturer

SATMANF1 General Description

The satellite manufacturer (SATMAN) provides support to Galileo operational activities through:

- Support to on-board failure analyses including reception of supporting data from the GCC (eg TC/TM extracts, on-board software dumps, reports etc);
- Provision of on-board software patches as required and in response to failure analyses;
- Provision of data to the GCC as required for the correct operation of the satellites (eg operational procedures, TC/TM databases, flight dynamics data etc)

SATMANF1 Outputs

This function is responsible for producing the following outputs for each spacecraft:

- Validated operational procedures (platform, payload, and OBCP)
- On-board software (OBSW) patches (for upload to the spacecraft)
- Avionics software (ASW) for use in the simulator function in the GCS
- Spacecraft TC/TM databases
- Flight dynamics data for input to the flight dynamics functions in the GCS
- Configuration data defining which versions of the spacecraft databases, operational procedures and on-board software have been validated together as a packaged dataset
- Satellite flight operations manual (ie satellite user manual)

SATMANF1 Functional Block Diagram

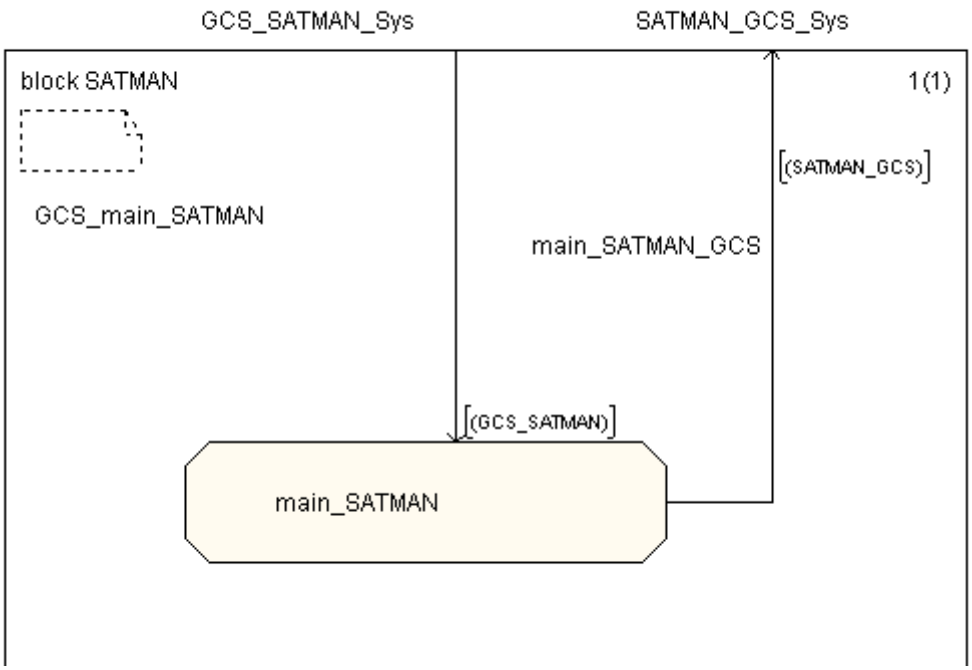


Figure 231: SATMAN SDL Functional Block Diagram

5.6.12 SC Functions

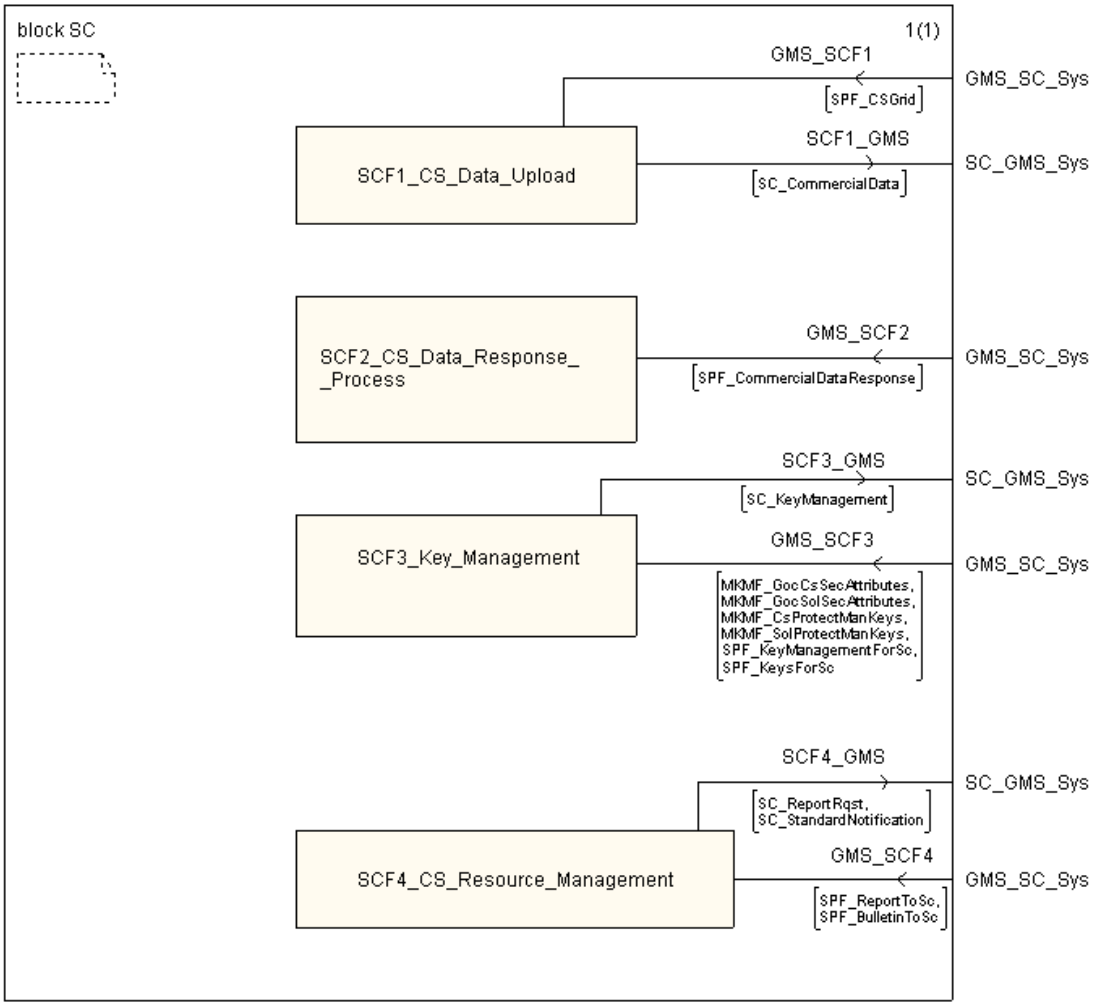


Figure 232: SC SDL Functional Block Diagram

5.6.12.1 SCF1 CS Data Upload

SCF1 Description

The CS Data Upload function inside GSC is in charge of generation of Commercial Service Data in ASN.1 basic aligned PER format to be sent to GMS in order to be included in the Commercial Service signals of Galileo. This CS data is preformatted in order to facilitate its uplink and downlink to/from Galileo satellites. GSC will include in this CS Data a unique ID field for the CS Data. This identifier is used for dissemination feedback purposes.

This function is also in charge of receiving and process the CS Grid in order to generate the Satellite ID required by the Commercial Service. Inside of CS Grid are defined which satellites should be accessed to disseminate information on a particular CS region, so based on this contact plan provided by GMS the GSC can select the required satellites to use for each Commercial Service Data message.

SCF1 Functional Block Diagram

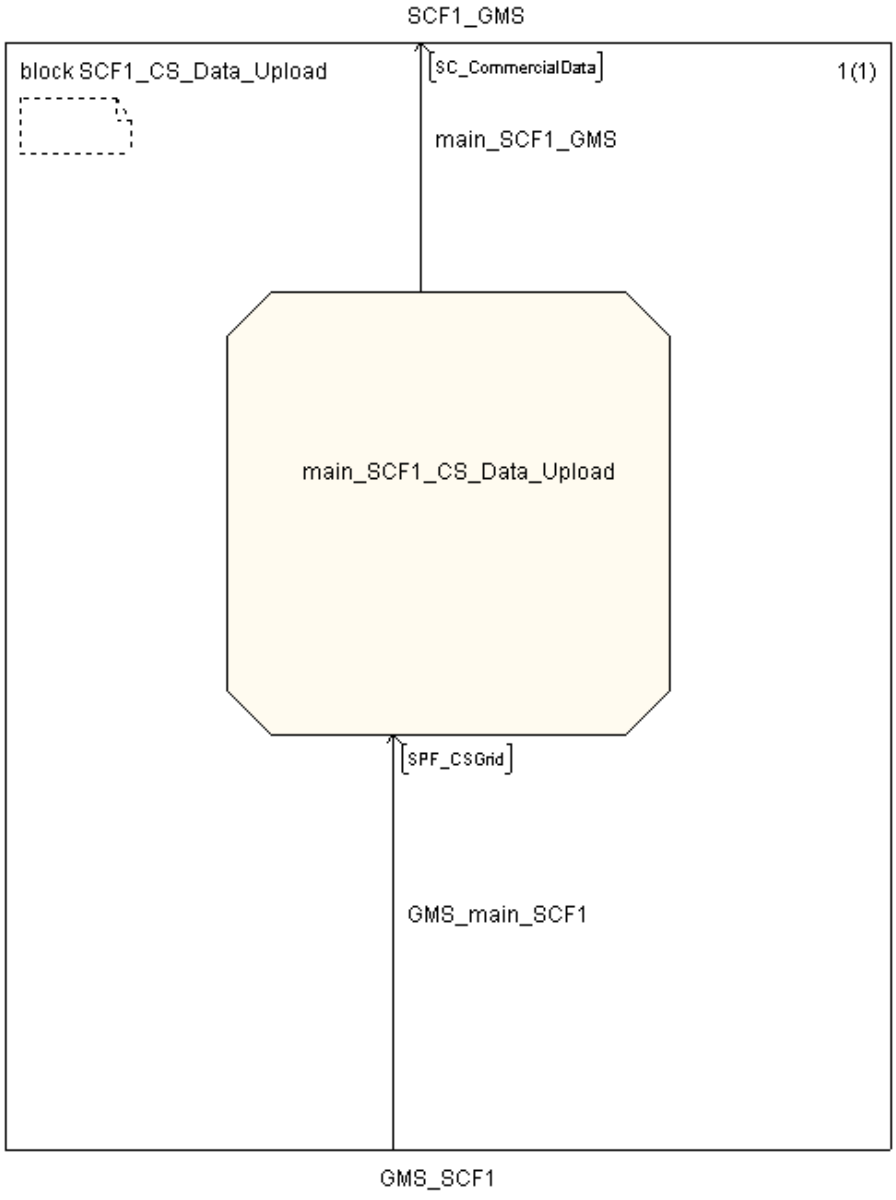


Figure 233: SCF1 SDL Functional Block Diagram

5.6.12.2 SCF2 CS Data Response Process

SCF2 Description

The CS Data Response Process function inside GSC is in charge of receiving the status of the commercial data stream which reports the status of CS Data messages scheduled for uplink in the preceding 30 minutes. With this information GSC is aware of the status of CS Data dissemination and is able to decide which information is required to be included in the next CS Data message.
This function also receives the evolution of the ULS sub-frame availability monitor parameter (overall single flag for ULS health status that will be used by GSC to rise a local alarm and allow the GSC operator to request a new CS Grid).

SCF2 Functional Block Diagram

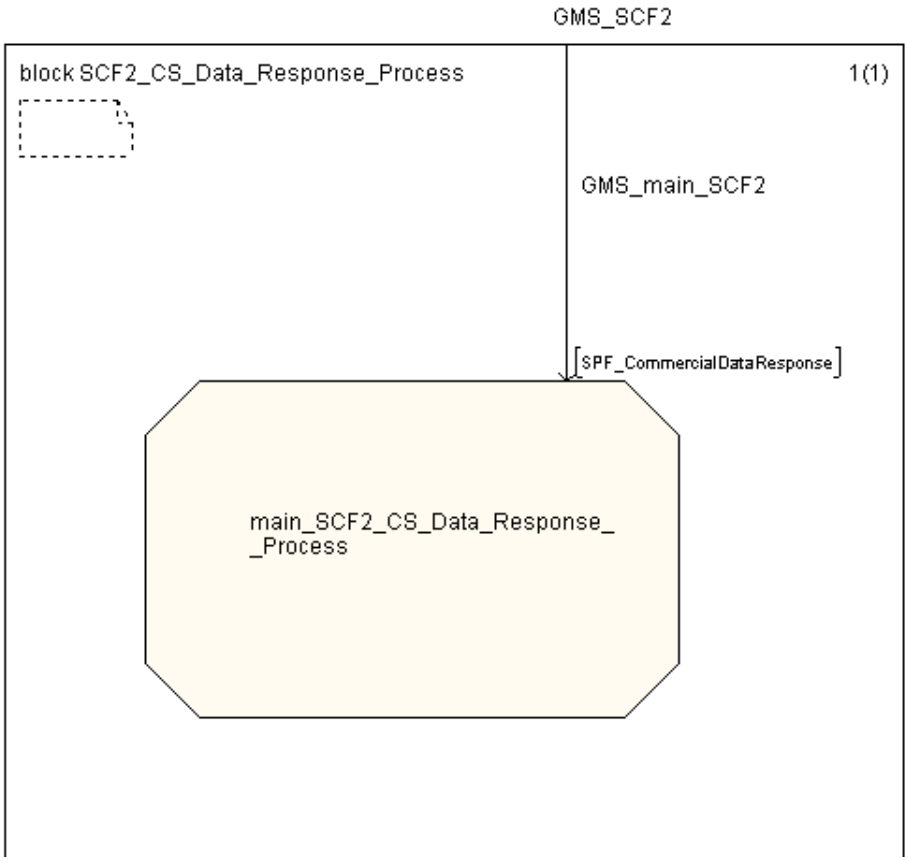


Figure 234: SCF2 SDL Functional Block Diagram

5.6.12.3 SCF3 Key Management

SCF3 Description

The Key Management function inside GSC is in charge of receiving and managing the distribution to users of keys for accessing commercial services on C/NAV navigation signal.
This function also is responsible of:
a) Generate requests for key or for changing the key used for the protection of C/NAV ranging code
b) Receive from GMS notice that the key for ranging code protection is gong to change and the new keys.

SCF3 Functional Block Diagram

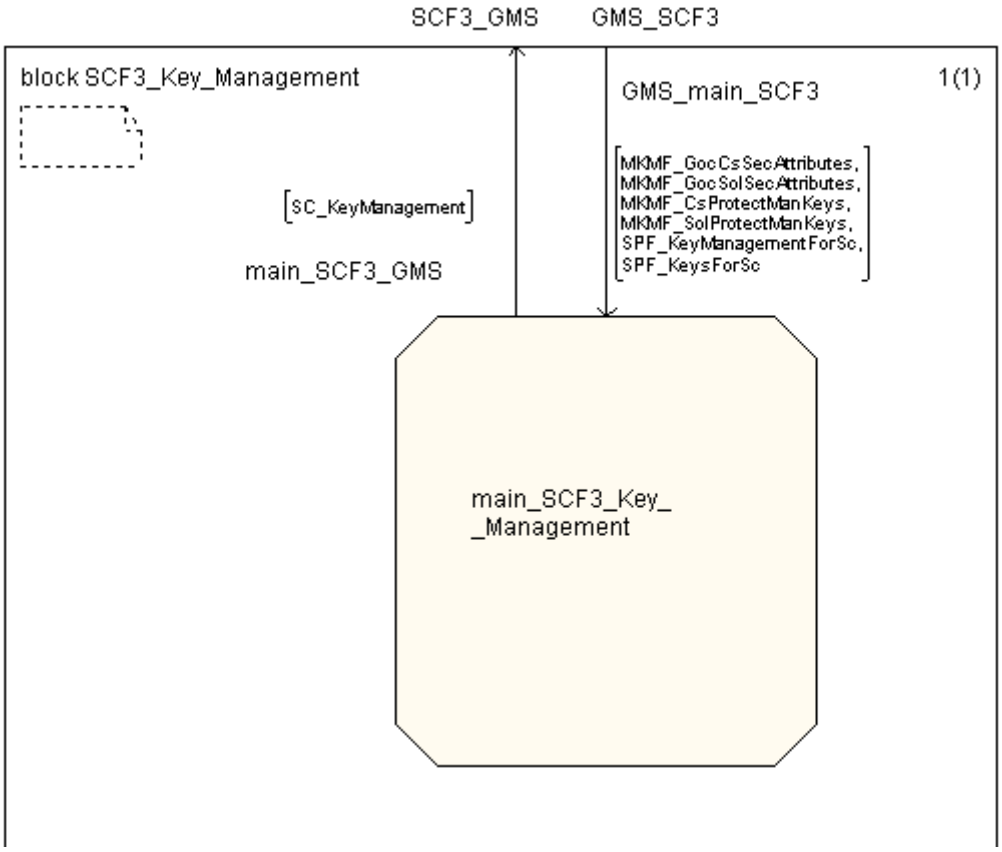


Figure 235: SCF3 SDL Functional Block Diagram

5.6.12.4 SCF4 CS Resource Management

SCF4 Description

The CS Resource Management function inside GSC is in charge of receiving different products obtained from GMS. These products include service and system performance reports, status, almanac, NAGUs, ERIS status, Galileo observables and navigation message, and other archived information from GMS. Based on these reports provided GSC is able to generate other Galileo products and value added services to other Service Centers or subscribers in the user community.

This function is also responsible to generate and send request to GMS in order to obtain particular data/reports from GMS or to retrieve old reports or introduce new products in the catalogue.

This function is also responsible to receive from GMS and process the maintenance bulletins and forecast performance bulletins. These bulletins contain information related to all maintenance activities on all system resources within the current planning time-frame, including scope of maintenance, duration and concerned elements, results from the comparison between the required performances and the system performances assessed from the data of the validation function, planning reports from log data, and NAGUs taking into account satellite control information provided by the GCS.

Remark: *the automated NAGU generation concept is currently not consolidated in the STRB2.0 baseline. It is expected that the content of the signal INF_NoticeAdvisoryToUsers will be refined in close coordination with GSC in the frame of GSC operational concept consolidation. In the meanwhile, the signal INF_NoticeAdvisoryToUsers will not be disseminated to the external entity and the NAGU will be generated manually and sent to the GSC. In both cases (automated and manual NAGU processes) the authorization for dissemination to the public is granted by the corresponding formal process at GSC level.*

All the information received by this function is not intended for public distribution or publication, intended only for GSC internal use.

Finally, this function is also responsible to process notifications obtained from Service Centre User Anomaly Reports and generate and send to GMS the Anomaly Notification message which contains the standard anomaly notifications composed from the collection and filtering of User Anomaly Reports.

SCF4 Functional Block Diagram

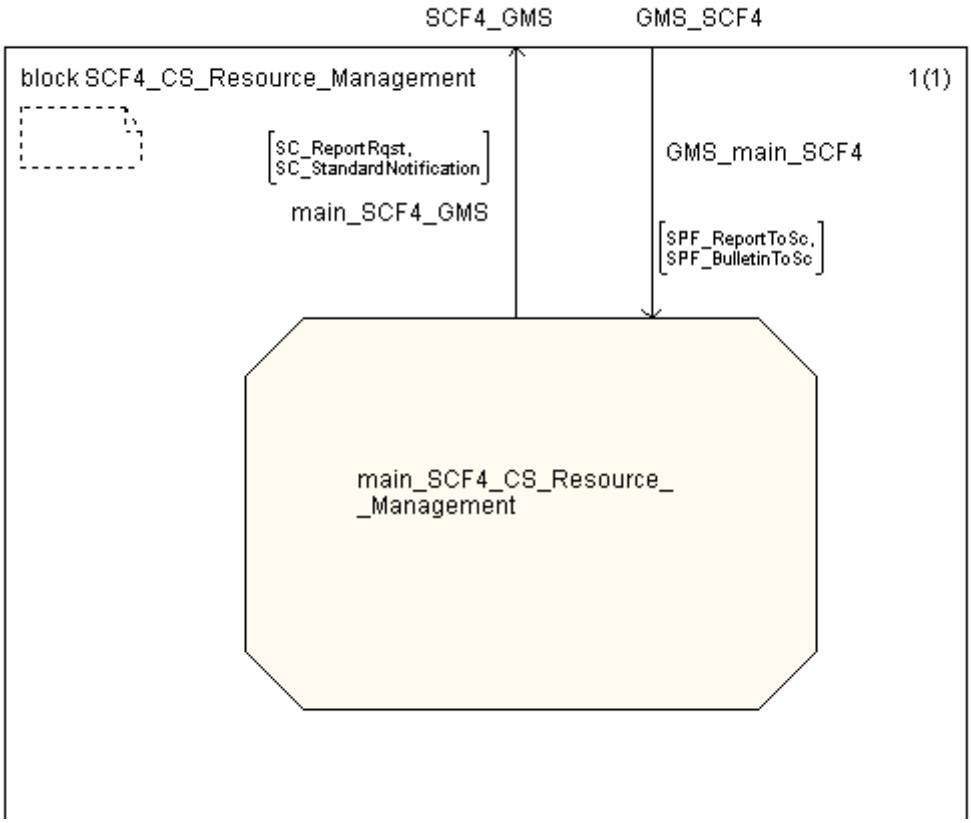


Figure 236: SCF4 SDL Functional Block Diagram

5.6.13 TSP Functions

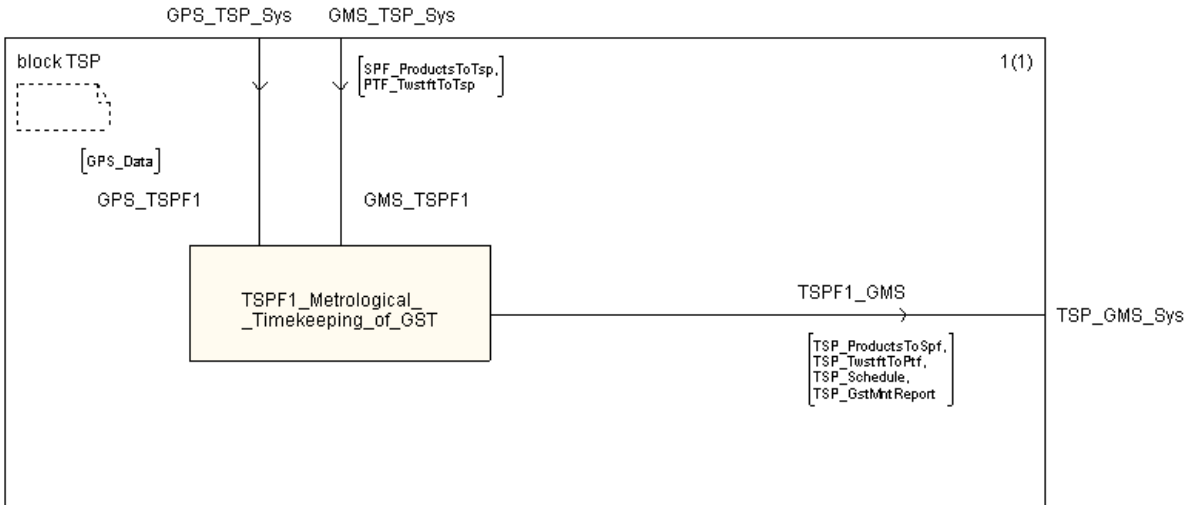


Figure 237: TSP SDL Functional Block Diagram

5.6.13.1 TSPF1 Metrological Timekeeping of GST

TSPF1 Description

This is a function allocated to Time Service Provider (TSP) the entity external to the Galileo Core. TSP in order to fulfill its functionality will rely on data supplied by BIPM, TSP/UTC(k) laboratories (4) and PTF's. TSP is required to deploy interfaces (TSPF1.1) and to deliver to GMS the data/products (TSPF1.2) as per specification given in the GMSEIFREQ and GMS to TSP-ICD. TSP's main task is to provide the GST to TAI steering products which form the basis for the Metrological timekeeping.

TSPF1 Functional Block Diagram

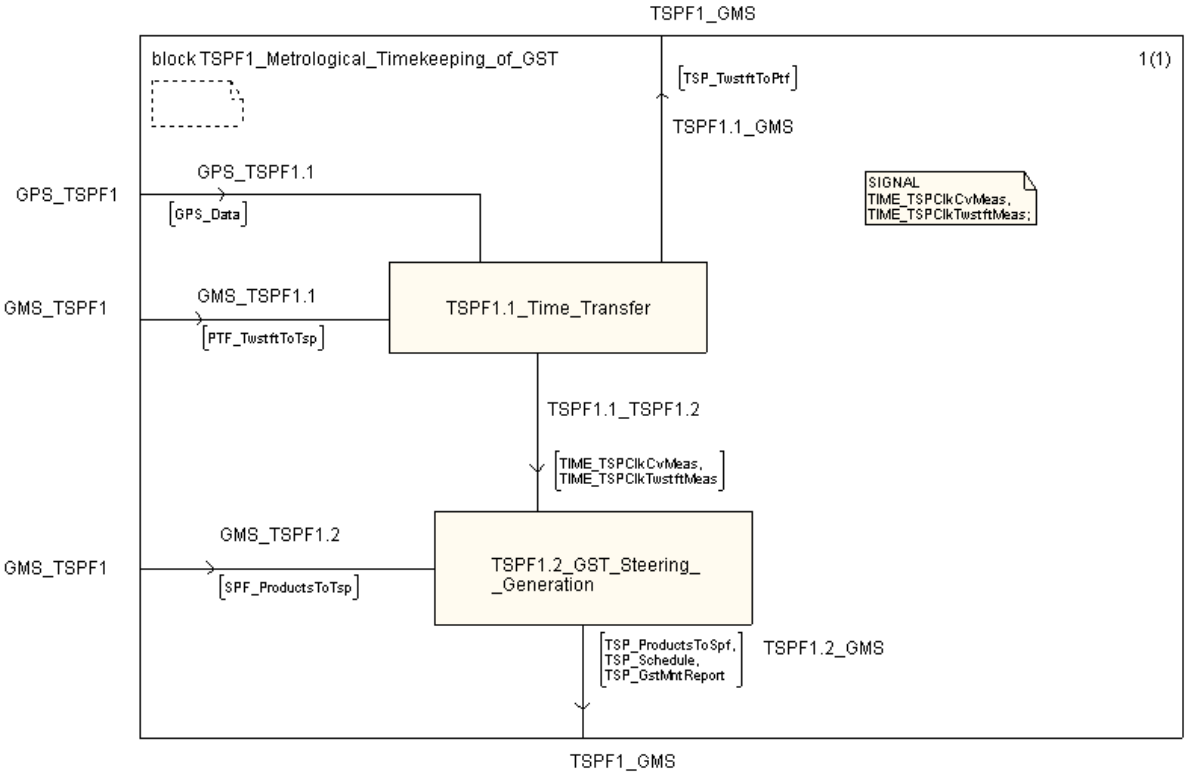


Figure 238: TSPF1 SDL Functional Block Diagram

5.6.13.1.1 TSPF1.1 TimeTransfer

TSPF1.1 Description

The TSP time transfer function is responsible for the deployment and the operation of the TWSTFT and GPS/Galileo-CV time transfer links between the PTF's and TSP/UTC(K) laboratories (4). TSP has to rely on the time transfer and clock data obtained from the TSP/UTC(k) laboratories as TSP itself is mainly a processing facility and as such TSP does not posses any time transfer hardware nor related facilities. The Time Transfer function feeds the TSPF1.2 with the data which are together with data from GMS and BIPM processed to generate the steering corrections and other timing products specified in GMSEIFREQ and GMS to TSP-ICD.

TSPF1.1 Functional Block Diagram

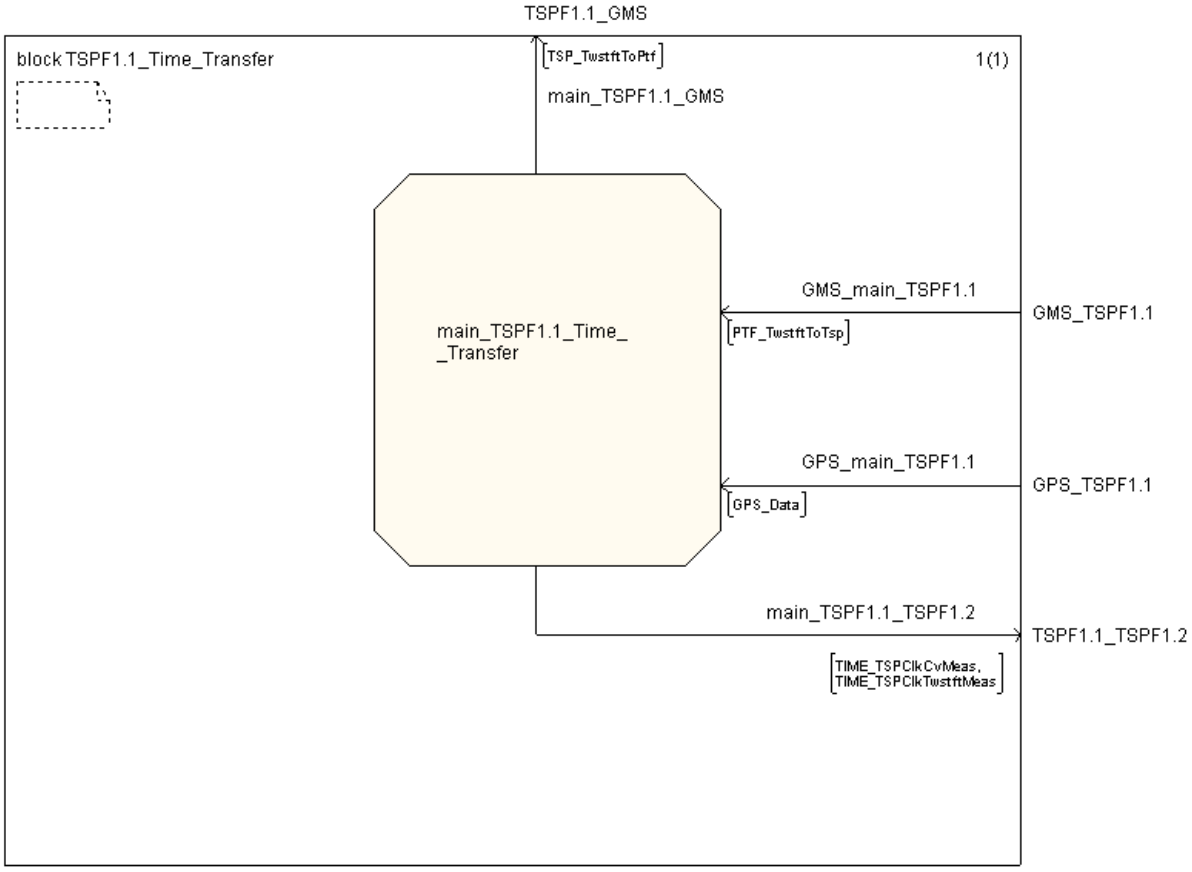


Figure 239: TSPF1.1 SDL Functional Block Diagram

5.6.13.1.2 TSPF1.2 GST Steering Generation

TSPF1.2 Description

This is the PTF's processing function. This function deals with all of the data supplied by GMS/PTF'S, BIPM and UTC(k). Within the scope of this function the TSP will generate and deliver to GMS/PTF following products:

- GST-TAI steering corrections
- UTC-TAI offset
- Reports on quality of GST
- Prediction of GST-TAI offset

Any other products as specified for the TSP interface in the GMSEIFREQ and GMS to TSP ICD.

TSPF1.2 Functional Block Diagram

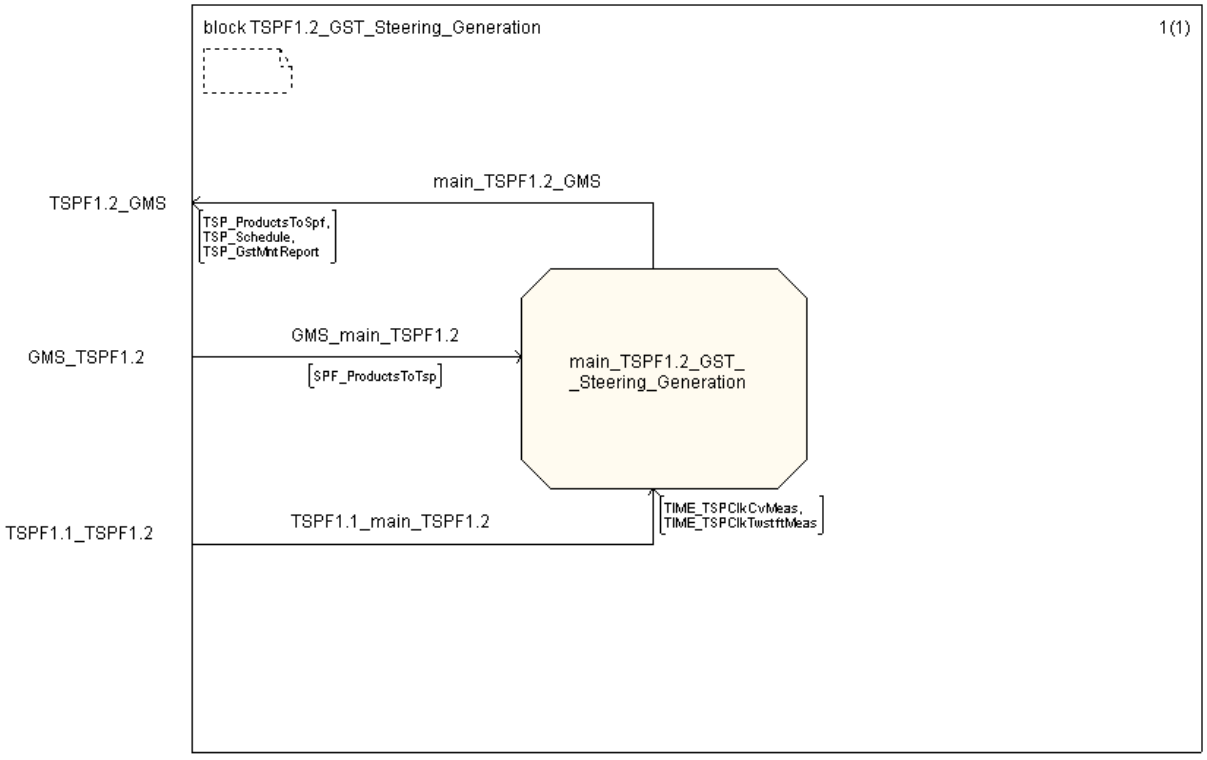


Figure 240: TSPF1.2 SDL Functional Block Diagram

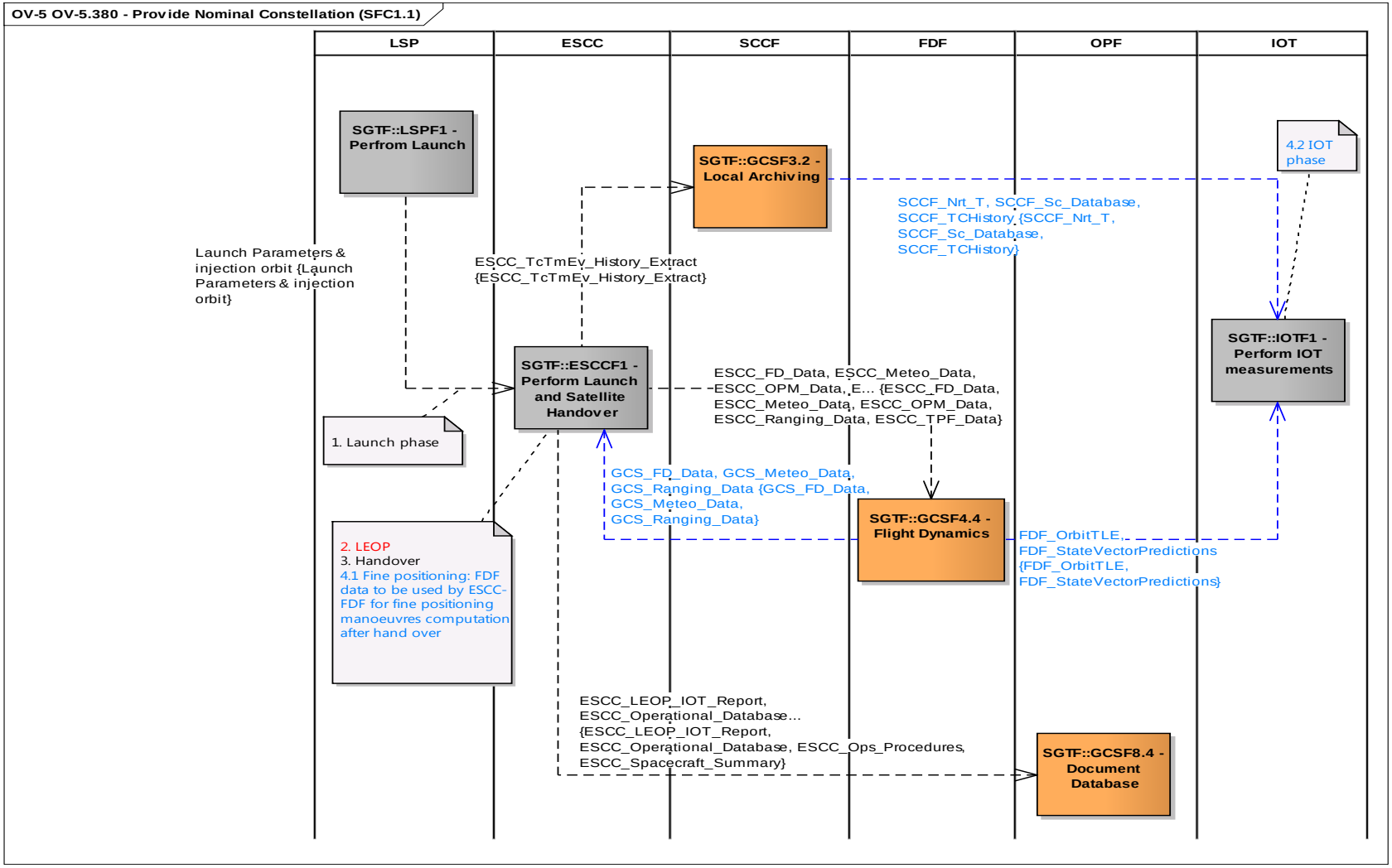
6. FUNCTIONAL ARCHITECTURE ALLOCATION

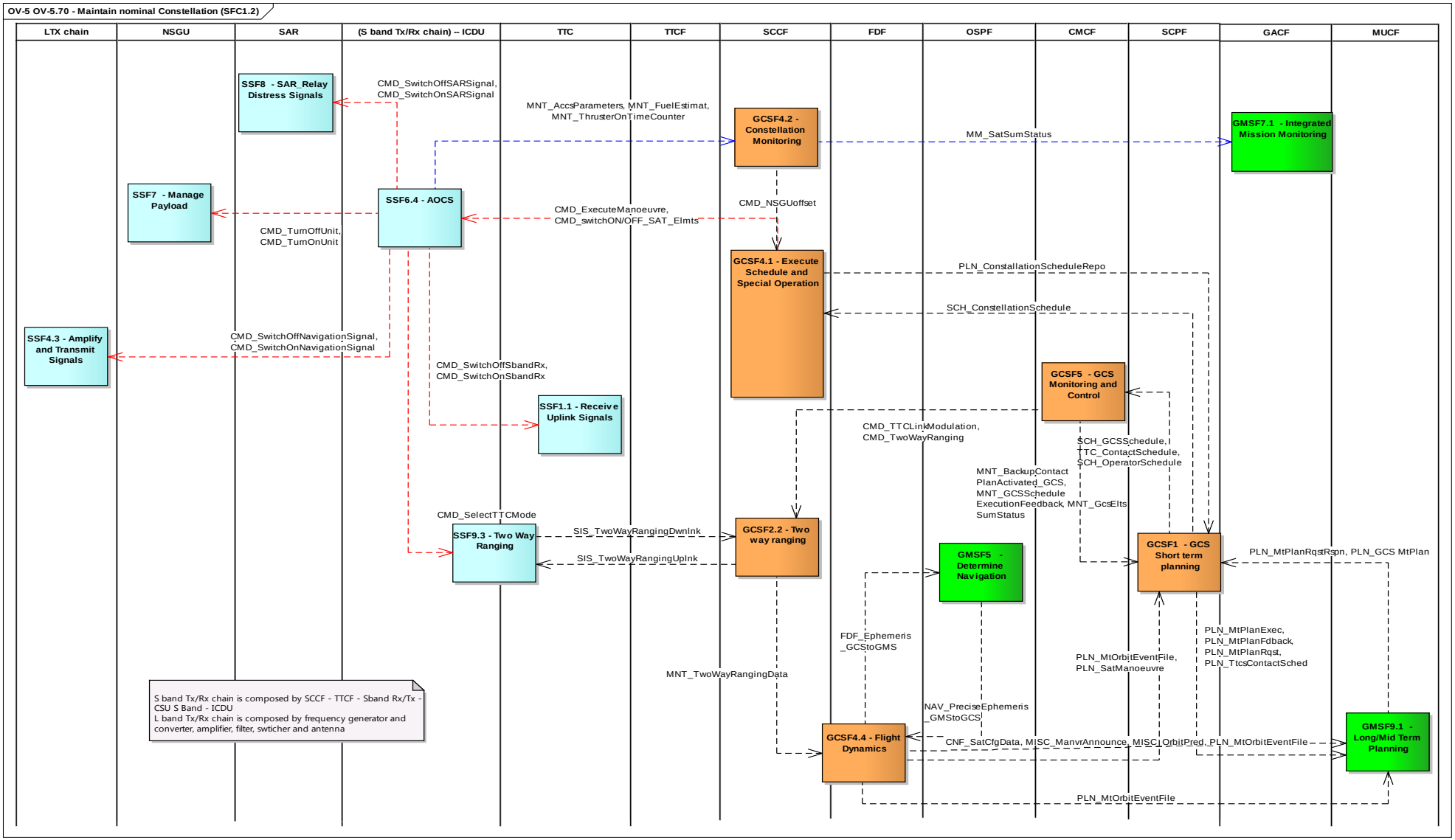
The section provides a different view on the SFC described in chapter 4 in the form of charts where the functions of the system functional architecture (as per chaper 5) are allocated to the physical system architecture. All charts have been generated with the Enterprise Architect tool.

Every column shown in the diagrams below represent a physical entity (physical element) of the system.

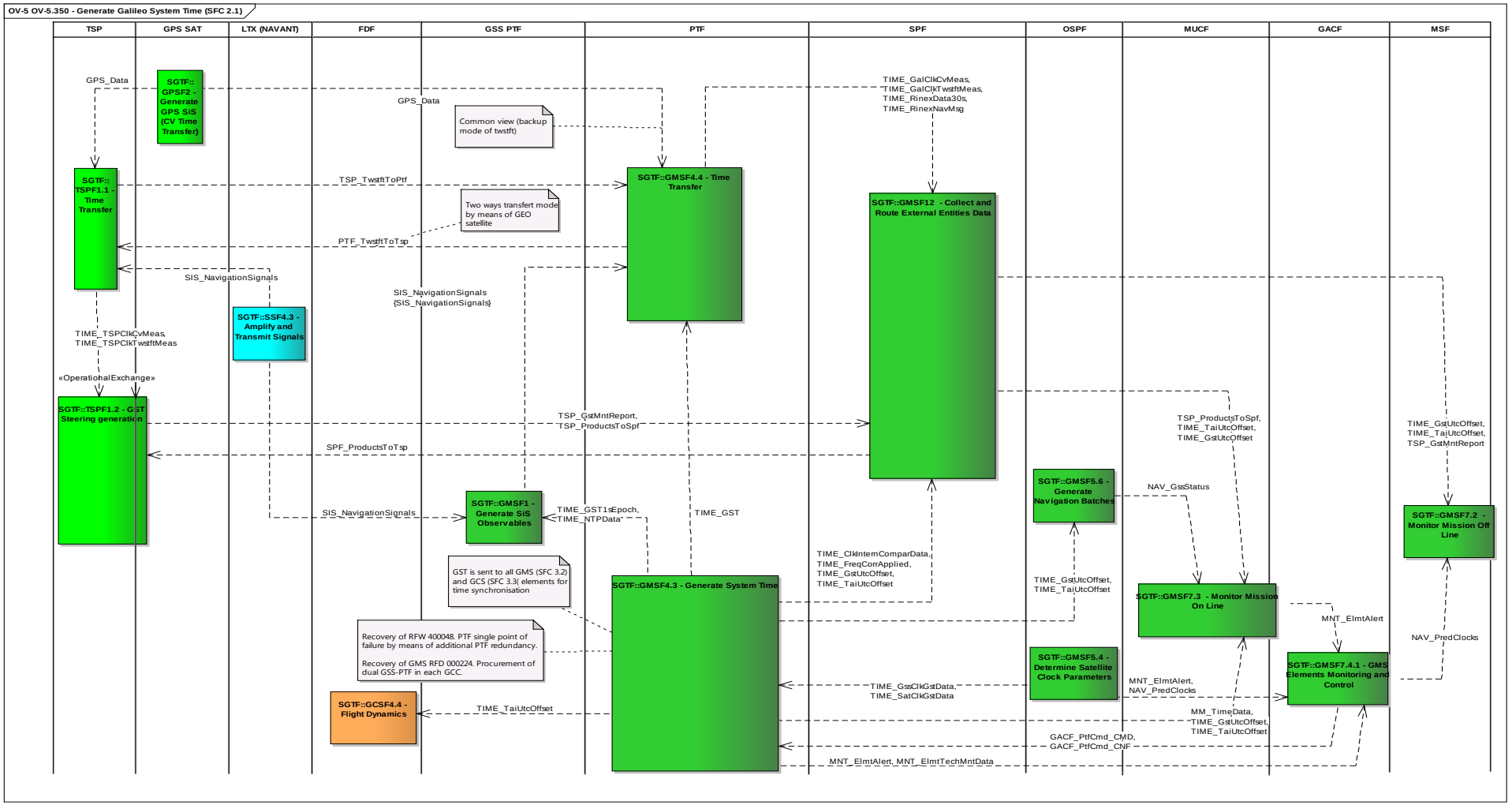
THALES
EU Proprietary information. Unauthorised distribution, dissemination or disclosure not allowed Compliant to 100181547K-EN-1
UNCLASSIFIED– COMMERCIAL IN CONFIDENCE

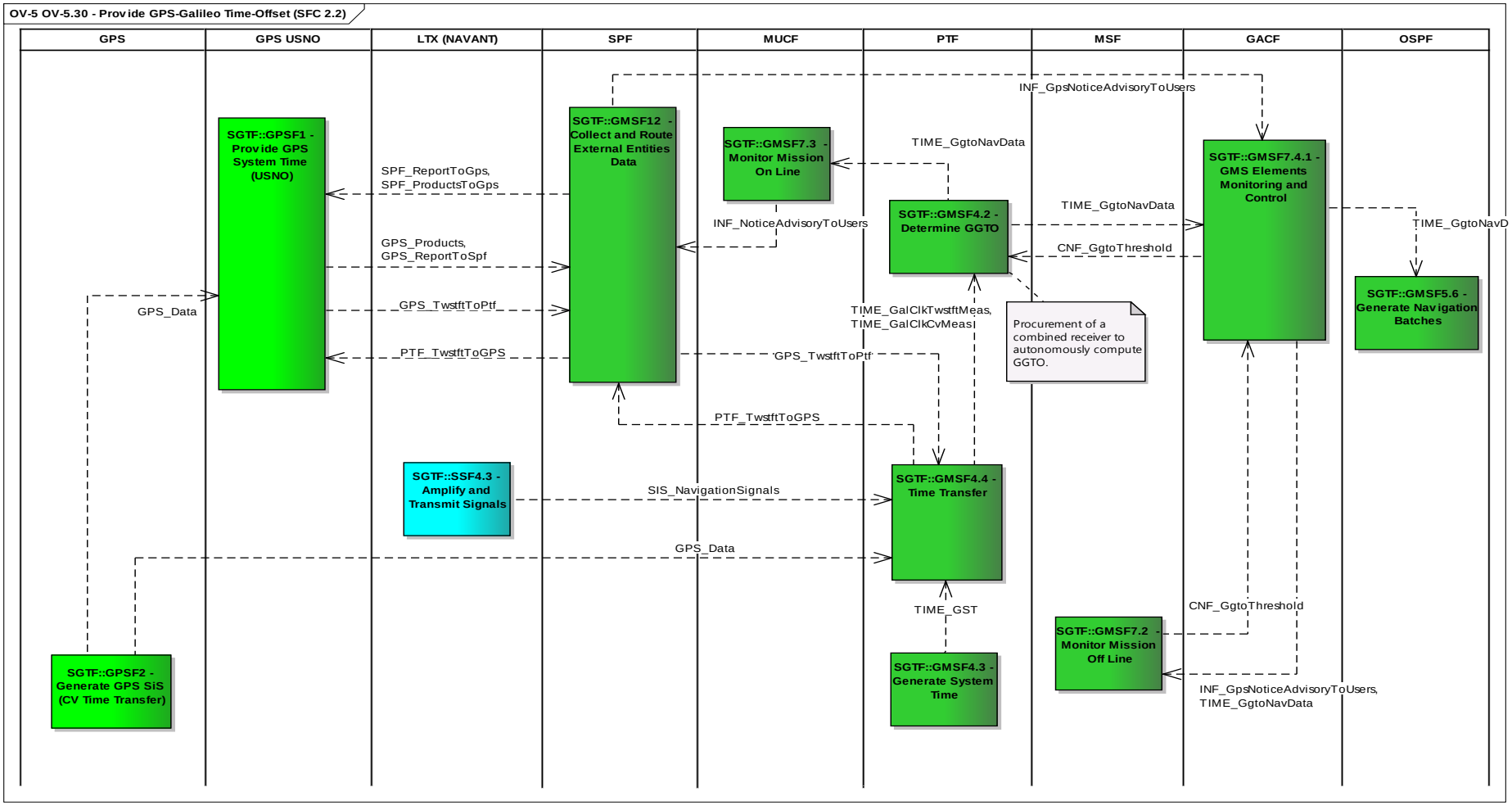
6.1 SFC1 Provide and Maintain Galileo Constellation

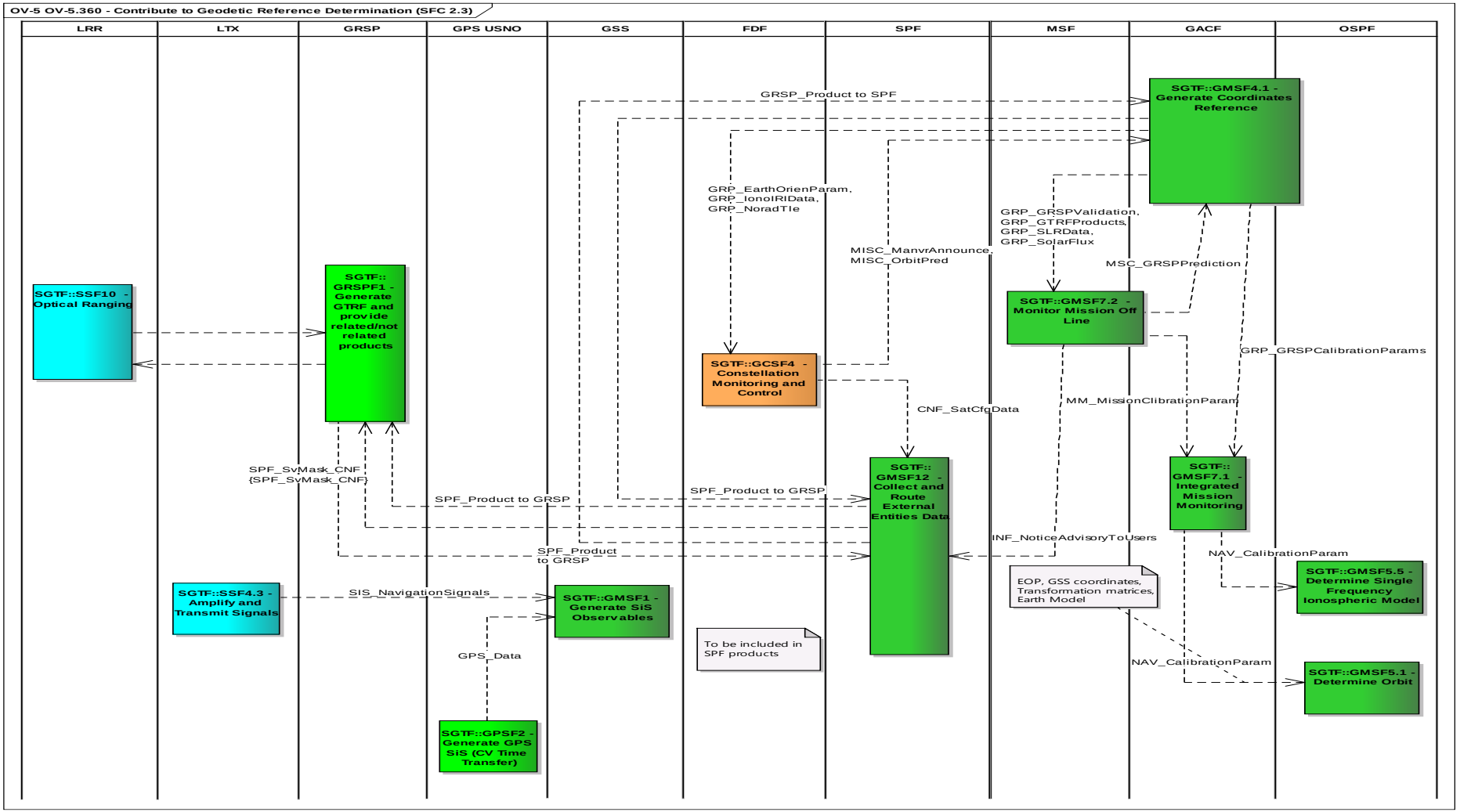



OV-5 OV-5.70 - Maintain nominal Constellation (SFC1.2)
LTX chain
NSGU
SAR
(S band Tx/Rx chain) – ICDU
TTC
TTCF
SCCF
FDF
OSPF
CMCF
SCPF
GACF
MUCF
SSF8 - SAR_Relay Distress Signals
CMD_SwitchOffSARSignal, CMD_SwitchOnSARSignal
MNT_AccsParameters, MNT_FuelEstimat, MNT_ThrusterOnTimeCounter
GCSF4.2 - Constellation Monitoring
MM_SatSumStatus
GMSF7.1 - Integrated Mission Monitoring
SSF7 - Manage Payload
SSF6.4 - AOCS
CMD_ExecuteManoeuvre, CMD_switchON/OFF_SAT_Elmts
CMD_NSGUoffset
CMD_TurnOffUnit, CMD_TurnOnUnit
GCSF4.1 - Execute Schedule and Special Operation
PLN_ConstallationScheduleRepo
SCH_ConstellationSchedule
SSF4.3 - Amplify and Transmit Signals
CMD_SwitchOffNavigationSignal, CMD_SwitchOnNavigationSignal
CMD_SwitchOffSbandRx, CMD_SwitchOnSbandRx
GCSF5 - GCS Monitoring and Control
SSF1.1 - Receive Uplink Signals
CMD_TTCLinkModulation, CMD_TwoWayRanging
SCH_GCSSchedule, TTC_ContactSchedule, SCH_OperatorSchedule
MNT_BackupContact PlanActivated_GCS, MNT_GCSSchedule ExecutionFeedback, MNT_GcsElts SumStatus
CMD_SelectTTCMode
SSF9.3 - Two Way Ranging
SIS_TwoWayRangingDwnlnk
SIS_TwoWayRangingUplnk
GCSF2.2 - Two way ranging
GMSF5 - Determine Navigation
GCSF1 - GCS Short term planning
PLN_MtPlanRqstRspn, PLN_GCS MtPlan
FDF_Ephemeris _GCStoGMS
PLN_MtPlanExec, PLN_MtPlanFdback, PLN_MtPlanRqst, PLN_TtcsContactSched
PLN_MtOrbitEventFile, PLN_SatManoeuvre
MNT_TwoWayRangingData
S band Tx/Rx chain is composed by SCCF - TTCF - Sband Rx/Tx - CSU S Band - ICDU
L band Tx/Rx chain is composed by frequency generator and converter, amplifier, filter, swticher and antenna
NAV_PreciseEphemeris _GMStoGCS
GCSF4.4 - Flight Dynamics
CNF_SatCfgData, MISC_ManvrAnnounce, MISC_OrbitPred, PLN_MtOrbitEventFile
GMSF9.1 - Long/Mid Term Planning
PLN_MtOrbitEventFile

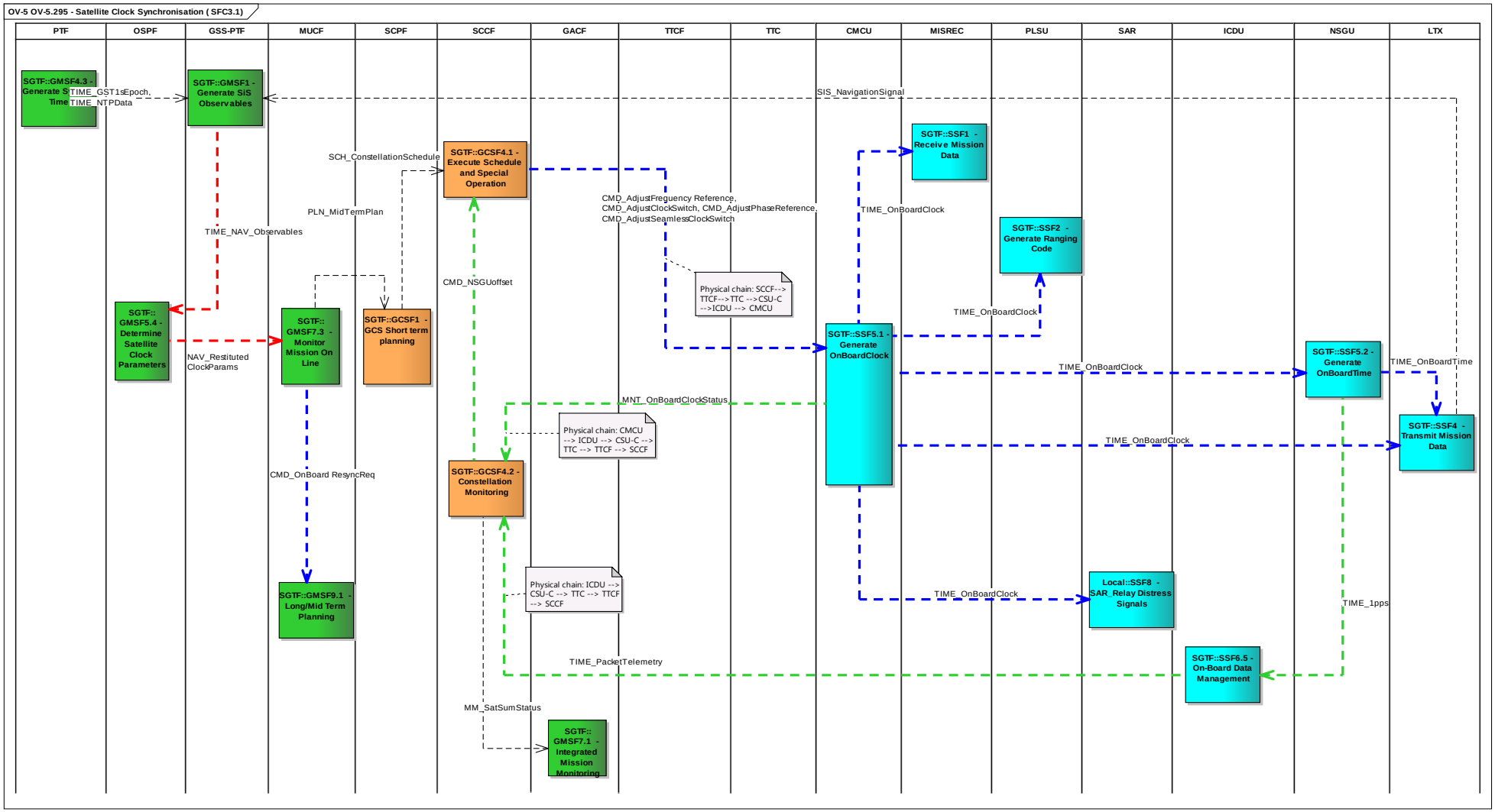
6.2 SFC2 Generate Galileo System Time

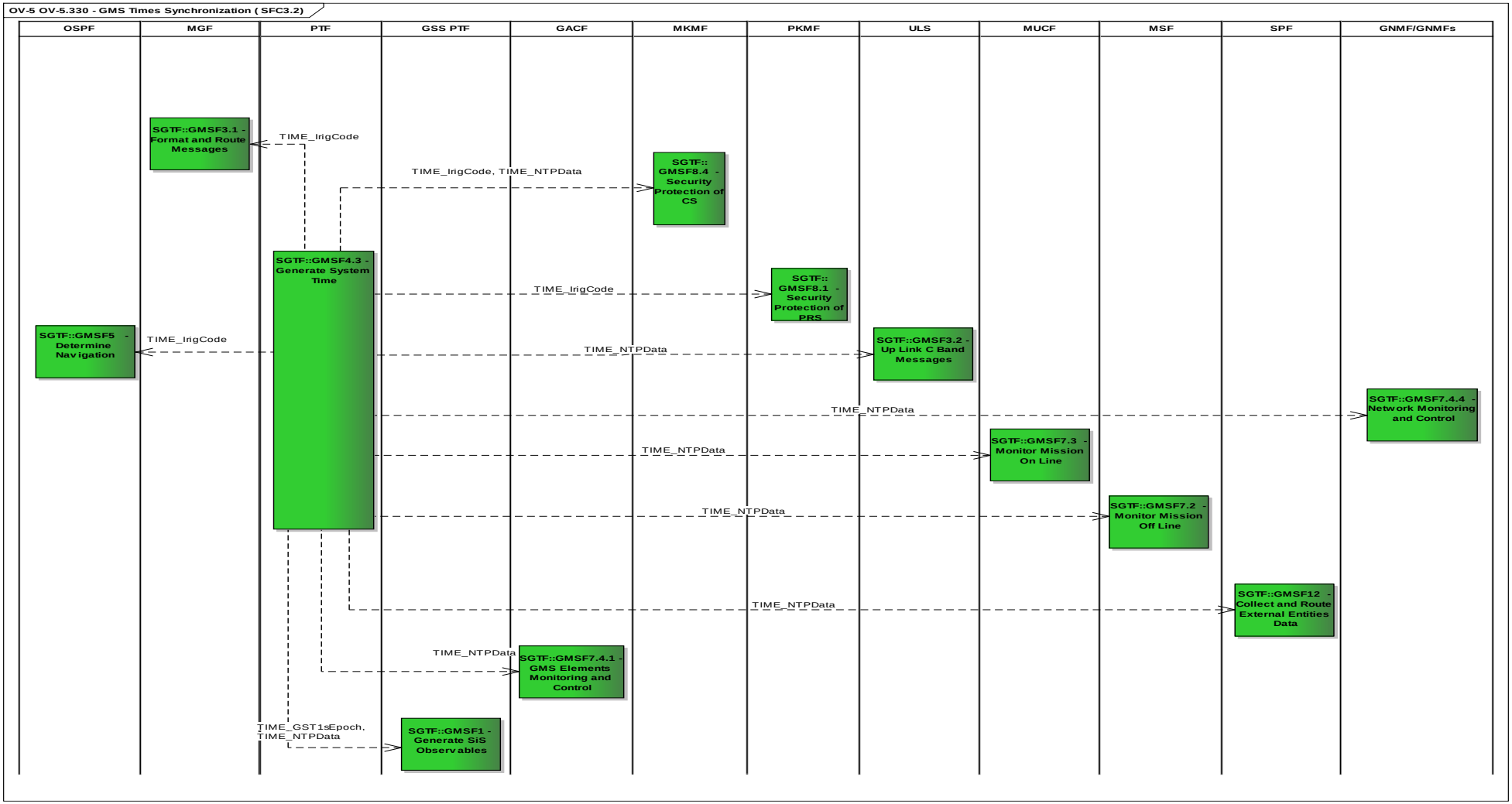


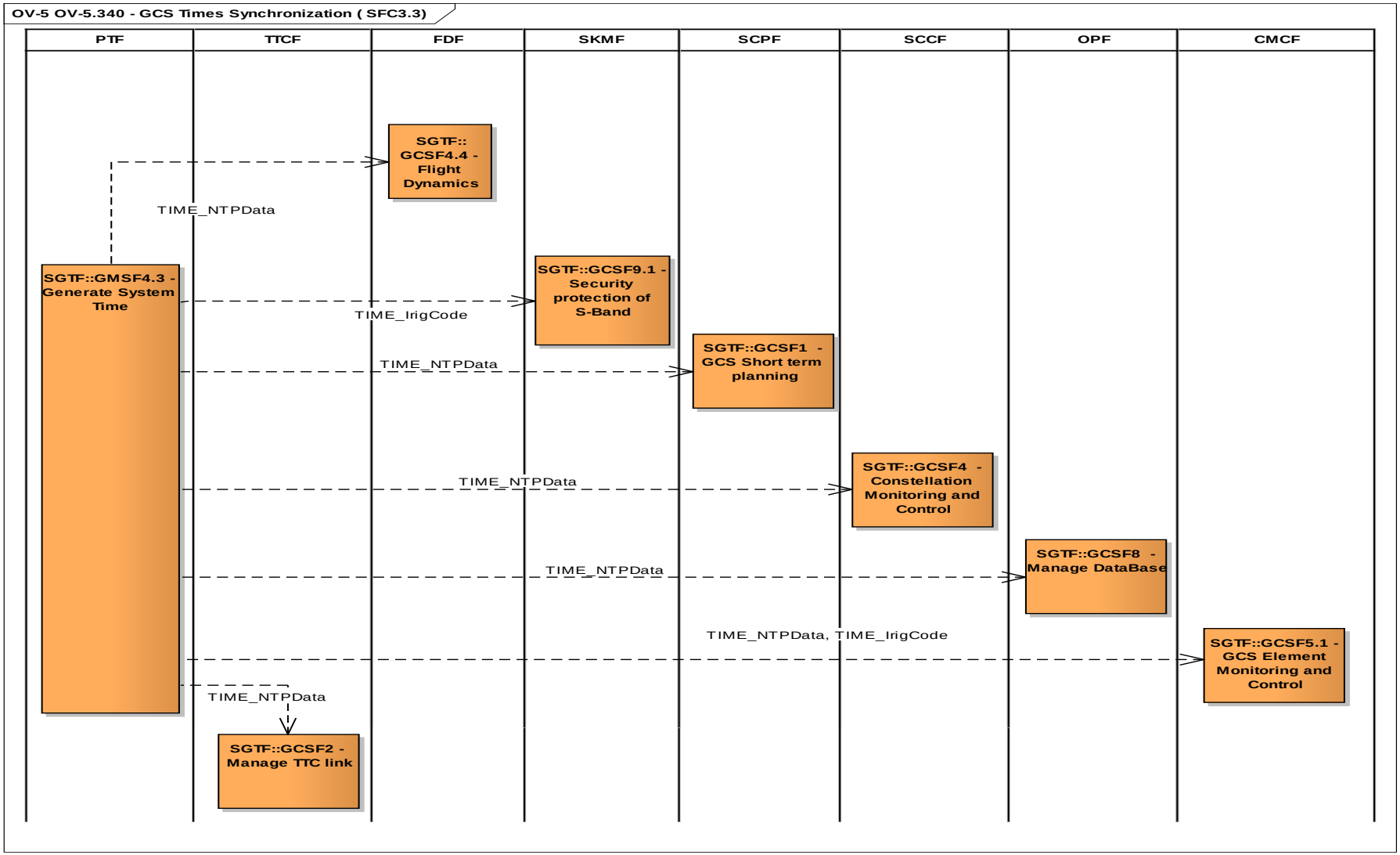

OV-5 OV-5.30 - Provide GPS-Galileo Time-Offset (SFC 2.2)
GPS
GPS USNO
LTX (NAVANT)
SPF
MUCF
PTF
MSF
GACF
OSPF
INF_GpsNoticeAdvisoryToUsers
SGTF::GPSF1 - Provide GPS System Time (USNO)
SGTF::GMSF12 - Collect and Route External Entities Data
SGTF::GMSF7.3 - Monitor Mission On Line
TIME_GgtoNavData
SGTF::GMSF7.4.1 - GMS Elements Monitoring and Control
SPF_ReportToGps, SPF_ProductsToGps
INF_NoticeAdvisoryToUsers
SGTF::GMSF4.2 - Determine GGTO
TIME_GgtoNavData
TIME_GgtoNavD
GPS_Products, GPS_ReportToSpf
CNF_GgtoThreshold
SGTF::GMSF5.6 - Generate Navigation Batches
GPS_Data
GPS_TwstftToPtf
TIME_GalClkTwstftMeas, TIME_GalClkCvMeas
Procurement of a combined receiver to autonomously compute GGTO.
PTF_TwstftToGPS
GPS_TwstftToPtf
PTF_TwstftToGPS
SGTF::SSF4.3 - Amplify and Transmit Signals
SIS_NavigationSignals
SGTF::GMSF4.4 - Time Transfer
GPS_Data
TIME_GST
CNF_GgtoThreshold
SGTF::GMSF7.2 - Monitor Mission Off Line
SGTF::GMSF4.3 - Generate System Time
SGTF::GPSF2 - Generate GPS SiS (CV Time Transfer)
INF_GpsNoticeAdvisoryToUsers, TIME_GgtoNavData


OV-5 OV-5.360 - Contribute to Geodetic Reference Determination (SFC 2.3)
LRR
LTX
GRSP
GPS USNO
GSS
FDF
SPF
MSF
GACF
OSPF
SGTF::SSF10 - Optical Ranging
SGTF:: GRSPF1 - Generate GTRF and provide related/not related products
SGTF::SSF4.3 - Amplify and Transmit Signals
SGTF::GMSF1 - Generate SiS Observables
SGTF::GPSF2 - Generate GPS SiS (CV Time Transfer)
SGTF::GCSF4 - Constellation Monitoring and Control
SGTF:: GMSF12 - Collect and Route External Entities Data
SGTF::GMSF4.1 - Generate Coordinates Reference
SGTF::GMSF7.2 - Monitor Mission Off Line
SGTF:: GMSF7.1 - Integrated Mission Monitoring
SGTF::GMSF5.5 - Determine Single Frequency Ionospheric Model
SGTF::GMSF5.1 - Determine Orbit
GRSP_Product to SPF
GRP_EarthOrienParam, GRP_IonoIRIData, GRP_NoradTle
GRP_GRSPValidation, GRP_GTRFProducts, GRP_SLRData, GRP_SolarFlux
MISC_ManvrAnnounce, MISC_OrbitPred
MSC_GRSPPrediction
GRP_GRSPCalibrationParams
CNF_SatCfgData
MM_MissionClibrationParam
SPF_SvMask_CNF {SPF_SvMask_CNF}
SPF_Product to GRSP
SPF_Product to GRSP
SPF_Product to GRSP
INF_NoticeAdvisoryToUsers
NAV_CalibrationParam
SIS_NavigationSignals
EOP, GSS coordinates, Transformation matrices, Earth Model
GPS_Data
To be included in SPF products
NAV_CalibrationParam

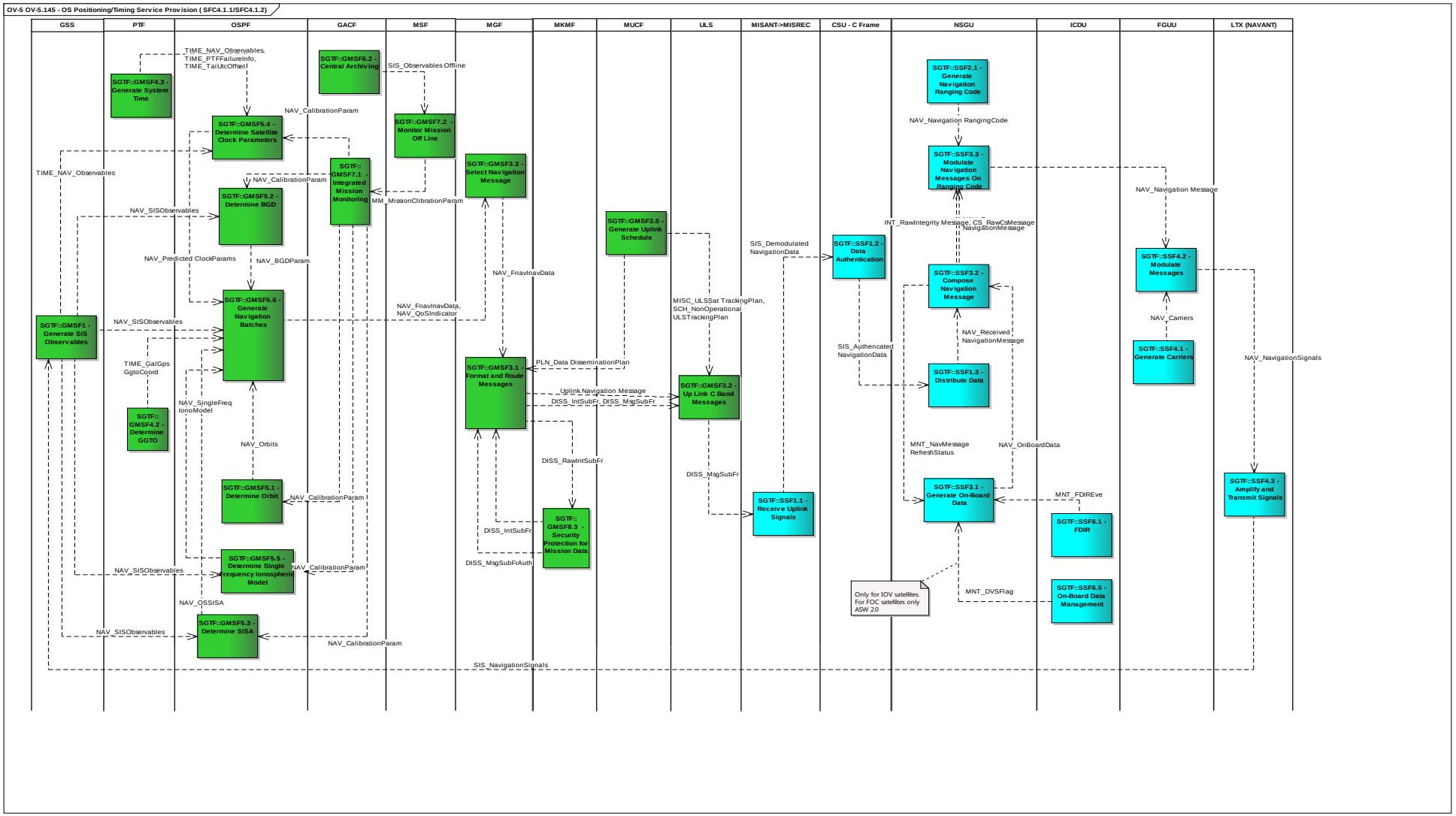
6.3 SFC3 Maintain Overall System Time Synchronization

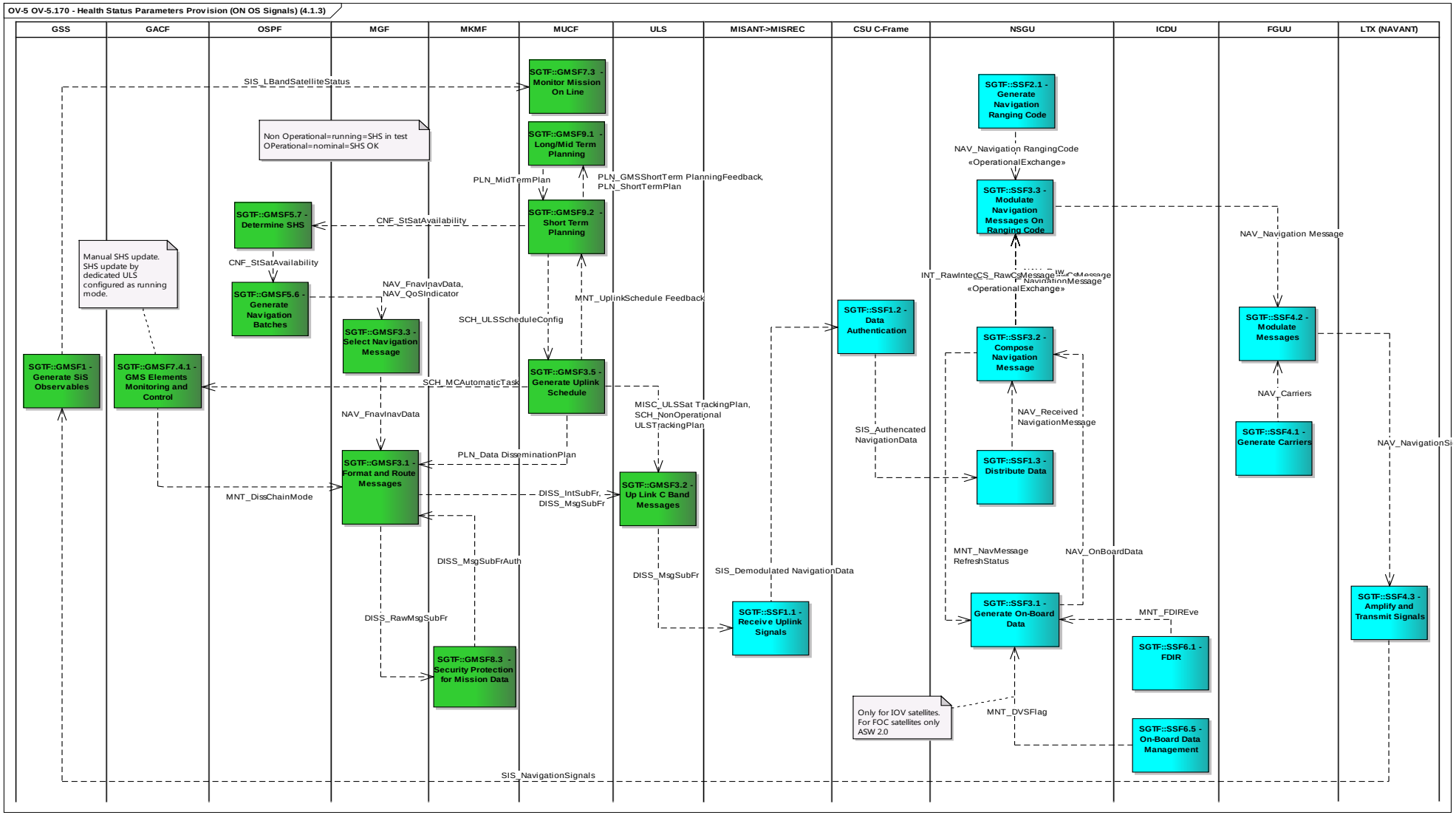


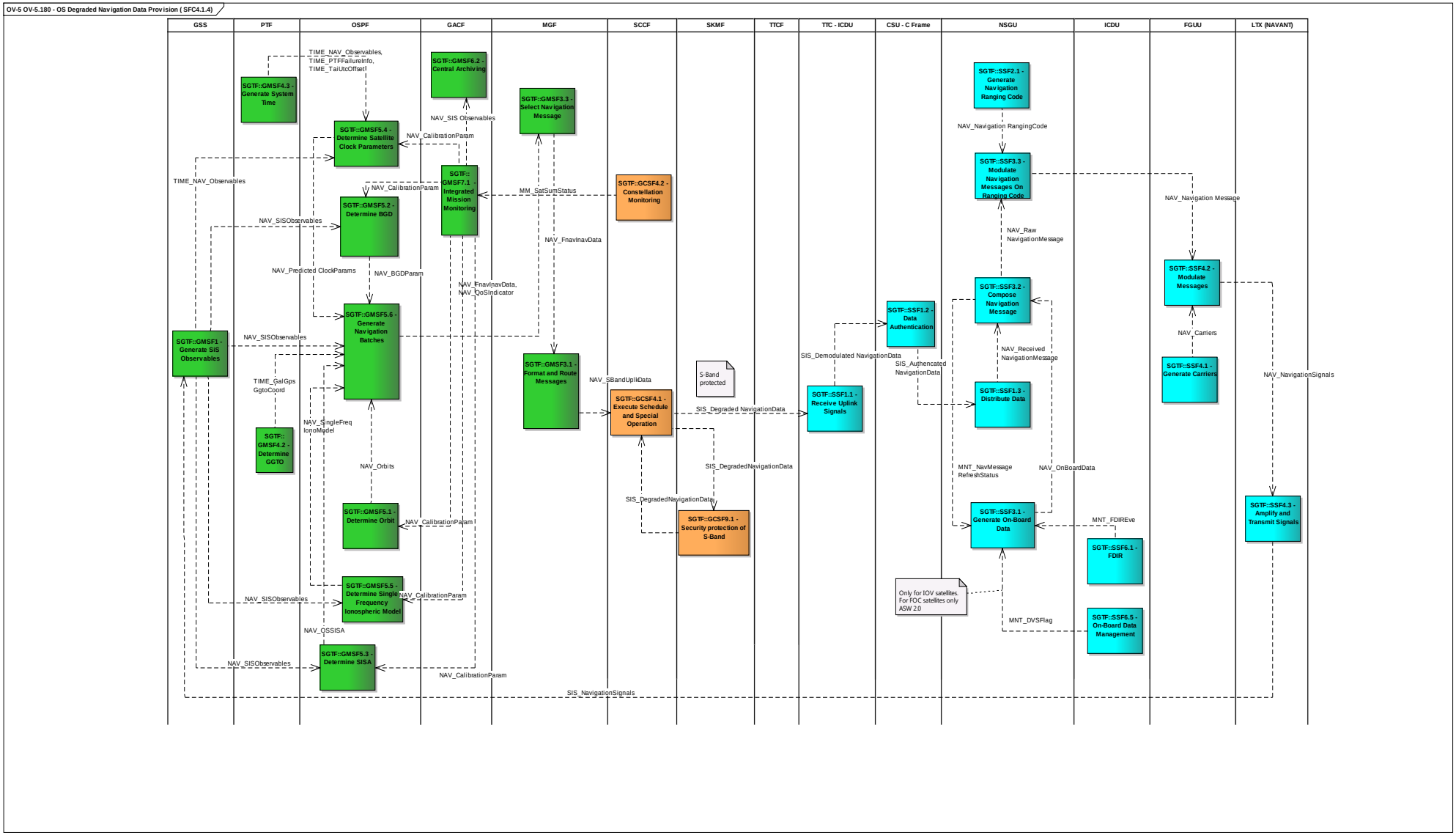

OV-5 OV-5.330 - GMS Times Synchronization (SFC3.2)
OSPF
MGF
PTF
GSS PTF
GACF
MKMF
PKMF
ULS
MUCF
MSF
SPF
GNMF/GNMFs
SGTF::GMSF3.1 - Format and Route Messages
TIME_IrigCode
TIME_IrigCode, TIME_NTPData
SGTF:: GMSF8.4 - Security Protection of CS
SGTF::GMSF4.3 - Generate System Time
TIME_IrigCode
SGTF:: GMSF8.1 - Security Protection of PRS
SGTF::GMSF5 - Determine Navigation
TIME_IrigCode
TIME_NTPData
SGTF::GMSF3.2 - Up Link C Band Messages
TIME_NTPData
SGTF::GMSF7.4.4 - Network Monitoring and Control
TIME_NTPData
SGTF::GMSF7.3 - Monitor Mission On Line
TIME_NTPData
SGTF::GMSF7.2 - Monitor Mission Off Line
TIME_NTPData
SGTF::GMSF12 - Collect and Route External Entities Data
TIME_NTPData
SGTF::GMSF7.4.1 - GMS Elements Monitoring and Control
TIME_GST1sEpoch, TIME_NTPData
SGTF::GMSF1 - Generate SiS Observables

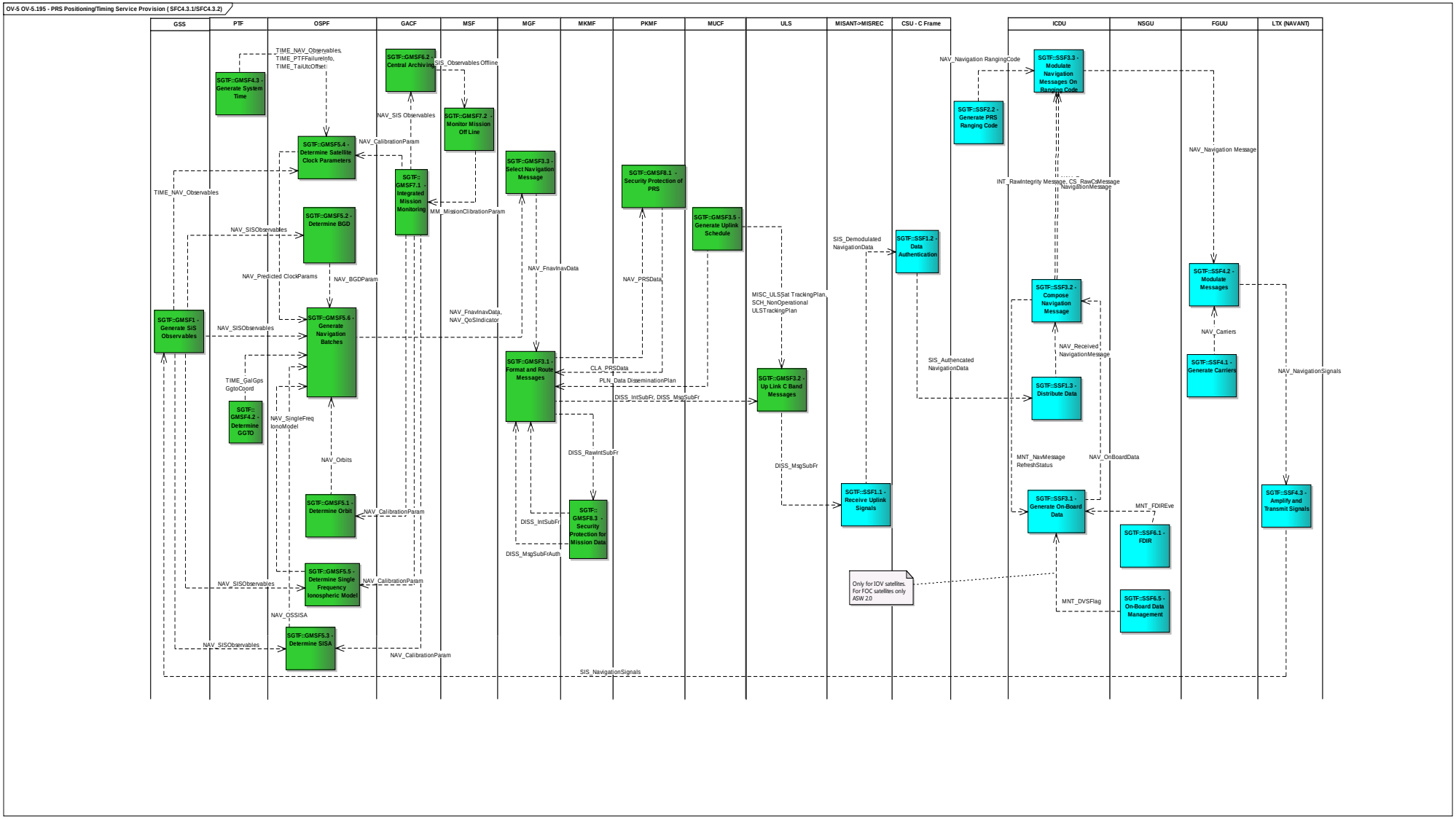

OV-5 OV-5.340 - GCS Times Synchronization (SFC3.3)
PTF
TTCF
FDF
SKMF
SCPF
SCCF
OPF
CMCF
SGTF:: GCSF4.4 - Flight Dynamics
TIME_NTPData
SGTF::GMSF4.3 - Generate System Time
SGTF::GCSF9.1 - Security protection of S-Band
TIME_IrigCode
SGTF::GCSF1 - GCS Short term planning
TIME_NTPData
SGTF::GCSF4 - Constellation Monitoring and Control
TIME_NTPData
SGTF::GCSF8 - Manage DataBase
TIME_NTPData
TIME_NTPData, TIME_IrigCode
SGTF::GCSF5.1 - GCS Element Monitoring and Control
TIME_NTPData
SGTF::GCSF2 - Manage TTC link

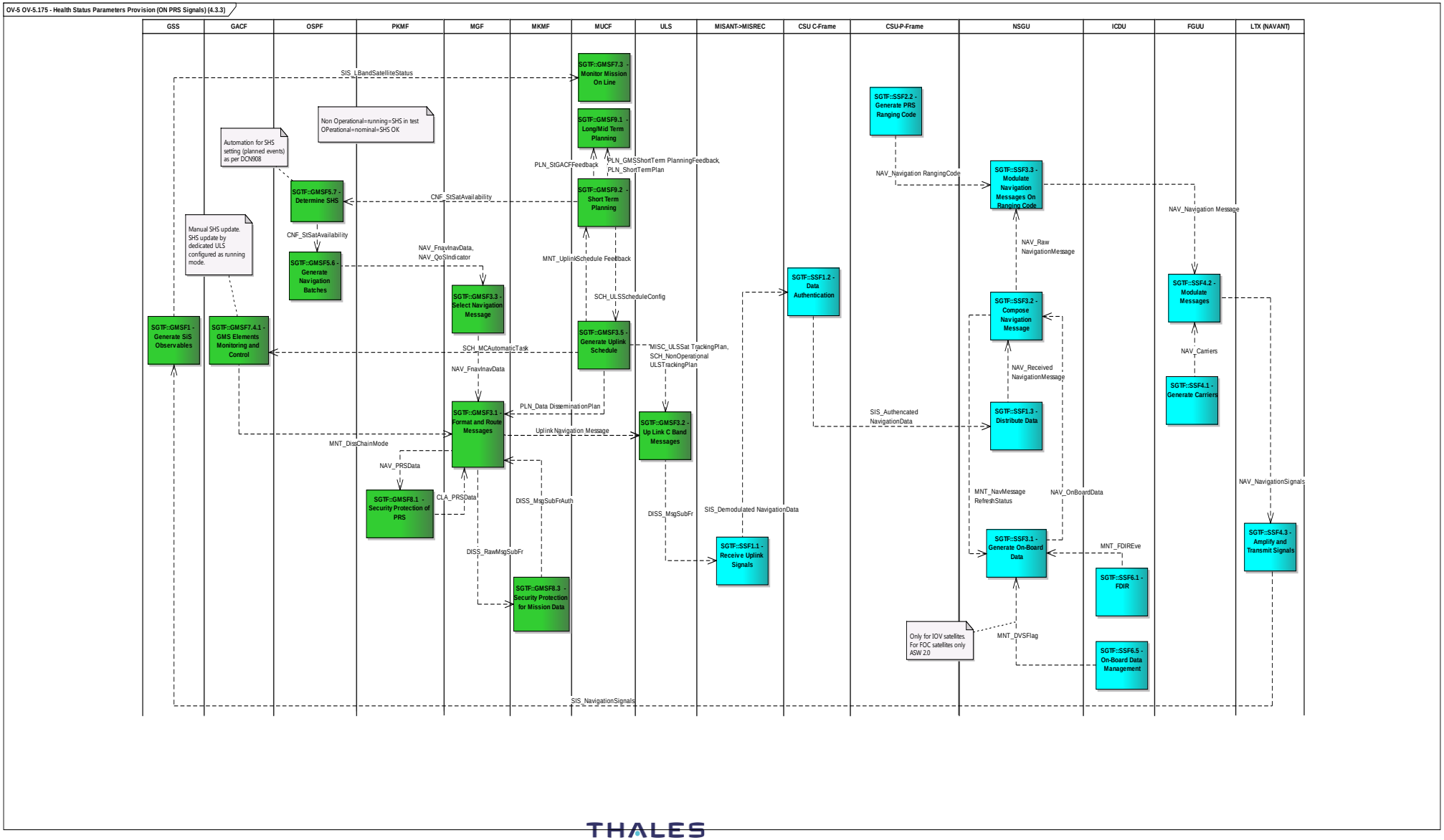
6.4 SFC4 Provide Mission Service (OS, PRS)

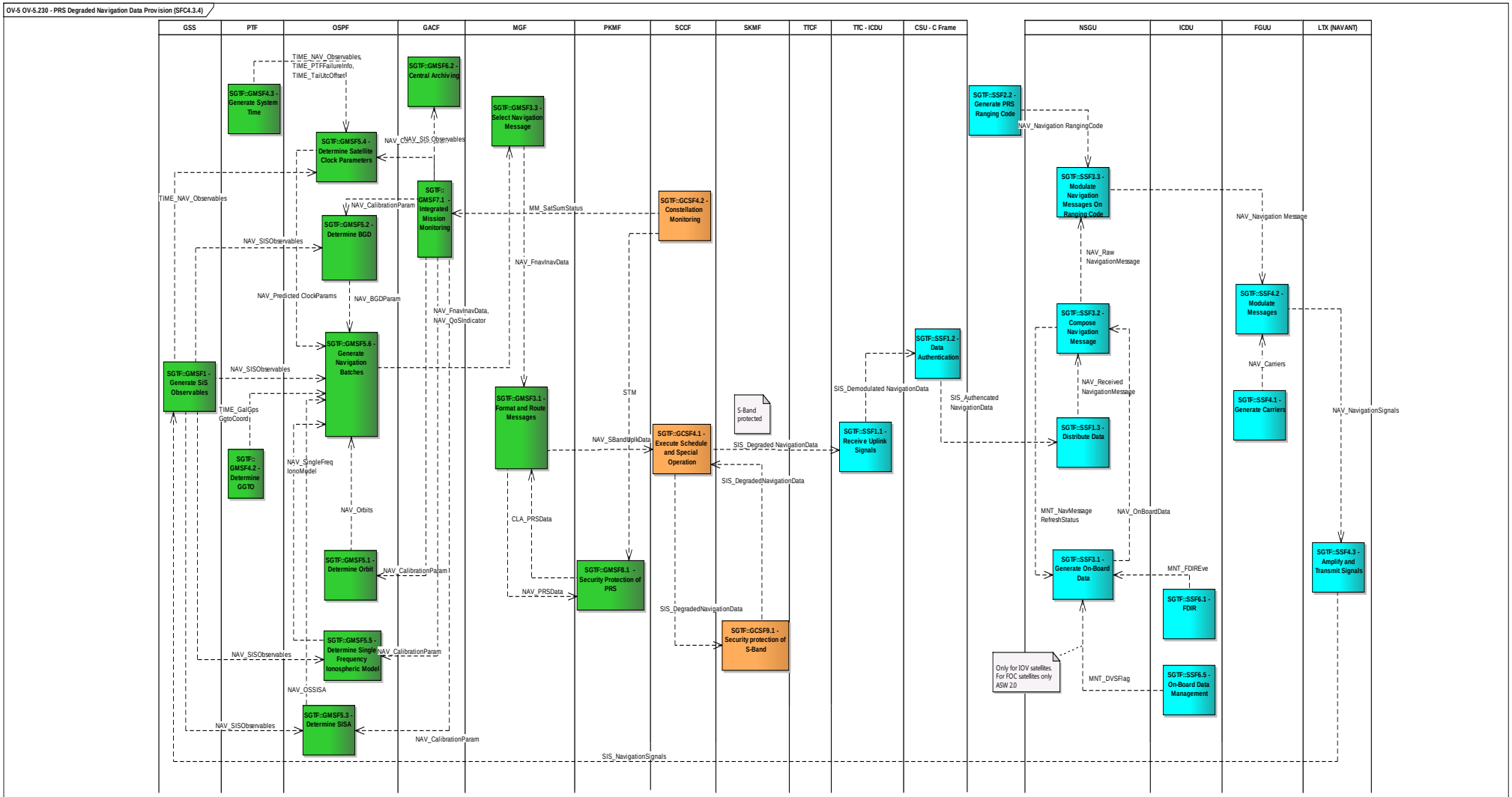



OV-5 OV-5.170 - Health Status Parameters Provision (ON OS Signals) (4.1.3)
GSS
GACF
OSPF
MGF
MKMF
MUCF
ULS
MISANT->MISREC
CSU C-Frame
NSGU
ICDU
FGUU
LTX (NAVANT)
SIS_LBandSatelliteStatus
SGTF::GMSF7.3 - Monitor Mission On Line
Non Operational=running=SHS in test
OPerational=nominal=SHS OK
SGTF::GMSF9.1 - Long/Mid Term Planning
PLN_MidTermPlan
PLN_GMSShortTerm PlanningFeedback, PLN_ShortTermPlan
SGTF::GMSF5.7 - Determine SHS
CNF_StSatAvailability
SGTF::GMSF9.2 - Short Term Planning
Manual SHS update. SHS update by dedicated ULS configured as running mode.
CNF_StSatAvailability
SGTF::GMSF5.6 - Generate Navigation Batches
NAV_FnavInavData, NAV_QoSIndicator
MNT_UplinkSchedule Feedback
SGTF::GMSF3.3 - Select Navigation Message
SCH_ULSScheduleConfig
SGTF::GMSF1 - Generate SiS Observables
SGTF::GMSF7.4.1 - GMS Elements Monitoring and Control
SCH_MCAutomaticTask
SGTF::GMSF3.5 - Generate Uplink Schedule
MISC_ULSSat TrackingPlan, SCH_NonOperational ULSTrackingPlan
NAV_FnavInavData
PLN_Data DisseminationPlan
SGTF::GMSF3.1 - Format and Route Messages
MNT_DissChainMode
DISS_IntSubFr, DISS_MsgSubFr
SGTF::GMSF3.2 - Up Link C Band Messages
DISS_MsgSubFrAuth
DISS_MsgSubFr
SIS_Demodulated NavigationData
DISS_RawMsgSubFr
SGTF::GMSF1.1 - Receive Uplink Signals
SGTF::GMSF8.3 - Security Protection for Mission Data
SGTF::SSF1.2 - Data Authentication
SIS_Authencated NavigationData
SGTF::SSF2.1 - Generate Navigation Ranging Code
NAV_Navigation RangingCode
«OperationalExchange»
SGTF::SSF3.3 - Modulate Navigation Messages On Ranging Code
INT_RawIntecCS_RawCsMessage, NavIonMessage
«OperationalExchange»
SGTF::SSF3.2 - Compose Navigation Message
NAV_Received NavigationMessage
SGTF::SSF1.3 - Distribute Data
MNT_NavMessage RefreshStatus
NAV_OnBoardData
SGTF::SSF3.1 - Generate On-Board Data
MNT_FDIREve
SGTF::SSF6.1 - FDIR
Only for IOV satellites. For FOC satellites only ASW 2.0
MNT_DVSFlag
SGTF::SSF6.5 - On-Board Data Management
NAV_Navigation Message
SGTF::SSF4.2 - Modulate Messages
NAV_Carriers
SGTF::SSF4.1 - Generate Carriers
SGTF::SSF4.3 - Amplify and Transmit Signals
SIS_NavigationSignals

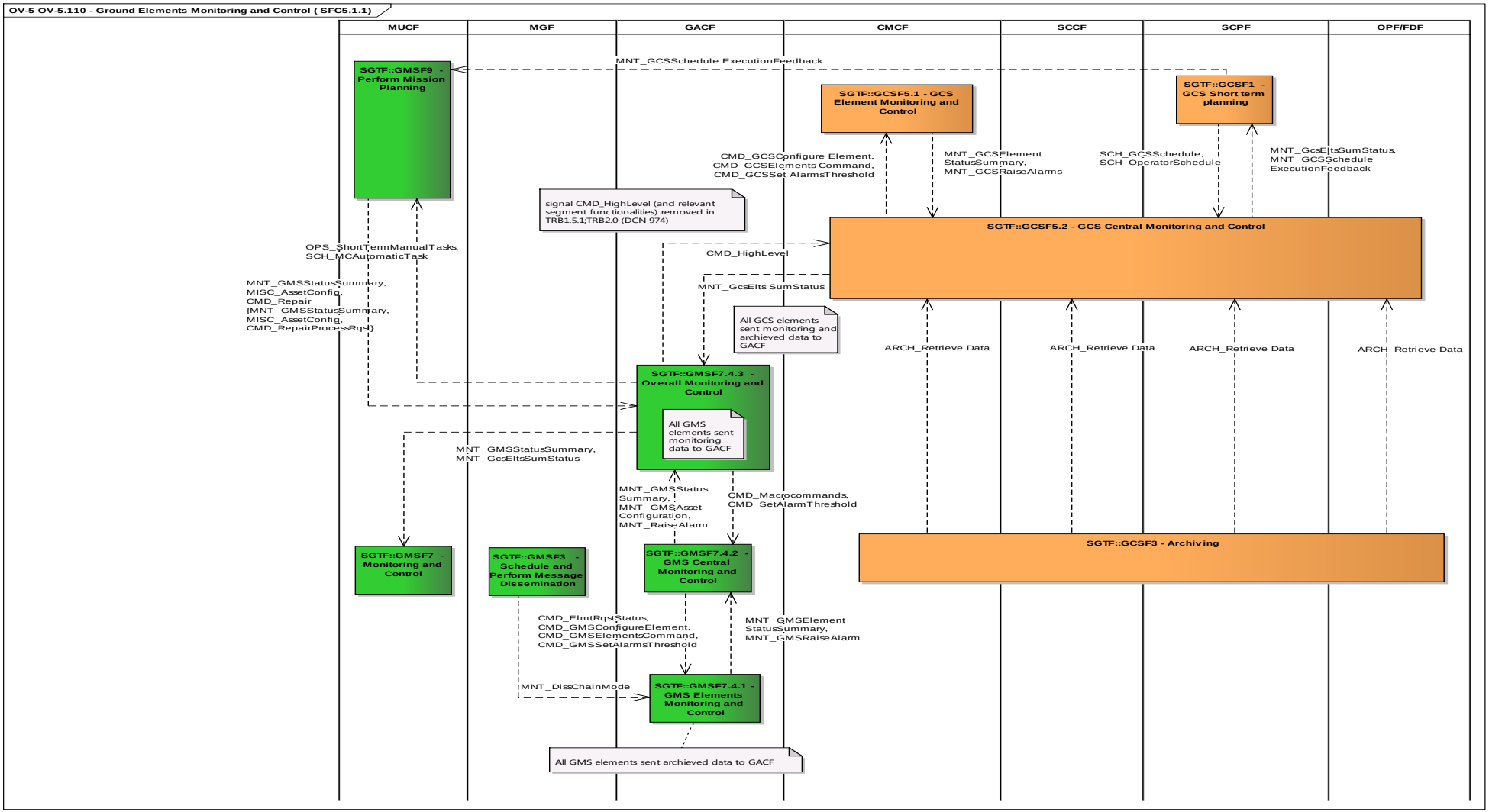

OV-5 OV-5.180 - OS Degraded Navigation Data Provision (SFC4.1.4)
GSS
PTF
OSPF
GACF
MGF
SCCF
SKMF
TTCF
TTC - ICDU
CSU - C Frame
NSGU
ICDU
FGUU
LTX (NAVANT)
SGTF::GMSF4.3 - Generate System Time
SGTF::GMSF6.2 - Central Archiving
SGTF::GMSF3.3 - Select Navigation Message
SGTF::GMSF5.4 - Determine Satellite Clock Parameters
SGTF::GMSF7.1 - Integrated Mission Monitoring
SGTF::GCSF4.2 - Constellation Monitoring
SGTF::GMSF5.2 - Determine BGD
SGTF::GMSF1 - Generate SIS Observables
SGTF::GMSF5.6 - Generate Navigation Batches
SGTF::GMSF3.1 - Format and Route Messages
S-Band protected
SGTF::GCSF4.1 - Execute Schedule and Special Operation
SGTF::SSF1.1 - Receive Uplink Signals
SGTF::SSF1.2 - Data Authentication
SGTF::SSF2.1 - Generate Navigation Ranging Code
SGTF::SSF3.3 - Modulate Navigation Messages On Ranging Code
SGTF::SSF3.2 - Compose Navigation Message
SGTF::SSF1.3 - Distribute Data
SGTF::SSF4.2 - Modulate Messages
SGTF::SSF4.1 - Generate Carriers
SGTF::SSF4.3 - Amplify and Transmit Signals
SGTF::GMSF4.2 - Determine GGTO
SGTF::GMSF5.1 - Determine Orbit
SGTF::GMSF5.5 - Determine Single Frequency Ionospheric Model
SGTF::GMSF5.3 - Determine SISA
SGTF::GCSF9.1 - Security protection of S-Band
SGTF::SSF3.1 - Generate On-Board Data
SGTF::SSF6.1 - FDIR
SGTF::SSF6.5 - On-Board Data Management
Only for IOV satellites. For FOC satellites only ASW 2.0
TIME_NAV_Observables, TIME_PTFFailureInfo, TIME_TaiUtcOffset
NAV_SIS Observables
NAV_CalibrationParam
TIME_NAV_Observables
MM_SatSumStatus
NAV_SISObservables
NAV_FnavInavData
NAV_Predicted ClockParams
NAV_BGDParam
NAV_FnavInavData, NAV_QoSIndicator
NAV_SISObservables
TIME_GalGps GgtoCoord
NAV_SBandUplData
SIS_Degraded NavigationData
NAV_SingleFreq IonoModel
NAV_Orbits
SIS_DegradedNavigationData
NAV_CalibrationParam
NAV_SISObservables
NAV_OSSISA
NAV_SISObservables
NAV_CalibrationParam
SIS_NavigationSignals
SIS_Demodulated NavigationData
SIS_Authenticated NavigationData
NAV_Navigation RangingCode
NAV_Raw NavigationMessage
NAV_Received NavigationMessage
MNT_NavMessage RefreshStatus
NAV_OnBoardData
MNT_FDIREve
MNT_DVSFlag
NAV_Navigation Message
NAV_Carriers
NAV_NavigationSignals

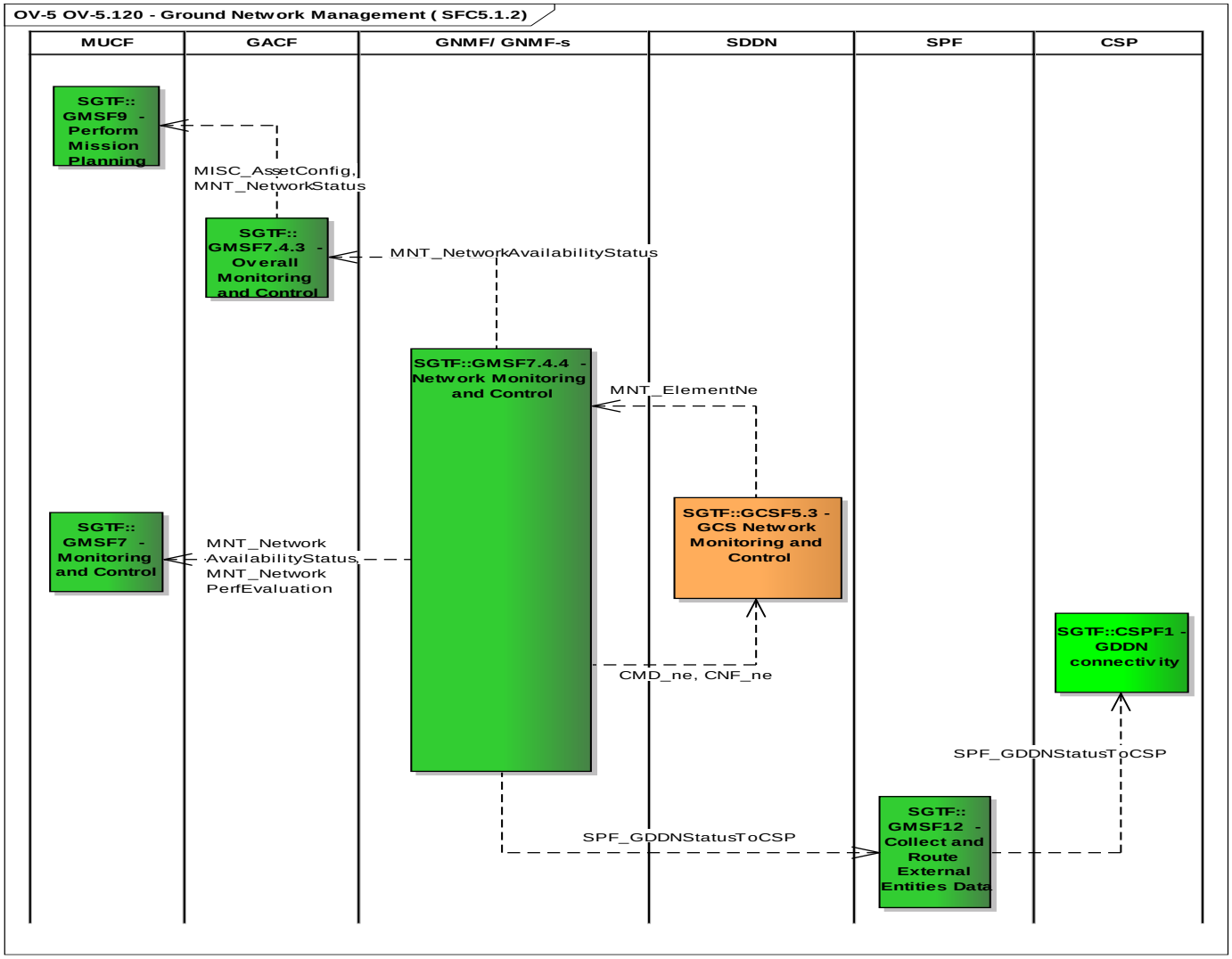

OV-5 OV-5.195 - PRS Positioning/Timing Service Provision (SFC4.3.1/SFC4.3.2)
GSS
PTF
OSPF
GACF
MSF
MGF
MKMF
PKMF
MUCF
ULS
MISANT>MISREC
CSU - C Frame
ICDU
NSGU
FGUU
LTX (NAVANT)
SGTF::GMSF4.3 - Generate System Time
SGTF::GMSF6.2 - Central Archiving
SGTF::GMSF7.2 - Monitor Mission Off Line
SGTF::GMSF5.4 - Determine Satellite Clock Parameters
SGTF::GMSF7.1 - Integrated Mission Monitoring
SGTF::GMSF3.3 - Select Navigation Message
SGTF::GMSF8.1 - Security Protection of PRS
SGTF::GMSF5.2 - Determine BGD
SGTF::GMSF3.5 - Generate Uplink Schedule
SGTF::SSF1.2 - Data Authentication
SGTF::GMSF1 - Generate SIS Observables
SGTF::GMSF5.6 - Generate Navigation Batches
SGTF::GMSF3.1 - Format and Route Messages
SGTF::GMSF3.2 - Up Link C Band Messages
SGTF::GMSF4.2 - Determine GGTO
SGTF::GMSF5.1 - Determine Orbit
SGTF::GMSF8.3 - Security Protection for Mission Data
SGTF::SSF1.1 - Receive Uplink Signals
SGTF::GMSF5.5 - Determine Single Frequency Ionospheric Model
SGTF::GMSF5.3 - Determine SISA
Only for IOV satellites. For FOC satellites only ASW 2.0
SGTF::SSF2.2 - Generate PRS Ranging Code
SGTF::SSF3.3 - Modulate Navigation Messages On Ranging Code
SGTF::SSF3.2 - Compose Navigation Message
SGTF::SSF1.3 - Distribute Data
SGTF::SSF3.1 - Generate On-Board Data
SGTF::SSF6.1 - FDIR
SGTF::SSF6.5 - On-Board Data Management
SGTF::SSF4.2 - Modulate Messages
SGTF::SSF4.1 - Generate Carriers
SGTF::SSF4.3 - Amplify and Transmit Signals
TIME_NAV_Observables, TIME_PTFFailureInfo, TIME_TaiUtcOffset
SIS_Observables Offline
NAV_SIS Observables
NAV_CalibrationParam
MM_MissionClibrationParam
NAV_SISObservables
NAV_Predicted ClockParams
NAV_BGDParam
NAV_FnavInavData
NAV_PRSData
NAV_FnavInavData, NAV_QoSIndicator
MISC_ULSSat TrackingPlan, SCH_NonOperational ULSTrackingPlan
SIS_Demodulated NavigationData
SIS_Authencated NavigationData
TIME_GalGps GgtoCoord
NAV_SingleFreq IonoModel
CLA_PRSData
PLN_Data DisseminationPlan
DISS_IntSubFr, DISS_MsgSubFr
DISS_RawIntSubFr
DISS_IntSubFr
DISS_MsgSubFrAuth
DISS_MsgSubFr
NAV_Orbits
NAV_OSSISA
NAV_Navigation RangingCode
INT_RawIntegrity Message, CS_RawCsMessage NavigBomMessage
NAV_Navigation Message
NAV_Carriers
NAV_Received NavigationMessage
MNT_NavMessage RefreshStatus
NAV_OnBoardData
MNT_FDIREve
MNT_DVSFlag
NAV_NavigationSignals
SIS_NavigationSignals

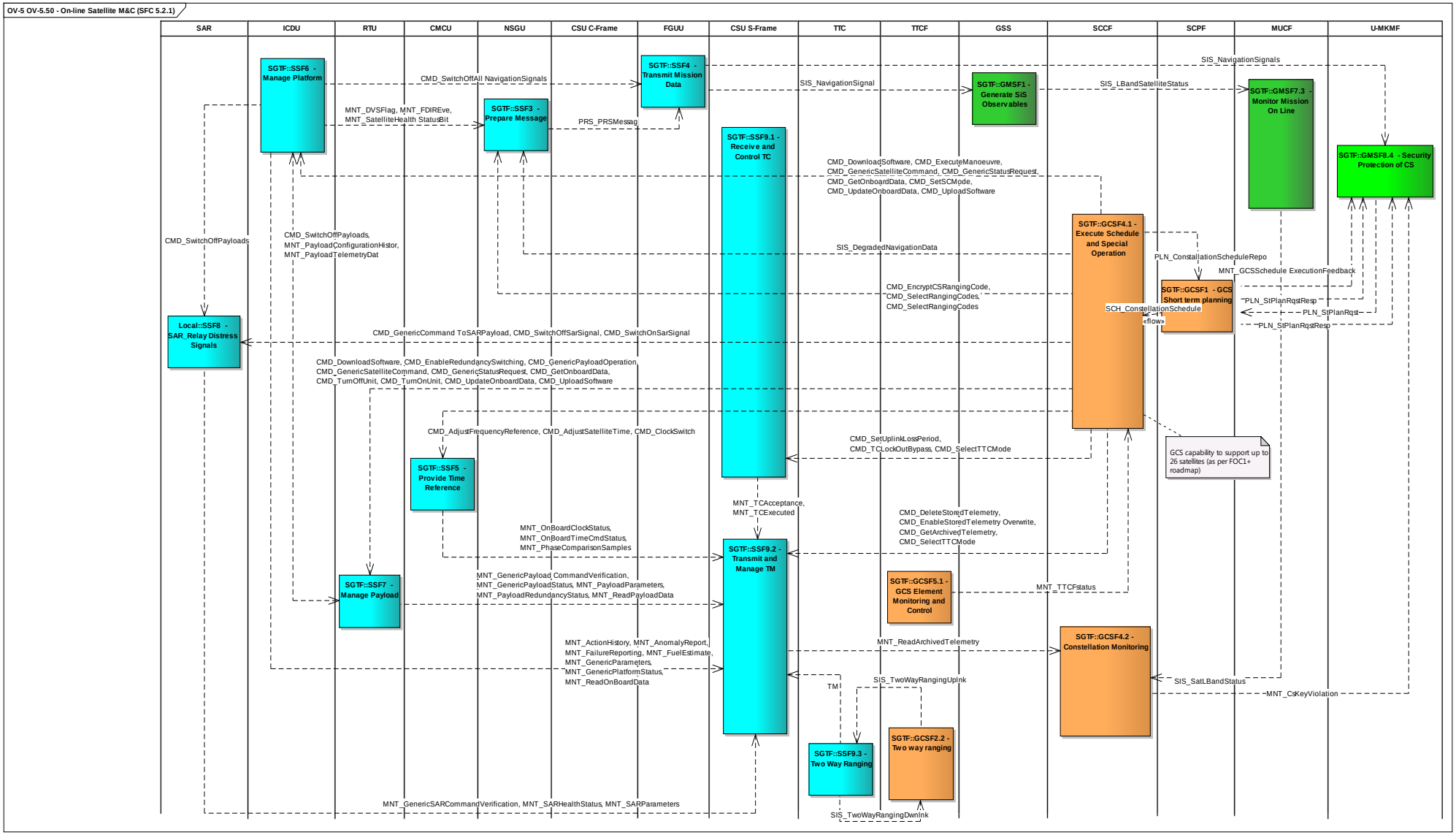

OV-5 OV-5.175 - Health Status Parameters Provision (ON PRS Signals) (4.3.3)
GSS
GACF
OSPF
PKMF
MGF
MKMF
MUCF
ULS
MISANT->MISREC
CSU C-Frame
CSU-P-Frame
NSGU
ICDU
FGUU
LTX (NAVANT)
SIS_LBandSatelliteStatus
SGTF::GMSF7.3 - Monitor Mission On Line
SGTF::SSF2.2 - Generate PRS Ranging Code
Non Operational=running=SHS in test OPerational=nominal=SHS OK
SGTF::GMSF9.1 - Long/Mid Term Planning
Automation for SHS setting (planned events) as per DCN908
PLN_StGACFFeedback
PLN_GMSShortTerm PlanningFeedback, PLN_ShortTermPlan
NAV_Navigation RangingCode
SGTF::SSF3.3 - Modulate Navigation Messages On Ranging Code
SGTF::GMSF5.7 - Determine SHS
SGTF::GMSF9.2 - Short Term Planning
CNF_StSatAvailability
NAV_Navigation Message
Manual SHS update. SHS update by dedicated ULS configured as running mode.
CNF_StSatAvailability
NAV_Raw NavigationMessage
NAV_FnavInavData, NAV_QoSIndicator
MNT_UplinkSchedule Feedback
SGTF::SSF1.2 - Data Authentication
SGTF::SSF4.2 - Modulate Messages
SGTF::GMSF5.6 - Generate Navigation Batches
SGTF::GMSF3.3 - Select Navigation Message
SCH_ULSScheduleConfig
SGTF::SSF3.2 - Compose Navigation Message
SGTF::GMSF1 - Generate SIS Observables
SGTF::GMSF7.4.1 - GMS Elements Monitoring and Control
SGTF::GMSF3.5 - Generate Uplink Schedule
SCH_MCAutomaticTask
MISC_ULSSat TrackingPlan, SCH_NonOperational ULSTrackingPlan
NAV_Carriers
NAV_Received NavigationMessage
NAV_FnavInavData
SGTF::GMSF4.1 - Generate Carriers
SGTF::GMSF3.1 - Format and Route Messages
PLN_Data DisseminationPlan
SGTF::GMSF3.2 - Up Link C Band Messages
SIS_Authencated NavigationData
SGTF::SSF1.3 - Distribute Data
Uplink Navigation Message
MNT_DissChainMode
NAV_PRSData
NAV_NavigationSignals
MNT_NavMessage RefreshStatus
NAV_OnBoardData
SGTF::GMSF8.1 - Security Protection of PRS
CLA_PRSData
DISS_MsgSubFrAuth
DISS_MsgSubFr
SIS_Demodulated NavigationData
SGTF::SSF3.1 - Generate On-Board Data
MNT_FDIREve
SGTF::SSF4.3 - Amplify and Transmit Signals
SGTF::SSF1.1 - Receive Uplink Signals
SGTF::SSF6.1 - FDIR
DISS_RawMsgSubFr
SGTF::GMSF8.3 - Security Protection for Mission Data
Only for IOV satellites. For FOC satellites only ASW 2.0
MNT_DVSFlag
SGTF::SSF6.5 - On-Board Data Management
SIS_NavigationSignals

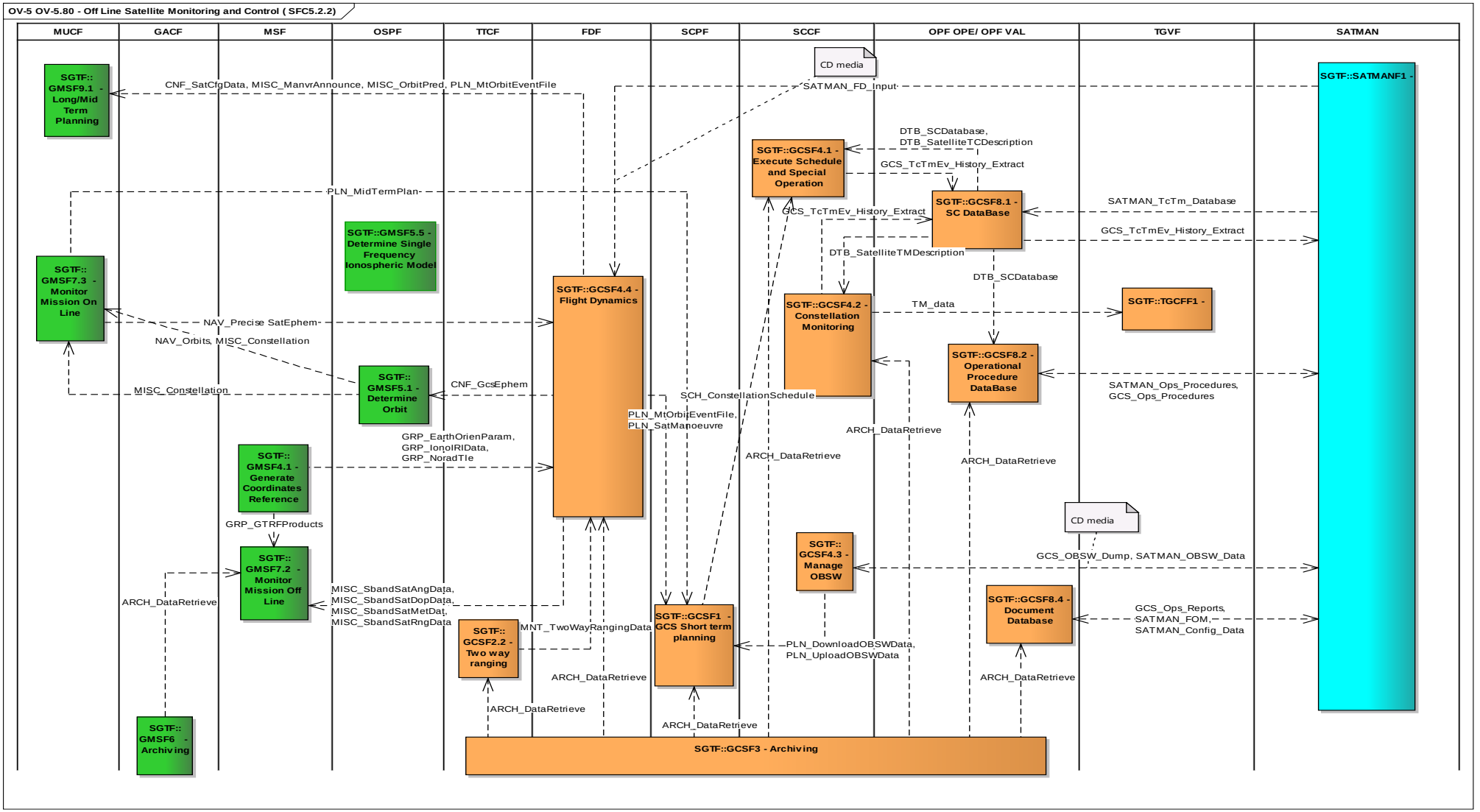

OV-5 OV-5.230 - PRS Degraded Navigation Data Provision (SFC4.3.4)
GSS
PTF
OSPF
GACF
MGF
PKMF
SCCF
SKMF
TTCF
TTC - ICDU
CSU - C Frame
NSGU
ICDU
FGUU
LTX (NAVANT)
SGTF::GMSF4.3 - Generate System Time
SGTF::GMSF6.2 - Central Archiving
SGTF::GMSF3.3 - Select Navigation Message
SGTF::SSF2.2 - Generate PRS Ranging Code
SGTF::GMSF5.4 - Determine Satellite Clock Parameters
SGTF::GMSF7.1 - Integrated Mission Monitoring
SGTF::GCSF4.2 - Constellation Monitoring
SGTF::SSF3.3 - Modulate Navigation Messages On Ranging Code
SGTF::GMSF5.2 - Determine BGD
SGTF::SSF4.2 - Modulate Messages
SGTF::SSF3.2 - Compose Navigation Message
SGTF::GMSF5.6 - Generate Navigation Batches
SGTF::SSF1.2 - Data Authentication
SGTF::GMSF1 - Generate SIS Observables
SGTF::GMSF3.1 - Format and Route Messages
SGTF::GCSF4.1 - Execute Schedule and Special Operation
SGTF::SSF1.1 - Receive Uplink Signals
SGTF::SSF1.3 - Distribute Data
SGTF::SSF4.1 - Generate Carriers
SGTF::GMSF4.2 - Determine GGTO
SGTF::GMSF5.1 - Determine Orbit
SGTF::GMSF8.1 - Security Protection of PRS
SGTF::SSF3.1 - Generate On-Board Data
SGTF::SSF4.3 - Amplify and Transmit Signals
SGTF::SSF6.1 - FDIR
SGTF::GCSF9.1 - Security protection of S-Band
SGTF::GMSF5.5 - Determine Single Frequency Ionospheric Model
SGTF::SSF6.5 - On-Board Data Management
SGTF::GMSF5.3 - Determine SISA
TIME_NAV_Observables, TIME_PTFFailureInfo, TIME_TaiUtcOffset
NAV_CNAV_SIS Observables
NAV_CalibrationParam
MM_SatSumStatus
NAV_SISObservables
NAV_FnavInavData
NAV_Predicted ClockParams
NAV_BGDParam
NAV_FnavInavData, NAV_QoSIndicator
NAV_Navigation RangingCode
NAV_Raw NavigationMessage
NAV_Navigation Message
NAV_Carriers
NAV_Received NavigationMessage
STM
S-Band protected
SIS_Demodulated NavigationData
SIS_Authencated NavigationData
NAV_SBandUplkData
SIS_Degraded NavigationData
SIS_DegradedNavigationData
TIME_GalGps GgtoCoord
NAV_SingleFreq IonoModel
NAV_Orbits
CLA_PRSData
NAV_PRSData
MNT_NavMessage RefreshStatus
NAV_OnBoardData
MNT_FDIREve
NAV_NavigationSignals
NAV_OSSISA
Only for IOV satellites. For FOC satellites only ASW 2.0
MNT_DVSFlag
SIS_NavigationSignals

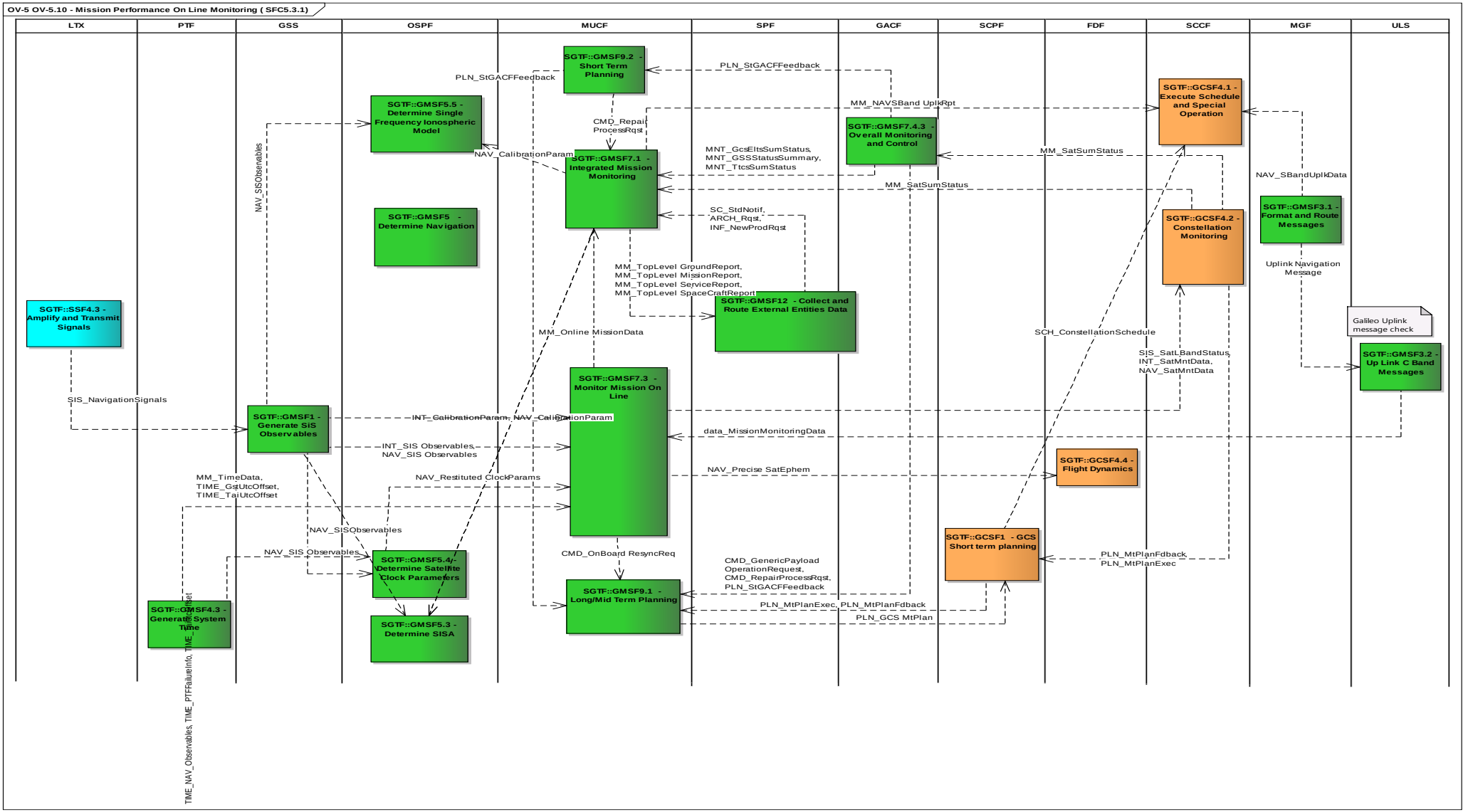
6.5 SFC5 Monitoring & Control

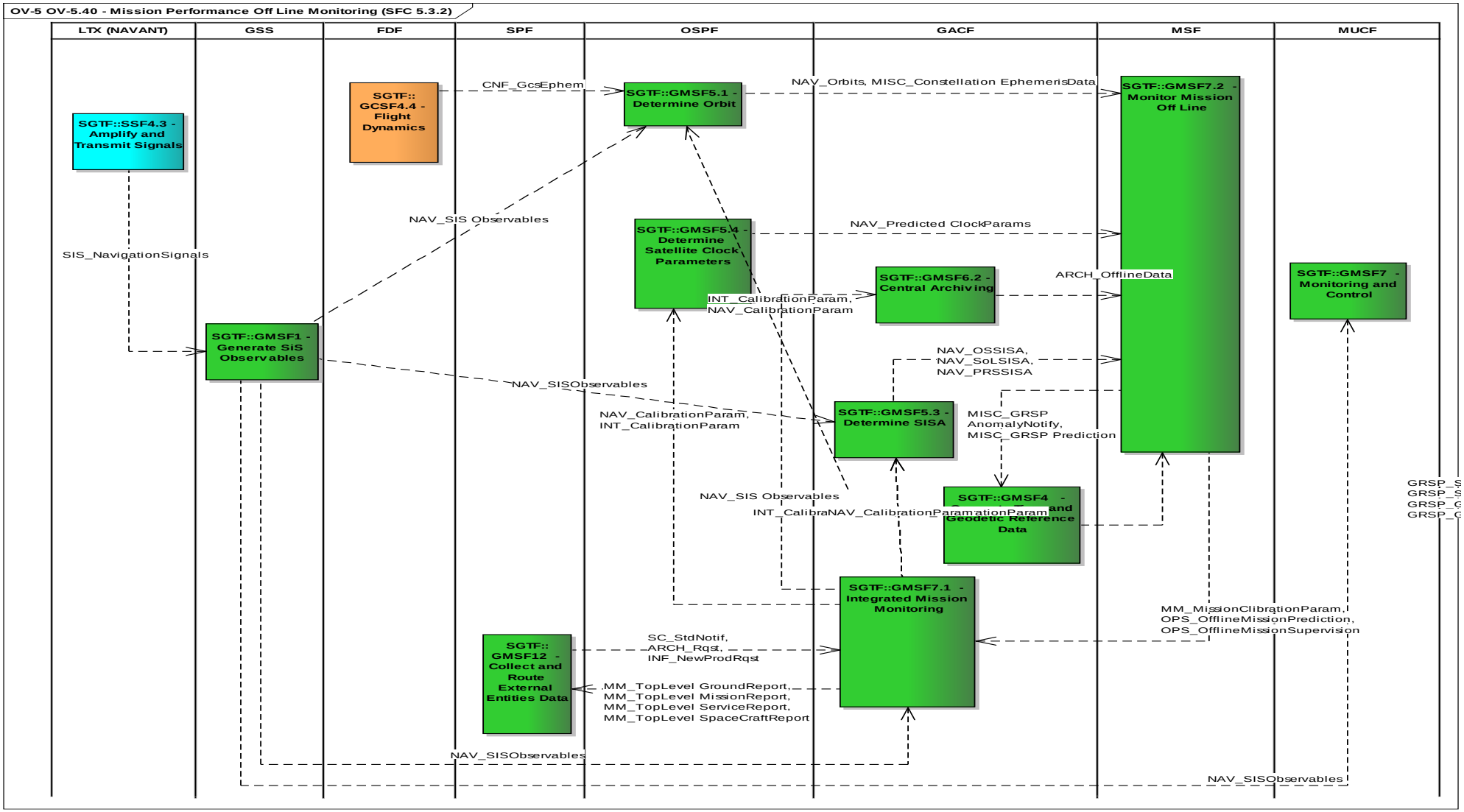


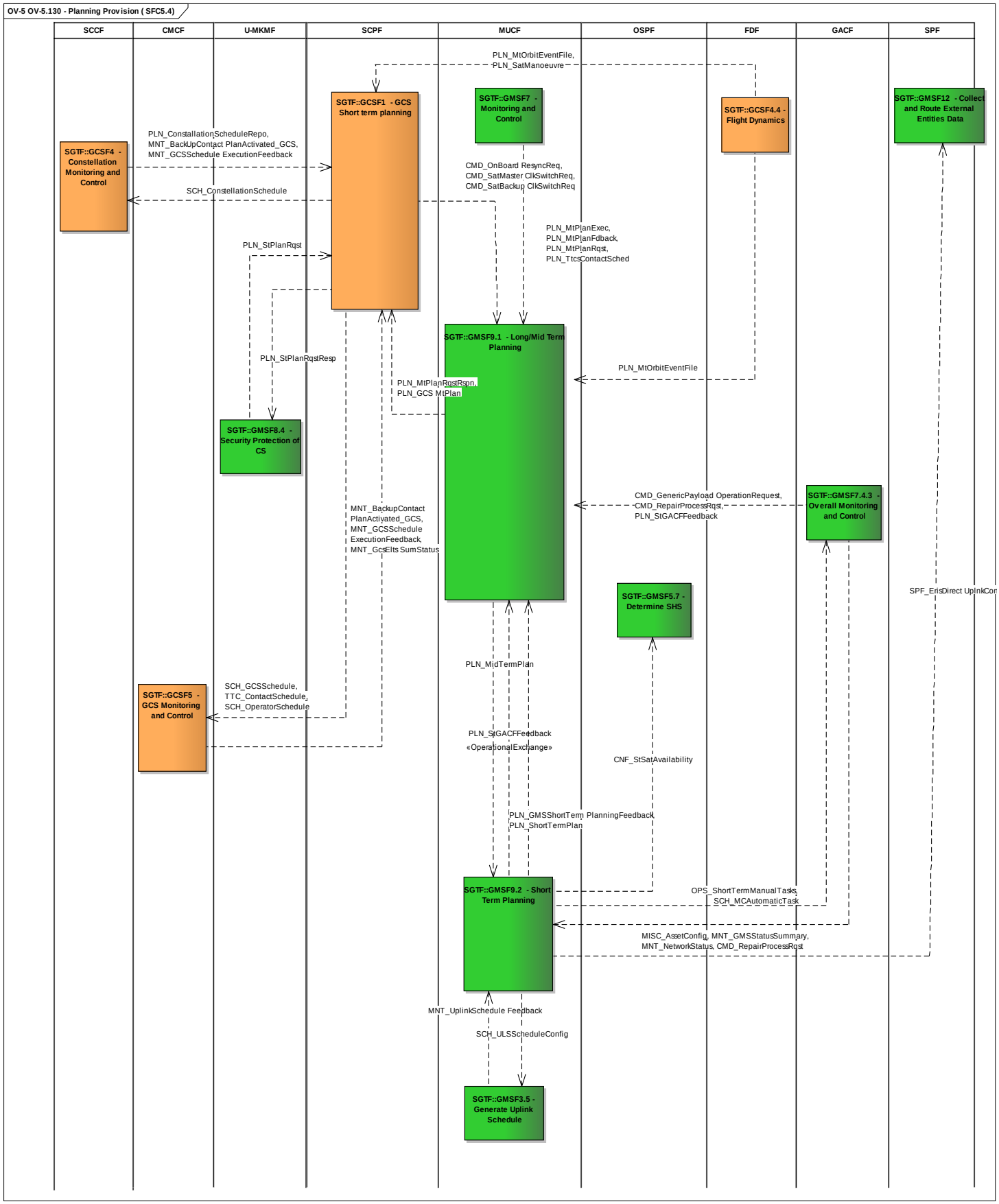

OV-5 OV-5.120 - Ground Network Management (SFC5.1.2)
MUCF
GACF
GNMF/ GNMF-s
SDDN
SPF
CSP
SGTF:: GMSF9 - Perform Mission Planning
MISC_AssetConfig, MNT_NetworkStatus
SGTF:: GMSF7.4.3 - Overall Monitoring and Control
MNT_NetworkAvailabilityStatus
SGTF::GMSF7.4.4 - Network Monitoring and Control
MNT_ElementNe
SGTF:: GMSF7 - Monitoring and Control
MNT_Network AvailabilityStatus, MNT_Network PerfEvaluation
SGTF::GCSF5.3 - GCS Network Monitoring and Control
CMD_ne, CNF_ne
SGTF::CSPF1 - GDDN connectivity
SPF_GDDNStatusToCSP
SGTF:: GMSF12 - Collect and Route External Entities Data
SPF_GDDNStatusToCSP


OV-5 OV-5.50 - On-line Satellite M&C (SFC 5.2.1)
SAR
ICDU
RTU
CMCU
NSGU
CSU C-Frame
FGUU
CSU S-Frame
TTC
TTCF
GSS
SCCF
SCPF
MUCF
U-MKMF
SGTF::SSF6 - Manage Platform
SGTF::SSF4 - Transmit Mission Data
SGTF::SSF3 - Prepare Message
SGTF::SSF9.1 - Receive and Control TC
SGTF::GMSF1 - Generate SiS Observables
SGTF::GMSF7.3 - Monitor Mission On Line
SGTF::GMSF8.4 - Security Protection of CS
SGTF::GCSF4.1 - Execute Schedule and Special Operation
SGTF::GCSF1 - GCS Short term planning
Local::SSF8 - SAR_Relay Distress Signals
SGTF::SSF5 - Provide Time Reference
SGTF::SSF9.2 - Transmit and Manage TM
SGTF::SSF7 - Manage Payload
SGTF::GCSF5.1 - GCS Element Monitoring and Control
SGTF::GCSF4.2 - Constellation Monitoring
SGTF::SSF9.3 - Two Way Ranging
SGTF::GCSF2.2 - Two way ranging
SIS_NavigationSignals
CMD_SwitchOffAll NavigationSignals
SIS_NavigationSignal
SIS_LBandSatelliteStatus
MNT_DVSFlag, MNT_FDIREve, MNT_SatelliteHealth StatusBit
PRS_PRSMessag
CMD_DownloadSoftware, CMD_ExecuteManoeuvre, CMD_GenericSatelliteCommand, CMD_GenericStatusRequest, CMD_GetOnboardData, CMD_SetSCMode, CMD_UpdateOnboardData, CMD_UploadSoftware
CMD_SwitchOffPayloads
CMD_SwitchOffPayloads, MNT_PayloadConfigurationHistor, MNT_PayloadTelemetryDat
SIS_DegradedNavigationData
PLN_ConstallationScheduleRepo
MNT_GCSSchedule ExecutionFeedback
CMD_EncryptCSRangingCode, CMD_SelectRangingCodes, CMD_SelectRangingCodes
SCH_ConstellationSchedule
PLN_StPlanRqstResp
PLN_StPlanRqst
CMD_GenericCommand ToSARPayload, CMD_SwitchOffSarSignal, CMD_SwitchOnSarSignal
CMD_DownloadSoftware, CMD_EnableRedundancySwitching, CMD_GenericPayloadOperation, CMD_GenericSatelliteCommand, CMD_GenericStatusRequest, CMD_GetOnboardData, CMD_TurnOffUnit, CMD_TurnOnUnit, CMD_UpdateOnboardData, CMD_UploadSoftware
CMD_AdjustFrequencyReference, CMD_AdjustSatelliteTime, CMD_ClockSwitch
CMD_SetUplinkLossPeriod, CMD_TCLockOutBypass, CMD_SelectTTCMode
GCS capability to support up to 26 satellites (as per FOC1+ roadmap)
MNT_TCAcceptance, MNT_TCExecuted
MNT_OnBoardClockStatus, MNT_OnBoardTimeCmdStatus, MNT_PhaseComparisonSamples
CMD_DeleteStoredTelemetry, CMD_EnableStoredTelemetry Overwrite, CMD_GetArchivedTelemetry, CMD_SelectTTCMode
MNT_GenericPayload CommandVerification, MNT_GenericPayloadStatus, MNT_PayloadParameters, MNT_PayloadRedundancyStatus, MNT_ReadPayloadData
MNT_TTCFstatus
MNT_ActionHistory, MNT_AnomalyReport, MNT_FailureReporting, MNT_FuelEstimate, MNT_GenericParameters, MNT_GenericPlatformStatus, MNT_ReadOnBoardData
MNT_ReadArchivedTelemetry
SIS_TwoWayRangingUplink
TM
SIS_SatLBandStatus
MNT_CsKeyViolation
MNT_GenericSARCommandVerification, MNT_SARHealthStatus, MNT_SARParameters
SIS_TwoWayRangingDwnlink


OV-5 OV-5.80 - Off Line Satellite Monitoring and Control (SFC5.2.2)
MUCF
GACF
MSF
OSPF
TTCF
FDF
SCPF
SCCF
OPF OPE/ OPF VAL
TGVF
SATMAN
SGTF:: GMSF9.1 - Long/Mid Term Planning
CNF_SatCfgData, MISC_ManvrAnnounce, MISC_OrbitPred, PLN_MtOrbitEventFile
CD media
SATMAN_FD_Input
SGTF::SATMANF1 -
SGTF::GCSF4.1 - Execute Schedule and Special Operation
DTB_SCDatabase, DTB_SatelliteTCDescription
GCS_TcTmEv_History_Extract
PLN_MidTermPlan
SGTF::GCSF8.1 - SC DataBase
SATMAN_TcTm_Database
GCS_TcTmEv_History_Extract
SGTF::GMSF5.5 - Determine Single Frequency Ionospheric Model
DTB_SatelliteTMDescription
DTB_SCDatabase
SGTF:: GMSF7.3 - Monitor Mission On Line
SGTF::GCSF4.4 - Flight Dynamics
SGTF::GCSF4.2 - Constellation Monitoring
TM_data
SGTF::TGCFF1 -
NAV_Precise SatEphem
NAV_Orbits, MISC_Constellation
SGTF:: GMSF5.1 - Determine Orbit
SGTF::GCSF8.2 - Operational Procedure DataBase
SATMAN_Ops_Procedures, GCS_Ops_Procedures
MISC_Constellation
CNF_GcsEphem
SCH_ConstellationSchedule
PLN_MtOrbitEventFile, PLN_SatManoeuvre
ARCH_DataRetrieve
GRP_EarthOrienParam, GRP_IonoIRIData, GRP_NoradTle
ARCH_DataRetrieve
ARCH_DataRetrieve
SGTF:: GMSF4.1 - Generate Coordinates Reference
CD media
GRP_GTRFProducts
SGTF:: GCSF4.3 - Manage OBSW
GCS_OBSW_Dump, SATMAN_OBSW_Data
SGTF:: GMSF7.2 - Monitor Mission Off Line
MISC_SbandSatAngData, MISC_SbandSatDopData, MISC_SbandSatMetDat, MISC_SbandSatRngData
SGTF::GCSF8.4 - Document Database
GCS_Ops_Reports, SATMAN_FOM, SATMAN_Config_Data
ARCH_DataRetrieve
SGTF:: GCSF2.2 - Two way ranging
MNT_TwoWayRangingData
SGTF::GCSF1 - GCS Short term planning
PLN_DownloadOBSWData, PLN_UploadOBSWData
ARCH_DataRetrieve
ARCH_DataRetrieve
ARCH_DataRetrieve
SGTF:: GMSF6 - Archiving
SGTF::GCSF3 - Archiving

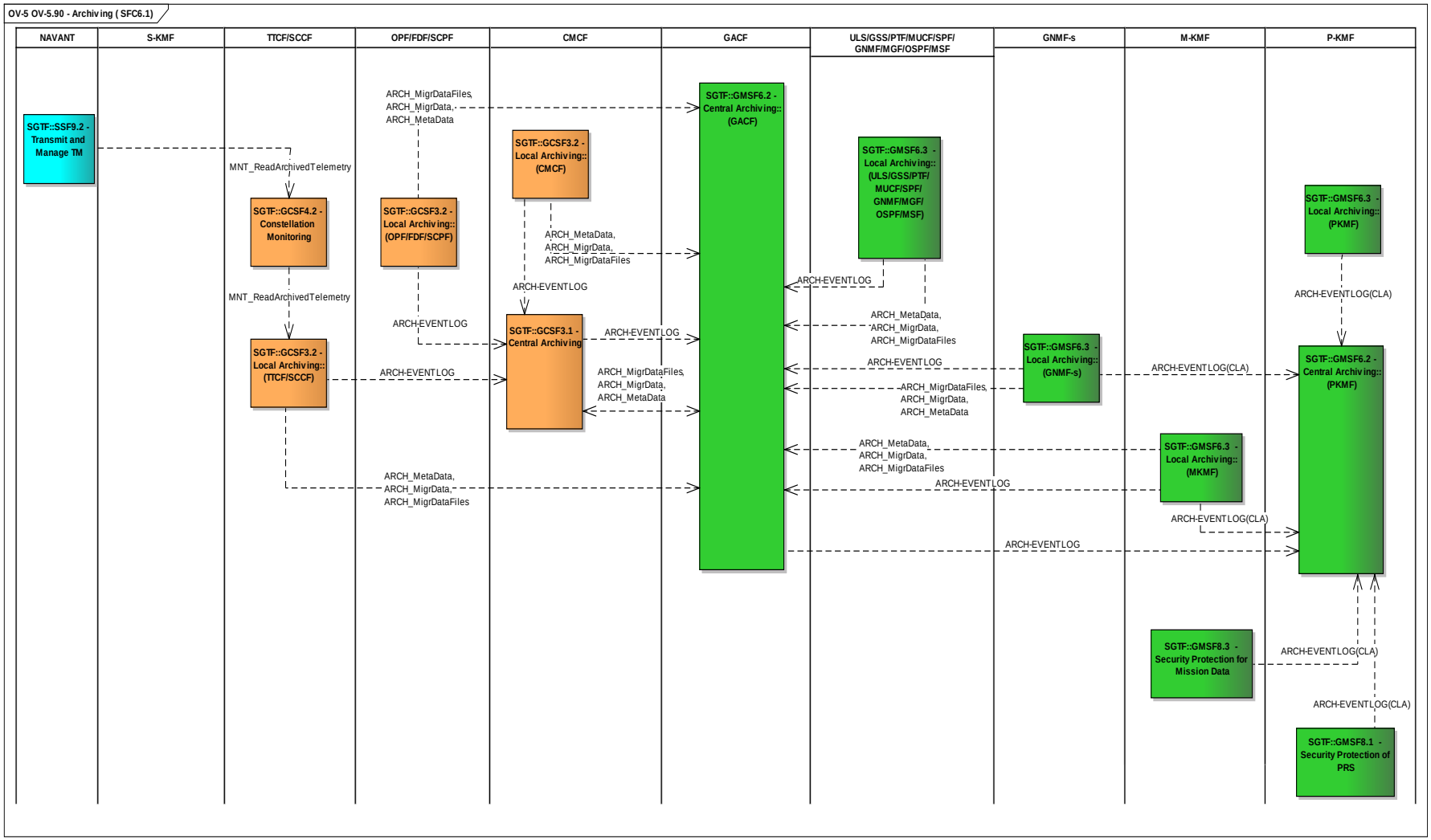

OV-5 OV-5.10 - Mission Performance On Line Monitoring (SFC5.3.1)
LTX
PTF
GSS
OSPF
MUCF
SPF
GACF
SCPF
FDF
SCCF
MGF
ULS
SGTF::GMSF9.2 - Short Term Planning
SGTF::GMSF5.5 - Determine Single Frequency Ionospheric Model
SGTF::GMSF7.1 - Integrated Mission Monitoring
SGTF::GMSF5 - Determine Navigation
SGTF::GMSF12 - Collect and Route External Entities Data
SGTF::GMSF7.3 - Monitor Mission On Line
SGTF::GMSF9.1 - Long/Mid Term Planning
SGTF::GMSF7.4.3 - Overall Monitoring and Control
SGTF::GCSF4.1 - Execute Schedule and Special Operation
SGTF::GCSF4.2 - Constellation Monitoring
SGTF::GCSF1 - GCS Short term planning
SGTF::GCSF4.4 - Flight Dynamics
SGTF::GMSF3.1 - Format and Route Messages
SGTF::GMSF3.2 - Up Link C Band Messages
SGTF::SSF4.3 - Amplify and Transmit Signals
SGTF::GMSF1 - Generate SIS Observables
SGTF::GMSF5.4 - Determine Satellite Clock Parameters
SGTF::GMSF5.3 - Determine SISA
SGTF::GMSF4.3 - Generate System Time
Galileo Uplink message check
PLN_StGACFFeedback
MM_NAVSBand UplkRpt
CMD_RepairProcessRqst
NAV_CalibrationParam
MNT_GcsEltsSumStatus, MNT_GSSStatusSummary, MNT_TtcsSumStatus
MM_SatSumStatus
NAV_SBandUplkData
NAV_SISObservables
SC_StdNotif, ARCH_Rqst, INF_NewProdRqst
MM_TopLevel GroundReport, MM_TopLevel MissionReport, MM_TopLevel ServiceReport, MM_TopLevel SpaceCraftReport
Uplink Navigation Message
MM_Online MissionData
SCH_ConstellationSchedule
SIS_SatL_BandStatus, INT_SatMntData, NAV_SatMntData
SIS_NavigationSignals
INT_CalibrationParam, NAV_CalibrationParam
data_MissionMonitoringData
INT_SIS Observables, NAV_SIS Observables
NAV_Precise SatEphem
MM_TimeData, TIME_GstUtcOffset, TIME_TaiUtcOffset
NAV_Restituted ClockParams
NAV_SISObservables
NAV_SIS Observables
CMD_OnBoard ResyncReq
CMD_GenericPayload OperationRequest, CMD_RepairProcessRqst, PLN_StGACFFeedback
PLN_MtPlanFdback PLN_MtPlanExec
PLN_MtPlanExec, PLN_MtPlanFdback
PLN_GCS MtPlan
TIME_NAV_Observables, TIME_PTFFailureInfo, TIME_

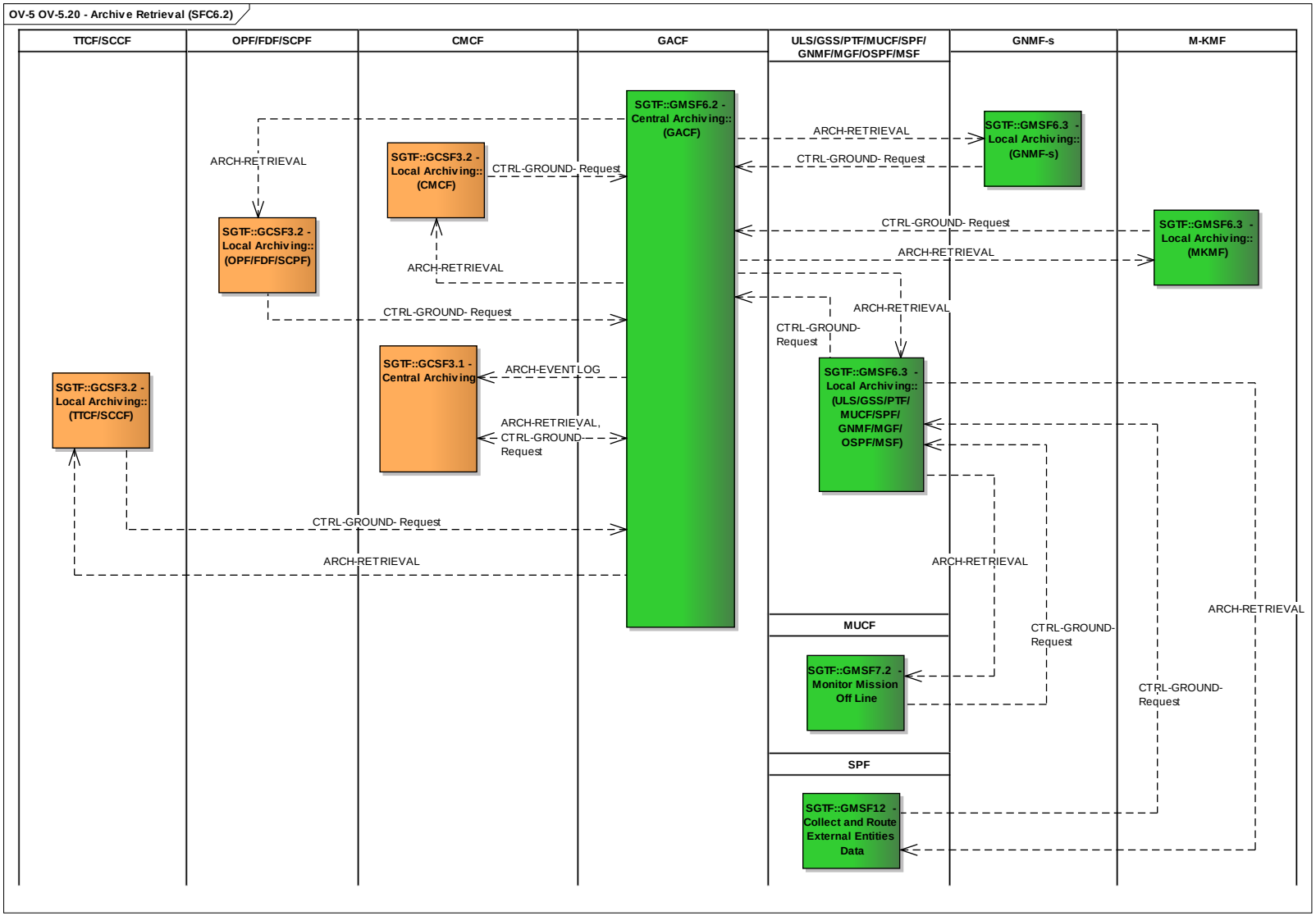

OV-5 OV-5.40 - Mission Performance Off Line Monitoring (SFC 5.3.2)
LTX (NAVANT)
GSS
FDF
SPF
OSPF
GACF
MSF
MUCF
SGTF::SSF4.3 - Amplify and Transmit Signals
SGTF:: GCSF4.4 - Flight Dynamics
SGTF::GMSF5.1 - Determine Orbit
SGTF::GMSF7.2 - Monitor Mission Off Line
SGTF::GMSF5.4 - Determine Satellite Clock Parameters
SGTF::GMSF6.2 - Central Archiving
SGTF::GMSF7 - Monitoring and Control
SGTF::GMSF1 - Generate SiS Observables
SGTF::GMSF5.3 - Determine SISA
SGTF::GMSF4 - Geodetic Reference Data
SGTF::GMSF7.1 - Integrated Mission Monitoring
SGTF:: GMSF12 - Collect and Route External Entities Data
CNF_GcsEphem
NAV_Orbits, MISC_Constellation EphemerisData
NAV_SIS Observables
NAV_Predicted ClockParams
SIS_NavigationSignals
ARCH_OfflineData
INT_CalibrationParam, NAV_CalibrationParam
NAV_OSSISA, NAV_SoLSISA, NAV_PRSSISA
NAV_SISObservables
NAV_CalibrationParam, INT_CalibrationParam
MISC_GRSP AnomalyNotify, MISC_GRSP Prediction
NAV_SIS Observables
INT_CalibraNAV_CalibrationParamationParam
MM_MissionClibrationParam, OPS_OfflineMissionPrediction, OPS_OfflineMissionSupervision
SC_StdNotif, ARCH_Rqst, INF_NewProdRqst
MM_TopLevel GroundReport, MM_TopLevel MissionReport, MM_TopLevel ServiceReport, MM_TopLevel SpaceCraftReport
NAV_SISObservables
NAV_SISObservables


OV-5 OV-5.130 - Planning Provision (SFC5.4)
SCCF
CMCF
U-MKMF
SCPF
MUCF
OSPF
FDF
GACF
SPF
PLN_MtOrbitEventFile,
PLN_SatManoeuvre
SGTF::GCSF1 - GCS Short term planning
SGTF::GMSF7 - Monitoring and Control
SGTF::GCSF4.4 - Flight Dynamics
SGTF::GMSF12 - Collect and Route External Entities Data
SGTF::GCSF4 - Constellation Monitoring and Control
PLN_ConstallationScheduleRepo,
MNT_BackUpContact PlanActivated_GCS,
MNT_GCSSchedule ExecutionFeedback
CMD_OnBoard ResyncReq,
CMD_SatMaster ClkSwitchReq,
CMD_SatBackup ClkSwitchReq
SCH_ConstellationSchedule
PLN_MtPlanExec,
PLN_MtPlanFdback,
PLN_MtPlanRqst,
PLN_TtcsContactSched
PLN_StPlanRqst
SGTF::GMSF9.1 - Long/Mid Term Planning
PLN_StPlanRqstResp
PLN_MtOrbitEventFile
PLN_MtPlanRqstRspn,
PLN_GCS MtPlan
SGTF::GMSF8.4 - Security Protection of CS
CMD_GenericPayload OperationRequest,
CMD_RepairProcessRqst,
PLN_StGACFFeedback
SGTF::GMSF7.4.3 - Overall Monitoring and Control
MNT_BackupContact
PlanActivated_GCS,
MNT_GCSSchedule
ExecutionFeedback,
MNT_GcsElts SumStatus
SGTF::GMSF5.7 - Determine SHS
SPF_EnsDirect UplnkCon
PLN_MidTermPlan
SGTF::GCSF5 - GCS Monitoring and Control
SCH_GCSSchedule,
TTC_ContactSchedule,
SCH_OperatorSchedule
PLN_StGACFFeedback
«OperationalExchange»
CNF_StSatAvailability
PLN_GMSShortTerm PlanningFeedback
PLN_ShortTermPlan
SGTF::GMSF9.2 - Short Term Planning
OPS_ShortTermManualTasks,
SCH_MCAutomaticTask
MISC_AssetConfig, MNT_GMSStatusSummary,
MNT_NetworkStatus, CMD_RepairProcessRqst
MNT_UplinkSchedule Feedback
SCH_ULSScheduleConfig
SGTF::GMSF3.5 - Generate Uplink Schedule

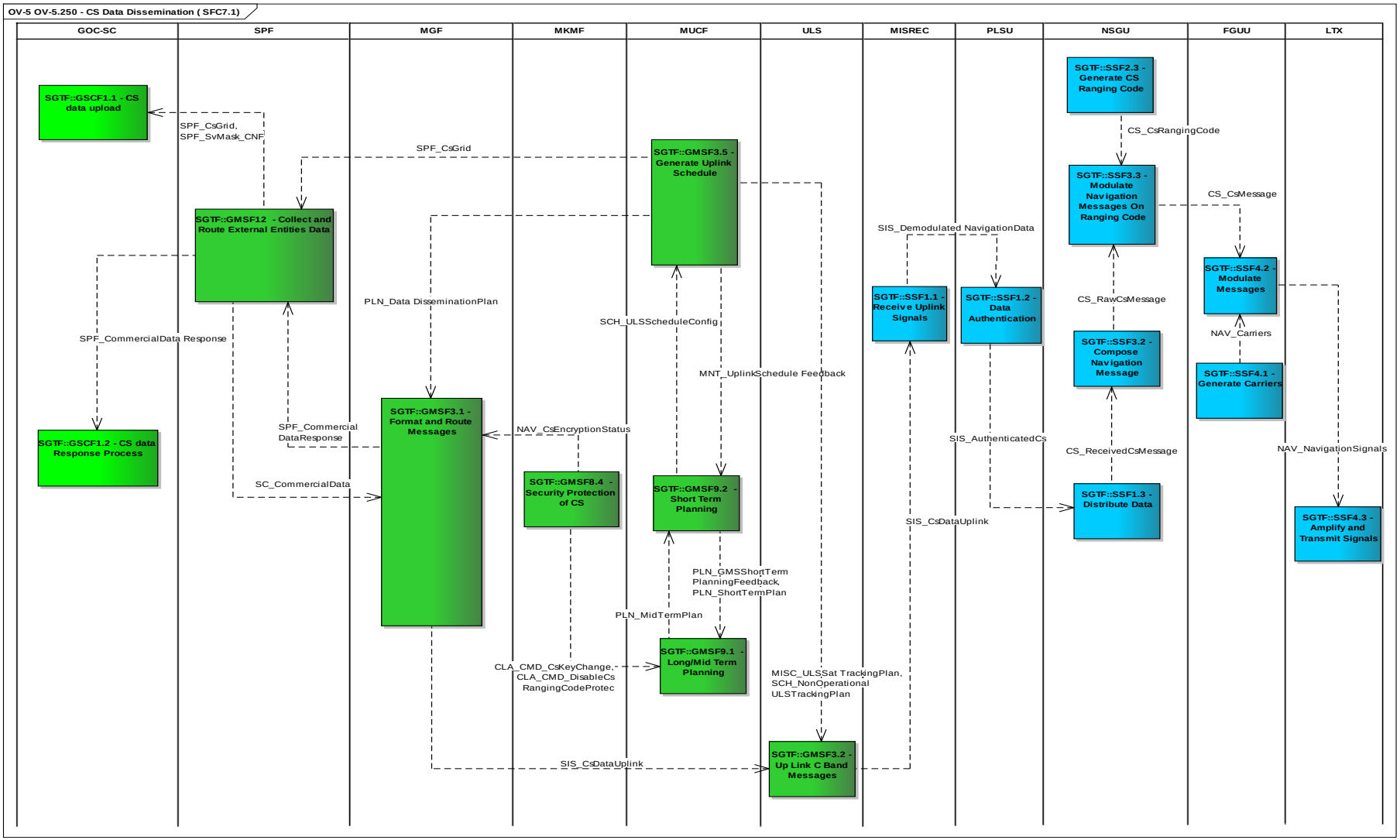
THALES
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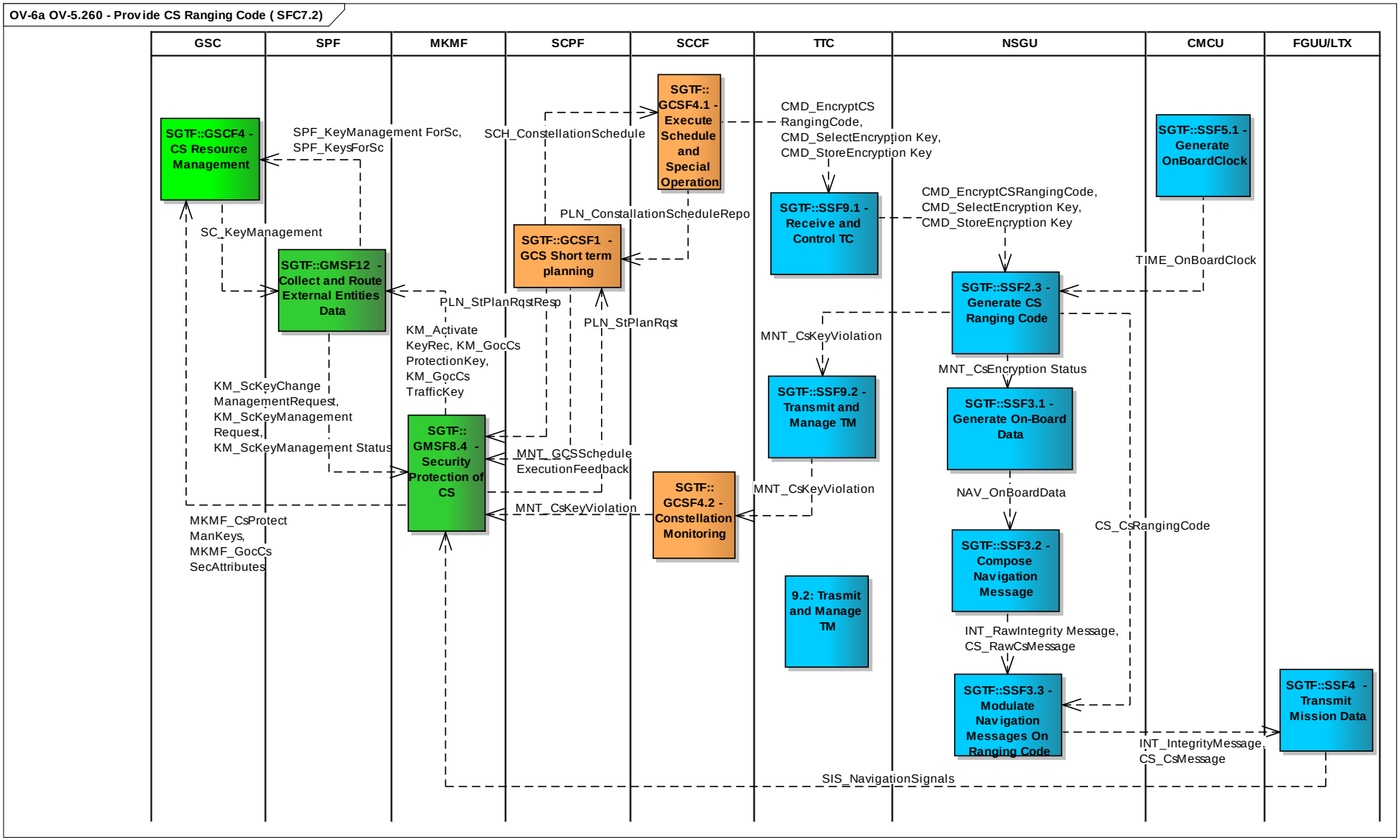
6.6 SFC6 Archiving

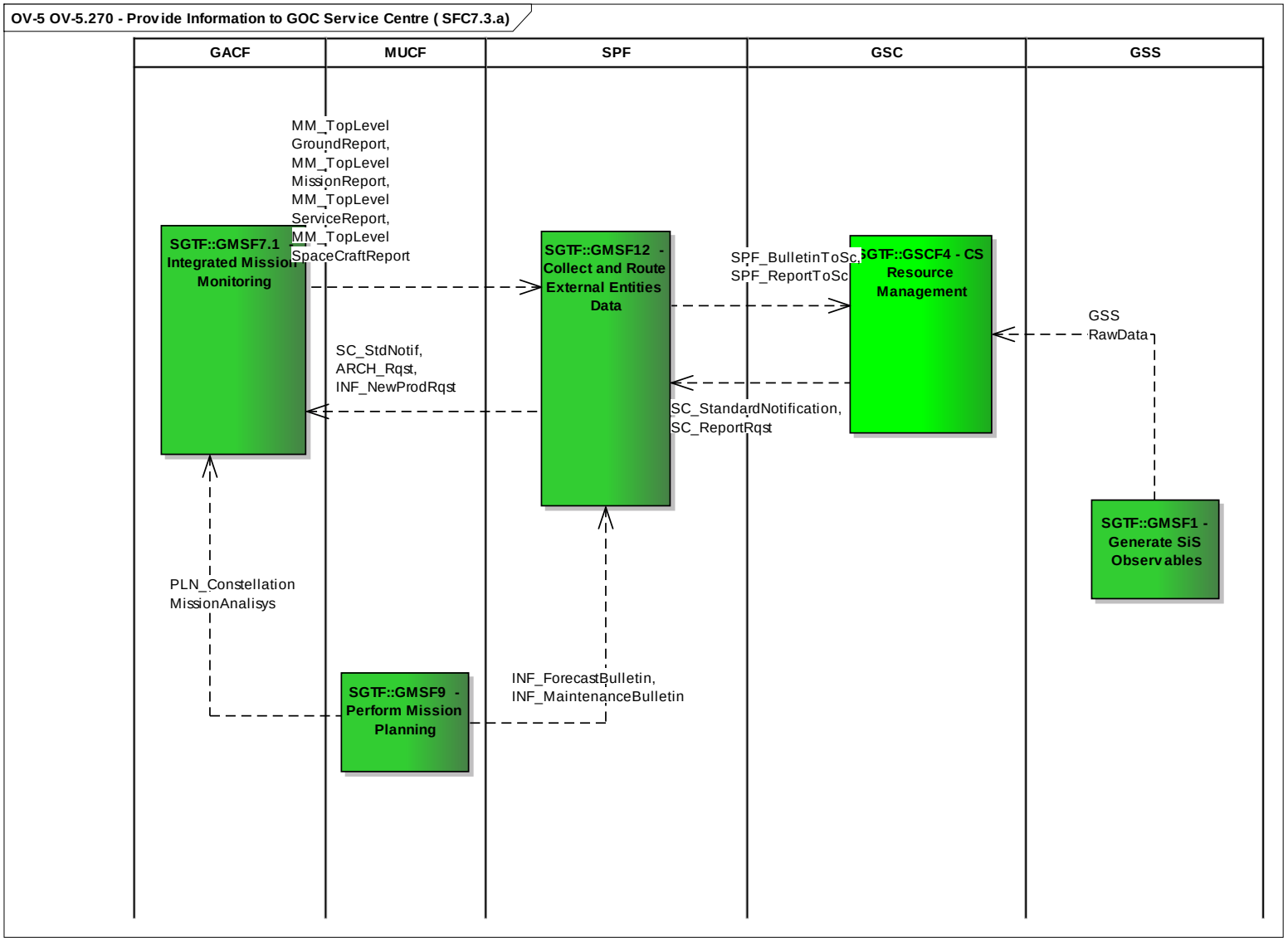


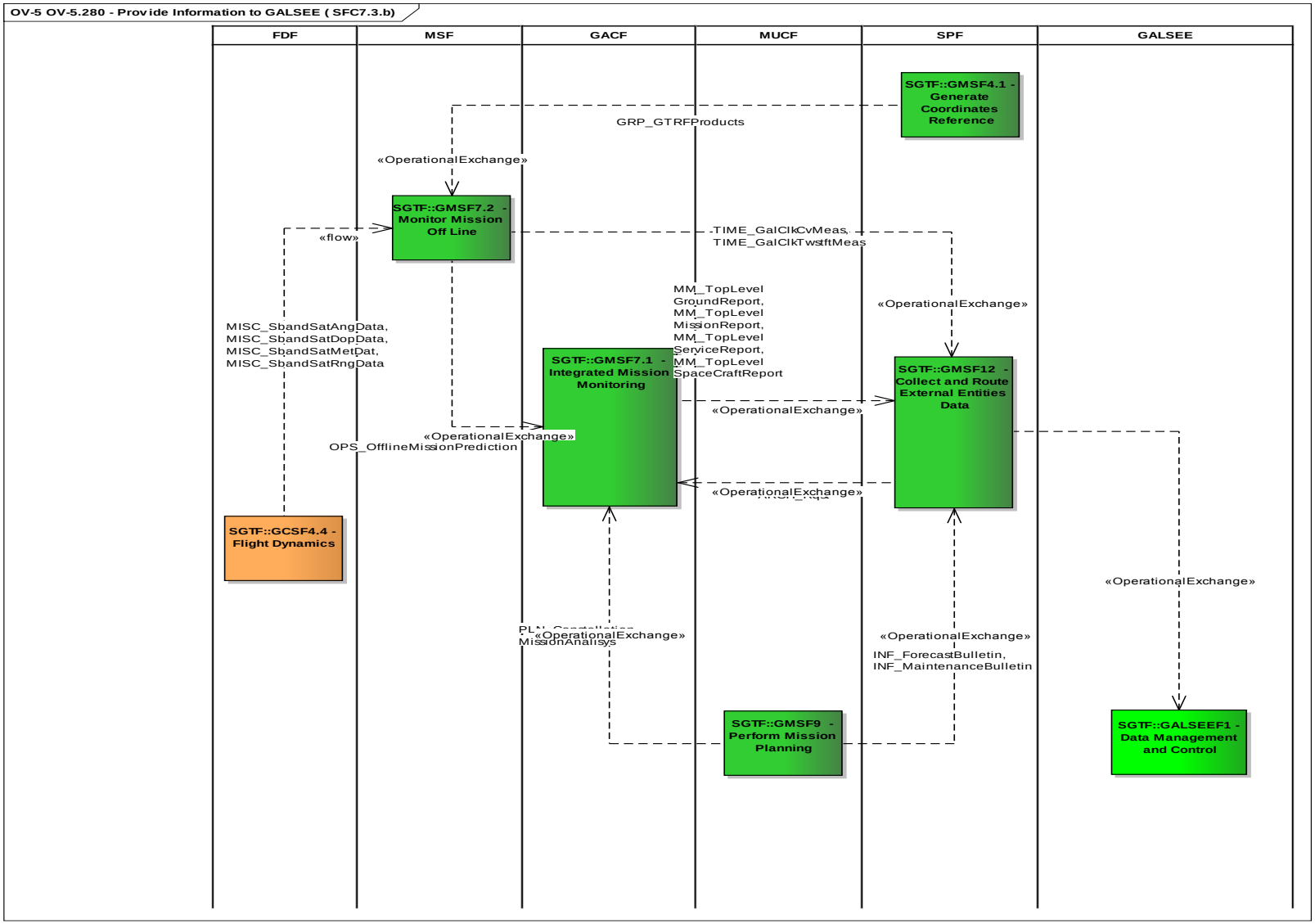

OV-5 OV-5.20 - Archive Retrieval (SFC6.2)
TTCF/SCCF
OPF/FDF/SCPF
CMCF
GACF
ULS/GSS/PTF/MUCF/SPF/ GNMF/MGF/OSPF/MSF
GNMF-s
M-KMF
SGTF::GMSF6.2 - Central Archiving:: (GACF)
SGTF::GMSF6.3 - Local Archiving:: (GNMF-s)
SGTF::GCSF3.2 - Local Archiving:: (CMCF)
SGTF::GCSF3.2 - Local Archiving:: (OPF/FDF/SCPF)
SGTF::GMSF6.3 - Local Archiving:: (MKMF)
SGTF::GCSF3.2 - Local Archiving:: (TTCF/SCCF)
SGTF::GCSF3.1 - Central Archiving
SGTF::GMSF6.3 - Local Archiving:: (ULS/GSS/PTF/ MUCF/SPF/ GNMF/MGF/ OSPF/MSF)
MUCF
SGTF::GMSF7.2 - Monitor Mission Off Line
SPF
SGTF::GMSF12 - Collect and Route External Entities Data
ARCH-RETRIEVAL
CTRL-GROUND- Request
ARCH-EVENTLOG
ARCH-RETRIEVAL, CTRL-GROUND- Request
CTRL-GROUND-Request

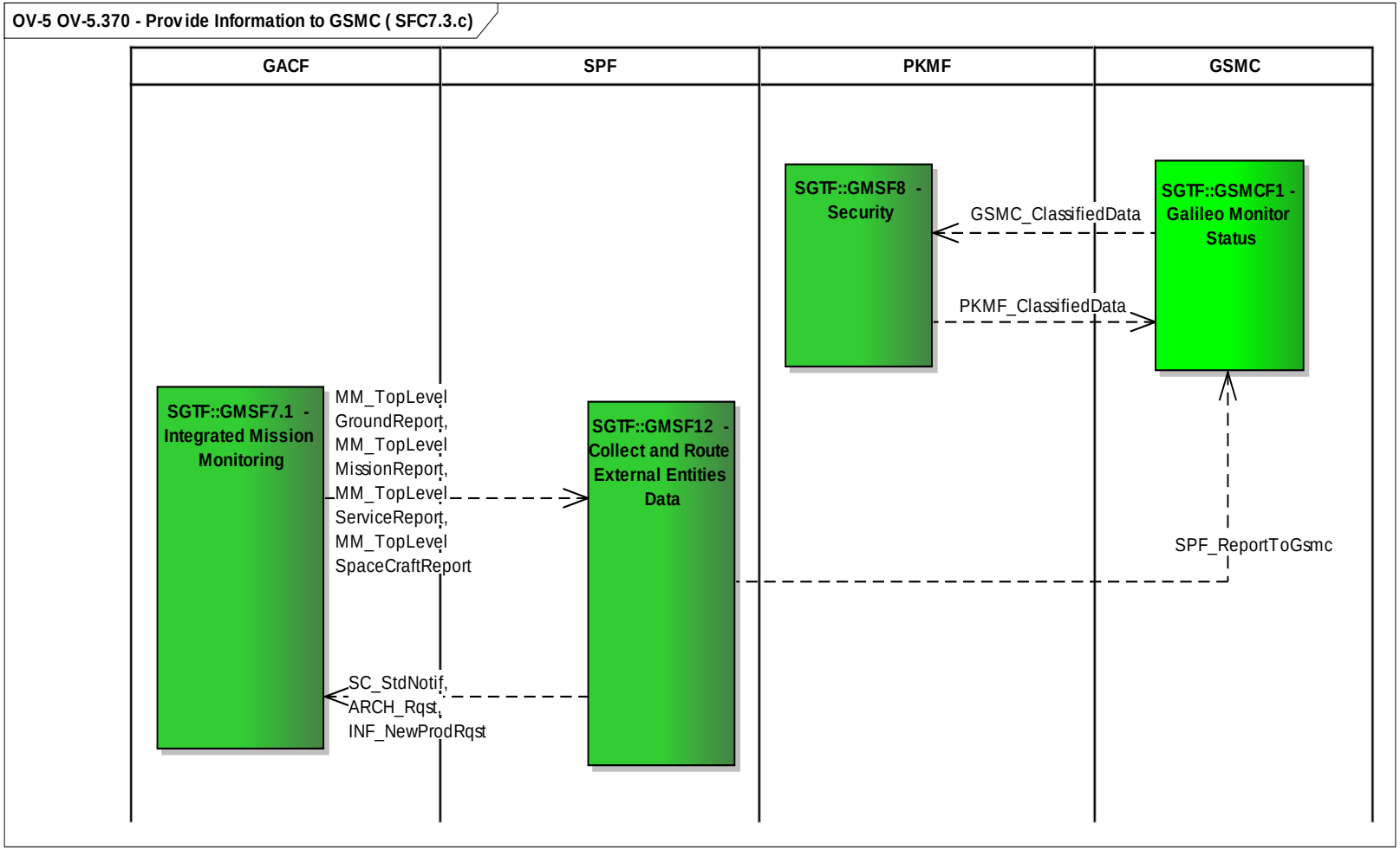
6.7 SFC7 Support Commercial Service



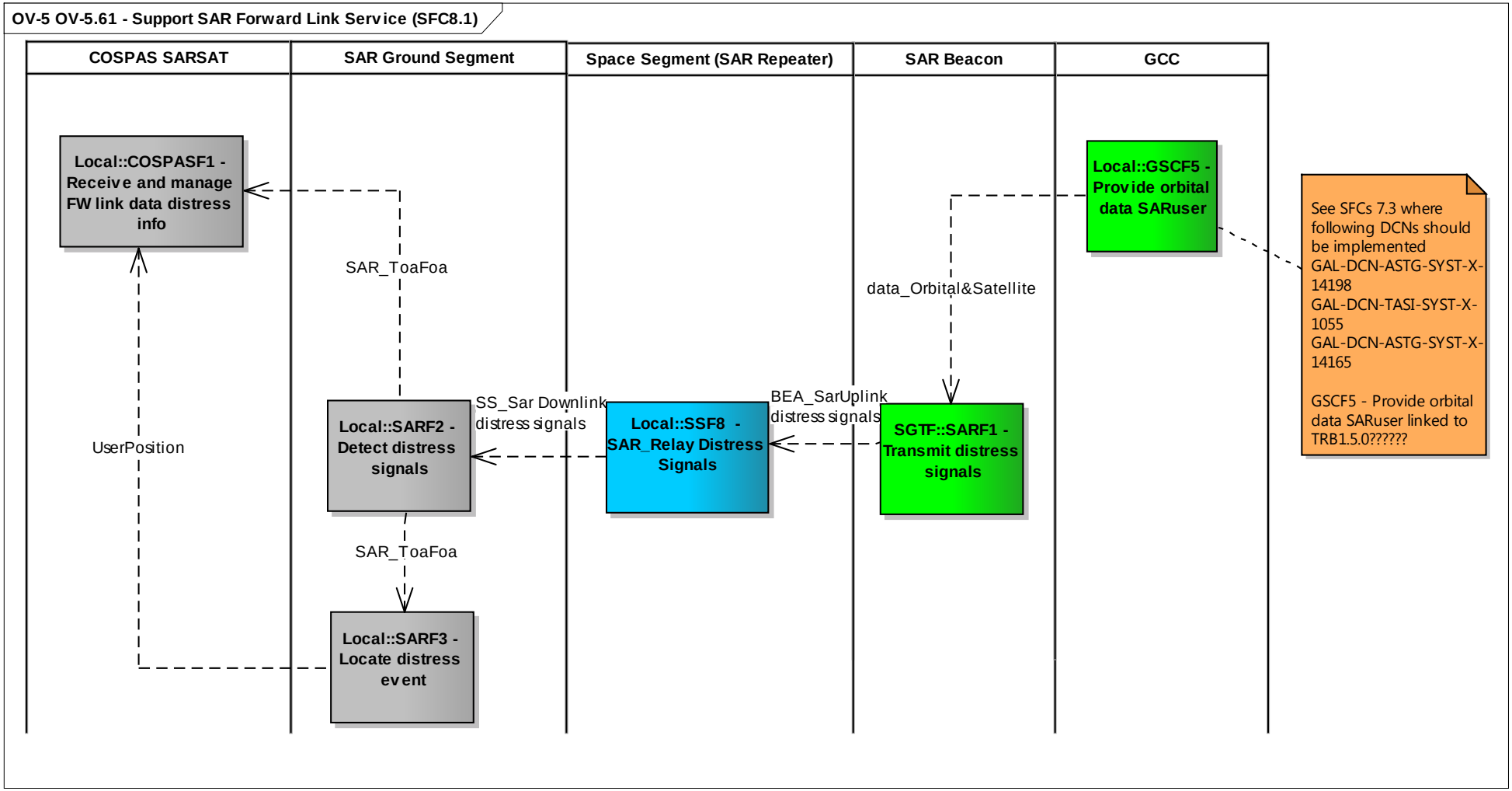

OV-6a OV-5.260 - Provide CS Ranging Code (SFC7.2)
GSC
SPF
MKMF
SCPF
SCCF
TTC
NSGU
CMCU
FGUU/LTX
SGTF::GSCF4 - CS Resource Management
SPF_KeyManagement ForSc, SPF_KeysForSc
SC_KeyManagement
SGTF::GMSF12 - Collect and Route External Entities Data
KM_ScKeyChange ManagementRequest, KM_ScKeyManagement Request, KM_ScKeyManagement Status
MKMF_CsProtect ManKeys, MKMF_GocCs SecAttributes
SGTF:: GMSF8.4 - Security Protection of CS
KM_Activate KeyRec, KM_GocCs ProtectionKey, KM_GocCs TrafficKey
PLN_StPlanRqstResp
SCH_ConstellationSchedule
SGTF::GCSF1 - GCS Short term planning
PLN_ConstallationScheduleRepo
PLN_StPlanRqst
SGTF:: GCSF4.1 - Execute Schedule and Special Operation
MNT_GCSSchedule ExecutionFeedback
MNT_CsKeyViolation
SGTF:: GCSF4.2 - Constellation Monitoring
CMD_EncryptCS RangingCode, CMD_SelectEncryption Key, CMD_StoreEncryption Key
SGTF::SSF9.1 - Receive and Control TC
MNT_CsKeyViolation
SGTF::SSF9.2 - Transmit and Manage TM
MNT_CsKeyViolation
9.2: Trasmit and Manage TM
CMD_EncryptCSRangingCode, CMD_SelectEncryption Key, CMD_StoreEncryption Key
SGTF::SSF2.3 - Generate CS Ranging Code
MNT_CsEncryption Status
SGTF::SSF3.1 - Generate On-Board Data
NAV_OnBoardData
SGTF::SSF3.2 - Compose Navigation Message
INT_RawIntegrity Message, CS_RawCsMessage
SGTF::SSF3.3 - Modulate Navigation Messages On Ranging Code
CS_CsRangingCode
SGTF::SSF5.1 - Generate OnBoardClock
TIME_OnBoardClock
INT_IntegrityMessage, CS_CsMessage
SGTF::SSF4 - Transmit Mission Data
SIS_NavigationSignals

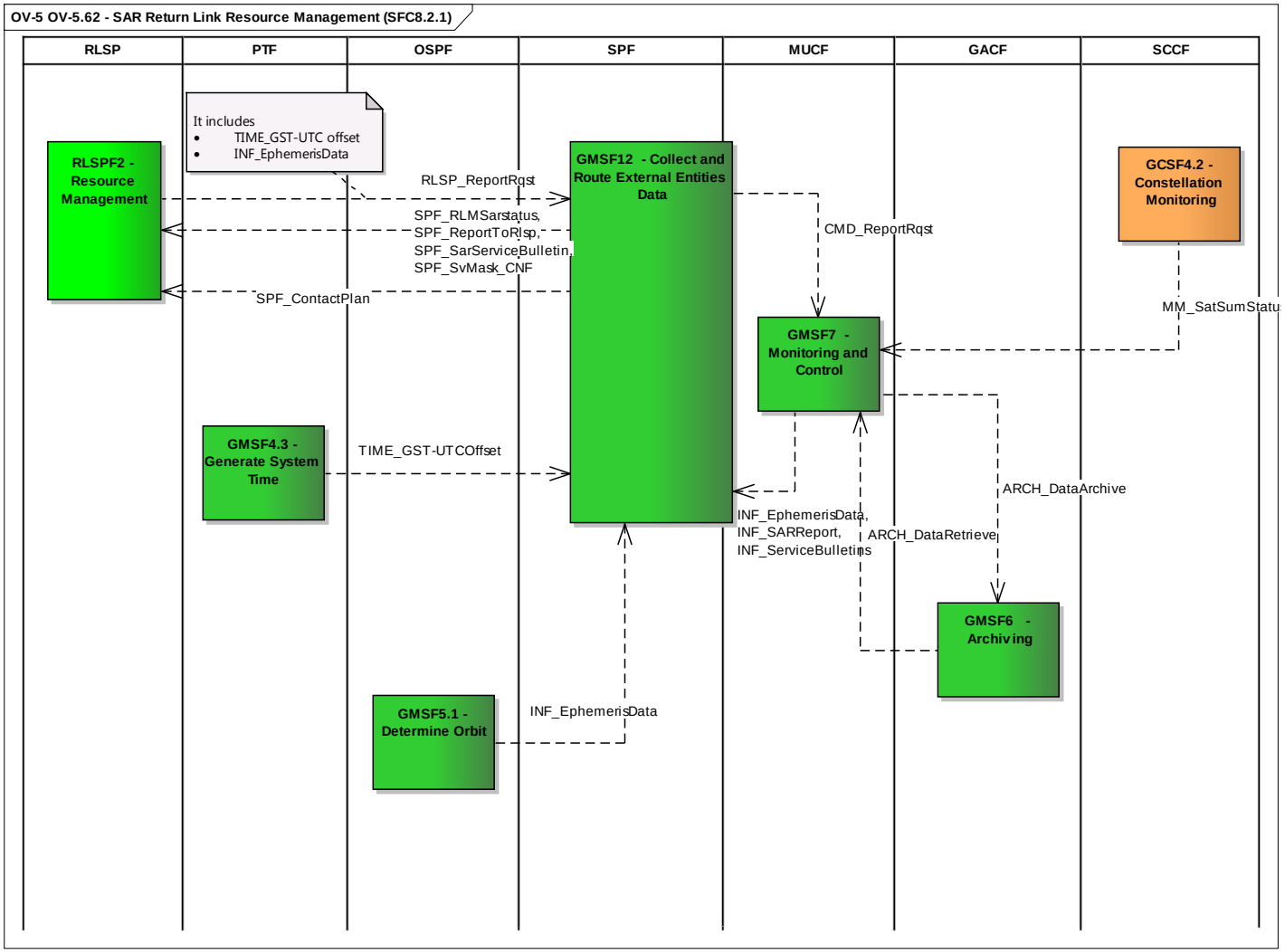

OV-5 OV-5.270 - Provide Information to GOC Service Centre (SFC7.3.a)
GACF
MUCF
SPF
GSC
GSS
MM_TopLevel GroundReport, MM_TopLevel MissionReport, MM_TopLevel ServiceReport, MM_TopLevel SpaceCraftReport
SGTF::GMSF7.1 Integrated Mission Monitoring
SGTF::GMSF12 - Collect and Route External Entities Data
SPF_BulletinToSc, SPF_ReportToSc
SGTF::GSCF4 - CS Resource Management
GSS RawData
SC_StdNotif, ARCH_Rqst, INF_NewProdRqst
SC_StandardNotification, SC_ReportRqst
SGTF::GMSF1 - Generate SiS Observables
PLN_Constellation MissionAnalisys
SGTF::GMSF9 - Perform Mission Planning
INF_ForecastBulletin, INF_MaintenanceBulletin

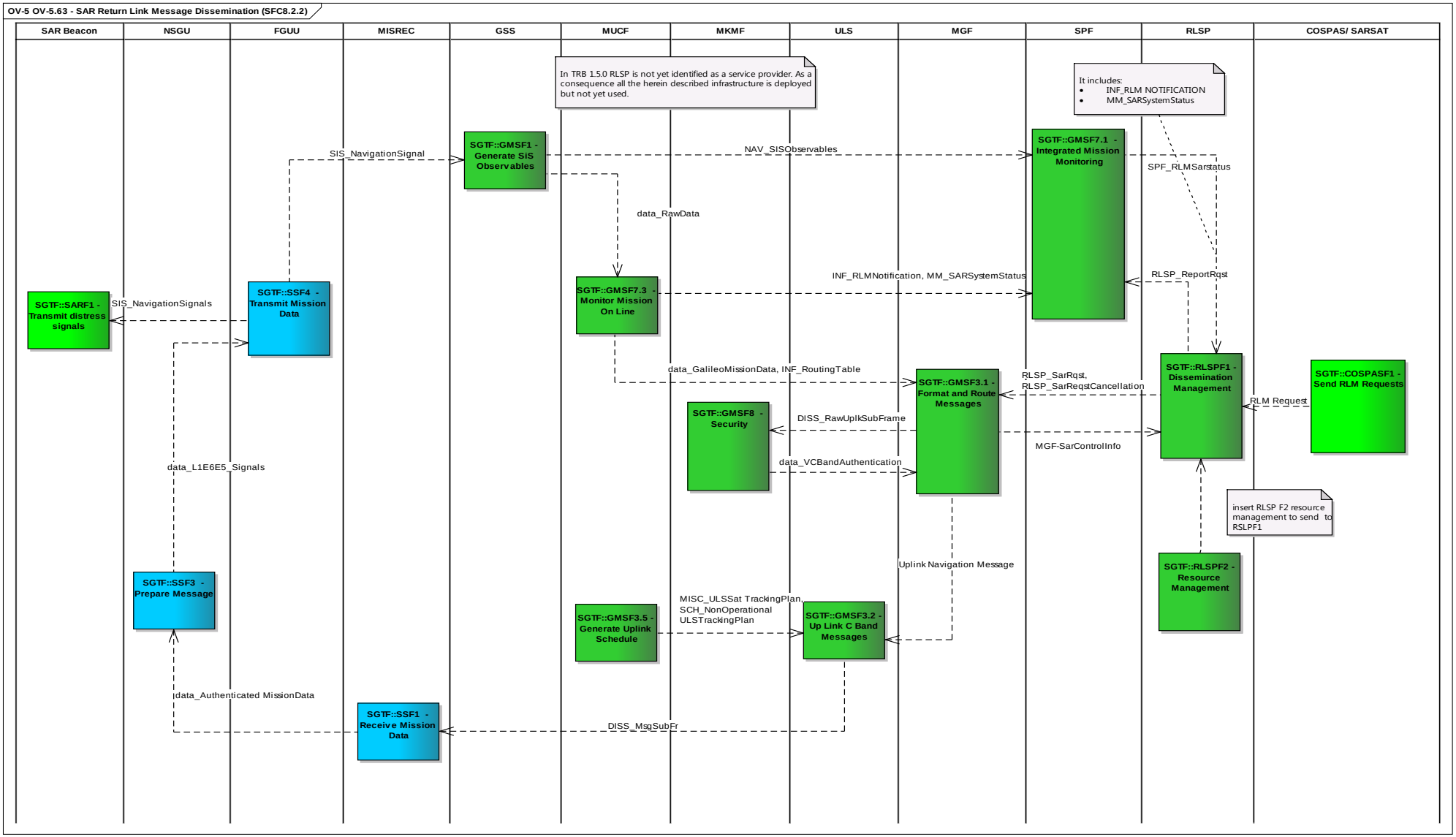

OV-5 OV-5.280 - Provide Information to GALSEE (SFC7.3.b)
FDF
MSF
GACF
MUCF
SPF
GALSEE
SGTF::GMSF4.1 - Generate Coordinates Reference
GRP_GTRFProducts
«OperationalExchange»
SGTF::GMSF7.2 - Monitor Mission Off Line
«flow»
TIME_GalClkCvMeas,
TIME_GalClkTwstftMeas
MM_TopLevel GroundReport, MM_TopLevel MissionReport, MM_TopLevel ServiceReport, MM_TopLevel SpaceCraftReport
«OperationalExchange»
MISC_SbandSatAngData, MISC_SbandSatDopData, MISC_SbandSatMetDat, MISC_SbandSatRngData
SGTF::GMSF7.1 - Integrated Mission Monitoring
SGTF::GMSF12 - Collect and Route External Entities Data
«OperationalExchange»
«OperationalExchange»
OPS_OfflineMissionPrediction
«OperationalExchange»
SGTF::GCSF4.4 - Flight Dynamics
«OperationalExchange»
«OperationalExchange»
MissionAnalisys
«OperationalExchange»
INF_ForecastBulletin, INF_MaintenanceBulletin
SGTF::GMSF9 - Perform Mission Planning
SGTF::GALSEEF1 - Data Management and Control


OV-5 OV-5.370 - Provide Information to GSMC (SFC7.3.c)
GACF
SPF
PKMF
GSMC
SGTF::GMSF8 - Security
SGTF::GSMCF1 - Galileo Monitor Status
GSMC_ClassifiedData
PKMF_ClassifiedData
SGTF::GMSF7.1 - Integrated Mission Monitoring
MM_TopLevel GroundReport, MM_TopLevel MissionReport, MM_TopLevel ServiceReport, MM_TopLevel SpaceCraftReport
SGTF::GMSF12 - Collect and Route External Entities Data
SPF_ReportToGsmc
SC_StdNotif, ARCH_Rqst, INF_NewProdRqst

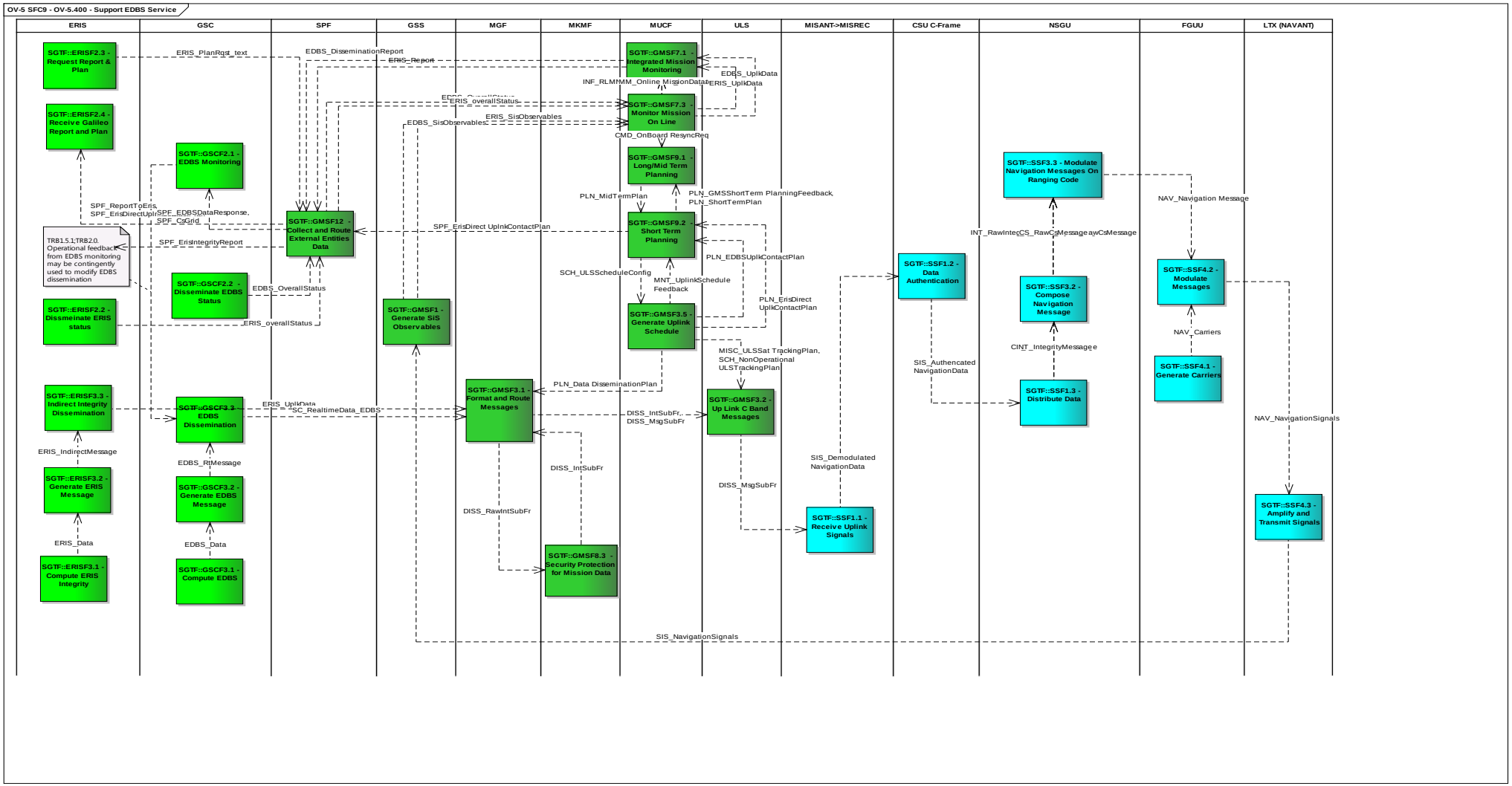
6.8 SFC8 Provide SAR Service




OV-5 OV-5.62 - SAR Return Link Resource Management (SFC8.2.1)
RLSP
PTF
OSPF
SPF
MUCF
GACF
SCCF
It includes
TIME_GST-UTC offset
INF_EphemerisData
RLSPF2 - Resource Management
GMSF12 - Collect and Route External Entities Data
GCSF4.2 - Constellation Monitoring
RLSP_ReportRqst
SPF_RLMSarstatus,
SPF_ReportToRlsp,
SPF_SarServiceBulletin,
SPF_SvMask_CNF
SPF_ContactPlan
CMD_ReportRqst
MM_SatSumStatu
GMSF7 - Monitoring and Control
GMSF4.3 - Generate System Time
TIME_GST-UTCOffset
ARCH_DataArchive
INF_EphemerisData,
INF_SARReport,
INF_ServiceBulletins
ARCH_DataRetrieve
GMSF6 - Archiving
GMSF5.1 - Determine Orbit
INF_EphemerisData


OV-5 OV-5.63 - SAR Return Link Message Dissemination (SFC8.2.2)
SAR Beacon
NSGU
FGUU
MISREC
GSS
MUCF
MKMF
ULS
MGF
SPF
RLSP
COSPAS/ SARSAT
In TRB 1.5.0 RLSP is not yet identified as a service provider. As a consequence all the herein described infrastructure is deployed but not yet used.
It includes:
INF_RLM NOTIFICATION
MM_SARSystemStatus
SGTF::GMSF1 - Generate SiS Observables
SGTF::GMSF7.1 - Integrated Mission Monitoring
SIS_NavigationSignal
NAV_SISObservables
SPF_RLMSarstatus
data_RawData
SGTF::GMSF7.3 - Monitor Mission On Line
INF_RLMNotification, MM_SARSystemStatus
RLSP_ReportRqst
SGTF::SARF1 - Transmit distress signals
SIS_NavigationSignals
SGTF::SSF4 - Transmit Mission Data
data_GalileoMissionData, INF_RoutingTable
SGTF::GMSF3.1 - Format and Route Messages
RLSP_SarRqst, RLSP_SarReqstCancellation
SGTF::RLSPF1 - Dissemination Management
SGTF::COSPASF1 - Send RLM Requests
RLM Request
SGTF::GMSF8 - Security
DISS_RawUplkSubFrame
MGF-SarControlInfo
data_VCBandAuthentication
data_L1E6E5_Signals
insert RLSP F2 resource management to send to RSLPF1
Uplink Navigation Message
SGTF::RLSPF2 - Resource Management
SGTF::SSF3 - Prepare Message
MISC_ULSSat TrackingPlan, SCH_NonOperational ULSTrackingPlan
SGTF::GMSF3.5 - Generate Uplink Schedule
SGTF::GMSF3.2 - Up Link C Band Messages
data_Authenticated MissionData
SGTF::SSF1 - Receive Mission Data
DISS_MsgSubFr

6.9 SFC9 Support EDBS Service



END OF DOCUMENT